Eclipses by Appointment - Supplement

Volume 3 of 4

R410-R539

Original continuous pages 1948-2450 of 3269

Navigation: each volume adds this one-page guide.

A page in this volume = original page minus 1947 plus 1.

The public supplement-review-pages.csv maps every R key to volume

and exact page. The scientific-review-register.csv maps status to report.

Volume 1: Original A catalogue and R001-R216 (original pages 1-1186).

Volume 2: Fixed magnitude comparison and R217-R409 (original pages 1187-1947).

Volume 3: R410-R539 (original pages 1948-2450).

Volume 4: R540-R675 and remaining reviewed evidence (original pages 2451-3269).

Original illustrated A entries and all R full reports remain in the

continuous source HTML and the public reports/evidence archive.

Cross-volume references use the page map above.

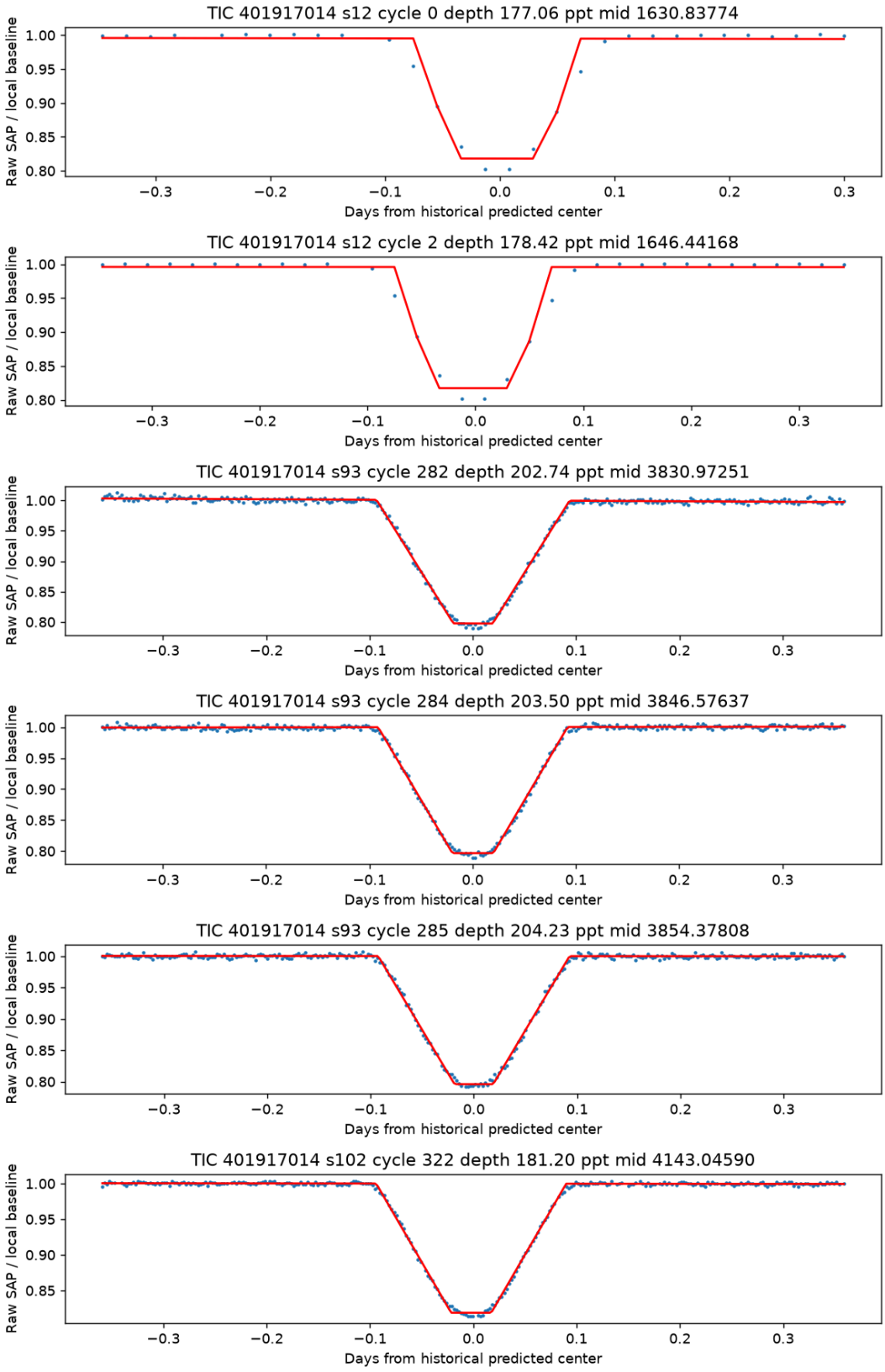
R410. TIC 401917014

physical_verdict: CONFIRMED

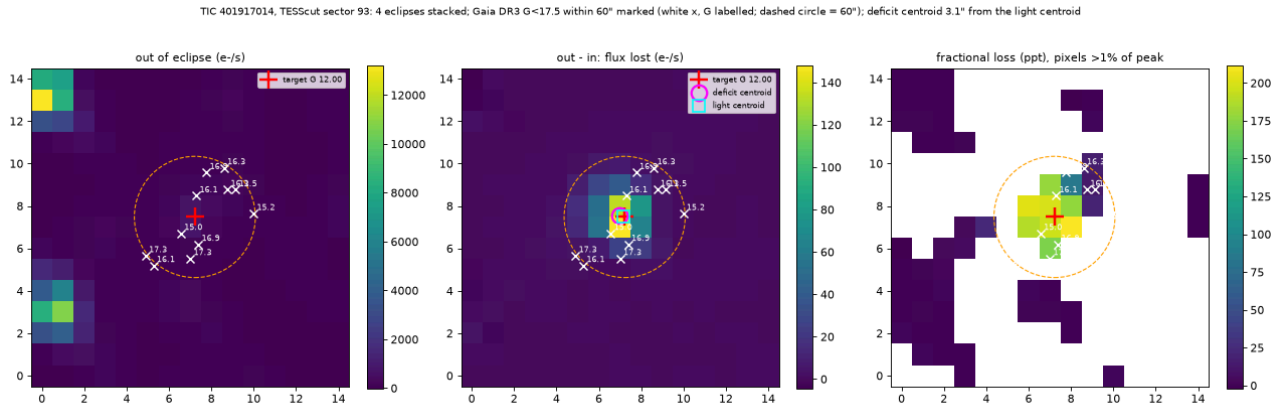
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R410; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R410; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 401917014: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured eclipses in representative sectors [12, 93, 102]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 8 actual midpoints yield P = 7.8018659062 ± 0.0000181348 d and reference epoch BTJD 1630.8460077 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 93 cutout puts the deficit 4.91 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.577; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 5796293562665379456: separation 46.11 arcsec, G 13.487, approximate intrinsic eclipse requirement 0.923. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 196.20/192.24/161.00 ppt in 4 stacked events; median selected QLP raw fit depth 191.97 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5796293459586158976 has orbital P = 7.8019703401 d, e = 0.02120; closest edge-on conjunction offset at the measured epoch is -0.02604 d (-0.00334 phase), with 0.13110 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.66 h versus observed median 4.41 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Eight deep raw eclipses across S12/93/102; capable G-band neighbor 46.1 arcsec away requires an extreme intrinsic loss, while the deficit core is target-centered. Pixel residual limits subpixel claims.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	0	1630.8377438	0.010417	177.06	used
12	2	1646.4416770	0.010417	178.42	used
93	282	3830.9725084	0.001157	202.74	used
93	283	3838.7746987	0.001157	205.41	used
93	284	3846.5763723	0.001157	203.50	used
93	285	3854.3780807	0.001157	204.23	used
102	321	4135.2443979	0.001157	180.80	used
102	322	4143.0459048	0.001157	181.20	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s93.json, historical/diff_s93.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R410. Public evidence files: diff_s93.json, diff_s93.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

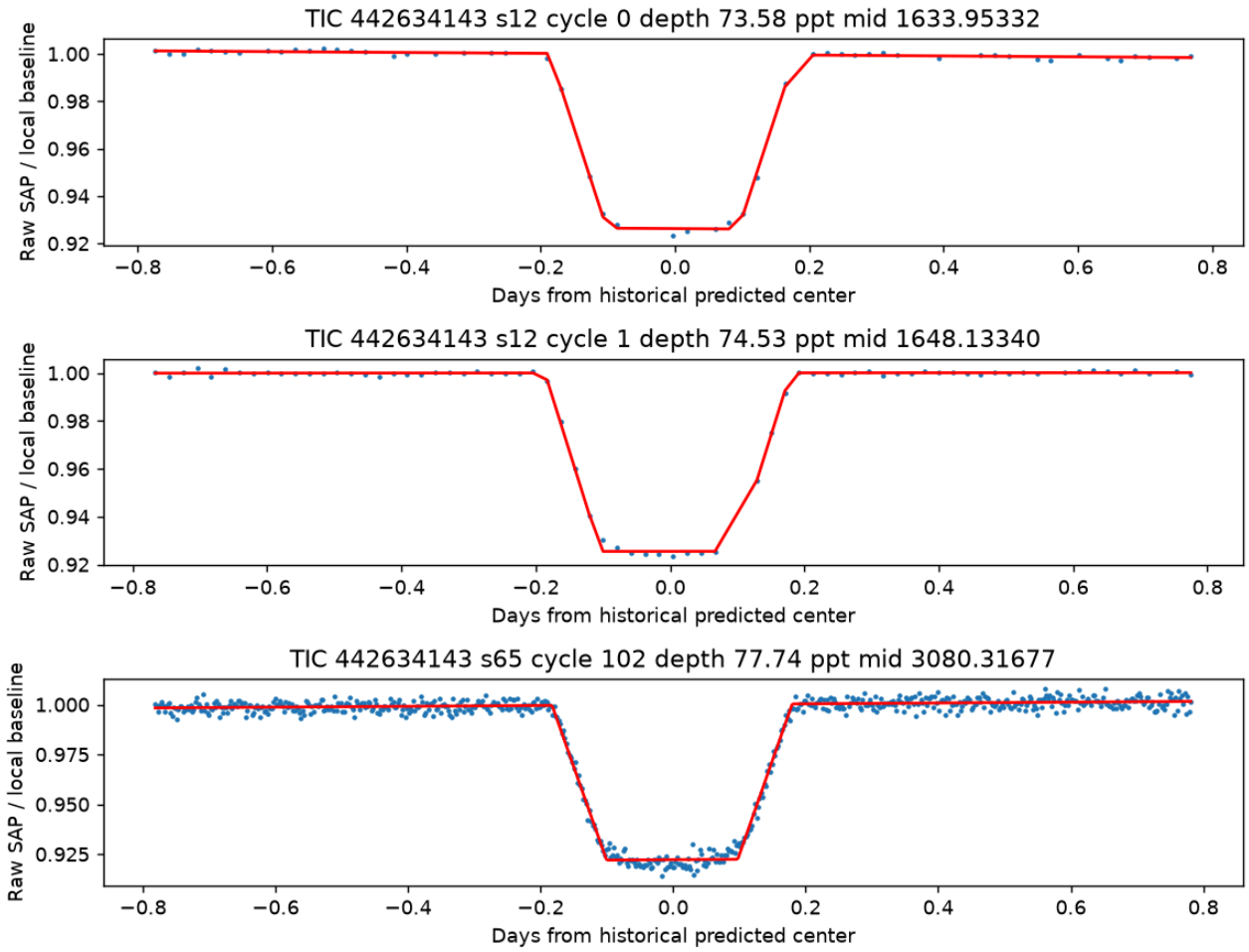
R411. TIC 442634143

physical_verdict: CONFIRMED

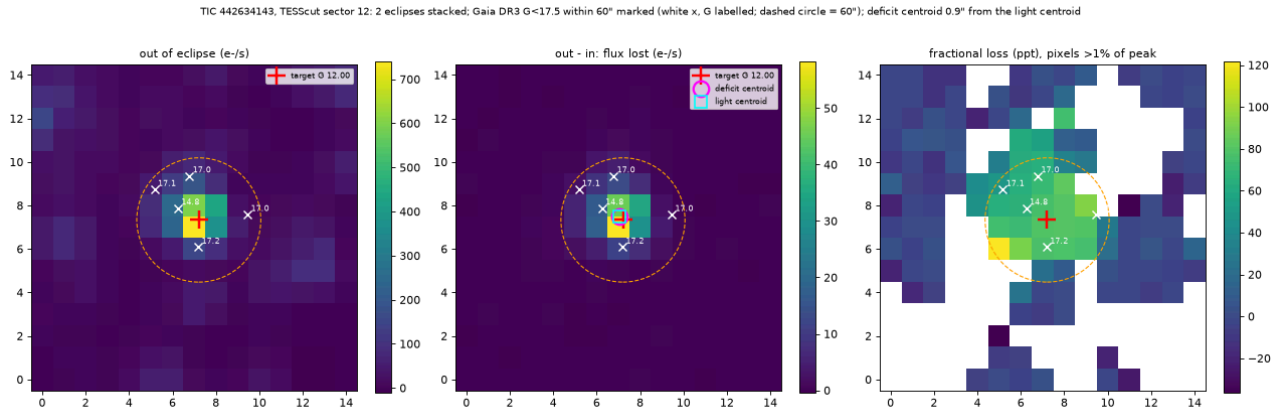
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R411; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R411; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 442634143: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 3 timing-usable measured eclipses in representative sectors [12, 65]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 3 actual midpoints yield P = 14.1800335439 ± 0.0000734560 d and reference epoch BTJD 1633.9533458 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 12 cutout puts the deficit 4.58 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.123; this is not a calibrated localization error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 78.72/77.08/75.71 ppt in 2 stacked events; median selected QLP raw fit depth 74.53 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 6040478190203632640 has orbital P = 14.1830337728 d, e = 0.14754; closest edge-on conjunction offset at the measured epoch is -0.20084 d (-0.01416 phase), with 0.38892 d linearly propagated full-SB1-covariance timing error. The rough central duration is 6.68 h versus observed median 8.78 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Three deep raw eclipses in S12/65. Observed duration exceeds a rough central radius/minimum-mass estimate; the estimate is model dependent and does not resolve detailed binary geometry.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	0	1633.9533229	0.010416	73.58	used
12	1	1648.1334025	0.010416	74.53	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
65	102	3080.3167673	0.001157	77.74	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s12.json, historical/diff_s12.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R411. Public evidence files: diff_s12.json, diff_s12.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

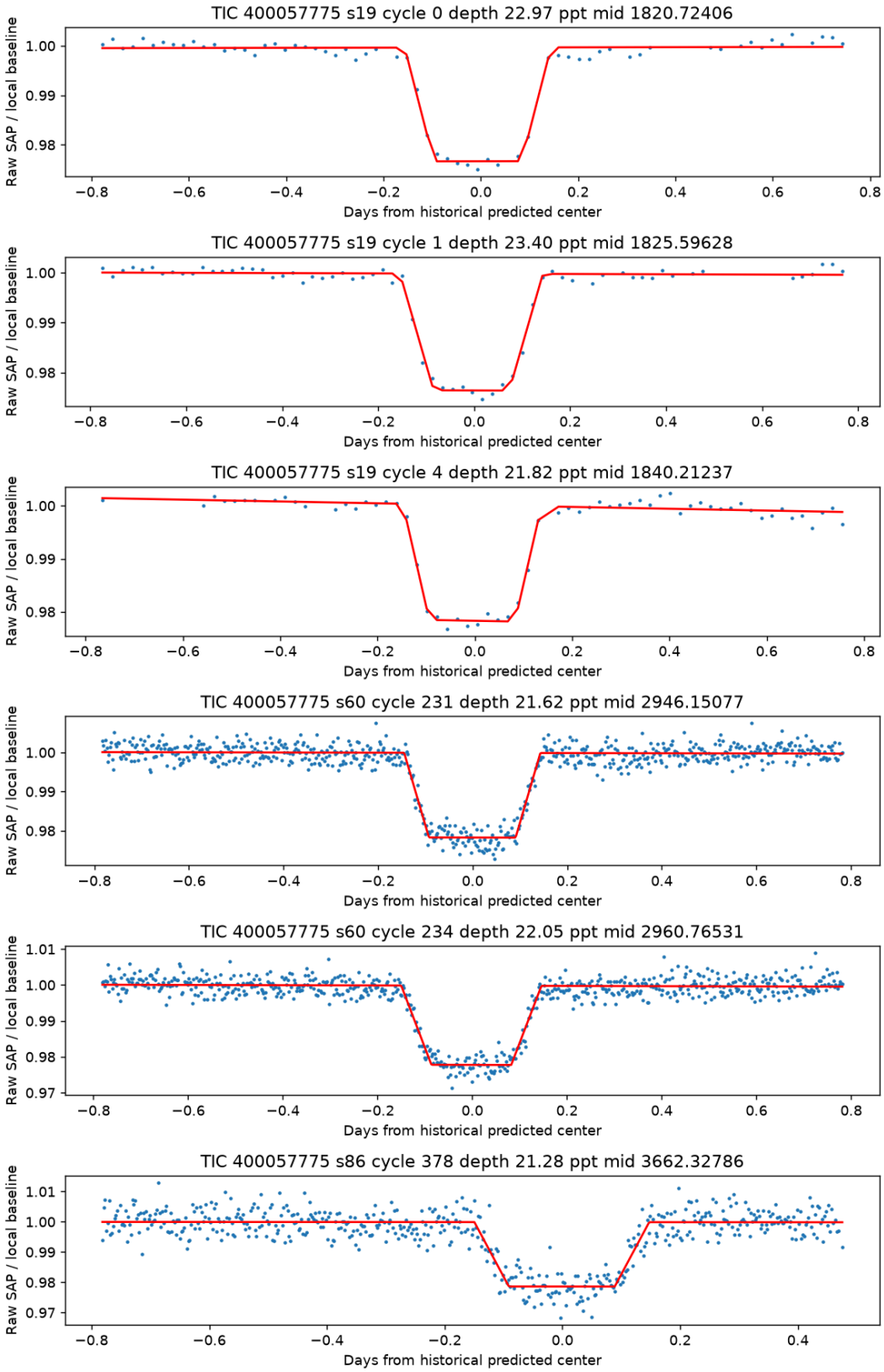
R412. TIC 400057775

physical_verdict: CONFIRMED

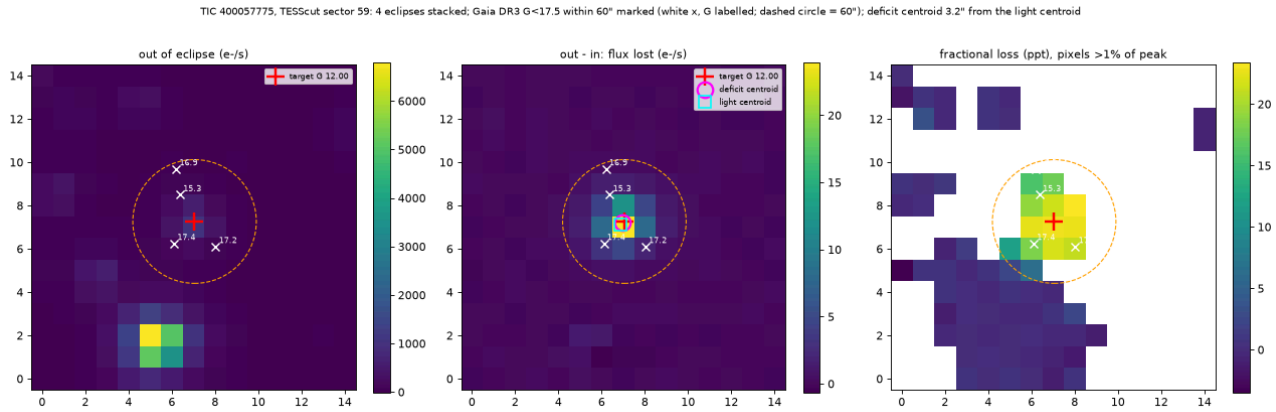
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R412; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R412; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 400057775: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 10 timing-usable measured eclipses in representative sectors [19, 60, 86]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 10 actual midpoints yield P = 4.8719600113 ± 0.0000070752 d and reference epoch BTJD 1820.7274949 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 59 cutout puts the deficit 1.73 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.822; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 480689479890567936: separation 29.38 arcsec, G 15.294, approximate intrinsic eclipse requirement 0.466. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 22.99/22.01/20.61 ppt in 4 stacked events; median selected QLP raw fit depth 21.93 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 480689509953048960 has orbital P = 4.8719149116 d, e = 0.08656; closest edge-on conjunction offset at the measured epoch is -0.01917 d (-0.00393 phase), with 0.12594 d linearly propagated full-SB1-covariance timing error. The rough central duration is 8.83 h versus observed median 7.09 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Ten repeated raw eclipses S19/60/86. A capable neighbor 29.4 arcsec away is distinct from the target-centered deficit; high image-model residual limits precision.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
19	0	1820.7240571	0.010417	22.97	used
19	1	1825.5962845	0.010417	23.40	used
19	3	1835.3436536	0.010417	23.15	used
19	4	1840.2123729	0.010417	21.82	used
60	230	2941.2790493	0.001157	21.44	used
60	231	2946.1507725	0.001157	21.62	used
60	233	2955.8940093	0.001157	22.42	used
60	234	2960.7653054	0.001157	22.05	used
86	375	3647.7126339	0.001157	21.72	used
86	378	3662.3278567	0.001562	21.28	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s59.json, historical/diff_s59.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R412. Public evidence files: diff_s59.json, diff_s59.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

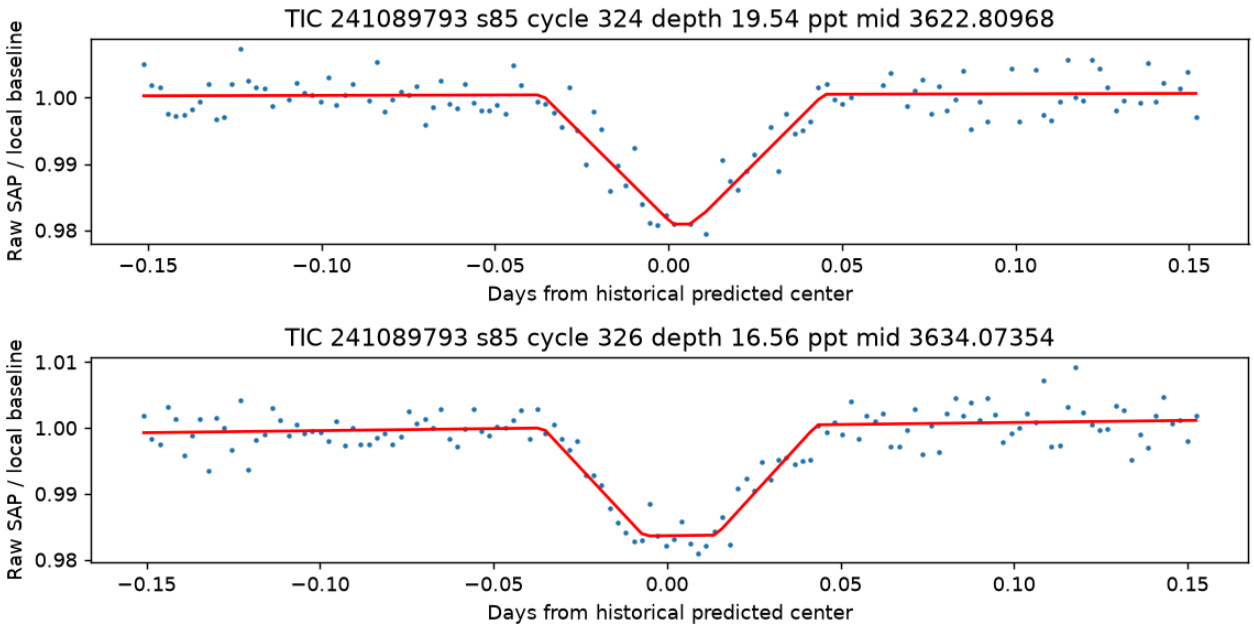
R413. TIC 241089793

physical_verdict: CONFIRMED

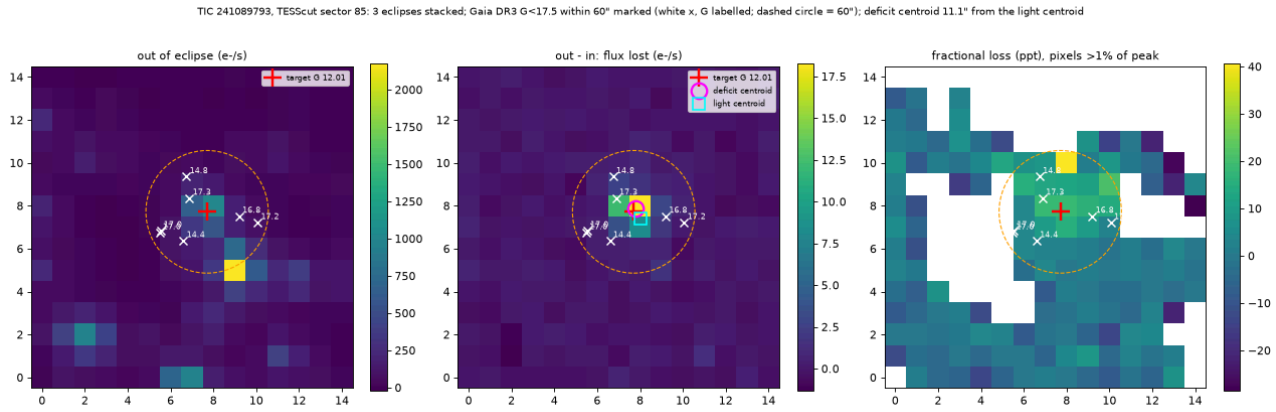
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R413; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R413; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 241089793: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured eclipses in representative sectors [18, 85]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 5 actual midpoints yield P = 5.6321610433 ± 0.0000186954 d and reference epoch BTJD 1792.3571061 (repeating eclipse; primary by convention). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear fit to independently measured raw S18 and S85 midpoints, with max(formal center covariance, half cadence) per event and scatter inflation; formal fit error, not total calibrated uncertainty. Historical period/cycle predictions guided extraction only. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 85 cutout puts the deficit 3.60 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.847; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 408289628502534144: separation 36.59 arcsec, G 14.375, approximate intrinsic eclipse requirement 0.166; Gaia 408289727280756608: separation 39.76 arcsec, G 14.819, approximate intrinsic eclipse requirement 0.242. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 18.09/15.54/10.73 ppt in 3 stacked events; median selected QLP raw fit depth 17.95 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 408289727280589312 has orbital P = 5.6321186642 d, e = 0.02967; closest edge-on conjunction offset at the measured epoch is -0.03436 d (-0.00610 phase), with 0.12821 d linearly propagated full-SB1-covariance timing error. The rough central duration is 4.30 h versus observed median 1.88 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: S18 clean raw eclipses at successive cycles -1 and 0 plus cycle 2 test the 5.632-day recurrence; measured S85 cycles 324/326 extend the timing baseline. S85 cycle 325 and S18 cycle 1 are flagged, not clean nulls. S18 partial cycle 3 is excluded from timing. The formal period error is not total calibrated uncertainty.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
18	-1	1792.3580780	0.010420	16.04	used
18	0	1797.9886680	0.010420	17.95	used
18	2	1809.2530956	0.010420	18.69	used
18	3	1814.8824888	0.010420	19.46	excluded: Partial coverage; excluded from joint timing fit.
85	324	3622.8096759	0.001157	19.54	used
85	326	3634.0735380	0.001157	16.56	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s85.json, historical/diff_s85.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). This target also has ../241089793-refined-timing.json, ../241089793-alias-coverage.json and s18-cycle-check.png. Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R413. Public evidence files: diff_s85.json, diff_s85.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv, s18-cycle-check.png. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

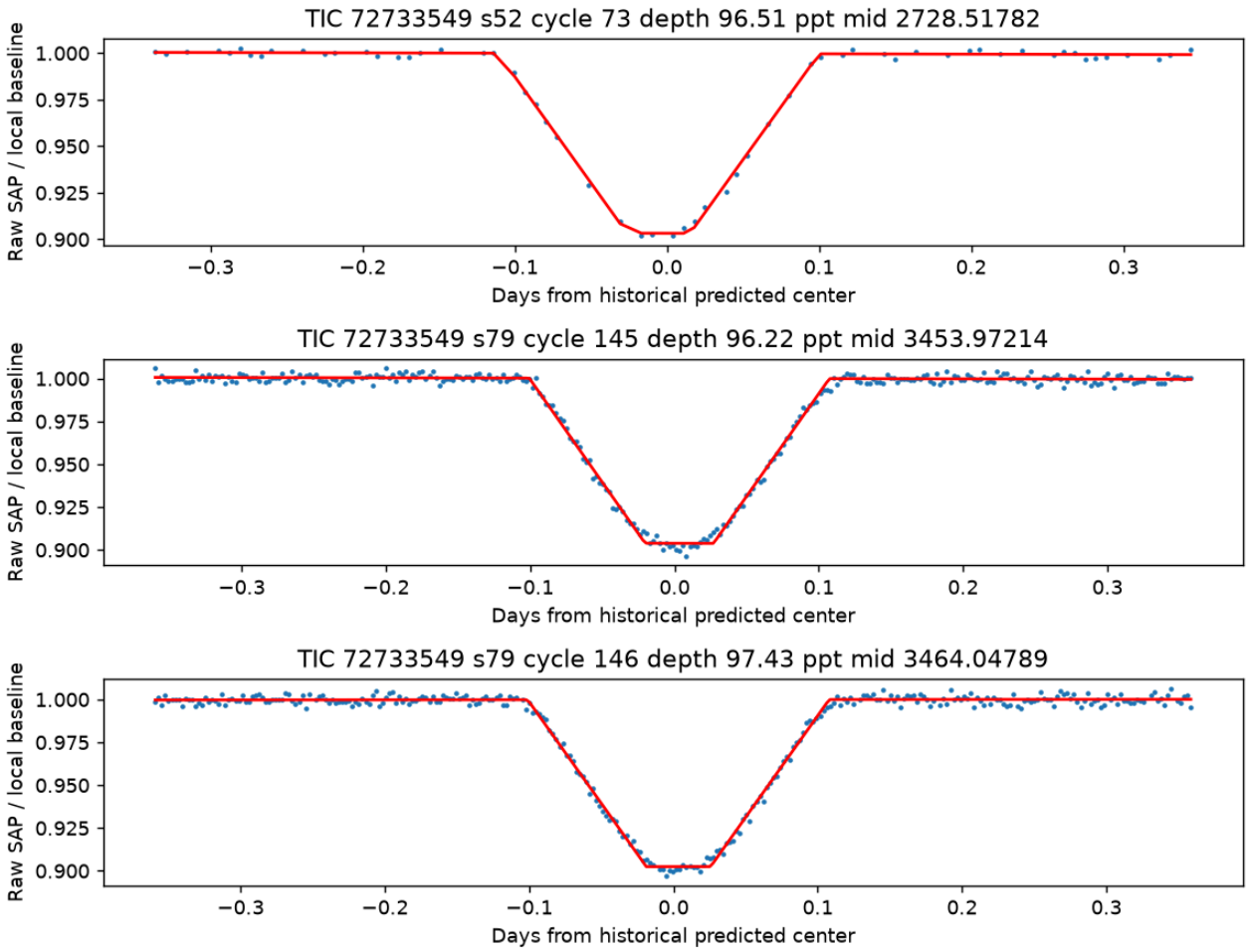
R414. TIC 72733549

physical_verdict: CONFIRMED

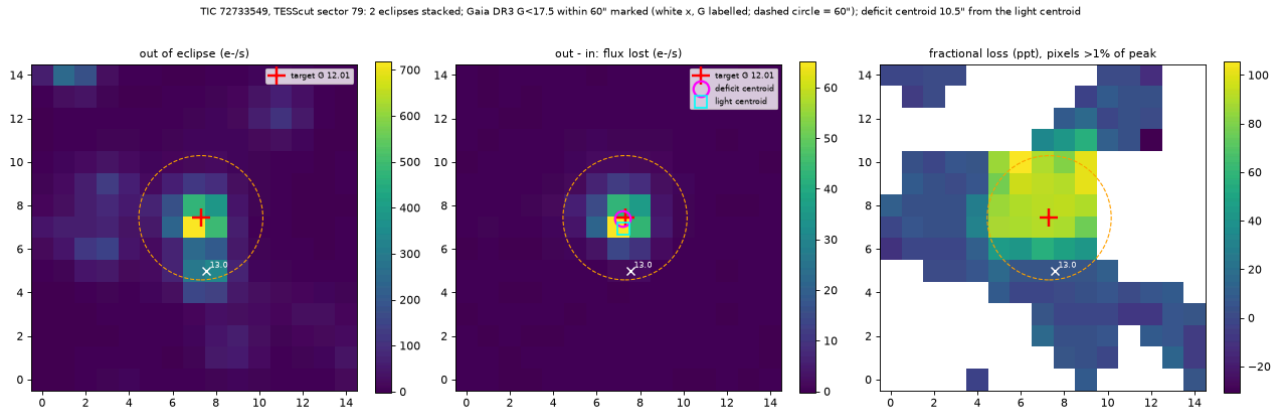
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R414; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R414; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 72733549: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 3 timing-usable measured eclipses in representative sectors [52, 79]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 3 actual midpoints yield P = 10.0757543600 ± 0.0000962802 d and reference epoch BTJD 2728.5178240 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 79 cutout puts the deficit 2.70 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.430; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 1351135967600649984: separation 48.12 arcsec, G 12.966, approximate intrinsic eclipse requirement 0.312. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 90.72/83.62/69.70 ppt in 2 stacked events; median selected QLP raw fit depth 96.51 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1351133729921222400 has orbital P = 10.0743628417 d, e = 0.01363; closest edge-on conjunction offset at the measured epoch is 0.28926 d (0.02871 phase), with 0.46418 d linearly propagated full-SB1-covariance timing error. The rough central duration is 7.25 h versus observed median 5.01 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Three deep raw eclipses S52/79, including successive S79 cycles. A capable neighbor 48.1 arcsec away is distinct from the target-centered deficit.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
52	73	2728.5178238	0.006944	96.51	used
79	145	3453.9721385	0.001157	96.22	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
79	146	3464.0478918	0.001157	97.43	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s79.json, historical/diff_s79.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R414. Public evidence files: diff_s79.json, diff_s79.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

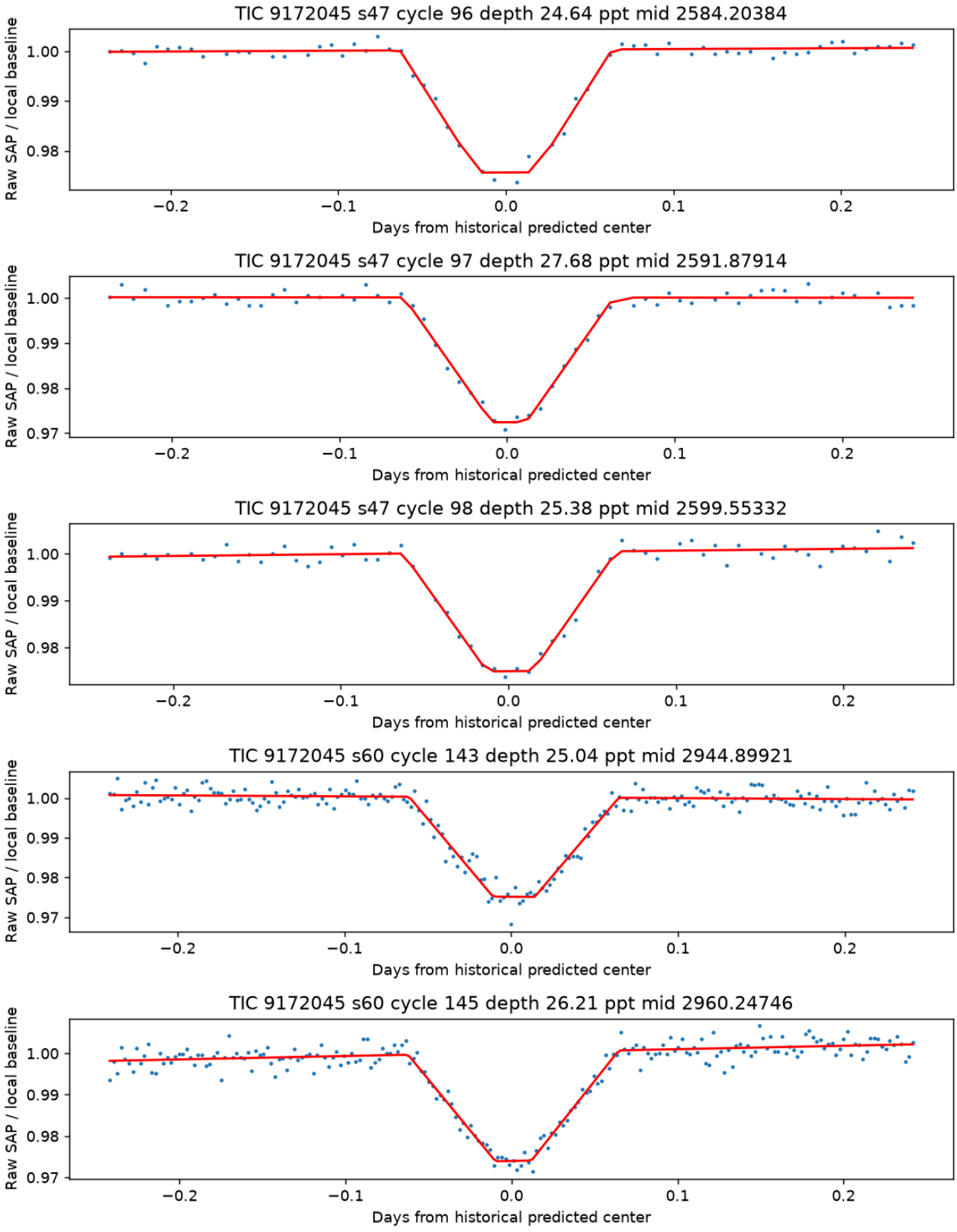
R415. TIC 9172045

physical_verdict: CONFIRMED

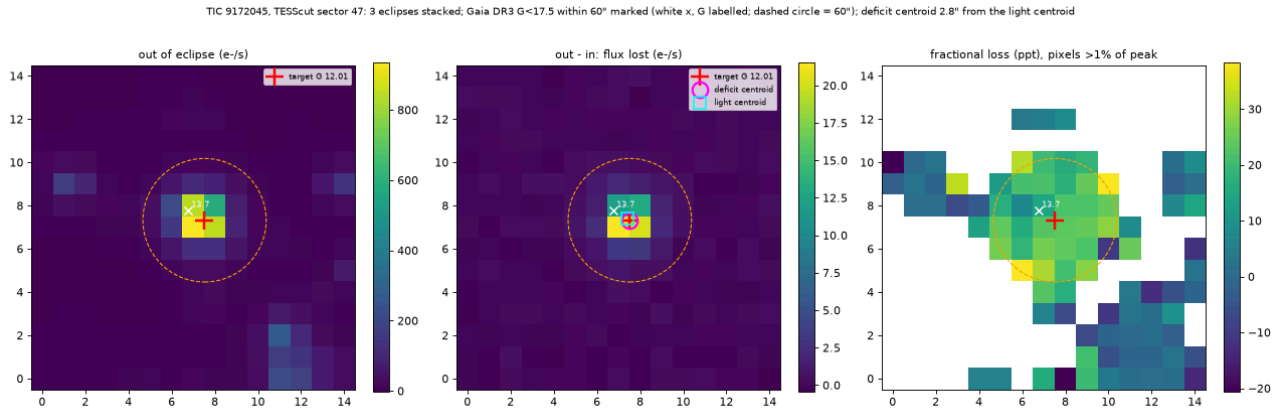
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R415; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R415; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 9172045: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured eclipses in representative sectors [47, 60]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 5 actual midpoints yield P = 7.6743519360 ± 0.0000459896 d and reference epoch BTJD 2584.2044385 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 47 cutout puts the deficit 0.24 arcsec from the target coordinate; peak [7, 7] versus target pixel [8, 7]. Image-model fractional residual 0.213; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 919146748219604736: separation 18.80 arcsec, G 13.721, approximate intrinsic eclipse requirement 0.139. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 23.77/21.23/20.42 ppt in 3 stacked events; median selected QLP raw fit depth 25.38 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 919146679500127616 has orbital P = 7.6744160954 d, e = 0.24606; closest edge-on conjunction offset at the measured epoch is -0.07616 d (-0.00992 phase), with 0.15053 d linearly propagated full-SB1-covariance timing error. The rough central duration is 6.62 h versus observed median 3.04 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Five repeated raw eclipses S47/60. A capable neighbor 18.8 arcsec away can in principle produce the blended loss, but the inspected deficit core falls on the target and shows no distinct neighbor peak; attribution is favored at TESS resolution, not proven subpixel.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
47	96	2584.2038405	0.003472	24.64	used
47	97	2591.8791377	0.003472	27.68	used
47	98	2599.5533210	0.003472	25.38	used
60	143	2944.8992147	0.001157	25.04	used
60	145	2960.2474561	0.001157	26.21	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s47.json, historical/diff_s47.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R415. Public evidence files: diff_s47.json, diff_s47.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

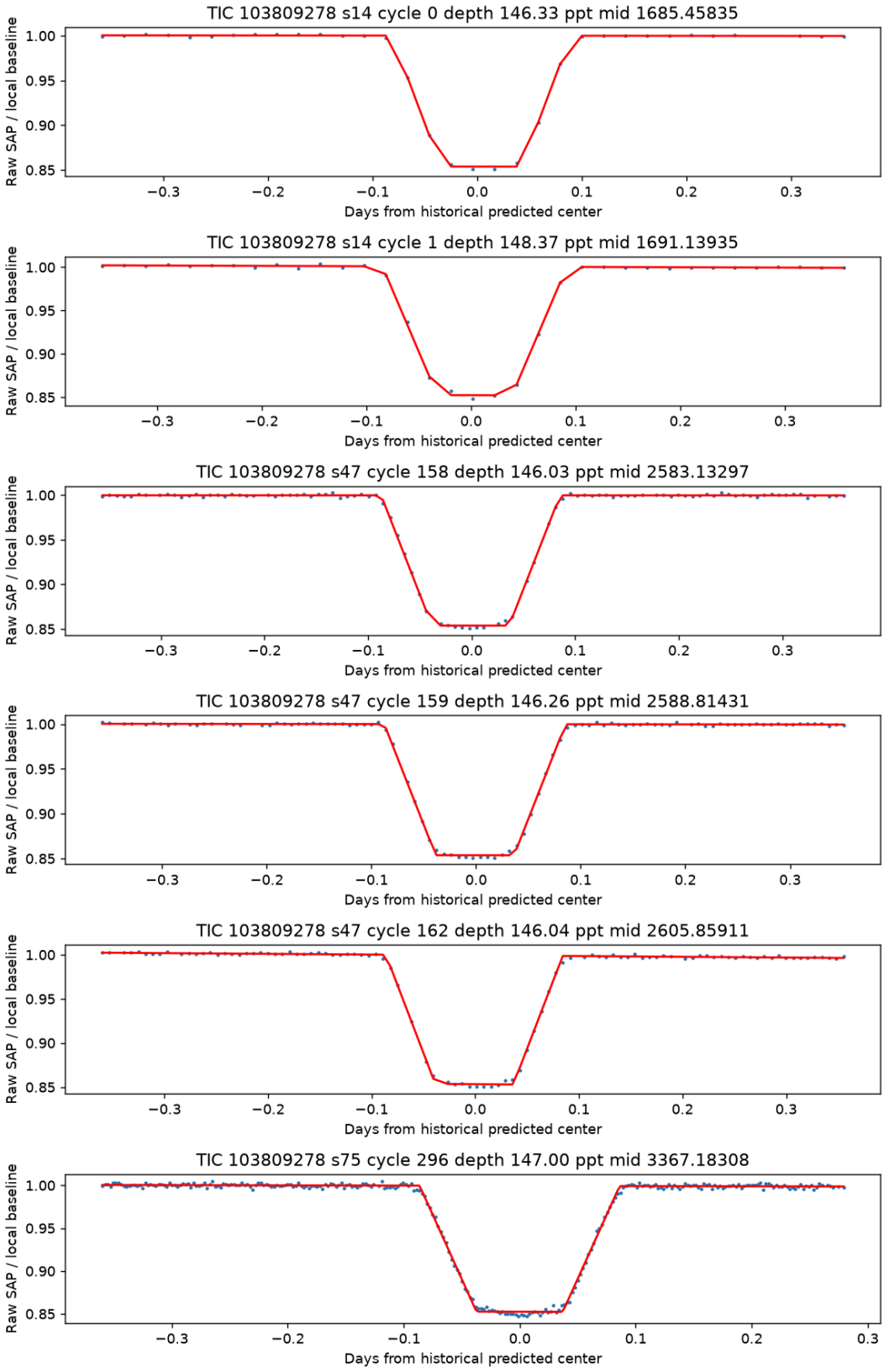
R416. TIC 103809278

physical_verdict: CONFIRMED

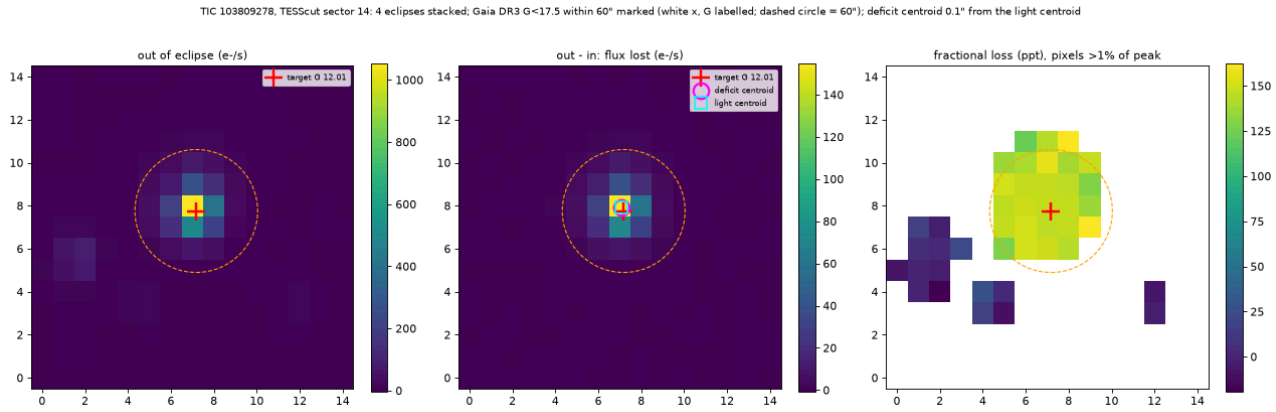
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R416; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R416; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 103809278: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 9 timing-usable measured eclipses in representative sectors [14, 47, 75]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 9 actual midpoints yield P = 5.6815171354 ± 0.0000120548 d and reference epoch BTJD 1685.4540122 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 14 cutout puts the deficit 3.79 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.023; this is not a calibrated localization error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 147.02/147.02/146.82 ppt in 4 stacked events; median selected QLP raw fit depth 146.26 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1059801184791364480 has orbital P = 5.6814017817 d, e = 0.08033; closest edge-on conjunction offset at the measured epoch is 0.13967 d (0.02458 phase), with 0.10317 d linearly propagated full-SB1-covariance timing error. The rough central duration is 4.38 h versus observed median 4.14 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Nine deep raw eclipses across S14/47/75; target-centered difference core and no capable Gaia neighbor under the brightness screen.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
14	0	1685.4583462	0.010416	146.33	used
14	1	1691.1393536	0.010416	148.37	used
14	3	1702.5021041	0.010416	141.24	used
47	158	2583.1329651	0.003472	146.03	used
47	159	2588.8143143	0.003472	146.26	used
47	161	2600.1777778	0.003472	146.26	used
47	162	2605.8591093	0.003472	146.04	used
75	293	3350.1387125	0.001157	147.35	used
75	296	3367.1830767	0.001157	147.00	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s14.json, historical/diff_s14.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R416. Public evidence files: diff_s14.json, diff_s14.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

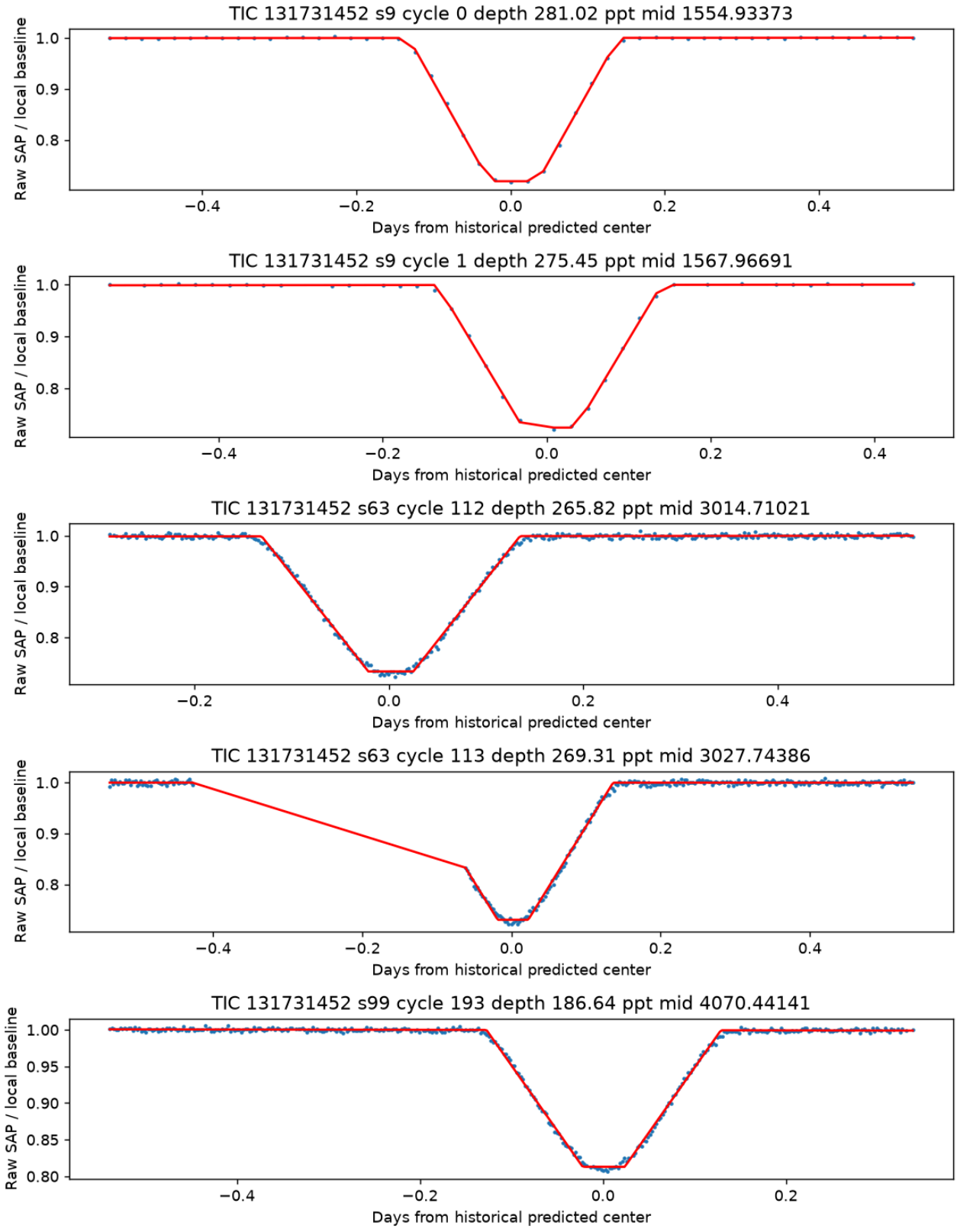
R417. TIC 131731452

physical_verdict: CONFIRMED

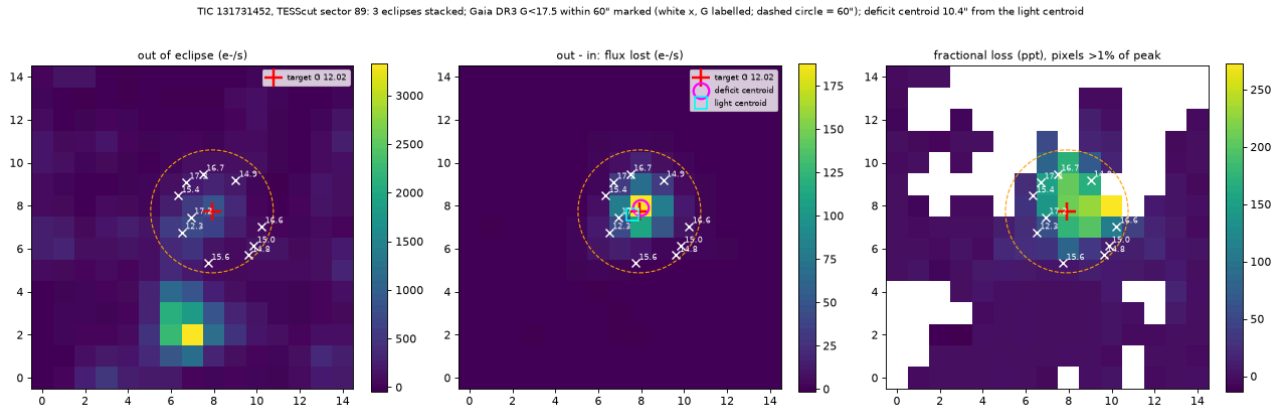
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R417; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R417; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 131731452: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured eclipses in representative sectors [9, 63, 99]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 5 actual midpoints yield P = 13.0337190439 ± 0.0000167190 d and reference epoch BTJD 1554.9336392 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 89 cutout puts the deficit 3.67 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.764; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 5406501911631243520: separation 32.41 arcsec, G 12.282, approximate intrinsic eclipse requirement 0.553. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 210.62/164.58/119.31 ppt in 3 stacked events; median selected QLP raw fit depth 269.31 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5406501915941740928 has orbital P = 13.0317563802 d, e = 0.15286; closest edge-on conjunction offset at the measured epoch is -0.06192 d (-0.00475 phase), with 0.15776 d linearly propagated full-SB1-covariance timing error. The rough central duration is 7.22 h versus observed median 6.44 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Five deep raw eclipses S9/63/99. A capable neighbor 32.4 arcsec away is distinct from the target-centered core; high pixel-model residual and changing aperture dilution limit intrinsic depth interpretation.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
9	0	1554.9337346	0.010417	281.02	used
9	1	1567.9669072	0.010417	275.45	used
63	112	3014.7102093	0.001157	265.82	used
63	113	3027.7438640	0.001157	269.31	used
99	193	4070.4414090	0.001157	186.64	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s89.json, historical/diff_s89.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R417. Public evidence files: diff_s89.json, diff_s89.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

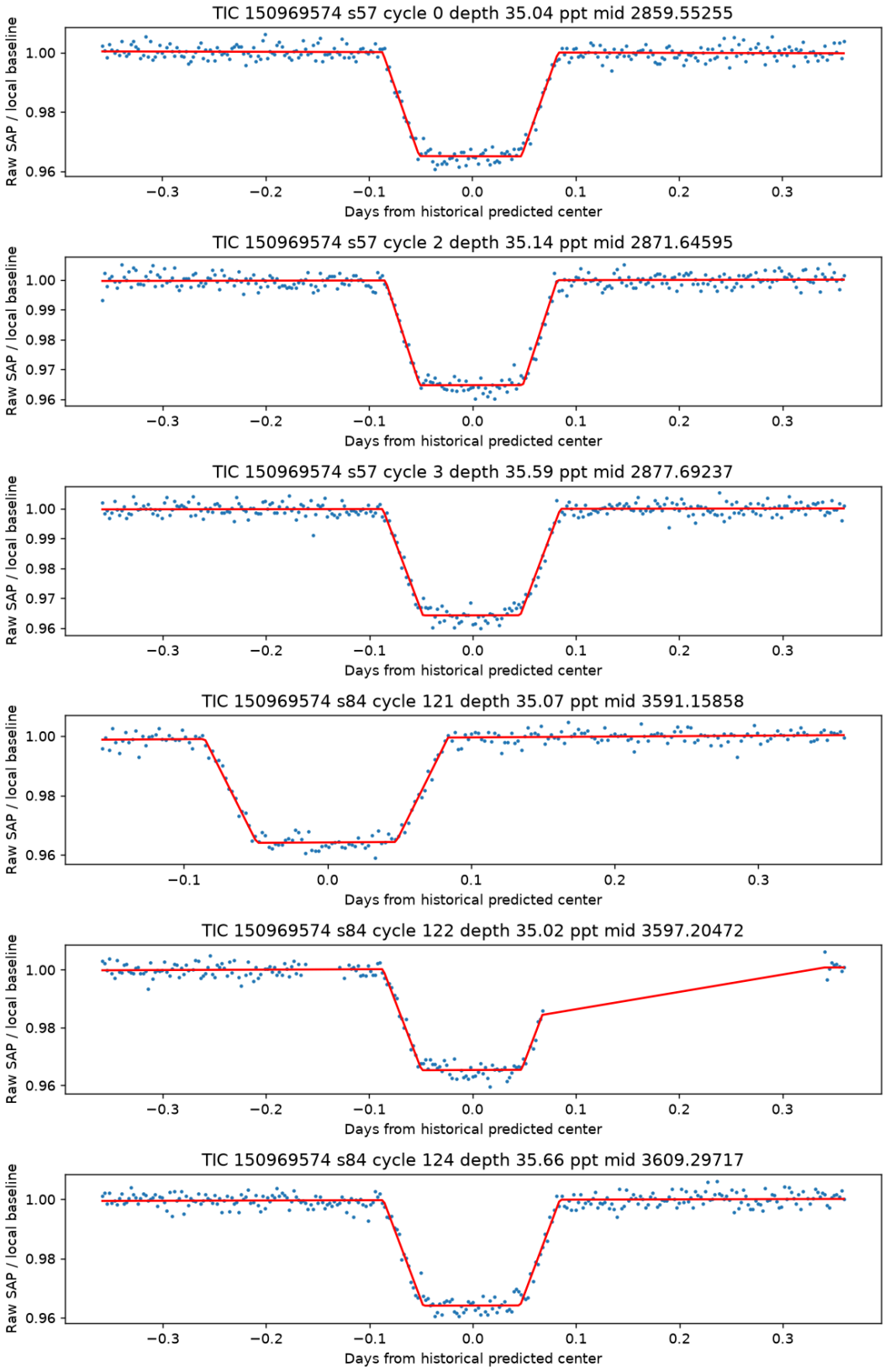
R418. TIC 150969574

physical_verdict: CONFIRMED

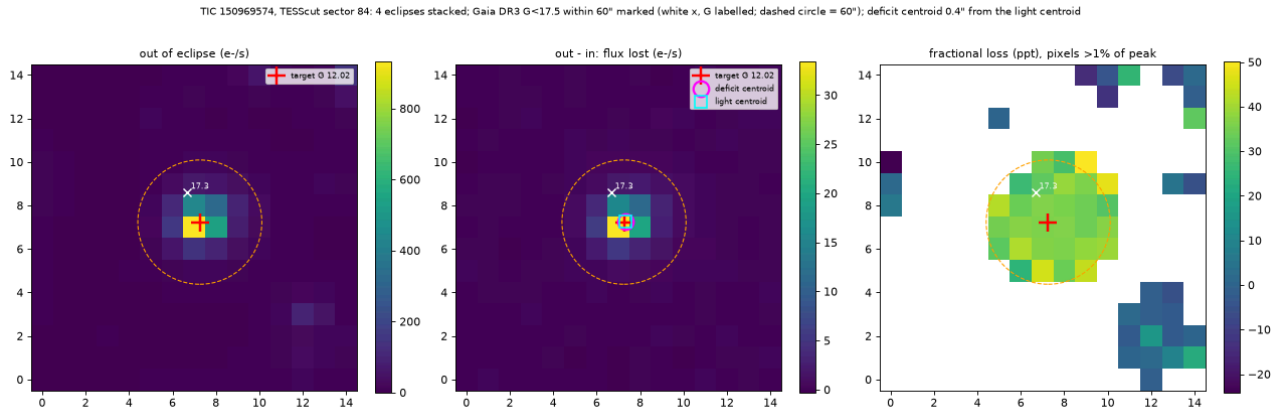
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R418; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R418; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 150969574: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured eclipses in representative sectors [57, 84]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 8 actual midpoints yield P = 6.0463235428 ± 0.0000070222 d and reference epoch BTJD 2859.5530800 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical sector 84 cutout puts the deficit 1.35 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.031; this is not a calibrated localization error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 35.85/36.10/35.95 ppt in 4 stacked events; median selected QLP raw fit depth 35.32 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 2799480161937494912 has orbital P = 6.0462607078 d, e = 0.08689; closest edge-on conjunction offset at the measured epoch is 0.12220 d (0.02021 phase), with 0.45776 d linearly propagated full-SB1-covariance timing error. The rough central duration is 4.48 h versus observed median 4.08 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Eight repeated raw eclipses S57/84; target-centered difference core and no capable Gaia neighbor under the brightness screen.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
57	0	2859.5525475	0.001157	35.04	used
57	2	2871.6459482	0.001157	35.14	used
57	3	2877.6923742	0.001157	35.59	used
84	120	3585.1115748	0.001157	35.53	used
84	121	3591.1585797	0.001157	35.07	used
84	122	3597.2047197	0.001157	35.02	used
84	123	3603.2507075	0.001157	35.51	used
84	124	3609.2971671	0.001157	35.66	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s84.json, historical/diff_s84.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R418. Public evidence files: diff_s84.json, diff_s84.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

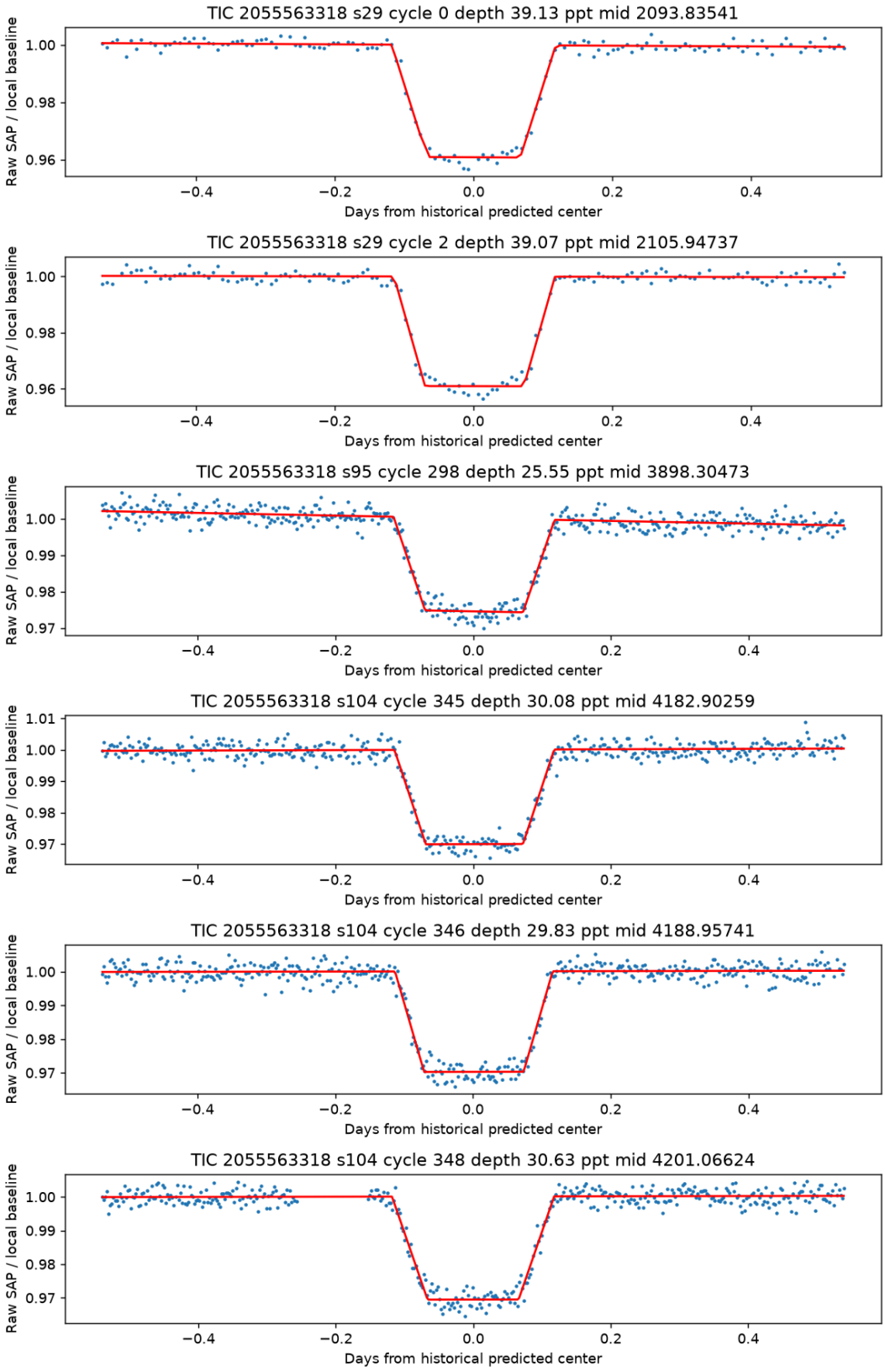
R419. TIC 2055563318

physical_verdict: PROBABLE

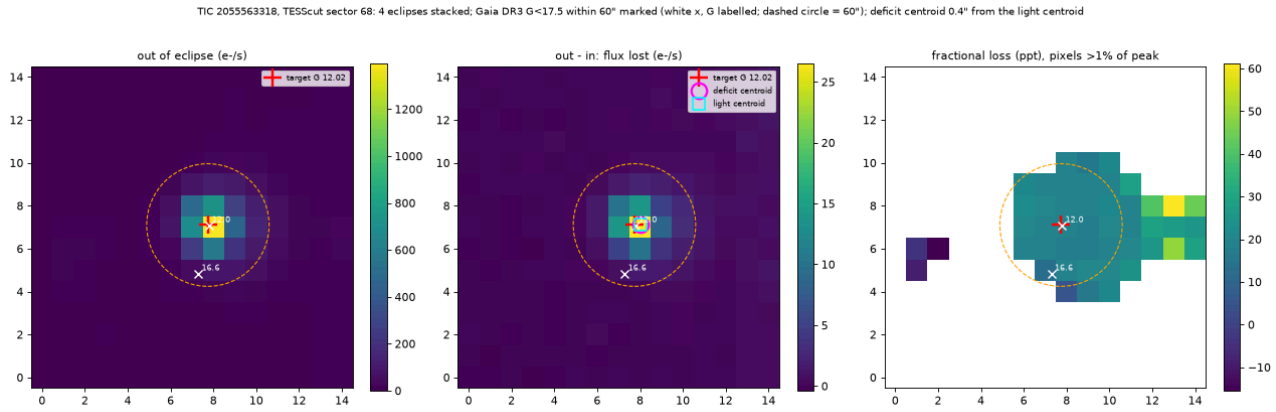
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry / multi-source PRF localization



R419; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R419; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 2055563318: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured eclipses in representative sectors [29, 95, 104]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 8 actual midpoints yield P = 6.0552614725 ± 0.0000071310 d and reference epoch BTJD 2093.8367290 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: unresolved after completed check.** Historical sector 68 cutout puts the deficit 5.79 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.044; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 6524911983920449664: separation 1.11 arcsec, G 11.971, approximate intrinsic eclipse requirement 0.059. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 19.01/19.01/19.12 ppt in 4 stacked events; median selected QLP raw fit depth 30.23 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 6524911983916150016 has orbital P = 6.0544220136 d, e = 0.14861; closest edge-on conjunction offset at the measured epoch is 0.12018 d (0.01985 phase), with 0.29407 d linearly propagated full-SB1-covariance timing error. The rough central duration is 3.87 h versus observed median 5.59 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto or proof of which unresolved neighbor eclipses.

Deciding interpretation and limitation: Eight repeated raw eclipses S29/95/104 show a real recurring signal, but a capable Gaia neighbor only 1.1 arcsec away shares the central TESS pixel. The image cannot uniquely identify the eclipsing star; geometry agrees with the selected Gaia orbit but does not cure this attribution limit.

Next check: Resolved eclipse photometry or multi-source PRF localization of the 1.1-arcsec pair.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
29	0	2093.8354098	0.003472	39.13	used
29	2	2105.9473696	0.003472	39.07	used
95	298	3898.3047275	0.001157	25.55	used
95	299	3904.3607308	0.001157	25.95	used
104	345	4182.9025932	0.001157	30.08	used
104	346	4188.9574118	0.001157	29.83	used
104	347	4195.0123085	0.001157	30.38	used
104	348	4201.0662356	0.001157	30.63	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s68.json, historical/diff_s68.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R419. Public evidence files: diff_s68.json, diff_s68.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

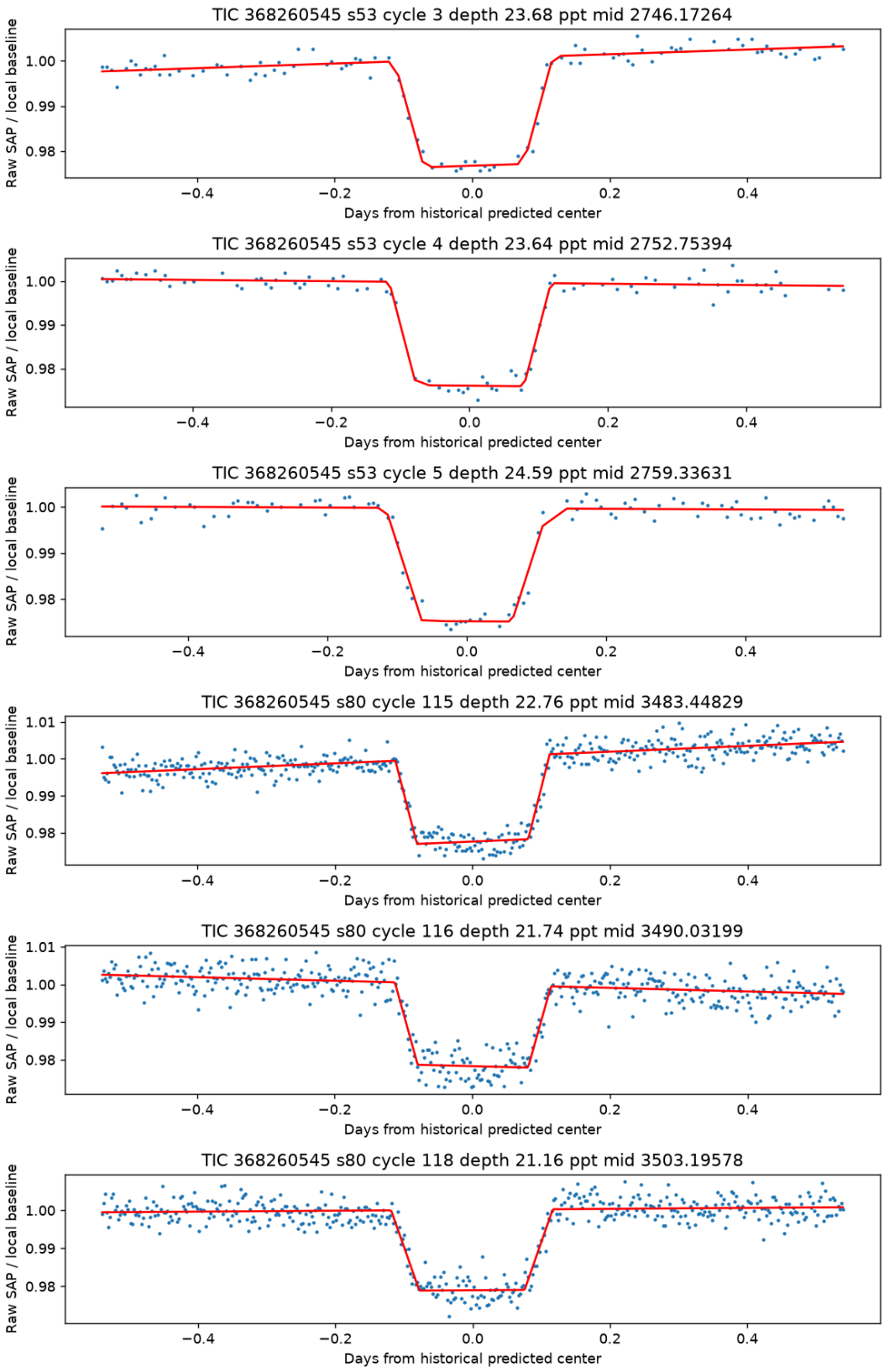
R420. TIC 368260545

physical_verdict: CONFIRMED

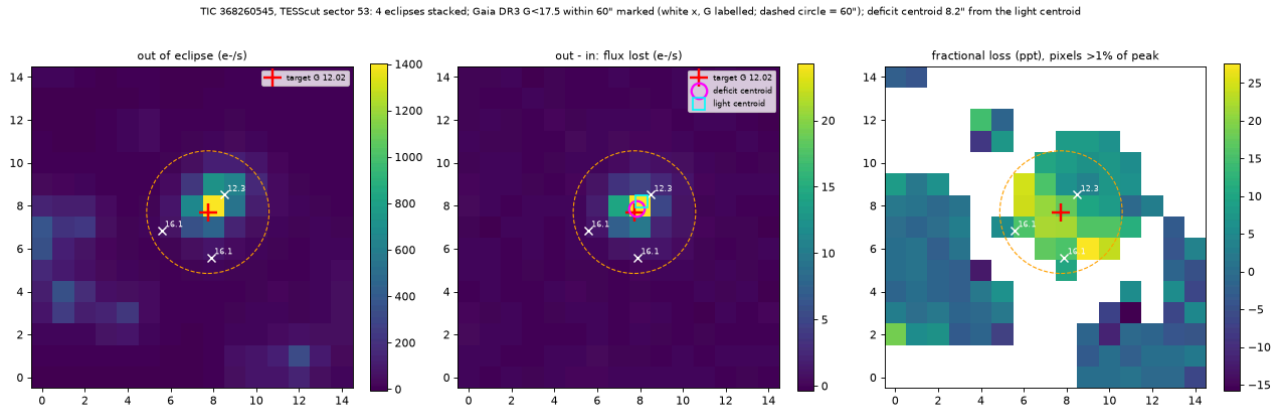
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional pixel orientations or spatially resolved photometry can strengthen attribution and refine the ephemeris.



R420; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R420; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 368260545: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 11; Gaia DR3 source 4556083135603641856; actual DR3 G=12.021002. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 8 usable measured eclipse centers across sectors 53, 80; 8 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 23.20 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=6.582836723 ± 0.000016330 d and reference center BTJD 2746.170982. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=6.583413245 d; difference=-0.000577 d.
- Pixel attribution and alternative reduction: historical TESScut sector 53 stacked 4 eclipse windows; the flux-deficit centroid is 0.206 pixel from the Gaia target and the peak deficit pixel is [8, 8] versus target pixel [8, 8]. Custom 1×1/3×3/5×5 pixel apertures give 17.3/14.4/14.0 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 18 Gaia neighbors within 60 arcsec were tested for flux capability; 1 is capable. Nearest capable: G=12.30, 22.87 arcsec, requiring 5.4% intrinsic eclipse, 0.92 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset -0.02615 in orbital phase, linear covariance scale 0.220 d. Approximate central duration 5.80 h versus observed median 5.57 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

A similarly bright G=12.30 Gaia neighbor 22.87 arcsec away is flux-capable, but the deficit is 0.206 pixel from the target versus 0.92 pixel from that neighbor and peaks on the target pixel. A targeted recomputation of all four stacked windows puts every event deficit closer to the target (0.13–0.47 pixel) than the neighbor (0.75–1.07 pixel). Positional uncertainty remains descriptive rather than calibrated.

Additional pixel orientations or spatially resolved photometry can strengthen attribution and refine the ephemeris.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/368260545/diff_s53.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet368260545/tesscut_s053.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`. Targeted four-window centroid audit: `pixel-event-audit.json`, reproduced by `../audit_368_pixels.py`.

Stable review key: R420. Public evidence files: diff_s53.json, diff_s53.png, gaia.json, measurements.json, pixel-event-audit.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

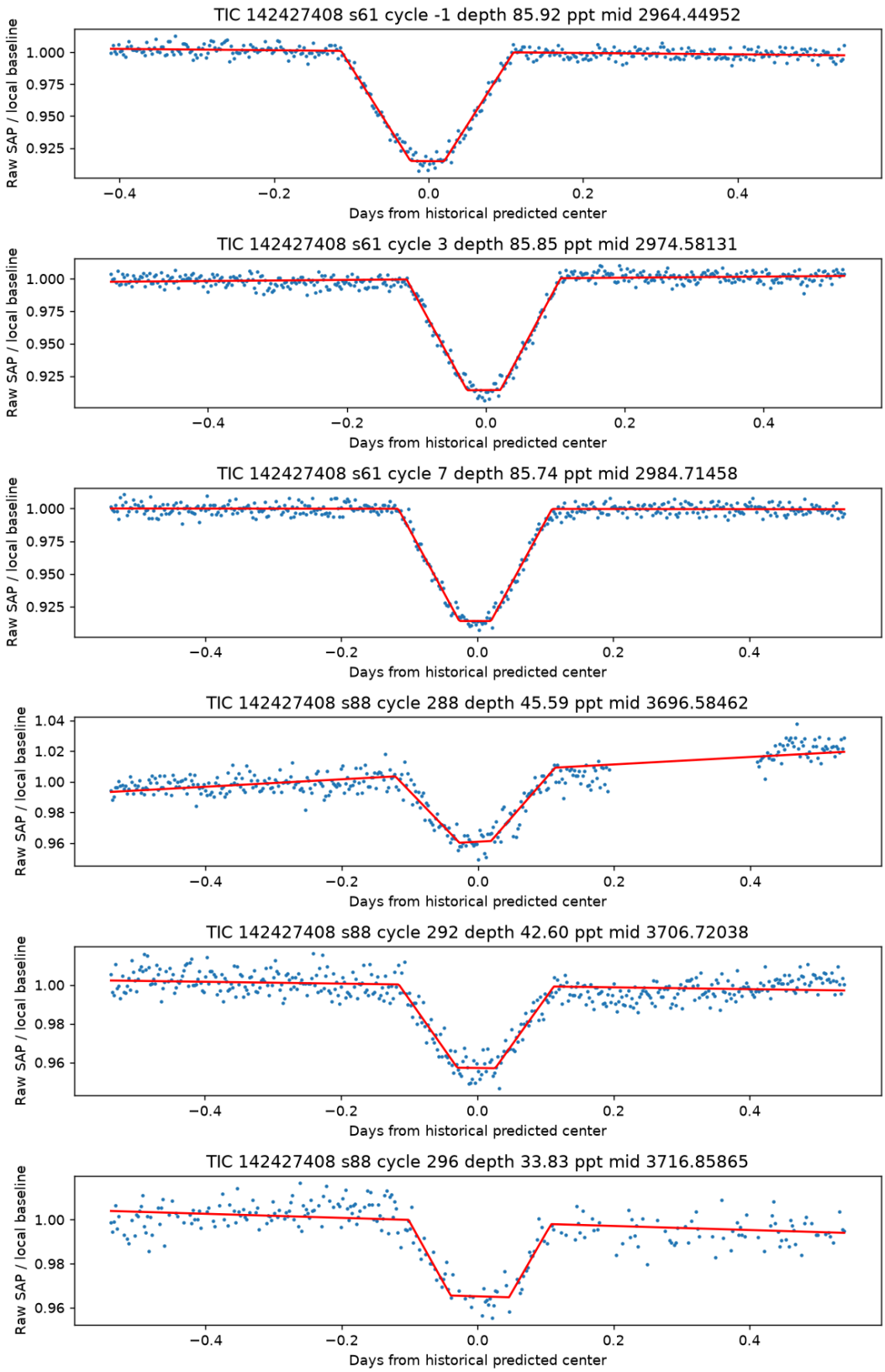
R421. TIC 142427408

physical_verdict: CONFIRMED

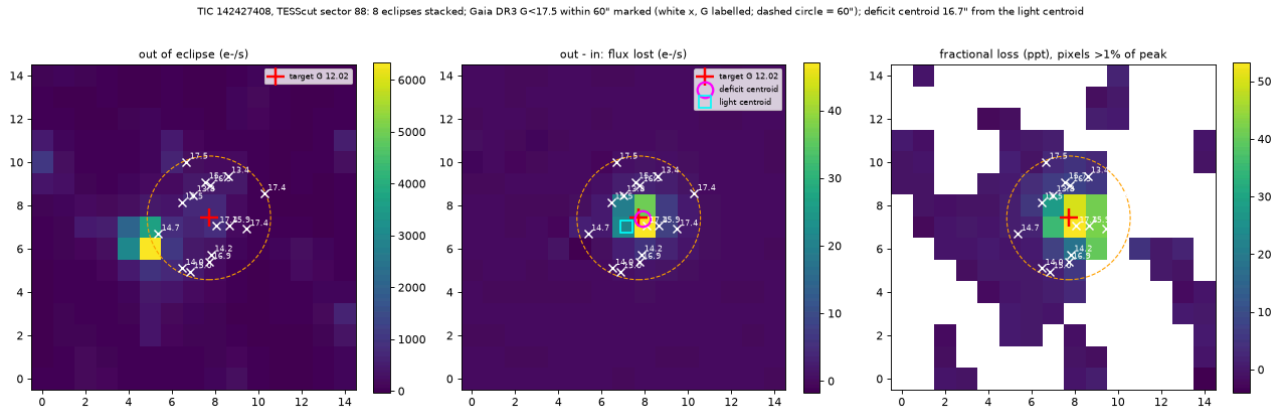
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R421; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R421; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 142427408: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 12; Gaia DR3 source 5711508945433894656; actual DR3 G=12.021337. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 19 usable measured eclipse centers across sectors 61, 88; 9 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 85.74 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=5.066684108 ± 0.000005802 d and reference center BTJD 2964.448321. Epoch kind: Selected deeper repeating eclipse; odd historical half-period cycles. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=5.066415582 d; difference=+0.000269 d.
- Pixel attribution and alternative reduction: historical TESScut sector 88 stacked 8 eclipse windows; the flux-deficit centroid is 0.172 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 53.3/37.1/18.8 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 71 Gaia neighbors within 60 arcsec were tested for flux capability; 7 are capable. Nearest capable: G=13.77, 27.00 arcsec, requiring 23.3% intrinsic eclipse, 1.42 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset +0.03100 in orbital phase, linear covariance scale 0.240 d. Approximate central duration 8.26 h versus observed median 5.38 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage. Alternating half-period events were separated by eclipse kind before orbital timing. Crowded pixel PSF fit is poor, but all flux-capable neighbors are at least 27 arcsec from the target and farther from the deficit.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/142427408/diff_s88.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet142427408/tesscut_s088.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R421. Public evidence files: diff_s88.json, diff_s88.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

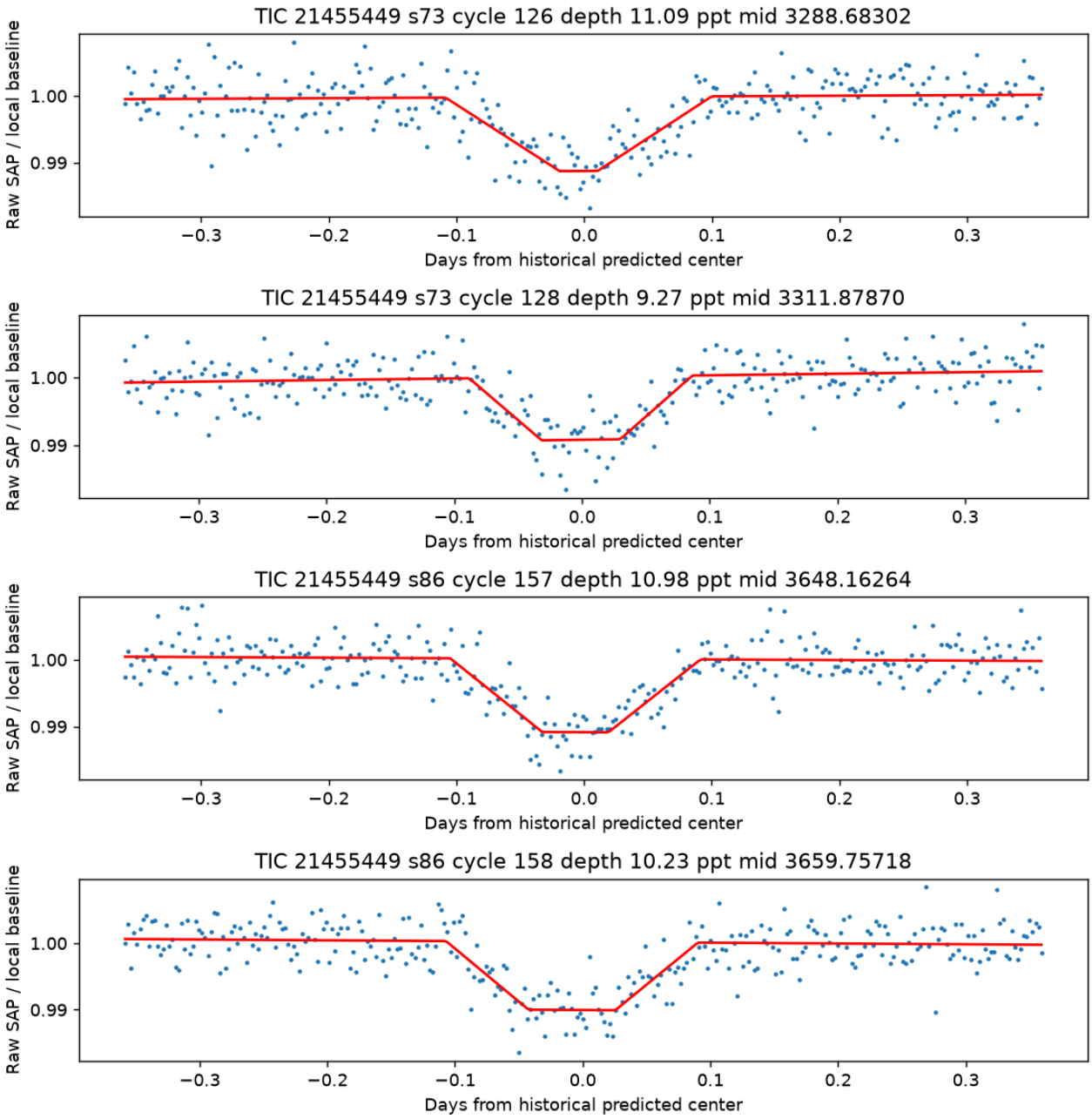
R422. TIC 21455449

physical_verdict: CONFIRMED

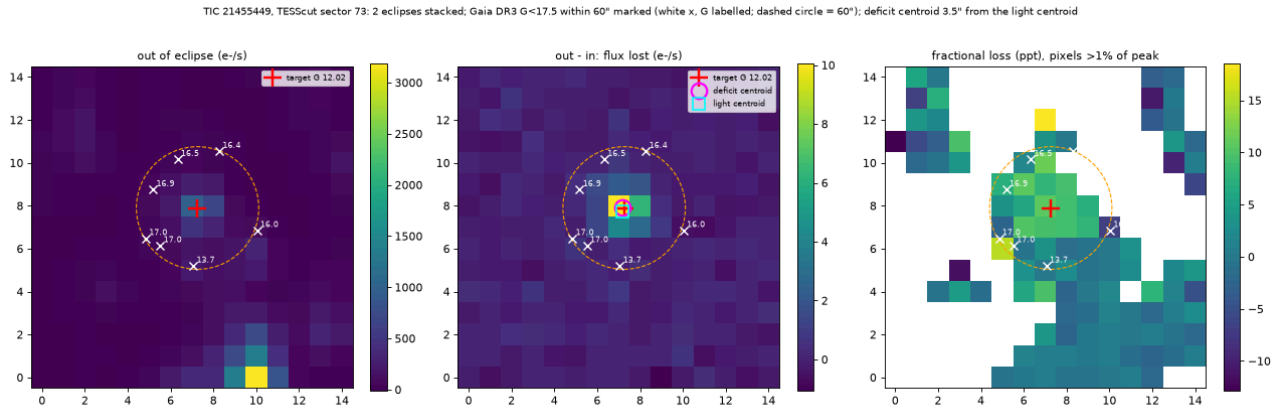
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R422; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R422; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 21455449: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 13; Gaia DR3 source 254615934869825920; actual DR3 G=12.021999. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 4 usable measured eclipse centers across sectors 73, 86; 4 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 10.61 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=11.596027763 ± 0.000084033 d and reference center BTJD 3288.685159. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=11.595182256 d; difference=+0.000846 d.
- Pixel attribution and alternative reduction: historical TESScut sector 73 stacked 2 eclipse windows; the flux-deficit centroid is 0.080 pixel from the Gaia target and the peak deficit pixel is [7, 8] versus target pixel [7, 8]. Custom 1×1/3×3/5×5 pixel apertures give 10.6/9.4/8.0 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 43 Gaia neighbors within 60 arcsec were tested for flux capability; 7 are capable. Nearest capable: G=16.92, 44.98 arcsec, requiring 87.2% intrinsic eclipse, 2.17 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset +0.01425 in orbital phase, linear covariance scale 0.610 d. Approximate central duration 14.66 h versus observed median 4.72 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage. Gaia duration estimate is crude and exceeds the observed narrow event; geometry phase and period remain consistent.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/21455449/diff_s73.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet21455449/tesscut_s073.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R422. Public evidence files: diff_s73.json, diff_s73.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

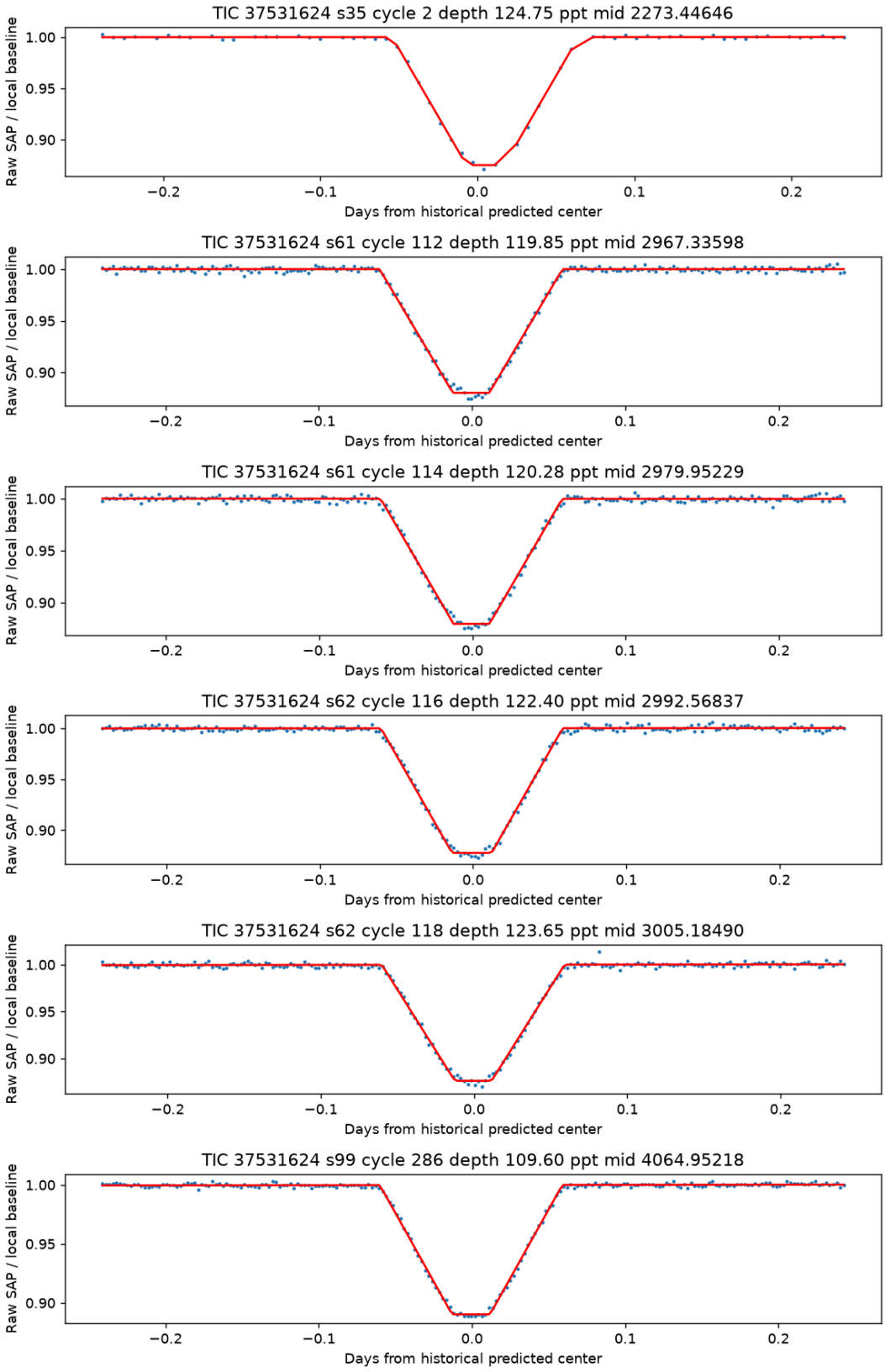
R423. TIC 37531624

physical_verdict: CONFIRMED

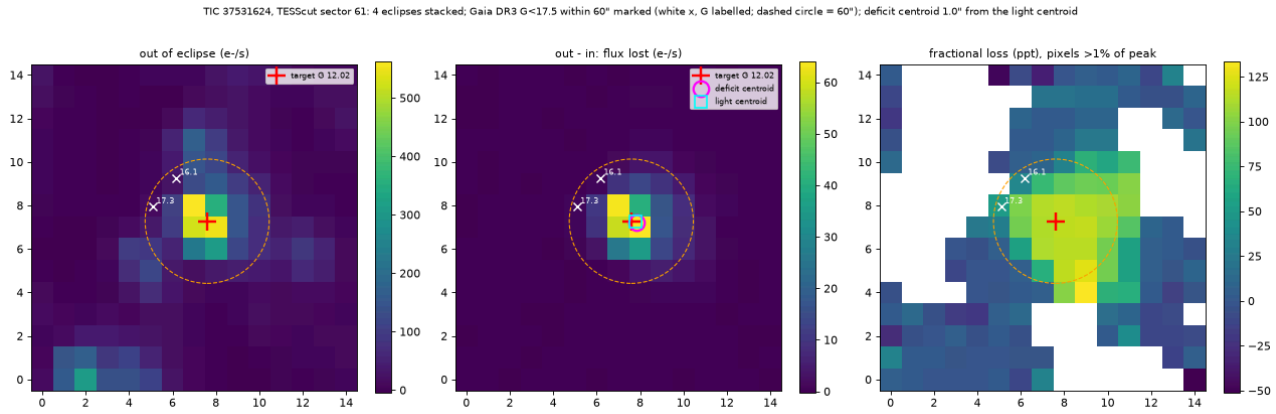
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R423; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R423; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 37531624: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 14; Gaia DR3 source 5654299985347511552; actual DR3 G=12.022275. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 12 usable measured eclipse centers across sectors 35, 61, 62, 99; 12 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 121.06 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=6.308135631 ± 0.000005140 d and reference center BTJD 2273.441351. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=6.308348116 d; difference=-0.000212 d.
- Pixel attribution and alternative reduction: historical TESScut sector 61 stacked 4 eclipse windows; the flux-deficit centroid is 0.224 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 116.4/114.2/107.2 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 13 Gaia neighbors within 60 arcsec were tested for flux capability; 0 are capable. No Gaia neighbor is flux-capable.
- Gaia geometry: selected edge-on conjunction offset -0.00871 in orbital phase, linear covariance scale 0.079 d. Approximate central duration 4.47 h versus observed median 2.85 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/37531624/diff_s61.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet37531624/tesscut_s061.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R423. Public evidence files: diff_s61.json, diff_s61.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

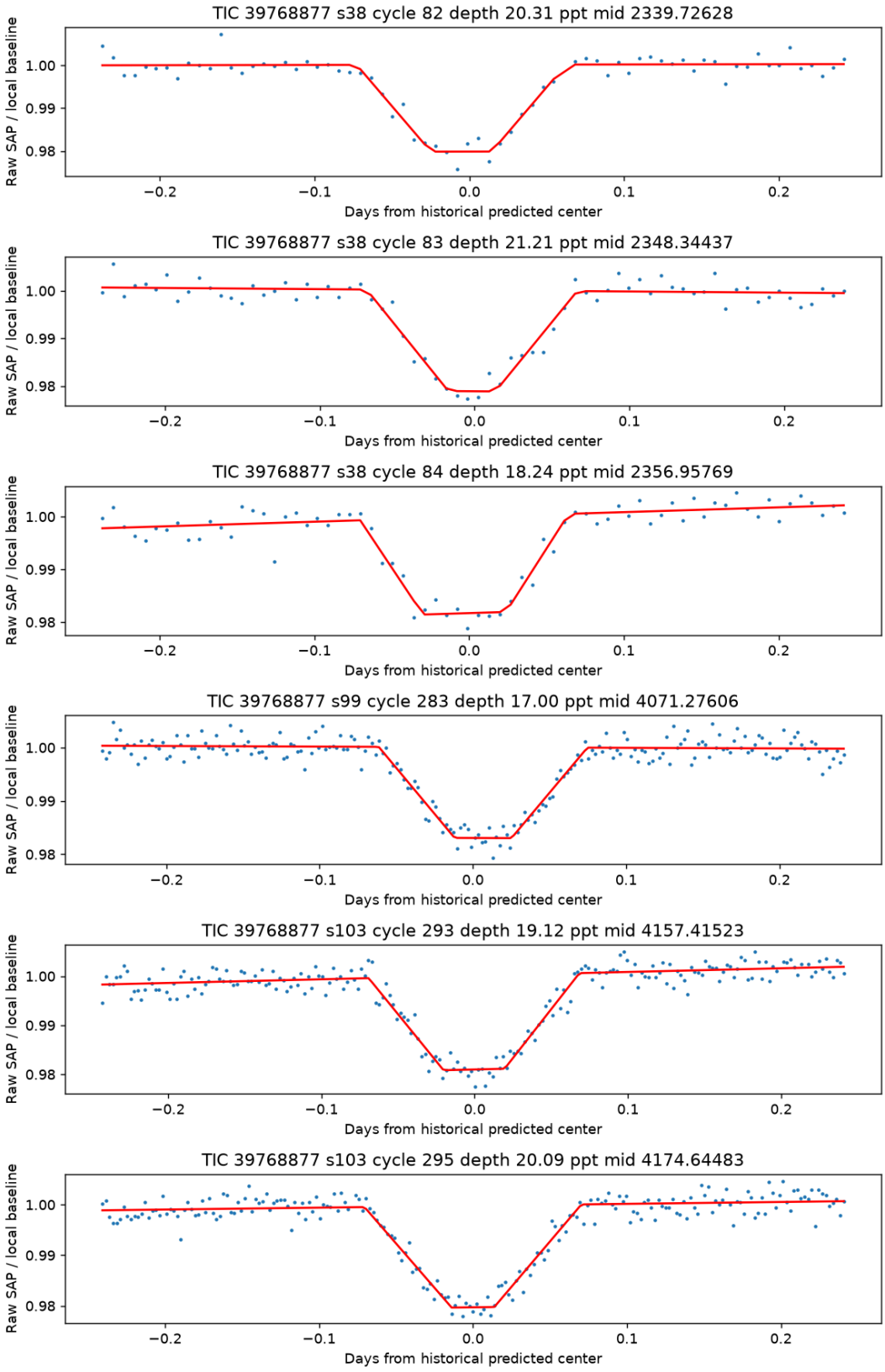
R424. TIC 39768877

physical_verdict: CONFIRMED

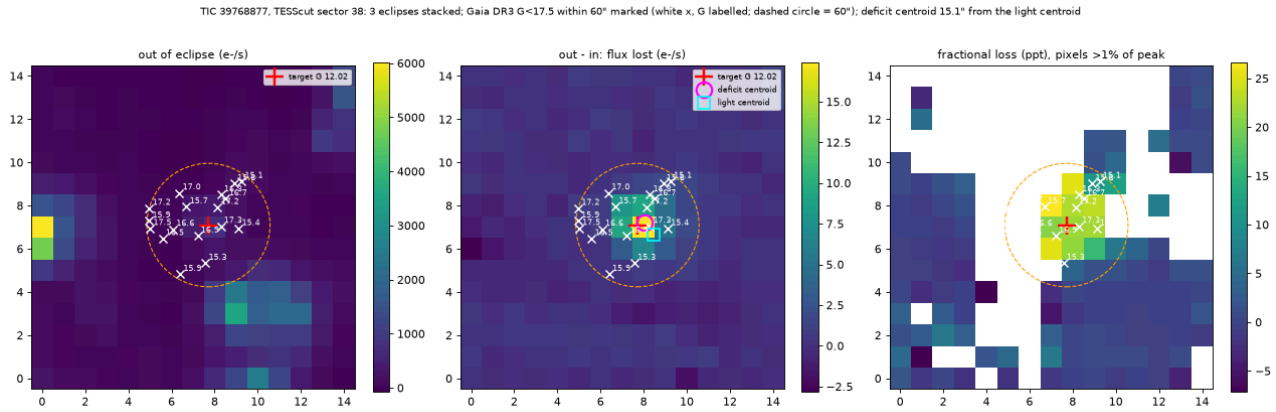
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R424; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R424; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 39768877: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 15; Gaia DR3 source 5887384768167350784; actual DR3 G=12.022689. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 8 usable measured eclipse centers across sectors 38, 99, 103; 8 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 18.97 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=8.614638525 ± 0.000026357 d and reference center BTJD 2339.729538. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=8.614908563 d; difference=-0.000270 d.
- Pixel attribution and alternative reduction: historical TESScut sector 38 stacked 3 eclipse windows; the flux-deficit centroid is 0.342 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 21.8/22.0/16.5 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 186 Gaia neighbors within 60 arcsec were tested for flux capability; 8 are capable. Nearest capable: G=15.69, 27.54 arcsec, requiring 57.4% intrinsic eclipse, 1.53 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset -0.00312 in orbital phase, linear covariance scale 0.159 d. Approximate central duration 6.91 h versus observed median 3.28 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/39768877/diff_s38.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet39768877/tesscut_s038.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R424. Public evidence files: diff_s38.json, diff_s38.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

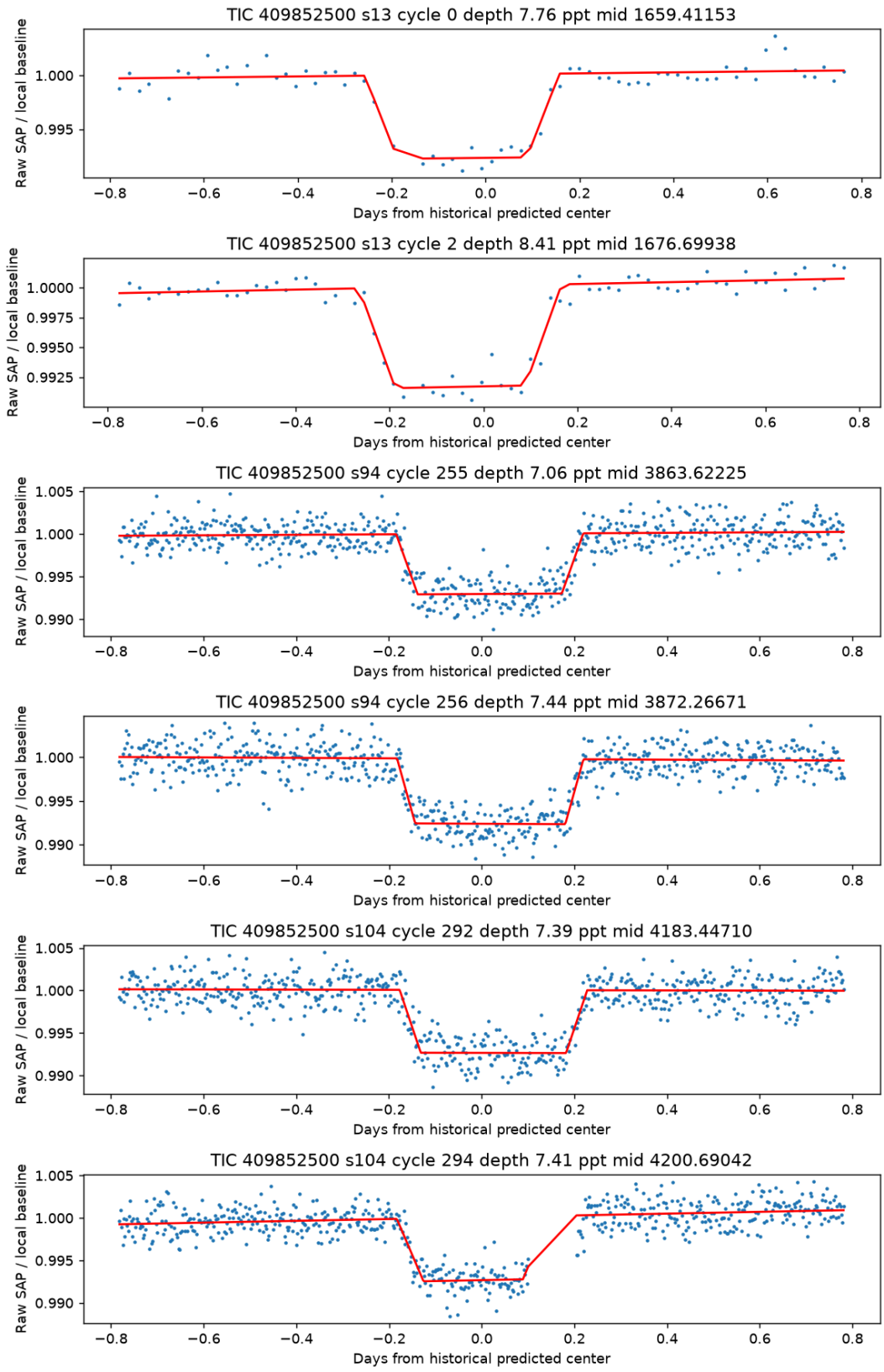
R425. TIC 409852500

physical_verdict: PROBABLE

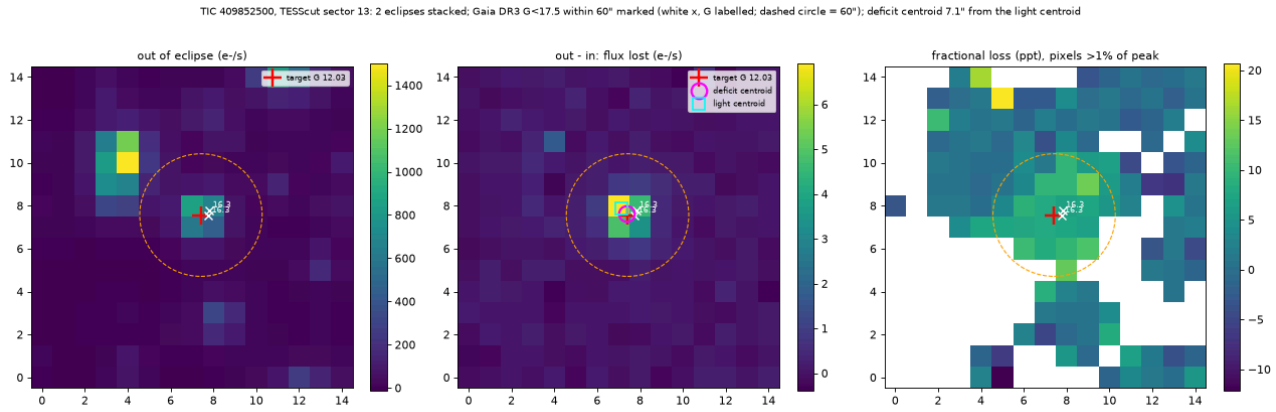
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Obtain spatially resolved eclipse photometry, or a stronger difference image in a suitable sector.



R425; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R425; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 409852500: batch 02 review

Physical verdict: probable. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 16; Gaia DR3 source 6445307372129779072; actual DR3 G=12.026384. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 5 usable measured eclipse centers across sectors 13, 94, 104; 5 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 7.44 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=8.643951250 ± 0.000025003 d and reference center BTJD 1659.414449. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=8.642794775 d; difference=+0.001156 d.
- Pixel attribution and alternative reduction: historical TESScut sector 13 stacked 2 eclipse windows; the flux-deficit centroid is 0.107 pixel from the Gaia target and the peak deficit pixel is [7, 8] versus target pixel [7, 8]. Custom 1×1/3×3/5×5 pixel apertures give 7.9/7.6/6.9 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 11 Gaia neighbors within 60 arcsec were tested for flux capability; 2 are capable. Nearest capable: G=16.33, 7.31 arcsec, requiring 41.0% intrinsic eclipse, 0.43 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset -0.00905 in orbital phase, linear covariance scale 0.421 d. Approximate central duration 11.27 h versus observed median 9.70 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Two G~16.3 neighbors at 7.3 and 9.5 arcsec are capable in flux and share a TESS pixel; the deficit centroid favors the target but does not securely resolve them. The sector-104 cycle-294 egress gap excludes that center from timing.

Obtain spatially resolved eclipse photometry, or a stronger difference image in a suitable sector.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/409852500/diff_s13.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet409852500/tesscut_s013.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R425. Public evidence files: diff_s13.json, diff_s13.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

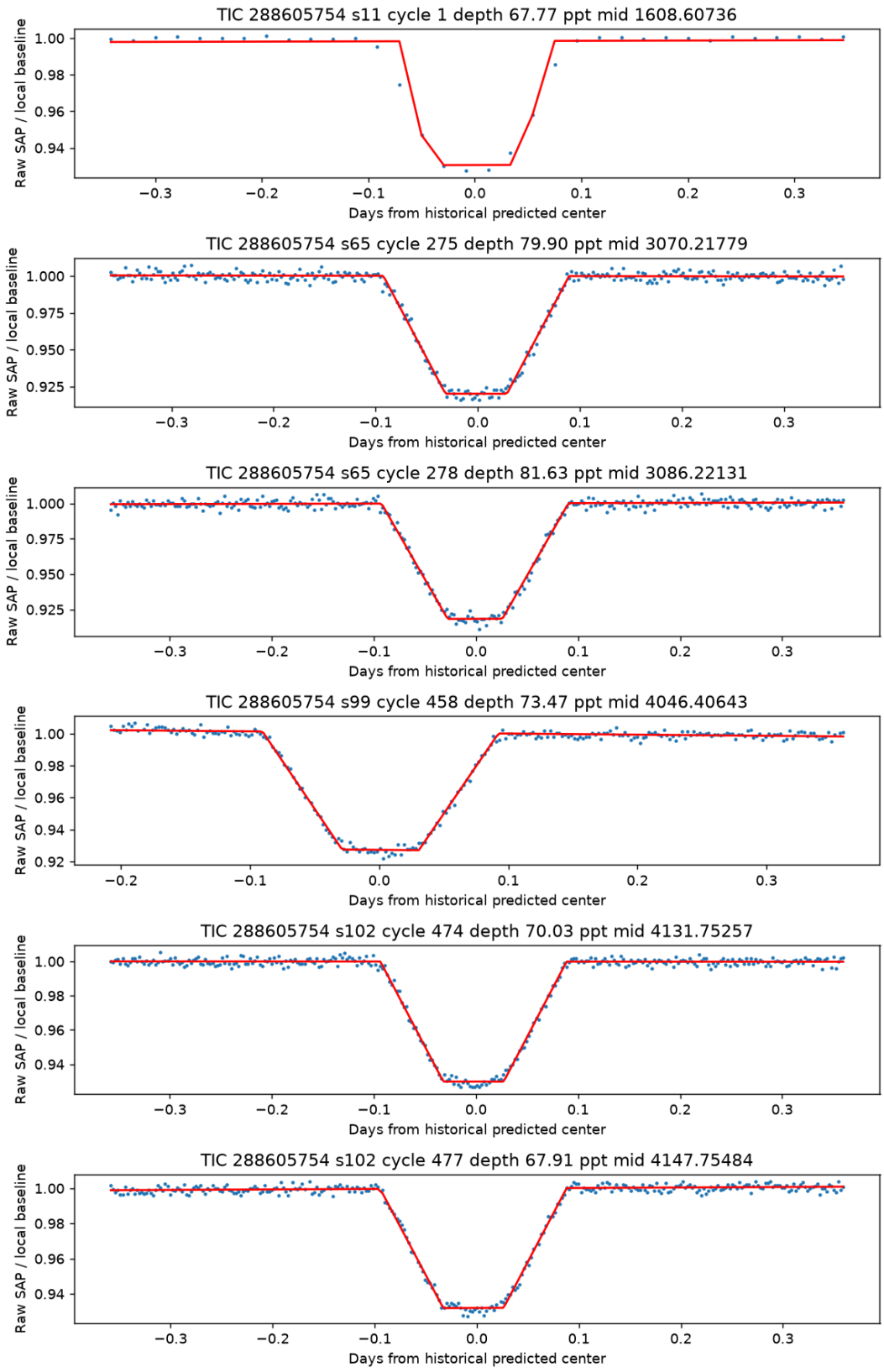
R426. TIC 288605754

physical_verdict: CONFIRMED

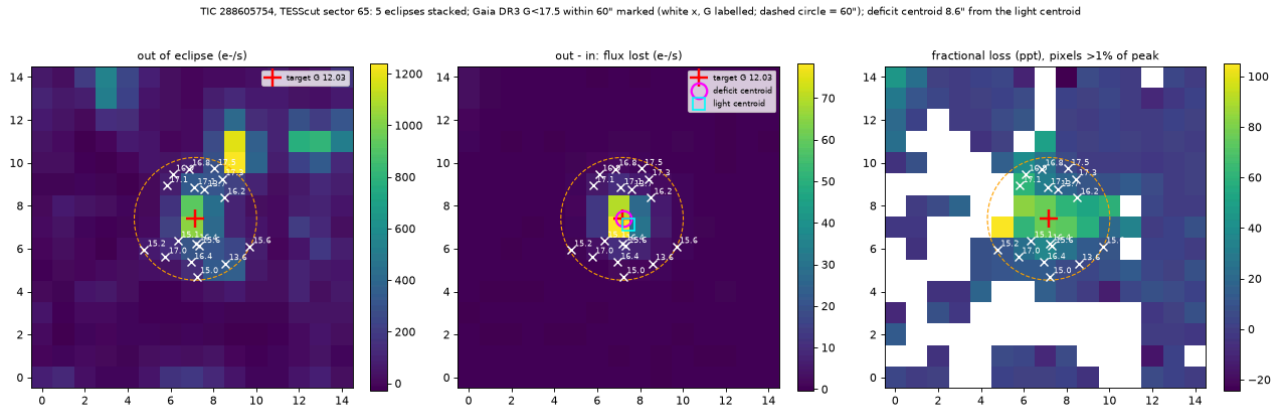
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R426; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R426; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 288605754: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 17; Gaia DR3 source 5894485208091053056; actual DR3 G=12.027896. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 14 usable measured eclipse centers across sectors 11, 65, 99, 102; 14 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 72.32 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=5.334347769 ± 0.000004228 d and reference center BTJD 1608.607034. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=5.334169975 d; difference=+0.000178 d.
- Pixel attribution and alternative reduction: historical TESScut sector 65 stacked 5 eclipse windows; the flux-deficit centroid is 0.059 pixel from the Gaia target and the peak deficit pixel is [7, 7] versus target pixel [7, 7]. Custom 1×1/3×3/5×5 pixel apertures give 76.6/64.3/47.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 164 Gaia neighbors within 60 arcsec were tested for flux capability; 2 are capable. Nearest capable: G=13.72, 29.39 arcsec, requiring 41.2% intrinsic eclipse, 1.38 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset +0.01007 in orbital phase, linear covariance scale 0.130 d. Approximate central duration 6.51 h versus observed median 4.38 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/288605754/diff_s65.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet288605754/tesscut_s065.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R426. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

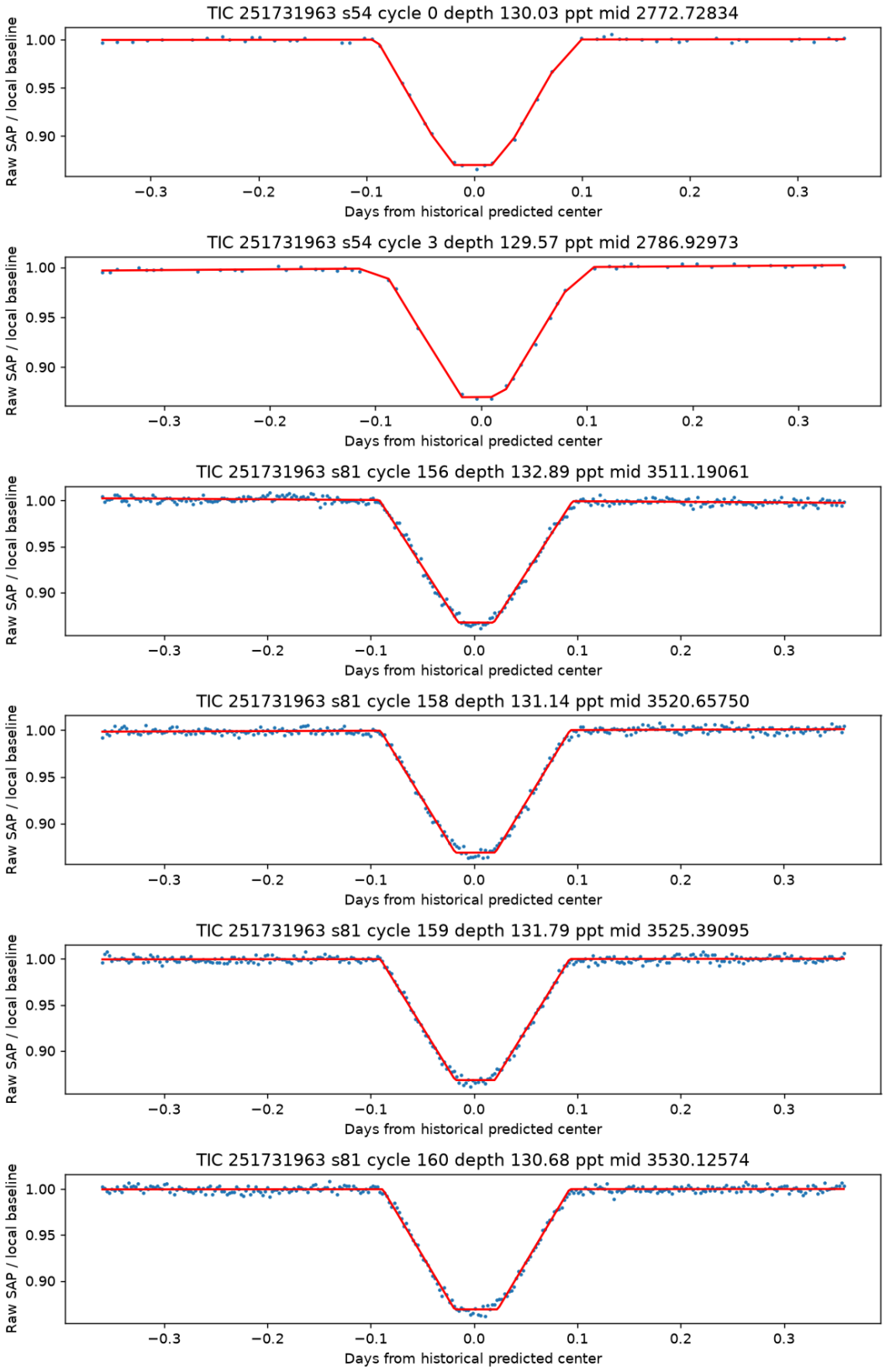
R427. TIC 251731963

physical_verdict: PROBABLE

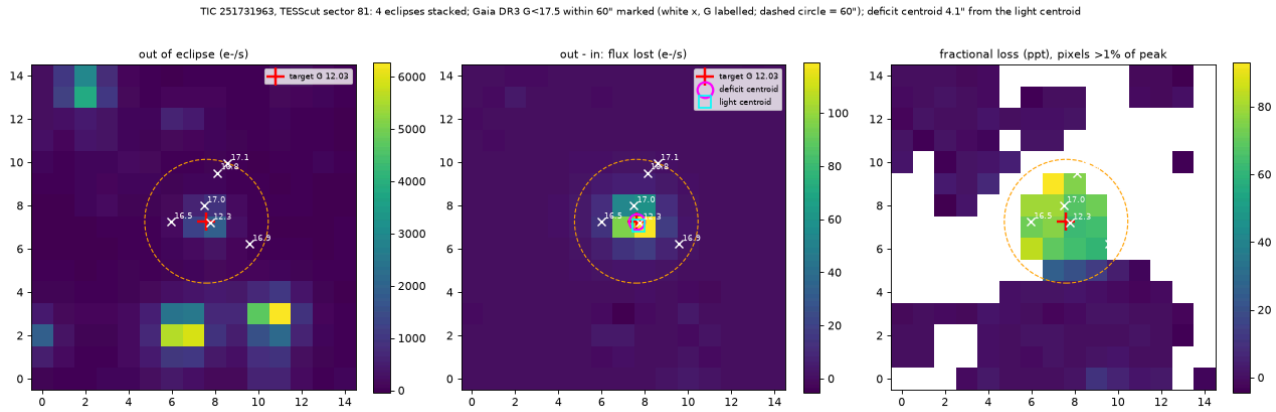
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolve the two stars in higher-spatial-resolution time-series photometry at predicted eclipse times.



R427; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R427; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 251731963: batch 02 review

Physical verdict: probable. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 18; Gaia DR3 source 4238560567095727744; actual DR3 G=12.032766. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 6 usable measured eclipse centers across sectors 54, 81; 6 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 130.91 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=4.733729823 ± 0.000031440 d and reference center BTJD 2772.728456. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=4.733742635 d; difference=-0.000013 d.
- Pixel attribution and alternative reduction: historical TESScut sector 81 stacked 4 eclipse windows; the flux-deficit centroid is 0.024 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 63.0/69.0/66.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 73 Gaia neighbors within 60 arcsec were tested for flux capability; 1 is capable. Nearest capable: G=12.27, 3.69 arcsec, requiring 27.5% intrinsic eclipse, 0.18 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset +0.01300 in orbital phase, linear covariance scale 0.108 d. Approximate central duration 6.03 h versus observed median 4.40 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

A G=12.27 Gaia star 3.69 arcsec away can produce the dip with a ~27% intrinsic eclipse; the TESS pixel centroid cannot resolve the pair.

Resolve the two stars in higher-spatial-resolution time-series photometry at predicted eclipse times.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/251731963/diff_s81.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet251731963/tesscut_s081.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R427. Public evidence files: diff_s81.json, diff_s81.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

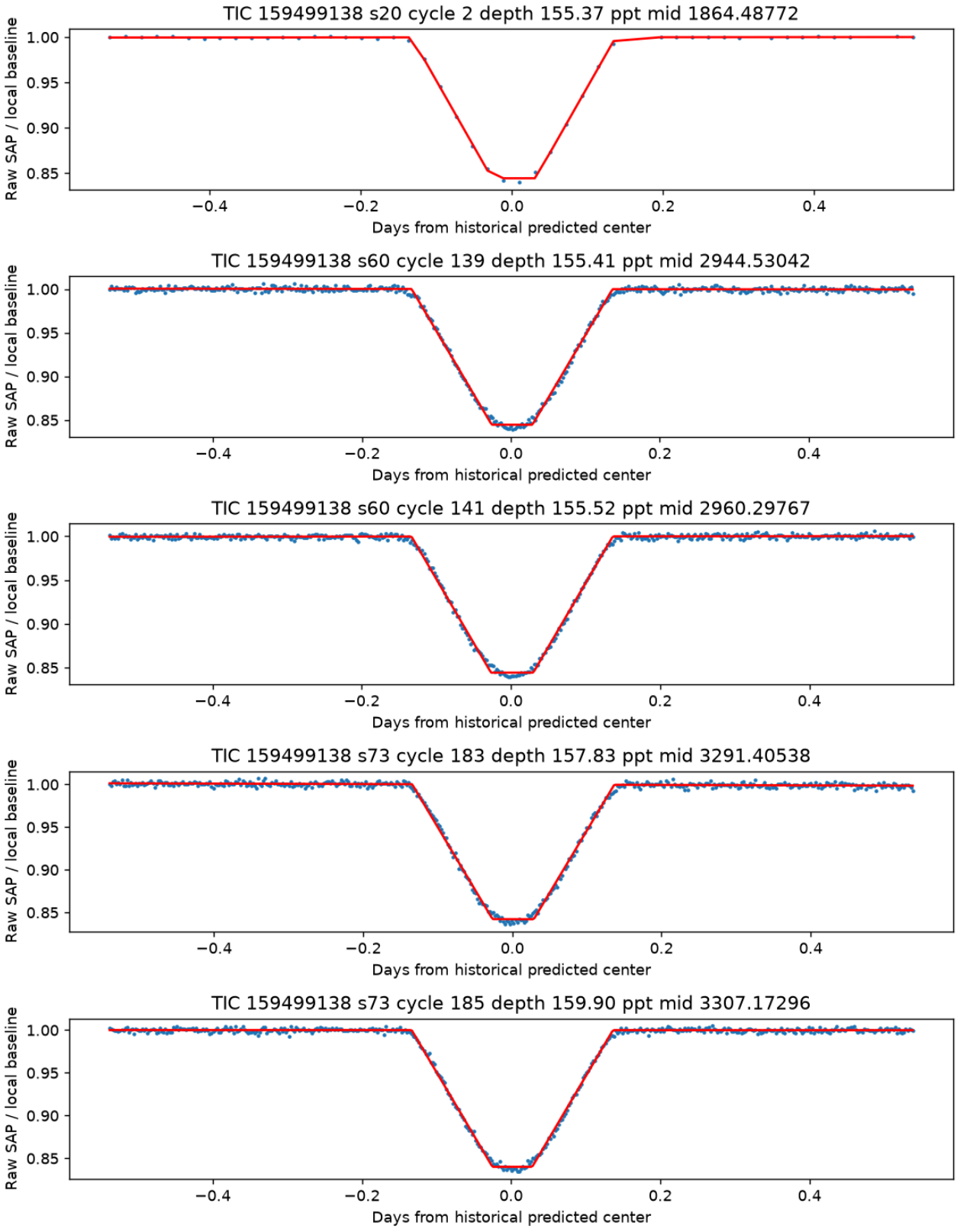
R428. TIC 159499138

physical_verdict: CONFIRMED

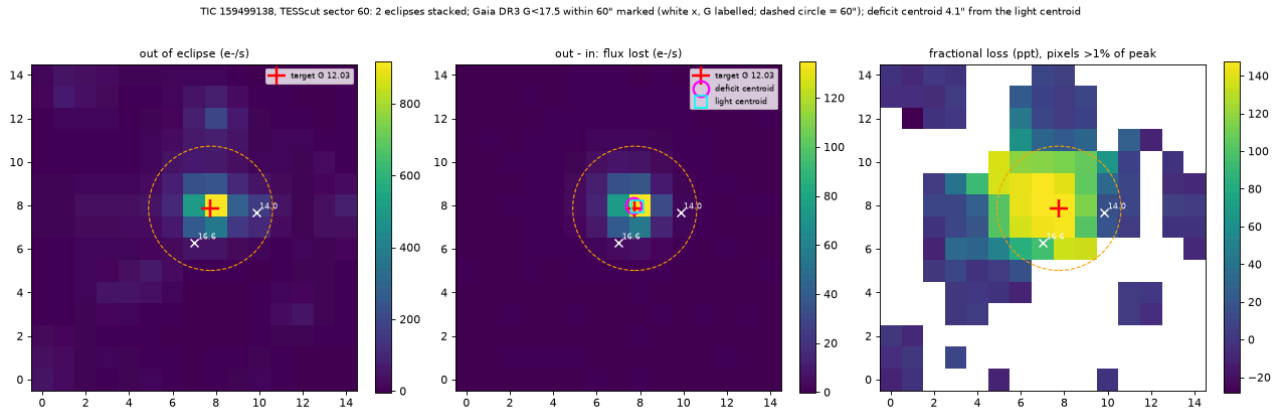
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R428; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R428; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 159499138: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 19; Gaia DR3 source 967967877695470848; actual DR3 G=12.033017. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 5 usable measured eclipse centers across sectors 20, 60, 73; 5 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 155.52 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=7.883525722 ± 0.000024373 d and reference center BTJD 1864.487494. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=7.883934523 d; difference=-0.000409 d.
- Pixel attribution and alternative reduction: historical TESScut sector 60 stacked 2 eclipse windows; the flux-deficit centroid is 0.135 pixel from the Gaia target and the peak deficit pixel is [8, 8] versus target pixel [8, 8]. Custom 1×1/3×3/5×5 pixel apertures give 147.0/139.6/127.9 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 10 Gaia neighbors within 60 arcsec were tested for flux capability; 0 are capable. No Gaia neighbor is flux-capable.
- Gaia geometry: selected edge-on conjunction offset -0.01428 in orbital phase, linear covariance scale 0.133 d. Approximate central duration 9.62 h versus observed median 6.48 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/159499138/diff_s60.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet159499138/tesscut_s060.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R428. Public evidence files: diff_s60.json, diff_s60.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

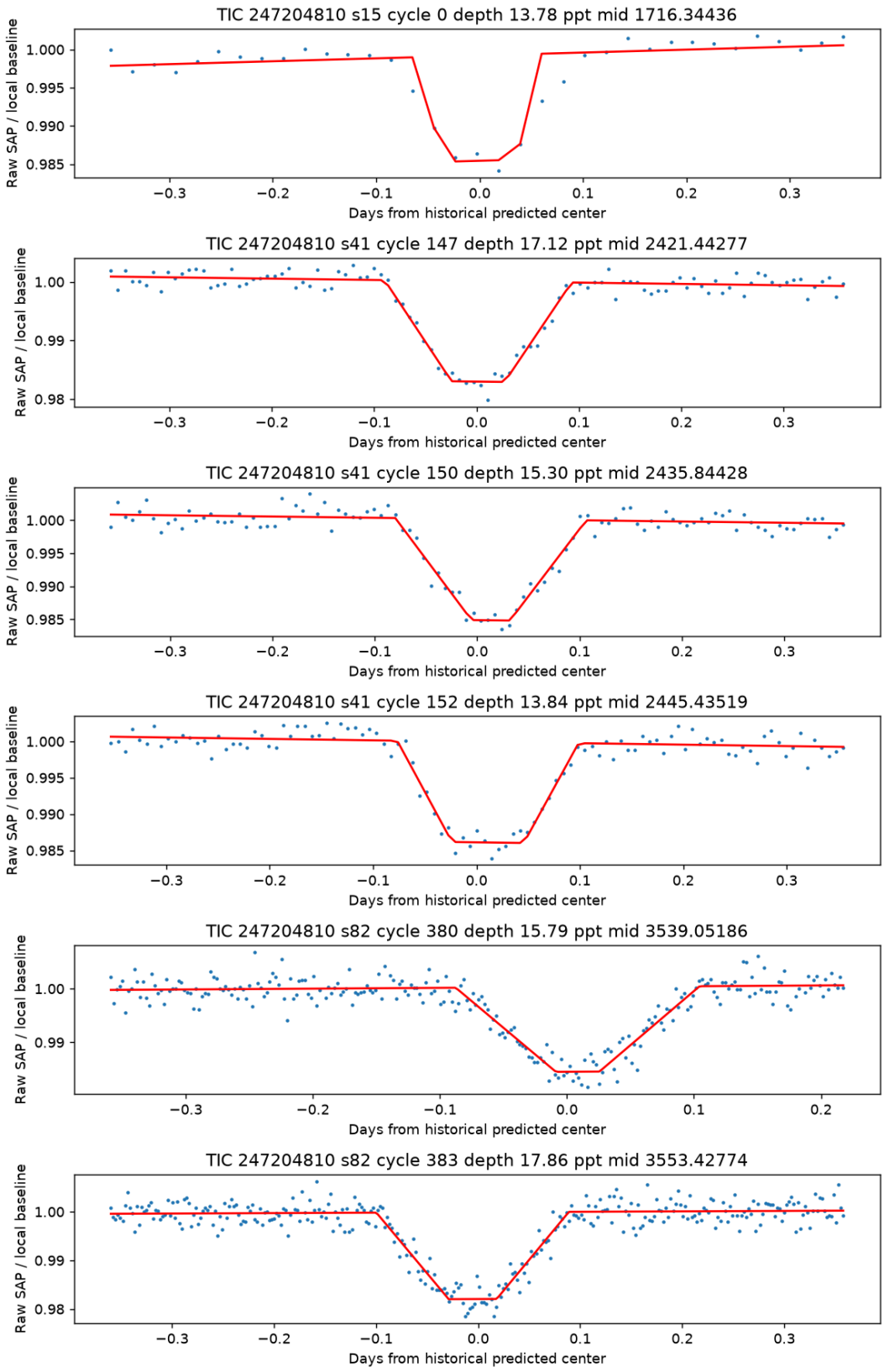
R429. TIC 247204810

physical_verdict: CONFIRMED

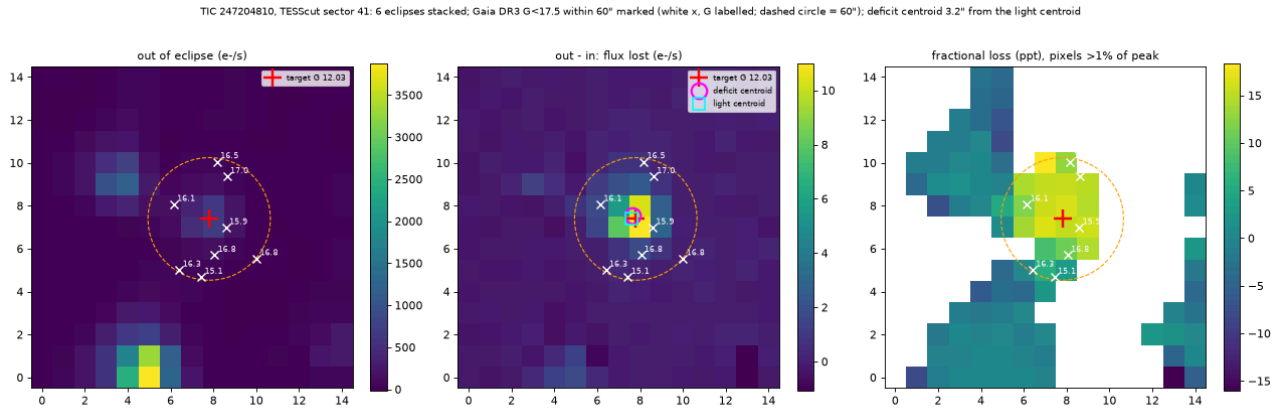
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional independent observations can refine the ephemeris and source attribution.



R429; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R429; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 247204810: batch 02 review

Physical verdict: confirmed. **Full bounded review:** complete. **Catalogue scope:** no identified prior EB; seven-group coverage complete. Catalogue status is separate from physical classification.

Frozen queue position 20; Gaia DR3 source 1842700536609008000; actual DR3 G=12.034502. No replacement was made.

Deciding evidence

- Raw QLP SAP flux: 12 usable measured eclipse centers across sectors 15, 41, 82; 7 selected same-kind centers enter the orbital timing fit. Sampled FITS and their hashes are in `../source-receipts.json`. Measured depths for the selected kind have median 17.16 ppt. `measurements.json`, `raw-windows.csv` and `raw-events.png` retain each fit and local flux.
- Timing: measured same-kind centers give P=9.593093220 ± 0.000017436 d and reference center BTJD 1730.740727. Epoch kind: Selected deeper repeating eclipse; odd historical half-period cycles. This is a linear timing fit to actual centers, with per-event max(formal trapezoid covariance error, half cadence) and covariance inflated for residual scatter. It does not include calibrated pipeline or shape systematics. Gaia SB1 P=9.594030212 d; difference=-0.000937 d.
- Pixel attribution and alternative reduction: historical TESScut sector 41 stacked 6 eclipse windows; the flux-deficit centroid is 0.192 pixel from the Gaia target and the peak deficit pixel is [8, 7] versus target pixel [8, 7]. Custom 1×1/3×3/5×5 pixel apertures give 15.4/15.3/14.4 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. All 42 Gaia neighbors within 60 arcsec were tested for flux capability; 5 are capable. Nearest capable: G=15.90, 17.29 arcsec, requiring 57.7% intrinsic eclipse, 1.07 pixel from the deficit.
- Gaia geometry: selected edge-on conjunction offset -0.01414 in orbital phase, linear covariance scale 0.111 d. Approximate central duration 10.53 h versus observed median 4.53 h. The duration estimate assumes TIC radius and a minimum-mass main-sequence companion; it is a plausibility check, not a fit.

Limitations and next check

Representative-sector timing and pixel evidence; period uncertainty is formal fit/cadence with scatter inflation where needed, not calibrated total coverage. Alternating half-period events were separated by eclipse kind before orbital timing.

Additional independent observations can refine the ephemeris and source attribution.

Source provenance

Historical screening: `Fable 5\made\sb1-scale\c6-catrows.md` and `[local path omitted] 5/made/sb1-scale/c6-vetsum-out.txt`. Historical pixel stack: `../../../../Fable 5/made/sb1-scale/diffimg/247204810/diff_s41.json` and adjacent PNG/Gaia JSON, computed by `diffstack.py` from `../../../../sky/data/sb1-scale-c3/vet247204810/tesscut_s041.fits`. Current measured evidence: this folder, `../gaia-geometry.json`, `../source-receipts.json`, `../download-receipts.json`; central scoped catalogue: `../../shared-catalogues/coverage-compact.csv`. Raw QLP URI and hashes appear in `../download-receipts.json`.

Stable review key: R429. Public evidence files: diff_s41.json, diff_s41.png, gaia.json, measurements.json, pixel-evidence.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

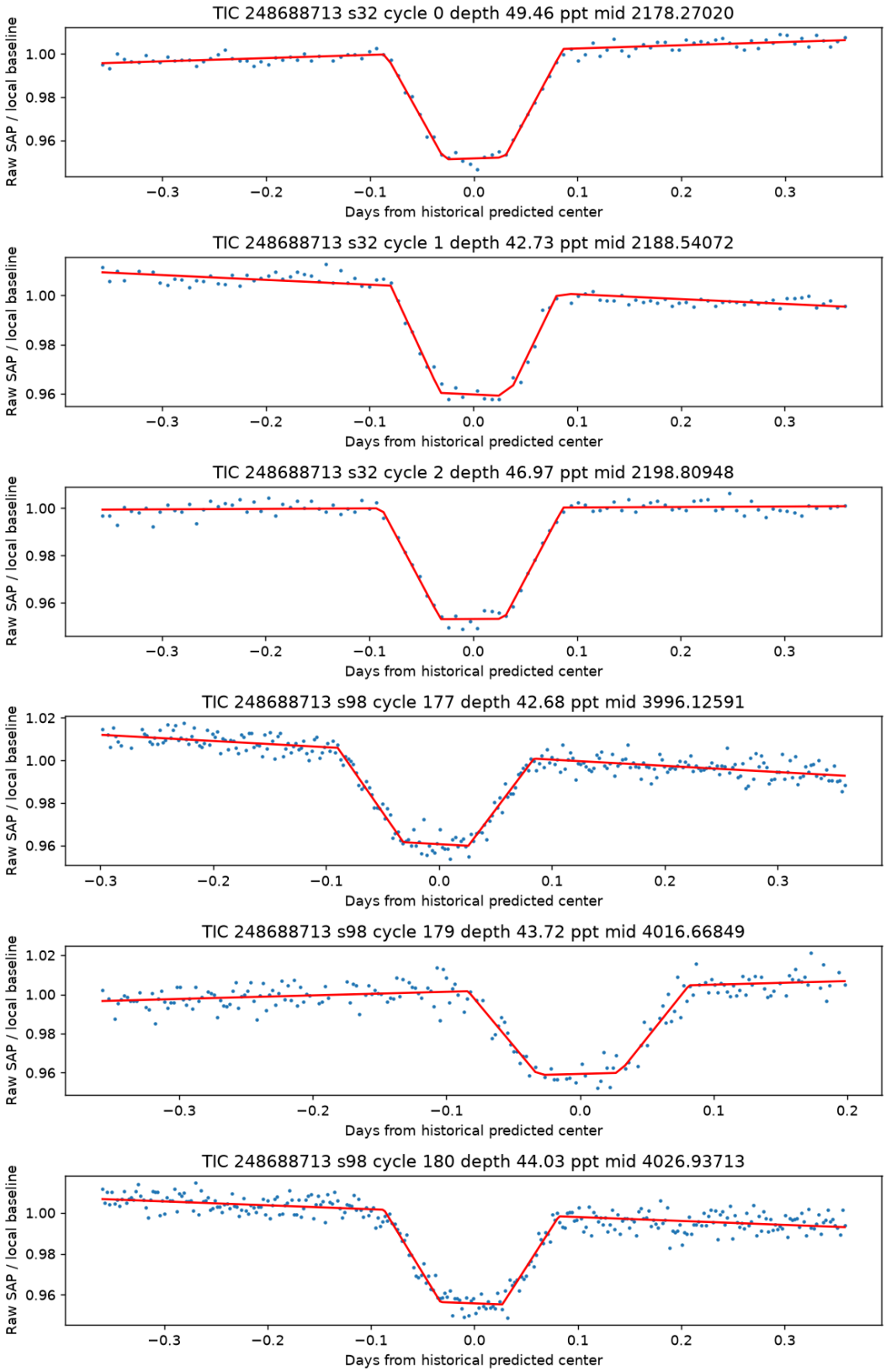
R430. TIC 248688713

physical_verdict: CONFIRMED

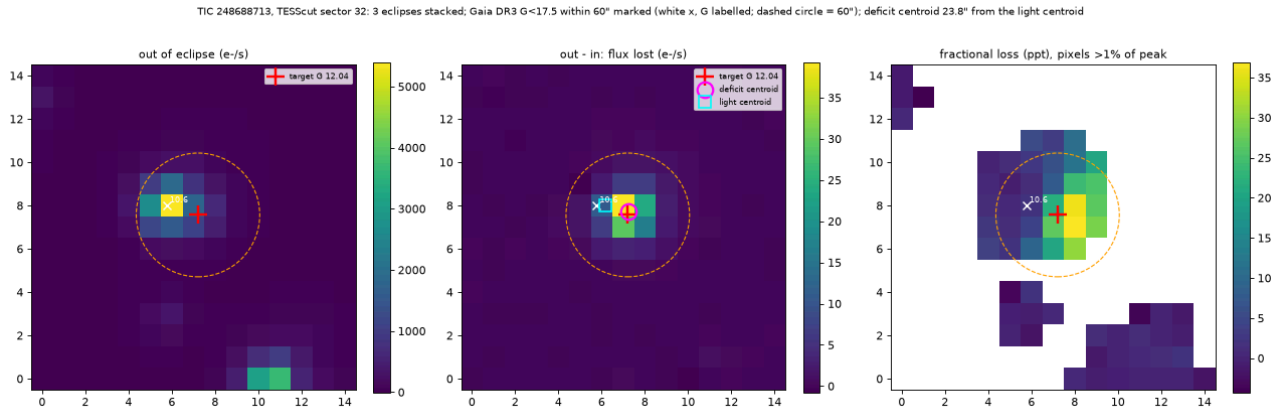
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R430; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R430; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 248688713: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured centers in sectors [32, 98]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 10.2703783974 ± 0.0000121037 d and reference epoch BTJD 2178.2697573 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 3 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 32 TESScut difference image puts the deficit 3.00 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.867 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 3205176248663357440 (31.57 arcsec, G 10.569, intrinsic loss approx. 0.054). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 22.05/10.14/8.07 ppt in 3 stacked historical events; representative QLP raw median fit depth is 43.88 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 3205176283023095424 has SB1 P = 10.2702471905 d and e = 0.09526. Closest edge-on conjunction offset is -0.01040 d (-0.00101 phase), with 0.82596 d linearly propagated full source-specific SB1 covariance. Rough central duration 5.62 h versus observed median fit 4.08 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Adjacent S32 cycles and S98 repeats establish recurrence. The brighter capable neighbor at 31.6 arcsec is distinct from the target-centered deficit; high image-model residual limits precision.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
32	0	2178.2702044	0.003472	49.46	used
32	1	2188.5407239	0.003472	42.73	used
32	2	2198.8094783	0.003472	46.97	used
98	177	3996.1259138	0.001736	42.68	used
98	179	4016.6684904	0.001197	43.72	used
98	180	4026.9371326	0.001314	44.03	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s32.json, historical/diff_s32.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R430. Public evidence files: diff_s32.json, diff_s32.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

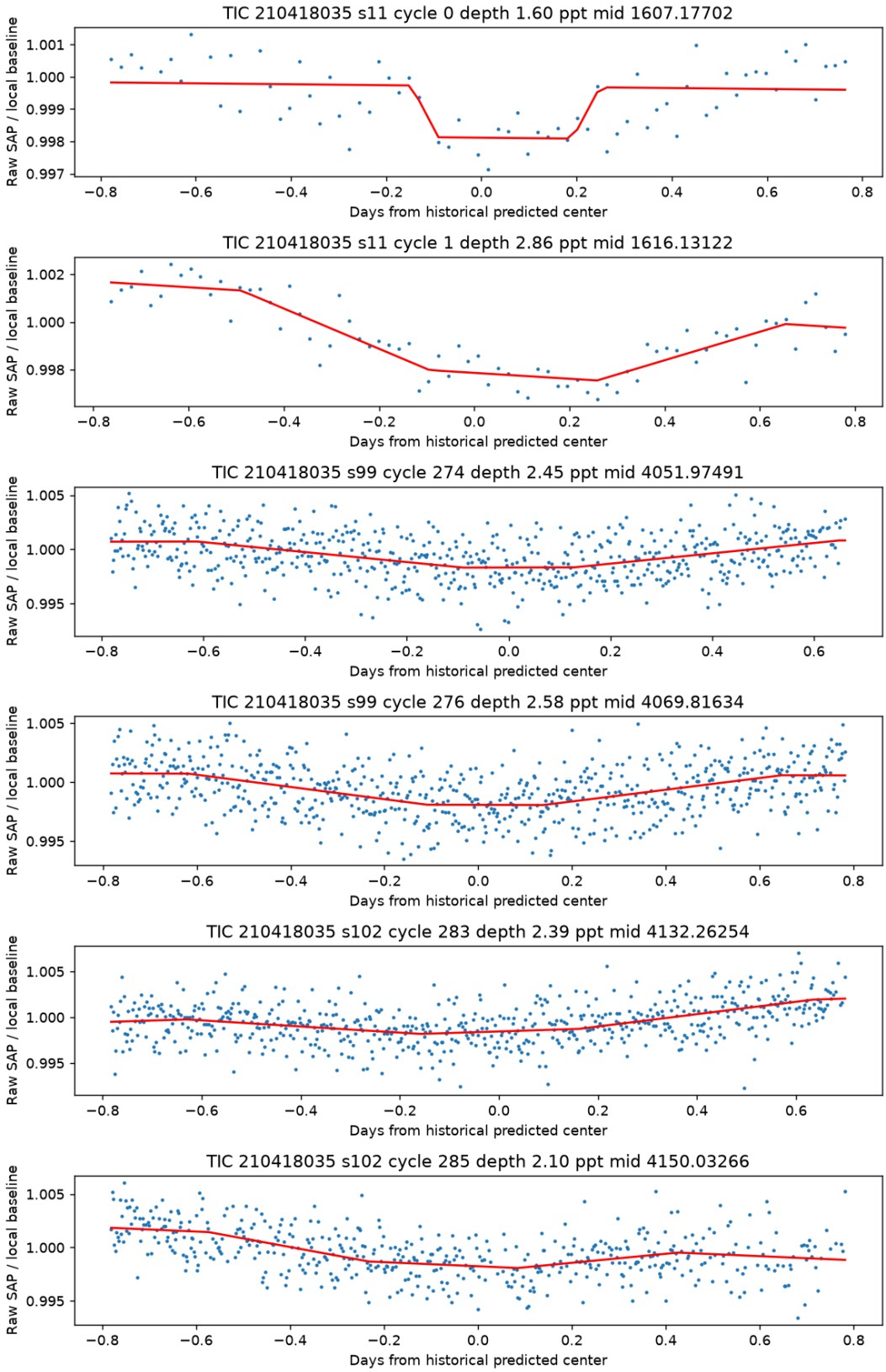
R431. TIC 210418035

physical_verdict: PLAUSIBLE

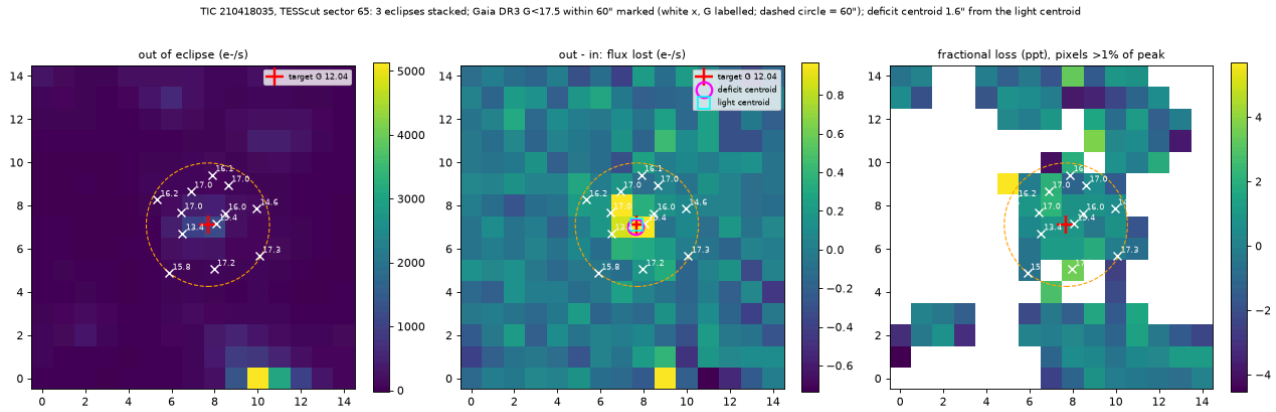
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved time-series photometry and a phase-resolved orbital model to distinguish broad variability from eclipses; repeat difference imaging in a cleaner sector.



R431; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R431; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 210418035: PLAUSIBLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 7 timing-usable measured centers in sectors [11, 99, 102]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. The morphology is eclipse-shaped but inconsistent across sectors.
2. **Timing and aliases.** Measured centers yield P = 8.9225385101 ± 0.0000978228 d and reference epoch BTJD 1607.1807945 (selected repeating phase; eclipse kind unestablished). There are 3 within-sector adjacent selected-phase pairs (../alias-audit.json). These are selected-phase adjacent pairs; the event kind is uncertain. The formal fit describes phase recurrence of a variable signal, not a secure eclipse ephemeris. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 65 TESScut difference image puts the deficit 2.12 arcsec from the target coordinate; peak [9, 0] versus target pixel [8, 7]. Image-model fractional residual 0.709 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 5896574623770523776 (7.68 arcsec, G 15.438, intrinsic loss approx. 0.021); Gaia 5896574619459900800 (9.18 arcsec, G 18.511, intrinsic loss approx. 0.346); Gaia 5896574623759948800 (10.27 arcsec, G 19.663, intrinsic loss approx. 0.999); Gaia 5896574623759957760 (15.97 arcsec, G 18.988, intrinsic loss approx. 0.537); Gaia 5896574619457099008 (18.58 arcsec, G 18.799, intrinsic loss approx. 0.451); Gaia 5896574623759957888 (19.52 arcsec, G 18.876, intrinsic loss approx. 0.485); Gaia 5896574619455713664 (19.90 arcsec, G 15.984, intrinsic loss approx. 0.035); Gaia 5896574619457093248 (21.97 arcsec, G 18.749, intrinsic loss approx. 0.431); Gaia 5896574619457101952 (22.44 arcsec, G 18.796, intrinsic loss approx. 0.450); Gaia 5896574619457093376 (23.21 arcsec, G 19.170, intrinsic loss approx. 0.635); Gaia 5896574623759952384 (23.80 arcsec, G 19.118, intrinsic loss approx. 0.605); Gaia 5896574623759938432 (24.48 arcsec, G 19.572, intrinsic loss approx. 0.920); Gaia 5896574619457102464 (24.49 arcsec, G 19.161, intrinsic loss approx. 0.630); Gaia

5896574619457548288 (25.42 arcsec, G 19.630, intrinsic loss approx. 0.969); Gaia
5896574623759979136 (25.77 arcsec, G 19.080, intrinsic loss approx. 0.585); Gaia
5896574413297283200 (25.91 arcsec, G 13.377, intrinsic loss approx. 0.004); Gaia
5896574623759954304 (26.82 arcsec, G 19.487, intrinsic loss approx. 0.850); Gaia
5896574413297281408 (28.11 arcsec, G 17.001, intrinsic loss approx. 0.087); Gaia
5896574619457547776 (28.82 arcsec, G 19.221, intrinsic loss approx. 0.665); Gaia
5896574658119698304 (29.26 arcsec, G 19.457, intrinsic loss approx. 0.827); Gaia
5896574623759928192 (32.64 arcsec, G 19.418, intrinsic loss approx. 0.798); Gaia
5896574619457104768 (33.21 arcsec, G 18.480, intrinsic loss approx. 0.337); Gaia
5896574653816841728 (33.51 arcsec, G 19.571, intrinsic loss approx. 0.918); Gaia
5896574619457105152 (33.74 arcsec, G 19.092, intrinsic loss approx. 0.591); Gaia
5896574417612080000 (34.55 arcsec, G 18.078, intrinsic loss approx. 0.233); Gaia
5896574516376495872 (34.90 arcsec, G 17.875, intrinsic loss approx. 0.193); Gaia
5896574692490007680 (35.11 arcsec, G 19.516, intrinsic loss approx. 0.873); Gaia
5896574417612079488 (35.91 arcsec, G 16.963, intrinsic loss approx. 0.084); Gaia
5896574688176582528 (37.61 arcsec, G 17.814, intrinsic loss approx. 0.183); Gaia
5896574653816838016 (37.91 arcsec, G 18.545, intrinsic loss approx. 0.358); Gaia
5896574516377875328 (37.91 arcsec, G 19.489, intrinsic loss approx. 0.852); Gaia
5896574417612078592 (38.25 arcsec, G 17.853, intrinsic loss approx. 0.189); Gaia
5896574658119723648 (38.42 arcsec, G 19.538, intrinsic loss approx. 0.891); Gaia
5896574585097358208 (38.53 arcsec, G 18.301, intrinsic loss approx. 0.286); Gaia
5896574413297282304 (38.64 arcsec, G 17.838, intrinsic loss approx. 0.187); Gaia
5896574692479462528 (38.71 arcsec, G 19.535, intrinsic loss approx. 0.888); Gaia
5896574688176583296 (40.38 arcsec, G 19.346, intrinsic loss approx. 0.747); Gaia
5896574413297279616 (40.88 arcsec, G 17.527, intrinsic loss approx. 0.141); Gaia
5896574653824506368 (42.90 arcsec, G 17.239, intrinsic loss approx. 0.108); Gaia
5896574520691300352 (43.04 arcsec, G 17.019, intrinsic loss approx. 0.088); Gaia
5896574688176583936 (45.47 arcsec, G 19.544, intrinsic loss approx. 0.896); Gaia
5896574413298657280 (45.69 arcsec, G 19.247, intrinsic loss approx. 0.681); Gaia
5896574589400232192 (46.92 arcsec, G 18.166, intrinsic loss approx. 0.252); Gaia
5896574447658407040 (47.01 arcsec, G 19.215, intrinsic loss approx. 0.662); Gaia
5896574688182055552 (47.10 arcsec, G 19.232, intrinsic loss approx. 0.672); Gaia
5896574413298659712 (47.19 arcsec, G 18.123, intrinsic loss approx. 0.243); Gaia
5896574413298662528 (47.26 arcsec, G 18.059, intrinsic loss approx. 0.229); Gaia
5896574653816845056 (47.66 arcsec, G 17.986, intrinsic loss approx. 0.214); Gaia
5896574520691294336 (48.26 arcsec, G 16.112, intrinsic loss approx. 0.039); Gaia
5896574722536323456 (48.27 arcsec, G 18.348, intrinsic loss approx. 0.298); Gaia
5896574417601476096 (48.55 arcsec, G 19.477, intrinsic loss approx. 0.842); Gaia
5896574585095976960 (49.70 arcsec, G 14.577, intrinsic loss approx. 0.010); Gaia
5896574314522261248 (50.06 arcsec, G 19.479, intrinsic loss approx. 0.844); Gaia
5896574692493101056 (50.92 arcsec, G 17.866, intrinsic loss approx. 0.192); Gaia
5896574688176585216 (51.45 arcsec, G 19.374, intrinsic loss approx. 0.766); Gaia
5896574451961279616 (52.23 arcsec, G 19.453, intrinsic loss approx. 0.824); Gaia
5896574589400233472 (52.75 arcsec, G 18.864, intrinsic loss approx. 0.479); Gaia
5896574447658408064 (52.75 arcsec, G 19.371, intrinsic loss approx. 0.764); Gaia
5896574447658406528 (53.28 arcsec, G 17.713, intrinsic loss approx. 0.167); Gaia
5896574413297279232 (54.11 arcsec, G 16.211, intrinsic loss approx. 0.042); Gaia
5896574722536324480 (55.11 arcsec, G 18.205, intrinsic loss approx. 0.262); Gaia

5896574585097810816 (55.12 arcsec, G 19.395, intrinsic loss approx. 0.781); Gaia 5896574413298655104 (55.31 arcsec, G 19.128, intrinsic loss approx. 0.611); Gaia 5896574692479480192 (55.72 arcsec, G 19.163, intrinsic loss approx. 0.631); Gaia 5896574310219439616 (55.76 arcsec, G 18.804, intrinsic loss approx. 0.453); Gaia 5896574413298655488 (56.70 arcsec, G 18.386, intrinsic loss approx. 0.309); Gaia 5896574692490013440 (57.19 arcsec, G 17.313, intrinsic loss approx. 0.116); Gaia 5896574417612073856 (59.74 arcsec, G 18.966, intrinsic loss approx. 0.527); Gaia 5896574451971834240 (59.95 arcsec, G 15.806, intrinsic loss approx. 0.029). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.

4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 0.83/0.97/0.78 ppt in 3 stacked historical events; representative QLP raw median fit depth is 2.39 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 5896574619455481216 has SB1 P = 8.9213571665 d and e = 0.27356. Closest edge-on conjunction offset is -0.15304 d (-0.01715 phase), with 0.15937 d linearly propagated full source-specific SB1 covariance. Rough central duration 5.84 h versus observed median fit 27.45 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Broad, changing 0.96–2.86 ppt depressions repeat near the selected phase, but fitted duration is 27.45 h versus a rough 5.84 h central eclipse duration. S11 includes a narrow-looking dip while S99/S102 are broad. A phase-dependent orbital variable, blend or extraction structure remains possible. The timing fit does not establish stable eclipses; crowded S65 pixels are noisy.

Next check: Resolved time-series photometry and a phase-resolved orbital model to distinguish broad variability from eclipses; repeat difference imaging in a cleaner sector.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
11	0	1607.1770181	0.017043	1.60	used
11	1	1616.1312173	0.048568	2.86	used
99	274	4051.9749060	0.047515	2.45	used
99	276	4069.8163427	0.027080	2.58	used
102	283	4132.2625396	0.031026	2.39	used
102	284	4141.2054083	0.018705	0.96	used
102	285	4150.0326581	0.026959	2.10	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s65.json, historical/diff_s65.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R431. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

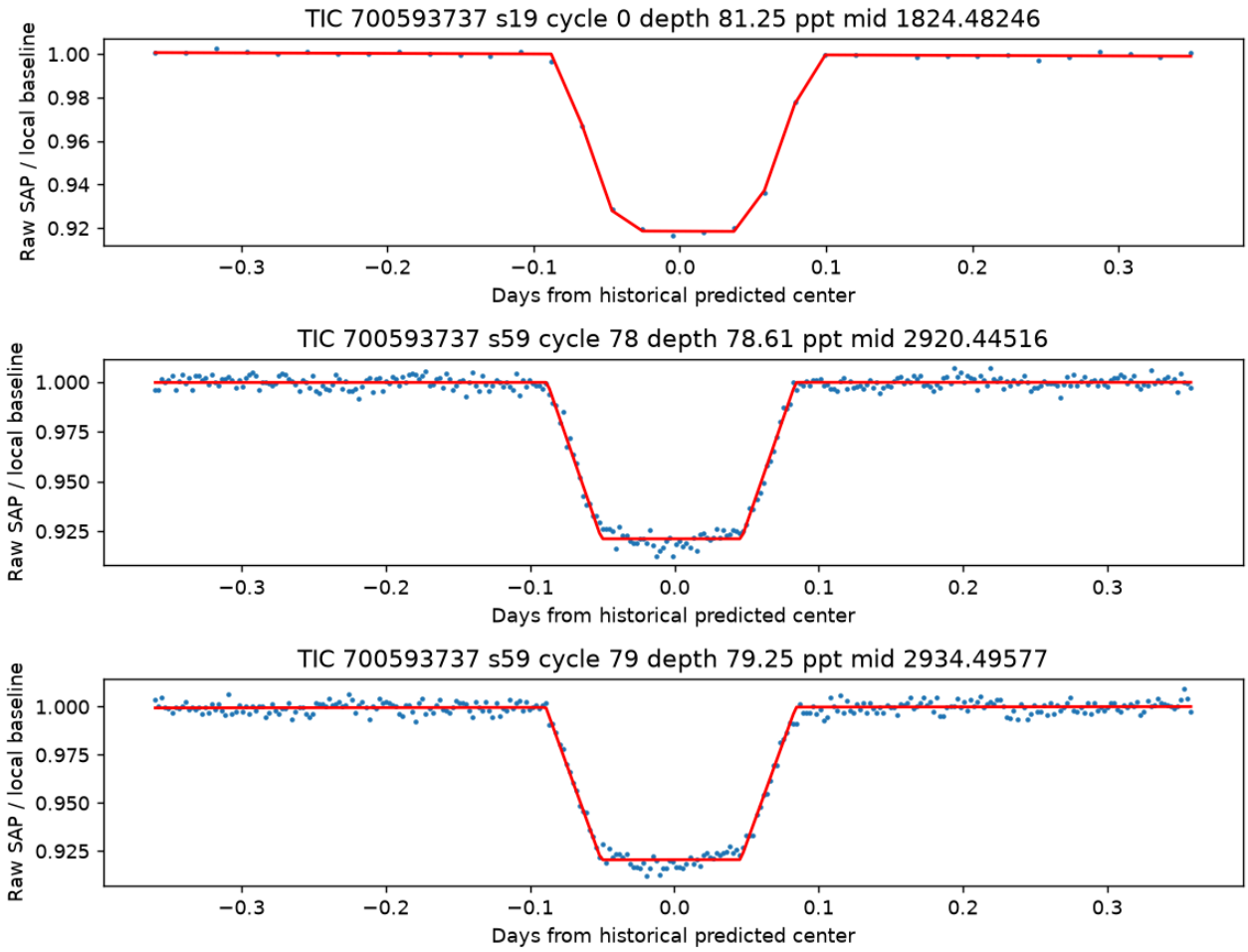
R432. TIC 700593737

physical_verdict: PROBABLE

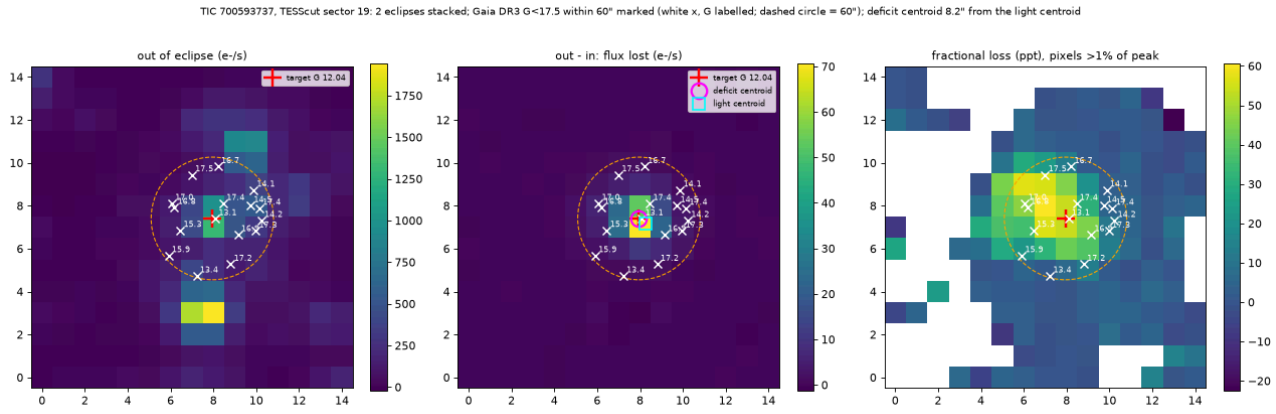
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PSF localization of the 2.8 arcsec pair.



R432; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R432; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 700593737: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 3 timing-usable measured centers in sectors [19, 59]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 14.0508013034 ± 0.0001326694 d and reference epoch BTJD 1824.4825619 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 19 TESScut difference image puts the deficit 0.69 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.866 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 190955384147342592 (2.80 arcsec, G 13.139, intrinsic loss approx. 0.301); Gaia 190955384151069696 (38.88 arcsec, G 14.451, intrinsic loss approx. 0.819); Gaia 190956857322461184 (47.10 arcsec, G 14.240, intrinsic loss approx. 0.688); Gaia 190956861618883840 (48.54 arcsec, G 14.129, intrinsic loss approx. 0.629); Gaia 190955349790083200 (58.08 arcsec, G 13.440, intrinsic loss approx. 0.371). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 54.10/49.36/34.54 ppt in 2 stacked historical events; representative QLP raw median fit depth is 79.25 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 190955379851574912 has SB1 P = 14.0495286778 d and e = 0.40569. Closest edge-on conjunction offset is 0.01921 d (0.00137 phase), with 0.42354 d linearly propagated full source-specific SB1 covariance. Rough central duration 3.87 h versus observed median fit 4.16 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: A raw S19 adjacent-cycle pair confirms repetition. A capable 2.8 arcsec Gaia companion shares the central TESS core; source attribution is unresolved.

Next check: Resolved eclipse photometry or multi-source PSF localization of the 2.8 arcsec pair.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
19	0	1824.4824625	0.010417	81.25	used
59	78	2920.4451605	0.001157	78.61	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
59	79	2934.4957691	0.001157	79.25	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s19.json, historical/diff_s19.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R432. Public evidence files: diff_s19.json, diff_s19.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

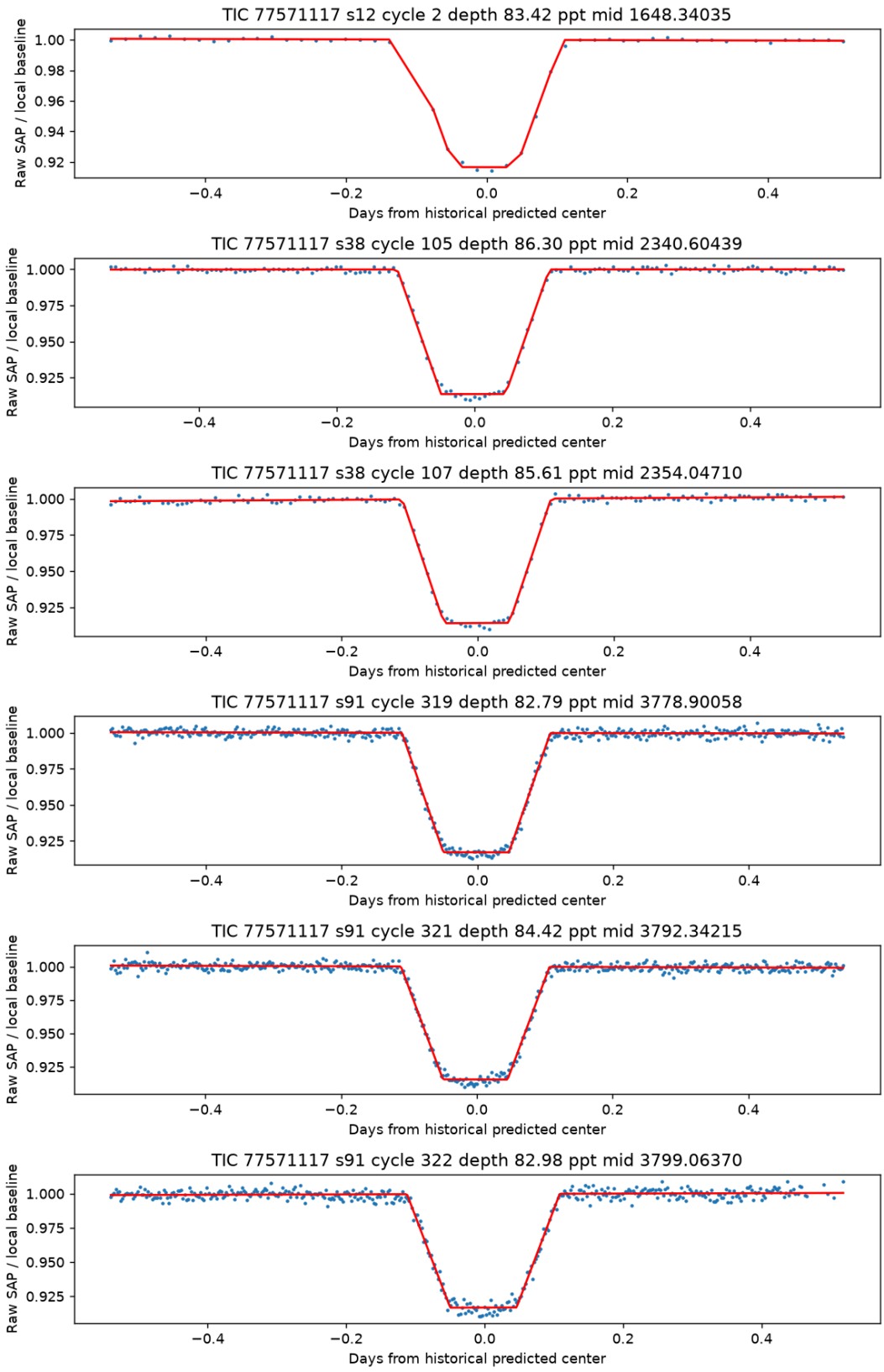
R433. TIC 77571117

physical_verdict: CONFIRMED

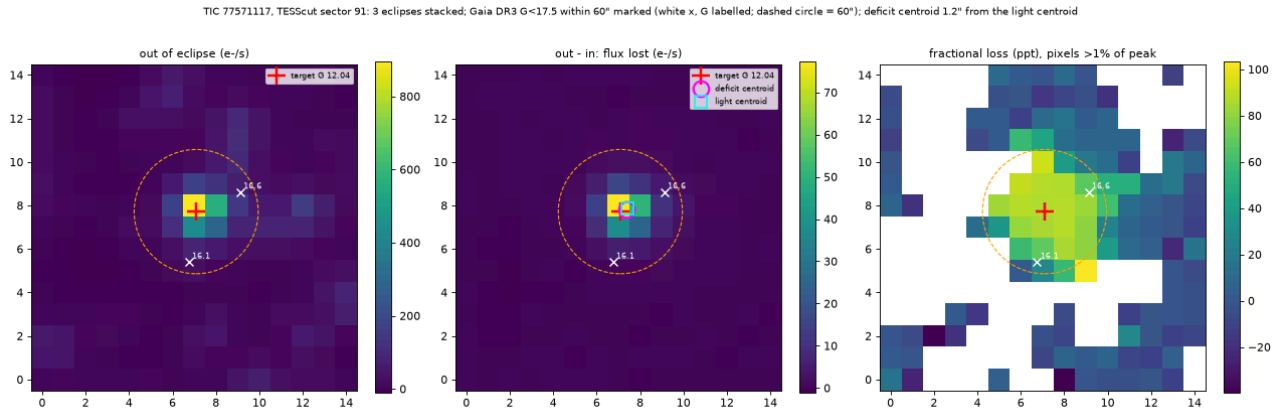
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R433; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R433; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 77571117: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured centers in sectors [12, 38, 91]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 6.7210082588 ± 0.0000112025 d and reference epoch BTJD 1648.3408444 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 91 TESScut difference image puts the deficit 5.93 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.176 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under approximate G-band flux budget. Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 86.36/86.68/82.50 ppt in 3 stacked historical events; representative QLP raw median fit depth is 83.92 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 6041559422449793792 has SB1 P = 6.7218718431 d and e = 0.00991. Closest edge-on conjunction offset is -0.13771 d (-0.02049 phase), with 0.25422 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.62 h versus observed median fit 5.26 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Six raw repeats span three sectors. The deficit is target-centered and no Gaia neighbor is capable under the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	2	1648.3403506	0.010416	83.42	used
38	105	2340.6043885	0.003472	86.30	used
38	107	2354.0471022	0.003472	85.61	used
91	319	3778.9005767	0.001157	82.79	used
91	321	3792.3421496	0.001157	84.42	used
91	322	3799.0636992	0.001157	82.98	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s91.json, historical/diff_s91.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R433. Public evidence files: diff_s91.json, diff_s91.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

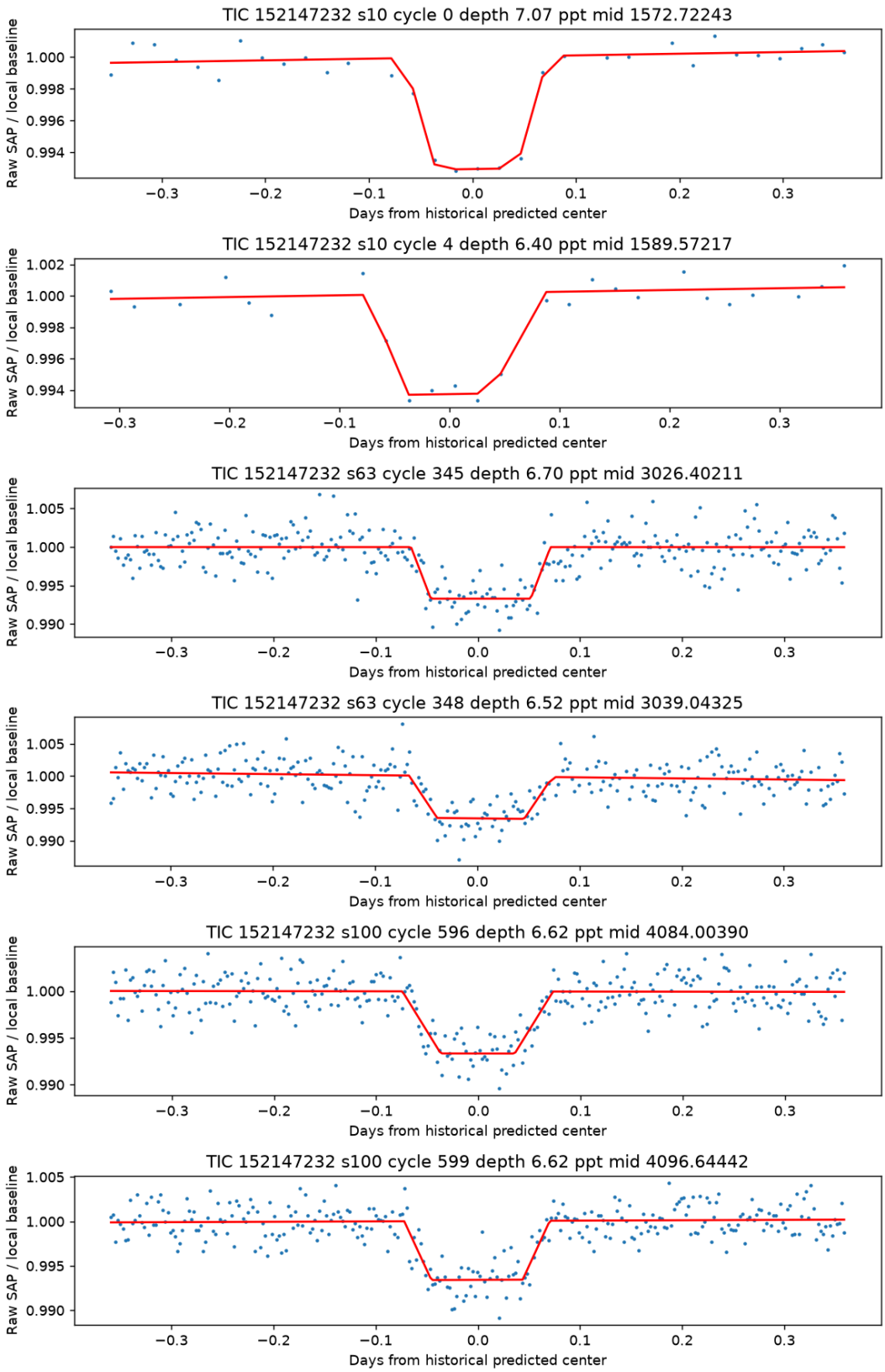
R434. TIC 152147232

physical_verdict: PROBABLE

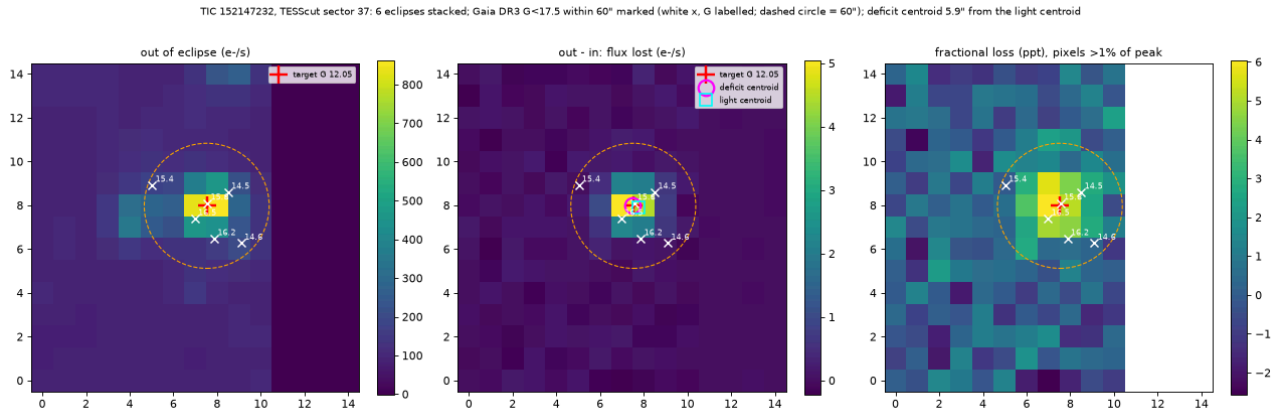
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PSF localization of the 2.7 arcsec pair.



R434; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R434; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 152147232: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 15 timing-usable measured centers in sectors [10, 63, 100]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 4.2135536598 ± 0.0000034278 d and reference epoch BTJD 1572.7251990 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 10 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 37 TESScut difference image puts the deficit 0.69 arcsec from the target coordinate; peak [7, 8] versus target pixel [8, 8]. Image-model fractional residual 0.531 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 5395959695358663040 (2.69 arcsec, G 15.822, intrinsic loss approx. 0.216); Gaia 5395959695356170624 (15.98 arcsec, G 16.524, intrinsic loss approx. 0.407); Gaia 5395959386121017472 (23.16 arcsec, G 14.509, intrinsic loss approx. 0.069); Gaia 5395959695358664832 (31.29 arcsec, G 16.245, intrinsic loss approx. 0.316); Gaia 5395959764078141952 (46.20 arcsec, G 14.642, intrinsic loss approx. 0.077); Gaia 5395959489200231936 (52.26 arcsec, G 15.438, intrinsic loss approx. 0.154). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 5.19/4.77/3.58 ppt in 6 stacked historical events; representative QLP raw median fit depth is 6.70 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 5395959695358663296 has SB1 P = 4.2132911086 d and e = 0.09094. Closest edge-on conjunction offset is -0.03876 d (-0.00920 phase), with 0.37454 d linearly

propagated full source-specific SB1 covariance. Rough central duration 4.84 h versus observed median fit 3.35 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Fifteen raw repeats include many within-sector pairs. A capable 2.7 arcsec Gaia neighbor shares the TESS core and cannot be resolved.

Next check: Resolved eclipse photometry or multi-source PSF localization of the 2.7 arcsec pair.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
10	0	1572.7224256	0.010417	7.07	used
10	2	1581.1488039	0.010417	7.22	used
10	4	1589.5721707	0.010417	6.40	used
63	343	3017.9727299	0.001818	7.87	used
63	344	3022.1868246	0.002043	7.09	used
63	345	3026.4021096	0.001357	6.70	used
63	346	3030.6159883	0.001675	6.81	used
63	347	3034.8274841	0.001542	7.44	used
63	348	3039.0432524	0.002049	6.52	used
100	594	4075.5762282	0.001268	6.27	used
100	595	4079.7915930	0.001251	6.81	used
100	596	4084.0038984	0.001917	6.62	used
100	597	4088.2146792	0.001852	6.67	used
100	598	4092.4283531	0.001173	6.28	used
100	599	4096.6444202	0.001402	6.62	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s37.json, historical/diff_s37.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R434. Public evidence files: diff_s37.json, diff_s37.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

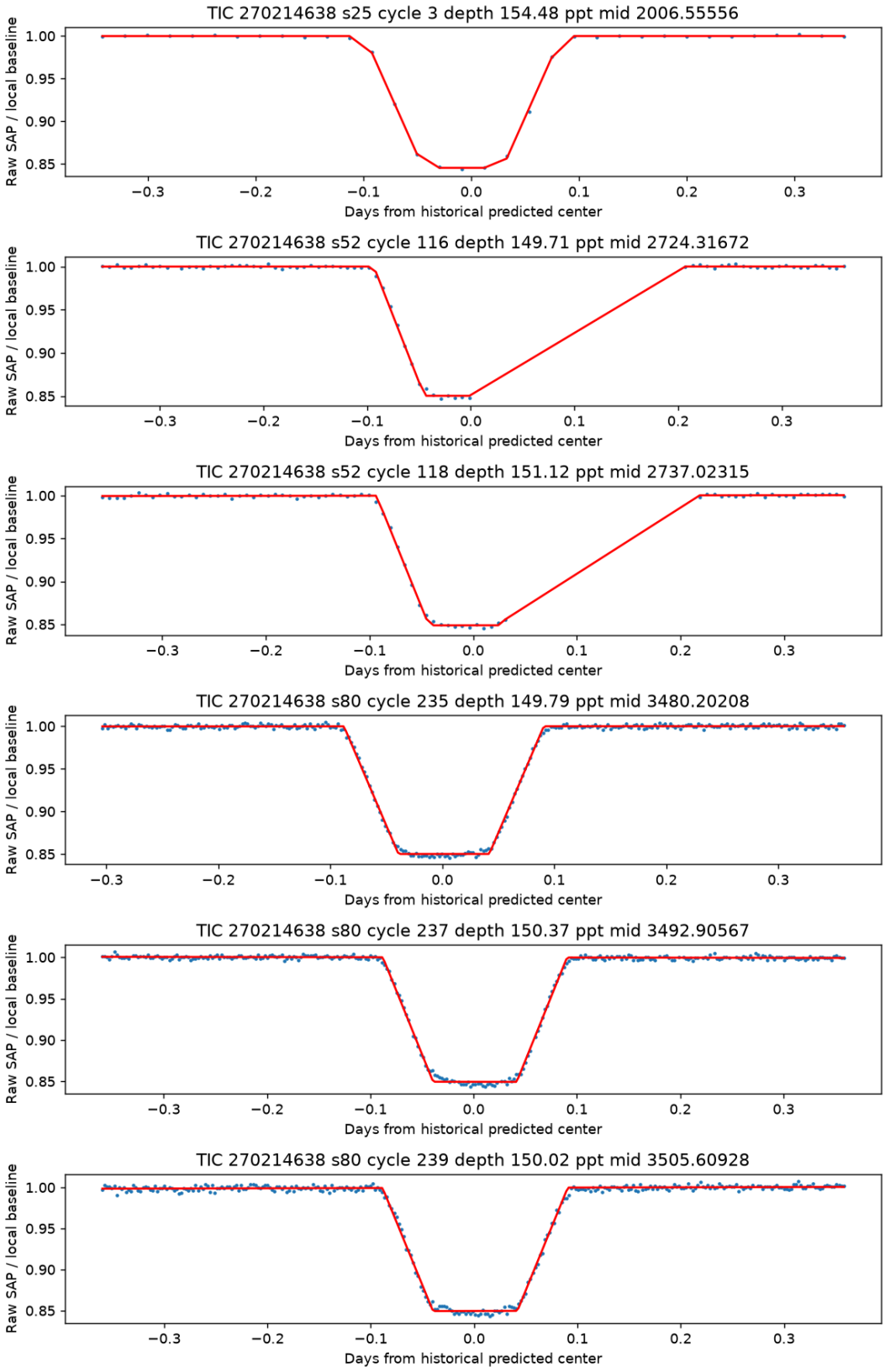
R435. TIC 270214638

physical_verdict: CONFIRMED

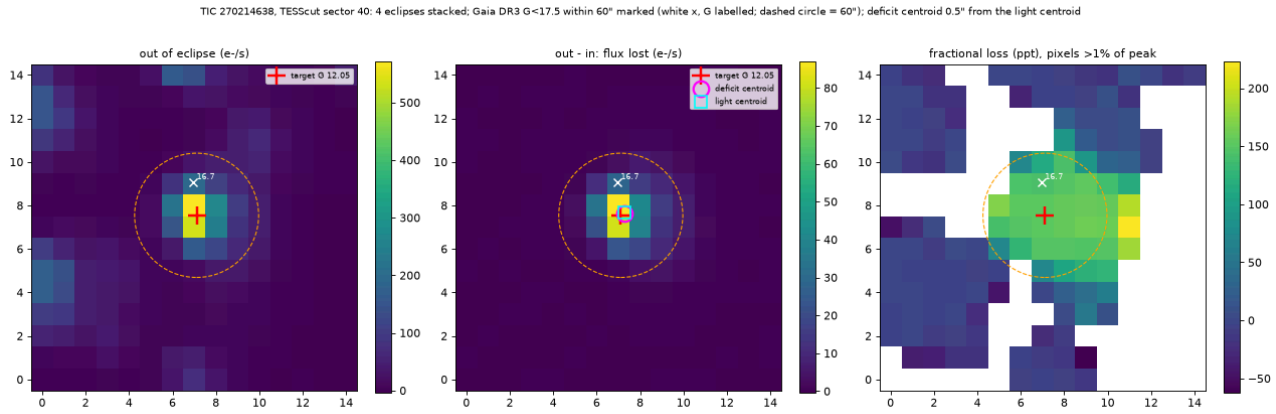
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R435; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R435; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 270214638: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured centers in sectors [25, 52, 80]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 6.3519439147 ± 0.0000246588 d and reference epoch BTJD 2006.5507448 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 40 TESScut difference image puts the deficit 4.28 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.143 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under approximate G-band flux budget. Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 152.27/151.57/149.77 ppt in 4 stacked historical events; representative QLP raw median fit depth is 150.54 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 1360991195262446720 has SB1 P = 6.3527030171 d and e = 0.00643. Closest edge-on conjunction offset is -0.19216 d (-0.03025 phase), with 0.19511 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.95 h versus observed median fit 4.28 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Six deep raw repeats span three sectors. The deficit is target-centered and no Gaia neighbor is capable under the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
25	3	2006.5555644	0.010417	154.48	used
52	116	2724.3167245	166474.921768	149.71	excluded: Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
52	118	2737.0231476	0.003472	151.12	used
80	235	3480.2020761	0.001157	149.79	used
80	237	3492.9056726	0.001157	150.37	used
80	238	3499.2574710	0.001157	150.70	used
80	239	3505.6092753	0.001157	150.02	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s40.json, historical/diff_s40.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R435. Public evidence files: diff_s40.json, diff_s40.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

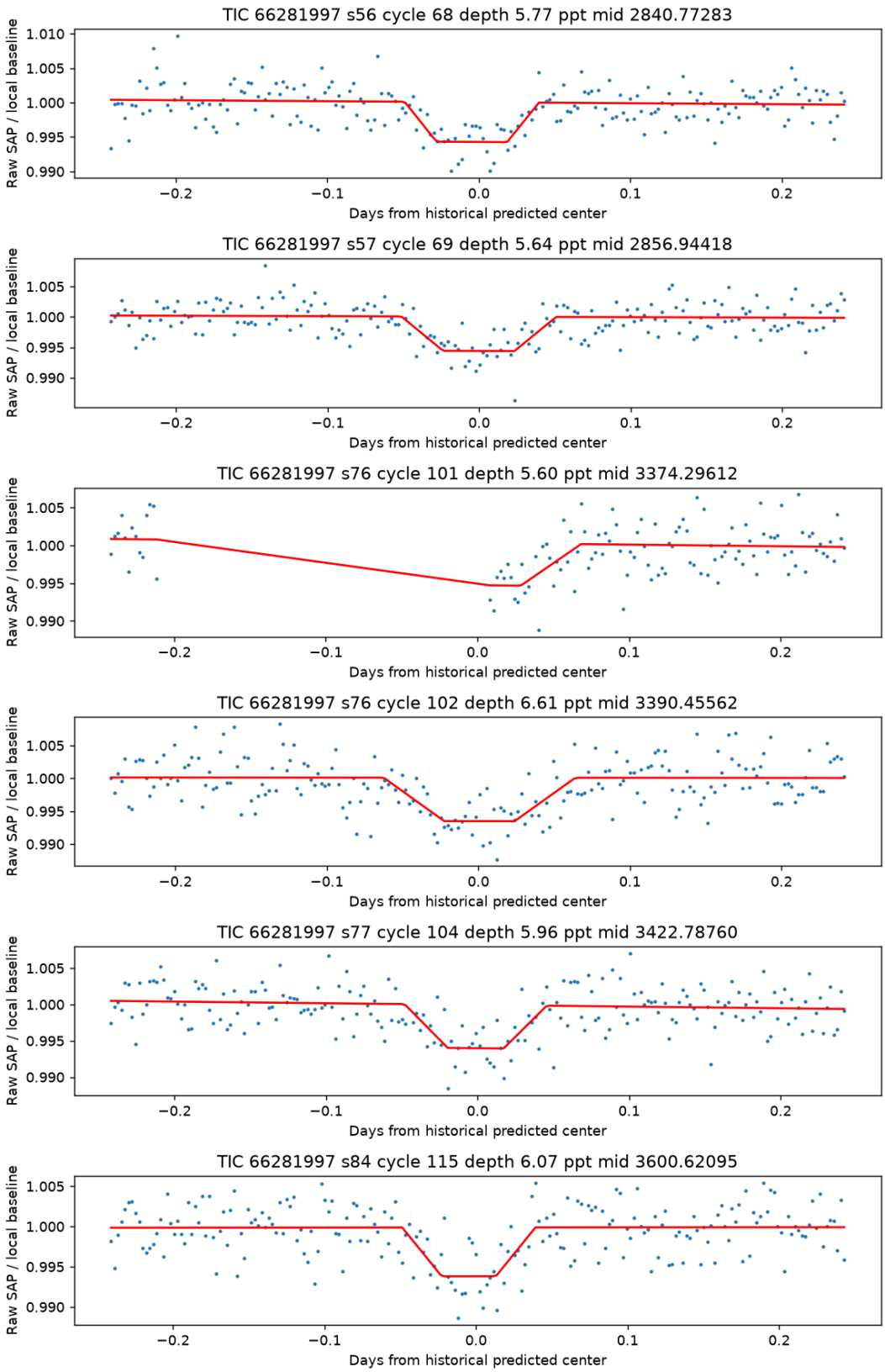
R436. TIC 66281997

physical_verdict: PROBABLE

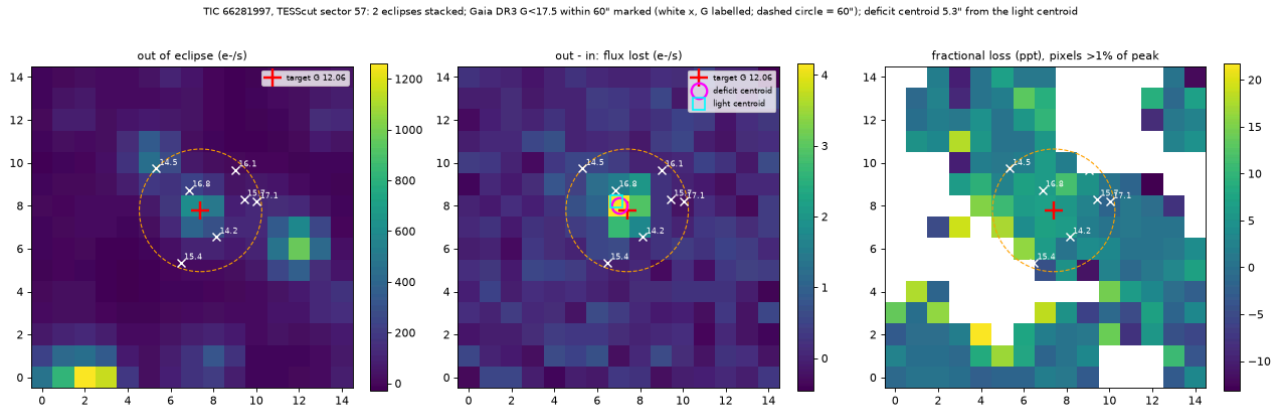
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Independent high-resolution eclipse photometry or multi-source PSF localization of the crowded deficit core.



R436; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R436; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 66281997: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured centers in sectors [56, 57, 76, 77, 83, 84]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 16.1669963889 ± 0.0000470947 d and reference epoch BTJD 2840.7742919 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 57 TESScut difference image puts the deficit 9.07 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.659 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 1988346111733826048 (21.78 arcsec, G 16.847, intrinsic loss approx. 0.477); Gaia 1988346180453300224 (28.72 arcsec, G 14.216, intrinsic loss approx. 0.047); Gaia 1988344634265076480 (42.81 arcsec, G 15.709, intrinsic loss approx. 0.171); Gaia 1988346111733827840 (51.33 arcsec, G 16.147, intrinsic loss approx. 0.253); Gaia 1988346936367541632 (53.51 arcsec, G 15.384, intrinsic loss approx. 0.128); Gaia 1988344634265076608 (54.24 arcsec, G 17.139, intrinsic loss approx. 0.622); Gaia 1988346146093565824 (56.84 arcsec, G 14.496, intrinsic loss approx. 0.060). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 6.83/6.71/5.38 ppt in 2 stacked historical events; representative QLP raw median fit depth is 5.95 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.

5. **Gaia geometry.** Matched DR3 source 1988346180453301632 has SB1 P = 16.1641569409 d and e = 0.40209. Closest edge-on conjunction offset is 0.43325 d (0.02680 phase), with 0.51578 d linearly propagated full source-specific SB1 covariance. Rough central duration 7.71 h versus observed median fit 2.48 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Eight usable raw centers lie in six sectors after all eight archived QLP sectors were examined; they include adjacent S56/S57 and S76/S77 cycles. The historical S57 deficit lies 9.1 arcsec from the target, toward a capable G16.8 star 21.8 arcsec away. Noisy pixels and other capable neighbors prevent unique attribution.

Next check: Independent high-resolution eclipse photometry or multi-source PSF localization of the crowded deficit core.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
56	68	2840.7728335	0.002230	5.77	used
57	69	2856.9441832	0.002354	5.64	used
57	70	2873.1056462	0.002773	5.92	used
76	101	3374.2961244	392421.524476	5.60	excluded: Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
76	102	3390.4556240	0.003173	6.61	used
77	103	3406.6183094	0.002544	6.55	used
77	104	3422.7876011	0.002693	5.96	used
83	113	3568.2897722	0.003193	5.94	used
84	115	3600.6209460	0.002463	6.07	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s57.json, historical/diff_s57.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R436. Public evidence files: diff_s57.json, diff_s57.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

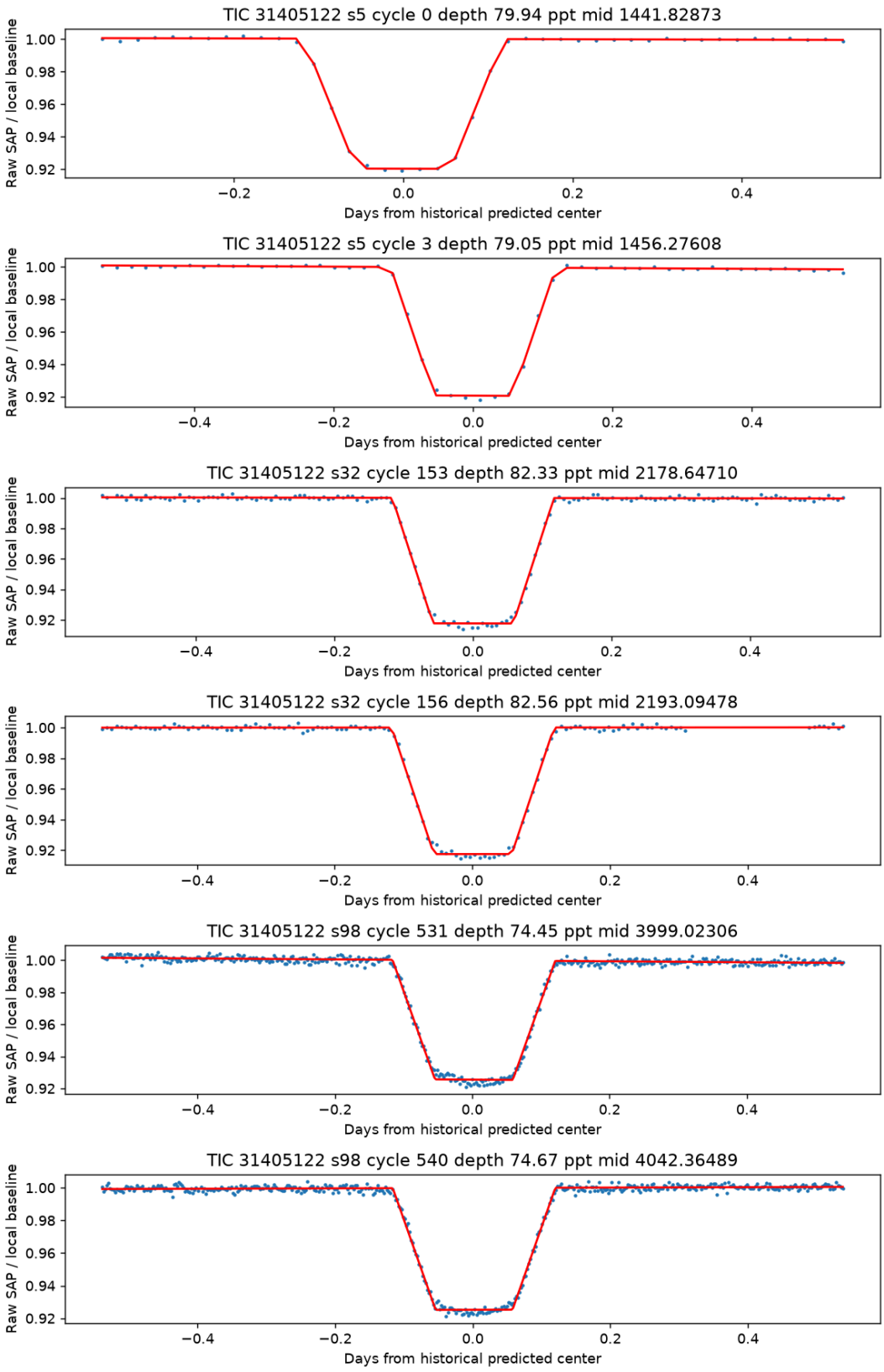
R437. TIC 31405122

physical_verdict: CONFIRMED

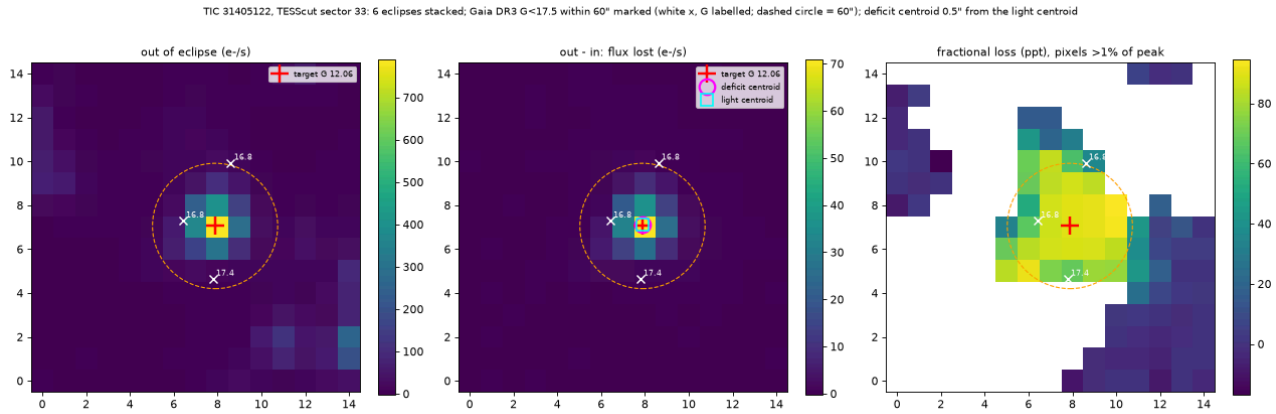
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R437; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R437; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 31405122: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 13 timing-usable measured centers in sectors [5, 32, 98]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 4.8158084158 ± 0.0000040354 d and reference epoch BTJD 1441.8286729 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 6 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 33 TESScut difference image puts the deficit 1.77 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.070 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under approximate G-band flux budget. Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 89.80/88.84/87.53 ppt in 6 stacked historical events; representative QLP raw median fit depth is 80.36 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 2909098108269909376 has SB1 P = 4.8156658314 d and e = 0.04267. Closest edge-on conjunction offset is 0.07549 d (0.01568 phase), with 0.06737 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.26 h versus observed median fit 5.67 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Thirteen deep raw repeats include six adjacent-cycle pairs. The deficit is target-centered and no Gaia neighbor is capable under the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
5	0	1441.8287271	0.010417	79.94	used
5	1	1446.6437397	0.010417	80.36	used
5	3	1456.2760786	0.010417	79.05	used
5	4	1461.0923054	0.010417	80.75	used
32	153	2178.6470952	0.003472	82.33	used
32	154	2183.4632085	0.003472	82.12	used
32	155	2188.2792504	0.003472	82.21	used
32	156	2193.0947793	0.003472	82.56	used
32	157	2197.9105536	0.003472	82.79	used
98	531	3999.0230649	0.001157	74.45	used
98	534	4013.4704924	0.001157	73.93	used
98	537	4027.9178740	0.001157	74.82	used
98	540	4042.3648909	0.001157	74.67	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s33.json, historical/diff_s33.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R437. Public evidence files: diff_s33.json, diff_s33.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

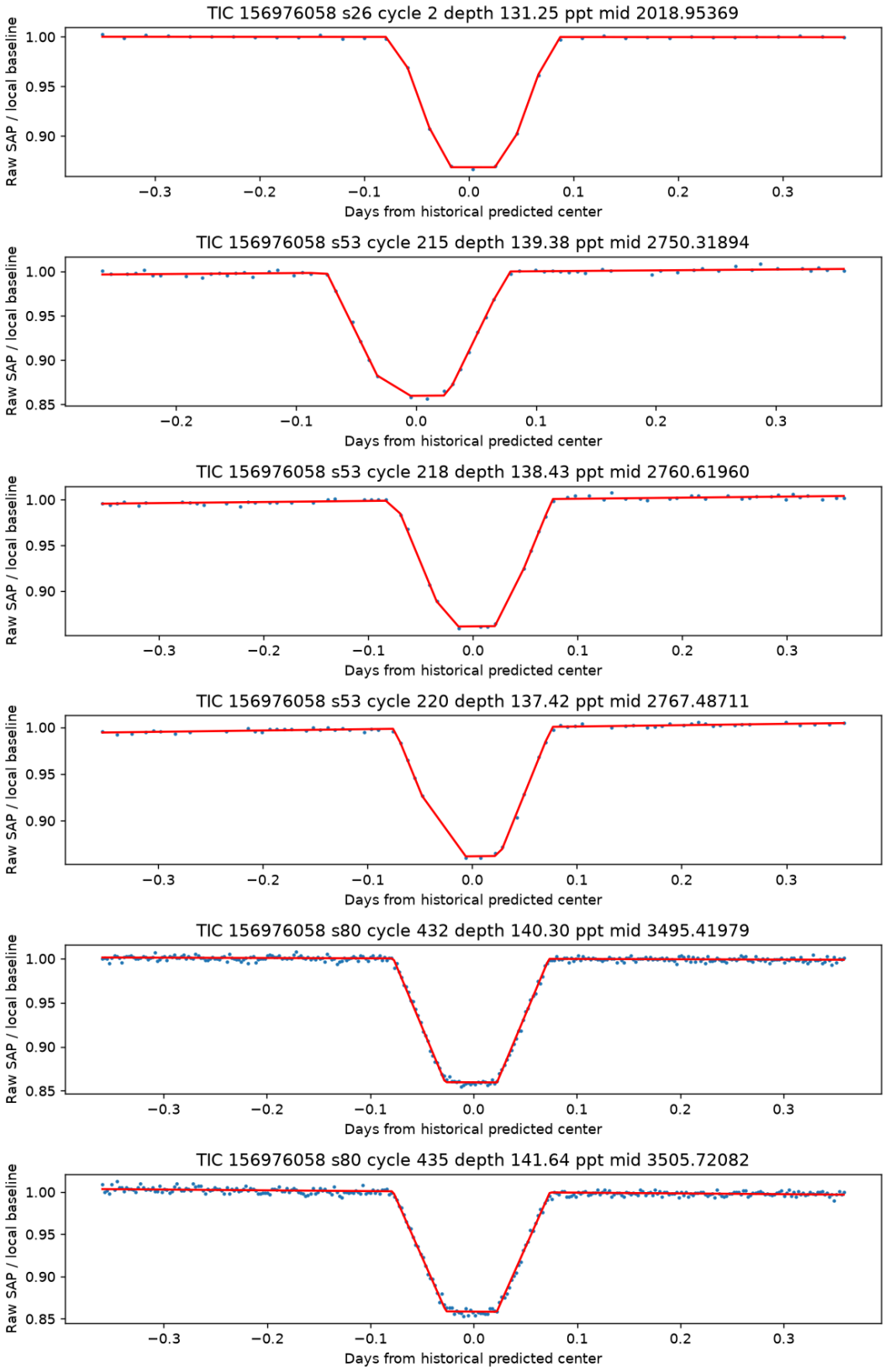
R438. TIC 156976058

physical_verdict: CONFIRMED

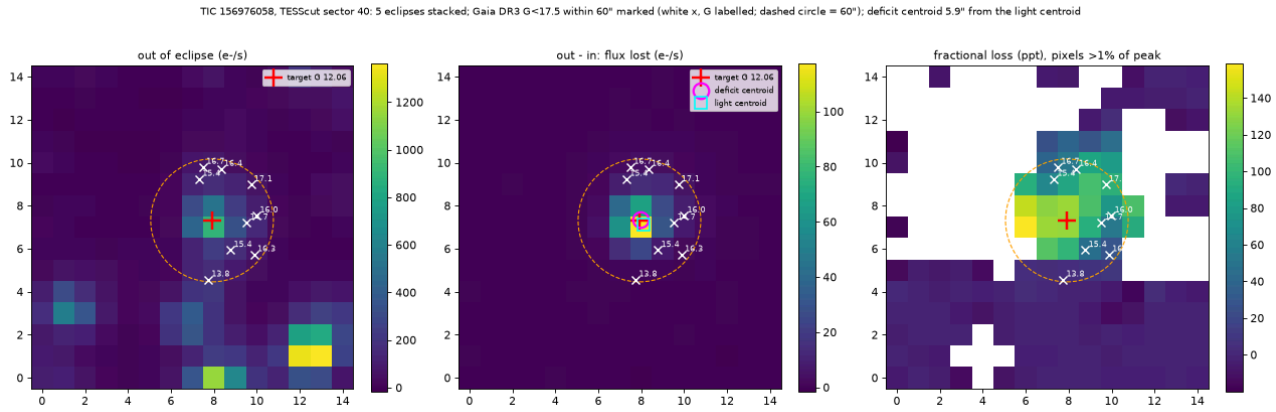
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R438; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R438; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 156976058: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 14 timing-usable measured centers in sectors [26, 53, 80]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 3.4336457438 ± 0.0000062829 d and reference epoch BTJD 2018.9522980 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 10 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 40 TESScut difference image puts the deficit 1.65 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.465 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 4523316211621183872 (55.81 arcsec, G 13.768, intrinsic loss approx. 0.794). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 132.09/118.89/100.11 ppt in 5 stacked historical events; representative QLP raw median fit depth is 138.82 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 4523316314700411136 has SB1 P = 3.4333917898 d and e = 0.03177. Closest edge-on conjunction offset is 0.07574 d (0.02206 phase), with 0.08230 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.49 h versus observed median fit 3.61 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Fourteen deep raw repeats include adjacent cycles in S53 and S80. The only nominally capable neighbor is 55.8 arcsec away, distinct from the target-centered deficit.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
26	2	2018.9536896	0.010417	131.25	used
53	214	2746.8856511	0.003472	136.41	used
53	215	2750.3189356	0.003472	139.38	used
53	216	2753.7522898	0.003472	138.79	used
53	217	2757.1857718	0.003472	137.72	used
53	218	2760.6195976	0.003472	138.43	used
53	219	2764.0532691	0.003472	137.64	used
53	220	2767.4871060	0.003472	137.42	used
80	428	3481.6852727	0.001157	138.85	used
80	431	3491.9865294	0.001157	140.49	used
80	432	3495.4197879	0.001157	140.30	used
80	433	3498.8537895	0.001157	139.67	used
80	434	3502.2872671	0.001157	139.86	used
80	435	3505.7208234	0.001157	141.64	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s40.json, historical/diff_s40.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R438. Public evidence files: diff_s40.json, diff_s40.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

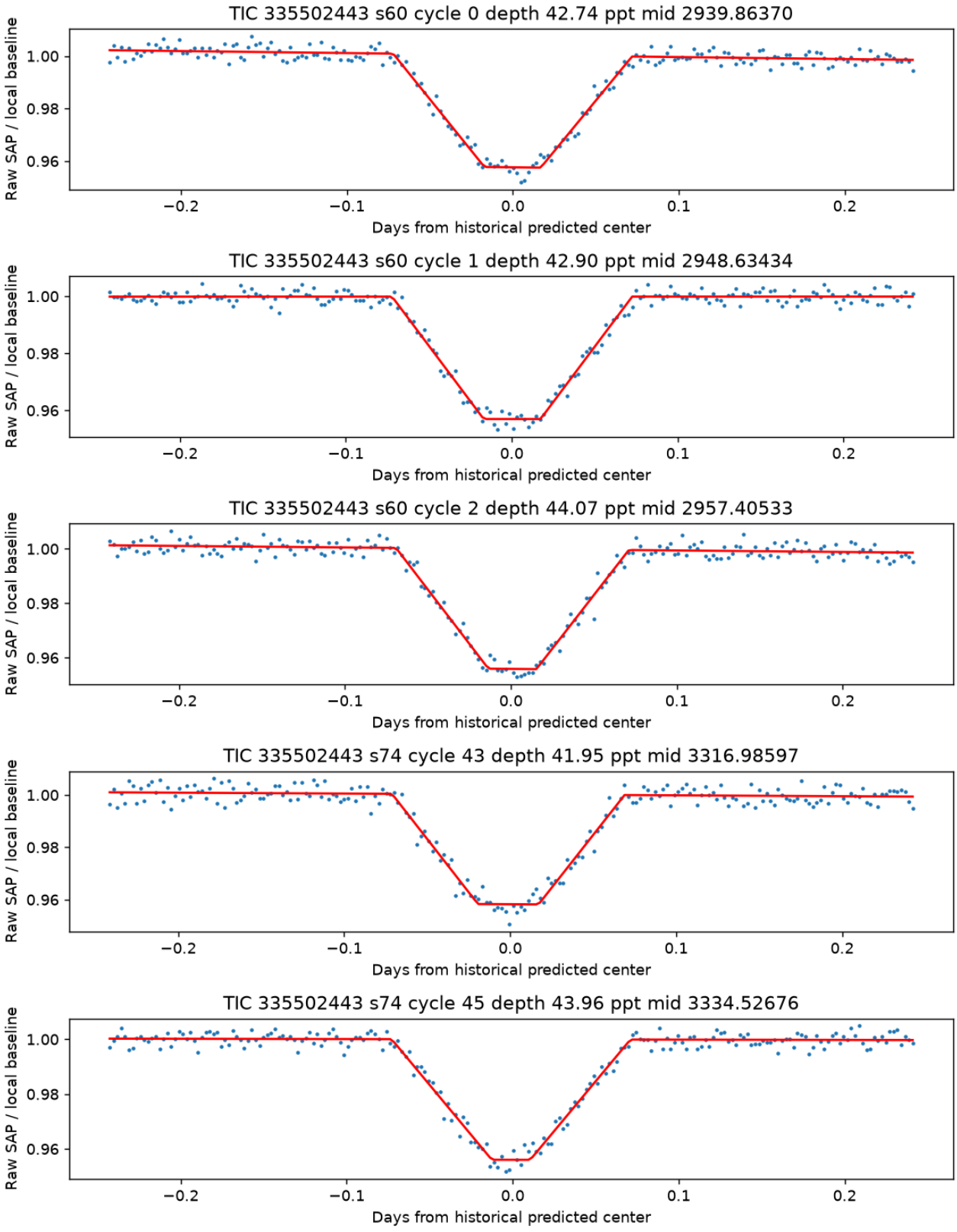
R439. TIC 335502443

physical_verdict: CONFIRMED

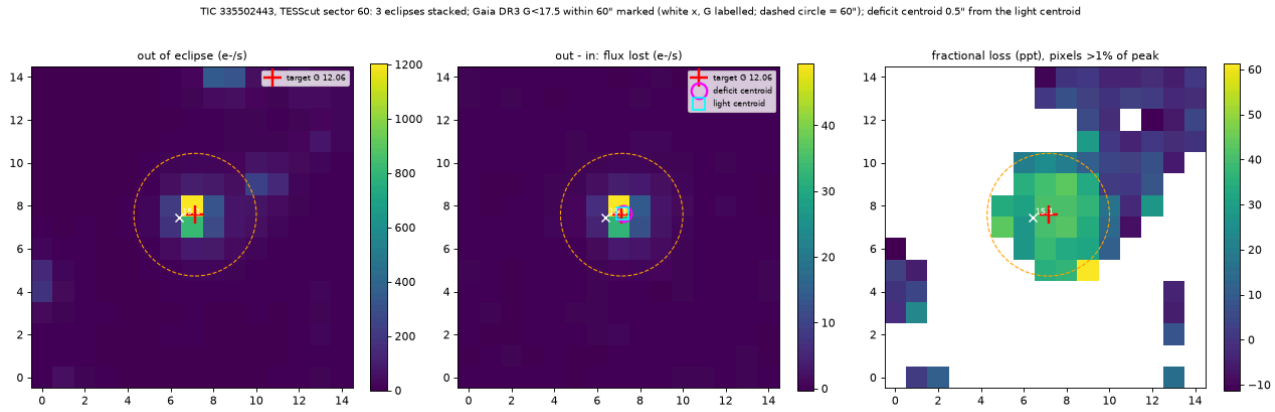
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R439; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R439; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 335502443: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured centers in sectors [60, 74]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative morphology shows repeated eclipses.
2. **Timing and aliases.** Measured centers yield P = 8.7702776381 ± 0.0000245490 d and reference epoch BTJD 2939.8641685 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector adjacent same-kind pairs (../alias-audit.json). Measured centers and same-kind pairs support the bounded period and alias check. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 60 TESScut difference image puts the deficit 1.55 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.193 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 1088529297616817152 (16.22 arcsec, G 15.075, intrinsic loss approx. 0.704). Intrinsic-loss requirements use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1x1/3x3/5x5 target apertures yield 41.06/39.39/38.17 ppt in 3 stacked historical events; representative QLP raw median fit depth is 42.90 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and event choice limit depth comparisons.
5. **Gaia geometry.** Matched DR3 source 1088529293321259648 has SB1 P = 8.7704426450 d and e = 0.03074. Closest edge-on conjunction offset is -0.01668 d (-0.00190 phase), with 0.24762 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.49 h versus observed median fit 3.45 h assumes TIC radius and a minimum-mass main-sequence companion, not a full binary model. Agreement is not source proof.

Interpretation and limitation: Five raw repeats include adjacent S60 cycles. A capable neighbor 16.2 arcsec away is distinct from the target-centered deficit core at TESS resolution.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
60	0	2939.8637048	0.001157	42.74	used
60	1	2948.6343381	0.001157	42.90	used
60	2	2957.4053258	0.001157	44.07	used
74	43	3316.9859743	0.001157	41.95	used
74	45	3334.5267647	0.001157	43.96	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s60.json, historical/diff_s60.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R439. Public evidence files: diff_s60.json, diff_s60.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

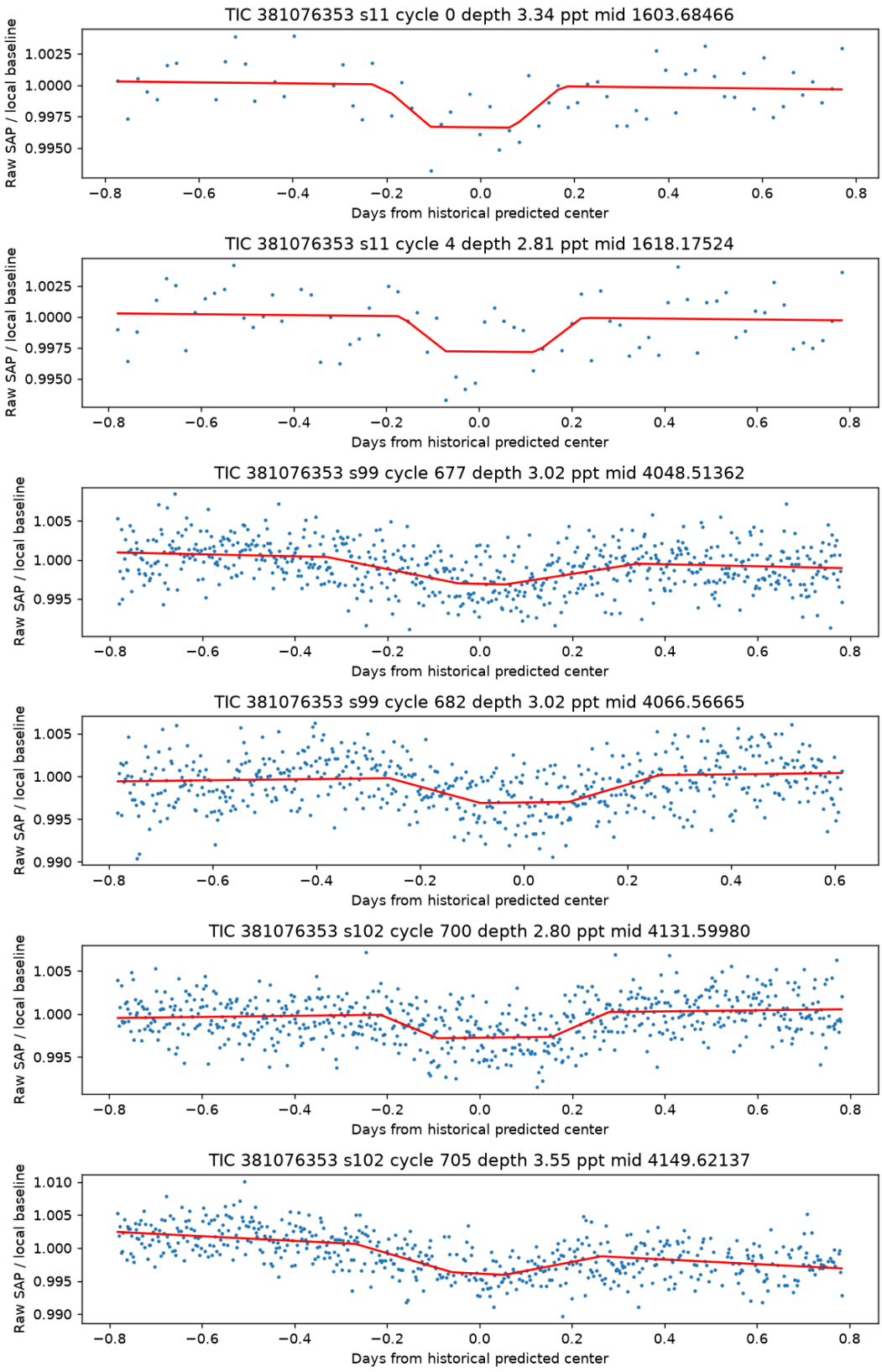
R440. TIC 381076353

physical_verdict: PLAUSIBLE

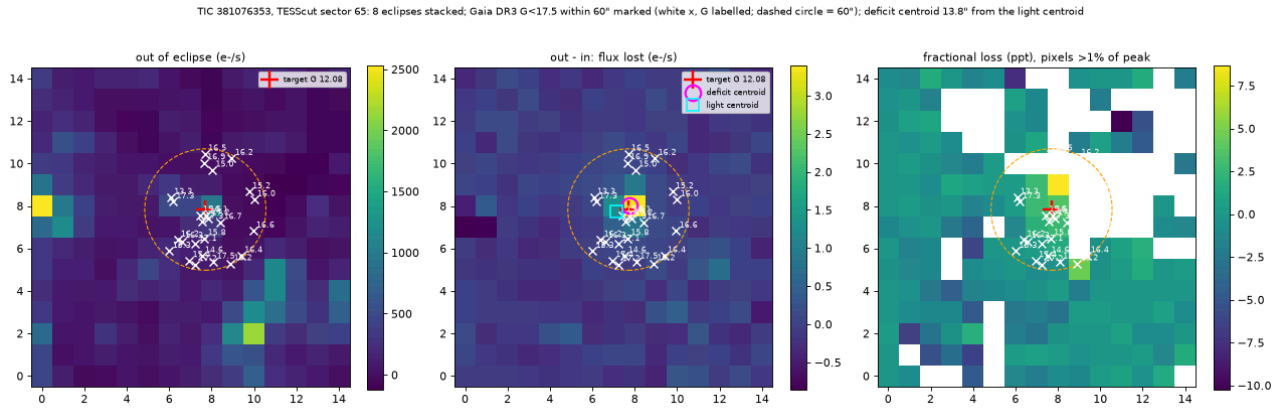
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

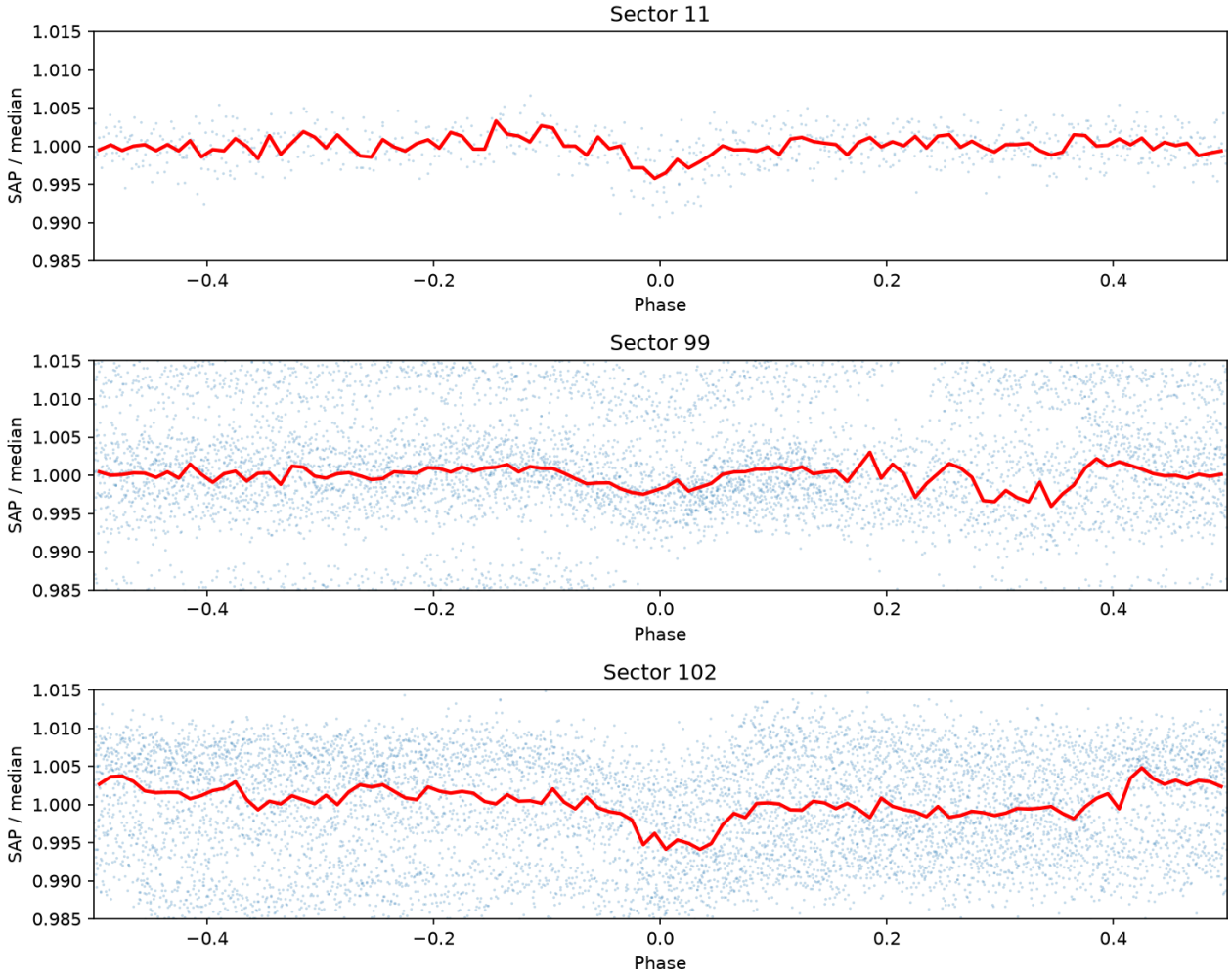
next_check: Test morphology and resolve source attribution



R440; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R440; Representative eclipse-deficit image; source attribution retains the report limits.



R440; Independent sector folds expose the variable eclipse morphology; this signal remains plausible and excluded.

Archived source: sanitized report text.

TIC 381076353: PLAUSIBLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 12 timing-selected candidate depressions in sectors [11, 99, 102]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured repeating depression centers give P=3.6112653345 d, fit error=0.0000376302 d, epoch BTJD 1603.6849541 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 consecutive within-sector same-kind cycle pairs. The inventory period was 3.61123 d and the historical SAP search branch was 3.6112336247 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: unresolved at TESS resolution.** Historical S65 TESScut deficit core is 3.71 arcsec from the target coordinate; peak pixel [8, 8] and target pixel [8, 8]. Fractional PSF model residual is 0.909, not a localization uncertainty. Capable Gaia neighbors: 1.9 arcsec, G=17.55, intrinsic-loss requirement ~0.48; 5.8 arcsec, G=16.08, intrinsic-loss requirement ~0.13; 6.8 arcsec, G=18.17, intrinsic-loss requirement ~0.85; 7.8 arcsec, G=15.47, intrinsic-loss requirement ~0.07; 9.2 arcsec, G=17.17, intrinsic-loss requirement ~0.34; 45 further capable sources in the 60 arcsec list. The full list and flux-budget calculation are in historical/diff_s65.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 3.60/3.63/2.38 ppt from 8 selected windows. Median selected QLP raw fit depth is 3.05 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 5864386386590001024 has P=3.6112336247 d and e=0.38173. Closest edge-on conjunction differs from the measured epoch by 0.00267 in phase (0.00964 d); linear propagation of the source-specific full SB1 covariance gives 0.07820 d as a descriptive timing scale. The rough central-duration estimate is 3.02 h versus trapezoid median 12.16 h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Recurrent ~3 ppt depressions are comparable to local scatter. Folded S11 and S102 show localized troughs and recovery, but S99 is rounded and broad variability is not decisively excluded; eclipse morphology remains uncertain. Fifty capable Gaia sources within 60 arcsec include several within 8 arcsec; the target-centered core has 0.909 image-model fractional residual, leaving attribution unresolved. Trapezoid durations and timing are shape-sensitive (up to 113 min timing residual).

Next check. Higher-resolution time-series photometry to test eclipse morphology and source attribution.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
11	0	1603.6846601	0.023909	3.34	used
11	1	1607.2888763	0.013657	3.93	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
11	4	1618.1752398	0.033312	2.81	used
11	5	1621.7413685	0.022295	2.20	used
99	677	4048.5136194	0.016278	3.02	used
99	678	4052.0649452	0.015813	3.08	used
99	682	4066.5666518	0.013352	3.02	used
99	683	4070.2579081	0.018299	5.10	used
102	700	4131.5997979	0.009527	2.80	used
102	701	4135.1379910	0.012265	4.25	used
102	703	4142.4034336	0.010024	2.88	used
102	705	4149.6213680	0.015593	3.55	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s65.json, historical/diff_s65.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R440. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

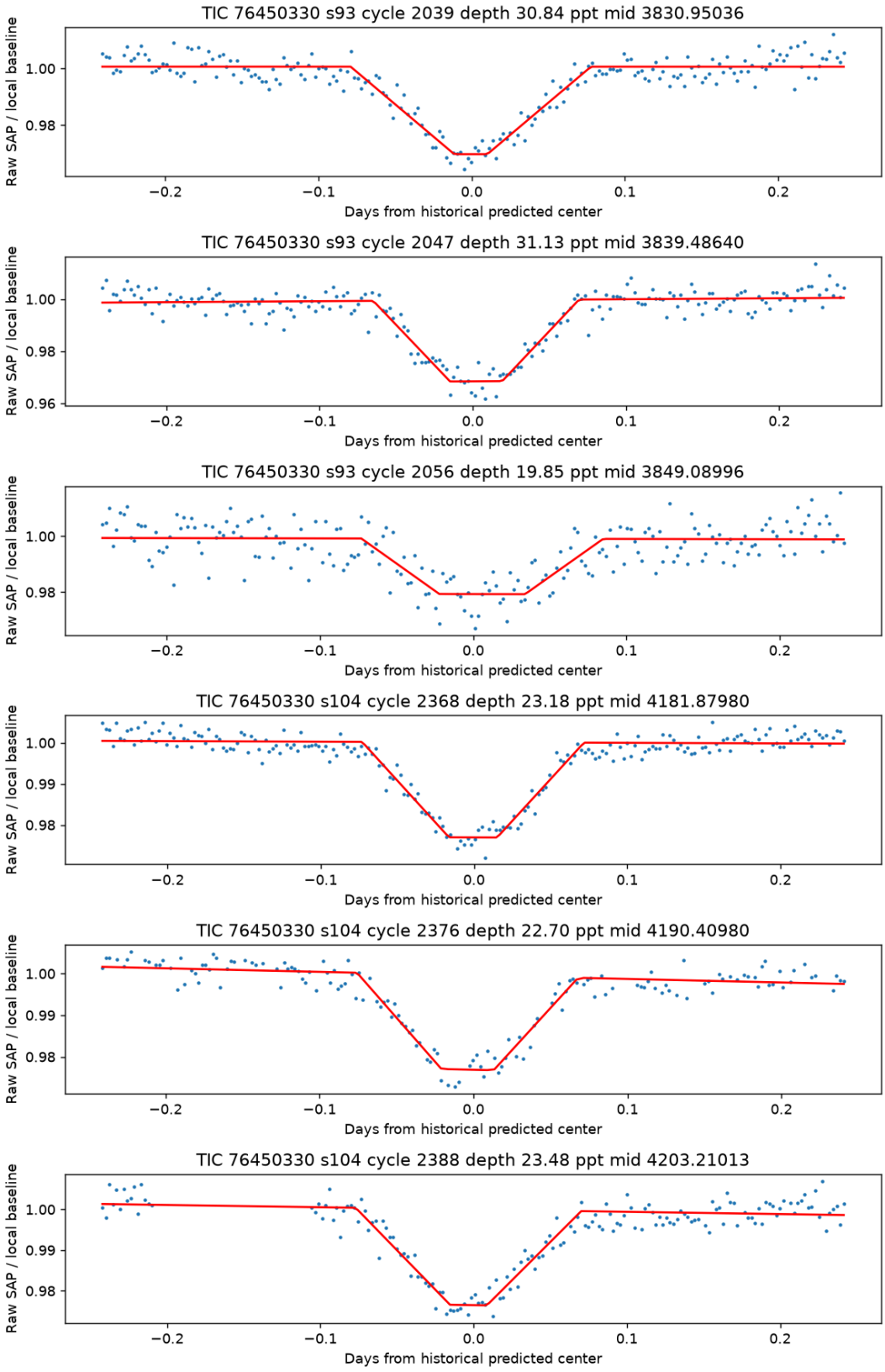
R441. TIC 76450330

physical_verdict: CONFIRMED

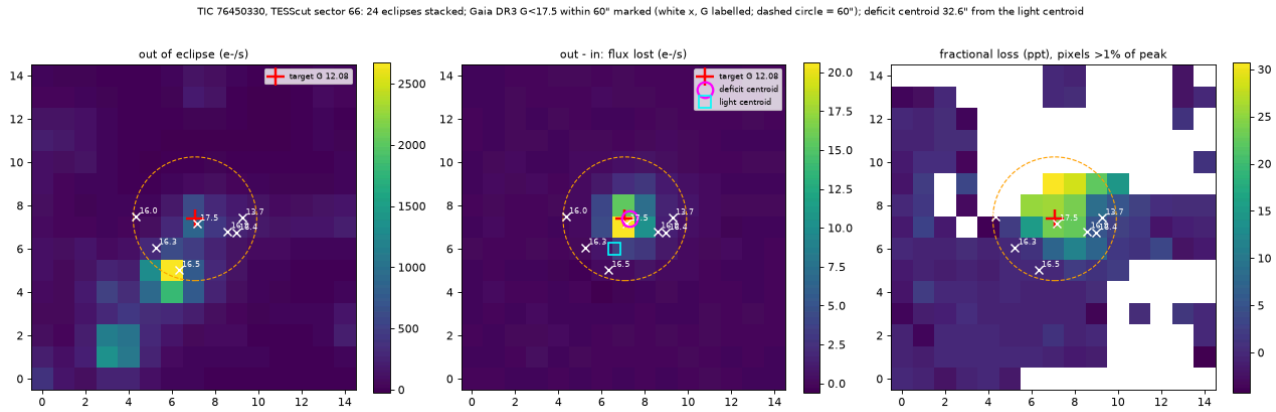
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R441; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R441; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 76450330: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 17 timing-selected measured eclipses in sectors [93, 104]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=2.1332813704 d, fit error=0.0000038241 d, epoch BTJD 3830.9530096 (Deeper repeating eclipse; historical half-period parity 1). There are 10 consecutive within-sector same-kind cycle pairs. The inventory period was 2.13331 d and the historical SAP search branch was 1.0666548545 d. Deeper parity 1 of historical half-period cycles; consecutive same-kind events have orbital-cycle count one. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S66 TESScut deficit core is 4.40 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Fractional PSF model residual is 0.952, not a localization uncertainty. Capable Gaia neighbors: 45.6 arcsec, G=13.73, intrinsic-loss requirement ~0.13; 56.6 arcsec, G=15.96, intrinsic-loss requirement ~0.83. The full list and flux-budget calculation are in historical/diff_s66.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 23.47/17.15/7.81 ppt from 24 selected windows. Median selected QLP raw fit depth is 31.53 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 5918304478211801728 has P=2.1333285880 d and e=0.04973. Closest edge-on conjunction differs from the measured epoch by -0.03250 in phase (-0.06933 d); linear propagation of the source-specific full SB1 covariance gives 0.07043 d as a descriptive timing scale. The rough central-duration estimate is 6.24 h versus trapezoid median 3.47

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Strong alternating depths establish the 2.1333 d orbital cycle; the historical SAP 1.06665 d branch is the half-period. The deeper parity is fitted for the reported epoch.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
93	2039	3830.9503554	0.001244	30.84	used
93	2040	3832.0188593	0.001403	23.02	other eclipse kind
93	2041	3833.0874435	0.001157	38.33	used
93	2042	3834.1488187	0.001425	25.77	other eclipse kind
93	2043	3835.2217774	0.001351	32.71	used
93	2045	3837.3508688	0.001157	34.70	used
93	2046	3838.4181594	0.001330	25.12	other eclipse kind
93	2047	3839.4864046	0.001157	31.13	used
93	2048	3840.5553168	0.001157	22.95	other eclipse kind
93	2051	3843.7553337	0.001602	29.37	used
93	2052	3844.8200618	0.001520	24.12	other eclipse kind
93	2053	3845.8862230	0.001157	33.14	used
93	2054	3846.9531815	0.001265	23.20	other eclipse kind
93	2055	3848.0204345	0.001252	30.82	used
93	2056	3849.0899565	0.002583	19.85	other eclipse kind
93	2058	3851.2177570	0.001737	23.20	other eclipse kind
93	2059	3852.2846859	0.001157	31.88	used
93	2060	3853.3545826	0.001509	23.26	other eclipse kind
93	2061	3854.4187693	0.001342	32.23	used
104	2366	4179.7433388	0.002615	24.03	other eclipse kind
104	2367	4180.8106391	0.001157	31.56	used
104	2368	4181.8797986	0.001157	23.18	other eclipse kind
104	2369	4182.9451359	0.001157	31.53	used
104	2370	4184.0114345	0.001157	22.49	other eclipse kind
104	2372	4186.1448417	0.001169	22.25	other eclipse kind
104	2373	4187.2111476	0.001157	31.41	used
104	2374	4188.2772217	0.001360	23.53	other eclipse kind
104	2375	4189.3448112	0.001157	30.98	used

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
104	2376	4190.4097993	0.001374	22.70	other eclipse kind
104	2380	4194.6770309	0.001425	24.09	other eclipse kind
104	2381	4195.7440795	0.001157	32.82	used
104	2382	4196.8101891	0.001319	25.07	other eclipse kind
104	2384	4198.9436754	0.001157	23.39	other eclipse kind
104	2385	4200.0099345	0.001157	31.36	used
104	2386	4201.0779941	0.001157	22.29	other eclipse kind
104	2387	4202.1439058	0.001157	30.93	used
104	2388	4203.2101329	0.001157	23.48	other eclipse kind

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s66.json, historical/diff_s66.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R441. Public evidence files: diff_s66.json, diff_s66.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

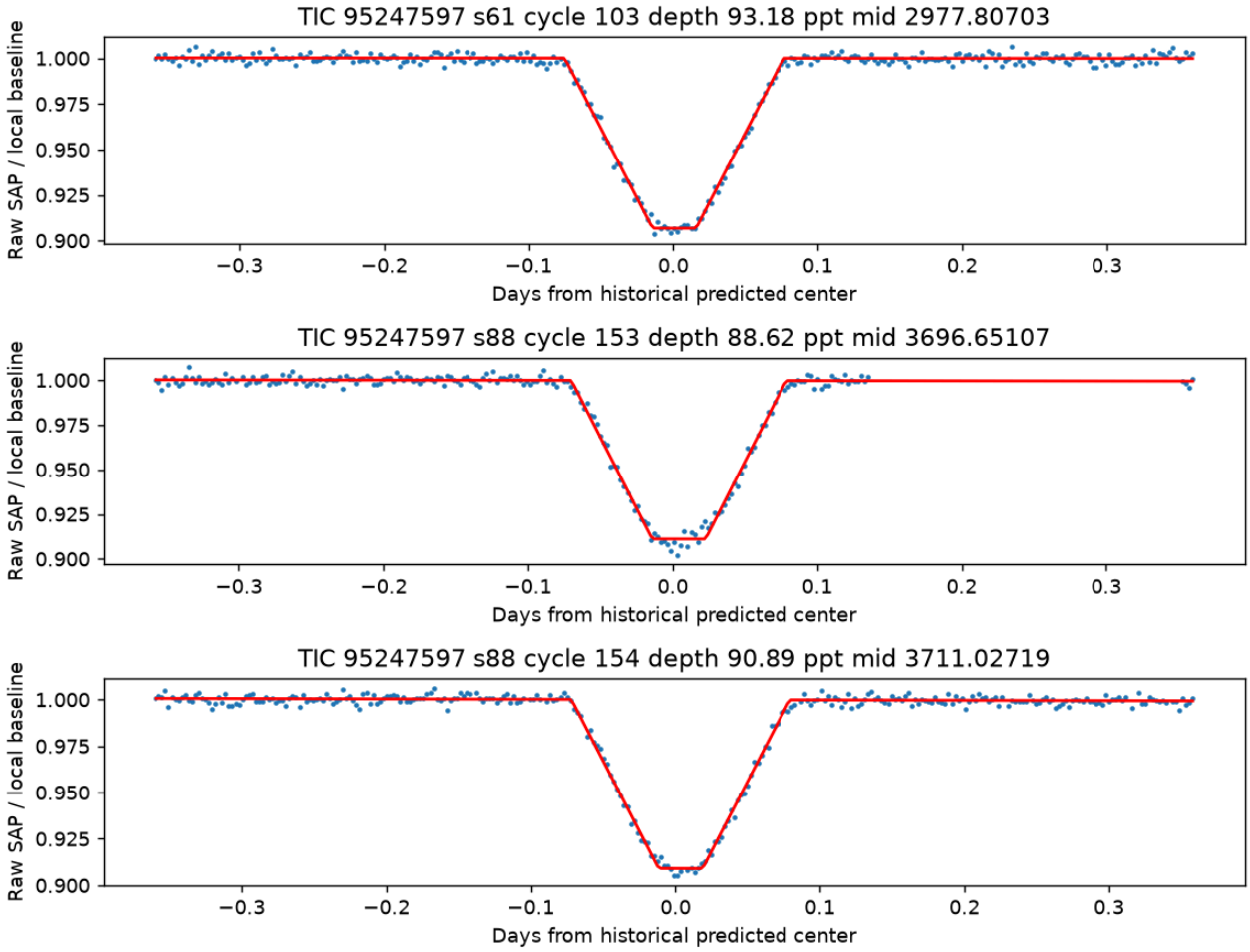
R442. TIC 95247597

physical_verdict: CONFIRMED

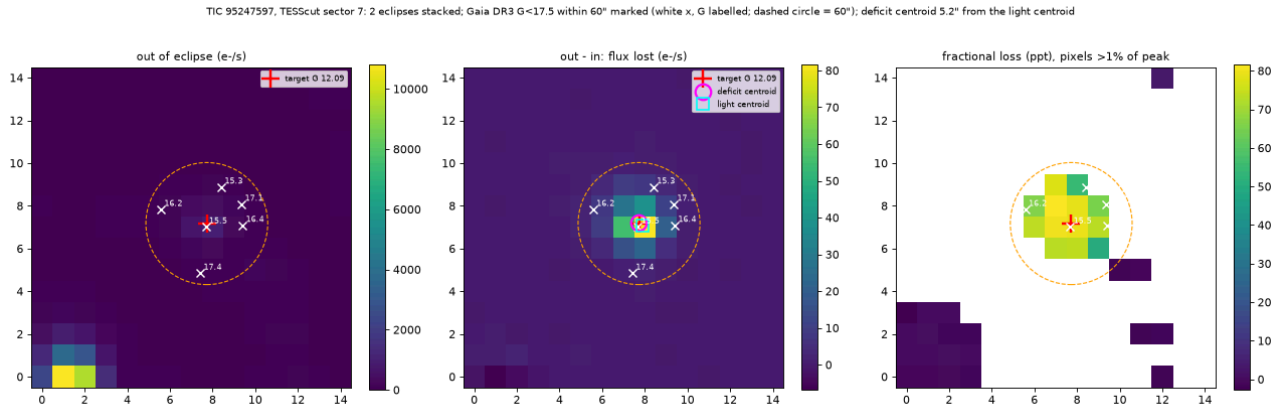
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R442; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R442; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 95247597: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 3 timing-selected measured eclipses in sectors [61, 88]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=14.3768729995 d, fit error=0.0000280654 d, epoch BTJD 2977.8070390 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 consecutive within-sector same-kind cycle pairs. The inventory period was 14.37681 d and the historical SAP search branch was 14.3768112970 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S7 TESScut deficit core is 1.48 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Fractional PSF model residual is 0.250, not a localization uncertainty. Capable Gaia neighbors: none under the approximate G-band flux budget. The full list and flux-budget calculation are in historical/diff_s07.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 79.56/76.84/68.07 ppt from 2 selected windows. Median selected QLP raw fit depth is 90.89 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 3042854243248092928 has P=14.3765198811 d and e=0.11457. Closest edge-on conjunction differs from the measured epoch by -0.02690 in phase (-0.38667 d); linear propagation of the source-specific full SB1 covariance gives 0.41907 d as a descriptive timing scale. The rough central-duration estimate is 6.15 h versus trapezoid median 3.63 h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. S88 has a consecutive 14.376 d same-kind recurrence and S61 adds a long baseline. Earlier S7 windows have only five quality-zero points each, so they are not clean nulls. Current MAST QLP availability includes S7/34/61/88; availability alone is not an eclipse detection.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
61	103	2977.8070316	0.001157	93.18	used
88	153	3696.6510671	0.001157	88.62	used
88	154	3711.0271913	0.001157	90.89	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s07.json, historical/diff_s07.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R442. Public evidence files: diff_s07.json, diff_s07.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

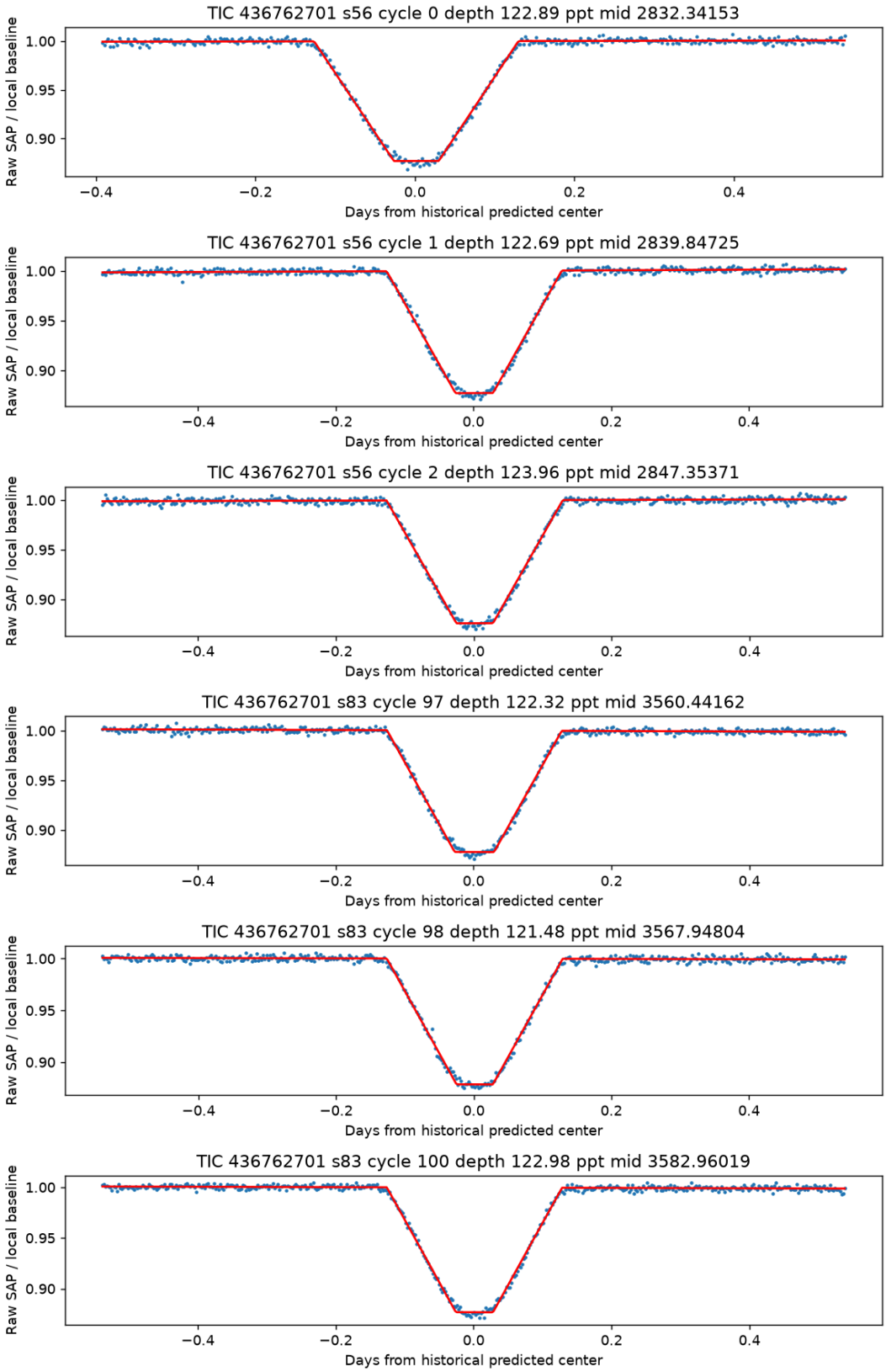
R443. TIC 436762701

physical_verdict: CONFIRMED

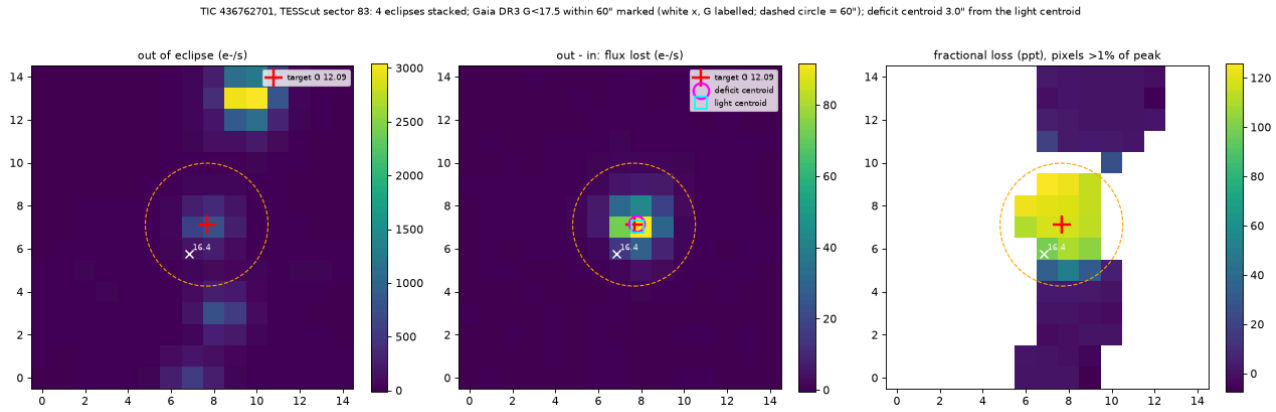
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R443; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R443; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 436762701: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 7 timing-selected measured eclipses in sectors [56, 83]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=7.5061902904 d, fit error=0.0000090645 d, epoch BTJD 2832.3413064 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 consecutive within-sector same-kind cycle pairs. The inventory period was 7.50619 d and the historical SAP search branch was 7.5061923952 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S83 TESScut deficit core is 3.16 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Fractional PSF model residual is 0.667, not a localization uncertainty. Capable Gaia neighbors: none under the approximate G-band flux budget. The full list and flux-budget calculation are in historical/diff_s83.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 117.56/115.42/108.68 ppt from 4 selected windows. Median selected QLP raw fit depth is 122.69 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 1885079155014227584 has P=7.5069767062 d and e=0.18531. Closest edge-on conjunction differs from the measured epoch by -0.02604 in phase (-0.19548 d); linear propagation of the source-specific full SB1 covariance gives 0.13174 d as a descriptive timing scale. The rough central-duration estimate is 9.36 h versus trapezoid median 6.11

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Seven deep raw events across S56 and S83 include consecutive cycles, with consistent depths and a target-centered deficit.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
56	0	2832.3415322	0.001157	122.89	used
56	1	2839.8472492	0.001157	122.69	used
56	2	2847.3537074	0.001157	123.96	used
83	97	3560.4416157	0.001157	122.32	used
83	98	3567.9480430	0.001157	121.48	used
83	99	3575.4543530	0.001157	121.25	used
83	100	3582.9601899	0.001157	122.98	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s83.json, historical/diff_s83.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R443. Public evidence files: diff_s83.json, diff_s83.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

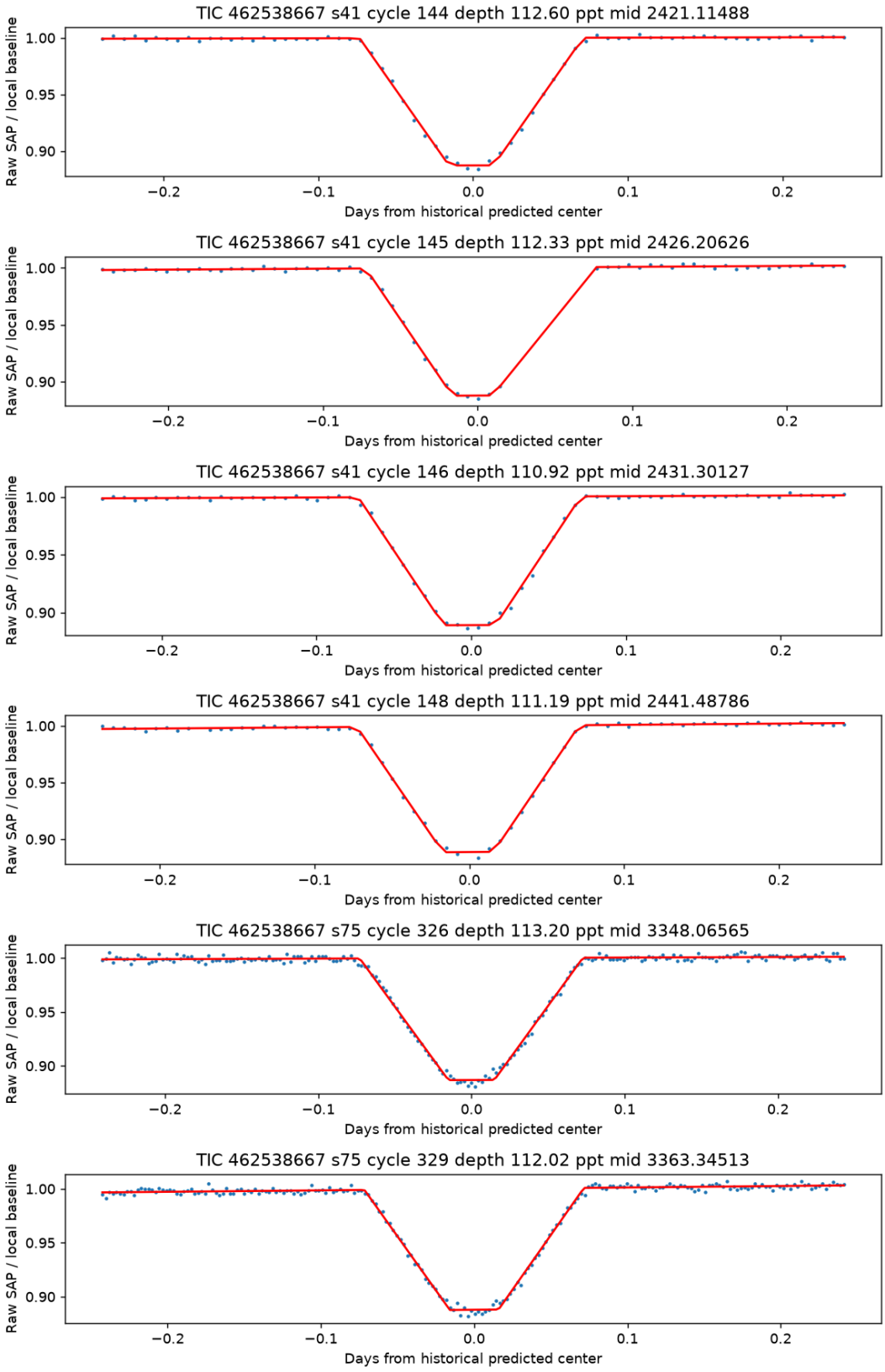
R444. TIC 462538667

physical_verdict: CONFIRMED

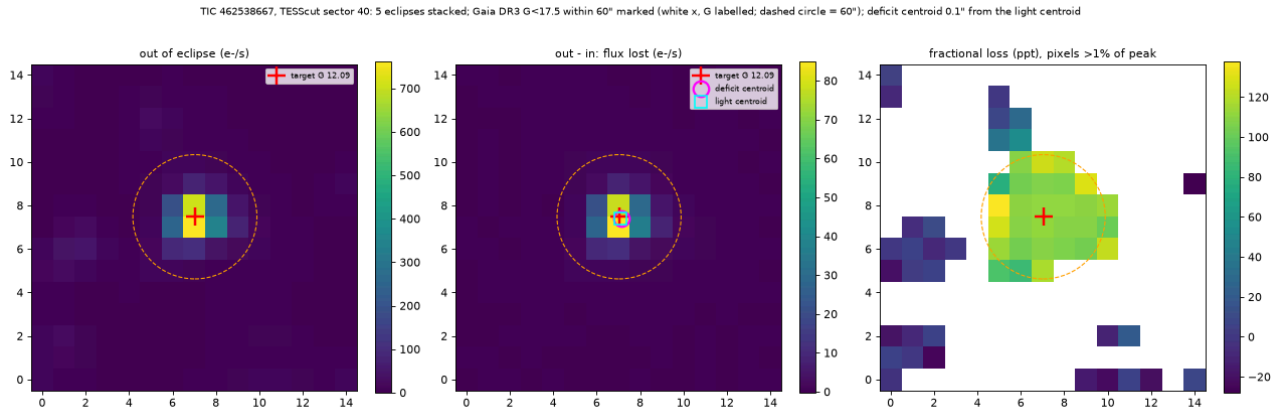
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R444; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R444; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 462538667: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 8 timing-selected measured eclipses in sectors [41, 75]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=5.0931372216 d, fit error=0.0000093209 d, epoch BTJD 2421.1146206 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 consecutive within-sector same-kind cycle pairs. The inventory period was 5.09313 d and the historical SAP search branch was 5.0931300597 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S40 TESScut deficit core is 2.62 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Fractional PSF model residual is 0.021, not a localization uncertainty. Capable Gaia neighbors: none under the approximate G-band flux budget. The full list and flux-budget calculation are in historical/diff_s40.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 111.52/110.09/109.81 ppt from 5 selected windows. Median selected QLP raw fit depth is 112.46 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 1688501177639000320 has P=5.0931642420 d and e=0.03904. Closest edge-on conjunction differs from the measured epoch by 0.02129 in phase (0.10842 d); linear propagation of the source-specific full SB1 covariance gives 0.13996 d as a descriptive timing scale. The rough central-duration estimate is 4.85 h versus trapezoid median 3.46

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Eight deep raw events across S41 and S75 include consecutive cycles, with a well-centered pixel deficit and stable depths.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
41	144	2421.1148783	0.003472	112.60	used
41	145	2426.2062636	0.003472	112.33	used
41	146	2431.3012696	0.003472	110.92	used
41	147	2436.3942317	0.003472	113.62	used
41	148	2441.4878561	0.003472	111.19	used
75	326	3348.0656501	0.001157	113.20	used
75	327	3353.1585480	0.001157	113.04	used
75	329	3363.3451330	0.001157	112.02	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s40.json, historical/diff_s40.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R444. Public evidence files: diff_s40.json, diff_s40.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

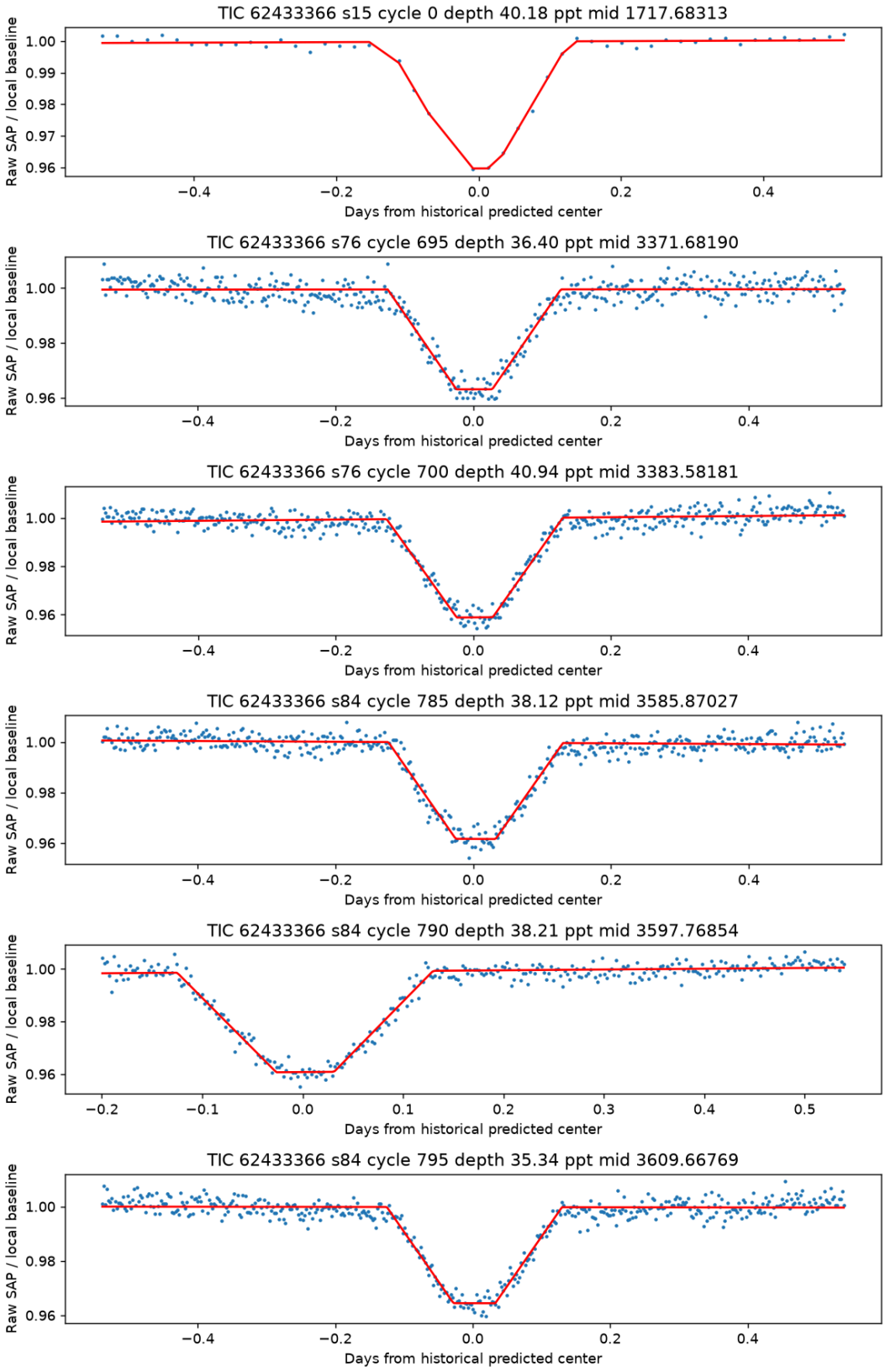
R445. TIC 62433366

physical_verdict: CONFIRMED

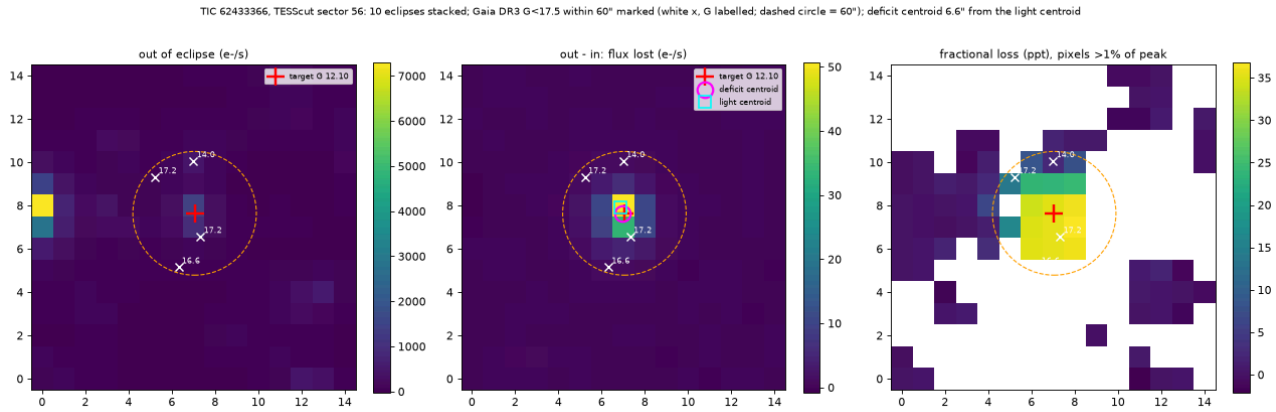
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R445; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R445; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 62433366: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 12 timing-selected measured eclipses in sectors [15, 76, 84]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=4.7597184350 d, fit error=0.0000117166 d, epoch BTJD 1717.6797337 (Deeper repeating eclipse; historical half-period parity 0). There are 6 consecutive within-sector same-kind cycle pairs. The inventory period was 4.75970 d and the historical SAP search branch was 2.3798505904 d. Deeper parity 0 of historical half-period cycles; consecutive same-kind events have orbital-cycle count one. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S56 TESScut deficit core is 1.91 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Fractional PSF model residual is 0.405, not a localization uncertainty. Capable Gaia neighbors: 51.0 arcsec, G=14.00, intrinsic-loss requirement ~0.23. The full list and flux-budget calculation are in historical/diff_s56.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 35.29/34.01/29.65 ppt from 10 selected windows. Median selected QLP raw fit depth is 40.07 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 2175713785143071360 has P=4.7596067442 d and e=0.01266. Closest edge-on conjunction differs from the measured epoch by 0.01077 in phase (0.05124 d); linear propagation of the source-specific full SB1 covariance gives 0.05299 d as a descriptive timing scale. The rough central-duration estimate is 13.51 h versus trapezoid median 6.17

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Even historical half-period cycles are modestly but persistently deeper than odd cycles in S76 and S84, supporting the 4.7597 d orbital period. The duration is only a rough trapezoid shape measure.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
15	0	1717.6831262	0.010417	40.18	used
15	4	1727.1993403	0.010417	38.05	used
15	5	1729.5828742	0.010417	35.04	other eclipse kind
15	6	1731.9623506	0.010417	38.58	used
76	694	3369.3007811	0.001157	39.46	used
76	695	3371.6818953	0.001157	36.40	other eclipse kind
76	696	3374.0525045	69105.867428	39.51	excluded: unidentifiable center covariance
76	697	3376.4414970	0.001157	36.42	other eclipse kind
76	698	3378.8210860	0.001157	41.23	used
76	699	3381.2015616	0.001183	36.31	other eclipse kind
76	700	3383.5818120	0.001157	40.94	used
76	701	3385.9613345	0.001157	37.25	other eclipse kind
76	702	3388.3405430	74615.274606	41.44	excluded: unidentifiable center covariance
76	703	3390.7214454	0.001157	37.46	other eclipse kind
76	704	3393.1009750	0.001157	41.70	used
84	785	3585.8702706	0.001157	38.12	other eclipse kind
84	786	3588.2496231	0.001157	40.09	used
84	787	3590.6295307	0.001157	36.54	other eclipse kind
84	788	3593.0082983	0.001157	40.05	used
84	789	3595.3870047	0.001157	35.52	other eclipse kind
84	790	3597.7685395	0.001157	38.21	used
84	791	3600.1486071	0.001157	35.99	other eclipse kind
84	792	3602.5287240	0.001157	39.26	used
84	793	3604.9075849	0.001157	37.18	other eclipse kind
84	794	3607.2879584	0.001157	41.04	used
84	795	3609.6676856	0.001157	35.34	other eclipse kind

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R445. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

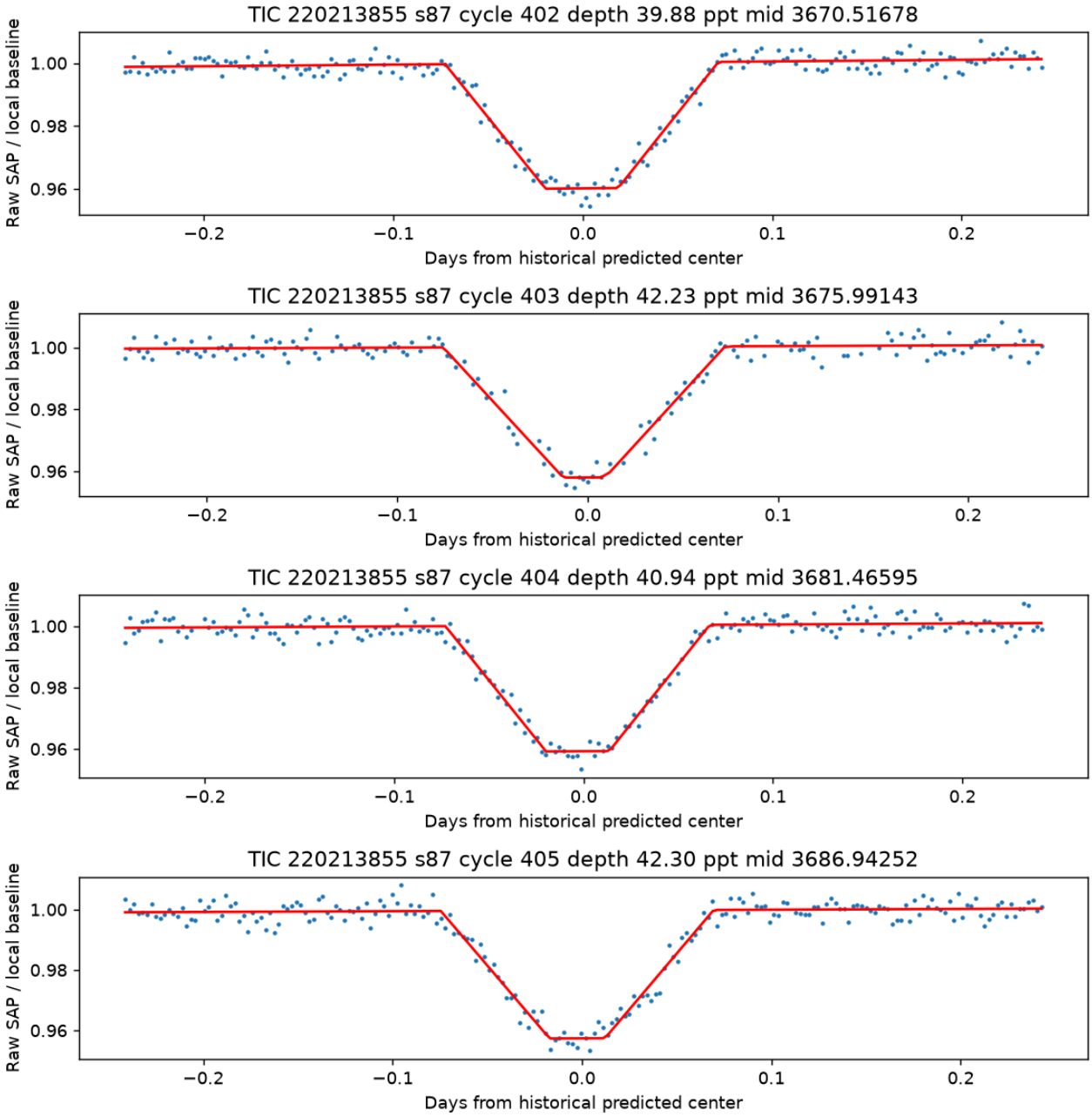
R446. TIC 220213855

physical_verdict: CONFIRMED

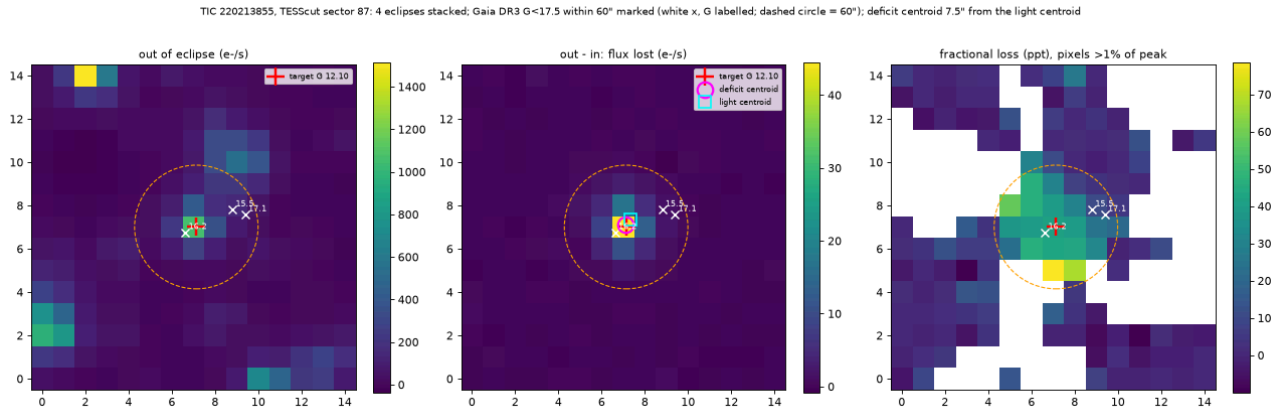
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R446; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R446; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 220213855: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 4 timing-selected measured eclipses in sectors [87]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=5.4751727504 d, fit error=0.0005176080 d, epoch BTJD 3670.5164102 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 3 consecutive within-sector same-kind cycle pairs. The inventory period was 5.47591 d and the historical SAP search branch was 5.4759094123 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S87 TESScut deficit core is 2.58 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Fractional PSF model residual is 0.608, not a localization uncertainty. Capable Gaia neighbors: 38.8 arcsec, G=15.53, intrinsic-loss requirement ~0.99. The full list and flux-budget calculation are in historical/diff_s87.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 41.78/40.53/35.53 ppt from 4 selected windows. Median selected QLP raw fit depth is 41.58 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 3326997734678765696 has P=5.4759465374 d and e=0.03177. Closest edge-on conjunction differs from the measured epoch by 0.01870 in phase (0.10240 d); linear propagation of the source-specific full SB1 covariance gives 0.34526 d as a descriptive timing scale. The rough central-duration estimate is 5.81 h versus trapezoid median 3.46

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Four consecutive raw eclipses in S87 establish recurrence across 16.43 d. S6 is available but its candidate windows have at most four quality-zero cadences inside half the search duration; no clean null is claimed. The fitted period error is limited by one sector and is not total ephemeris uncertainty.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
87	402	3670.5167841	0.001157	39.88	used
87	403	3675.9914259	0.001157	42.23	used
87	404	3681.4659477	0.001157	40.94	used
87	405	3686.9425194	0.001157	42.30	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s87.json, historical/diff_s87.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R446. Public evidence files: diff_s87.json, diff_s87.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

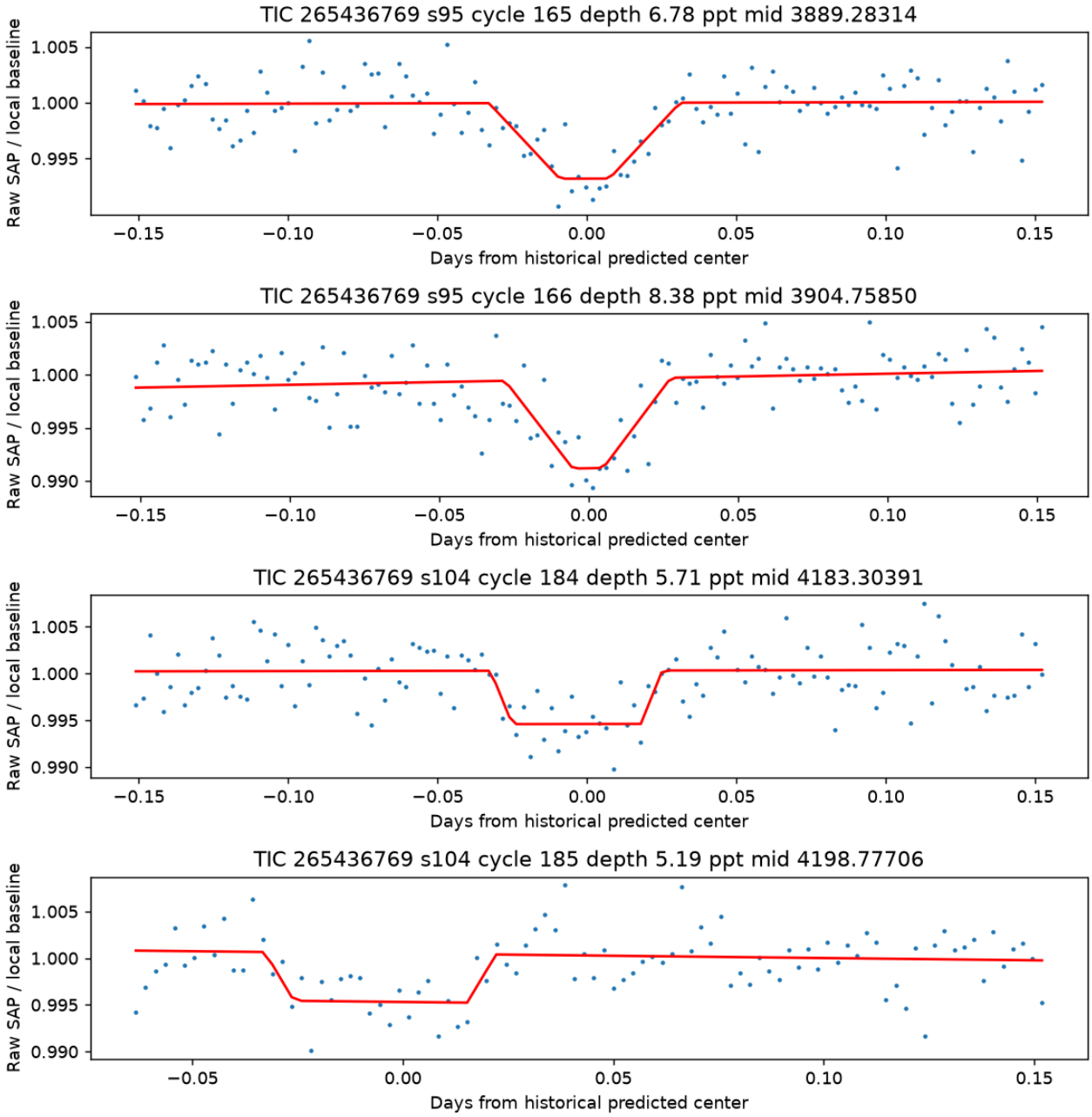
R447. TIC 265436769

physical_verdict: PROBABLE

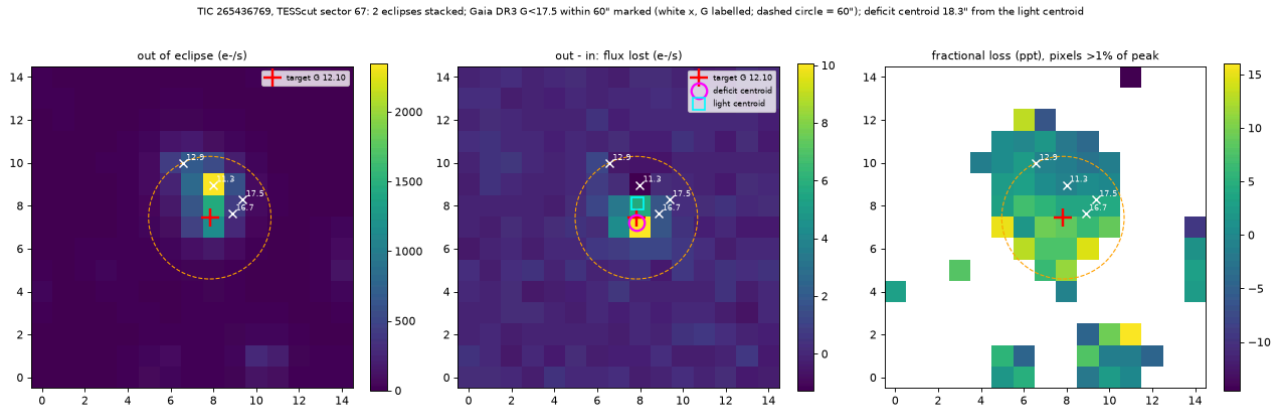
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved source localization



R447; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R447; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 265436769: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 4 timing-selected measured eclipses in sectors [95, 104]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=15.4747037950 d, fit error=0.0000777982 d, epoch BTJD 3889.2835333 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 consecutive within-sector same-kind cycle pairs. The inventory period was 15.47493 d and the historical SAP search branch was 15.4749291199 d. All selected historical photometric cycles used; no systematic alternating depth pattern requiring a double-period fit in representative data. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: unresolved at TESS resolution.** Historical S67 TESScut deficit core is 4.64 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Fractional PSF model residual is 0.870, not a localization uncertainty. Capable Gaia neighbors: 22.6 arcsec, G=16.71, intrinsic-loss requirement ~0.42; 32.1 arcsec, G=11.32, intrinsic-loss requirement ~0.01; 36.8 arcsec, G=17.48, intrinsic-loss requirement ~0.84; 59.2 arcsec, G=12.94, intrinsic-loss requirement ~0.02. The full list and flux-budget calculation are in historical/diff_s67.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 8.63/6.38/3.66 ppt from 2 selected windows. Median selected QLP raw fit depth is 6.24 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 6403358774444776960 has P=15.4729899462 d and e=0.21828. Closest edge-on conjunction differs from the measured epoch by 0.02964 in phase

(0.45867 d); linear propagation of the source-specific full SB1 covariance gives 0.48556 d as a descriptive timing scale. The rough central-duration estimate is 4.96 h versus trapezoid median 1.36 h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Four shallow raw events in S95 and S104 include same-sector consecutive cycles. The S67 difference core favors the target, yet a capable source 22.6 arcsec away and a high image-model residual (0.87) leave attribution unresolved. Early S1 windows are sparse or flagged, not clean nulls.

Next check. Resolved eclipse photometry or multi-source PRF localization for source attribution.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
95	165	3889.2831391	0.001679	6.78	used
95	166	3904.7584970	0.001422	8.38	used
104	184	4183.3039144	0.001691	5.71	used
104	185	4198.7770617	0.001266	5.19	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s67.json, historical/diff_s67.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R447. Public evidence files: diff_s67.json, diff_s67.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

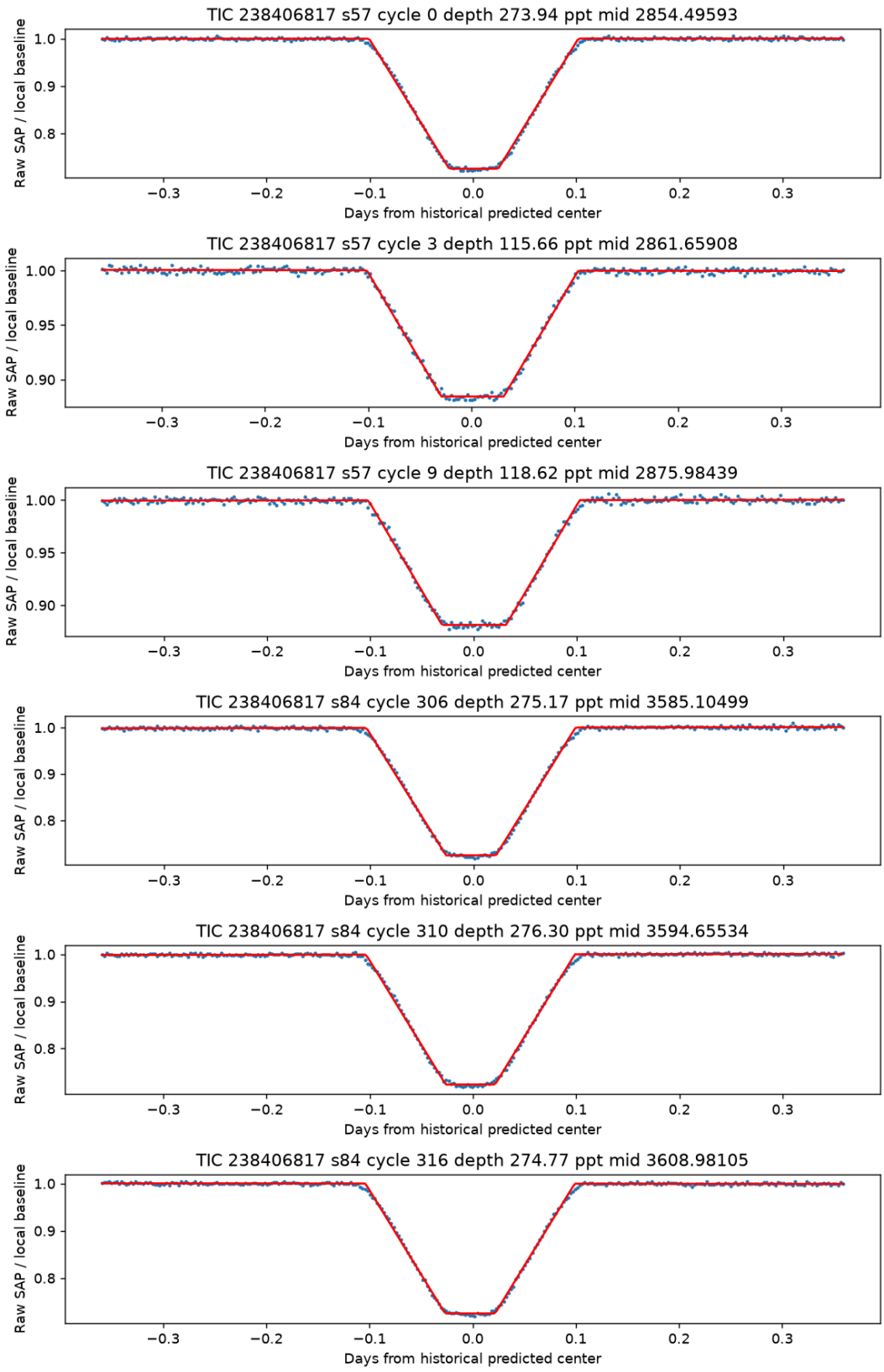
R448. TIC 238406817

physical_verdict: CONFIRMED

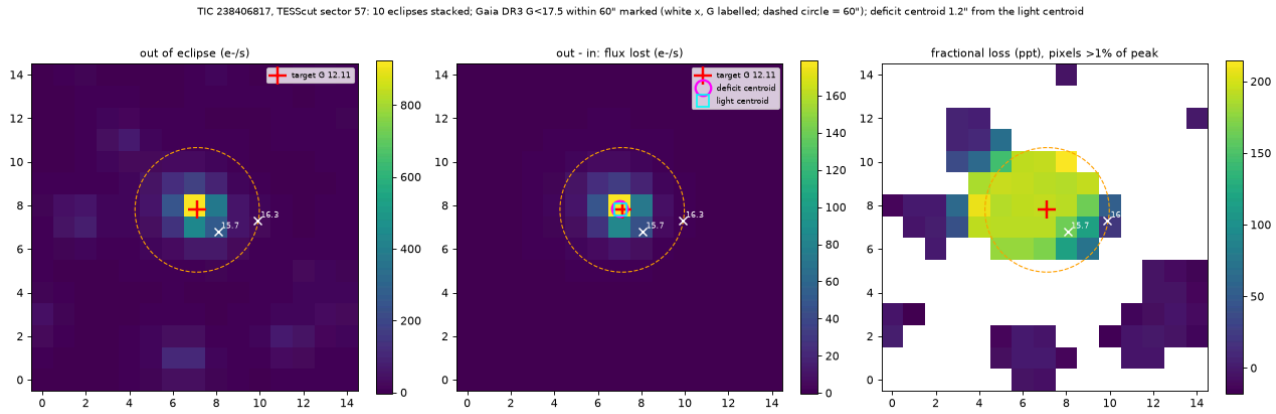
novelty_status: known prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R448; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R448; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 238406817: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 9 timing-selected measured eclipses in sectors [57, 84]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=4.7752225643 d, fit error=0.0000050930 d, epoch BTJD 2854.4959258 (Deeper repeating eclipse; historical half-period parity 0). There are 5 consecutive within-sector same-kind cycle pairs. The inventory period was 4.77524 d and the historical SAP search branch was 2.3876213686 d. Deeper parity 0 of historical half-period cycles; consecutive same-kind events have orbital-cycle count one. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: favors target at TESS resolution.** Historical S57 TESScut deficit core is 2.54 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Fractional PSF model residual is 0.089, not a localization uncertainty. Capable Gaia neighbors: none under the approximate G-band flux budget. The full list and flux-budget calculation are in historical/diff_s57.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 193.58/189.24/185.20 ppt from 10 selected windows. Median selected QLP raw fit depth is 274.77 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 2854017381783700352 has P=4.7751949852 d and e=0.01333. Closest edge-on conjunction differs from the measured epoch by 0.00195 in phase (0.00933 d); linear propagation of the source-specific full SB1 covariance gives 0.12817 d as a descriptive timing scale. The rough central-duration estimate is 5.98 h versus trapezoid median 4.87

h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Alternating ~275 and ~116 ppt eclipses identify the 4.7752 d orbital period; the historical 2.38762 d SAP branch is the half-period. The deeper parity defines the epoch.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
57	0	2854.4959299	0.001157	273.94	used
57	1	2856.8836580	0.001157	114.76	other eclipse kind
57	2	2859.2710510	0.001157	273.97	used
57	3	2861.6590785	0.001157	115.66	other eclipse kind
57	6	2868.8215384	0.001157	273.49	used
57	7	2871.2094824	0.001157	117.85	other eclipse kind
57	8	2873.5968356	0.001157	274.92	used
57	9	2875.9843939	0.001157	118.62	other eclipse kind
57	10	2878.3721685	0.001157	274.42	used
57	11	2880.7598585	0.001157	118.44	other eclipse kind
84	306	3585.1049933	0.001157	275.17	used
84	307	3587.4930233	0.001157	114.11	other eclipse kind
84	308	3589.8803134	0.001157	276.62	used
84	309	3592.2681450	0.001157	116.02	other eclipse kind
84	310	3594.6553399	0.001157	276.30	used
84	313	3601.8182245	0.001157	115.69	other eclipse kind
84	314	3604.2136516	53785.751904	275.92	excluded: unidentifiable center covariance
84	315	3606.5929309	0.001157	114.09	other eclipse kind
84	316	3608.9810451	0.001157	274.77	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s57.json, historical/diff_s57.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R448. Public evidence files: diff_s57.json, diff_s57.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

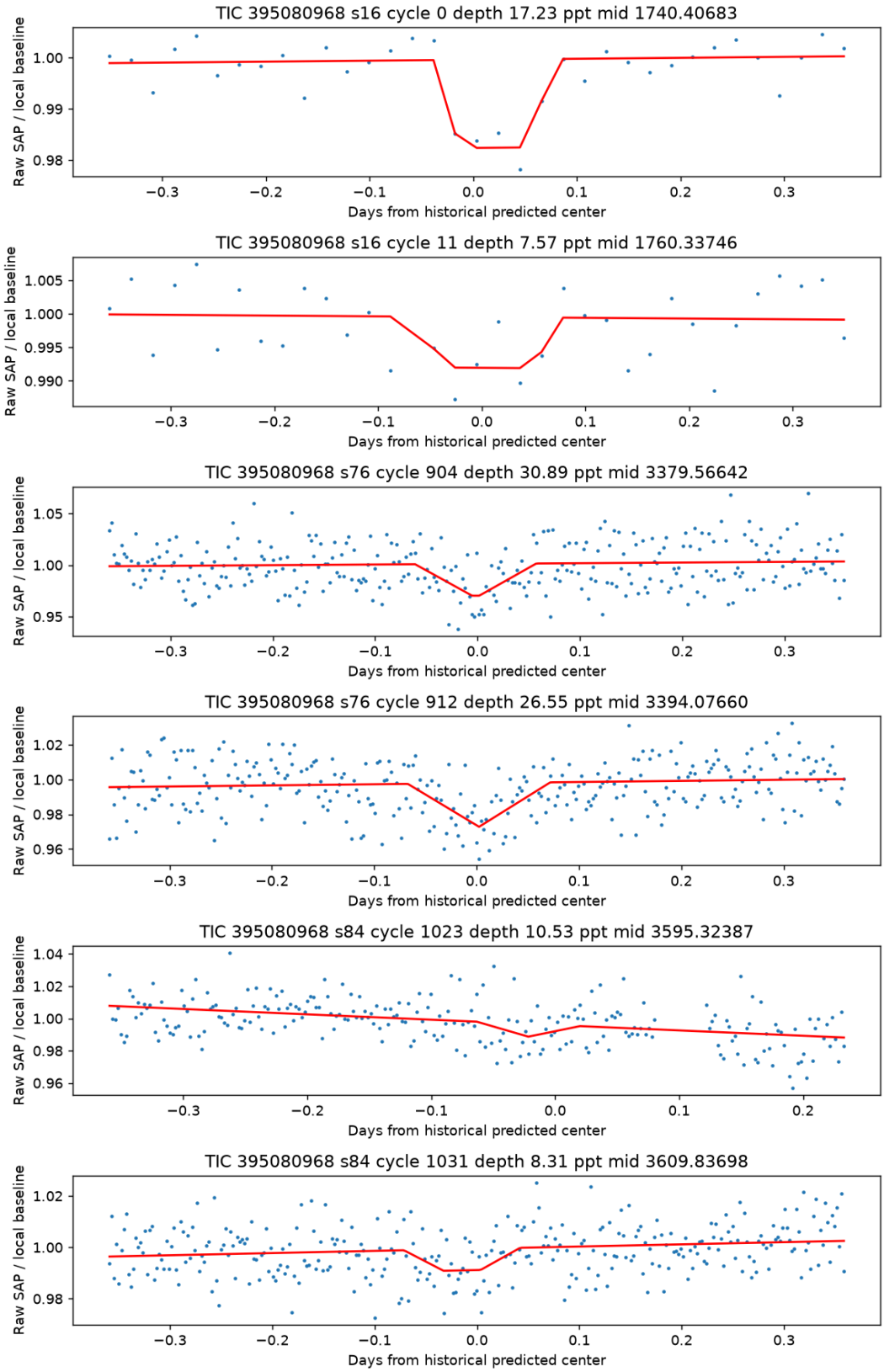
R449. TIC 395080968

physical_verdict: PROBABLE

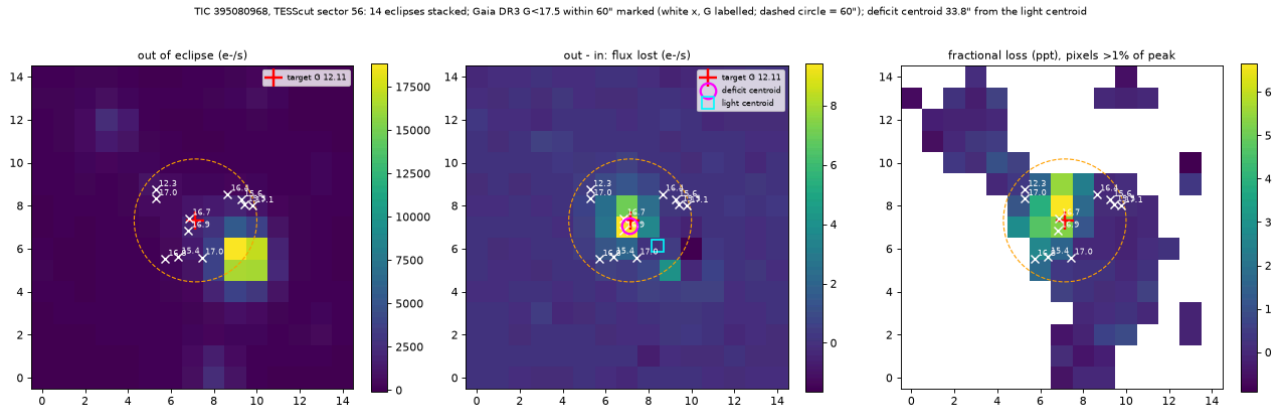
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved source localization



R449; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R449; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 395080968: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate. This report reviews the frozen batch04 target; historical screening is not an admission verdict.

1. **Raw events.** Quality-zero QLP SAP_FLUX shows 17 timing-selected measured eclipses in sectors [16, 76, 84]. Local trapezoid fits allow free centers and baseline. Predicted windows are search guides only; flagged or sparsely sampled windows are not null detections. Numeric windows and the raw plot retain the evidence.
2. **Timing and aliases.** Measured same-kind centers give P=3.6265035299 d, fit error=0.0000164528 d, epoch BTJD 1740.3942050 (Deeper repeating eclipse; historical half-period parity 0). There are 12 consecutive within-sector same-kind cycle pairs. The inventory period was 3.62651 d and the historical SAP search branch was 1.8132553250 d. Deeper parity 0 of historical half-period cycles; consecutive same-kind events have orbital-cycle count one. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The fit error is not calibrated total uncertainty. This bounded alias check is not an exhaustive blind period search.
3. **Pixel attribution: unresolved at TESS resolution.** Historical S56 TESScut deficit core is 4.65 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Fractional PSF model residual is 0.982, not a localization uncertainty. Capable Gaia neighbors: 6.3 arcsec, G=16.71, intrinsic-loss requirement ~0.75; 11.9 arcsec, G=16.91, intrinsic-loss requirement ~0.91; 35.7 arcsec, G=16.98, intrinsic-loss requirement ~0.96; 38.5 arcsec, G=15.37, intrinsic-loss requirement ~0.23; 40.1 arcsec, G=16.38, intrinsic-loss requirement ~0.56; 5 further capable sources in the 60 arcsec list. The full list and flux-budget calculation are in historical/diff_s56.json; the original cutout and stacked image are retained and hashed.
4. **Alternative reduction.** A target-centered, background-subtracted TESScut extraction gives 1x1/3x3/5x5 stacked depths 5.29/2.22/0.65 ppt from 14 selected windows. Median selected QLP raw fit depth is 18.54 ppt. Aperture dilution, event-kind mixing and model residuals limit direct depth comparisons. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument.
5. **Gaia orbit.** The matched DR3 NSS source 2171489770715391616 has P=3.6264741663 d and e=0.06738. Closest edge-on conjunction differs from the measured epoch by -0.00711 in phase

(-0.02580 d); linear propagation of the source-specific full SB1 covariance gives 0.07572 d as a descriptive timing scale. The rough central-duration estimate is 8.38 h versus trapezoid median 3.11 h; these assume a TIC radius and minimum-mass main-sequence companion and are not a full geometry fit or a sigma gate.

Deciding interpretation and limitation. Alternating depths support the 3.6265 d orbital period, but raw fits scatter strongly and residuals reach 75 min. The S56 alternative aperture stack mixes eclipse kinds; its depth is not an intrinsic primary depth. Capable neighbors at 6.3 and 11.9 arcsec share the TESS image core and the image-model residual is 0.982, so source attribution and precision ephemeris remain limited.

Next check. Resolved eclipse photometry or multi-source PRF localization for source attribution.

Catalogue novelty is evaluated centrally against seven scoped groups in ../../shared-catalogues/coverage-matrix.csv. This physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
16	0	1740.4068340	0.010417	17.23	used
16	1	1742.1753092	0.010417	15.29	other eclipse kind
16	2	1744.0135012	0.010417	21.36	used
16	7	1753.0596536	0.019758	5.08	other eclipse kind
16	8	1754.8942611	0.010417	21.08	used
16	9	1756.6989109	0.010417	9.32	other eclipse kind
16	11	1760.3374561	0.016615	7.57	other eclipse kind
76	898	3368.7464702	0.014812	16.95	used
76	899	3370.4767960	0.003860	7.00	other eclipse kind
76	900	3372.3198869	0.010680	18.60	used
76	902	3375.9517373	0.005173	22.92	used
76	903	3377.7228961	0.008759	16.70	other eclipse kind
76	904	3379.5664159	0.005774	30.89	used
76	906	3383.2096794	0.005946	20.55	used
76	907	3385.0113163	0.007454	8.18	other eclipse kind
76	908	3386.8231146	0.004738	17.53	used
76	909	3388.6222925	0.003181	54.45	other eclipse kind
76	910	3390.4504345	0.004672	14.95	used
76	911	3392.2404815	0.004192	46.66	other eclipse kind
76	912	3394.0765956	0.005304	26.55	used
84	1018	3586.2803727	0.004857	20.02	used
84	1019	3588.0793575	0.007339	11.74	other eclipse kind
84	1020	3589.9091733	0.004168	18.54	used
84	1021	3591.7124273	0.008708	11.84	other eclipse kind
84	1022	3593.5427983	0.004557	16.65	used
84	1023	3595.3238697	0.010836	10.53	other eclipse kind

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
84	1025	3598.9485479	0.005482	15.84	other eclipse kind
84	1026	3600.7852493	0.003297	15.68	used
84	1027	3602.5784254	0.006899	9.21	other eclipse kind
84	1028	3604.4200775	0.003981	18.16	used
84	1029	3606.2246957	0.008835	8.32	other eclipse kind
84	1030	3608.0478675	0.003847	16.65	used
84	1031	3609.8369787	0.007819	8.31	other eclipse kind

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json, original TESScut [local path omitted] and ../helper-audit.md. The original cutout and source files are SHA-256 recorded in ../source-hashes.json and ../SHA256SUMS.json.

Stable review key: R449. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

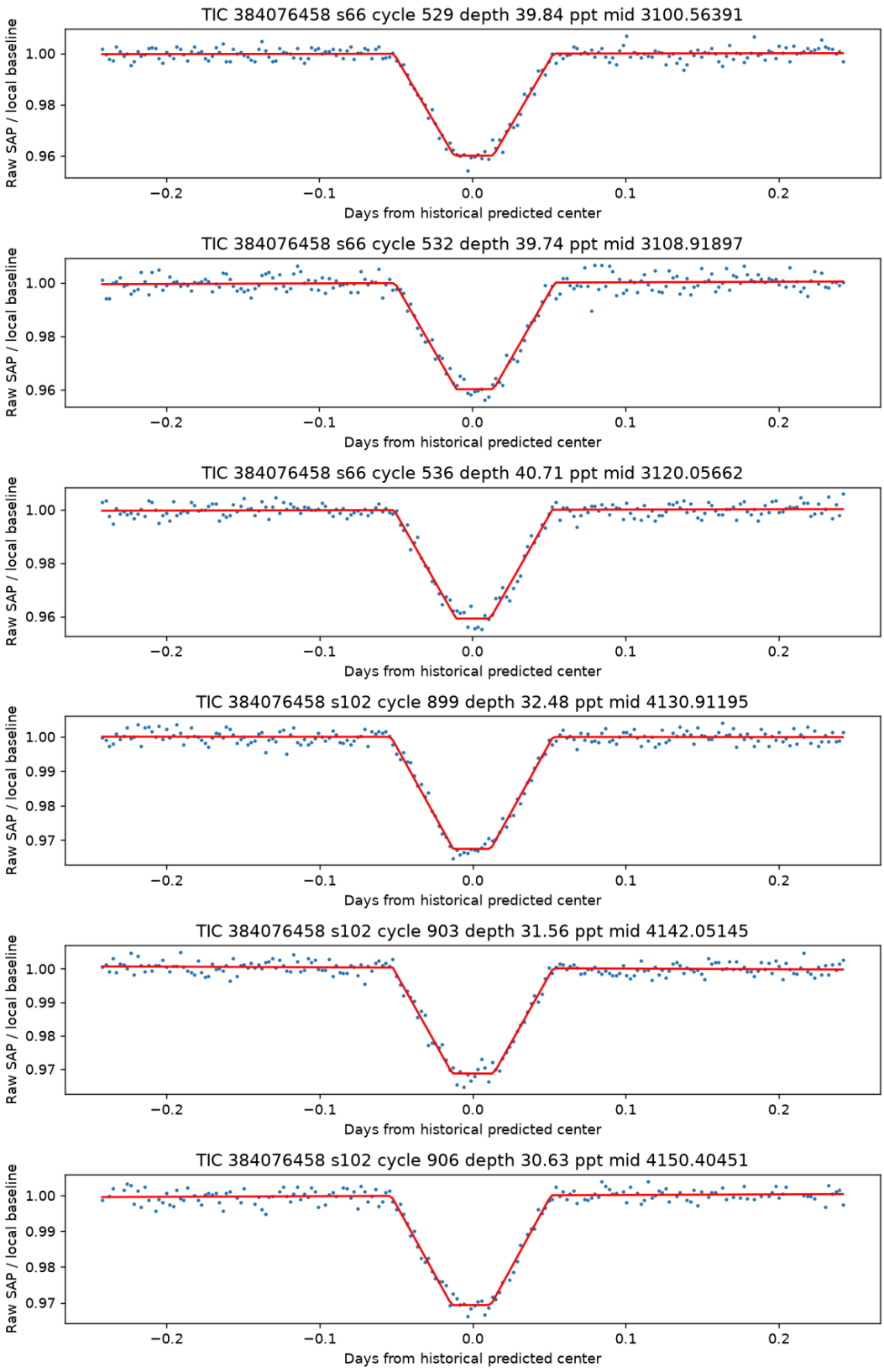
R450. TIC 384076458

physical_verdict: CONFIRMED

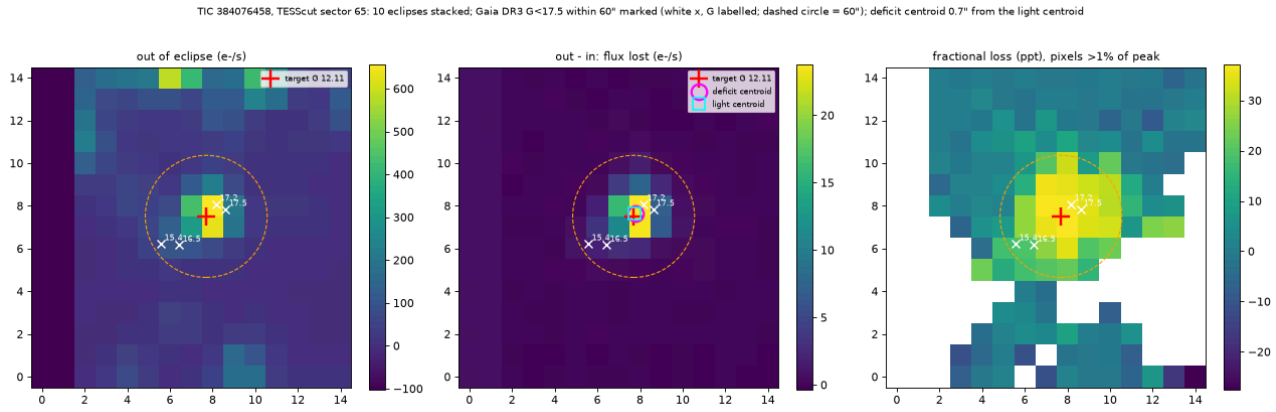
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R450; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R450; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 384076458: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 16 timing-selected measured centers in sectors [66, 102]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 2.7847243427 ± 0.0000015681 d and reference epoch BTJD 3100.5639723 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 12 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 2.78472813 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 65 TESScut difference image puts the deficit 3.54 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.204 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 5778272945037909632 (46.63 arcsec, G 15.372, intrinsic required loss about 0.728). Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 36.20/35.87/33.61 ppt in 10 stacked historical events; representative QLP raw median selected depth is 35.73 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 5778272841958695040 has SB1 P = 2.7847986876 d and e = 0.02887. Closest edge-on conjunction offset is -0.05097 d (-0.01830 phase), with 0.08625 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.37 h versus observed median fit 2.49 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Sixteen measured raw eclipses span sectors 66 and 102 with twelve adjacent-cycle pairs. The only approximately capable Gaia neighbor is 46.6 arcsec away from the target-centered deficit.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
66	529	3100.5639140	0.001157	39.84	used
66	530	3103.3478225	0.001157	38.49	used
66	531	3106.1333249	0.001157	39.70	used
66	532	3108.9189679	0.001157	39.74	used
66	534	3114.4877752	0.001157	39.69	used
66	535	3117.2728831	0.001157	39.11	used
66	536	3120.0566188	0.001157	40.71	used
66	537	3122.8416413	0.001157	38.09	used
102	898	4128.1276997	0.001157	32.10	used
102	899	4130.9119539	0.001157	32.48	used
102	900	4133.6967499	0.001157	32.44	used
102	901	4136.4814578	0.001157	32.36	used
102	903	4142.0514537	0.001157	31.56	used
102	904	4144.8357621	0.001157	33.36	used
102	905	4147.6196430	0.001157	33.10	used
102	906	4150.4045059	0.001157	30.63	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s65.json, historical/diff_s65.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R450. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

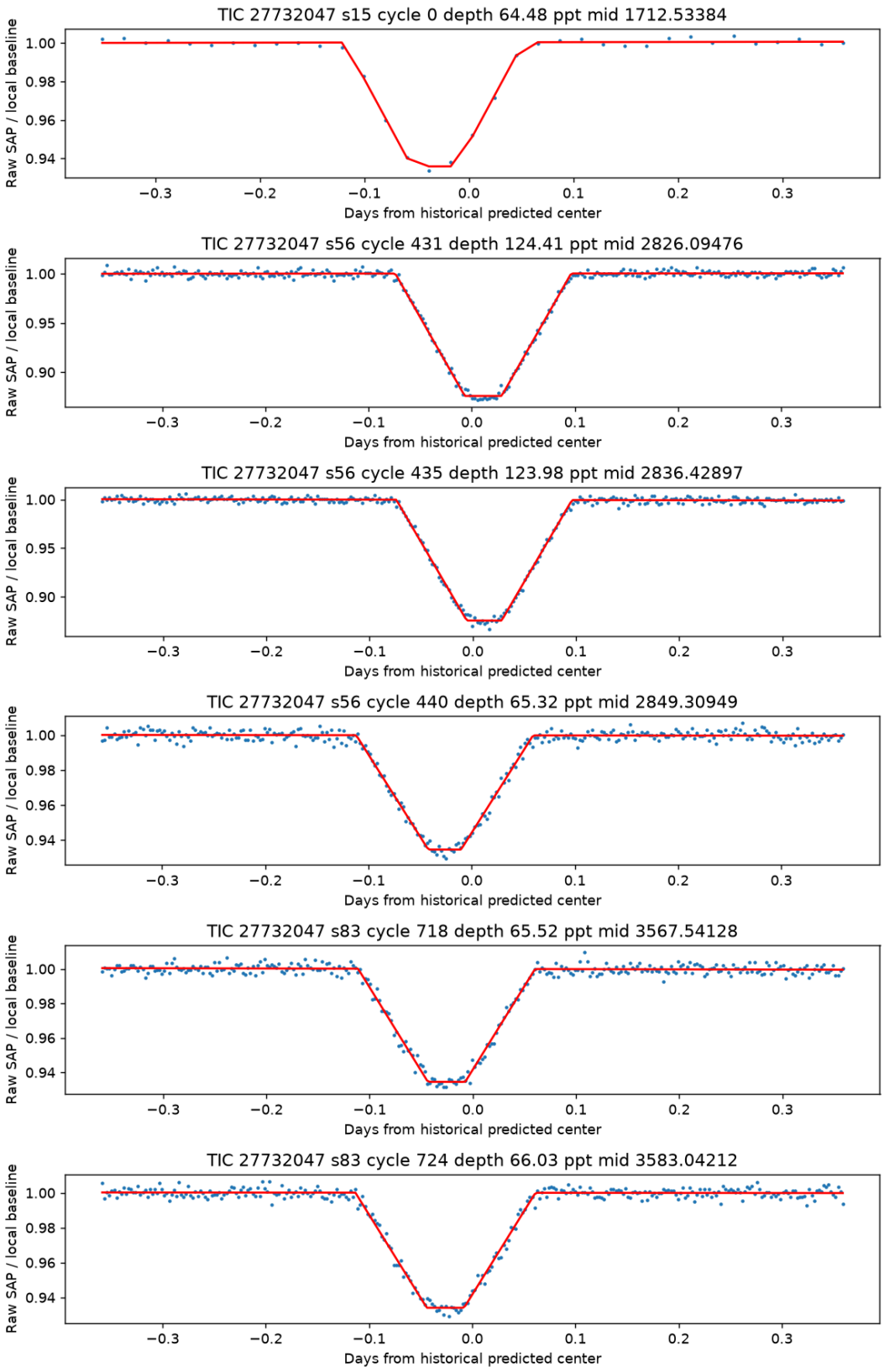
R451. TIC 27732047

physical_verdict: CONFIRMED

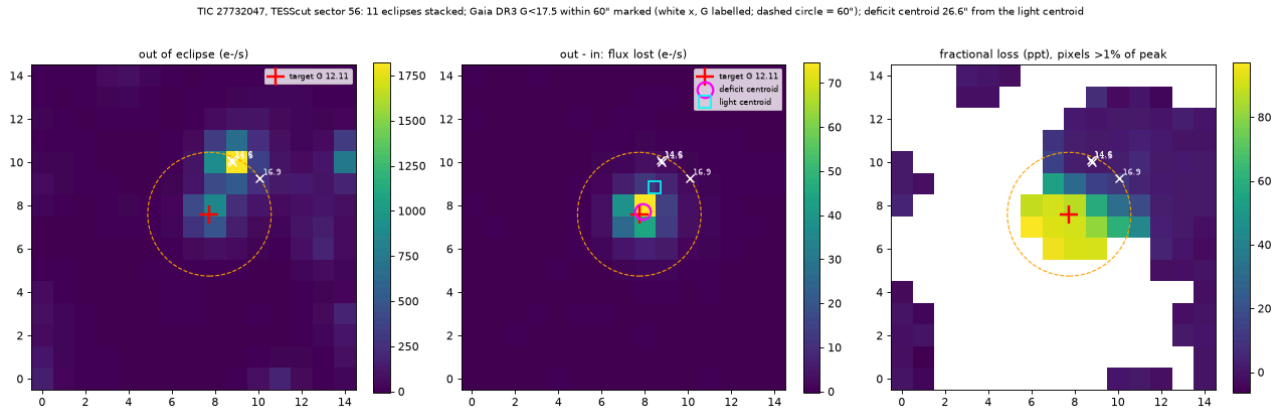
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R451; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R451; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 27732047: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 23 timing-selected measured centers in sectors [15, 56, 83]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 5.1671309470 ± 0.0000052202 d and reference epoch BTJD 1730.6628793 (Deeper repeating eclipse; historical half-period parity 1). There are 7 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 2.58356477 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 56 TESScut difference image puts the deficit 3.97 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.864 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 1893685410481942016 (54.40 arcsec, G 11.479, intrinsic required loss about 0.093); Gaia 1893686166394478208 (55.48 arcsec, G 14.610, intrinsic required loss about 0.655). Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 86.89/67.21/37.77 ppt in 11 stacked historical events; representative QLP raw median selected depth is 123.94 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 1893685406185047936 has SB1 P = 5.1675861098 d and e = 0.01880. Closest edge-on conjunction offset is -0.09725 d (-0.01882 phase), with 0.07040 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.54 h versus observed median fit 4.09 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Deep and shallow eclipses alternate at the 2.58356-d photometric search period. The 5.16713-d orbital ephemeris uses the eleven deeper odd-parity centers; shallow even-parity eclipses independently support the double-period interpretation. Two nominally capable Gaia neighbors lie beyond 54 arcsec; the source image is structured, but its strongest deficit is on the target core.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
15	0	1712.5338395	0.010417	64.48	used
15	2	1717.7013600	0.010417	65.30	used
15	6	1728.0365599	0.010417	65.75	used
15	7	1730.6575933	0.010417	124.34	used
56	431	2826.0947577	0.001157	124.41	used
56	432	2828.6404641	0.001157	65.54	used
56	433	2831.2618576	0.001157	124.37	used
56	434	2833.8087977	0.001157	65.96	used
56	435	2836.4289748	0.001157	123.98	used
56	436	2838.9748702	0.001157	64.57	used
56	437	2841.5959721	0.001157	125.20	used
56	438	2844.1422710	0.001157	64.04	used
56	439	2846.7629037	0.001157	123.38	used
56	440	2849.3094867	0.001157	65.32	used
56	441	2851.9304934	0.001157	123.68	used
83	715	3559.8273915	0.001157	123.43	used
83	716	3562.3739161	0.001157	64.78	used
83	718	3567.5412835	0.001157	65.52	used
83	719	3570.1615750	0.001157	123.76	used
83	720	3572.7081264	0.001157	65.19	used
83	721	3575.3285814	0.001157	123.94	used
83	723	3580.4954790	0.001157	122.52	used
83	724	3583.0421157	0.001157	66.03	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R451. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

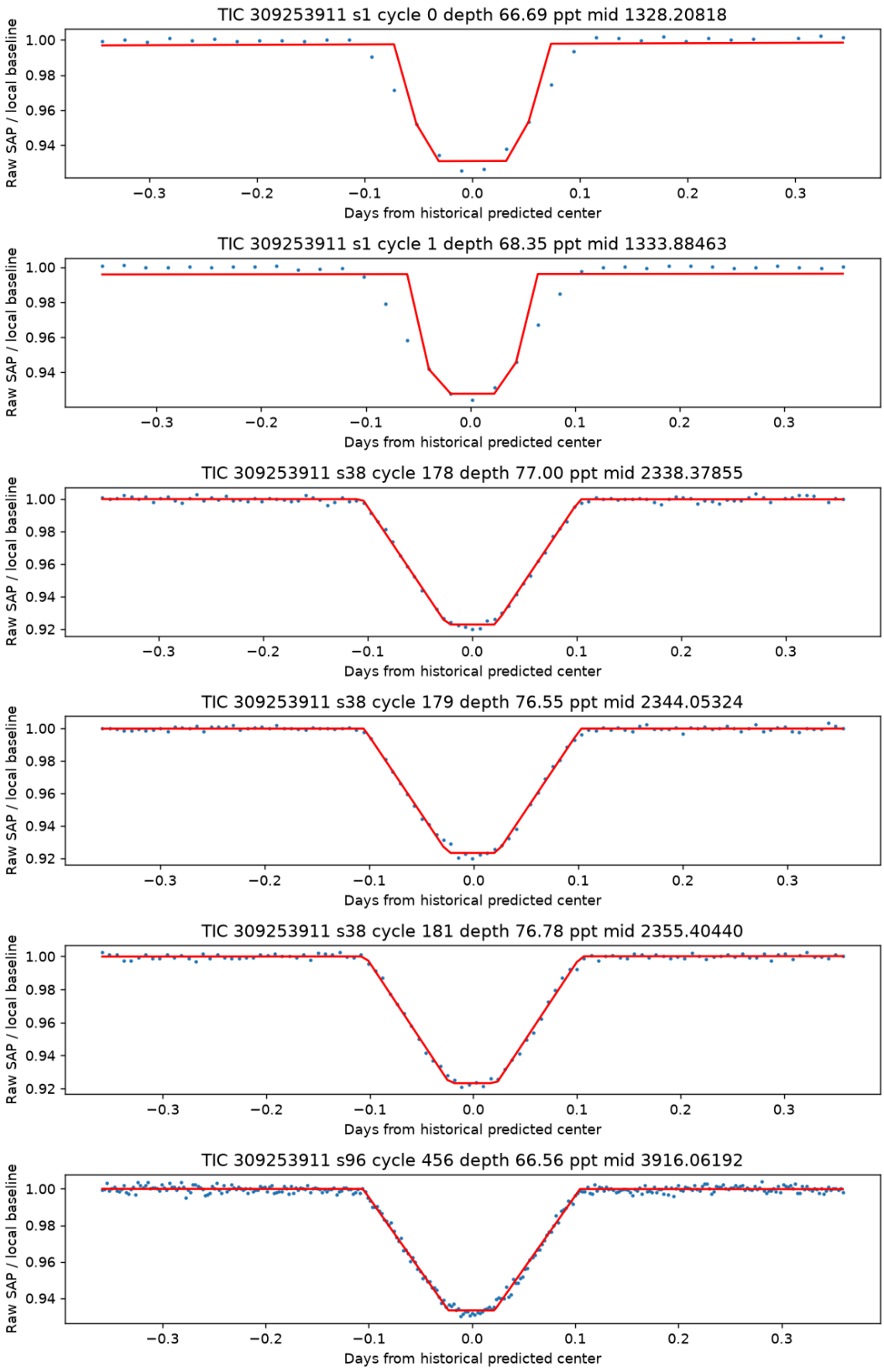
R452. TIC 309253911

physical_verdict: CONFIRMED

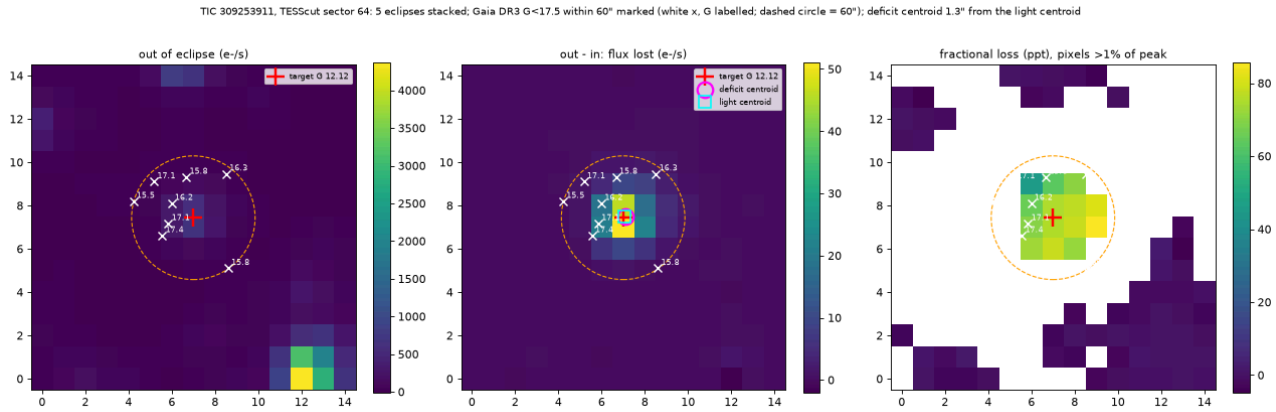
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R452; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R452; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 309253911: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 9 timing-selected measured centers in sectors [1, 38, 96]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 5.6751203957 ± 0.0000063088 d and reference epoch BTJD 1328.2070579 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 5.67511914 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 64 TESScut difference image puts the deficit 2.46 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.179 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 75.40/74.07/69.71 ppt in 5 stacked historical events; representative QLP raw median selected depth is 68.35 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 5276097362970850432 has SB1 P = 5.6754892774 d and e = 0.03670. Closest edge-on conjunction offset is -0.05998 d (-0.01057 phase), with 0.04432 d linearly propagated full source-specific SB1 covariance. Rough central duration 7.71 h versus observed median fit 4.98 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Nine measured eclipses in sectors 1, 38 and 96 include five adjacent-cycle pairs. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
1	0	1328.2081849	0.010416	66.69	used
1	1	1333.8846253	0.010416	68.35	used
1	3	1345.2296355	0.010416	66.37	used
38	178	2338.3785526	0.003472	77.00	used
38	179	2344.0532360	0.003472	76.55	used
38	180	2349.7283396	0.003472	77.20	used
38	181	2355.4044016	0.003472	76.78	used
96	455	3910.3868805	0.001157	65.77	used
96	456	3916.0619223	0.001157	66.56	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s64.json, historical/diff_s64.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R452. Public evidence files: diff_s64.json, diff_s64.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

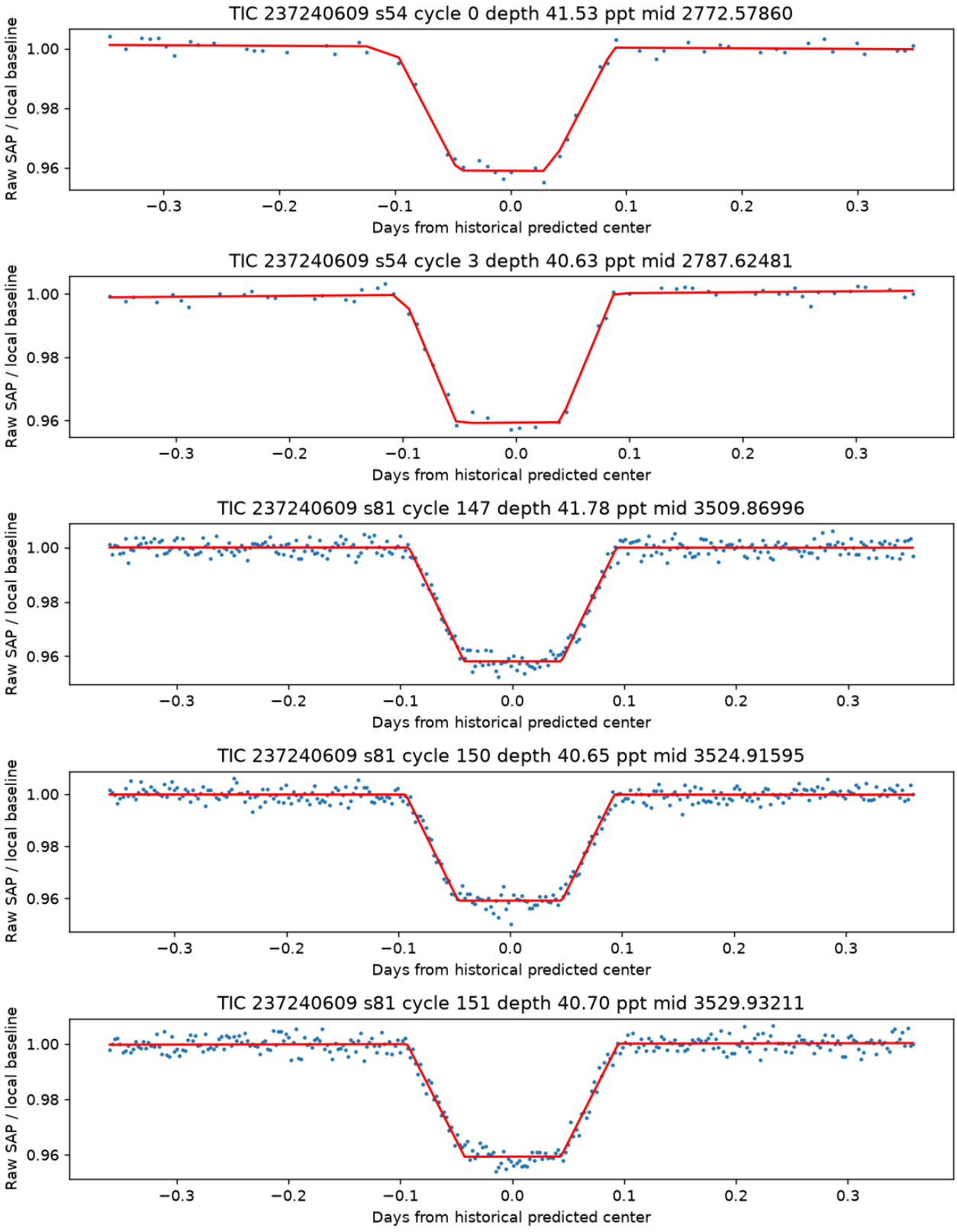
R453. TIC 237240609

physical_verdict: CONFIRMED

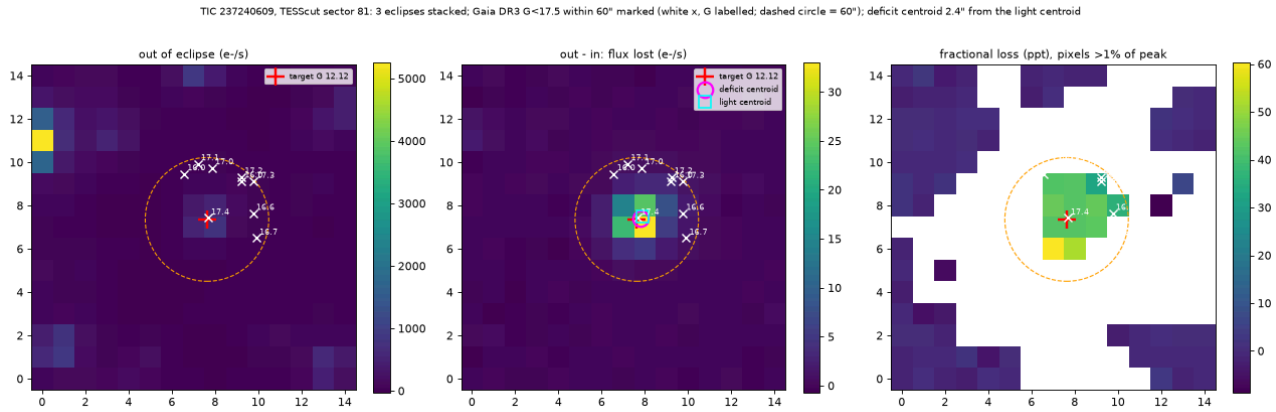
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R453; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R453; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 237240609: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 5 timing-selected measured centers in sectors [54, 81]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 5.0155864471 ± 0.0000333988 d and reference epoch BTJD 2772.5784287 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 5.01554178 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 81 TESScut difference image puts the deficit 3.87 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.262 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 42.53/43.97/44.53 ppt in 3 stacked historical events; representative QLP raw median selected depth is 40.70 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 4196687071198155520 has SB1 P = 5.0156410953 d and e = 0.01696. Closest edge-on conjunction offset is -0.03987 d (-0.00795 phase), with 0.31438 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.29 h versus observed median fit 4.47 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Five measured eclipses in sectors 54 and 81 include a sector-81 adjacent pair. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
54	0	2772.5785995	0.006944	41.53	used
54	3	2787.6248058	0.006944	40.63	used
81	147	3509.8699618	0.001157	41.78	used
81	150	3524.9159492	0.001157	40.65	used
81	151	3529.9321092	0.001157	40.70	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s81.json, historical/diff_s81.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R453. Public evidence files: diff_s81.json, diff_s81.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

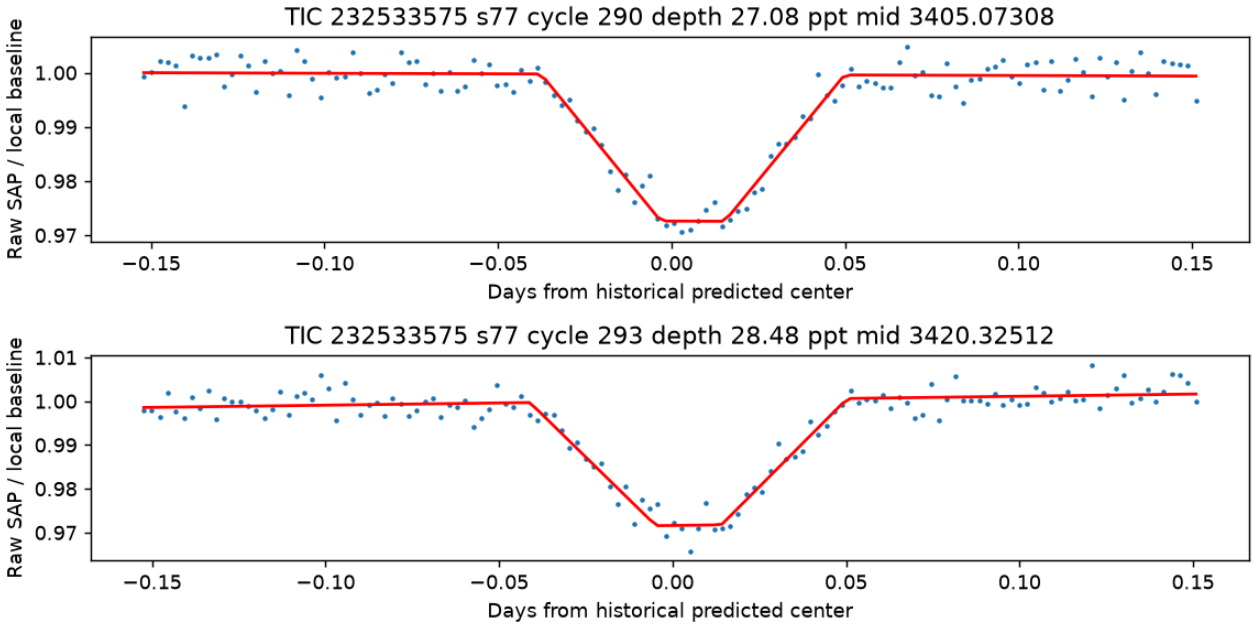
R454. TIC 232533575

physical_verdict: PLAUSIBLE

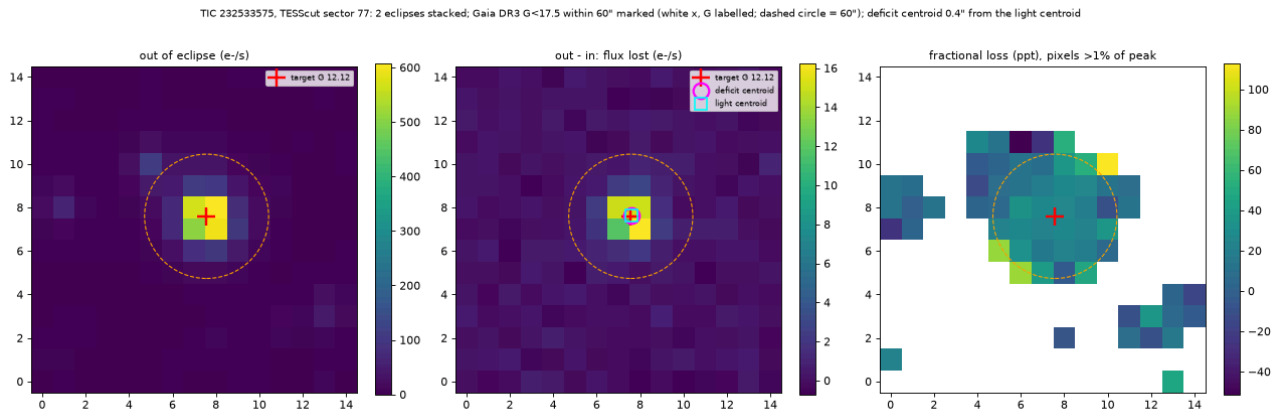
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: A third measured eclipse in future coverage to distinguish 15.252 d divided by three from other integer-cycle aliases.



R454; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R454; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 232533575: PLAUSIBLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 2 timing-selected measured centers in sectors [77]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Two observed eclipse-shaped events do not test the integer-cycle period.
2. **Timing and aliases.** Selected centers yield P = 5.0840126999 ± 0.0005456124 d and reference epoch BTJD 3405.0730806 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 0 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 5.08450258 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search. Live MAST timeseries availability query is retained in ../232533575-availability.json; availability is not an eclipse detection. The two intervening predicted cycles have no quality-zero or flagged QLP cadences within 0.25 d, so neither is a clean null.
3. **Source attribution.** Historical sector 77 TESScut difference image puts the deficit 0.98 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 8]. Image-model fractional residual 0.154 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 24.67/25.12/24.73 ppt in 2 stacked historical events; representative QLP raw median selected depth is 27.78 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 1285502678268839680 has SB1 P = 5.0852473937 d and e = 0.04997. Closest edge-on conjunction offset is -0.35556 d (-0.06992 phase), with 0.18332 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.50 h versus observed median fit 2.15 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Two target-centered eclipses in sector 77 are separated by 15.252 d. The Gaia-like 5.084-d period assigns three cycles, but cycles 291 and 292 fall in an observation gap. Sector 23 has no usable repeated center and current MAST lists only sectors 23 and 77. Thus the integer-cycle alias is untested by a third eclipse; the reported period is conditional on the Gaia-supported three-cycle assignment.

Next check: A third measured eclipse in future coverage to distinguish 15.252 d divided by three from other integer-cycle aliases.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
77	290	3405.0730806	0.001157	27.08	used
77	293	3420.3251187	0.001157	28.48	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s77.json, historical/diff_s77.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json).

Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R454. Public evidence files: diff_s77.json, diff_s77.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

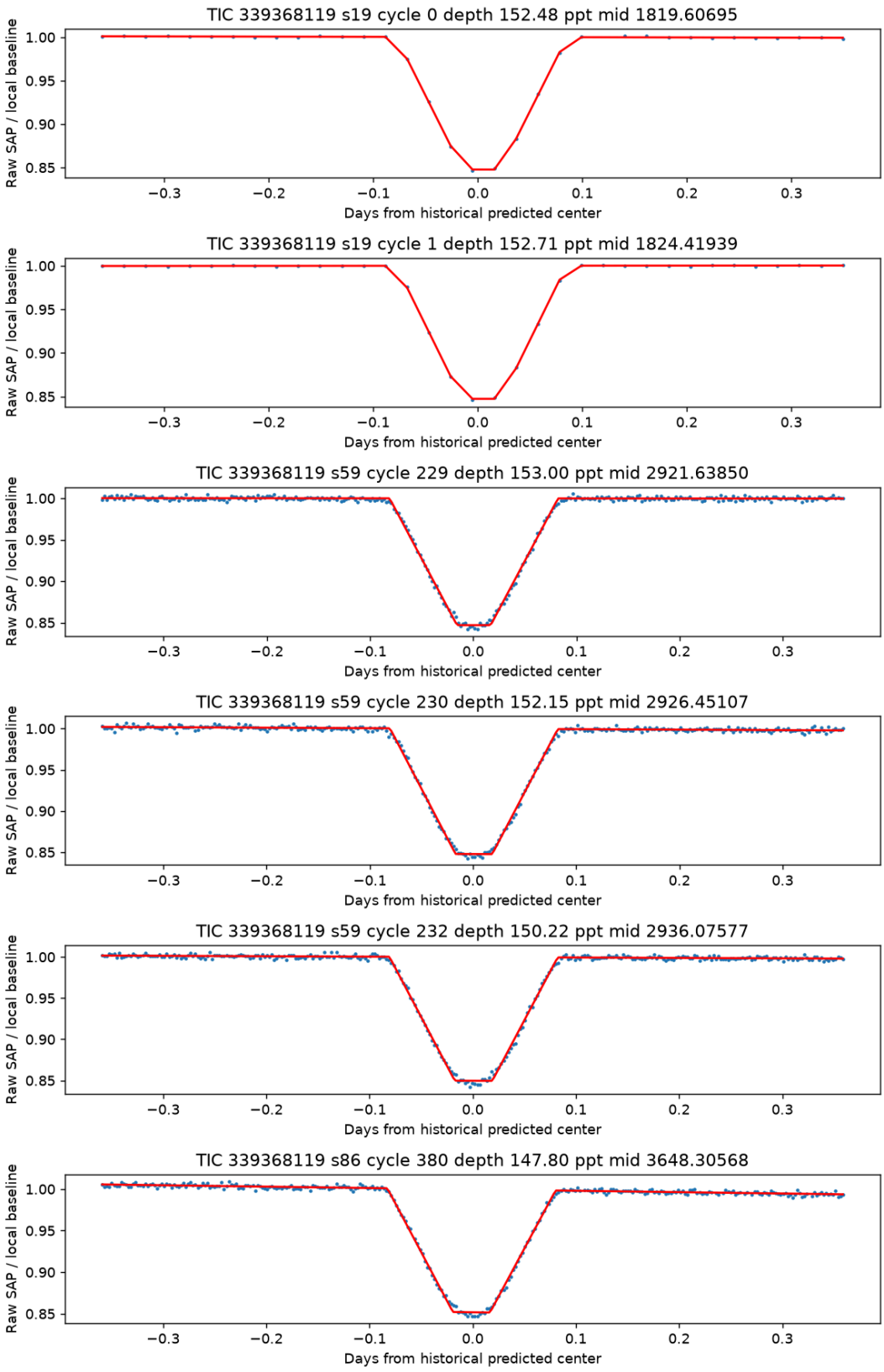
R455. TIC 339368119

physical_verdict: CONFIRMED

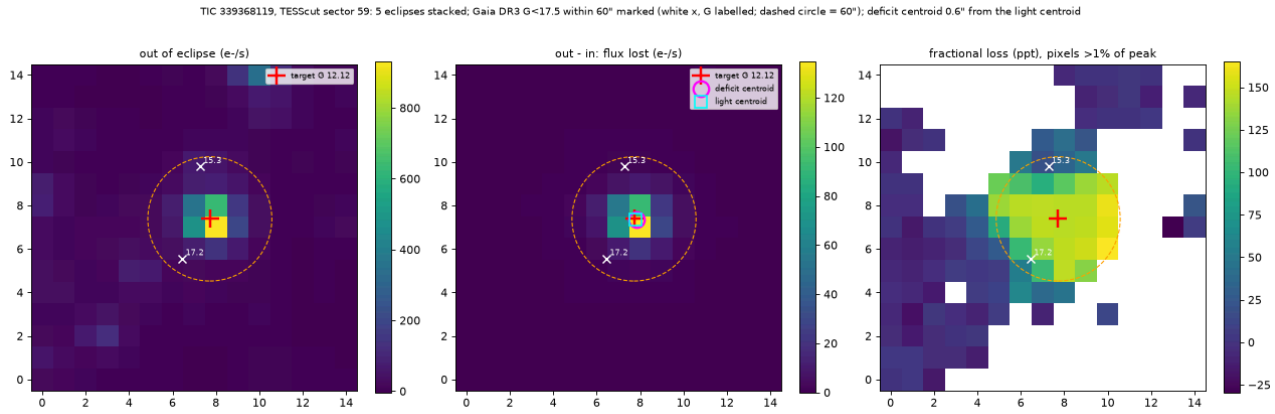
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R455; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R455; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 339368119: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 9 timing-selected measured centers in sectors [19, 59, 86]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 4.8123646667 ± 0.0000062908 d and reference epoch BTJD 1819.6071844 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 5 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 4.81237757 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 59 TESScut difference image puts the deficit 2.41 arcsec from the target coordinate; peak [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.110 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 144.77/145.89/142.74 ppt in 5 stacked historical events; representative QLP raw median selected depth is 152.15 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 287971617259251968 has SB1 P = 4.8121222282 d and e = 0.04395. Closest edge-on conjunction offset is 0.04761 d (0.00989 phase), with 0.06365 d linearly propagated full source-specific SB1 covariance. Rough central duration 4.47 h versus observed median fit 3.92 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Nine deep measured eclipses in sectors 19, 59 and 86 include five adjacent-cycle pairs. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
19	0	1819.6069525	0.010417	152.48	used
19	1	1824.4193945	0.010417	152.71	used
59	227	2912.0141496	0.001157	152.95	used
59	229	2921.6385015	0.001157	153.00	used
59	230	2926.4510671	0.001157	152.15	used
59	231	2931.2634482	0.001157	151.22	used
59	232	2936.0757667	0.001157	150.22	used
86	379	3643.4934662	0.001157	148.42	used
86	380	3648.3056804	0.001157	147.80	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s59.json, historical/diff_s59.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R455. Public evidence files: diff_s59.json, diff_s59.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

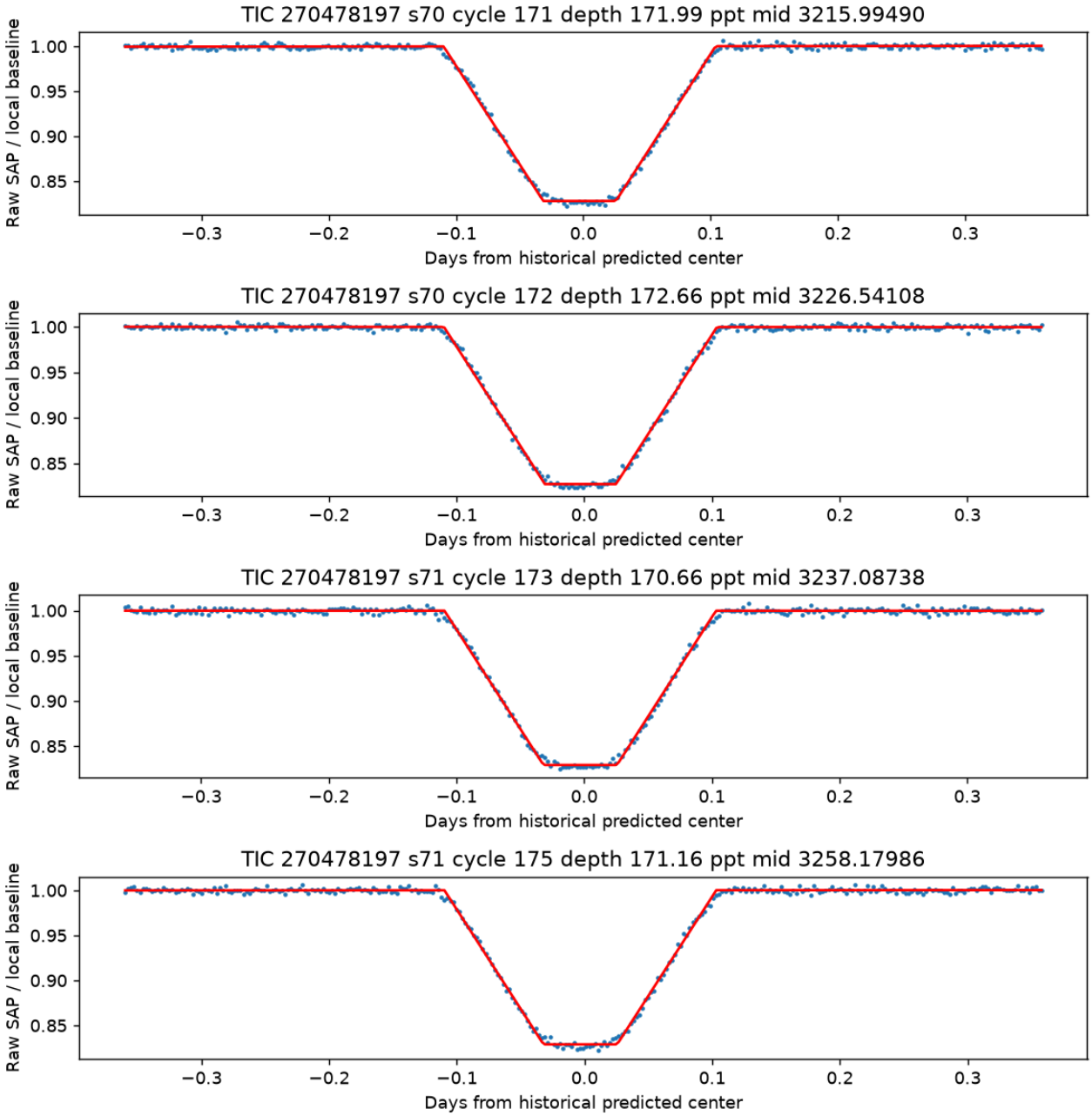
R456. TIC 270478197

physical_verdict: CONFIRMED

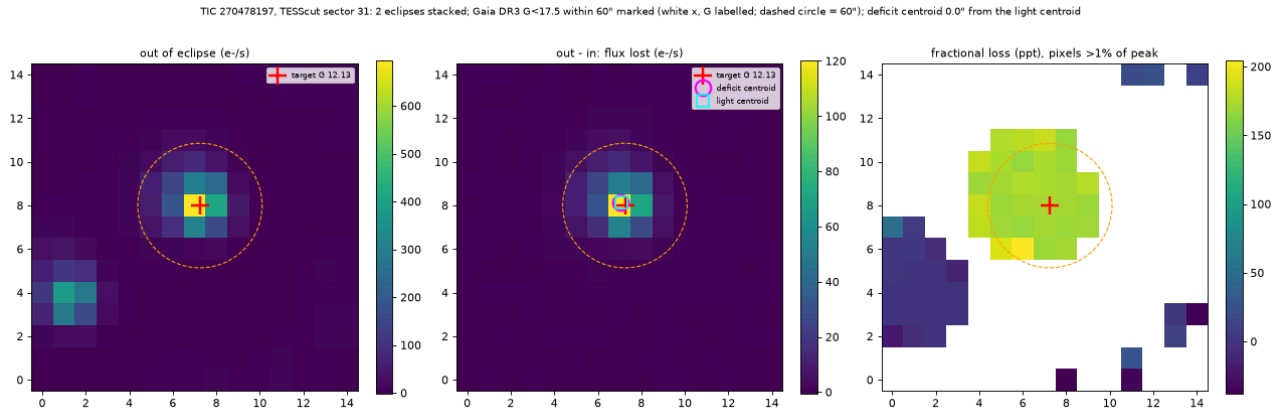
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R456; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R456; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 270478197: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 4 timing-selected measured centers in sectors [70, 71]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 10.5462440932 ± 0.0003912788 d and reference epoch BTJD 3215.9948789 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 10.54632873 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 31 TESScut difference image puts the deficit 5.02 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.036 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 172.94/172.35/172.38 ppt in 2 stacked historical events; representative QLP raw median selected depth is 171.57 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 2513059066528401664 has SB1 P = 10.5462213303 d and e = 0.09909. Closest edge-on conjunction offset is 0.19326 d (0.01833 phase), with 0.54444 d linearly propagated full source-specific SB1 covariance. Rough central duration 6.32 h versus observed median fit 5.12 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Four deep measured eclipses include an adjacent sector-70 pair and a sector-71 repeat. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
70	171	3215.9949044	0.001157	171.99	used
70	172	3226.5410800	0.001157	172.66	used
71	173	3237.0873804	0.001157	170.66	used
71	175	3258.1798593	0.001157	171.16	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s31.json, historical/diff_s31.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R456. Public evidence files: diff_s31.json, diff_s31.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

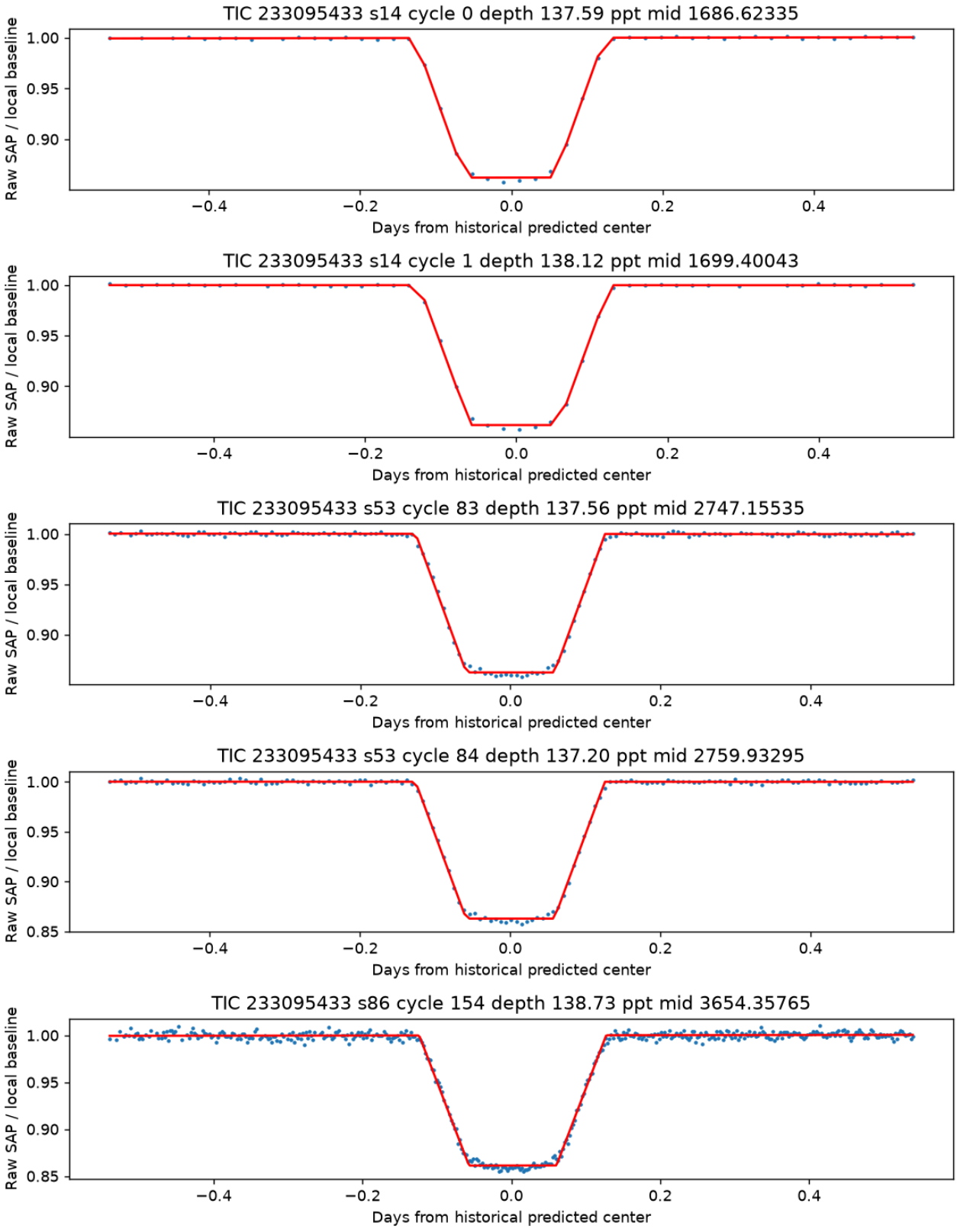
R457. TIC 233095433

physical_verdict: CONFIRMED

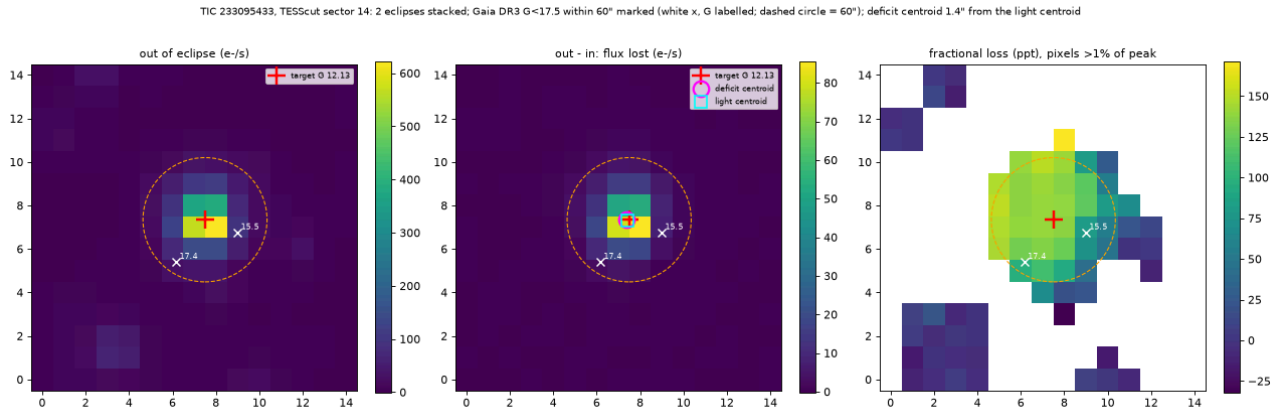
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R457; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R457; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 233095433: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 5 timing-selected measured centers in sectors [14, 53, 86]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 12.7774965312 ± 0.0000311255 d and reference epoch BTJD 1686.6231819 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 12.77746923 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 14 TESScut difference image puts the deficit 2.97 arcsec from the target coordinate; peak [8, 7] versus target pixel [7, 7]. Image-model fractional residual 0.067 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 139.98/137.52/132.99 ppt in 2 stacked historical events; representative QLP raw median selected depth is 137.59 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 2160869549233604608 has SB1 P = 12.7774692281 d and e = 0.04193. Closest edge-on conjunction offset is 0.24719 d (0.01935 phase), with 0.26052 d linearly propagated full source-specific SB1 covariance. Rough central duration 7.34 h versus observed median fit 6.05 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Five deep measured eclipses include adjacent pairs in sectors 14 and 53 and a sector-86 repeat. No Gaia neighbor within 60 arcsec meets the approximate flux budget.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
14	0	1686.6233507	0.010417	137.59	used
14	1	1699.4004320	0.010417	138.12	used
53	83	2747.1553542	0.003472	137.56	used
53	84	2759.9329495	0.003472	137.20	used
86	154	3654.3576465	0.001157	138.73	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s14.json, historical/diff_s14.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R457. Public evidence files: diff_s14.json, diff_s14.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

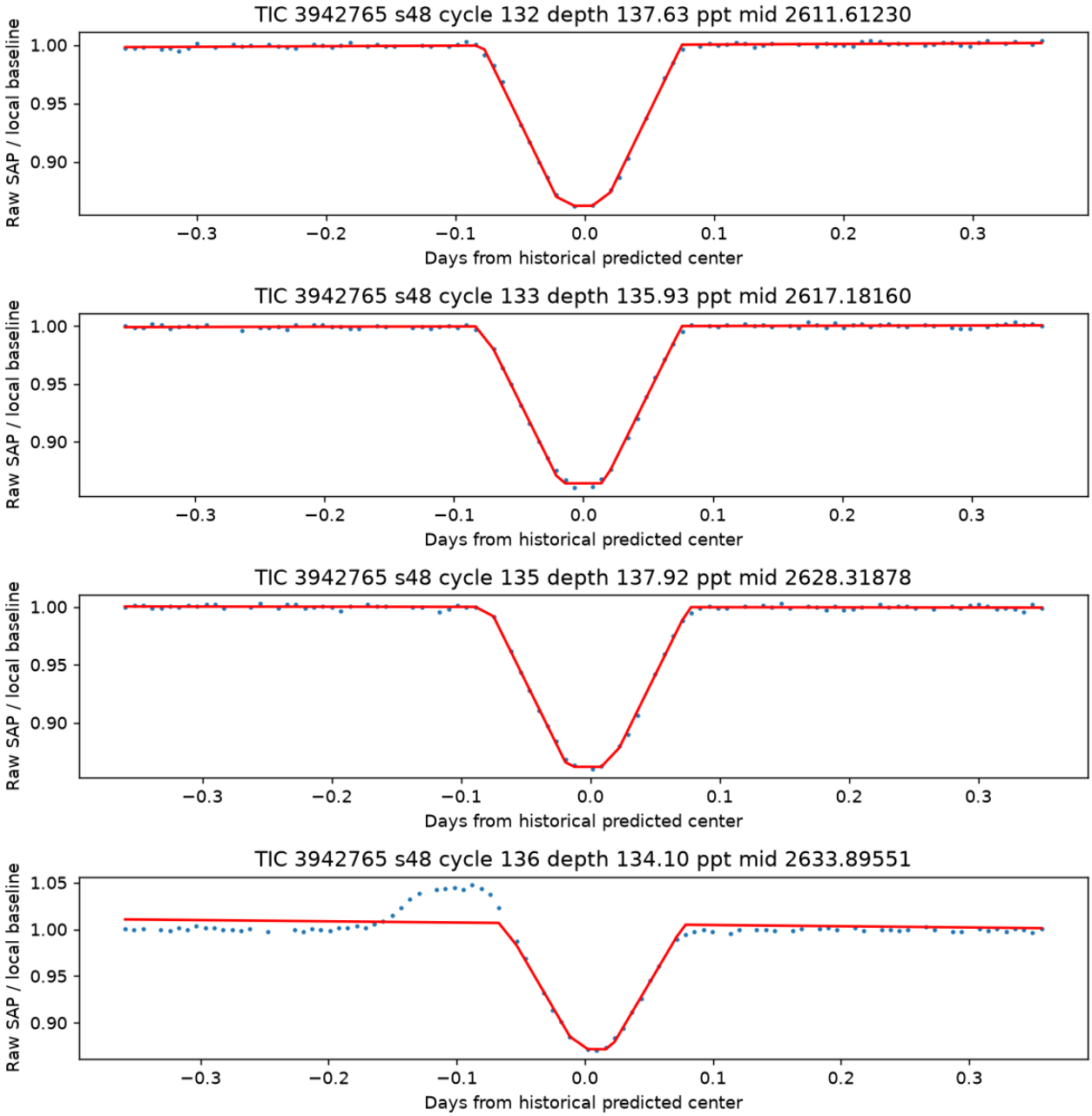
R458. TIC 3942765

physical_verdict: CONFIRMED

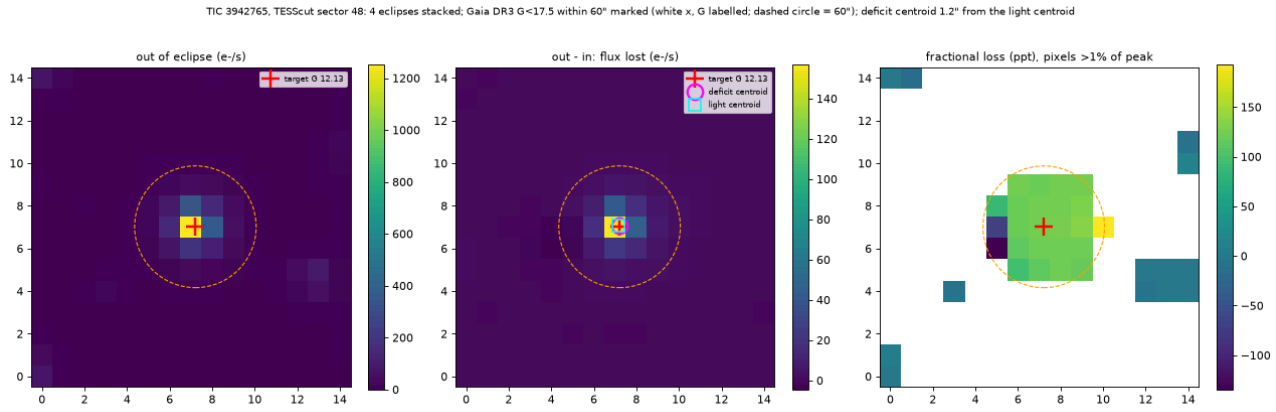
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R458; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R458; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 3942765: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 3 timing-selected measured centers in sectors [48]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 5.5687926067 ± 0.0016073409 d and reference epoch BTJD 2611.6125011 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 1 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 5.56862471 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 48 TESScut difference image puts the deficit 1.56 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.063 is not a localization error. Capable Gaia neighbors within 60 arcsec: none under the approximate G-band flux budget. Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 125.29/124.28/120.53 ppt in 4 stacked historical events; representative QLP raw median selected depth is 137.63 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 740708994716293120 has SB1 P = 5.5682939164 d and e = 0.01875. Closest edge-on conjunction offset is 0.12378 d (0.02223 phase), with 0.32903 d linearly propagated full source-specific SB1 covariance. Rough central duration 5.06 h versus observed median fit 3.70 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Four eclipse-shaped events occur in sector 48; three centers give the ephemeris. Cycle 136 has a positive pre-ingress excursion that biases its fitted baseline and midpoint, so it is retained as morphology and excluded from timing. The remaining adjacent pair tests recurrence; no capable Gaia neighbor lies within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
48	132	2611.6122989	0.003472	137.63	used
48	133	2617.1815969	0.003472	135.93	used
48	135	2628.3187778	0.003472	137.92	used
48	136	2633.8955128	0.003472	134.10	excluded: Pre-ingress positive flux excursion biases the local linear baseline and center; retained as eclipse morphology, excluded from timing fit.

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s48.json, historical/diff_s48.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R458. Public evidence files: diff_s48.json, diff_s48.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

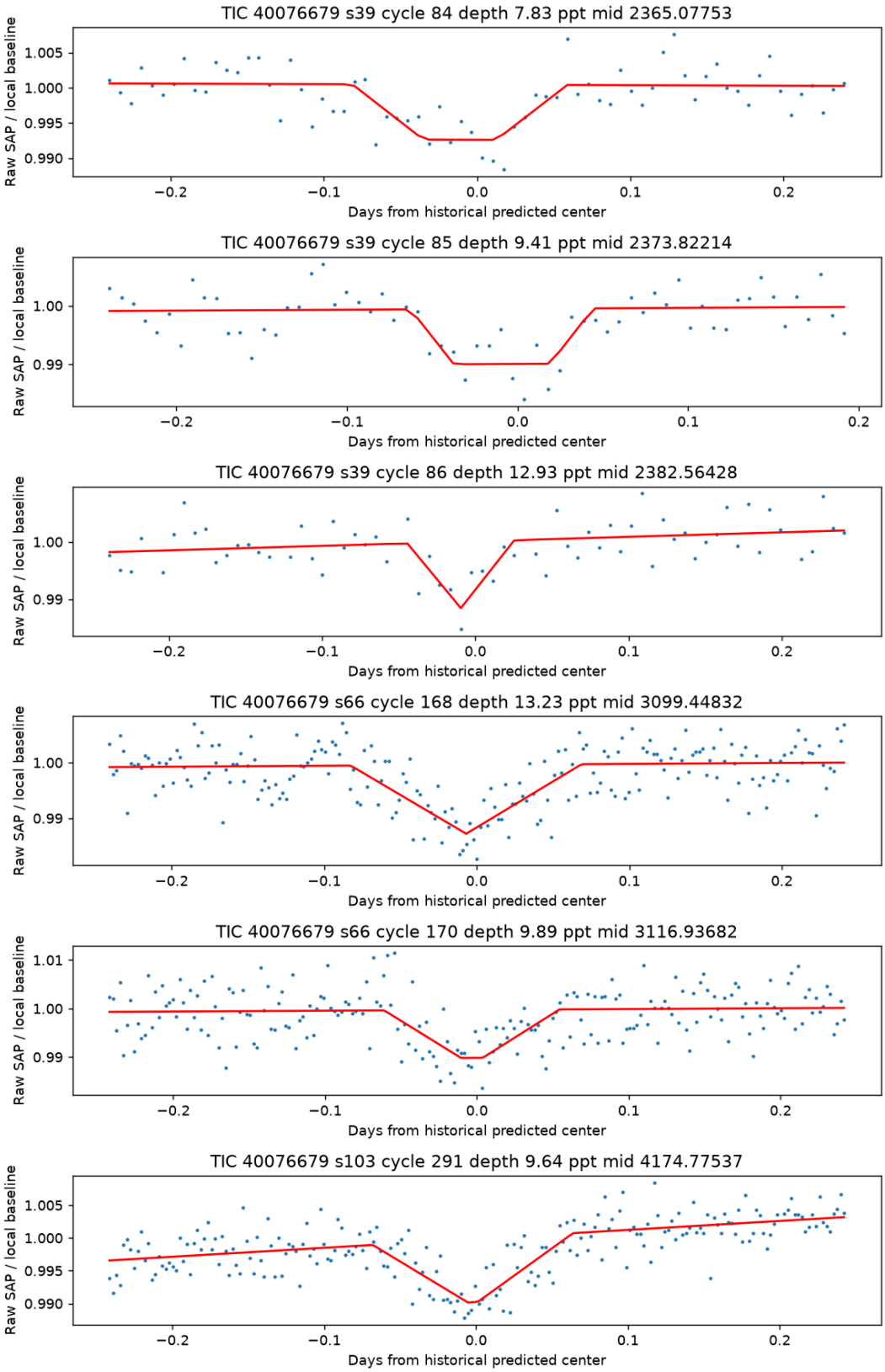
R459. TIC 40076679

physical_verdict: PROBABLE

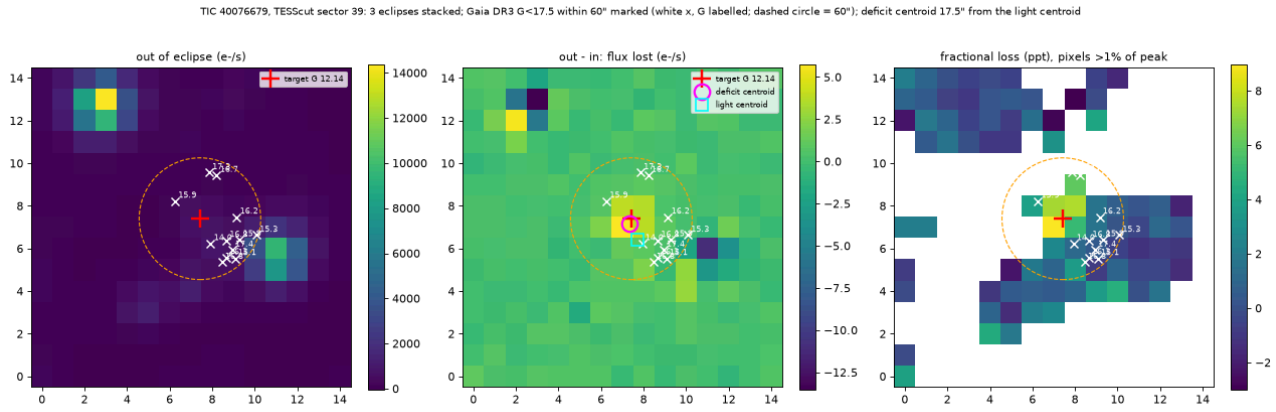
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved time-series photometry or multi-source pixel modeling during an eclipse to identify the variable source.



R459; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R459; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 40076679: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw evidence.** Quality-zero QLP SAP_FLUX gives 7 timing-selected measured centers in sectors [39, 66, 103]. Local trapezoids fit free centers, depths, durations and baselines. Raw plots and windows retain the evidence. Predicted centers guided extraction only; flagged or uncovered windows are not null detections. Representative raw morphology shows repeated eclipses.
2. **Timing and aliases.** Selected centers yield P = 8.7425051500 ± 0.0000139891 d and reference epoch BTJD 2365.0791284 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). There are 2 within-sector same-kind adjacent pairs after the specified parity selection. The historical search branch is 8.74245731 d. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This conditional formal error is not total calibrated uncertainty or an exhaustive alias search.
3. **Source attribution.** Historical sector 39 TESScut difference image puts the deficit 5.05 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.856 is not a localization error. Capable Gaia neighbors within 60 arcsec: Gaia 5937789026883670528 (25.82 arcsec, G 14.049, intrinsic required loss about 0.062); Gaia 5937789026883674624 (29.68 arcsec, G 15.869, intrinsic required loss about 0.291); Gaia 5937788270969425664 (32.98 arcsec, G 16.761, intrinsic required loss about 0.649); Gaia 5937794146484859520 (35.77 arcsec, G 16.205, intrinsic required loss about 0.393); Gaia 5937788270969425408 (43.72 arcsec, G 15.946, intrinsic required loss about 0.311); Gaia 5937794902399104128 (43.96 arcsec, G 16.674, intrinsic required loss about 0.600); Gaia 5937788270969424256 (44.85 arcsec, G 16.253, intrinsic required loss about 0.410); Gaia 5937788236609685632 (46.54 arcsec, G 16.804, intrinsic required loss about 0.675); Gaia 5937788270969423744 (51.00 arcsec, G 15.052, intrinsic required loss about 0.142); Gaia 5937788270969470208 (57.24 arcsec, G 15.251, intrinsic required loss about 0.168). Intrinsic required losses use an approximate flux budget and cannot be equated to blended pixel loss. A fresh sector-66 TESScut stack of two actual measured windows gives a 7.48-arcsec target offset, target-core peak, 1x1/3x3/5x5 depths 8.10/6.43/3.15 ppt and image-model residual 0.999; see historical/diff_s66.json and .png. These residuals prevent unique source attribution.

4. **Alternative reduction.** Background-subtracted TESScut target 1x1/3x3/5x5 apertures yield 8.97/5.31/3.45 ppt in 3 stacked historical events; representative QLP raw median selected depth is 9.64 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. aperture dilution and event choice limit direct depth comparison.
5. **Gaia geometry.** Matched DR3 source 5937789022547559296 has SB1 P = 8.7431794619 d and e = 0.06793. Closest edge-on conjunction offset is -0.13259 d (-0.01516 phase), with 0.31187 d linearly propagated full source-specific SB1 covariance. Rough central duration 8.16 h versus observed median fit 3.01 h assumes TIC radius and a minimum-mass main-sequence companion, not a complete binary model. Agreement does not prove source identity.

Interpretation and limitation: Seven shallow raw eclipses recur in sectors 39, 66 and 103; three adjacent sector-39 cycles directly test the period. Both sector-39 and newly measured sector-66 difference images have strong structured residuals in a crowded field. Ten Gaia neighbors within 60 arcsec can in principle produce the blended depth. The sector-66 deficit core is near the target, yet the pixel data cannot uniquely assign the signal.

Next check: Resolved time-series photometry or multi-source pixel modeling during an eclipse to identify the variable source.

Catalogue novelty: central seven-group result is ../../shared-catalogues/coverage-matrix.csv. A cone source alone is not an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
39	84	2365.0775254	0.004904	7.83	used
39	85	2373.8221439	0.003974	9.41	used
39	86	2382.5642846	0.003740	12.93	used
66	168	3099.4483213	0.003055	13.23	used
66	170	3116.9368236	0.003608	9.89	used
103	289	4157.2935634	0.002077	7.92	used
103	291	4174.7753719	0.003248	9.64	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s39.json, historical/diff_s39.png, historical/gaia.json and ../helper-audit.md. Original TESScut: [local path omitted] (SHA256 in ../source-hashes.json). Retained artifacts are in ../SHA256SUMS.json.

Stable review key: R459. Public evidence files: diff_s39.json, diff_s39.png, diff_s66.json, diff_s66.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

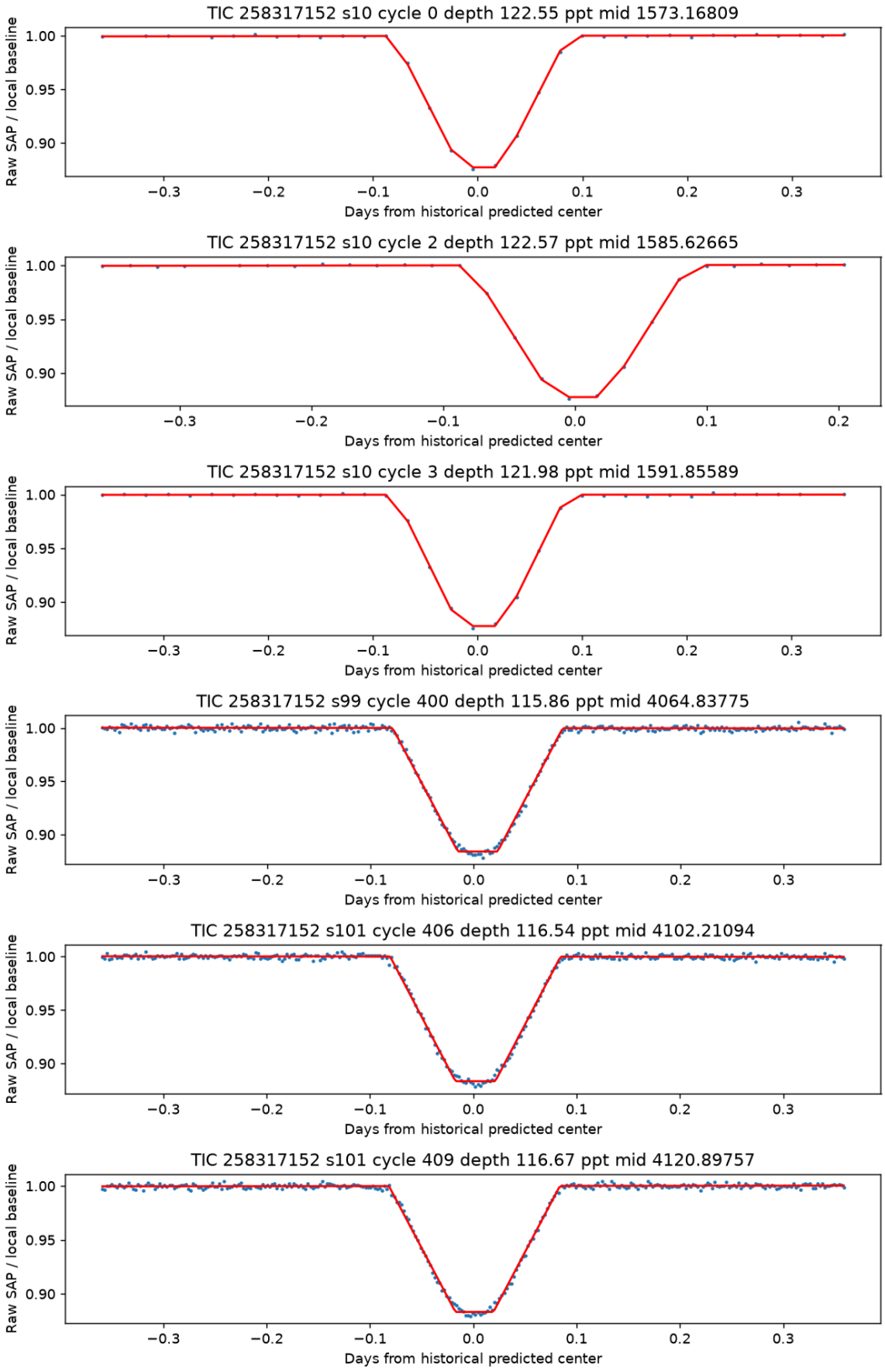
R460. TIC 258317152

physical_verdict: CONFIRMED

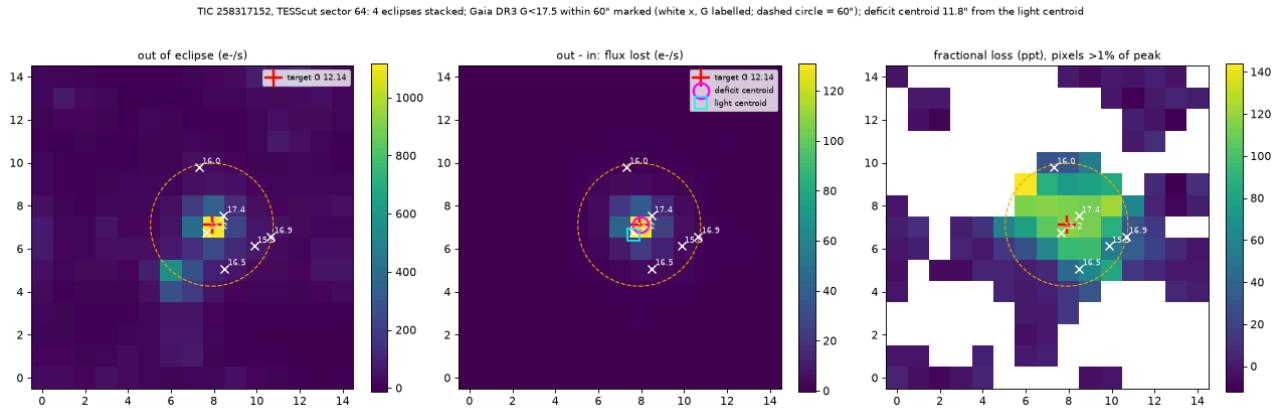
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R460; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R460; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 258317152: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 8 measured same-kind eclipses in representative QLP SAP_FLUX sectors [10, 99, 101]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 6.2291664080 ± 0.0000148911 d and reference BTJD 1573.1699570 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S64 TESScut deficit lies 1.37 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Image-model fractional residual 0.484 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 117.22/108.87/83.39 ppt in 4 stacked events; selected QLP raw depth is 116.70 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 6143372790427445888 has SB1 period 6.2285564049 d, e = 0.07165. The closest edge-on conjunction differs from the measured eclipse epoch by 0.08404 d (0.01349 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.15193 d. Rough central duration 5.14 h versus measured median 3.97 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Eight strong raw eclipses across S10/99/101. The historical S64 deficit peaks on the target, with no capable Gaia neighbor under the approximate flux budget; the image-model residual is large, so its centroid is only descriptive.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
10	0	1573.1680904	0.010417	122.55	used
10	2	1585.6266461	0.010417	122.57	used
10	3	1591.8558908	0.010417	121.98	used
99	398	4052.3797987	0.001157	115.32	used
99	400	4064.8377476	0.001157	115.86	used
101	406	4102.2109375	0.001157	116.54	used
101	407	4108.4399395	0.001157	116.73	used
101	409	4120.8975692	0.001157	116.67	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s64.json, historical/diff_s64.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R460. Public evidence files: diff_s64.json, diff_s64.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

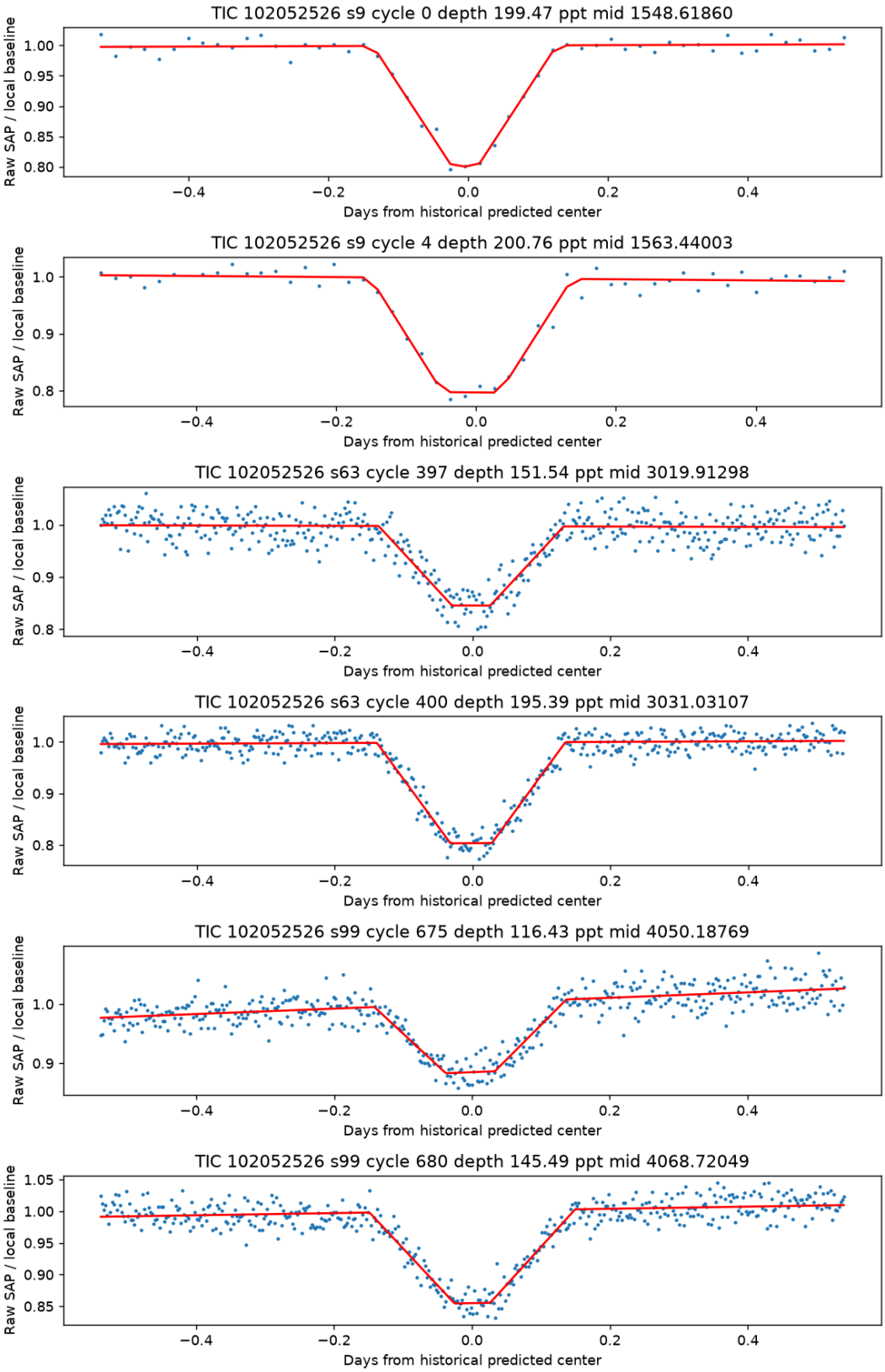
R461. TIC 102052526

physical_verdict: CONFIRMED

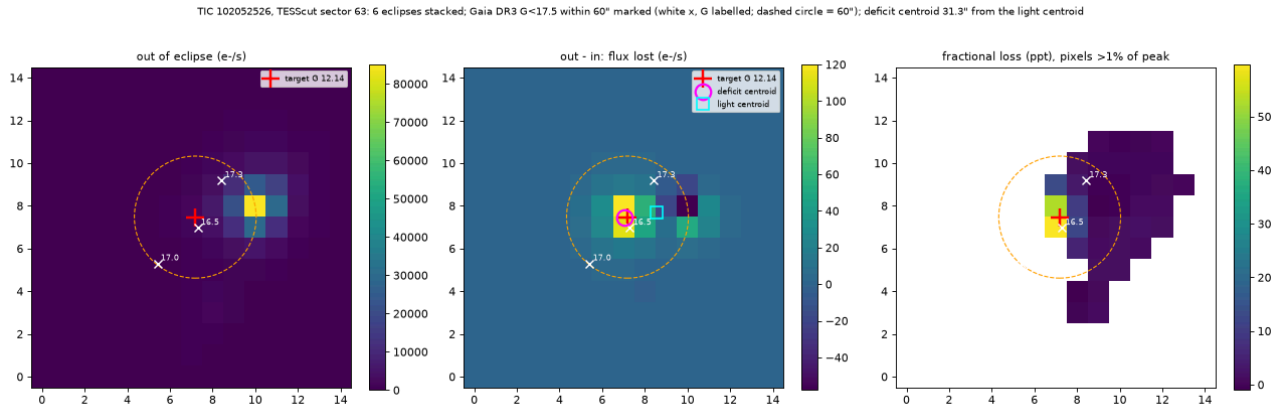
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R461; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R461; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 102052526: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 8 measured same-kind eclipses in representative QLP SAP_FLUX sectors [9, 63, 99]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 7.4120614461 ± 0.0000108580 d and reference BTJD 1548.6197006 (Deeper repeating eclipse; even half-period cycles). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S63 TESScut deficit lies 2.66 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 7]. Image-model fractional residual 0.997 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 59.76/24.22/6.56 ppt in 6 stacked events; selected QLP raw depth is 195.82 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 5416772041726772864 has SB1 period 7.4112548348 d, e = 0.01800. The closest edge-on conjunction differs from the measured eclipse epoch by 0.14870 d (0.02006 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.05727 d. Rough central duration 8.59 h versus measured median 6.72 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Deep and shallow events alternate in S9/63/99. The 3.706-day photometric half-period is not the orbital period: the deeper even parity recurs at 7.412 days. The historical S63 aperture depths are strongly diluted and its image-model residual is high, but the

deficit core is on the target and no capable Gaia neighbor is found.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
9	0	1548.6186004	0.010417	199.47	used
9	3	1559.7334572	0.010417	147.91	excluded from primary timing
9	4	1563.4400284	0.010417	200.76	used
9	5	1567.1466206	0.010417	122.32	excluded from primary timing
63	396	3016.2076632	0.001257	196.25	used
63	397	3019.9129756	0.002086	151.54	excluded from primary timing
63	398	3023.6202250	0.001414	207.16	used
63	399	3027.3144753	0.007049	118.26	excluded from primary timing
63	400	3031.0310660	0.001157	195.39	used
63	401	3034.7348220	0.002138	155.44	excluded from primary timing
63	402	3038.4460476	0.001594	186.63	used
99	675	4050.1876910	0.002411	116.43	excluded from primary timing
99	676	4053.8959706	0.003968	150.78	used
99	679	4065.0094469	0.001528	158.59	excluded from primary timing
99	680	4068.7204943	0.001582	145.49	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s63.json, historical/diff_s63.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R461. Public evidence files: diff_s63.json, diff_s63.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

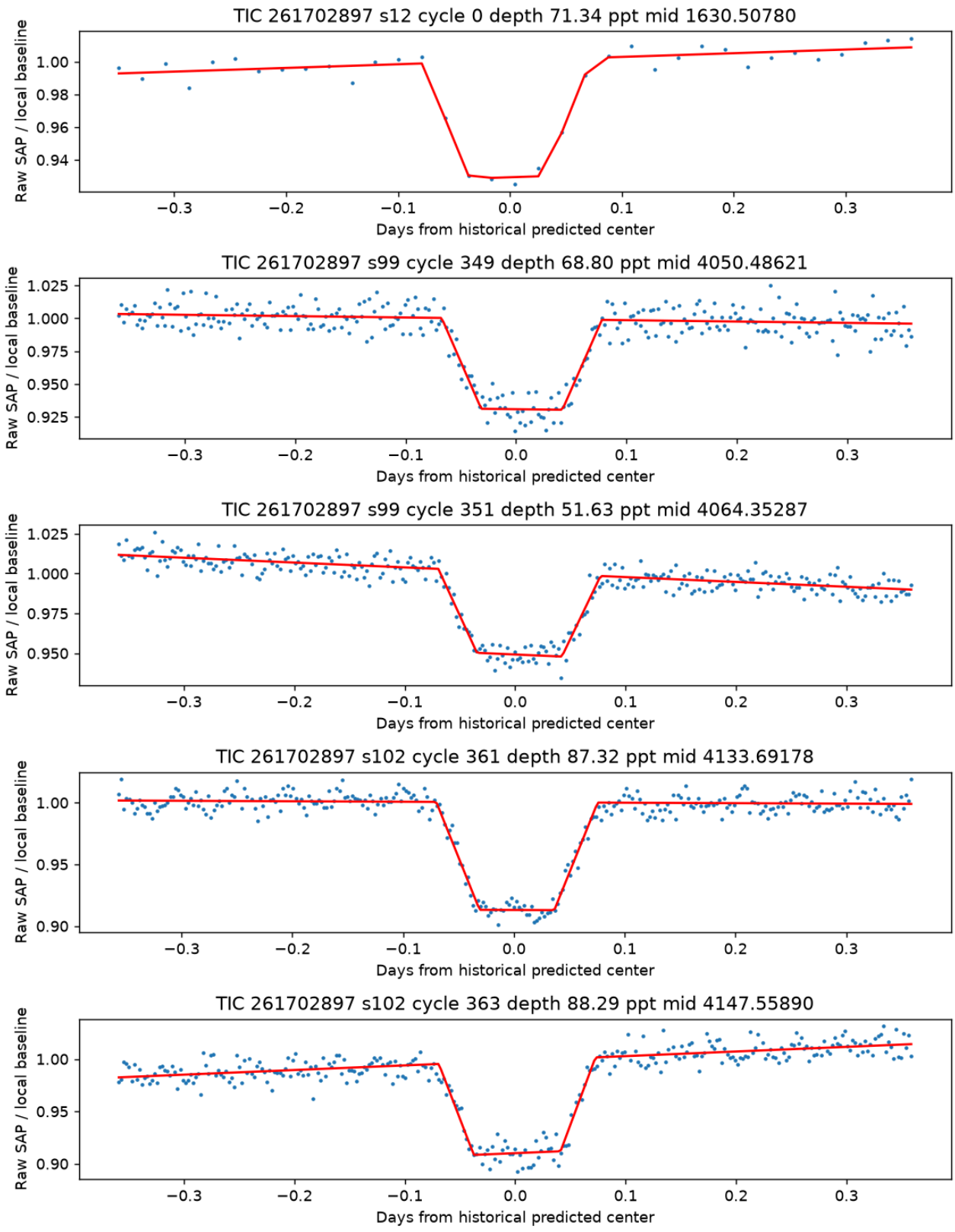
R462. TIC 261702897

physical_verdict: CONFIRMED

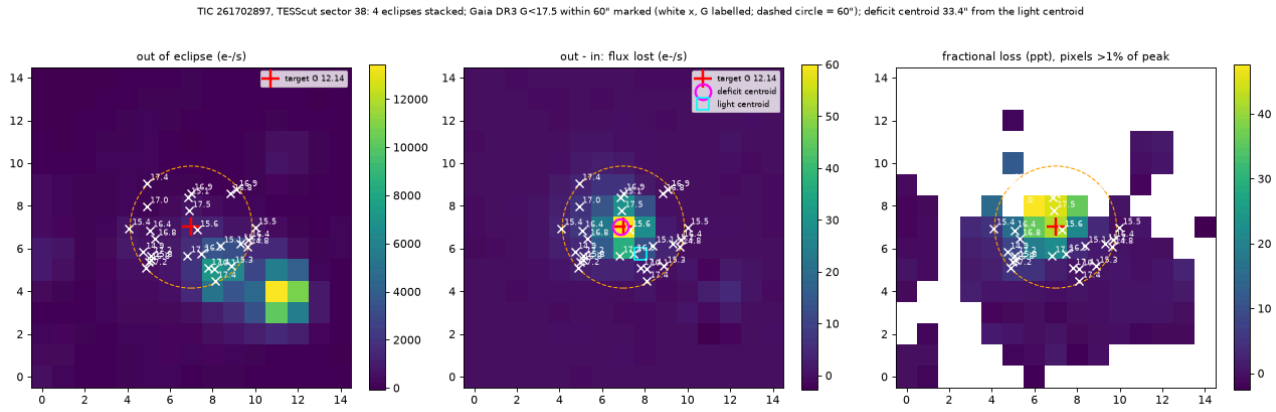
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R462; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R462; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 261702897: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 5 measured same-kind eclipses in representative QLP SAP_FLUX sectors [12, 99, 102]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 6.9340108453 ± 0.0000376792 d and reference BTJD 1630.5145301 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S38 TESScut deficit lies 3.09 arcsec from the target coordinate; peak pixel [7, 7] and target pixel [7, 7]. Image-model fractional residual 0.985 is not a calibrated position error. Capable Gaia neighbors: Gaia 5824727895463607808, 44.9 arcsec, G 10.06, approximate intrinsic requirement 0.09; Gaia 5824727517506486016, 55.3 arcsec, G 14.78, approximate intrinsic requirement 0.95. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 41.85/21.50/8.76 ppt in 4 stacked events; selected QLP raw depth is 71.34 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 5824728342140218880 has SB1 period 6.9344260455 d, e = 0.04238. The closest edge-on conjunction differs from the measured eclipse epoch by -0.06187 d (-0.00892 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.09078 d. Rough central duration 4.51 h versus measured median 3.50 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Five raw eclipses across S12/99/102. Two capable Gaia neighbors at about 45 and 55 arcsec require separate evaluation; the S38 deficit peak is at the target pixel, distinct from the bright southeast neighbor. The poor image-model fit limits centroid precision.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	0	1630.5077983	0.010417	71.34	used
99	349	4050.4862110	0.001157	68.80	used
99	351	4064.3528726	0.001319	51.63	used
102	361	4133.6917844	0.001157	87.32	used
102	363	4147.5589027	0.001157	88.29	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s38.json, historical/diff_s38.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R462. Public evidence files: diff_s38.json, diff_s38.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

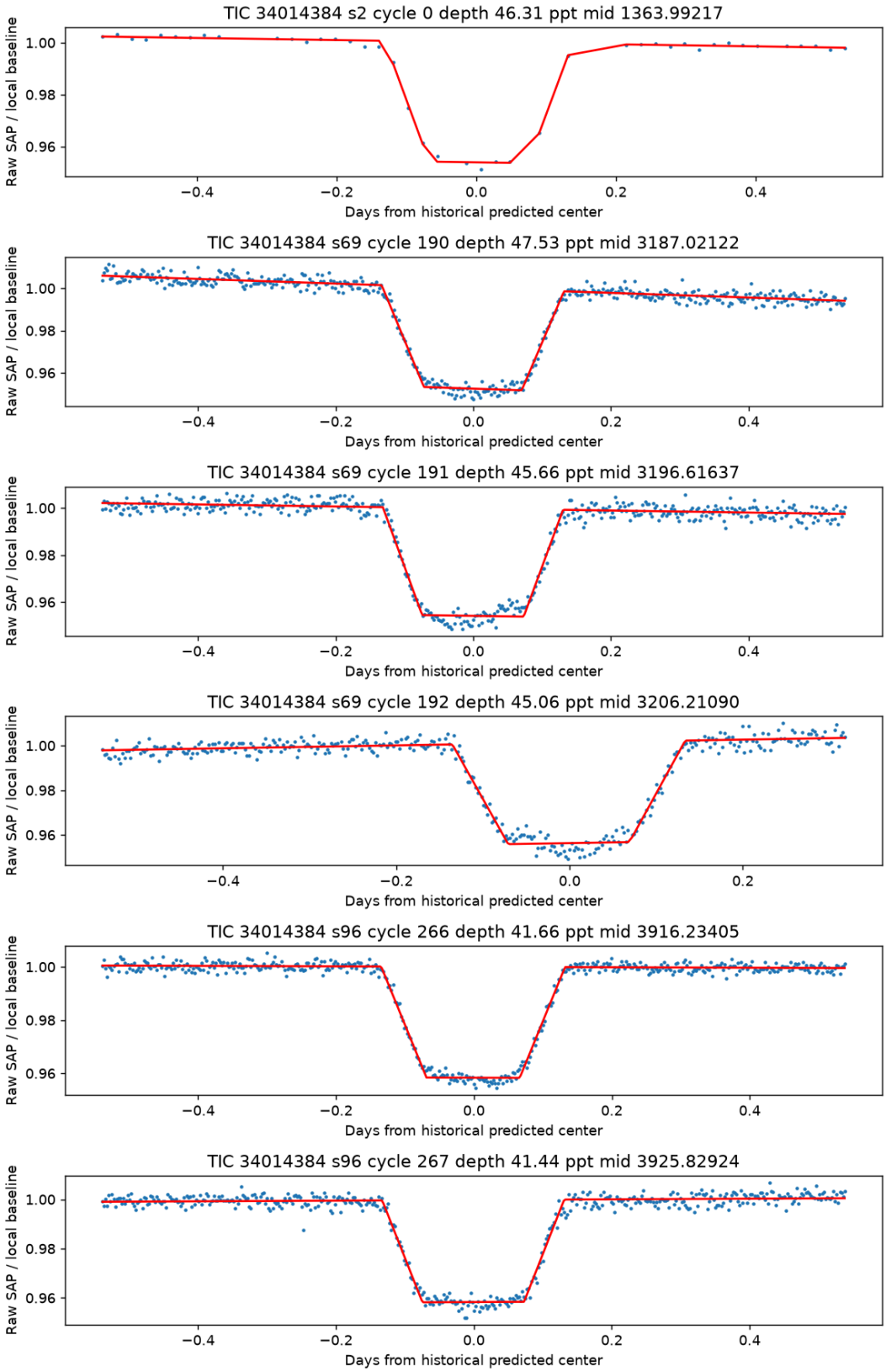
R463. TIC 34014384

physical_verdict: CONFIRMED

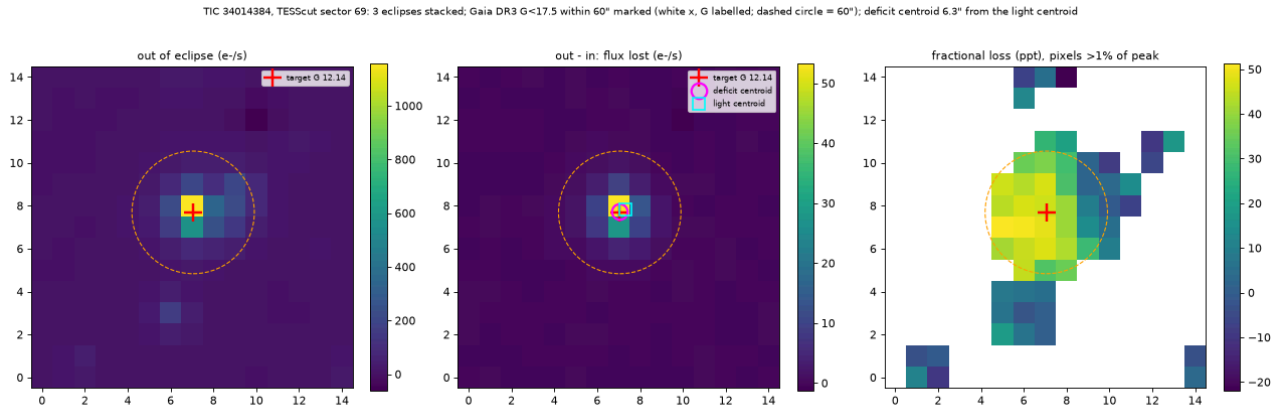
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R463; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R463; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 34014384: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 6 measured same-kind eclipses in representative QLP SAP_FLUX sectors [2, 69, 96]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 9.5949058113 ± 0.0000134136 d and reference BTJD 1363.9891996 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S69 TESScut deficit lies 0.80 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Image-model fractional residual 0.224 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 46.04/45.25/39.29 ppt in 3 stacked events; selected QLP raw depth is 45.36 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 2328077646884230912 has SB1 period 9.5946733453 d, e = 0.07996. The closest edge-on conjunction differs from the measured eclipse epoch by 0.19039 d (0.01984 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.52291 d. Rough central duration 7.88 h versus measured median 6.38 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Six raw eclipses across S2/69/96, including three consecutive S69 cycles. The S69 deficit peaks at the target and the aperture reductions retain the eclipse.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
2	0	1363.9921725	0.010417	46.31	used
69	190	3187.0212237	0.001157	47.53	used
69	191	3196.6163746	0.001157	45.66	used
69	192	3206.2109008	0.001157	45.06	used
96	266	3916.2340458	0.001157	41.66	used
96	267	3925.8292435	0.001157	41.44	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s69.json, historical/diff_s69.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R463. Public evidence files: diff_s69.json, diff_s69.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

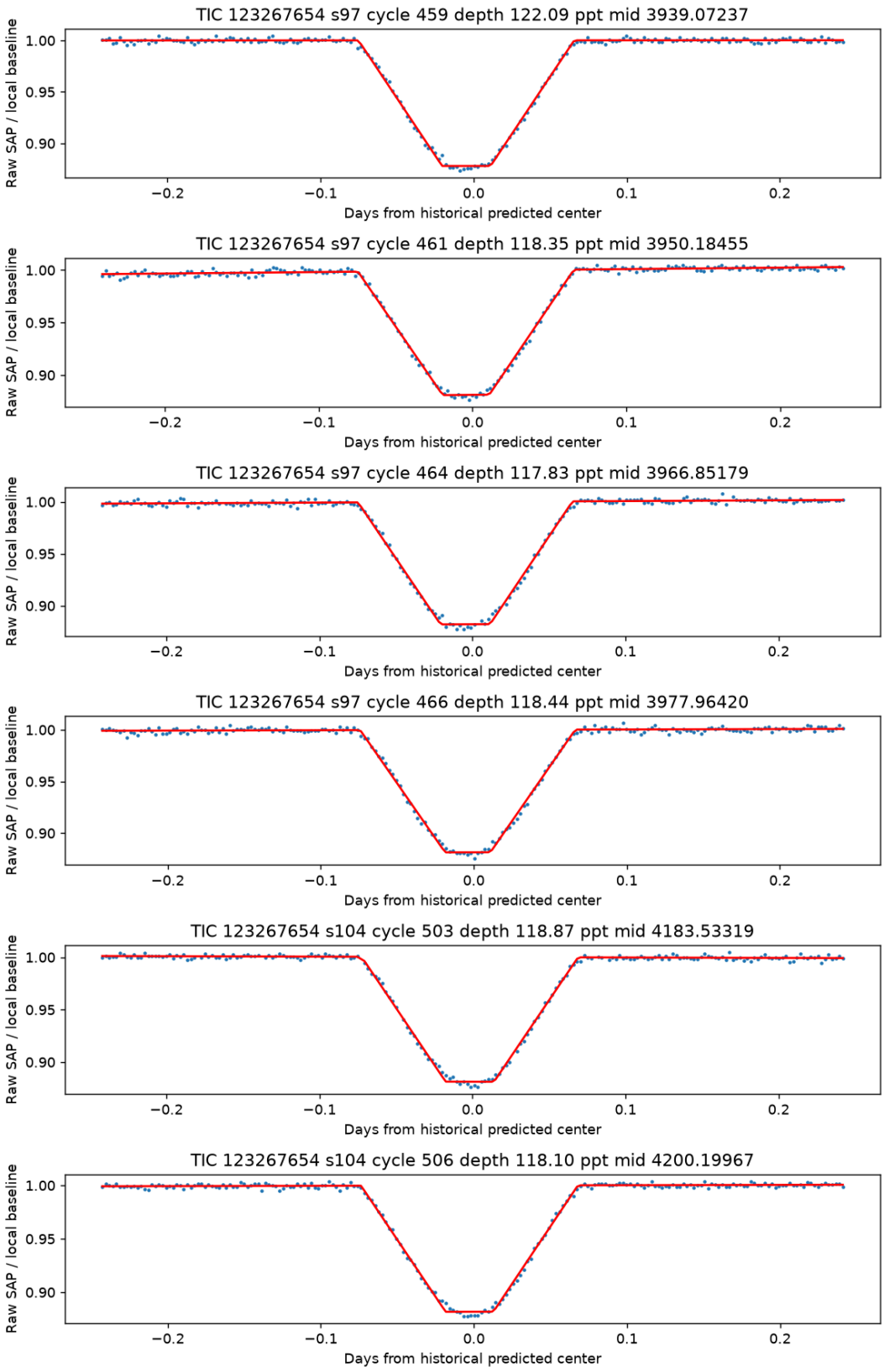
R464. TIC 123267654

physical_verdict: CONFIRMED

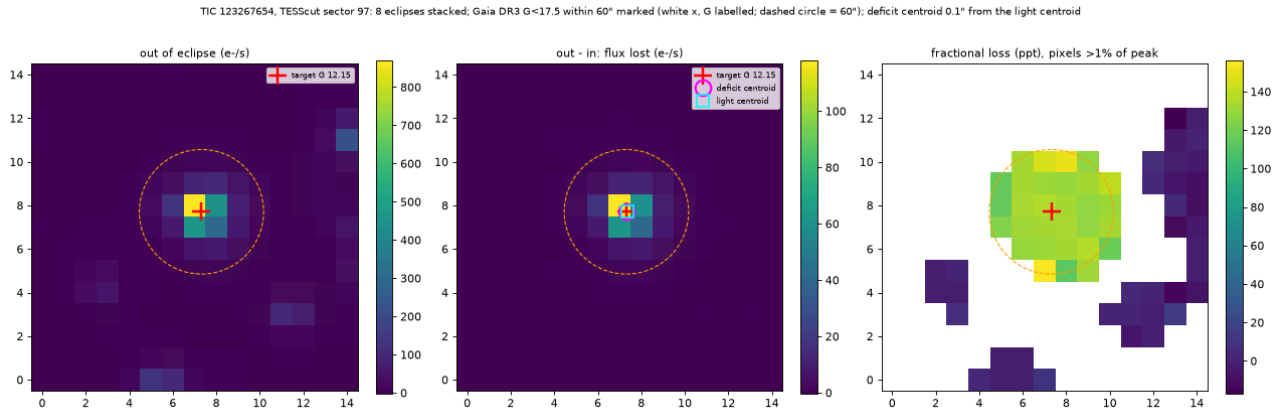
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R464; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R464; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 123267654: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 12 measured same-kind eclipses in representative QLP SAP_FLUX sectors [97, 104]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 5.5559075318 ± 0.0000169590 d and reference BTJD 3939.0725724 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S97 TESScut deficit lies 0.46 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Image-model fractional residual 0.060 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 135.68/133.72/133.25 ppt in 8 stacked events; selected QLP raw depth is 118.29 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 5018762885599910016 has SB1 period 5.5561122262 d, e = 0.07698. The closest edge-on conjunction differs from the measured eclipse epoch by -0.17143 d (-0.03085 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.25667 d. Rough central duration 4.65 h versus measured median 3.38 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Twelve raw eclipses across S97/104, with consecutive cycles and consistent depths. The S97 cutout has a clean target-centered deficit. The selected earlier S3 QLP file supplied no usable midpoint; it is not treated as a null eclipse.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
97	459	3939.0723654	0.001157	122.09	used
97	460	3944.6289202	0.001157	120.31	used
97	461	3950.1845485	0.001157	118.35	used
97	462	3955.7403975	0.001157	118.23	used
97	464	3966.8517920	0.001157	117.83	used
97	465	3972.4078429	0.001157	114.57	used
97	466	3977.9642012	0.001157	118.44	used
97	467	3983.5194413	0.001157	116.32	used
104	503	4183.5331930	0.001157	118.87	used
104	504	4189.0888142	0.001157	118.09	used
104	505	4194.6438920	0.001157	118.65	used
104	506	4200.1996719	0.001157	118.10	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s97.json, historical/diff_s97.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R464. Public evidence files: diff_s97.json, diff_s97.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

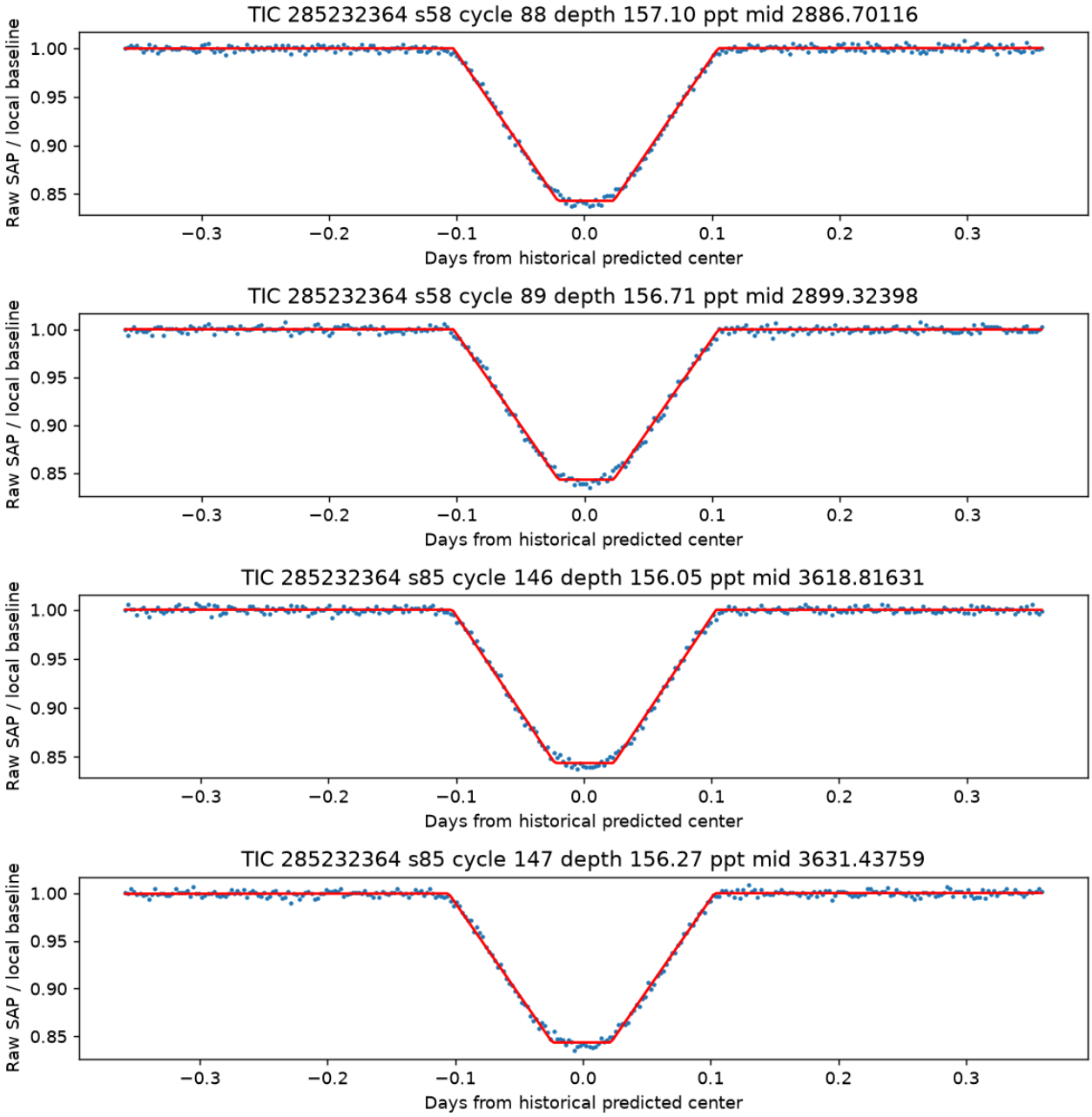
R465. TIC 285232364

physical_verdict: CONFIRMED

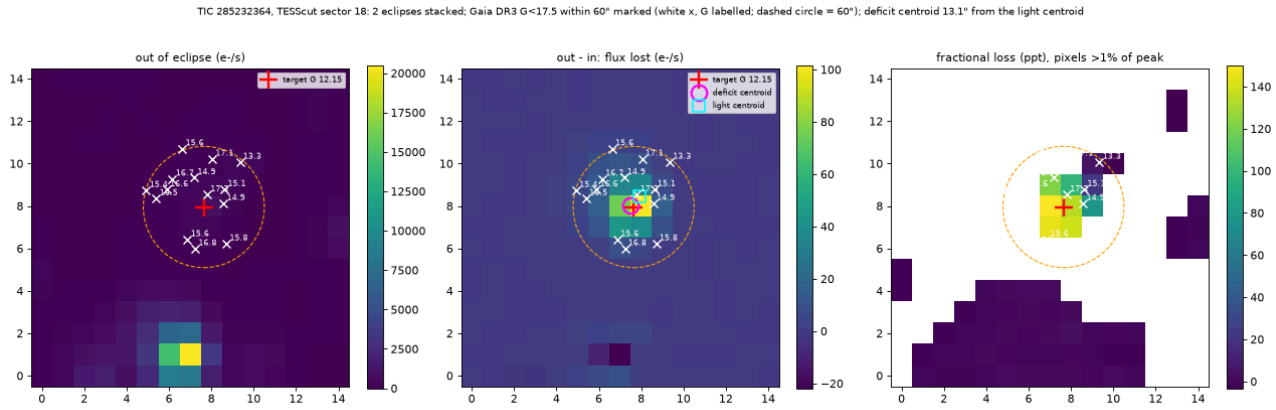
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R465; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R465; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 285232364: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 4 measured same-kind eclipses in representative QLP SAP_FLUX sectors [58, 85]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 12.6226616088 ± 0.0000199522 d and reference BTJD 2886.7012424 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S18 TESScut deficit lies 4.06 arcsec from the target coordinate; peak pixel [8, 8] and target pixel [8, 8]. Image-model fractional residual 0.758 is not a calibrated position error. Capable Gaia neighbors: Gaia 427362616196340608, 56.1 arcsec, G 13.35, approximate intrinsic requirement 0.61. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 134.81/117.84/92.18 ppt in 2 stacked events; selected QLP raw depth is 156.49 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 427362714972324736 has SB1 period 12.6184477876 d, e = 0.26083. The closest edge-on conjunction differs from the measured eclipse epoch by 0.63322 d (0.05018 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.33878 d. Rough central duration 6.67 h versus measured median 4.98 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Four broad raw eclipses in S58/85, two adjacent cycles per sector. A capable neighbor about 56 arcsec away is spatially distinct from the S18 target-centered deficit. The image fit is imperfect and aperture depths are diluted.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
58	88	2886.7011554	0.001157	157.10	used
58	89	2899.3239805	0.001157	156.71	used
85	146	3618.8163095	0.001157	156.05	used
85	147	3631.4375941	0.001157	156.27	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s18.json, historical/diff_s18.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R465. Public evidence files: diff_s18.json, diff_s18.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

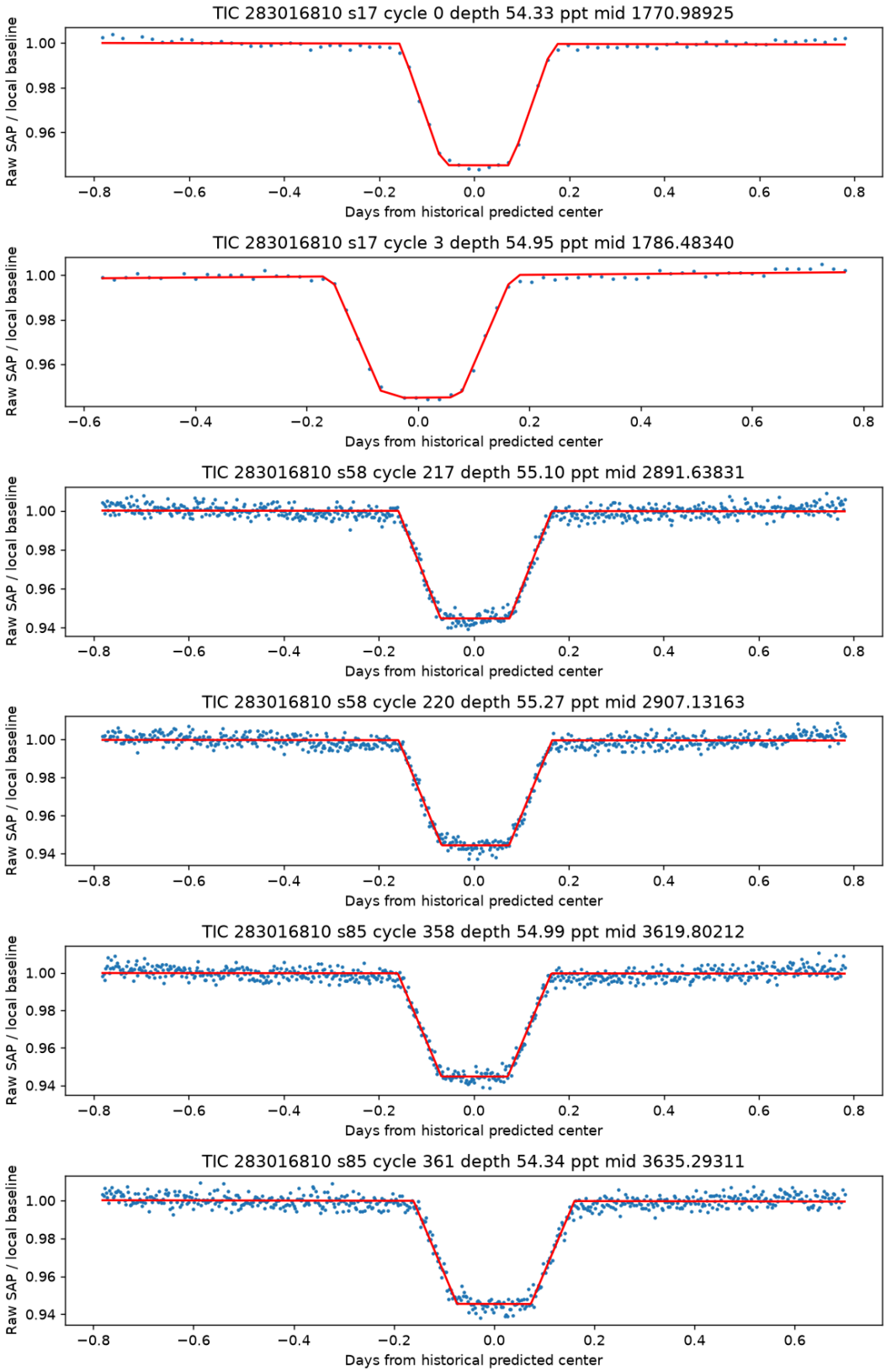
R466. TIC 283016810

physical_verdict: PROBABLE

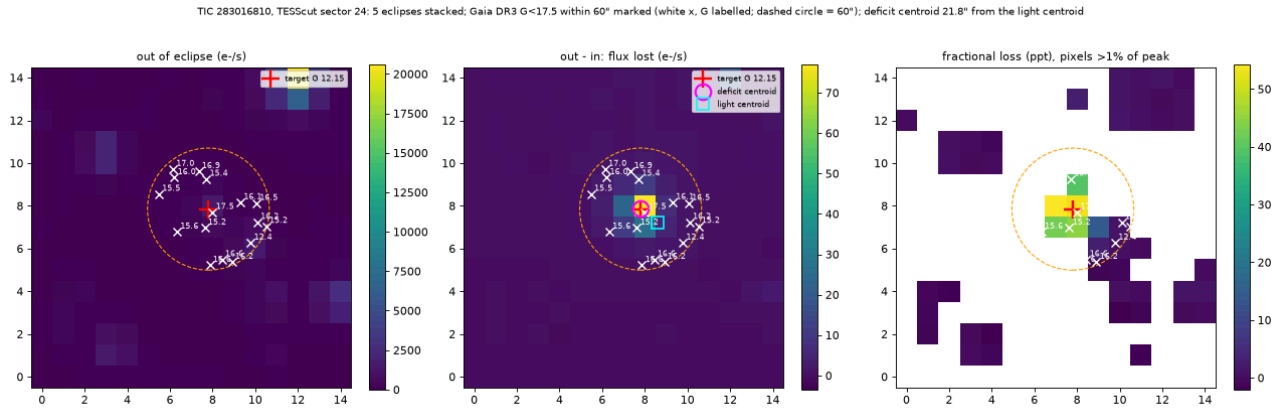
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization



R466; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R466; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 283016810: PROBABLE

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 13 measured same-kind eclipses in representative QLP SAP_FLUX sectors [17, 58, 85]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 5.1642748232 ± 0.0000050389 d and reference BTJD 1770.9909383 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S24 TESScut deficit lies 0.94 arcsec from the target coordinate; peak pixel [8, 8] and target pixel [8, 8]. Image-model fractional residual 0.808 is not a calibrated position error. Capable Gaia neighbors: Gaia 431751179422562816, 19.0 arcsec, G 15.24, approximate intrinsic requirement 0.95; Gaia 431751351221252480, 53.5 arcsec, G 12.36, approximate intrinsic requirement 0.12; Gaia 431751351221249280, 59.7 arcsec, G 15.23, approximate intrinsic requirement 0.95. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 53.08/45.84/26.48 ppt in 5 stacked events; selected QLP raw depth is 54.70 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 431751175113711872 has SB1 period 5.1640736469 d, e = 0.06043. The closest edge-on conjunction differs from the measured eclipse epoch by -0.01453 d (-0.00281 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.09489 d. Rough central duration 10.99 h versus measured median 7.72 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Thirteen raw eclipses across S17/58/85. Three capable Gaia neighbors include one about 19 arcsec away. The S24 deficit peak overlaps the target pixel, but TESS cannot resolve the nearby capable source within roughly one pixel. The image-model residual further limits attribution.

Next check: Resolved eclipse photometry or multi-source PRF localization of the capable nearby star.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
17	0	1770.9892475	0.010417	54.33	used
17	2	1781.3168663	0.010417	54.90	used
17	3	1786.4833952	0.010417	54.95	used
58	216	2886.4738673	0.001157	54.70	used
58	217	2891.6383068	0.001157	55.10	used
58	218	2896.8035472	0.001157	54.15	used
58	219	2901.9670358	0.001157	54.24	used
58	220	2907.1316295	0.001157	55.27	used
85	357	3614.6371395	0.001157	54.53	used
85	358	3619.8021171	0.001157	54.99	used
85	359	3624.9653593	0.001157	55.97	used
85	360	3630.1301941	0.001157	54.64	used
85	361	3635.2931091	0.001157	54.34	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s24.json, historical/diff_s24.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R466. Public evidence files: diff_s24.json, diff_s24.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

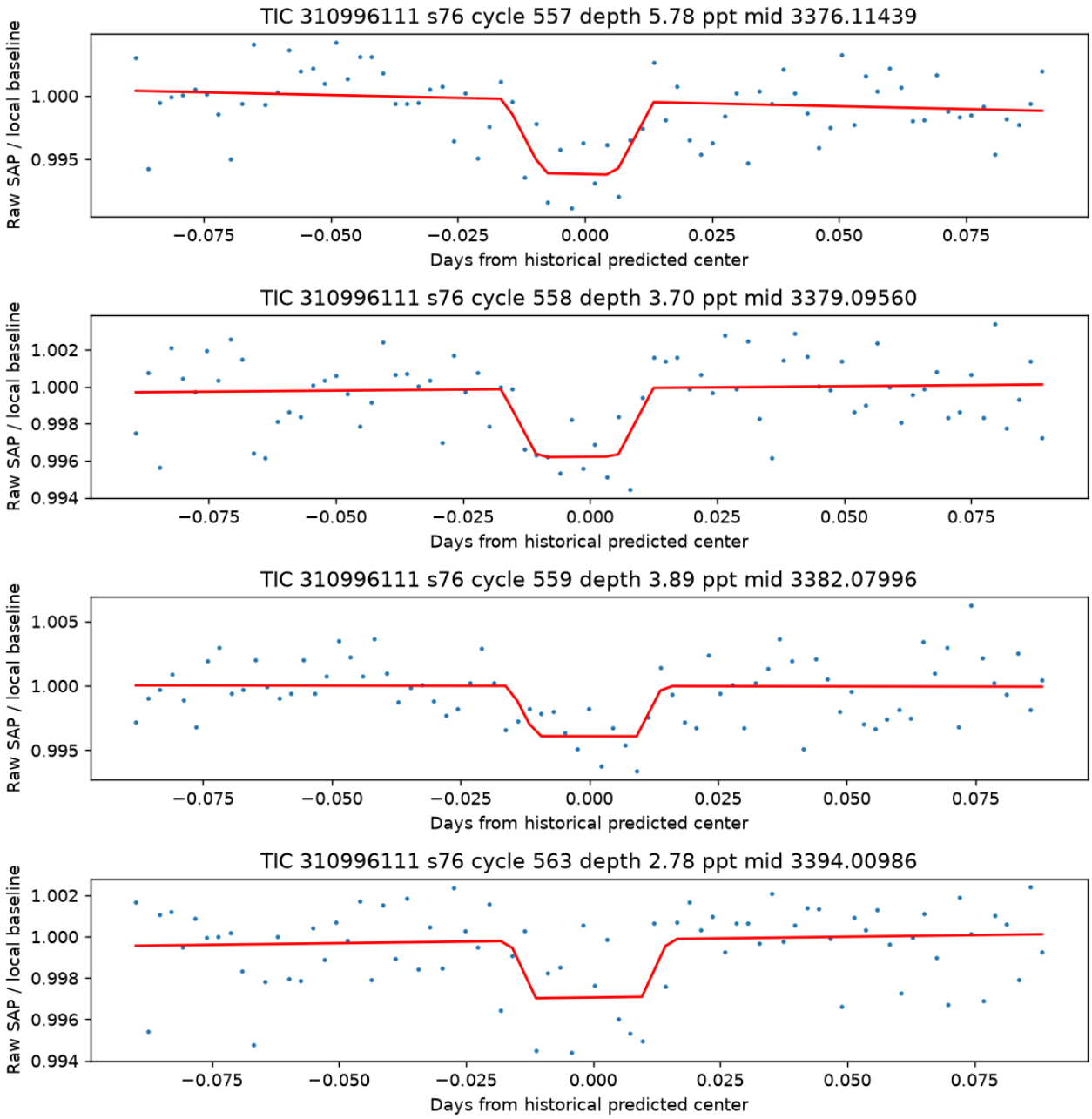
R467. TIC 310996111

physical_verdict: CONFIRMED

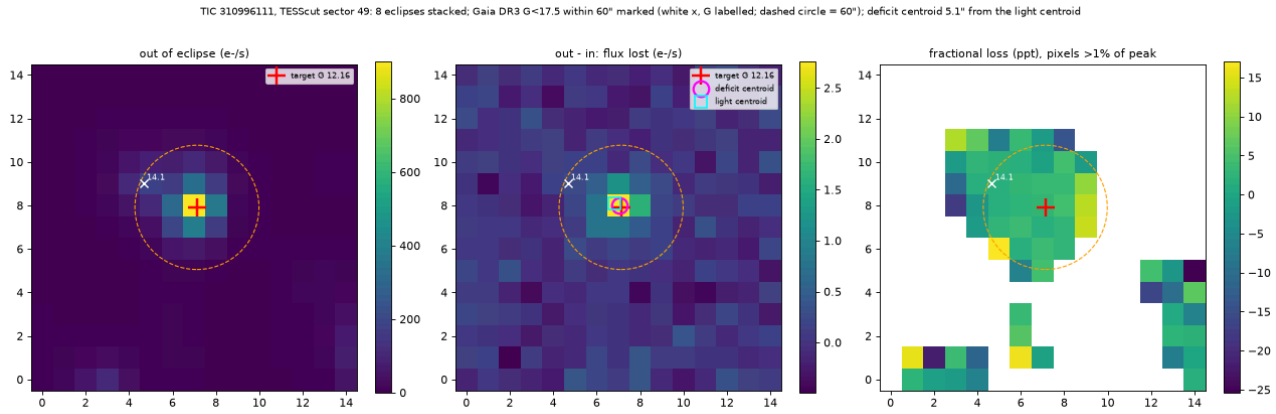
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R467; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R467; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 310996111: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 4 measured same-kind eclipses in representative QLP SAP_FLUX sectors [76]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 2.9826325236 ± 0.0003163772 d and reference BTJD 3376.1140790 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S49 TESScut deficit lies 2.75 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [7, 8]. Image-model fractional residual 0.424 is not a calibrated position error. Capable Gaia neighbors: Gaia 1559248294394682112, 52.0 arcsec, G 18.35, approximate intrinsic requirement 0.84; Gaia 1559154737124070528, 55.3 arcsec, G 14.06, approximate intrinsic requirement 0.02. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 3.06/3.09/3.02 ppt in 8 stacked events; selected QLP raw depth is 3.80 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 1559154664107634560 has SB1 period 2.9825457445 d, e = 0.07326. The closest edge-on conjunction differs from the measured eclipse epoch by -0.07548 d (-0.02531 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.11227 d. Rough central duration 2.56 h versus measured median 0.72 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Four shallow S76 raw events recur at the predicted phase, and the historical S49 aperture extraction retains a roughly 3 ppt target-pixel loss. S15/23 are 30-minute cadence: each predicted short event has at most one central sample, so their apparent weak or absent dips are not clean nulls. Capable neighbors about 52 and 55 arcsec away are distinct from the target-centered S49 deficit; the morphology and attribution remain limited by shallow depth.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
76	557	3376.1143874	0.001284	5.78	used
76	558	3379.0956038	0.001653	3.70	used
76	559	3382.0799617	0.001651	3.89	used
76	563	3394.0098566	0.001488	2.78	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s49.json, historical/diff_s49.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R467. Public evidence files: diff_s49.json, diff_s49.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

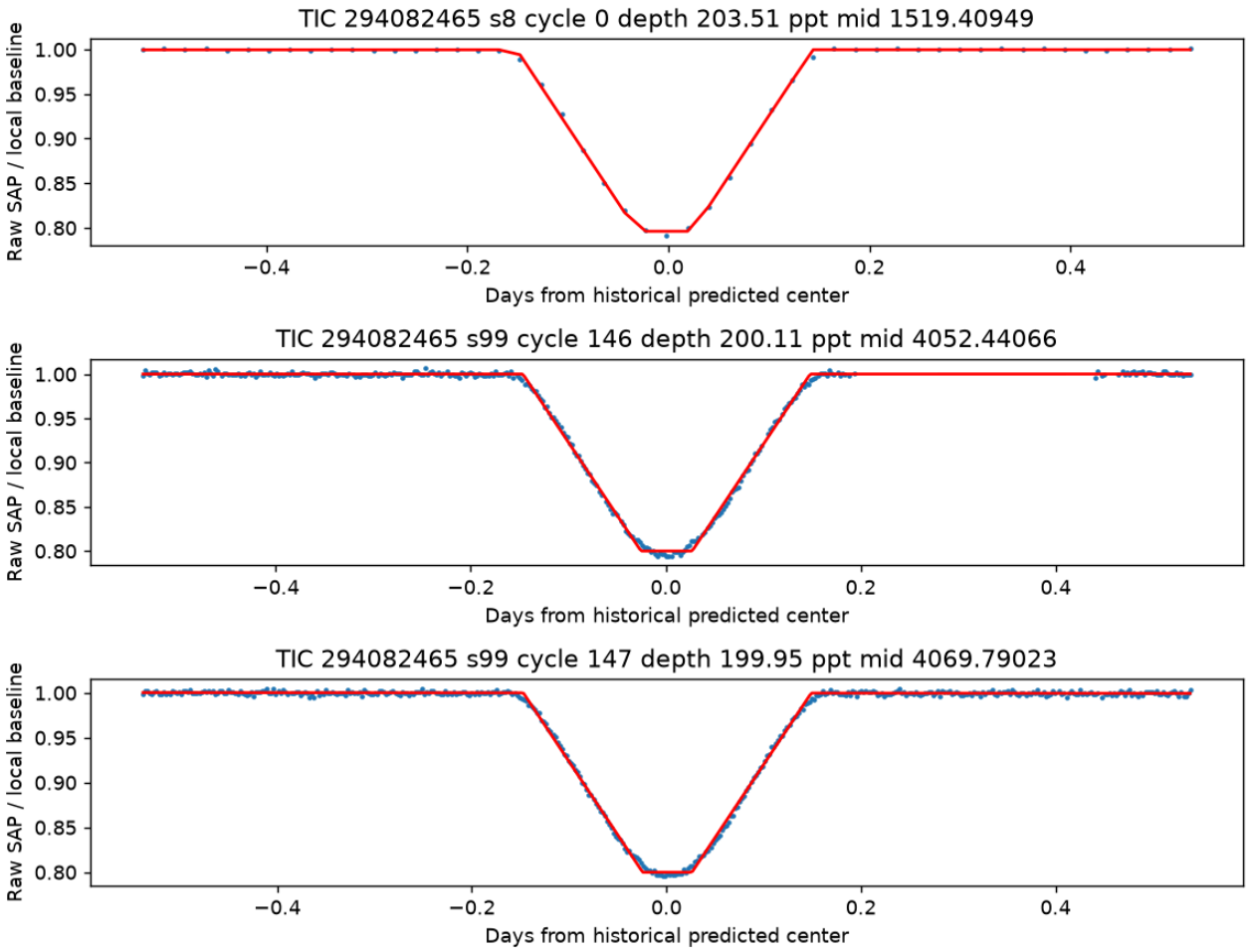
R468. TIC 294082465

physical_verdict: CONFIRMED

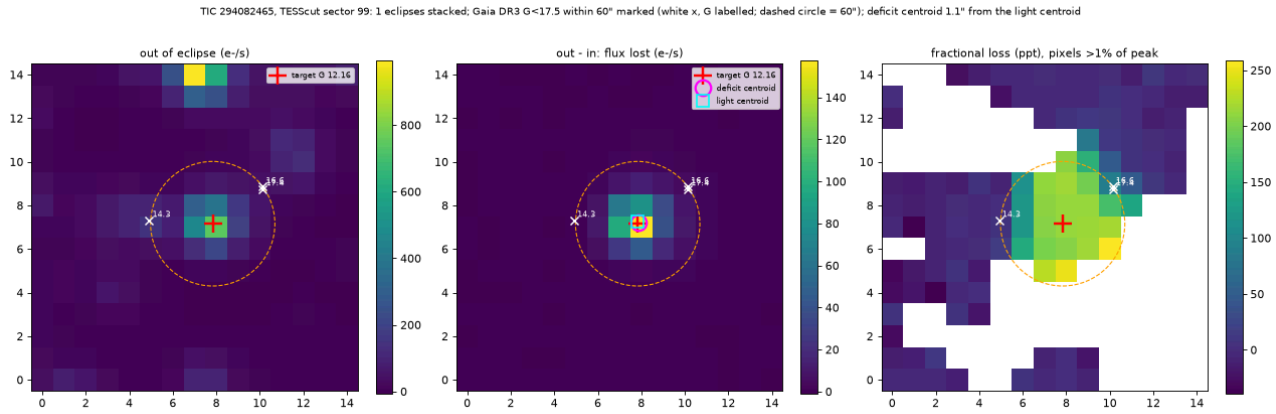
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R468; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R468; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 294082465: CONFIRMED

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 3 measured same-kind eclipses in representative QLP SAP_FLUX sectors [8, 99]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 17.3495287874 ± 0.0000712547 d and reference BTJD 1519.4094810 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S99 TESScut deficit lies 1.70 arcsec from the target coordinate; peak pixel [8, 7] and target pixel [8, 7]. Image-model fractional residual 0.296 is not a calibrated position error. Capable Gaia neighbors: None within 60 arcsec under the approximate G-band flux budget. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 212.51/210.43/203.81 ppt in 1 stacked events; selected QLP raw depth is 200.11 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 5705460089589085952 has SB1 period 17.3434002638 d, e = 0.43675. The closest edge-on conjunction differs from the measured eclipse epoch by 0.50447 d (0.02909 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.51559 d. Rough central duration 9.27 h versus measured median 7.07 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Three broad deep raw events in S8/99, including consecutive S99 cycles. The S99 target-centered deficit is supported by all three apertures, though that historical image stack contains one event only.

Next check: Optional resolved photometry or spectroscopy for precision binary parameters; no bounded admission check remains missing.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
8	0	1519.4094921	0.010417	203.51	used
99	146	4052.4406638	0.001157	200.11	used
99	147	4069.7902328	0.001157	199.95	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s99.json, historical/diff_s99.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R468. Public evidence files: diff_s99.json, diff_s99.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

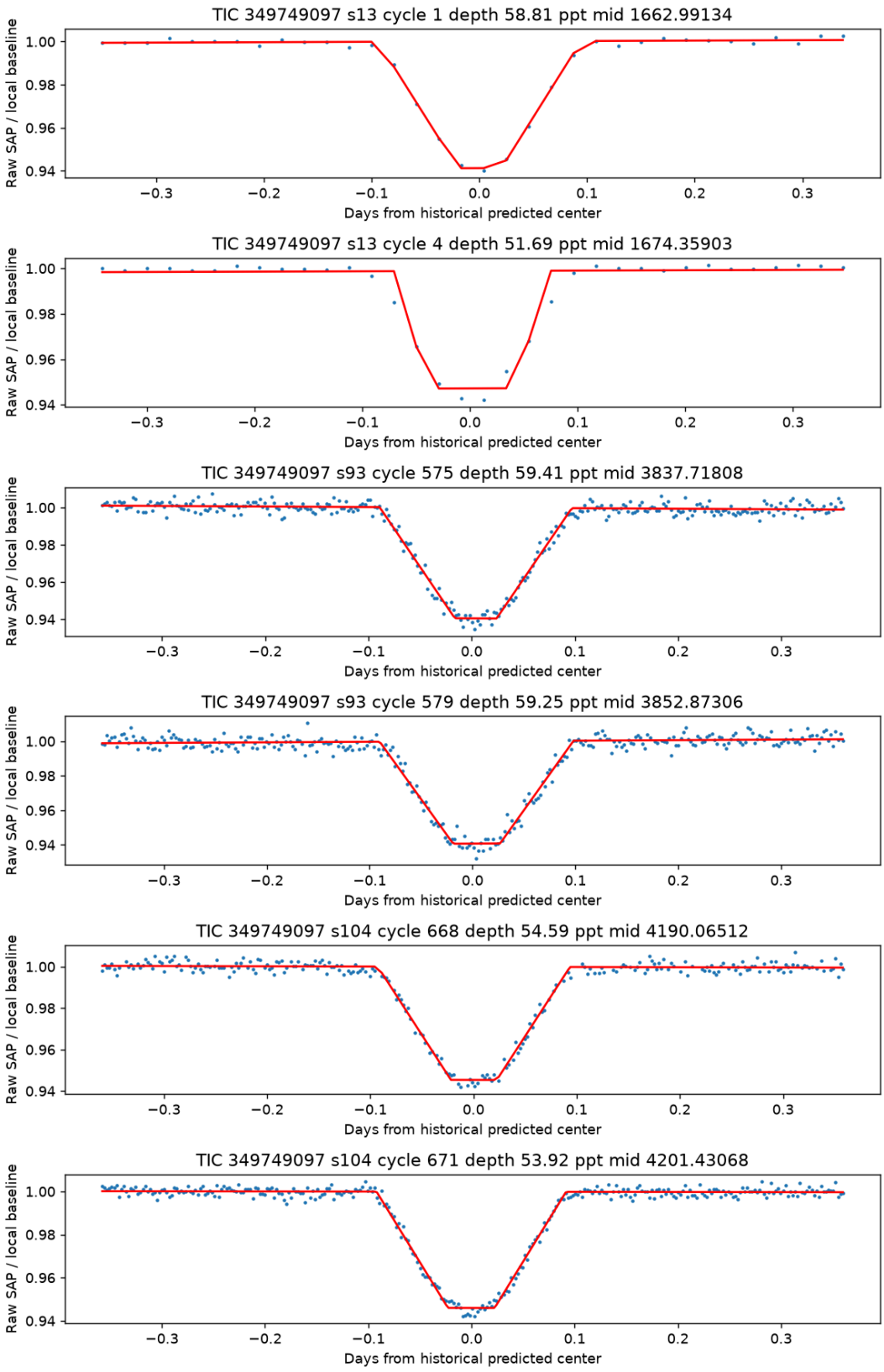
R469. TIC 349749097

physical_verdict: PROBABLE

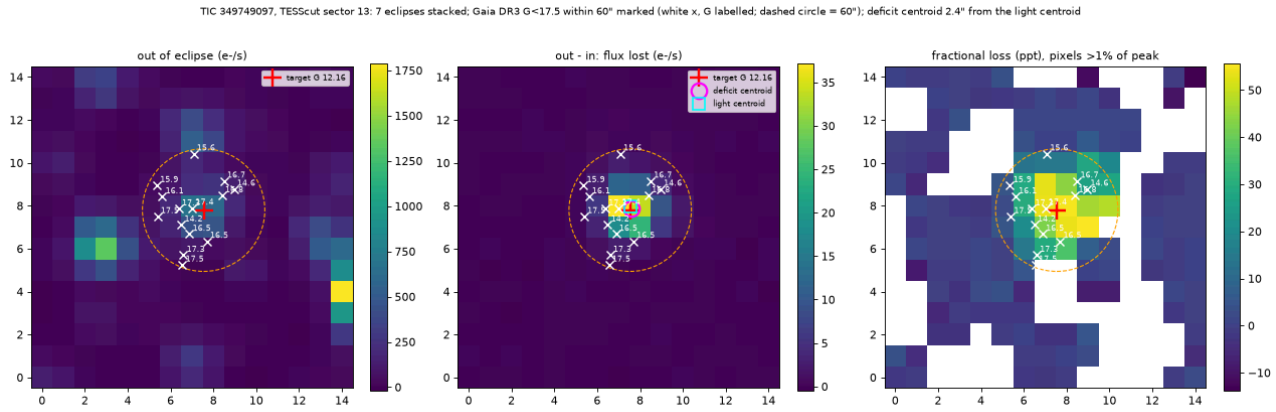
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization



R469; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R469; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 349749097: PROBABLE

Full bounded review: **COMPLETE**. This physical outcome is separate from catalogue novelty and from the historical screening label.

1. **Raw repeat:** 16 measured same-kind eclipses in representative QLP SAP_FLUX sectors [13, 93, 104]; raw flux, quality filtering, fitted windows and plots are retained. Fitted periodic depressions alone would be insufficient: the inspected windows have recurring eclipse morphology. Predicted centers were search guides, never timing points. Flagged or unresolved windows are not clean nulls.
2. **Timing and aliases:** measured midpoints give P = 3.7886980284 ± 0.0000059917 d and reference BTJD 1663.0043255 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind within-sector counts and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than a blind exhaustive search.
3. **Attribution:** historical S13 TESScut deficit lies 2.47 arcsec from the target coordinate; peak pixel [7, 8] and target pixel [8, 8]. Image-model fractional residual 0.655 is not a calibrated position error. Capable Gaia neighbors: Gaia 6702211498076393216, 24.4 arcsec, G 14.21, approximate intrinsic requirement 0.41; Gaia 6702211601155599616, 33.4 arcsec, G 14.55, approximate intrinsic requirement 0.54. The intrinsic neighbor requirement is not the same as the blended pixel loss.
4. **Alternative reduction:** independent target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture extraction gives 54.70/49.79/41.74 ppt in 7 stacked events; selected QLP raw depth is 55.44 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution, event-kind mixing and temporal changes limit a direct depth comparison.
5. **Gaia geometry:** matched DR3 source 6702211493775030784 has SB1 period 3.7889568790 d, e = 0.03612. The closest edge-on conjunction differs from the measured eclipse epoch by -0.06788 d (-0.01791 phase); the linear propagated timing scale from source-specific full SB1 covariance is 0.05803 d. Rough central duration 6.91 h versus measured median 4.46 h uses TIC radius and a minimum-mass main-sequence companion; it is not a physical fit or a sigma veto.

Deciding interpretation and limitations: Sixteen raw eclipses across S13/93/104. Two capable Gaia neighbors about 24 and 33 arcsec away lie in a crowded field. The S13 deficit core overlaps the target pixel, but the high image-model residual and roughly one-pixel separation leave source identity unresolved.

Next check: Resolved eclipse photometry or multi-source PRF localization of the capable nearby star.

Catalogue novelty: the central seven-group check in ../../shared-catalogues/coverage-matrix.csv reports complete scoped coverage and no identified prior eclipsing-binary match for this exact target. This is not exhaustive priority and does not resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
13	1	1662.9913433	0.010416	58.81	used
13	2	1666.7774115	0.010416	53.37	used
13	3	1670.5687257	0.010416	53.30	used
13	4	1674.3590344	0.010416	51.69	used
13	5	1678.1464556	0.010416	58.45	used
93	574	3833.9294581	0.001157	59.72	used
93	575	3837.7180765	0.001157	59.41	used
93	577	3845.2955559	0.001157	58.11	used
93	578	3849.0842975	0.001157	58.79	used
93	579	3852.8730619	0.001157	59.25	used
104	666	4182.4872216	0.001157	55.37	used
104	667	4186.2759269	0.001157	55.51	used
104	668	4190.0651155	0.001157	54.59	used
104	669	4193.8542070	0.001157	54.46	used
104	670	4197.6433183	0.001157	55.06	used
104	671	4201.4306811	0.001157	53.92	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s13.json, historical/diff_s13.png and historical/gaia.json. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R469. Public evidence files: diff_s13.json, diff_s13.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

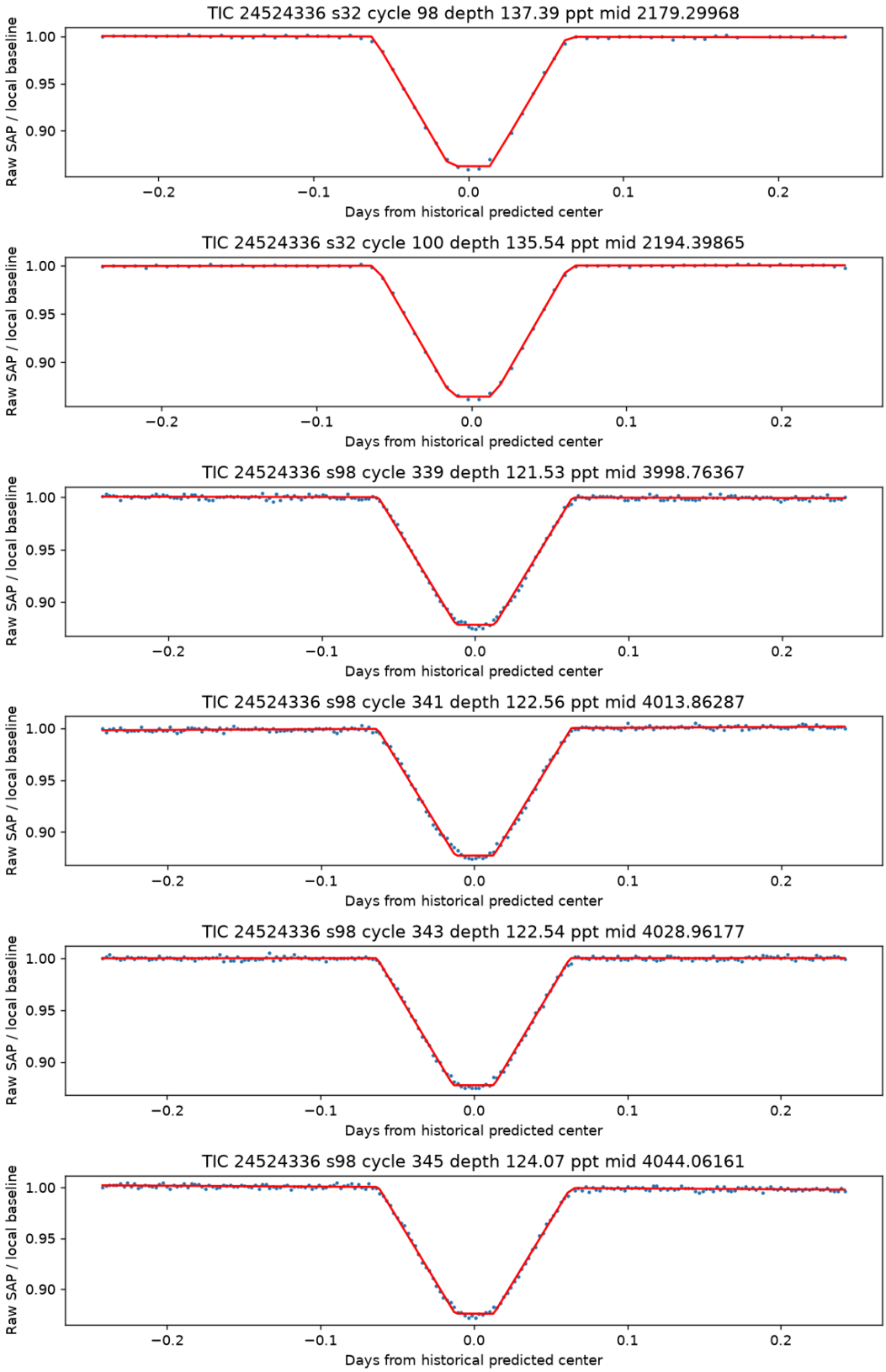
R470. TIC 24524336

physical_verdict: CONFIRMED

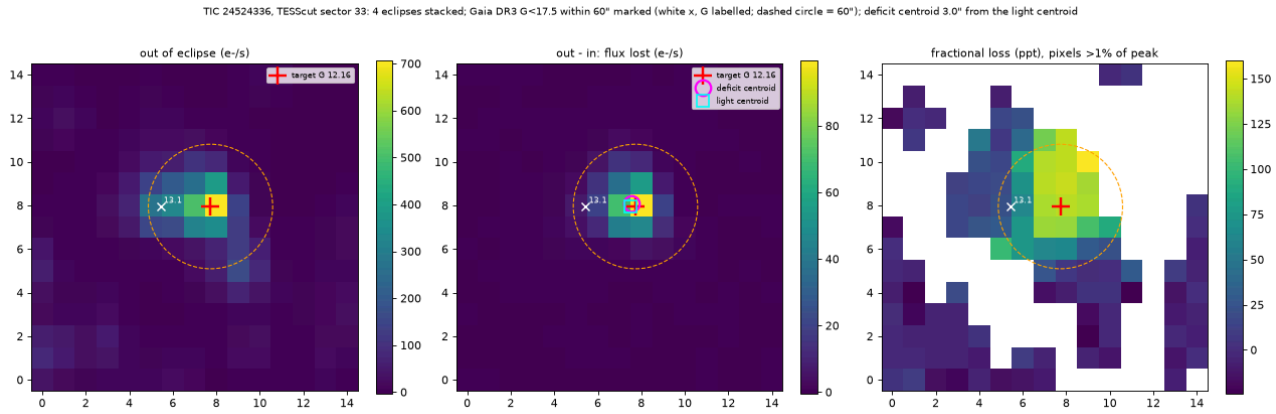
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R470; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R470; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 24524336: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.159077. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [32, 98] contains 6 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 7.5496432496 ± 0.0000103723 d and epoch BTJD 2179.2995260, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: []. Flagged intervening cycles are not negative detections; cross-sector parity and within-sector spacing discriminate the alias. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 33 TESScut difference image has deficit peak [8, 8] and target pixel [8, 8]; deficit centroid is 4.73 arcsec from the target coordinate. Image-model fractional residual 0.349 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 2901581709342471552 (44.6 arcsec, G 13.129, intrinsic loss needed ~0.45). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 141.0/137.2/117.4 ppt from 4 stacked events, versus median selected raw QLP depth 123.3 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 2901581915500902912 has P = 7.5496902503 d and e = 0.03847; the closer edge-on conjunction differs from the measured epoch by 0.04453 d (0.00590 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.12718 d,

descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 4.40 h against 3.01 h observed median; it is not a complete binary model.

Decision and limitation. S32 cycles 98/100 and S98 cycles 339/341/343/345 have opposite half-period parity. Intervening windows are quality-flagged, not clean nulls. The within-sector 15.099-day spacing together with cross-sector parity excludes a constant doubled period. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
32	98	2179.2996811	0.003472	137.4	yes
32	100	2194.3986515	0.003472	135.5	yes
98	339	3998.7636658	0.001157	121.5	yes
98	341	4013.8628696	0.001157	122.6	yes
98	343	4028.9617662	0.001157	122.5	yes
98	345	4044.0616145	0.001157	124.1	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s33.json, historical/diff_s33.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R470. Public evidence files: diff_s33.json, diff_s33.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

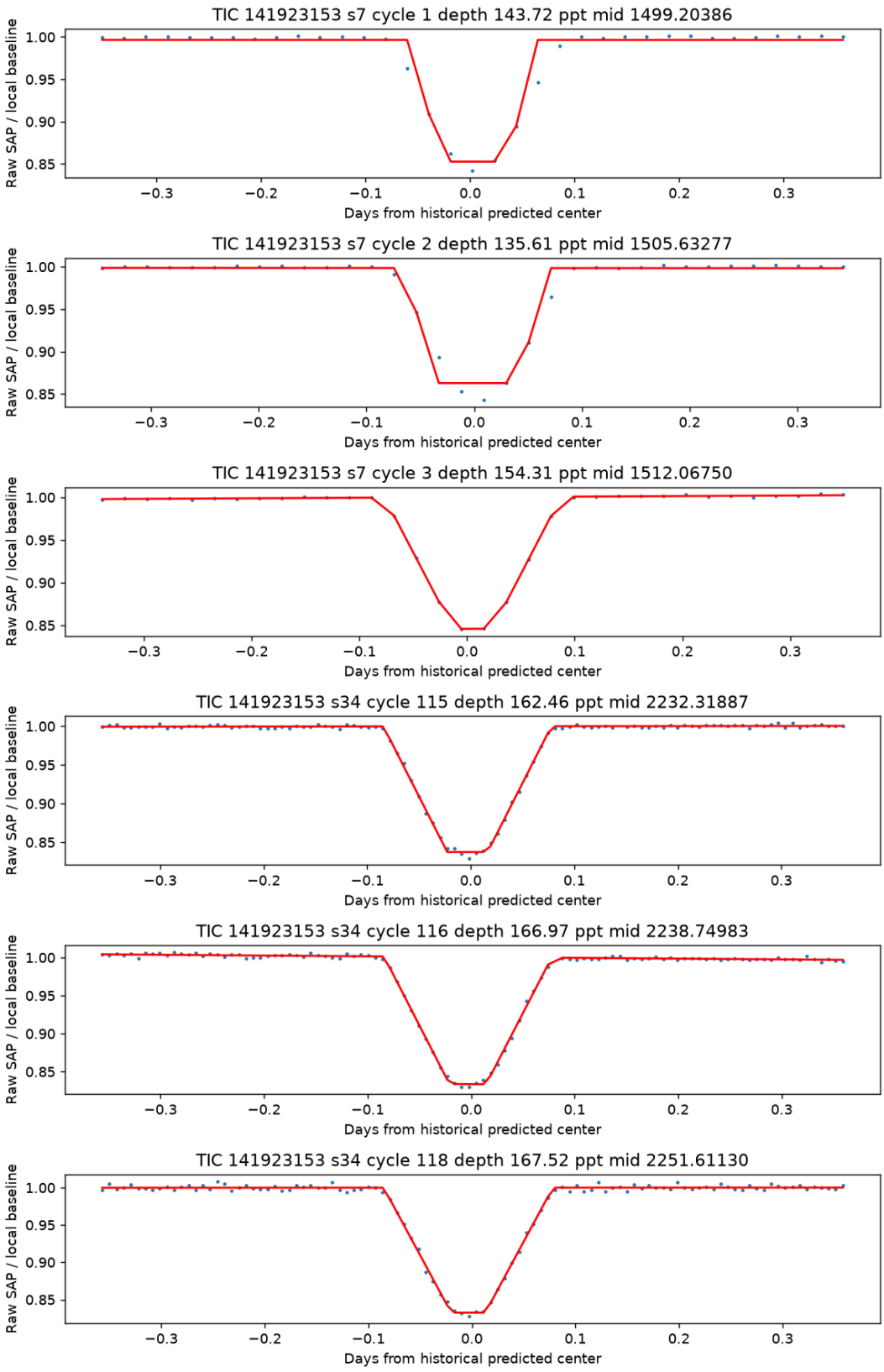
R471. TIC 141923153

physical_verdict: CONFIRMED

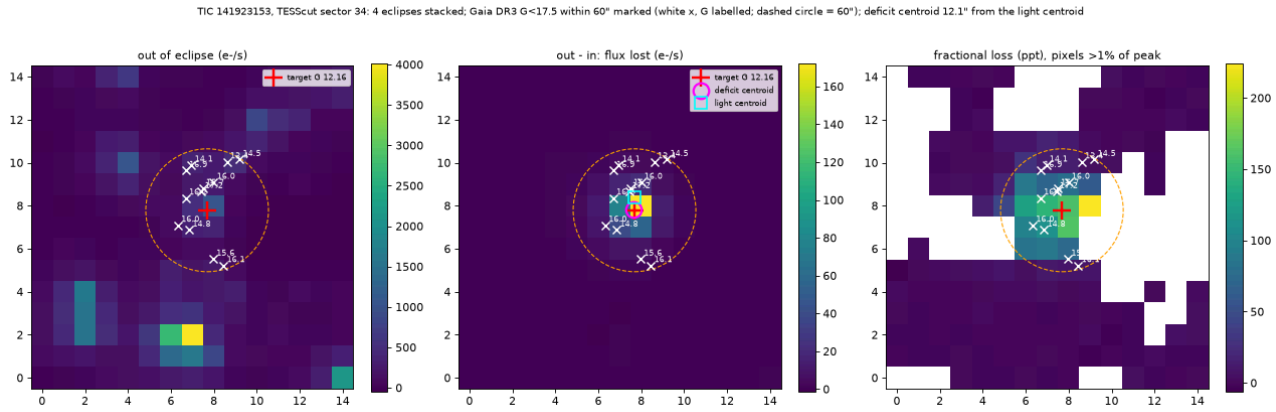
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R471; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R471; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 141923153: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.162264. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [7, 34] contains 7 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 6.4308327985 ± 0.0000546344 d and epoch BTJD 1499.2038780, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[7, 1, 2], [7, 2, 3], [34, 115, 116], [34, 116, 117], [34, 117, 118]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 34 TESScut difference image has deficit peak [8, 8] and target pixel [8, 8]; deficit centroid is 1.01 arcsec from the target coordinate. Image-model fractional residual 0.811 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 5710669575691823360 (51.6 arcsec, G 13.425, intrinsic loss needed ~0.64). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 159.4/137.3/101.6 ppt from 4 stacked events, versus median selected raw QLP depth 162.5 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 5710669537024684416 has P = 6.4302209846 d and e = 0.11073; the closer edge-on conjunction differs from the measured epoch by 0.05310 d (0.00826 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.11745 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 5.55 h against 3.89 h observed median; it is not a complete binary model.

Decision and limitation. Consecutive raw eclipses in S7 and S34 support recurrence; the capable 51.6-arcsec neighbor is separated from the target-centered deficit despite substantial image-model residual. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
7	1	1499.2038586	0.010417	143.7	yes
7	2	1505.6327667	0.010417	135.6	yes
7	3	1512.0675010	0.010417	154.3	yes
34	115	2232.3188695	0.003472	162.5	yes
34	116	2238.7498281	0.003472	167.0	yes
34	117	2245.1802637	0.003472	165.3	yes
34	118	2251.6113041	0.003472	167.5	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s34.json, historical/diff_s34.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R471. Public evidence files: diff_s34.json, diff_s34.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

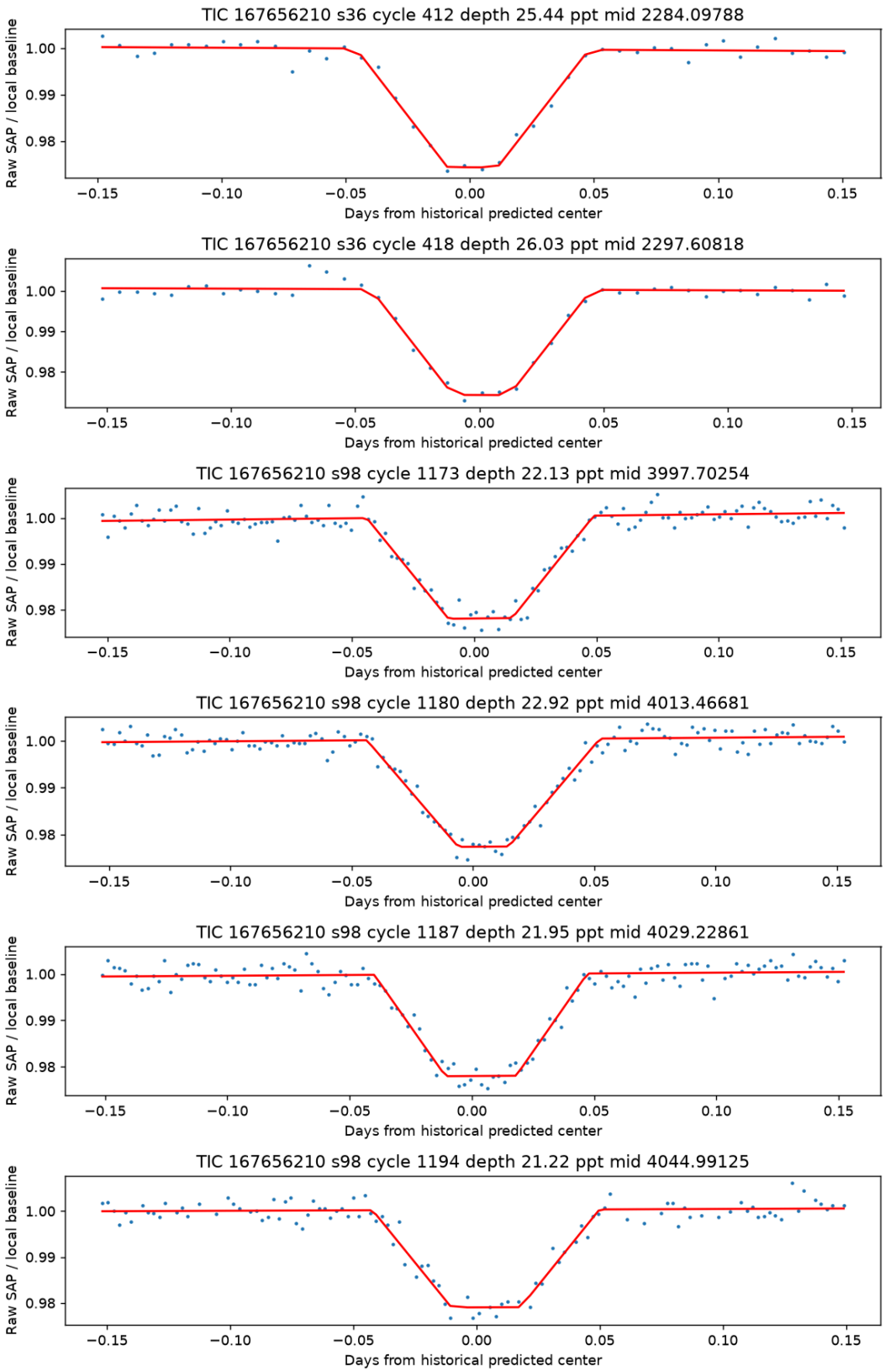
R472. TIC 167656210

physical_verdict: PROBABLE

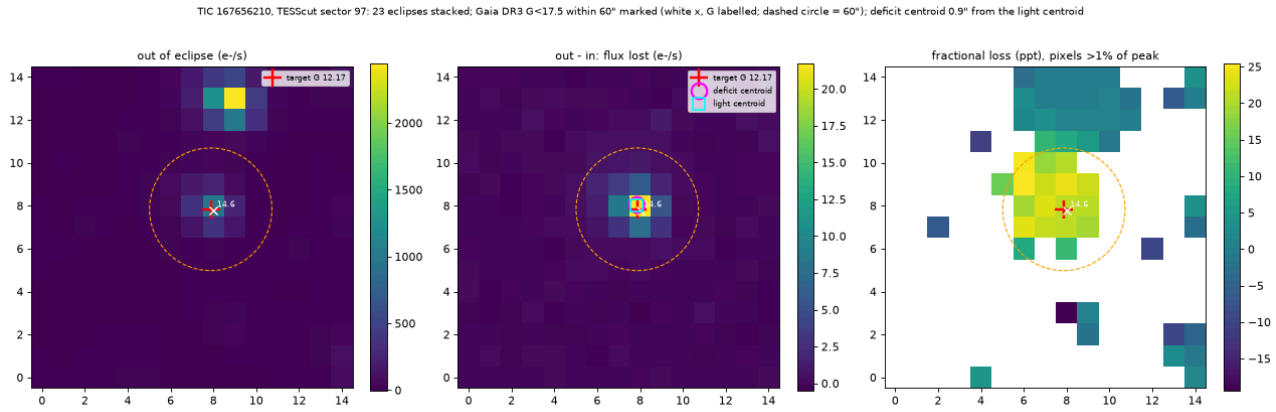
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved time-series photometry or multi-source PRF fitting of the 2.9-arcsec pair.



R472; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R472; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 167656210: PROBABLE

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.166932. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [36, 98] contains 22 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 2.2517814962 ± 0.0000017527 d and epoch BTJD 2284.0975046, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[36, 412, 413], [36, 413, 414], [36, 414, 415], [36, 420, 421], [98, 1172, 1173], [98, 1173, 1174], [98, 1174, 1175], [98, 1179, 1180], [98, 1180, 1181], [98, 1185, 1186], [98, 1186, 1187], [98, 1187, 1188], [98, 1191, 1192], [98, 1192, 1193], [98, 1193, 1194]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 97 TESScut difference image has deficit peak [8, 8] and target pixel [8, 8]; deficit centroid is 4.63 arcsec from the target coordinate. Image-model fractional residual 0.652 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 5480019042928976896 (2.9 arcsec, G 14.551, intrinsic loss needed ~0.24). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 21.2/21.7/21.5 ppt from 23 stacked events, versus median selected raw QLP depth 23.1 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 5480019038632380288 has P = 2.2517090090 d and e = 0.03953; the closer edge-on conjunction differs from the measured epoch by 0.04866 d (0.02161 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.09510 d,

descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 4.78 h against 2.22 h observed median; it is not a complete binary model.

Decision and limitation. A 2.9-arcsec capable Gaia neighbor shares the TESS core. The recurring eclipses are real, but these pixels cannot uniquely assign them to the selected Gaia star. Attribution remains unresolved after the completed pixel check.

Next check. Resolved time-series photometry or multi-source PRF fitting of the 2.9-arcsec pair.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
36	412	2284.0978758	0.003472	25.4	yes
36	413	2286.3495175	0.003472	25.4	yes
36	414	2288.6013787	0.003472	23.9	yes
36	415	2290.8530233	0.003472	24.8	yes
36	418	2297.6081832	0.003472	26.0	yes
36	420	2302.1110686	0.003472	24.3	yes
36	421	2304.3631005	0.003472	25.4	yes
98	1172	3995.4520614	0.001157	23.0	yes
98	1173	3997.7025415	0.001157	22.1	yes
98	1174	3999.9547706	0.001157	22.0	yes
98	1175	4002.2074201	0.001157	21.9	yes
98	1179	4011.2131845	0.001157	23.9	yes
98	1180	4013.4668097	0.001157	22.9	yes
98	1181	4015.7173665	0.001157	23.3	yes
98	1185	4024.7241694	0.001157	24.1	yes
98	1186	4026.9757790	0.001157	23.3	yes
98	1187	4029.2286132	0.001157	21.9	yes
98	1188	4031.4798496	0.001157	22.7	yes
98	1191	4038.2353658	0.001157	21.5	yes
98	1192	4040.4869257	0.001157	21.9	yes
98	1193	4042.7383739	0.001157	22.2	yes
98	1194	4044.9912545	0.001157	21.2	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s97.json, historical/diff_s97.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R472. Public evidence files: diff_s97.json, diff_s97.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

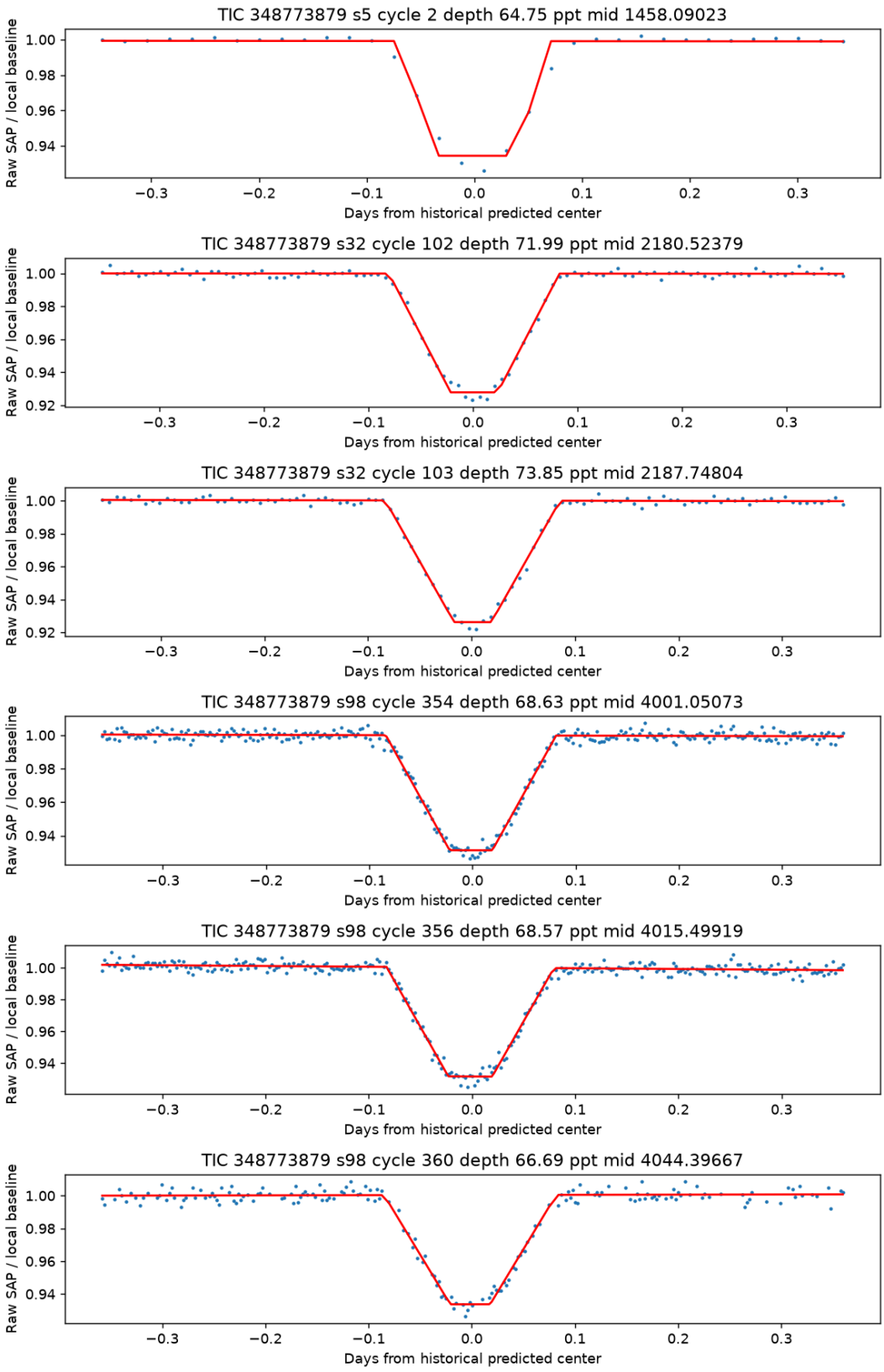
R473. TIC 348773879

physical_verdict: CONFIRMED

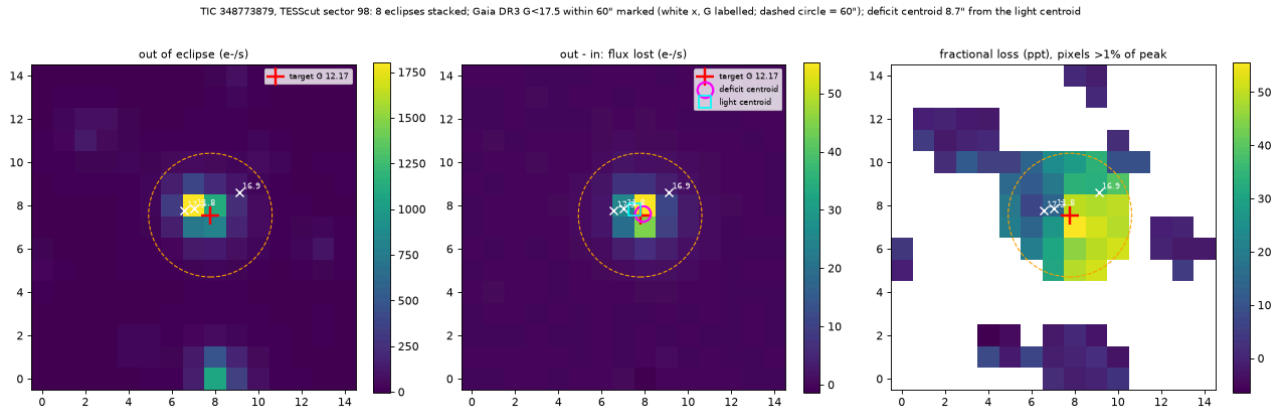
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R473; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R473; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 348773879: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.169457. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [5, 32, 98] contains 8 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 7.2243136689 ± 0.0000079385 d and epoch BTJD 1458.0924122, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[32, 102, 103], [32, 103, 104]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 98 TESScut difference image has deficit peak [8, 8] and target pixel [8, 8]; deficit centroid is 2.72 arcsec from the target coordinate. Image-model fractional residual 0.535 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 2987292965757749376 (17.0 arcsec, G 11.841, intrinsic loss needed ~0.11). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 49.9/32.8/30.9 ppt from 8 stacked events, versus median selected raw QLP depth 68.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 2987292862678534144 has P = 7.2246361222 d and e = 0.04277; the closer edge-on conjunction differs from the measured epoch by -0.09908 d (-0.01371 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.12913 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 5.95 h against 3.98 h observed median; it is not a complete binary model.

Decision and limitation. The capable 17.0-arcsec neighbor is about a pixel away; the deficit core falls much closer to the target. Attribution favors the target at TESS resolution, with uncalibrated subpixel precision. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
5	2	1458.0902329	0.010417	64.7	yes
32	102	2180.5237896	0.003472	72.0	yes
32	103	2187.7480351	0.003472	73.9	yes
32	104	2194.9728229	0.003472	75.0	yes
98	354	4001.0507308	0.001157	68.6	yes
98	356	4015.4991851	0.001157	68.6	yes
98	358	4029.9484542	0.001157	67.6	yes
98	360	4044.3966744	0.001157	66.7	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s98.json, historical/diff_s98.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R473. Public evidence files: diff_s98.json, diff_s98.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

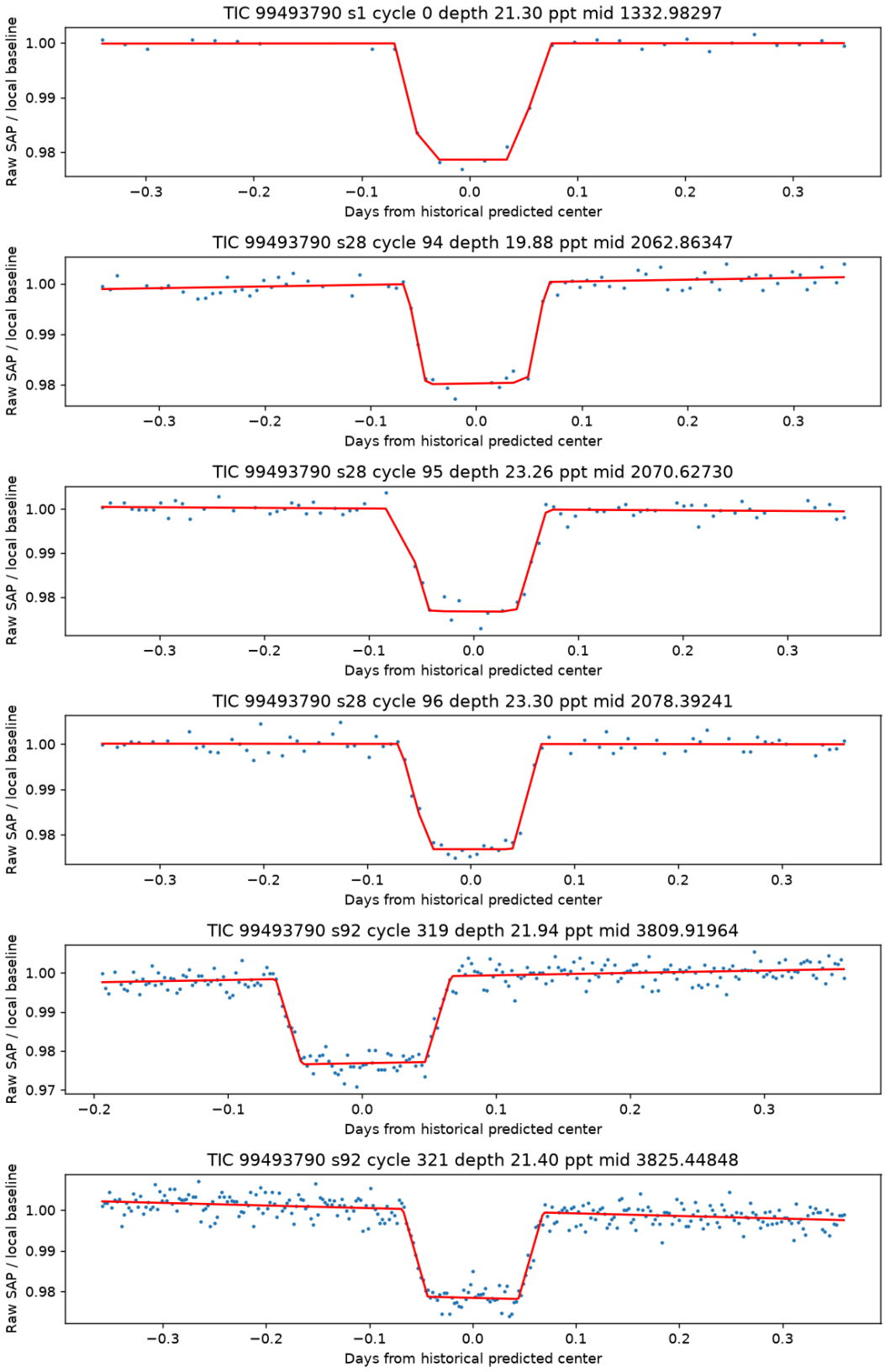
R474. TIC 99493790

physical_verdict: CONFIRMED

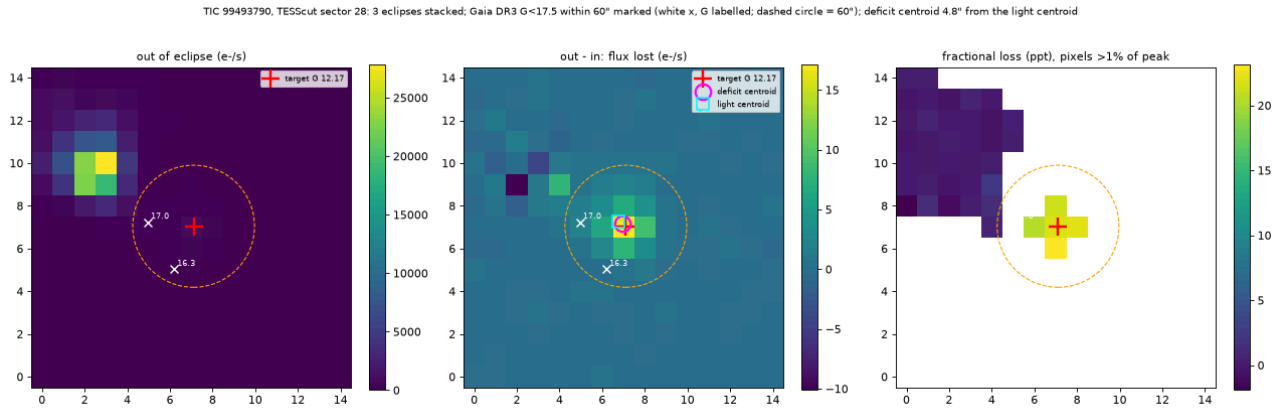
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R474; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R474; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 99493790: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.170427. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [1, 28, 92] contains 7 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 7.7646935289 ± 0.0000090716 d and epoch BTJD 1332.9819065, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[28, 94, 95], [28, 95, 96], [92, 319, 320], [92, 320, 321]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 28 TESScut difference image has deficit peak [7, 7] and target pixel [7, 7]; deficit centroid is 3.68 arcsec from the target coordinate. Image-model fractional residual 0.999 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: None under the approximate G-band brightness budget. The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 21.8/21.2/18.7 ppt from 3 stacked events, versus median selected raw QLP depth 21.9 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 6804708139515008640 has P = 7.7653086675 d and e = 0.08951; the closer edge-on conjunction differs from the measured epoch by 0.05671 d (0.00730 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.33323 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 3.89 h against 3.26 h observed median; it is not a complete binary model.

Decision and limitation. The target-centered deficit and absence of a brightness-capable Gaia neighbor favor attribution; the image-model residual is high. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
1	0	1332.9829705	0.010417	21.3	yes
28	94	2062.8634721	0.003472	19.9	yes
28	95	2070.6273013	0.003472	23.3	yes
28	96	2078.3924097	0.003472	23.3	yes
92	319	3809.9196418	0.001158	21.9	yes
92	320	3817.6833970	0.001158	23.3	yes
92	321	3825.4484769	0.001158	21.4	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s28.json, historical/diff_s28.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R474. Public evidence files: diff_s28.json, diff_s28.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

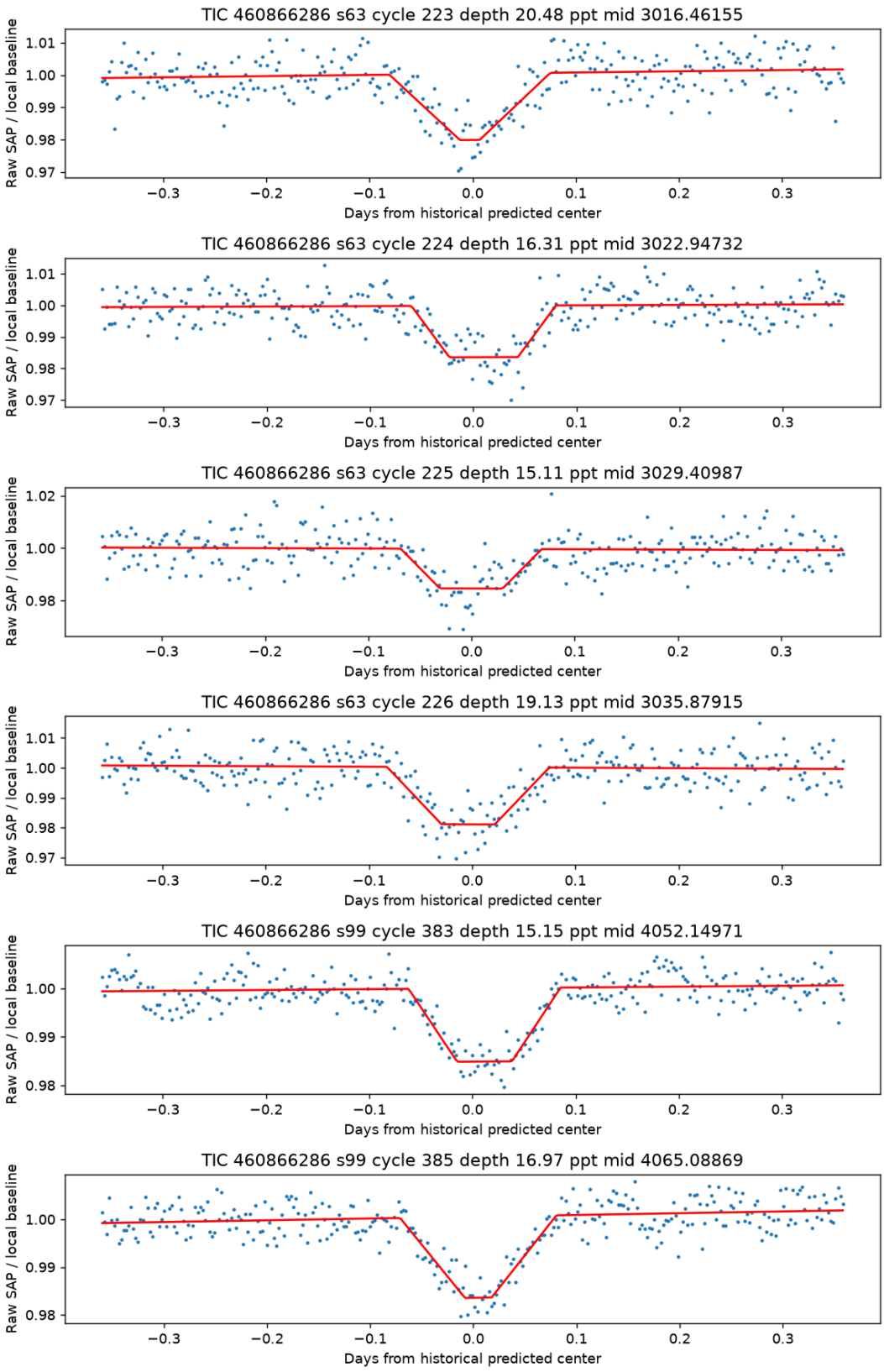
R475. TIC 460866286

physical_verdict: PROBABLE

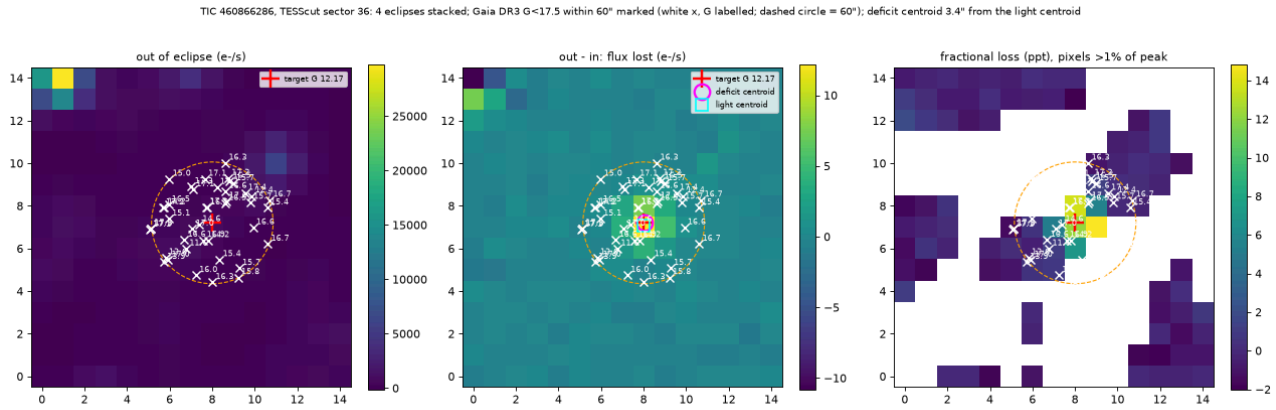
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Higher-resolution eclipse photometry or multi-source PRF fitting in this crowded field.



R475; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R475; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 460866286: PROBABLE

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.171505. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [63, 99] contains 6 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 6.4730069098 ± 0.0000307395 d and epoch BTJD 3016.4658767, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[63, 223, 224], [63, 224, 225], [63, 225, 226]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 36 TESScut difference image has deficit peak [8, 7] and target pixel [8, 7]; deficit centroid is 0.95 arcsec from the target coordinate. Image-model fractional residual 0.970 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 5254306927996818560 (12.3 arcsec, G 14.563, intrinsic loss needed ~0.16); Gaia 5254306932350664960 (17.3 arcsec, G 14.152, intrinsic loss needed ~0.11); Gaia 5254301812749381248 (21.9 arcsec, G 16.515, intrinsic loss needed ~0.86); Gaia 5254306932350668800 (30.4 arcsec, G 16.610, intrinsic loss needed ~0.94); Gaia 5254301812749372800 (32.3 arcsec, G 16.629, intrinsic loss needed ~0.95); Gaia 5254306966710413440 (34.4 arcsec, G 11.637, intrinsic loss needed ~0.02); Gaia 5254307722624660992 (36.0 arcsec, G 15.407, intrinsic loss needed ~0.32); Gaia 5254307683911072000 (38.1 arcsec, G 16.620, intrinsic loss needed ~0.95); Gaia 5254301812749375232 (40.0 arcsec, G 15.746, intrinsic loss needed ~0.43); Gaia 5254306932350657024 (40.7 arcsec, G 15.113, intrinsic loss needed ~0.25); Gaia 5254301808395792768 (41.9 arcsec, G 15.719, intrinsic loss needed ~0.42); Gaia 5254301812749364096 (42.7 arcsec, G 16.547, intrinsic loss needed ~0.89); Gaia

5254301808395353216 (44.3 arcsec, G 12.383, intrinsic loss needed ~0.03); Gaia 5254306932350647040 (44.6 arcsec, G 16.498, intrinsic loss needed ~0.85); Gaia 5254306932293814016 (47.3 arcsec, G 14.187, intrinsic loss needed ~0.11); Gaia 5254307718270821504 (50.7 arcsec, G 15.688, intrinsic loss needed ~0.41); Gaia 5254306966710433152 (52.3 arcsec, G 16.044, intrinsic loss needed ~0.56); Gaia 5254301808414672768 (52.6 arcsec, G 15.358, intrinsic loss needed ~0.31); Gaia 5254306966710944384 (54.1 arcsec, G 13.649, intrinsic loss needed ~0.08); Gaia 5254306966710435840 (57.2 arcsec, G 16.331, intrinsic loss needed ~0.73); Gaia 5254306966710944768 (58.0 arcsec, G 13.933, intrinsic loss needed ~0.09); Gaia 5254307718271123328 (58.2 arcsec, G 15.760, intrinsic loss needed ~0.44); Gaia 5254301778389613440 (58.5 arcsec, G 16.272, intrinsic loss needed ~0.69); Gaia 5254301056835128576 (58.7 arcsec, G 15.025, intrinsic loss needed ~0.23). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.

4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 12.3/7.7/3.9 ppt from 4 stacked events, versus median selected raw QLP depth 16.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 5254307683911065728 has P = 6.4725617010 d and e = 0.06549; the closer edge-on conjunction differs from the measured epoch by 0.10544 d (0.01629 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.18827 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 8.14 h against 3.59 h observed median; it is not a complete binary model.

Decision and limitation. The field has 24 brightness-capable Gaia neighbors within 60 arcsec, including one at 12.3 arcsec. The deficit core is near the target, but the weak signal and 0.97 image-model residual cannot uniquely resolve nearby alternatives. Attribution remains unresolved after the completed pixel check.

Next check. Higher-resolution eclipse photometry or multi-source PRF fitting in this crowded field.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
63	223	3016.4615526	0.002601	20.5	yes
63	224	3022.9473244	0.002086	16.3	yes
63	225	3029.4098721	0.002859	15.1	yes
63	226	3035.8791532	0.002305	19.1	yes
99	383	4052.1497109	0.001449	15.2	yes
99	385	4065.0886872	0.001840	17.0	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s36.json, historical/diff_s36.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R475. Public evidence files: diff_s36.json, diff_s36.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed;

source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

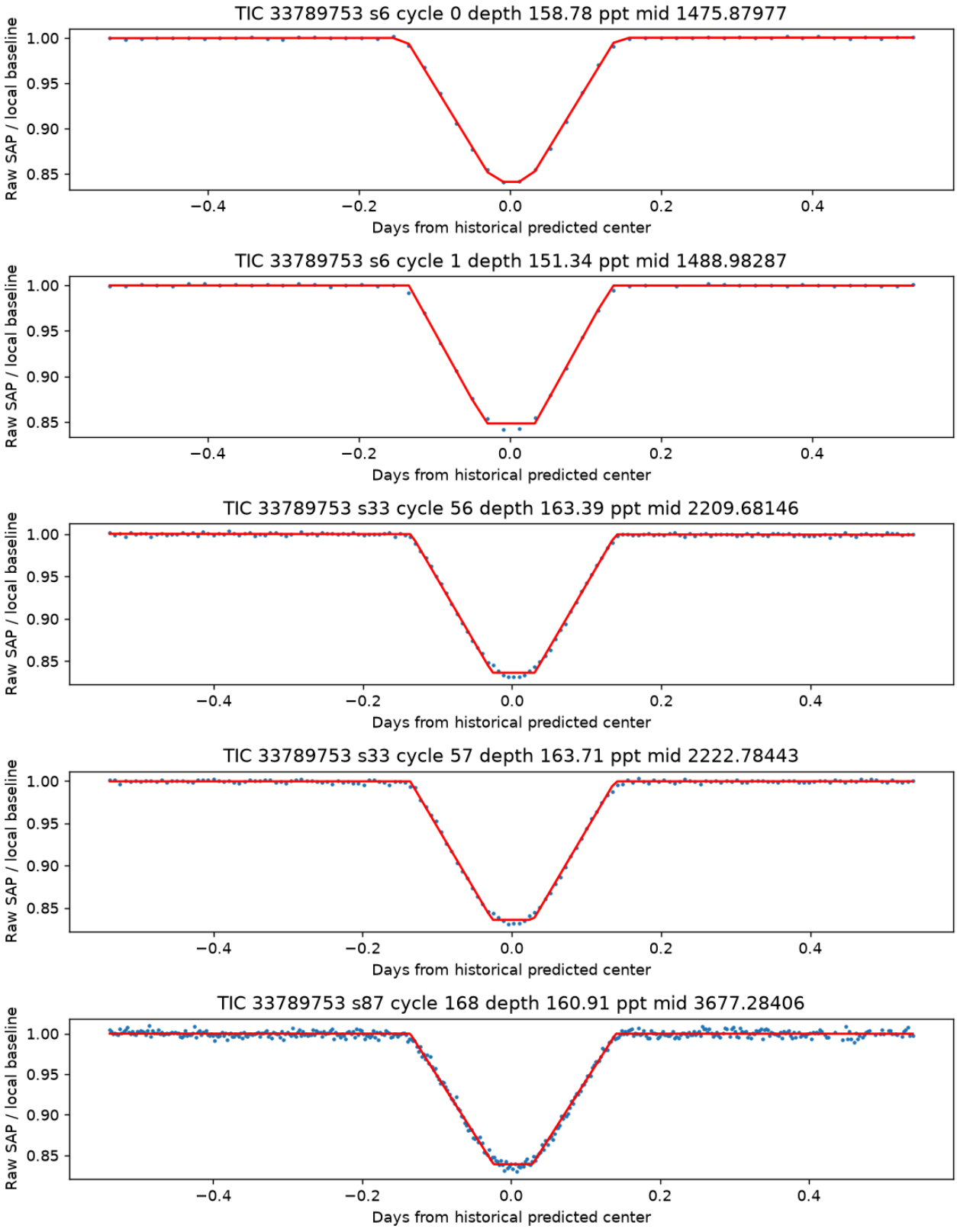
R476. TIC 33789753

physical_verdict: CONFIRMED

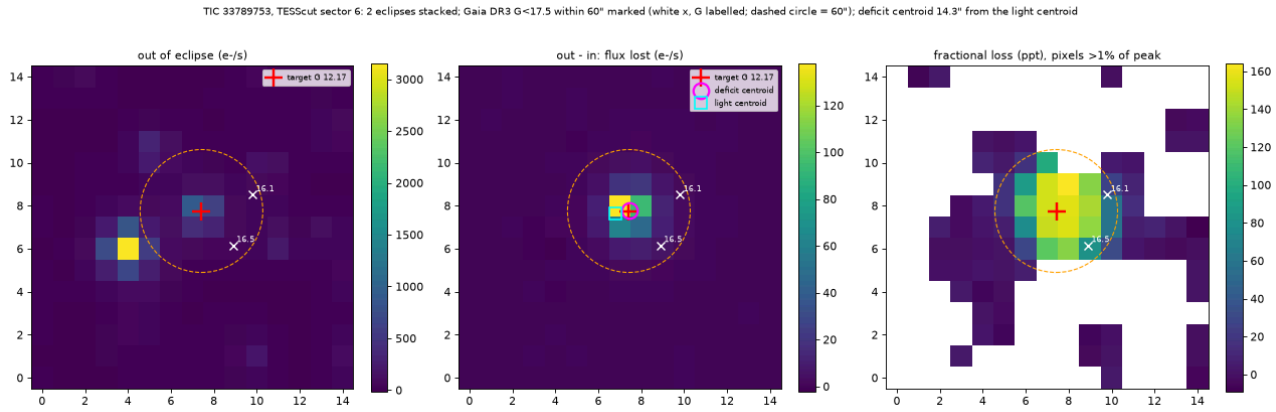
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R476; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R476; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 33789753: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.174696. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [6, 33, 87] contains 5 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 13.1035976171 ± 0.0000219313 d and epoch BTJD 1475.8796596, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[6, 0, 1], [33, 56, 57]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 6 TESScut difference image has deficit peak [7, 8] and target pixel [7, 8]; deficit centroid is 0.63 arcsec from the target coordinate. Image-model fractional residual 0.947 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: None under the approximate G-band brightness budget. The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 153.5/148.6/104.3 ppt from 2 stacked events, versus median selected raw QLP depth 160.9 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 2993617253561277056 has P = 13.1020129210 d and e = 0.41294; the closer edge-on conjunction differs from the measured epoch by 0.17799 d (0.01359 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.22281 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 10.15 h against 6.59 h observed median; it is not a complete binary model.

Decision and limitation. Five raw events in S6/S33/S87 agree in timing and shape. No brightness-capable Gaia neighbor appears within 60 arcsec. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
6	0	1475.8797672	0.010417	158.8	yes
6	1	1488.9828743	0.010417	151.3	yes
33	56	2209.6814606	0.003472	163.4	yes
33	57	2222.7844323	0.003472	163.7	yes
87	168	3677.2840579	0.001157	160.9	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s06.json, historical/diff_s06.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R476. Public evidence files: diff_s06.json, diff_s06.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

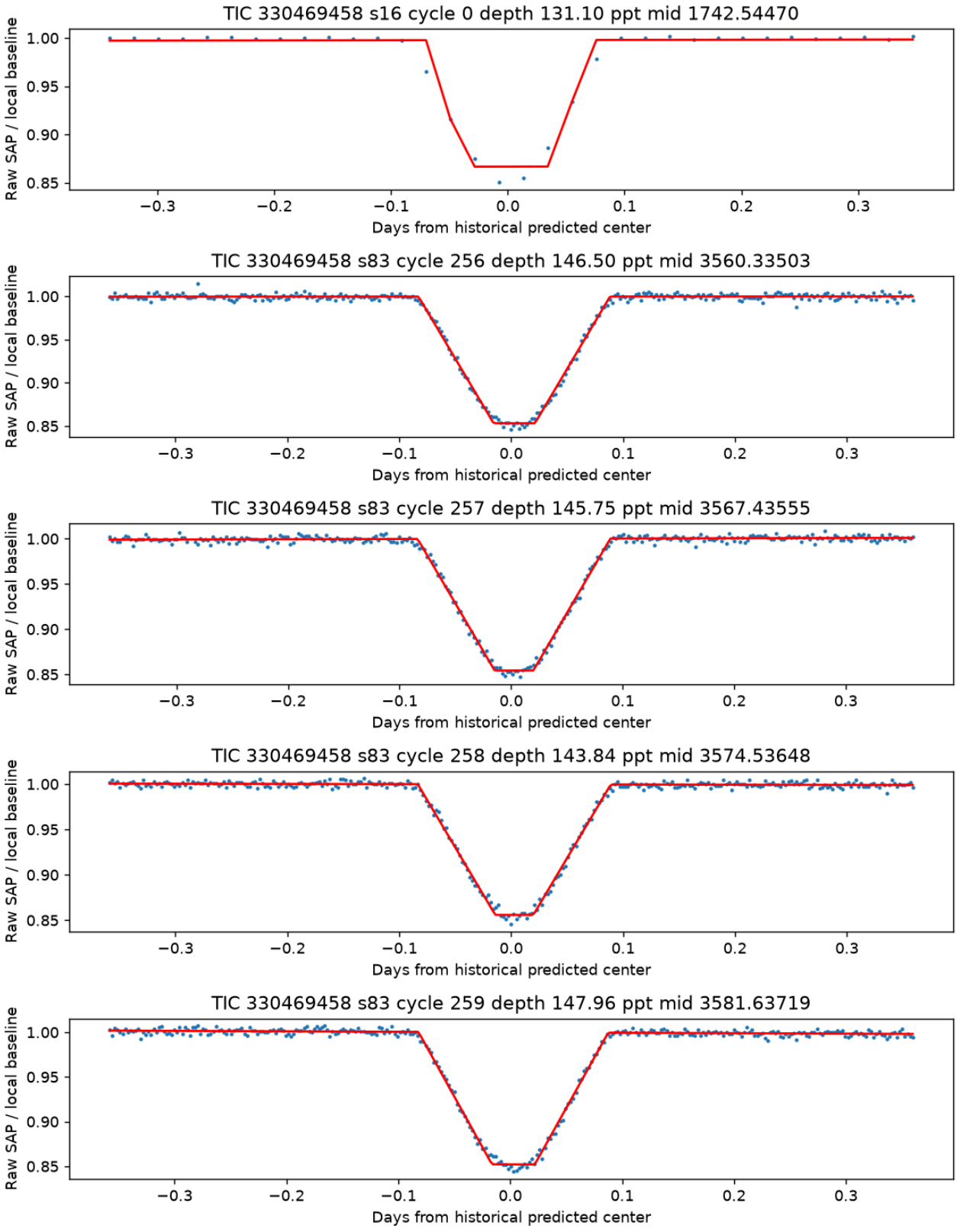
R477. TIC 330469458

physical_verdict: CONFIRMED

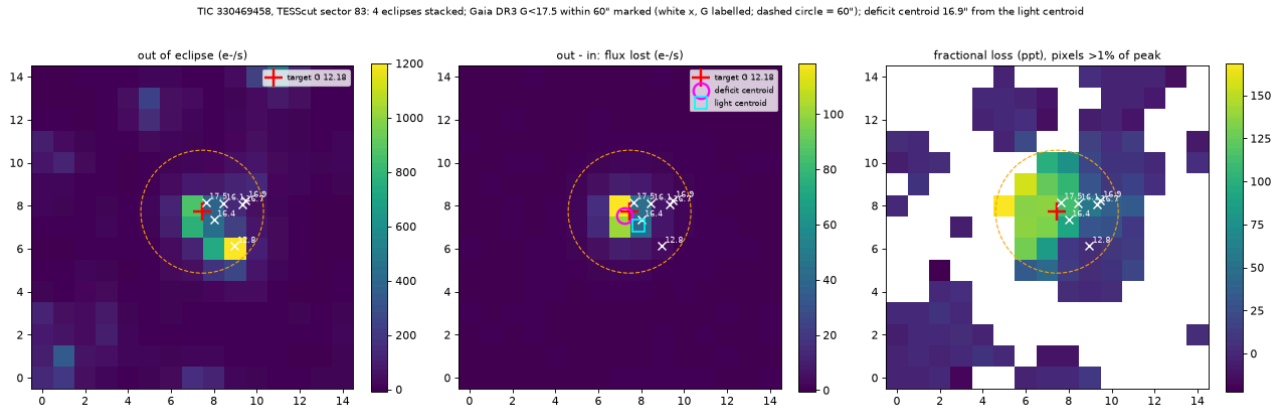
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R477; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R477; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 330469458: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.182403. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [16, 83] contains 5 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 7.1007431519 ± 0.0000403912 d and epoch BTJD 1742.5447026, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[83, 256, 257], [83, 257, 258], [83, 258, 259]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 83 TESScut difference image has deficit peak [7, 8] and target pixel [7, 8]; deficit centroid is 5.88 arcsec from the target coordinate. Image-model fractional residual 0.717 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 1906480221138868736 (46.3 arcsec, G 12.790, intrinsic loss needed ~0.37). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 136.4/123.7/73.9 ppt from 4 stacked events, versus median selected raw QLP depth 145.7 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 1906480289858343040 has P = 7.1014333035 d and e = 0.03164; the closer edge-on conjunction differs from the measured epoch by -0.13985 d (-0.01969 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.16156 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 5.78 h against 4.11 h observed median; it is not a complete binary model.

Decision and limitation. Five deep raw events in S16/S83 recur at the measured period. A capable neighbor 46.3 arcsec away is distinct from the target-centered deficit. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
16	0	1742.5446985	0.010416	131.1	yes
83	256	3560.3350331	0.001157	146.5	yes
83	257	3567.4355511	0.001157	145.7	yes
83	258	3574.5364814	0.001157	143.8	yes
83	259	3581.6371914	0.001157	148.0	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s83.json, historical/diff_s83.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R477. Public evidence files: diff_s83.json, diff_s83.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

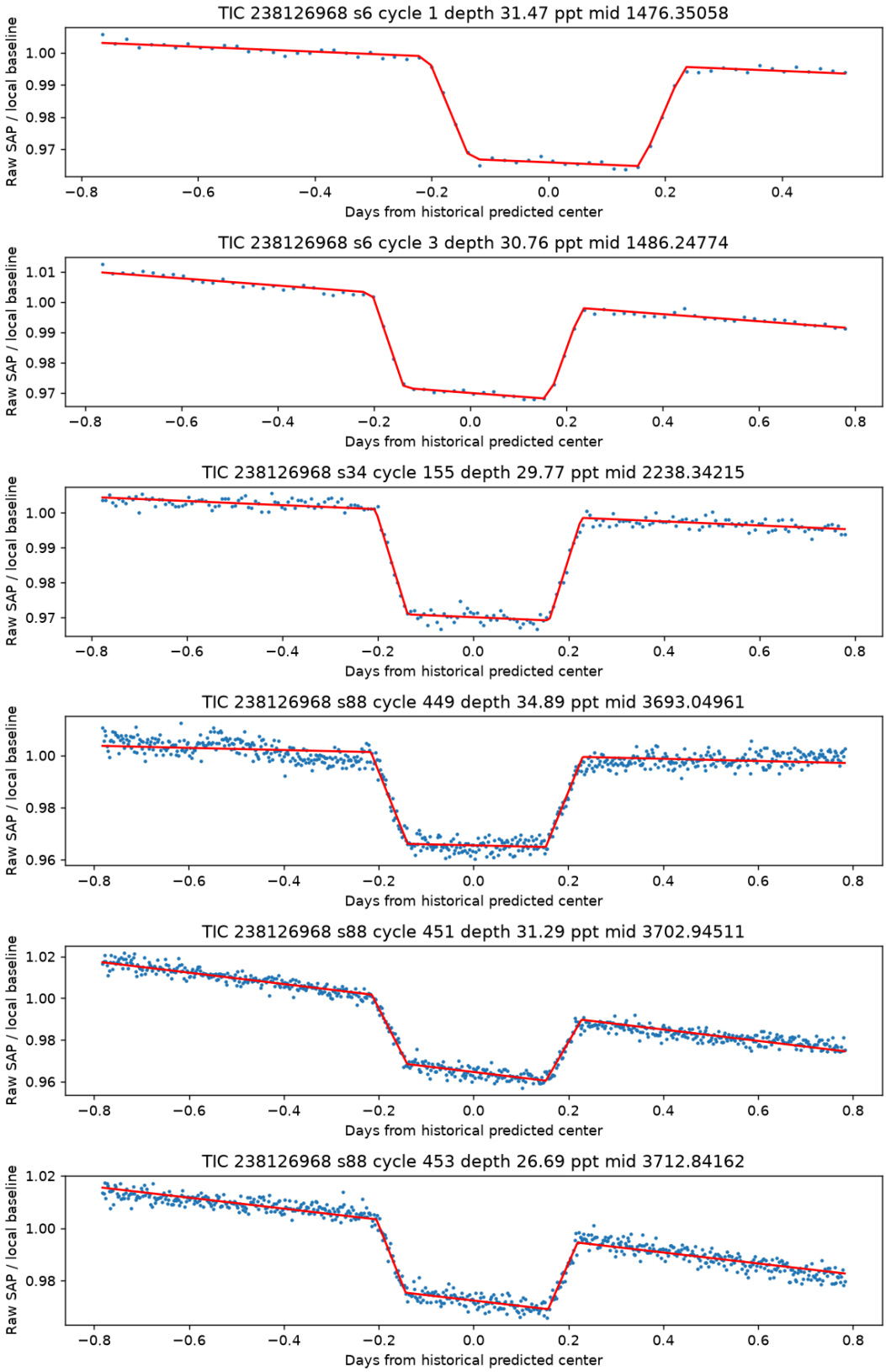
R478. TIC 238126968

physical_verdict: CONFIRMED

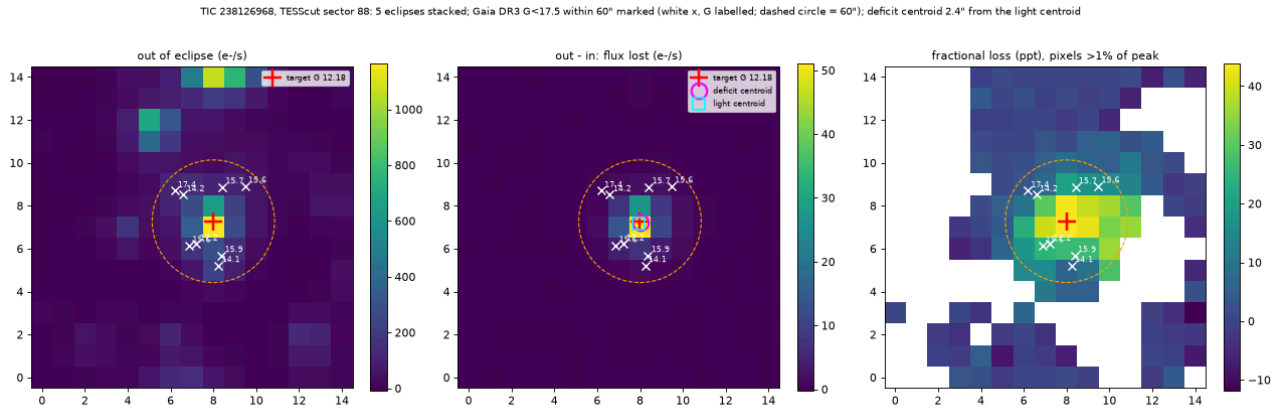
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R478; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R478; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 238126968: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.183063. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [6, 34, 88] contains 5 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 9.8959493292 ± 0.0000245311 d and epoch BTJD 1481.3039577, kind: Deeper repeating eclipse; even half-period cycles. Within-sector same-kind measured cycle pairs: [[34, 154, 156], [88, 450, 452]]. Only the deep even historical cycles enter the orbital fit; odd cycles are the shallower kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 88 TESScut difference image has deficit peak [8, 7] and target pixel [8, 7]; deficit centroid is 1.14 arcsec from the target coordinate. Image-model fractional residual 0.380 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: Gaia 5508449115088643712 (39.7 arcsec, G 14.211, intrinsic loss needed ~0.39); Gaia 5508448840210743424 (43.6 arcsec, G 14.129, intrinsic loss needed ~0.36). The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 43.9/38.6/32.0 ppt from 5 stacked events, versus median selected raw QLP depth 57.6 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 5508448904633378560 has P = 9.8961317032 d and e = 0.04783; the closer edge-on conjunction differs from the measured epoch by -0.04622 d (-0.00467 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.13469 d,

descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 10.68 h against 10.48 h observed median; it is not a complete binary model.

Decision and limitation. Shallow and deep eclipses alternate at the historical 4.94798-day search interval. Five deep even-cycle midpoints give the 9.89595-day orbital period; shallow odd cycles support the alternating pattern. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
6	1	1476.3505764	0.010417	31.5	no: other kind or timing failure
6	2	1481.2972292	0.010417	56.3	yes
6	3	1486.2477423	0.010417	30.8	no: other kind or timing failure
34	154	2233.4000168	0.005811	61.8	yes
34	155	2238.3421474	0.003472	29.8	no: other kind or timing failure
34	156	2243.2914492	0.005204	63.7	yes
88	449	3693.0496143	0.001157	34.9	no: other kind or timing failure
88	450	3697.9966498	0.001571	55.8	yes
88	451	3702.9451069	0.006515	31.3	no: other kind or timing failure
88	452	3707.8919600	0.003507	57.6	yes
88	453	3712.8416212	0.004982	26.7	no: other kind or timing failure

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s88.json, historical/diff_s88.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R478. Public evidence files: diff_s88.json, diff_s88.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

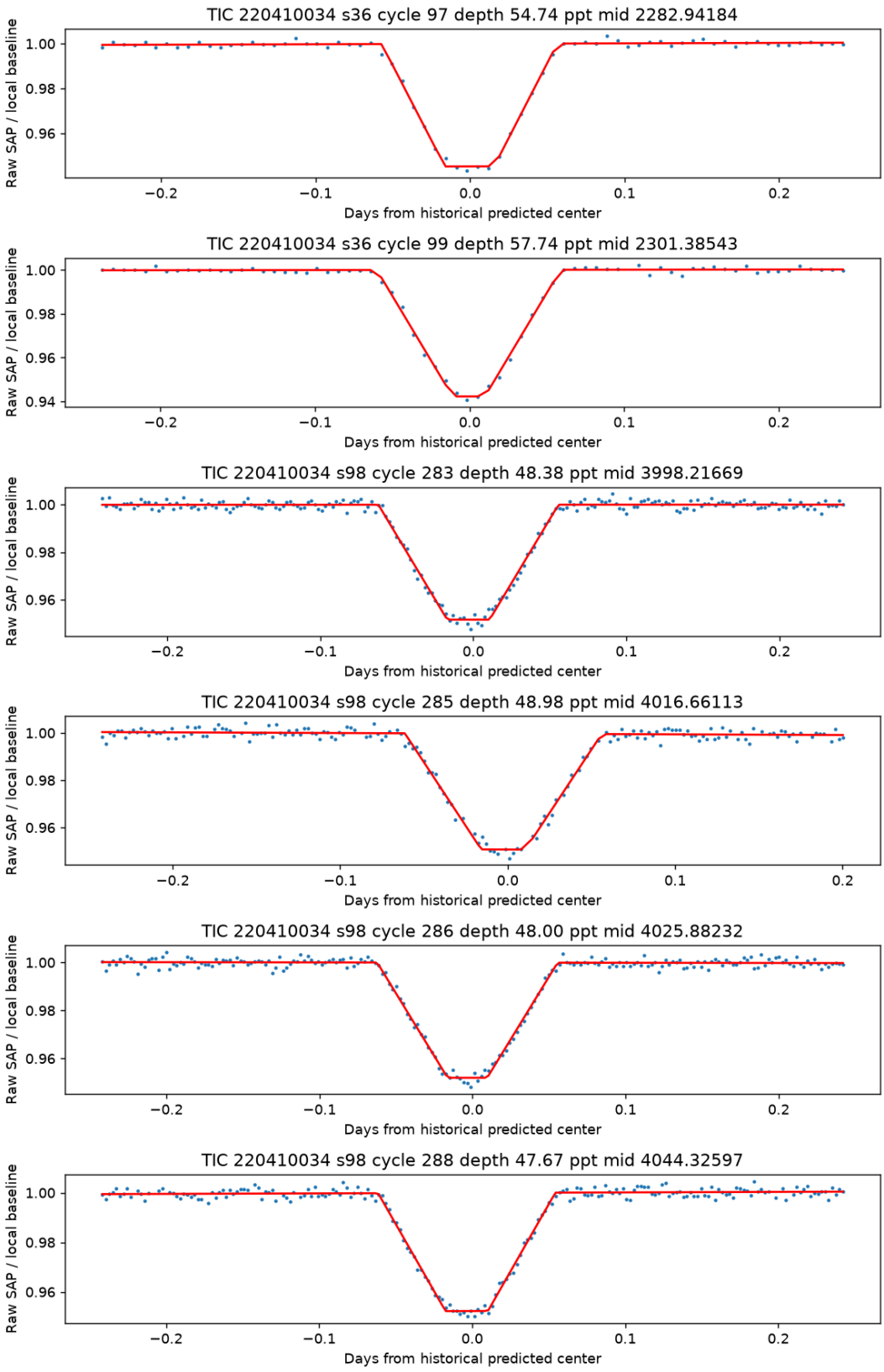
R479. TIC 220410034

physical_verdict: CONFIRMED

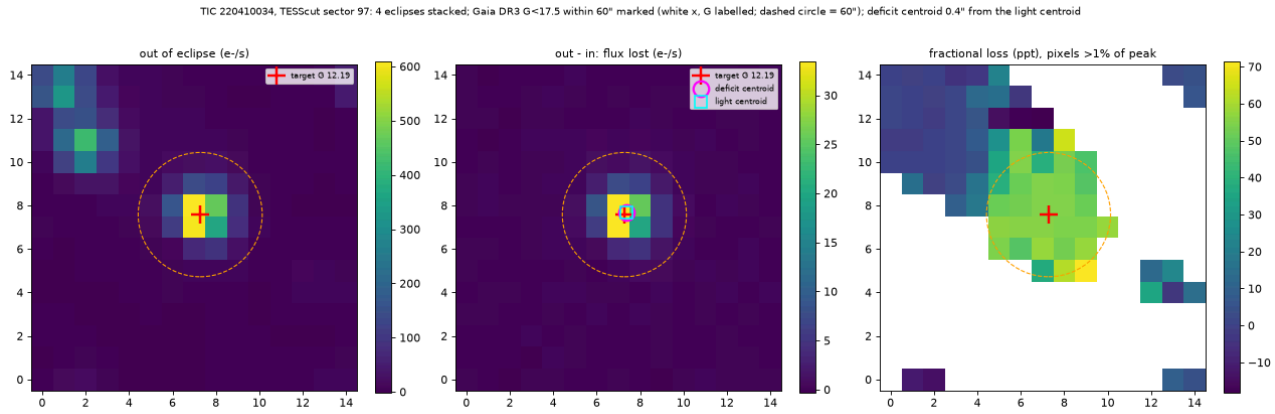
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization



R479; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R479; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 220410034: CONFIRMED

Full bounded review: **COMPLETE**. Actual Gaia DR3 G = 12.185472. The screening period and depth are historical selection values. Physical outcome and central catalogue novelty are separate.

1. **Raw repeated morphology.** Quality-zero QLP SAP_FLUX in representative downloaded sectors [36, 98] contains 6 selected timing events. Local trapezoids fit free midpoint, depth, duration and baseline; historical predictions only select windows. Numeric windows and a raw event plot accompany this report. Flagged intervals are never clean nulls.
2. **Measured timing and aliases.** Same-kind midpoint regression gives P = 9.2219086085 ± 0.0000134411 d and epoch BTJD 2282.9417514, kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Within-sector same-kind measured cycle pairs: [[98, 285, 286]]. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This bounded alias check does not claim an exhaustive all-period search.
3. **Target attribution.** Historical sector 97 TESScut difference image has deficit peak [7, 8] and target pixel [7, 8]; deficit centroid is 2.28 arcsec from the target coordinate. Image-model fractional residual 0.209 and centroid offset are descriptive, not calibrated localization uncertainties. Every brightness-capable Gaia neighbor within 60 arcsec under the historical screen: None under the approximate G-band brightness budget. The intrinsic-loss budget is approximate and cannot be compared directly with blended pixel loss.
4. **Alternative reduction.** Background-subtracted TESScut 1×1/3×3/5×5 aperture depths are 54.9/54.8/54.3 ppt from 4 stacked events, versus median selected raw QLP depth 48.7 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Dilution and mixed eclipse kinds can change depths. Original cutout identity and SHA256 are in ../source-hashes.json.
5. **Gaia geometry.** Matched DR3 NSS source 4777341154688327424 has P = 9.2219102799 d and e = 0.08597; the closer edge-on conjunction differs from the measured epoch by -0.21615 d (-0.02344 phase). Its linearly propagated full source-specific NSS-covariance timing error is 0.40158 d, descriptive rather than a veto threshold. An approximate radius/minimum-mass central duration is 4.68 h against 2.82 h observed median; it is not a complete binary model.

Decision and limitation. S98 contains measured eclipses in both parity classes of the historical period, excluding a simple double-period alias. No brightness-capable Gaia neighbor appears within 60 arcsec. The available checks favor the target, subject to the stated pixel and timing limits.

Next check. Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty is determined by the central seven-group and NSS coverage matrix at ../../shared-catalogues/coverage-matrix.csv; this physical review does not assert exhaustive literature priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Used in period fit
36	97	2282.9418398	0.003472	54.7	yes
36	99	2301.3854303	0.003472	57.7	yes
98	283	3998.2166875	0.001157	48.4	yes
98	285	4016.6611258	0.001157	49.0	yes
98	286	4025.8823212	0.001157	48.0	yes
98	288	4044.3259673	0.001157	47.7	yes

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s97.json, historical/diff_s97.png, historical/gaia.json, ../pixel-contact.png, ../source-hashes.json and ../helper-audit.md. Original TESScut FITS: [local path omitted] File hashes are frozen in ../SHA256SUMS.json.

Stable review key: R479. Public evidence files: diff_s97.json, diff_s97.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

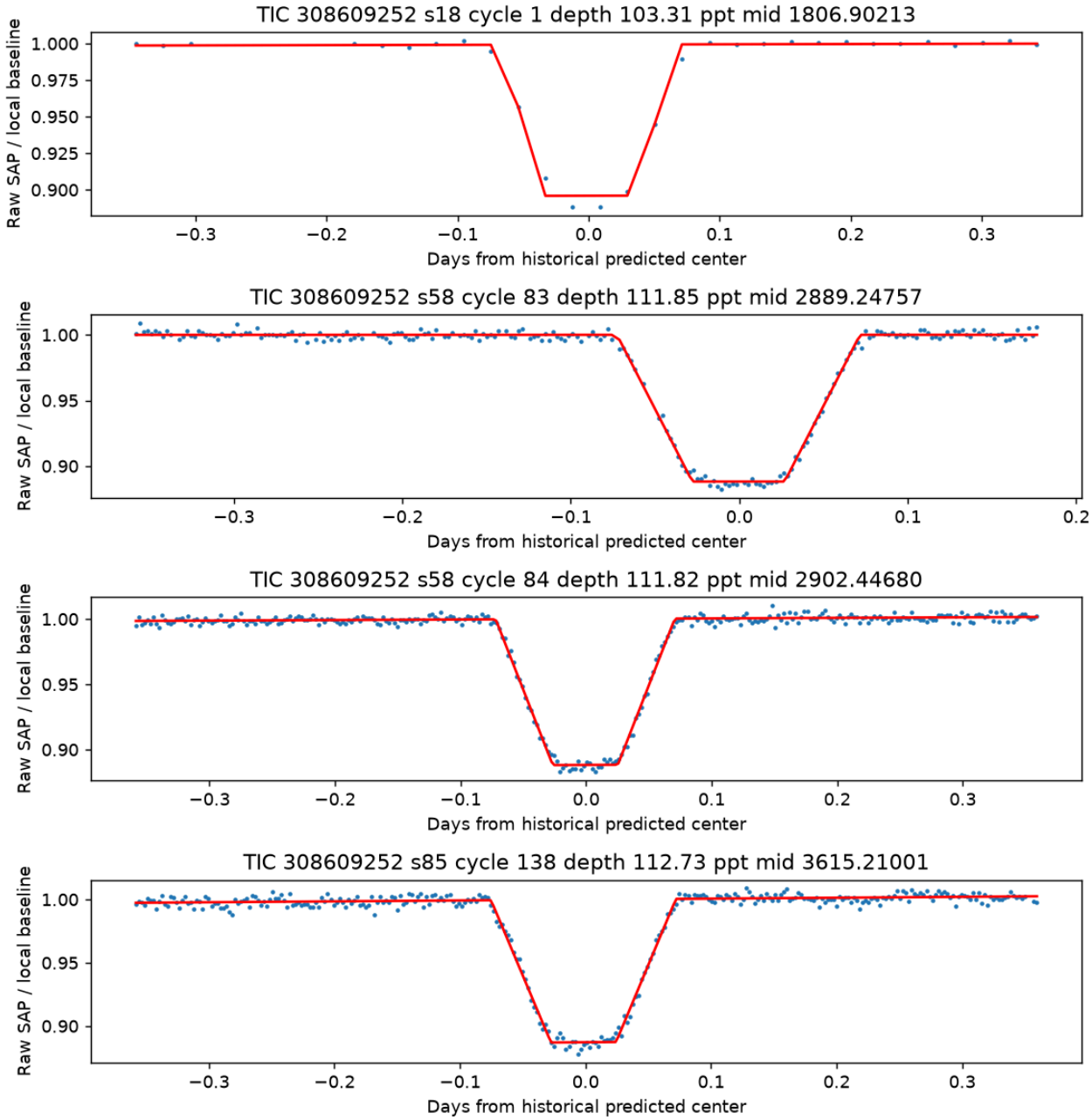
R480. TIC 308609252

physical_verdict: CONFIRMED

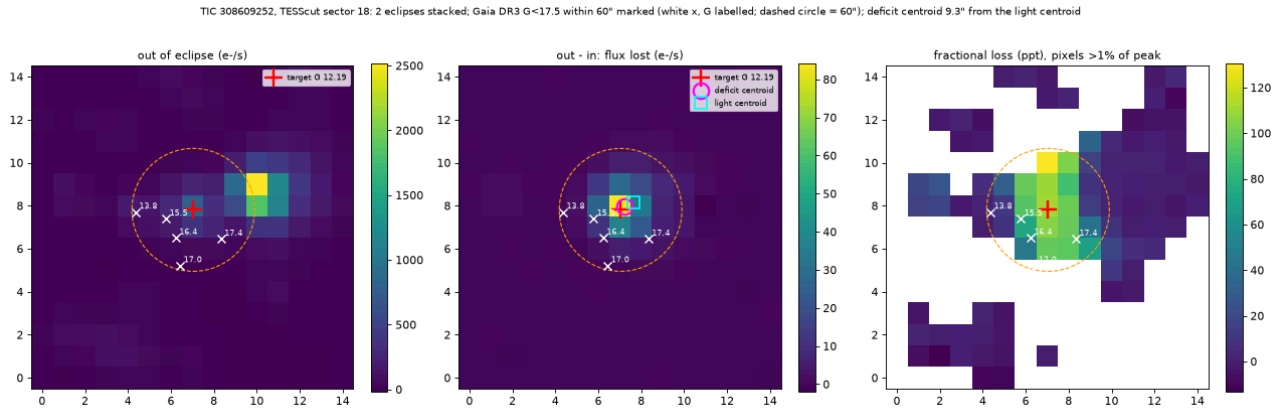
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R480; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R480; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 308609252: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 4 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 18, 58, 85. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 13.1993185849 ± 0.0000252249 d from actual midpoints, epoch BTJD 1806.9033855, cycle 1, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-18 TESScut difference centroid is 5.39 arcsec from the target coordinate; peak pixel [7, 8], target pixel [7, 8]. Fractional image-model residual 0.940 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 111.44/101.21/64.57 ppt across 2 stacked events, versus median raw QLP SAP fit depth 111.83 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 399670247610908160 has orbital P = 13.1964046632 d, e = 0.26890. Closest edge-on conjunction is offset by 0.20557 d (0.01558 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.25980 d. The rough central eclipse duration is 4.27 h versus observed median 3.44 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Four raw events include an adjacent sector-58 pair. The sector-18 difference core is centered near the target, while the only capable catalogued Gaia neighbor is 54.4 arcsec away. The 0.94 image-model residual limits centroid precision.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
18	1	1806.9021315	0.010417	103.31	used
58	83	2889.2475745	0.001157	111.85	used
58	84	2902.4468010	0.001157	111.82	used
85	138	3615.2100090	0.001157	112.73	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
3996731681886722256	54.37	13.801	0.590	60.5

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s18.json`, `historical/diff_s18.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R480. Public evidence files: diff_s18.json, diff_s18.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

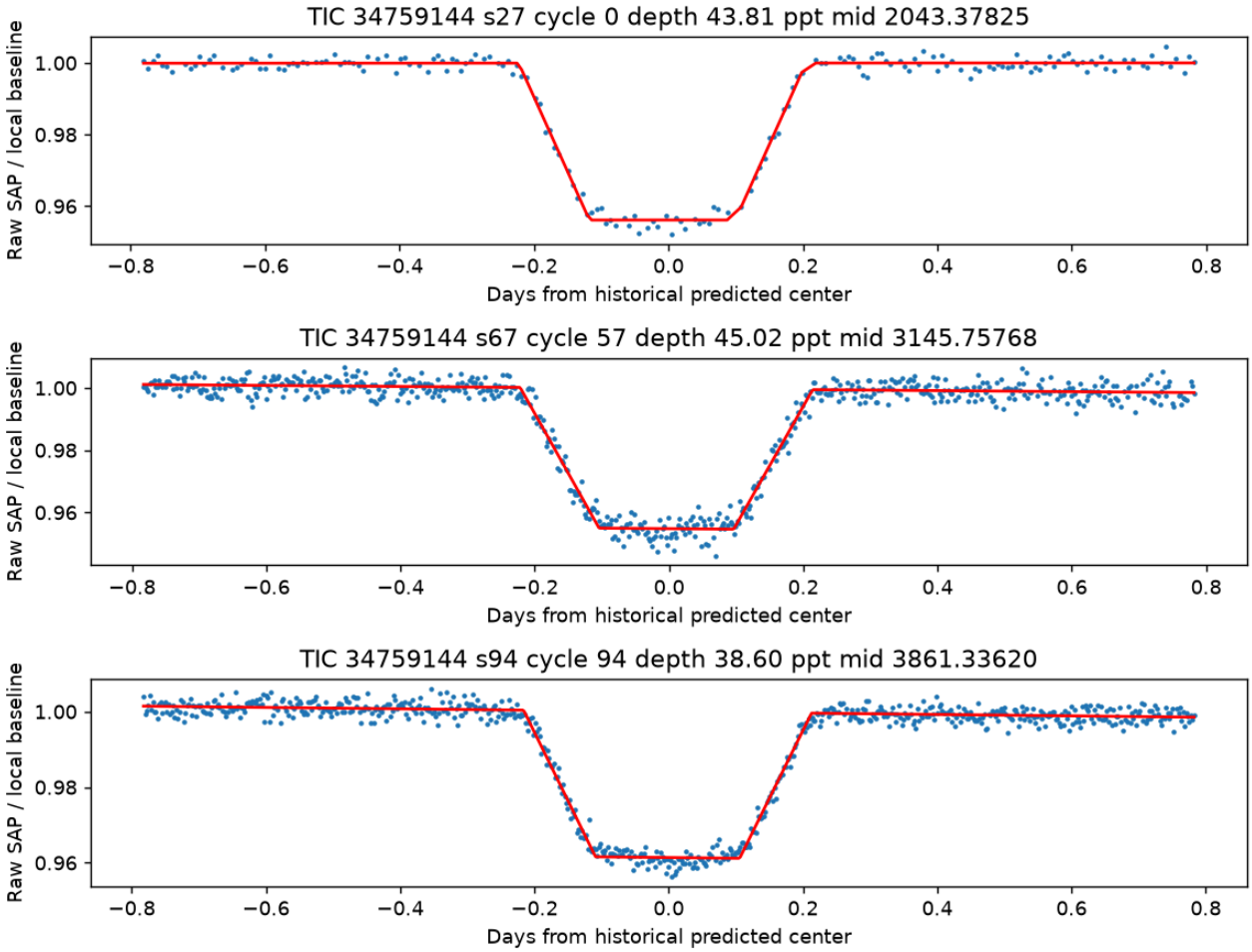
R481. TIC 34759144

physical_verdict: CONFIRMED

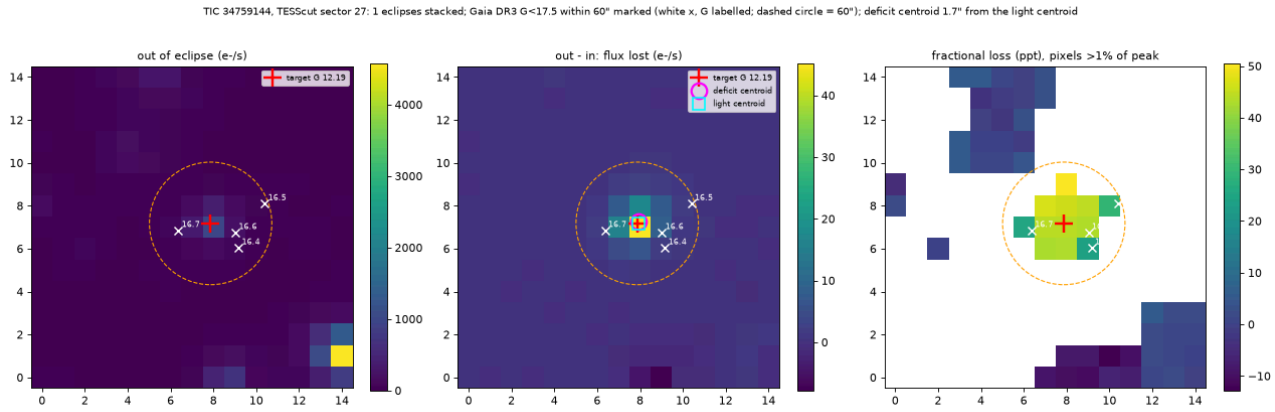
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R481; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R481; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 34759144: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 3 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 27, 67, 94. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 19.3399704780 ± 0.0000322940 d from actual midpoints, epoch BTJD 2043.3791201, cycle 0, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json` and the targeted `../34759144-alias-enumeration.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-27 TESScut difference centroid is 1.75 arcsec from the target coordinate; peak pixel [8, 7], target pixel [8, 7]. Fractional image-model residual 0.252 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 44.28/43.68/41.44 ppt across 1 stacked events, versus median raw QLP SAP fit depth 43.81 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 6696505582479751296 has orbital P = 19.3348568551 d, e = 0.21375. Closest edge-on conjunction is offset by 0.50738 d (0.02624 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.89805 d. The rough central eclipse duration is 14.22 h versus observed median 10.29 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Three clean raw eclipses occur one per selected sector. Enumerating neighboring integer cycle counts gives 57 then 37 cycles at 19.33997 d; the next-best pairs disagree by at least 0.0248 d between gaps. This tests nearby aliases, while subharmonics remain bounded by observed coverage.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
27	0	2043.3782485	0.003472	43.81	used
67	57	3145.7576833	0.001157	45.02	used
94	94	3861.3361958	0.001157	38.60	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s27.json`, `historical/diff_s27.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in` ../source-hashes.json`)`. `Source code and retained evidence are listed in` ../SHA256SUMS.json`.`

Stable review key: R481. Public evidence files: diff_s27.json, diff_s27.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

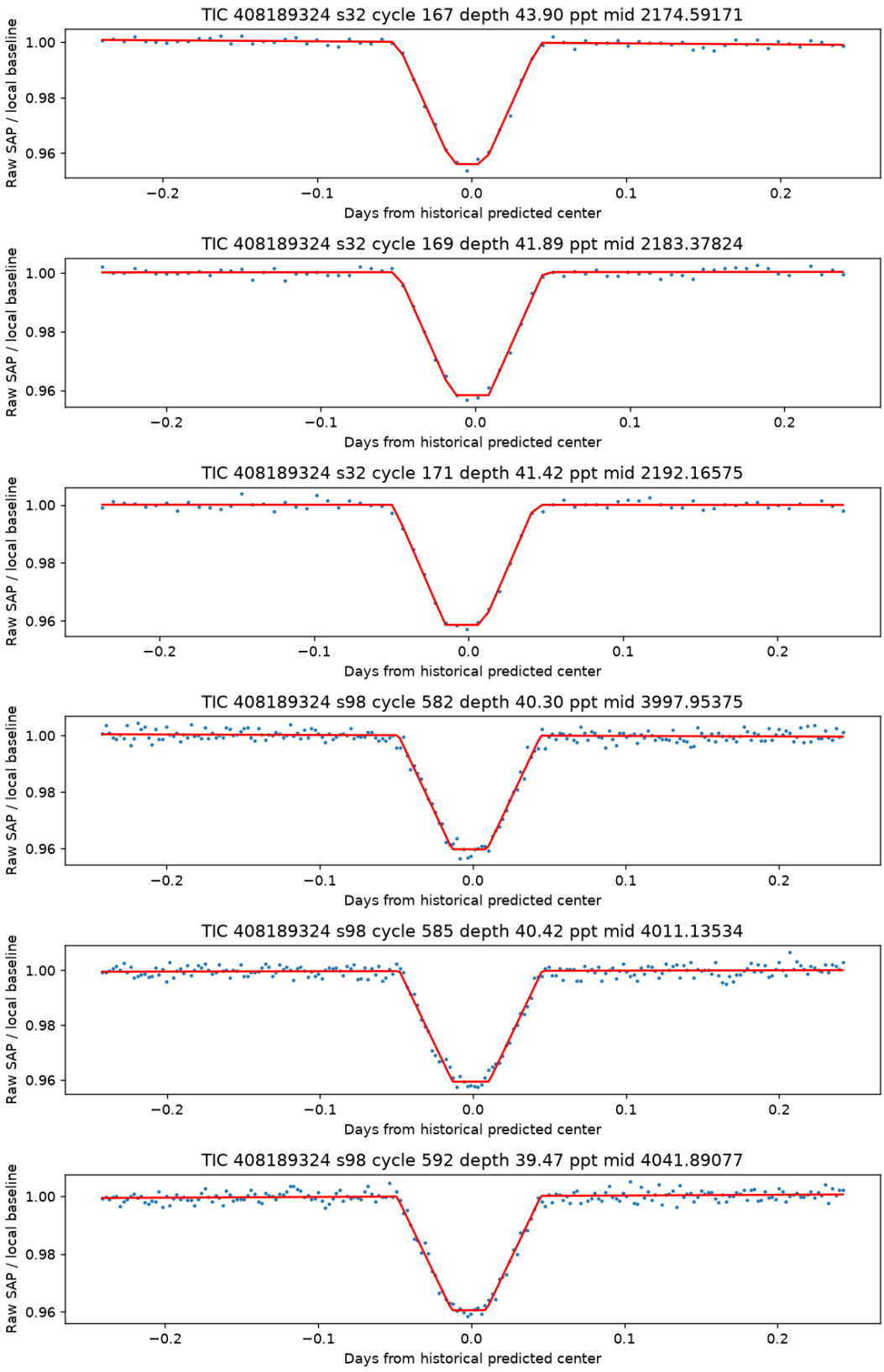
R482. TIC 408189324

physical_verdict: CONFIRMED

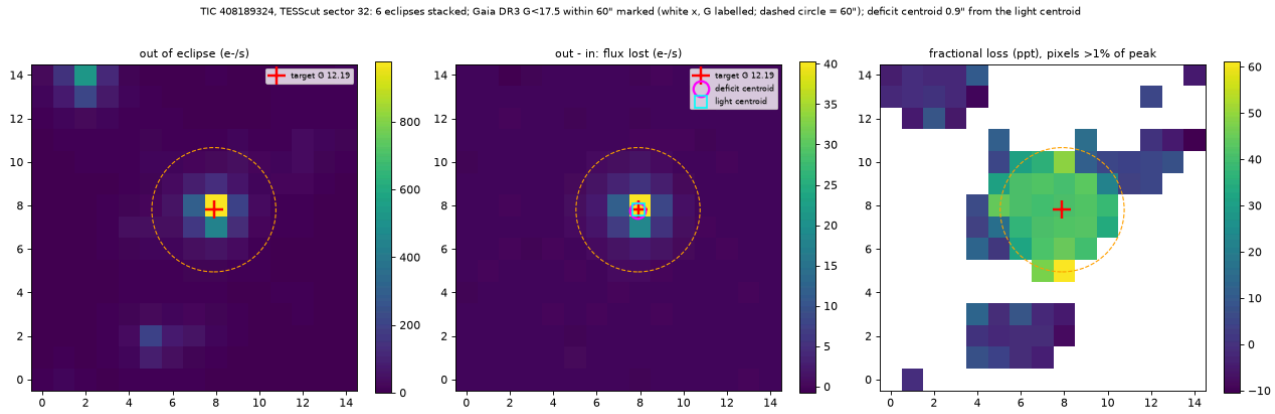
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R482; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R482; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 408189324: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 12 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 32, 98. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 4.3936455322 ± 0.0000035849 d from actual midpoints, epoch BTJD 2174.5912708, cycle 167, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-32 TESScut difference centroid is 0.48 arcsec from the target coordinate; peak pixel [8, 8], target pixel [8, 8]. Fractional image-model residual 0.077 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 41.21/41.38/40.39 ppt across 6 stacked events, versus median raw QLP SAP fit depth 40.93 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 2969339250844920832 has orbital P = 4.3934150548 d, e = 0.29683. Closest edge-on conjunction is offset by 0.03354 d (0.00763 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.05028 d. The rough central eclipse duration is 3.93 h versus observed median 2.22 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Six consecutive sector-32 events and six sector-98 events support the same 4.39365-day cycle. The deficit is sharply centered on the target and no capable Gaia neighbor is listed within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
32	167	2174.5917124	0.003472	43.90	used
32	168	2178.9850342	0.003472	43.13	used
32	169	2183.3782386	0.003472	41.89	used
32	170	2187.7723515	0.003472	42.88	used
32	171	2192.1657544	0.003472	41.42	used
32	172	2196.5592879	0.003472	43.67	used
98	582	3997.9537519	0.001157	40.30	used
98	583	4002.3475700	0.001157	39.77	used
98	585	4011.1353392	0.001157	40.42	used
98	586	4015.5288585	0.001157	39.92	used
98	589	4028.7098454	0.001157	40.44	used
98	592	4041.8907654	0.001157	39.47	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s32.json`, `historical/diff_s32.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json)`. `Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R482. Public evidence files: diff_s32.json, diff_s32.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

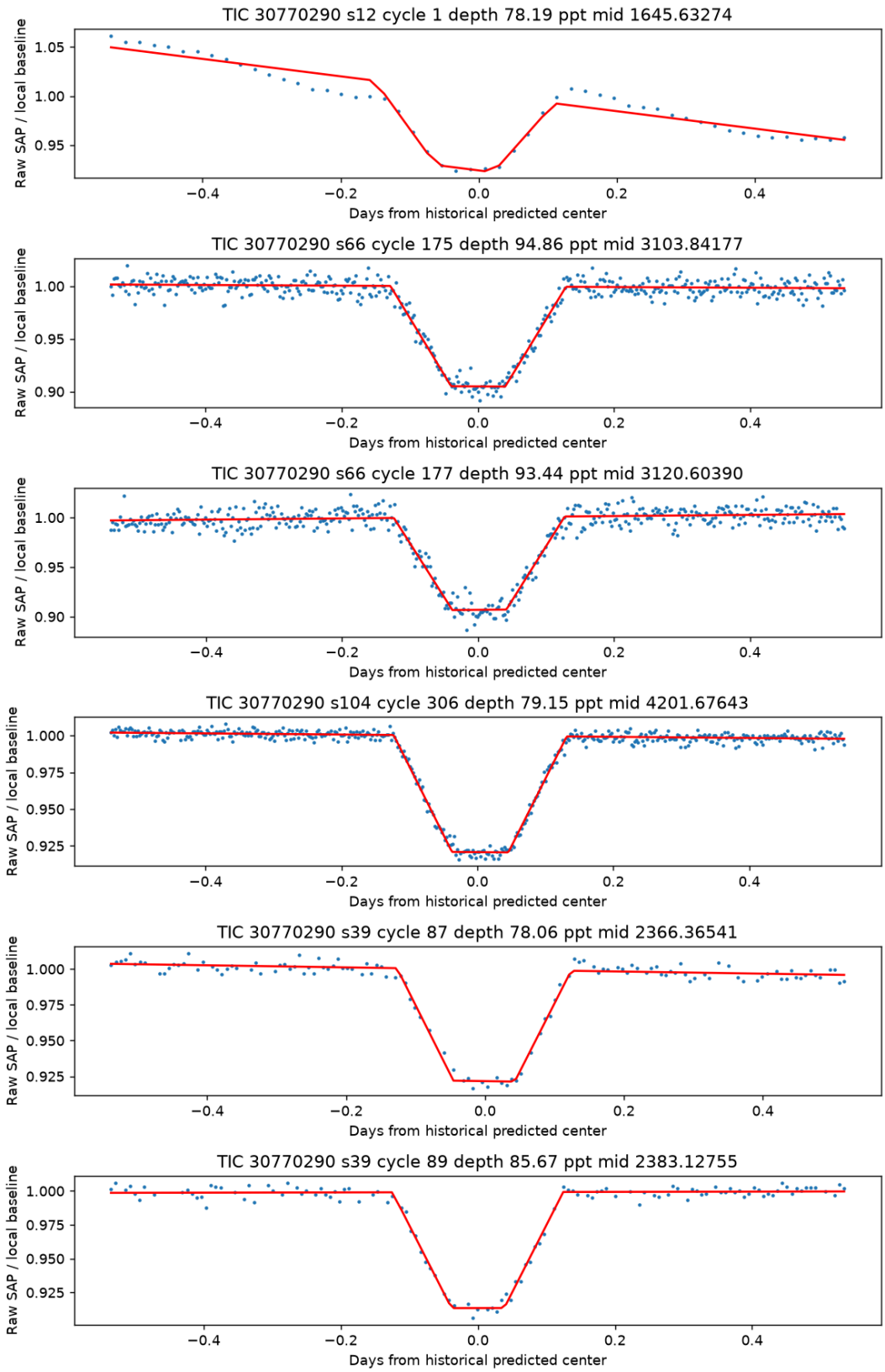
R483. TIC 30770290

physical_verdict: CONFIRMED

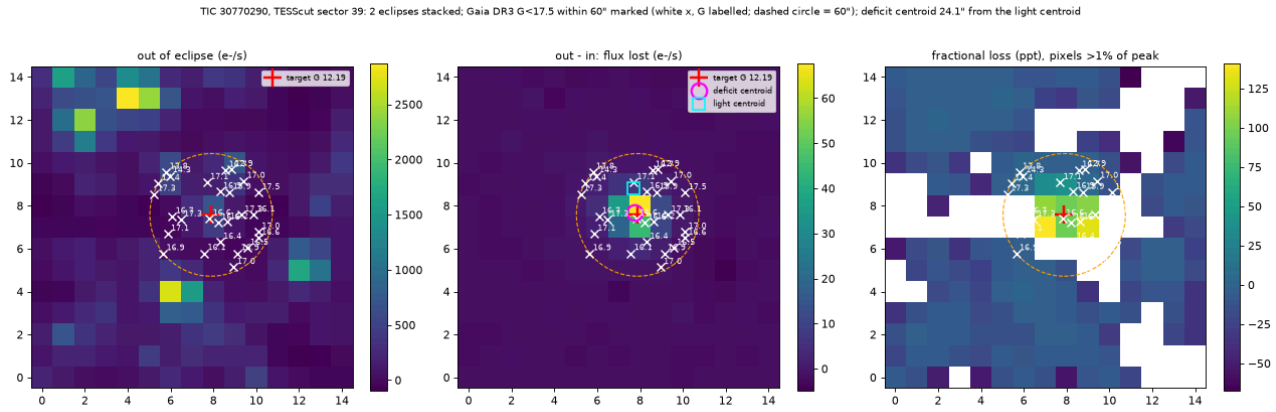
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R483; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R483; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 30770290: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 6 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 12, 39, 66, 104. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 8.3804132410 ± 0.0000094619 d from actual midpoints, epoch BTJD 1645.6504646, cycle 1, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-39 TESScut difference centroid is 3.33 arcsec from the target coordinate; peak pixel [8, 8], target pixel [8, 8]. Fractional image-model residual 0.988 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 96.05/80.93/45.08 ppt across 2 stacked events, versus median raw QLP SAP fit depth 82.41 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 5955349082236051072 has orbital P = 8.3812349333 d, e = 0.09409. Closest edge-on conjunction is offset by -0.18507 d (-0.02208 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.15637 d. The rough central eclipse duration is 7.78 h versus observed median 6.07 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Sector-39 and sector-66 measured pairs are separated by two historical cycles, but independent inspection finds a real ~69-ppt usable-core deficit at intervening sector-66 cycle 176. Its partial coverage prevents a reliable midpoint fit; it supports the 8.3804-day recurrence over a simple 16.76-day parity alias. Sector-39 cycle 88 is a gap, not a null. The high image-model residual limits localization precision.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	1	1645.6327354	0.021612	78.19	used
39	87	2366.3654077	0.003472	78.06	used
39	89	2383.1275530	0.003472	85.67	used
66	175	3103.8417724	0.001157	94.86	used
66	177	3120.6038978	0.001157	93.44	used
104	306	4201.6764331	0.001157	79.15	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5955349494552910848	48.59	12.925	0.251	48.3
5955349872514790784	54.94	14.307	0.678	52.7
5955349872510033024	59.62	13.836	0.469	57.4

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s39.json`, `historical/diff_s39.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R483. Public evidence files: diff_s39.json, diff_s39.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

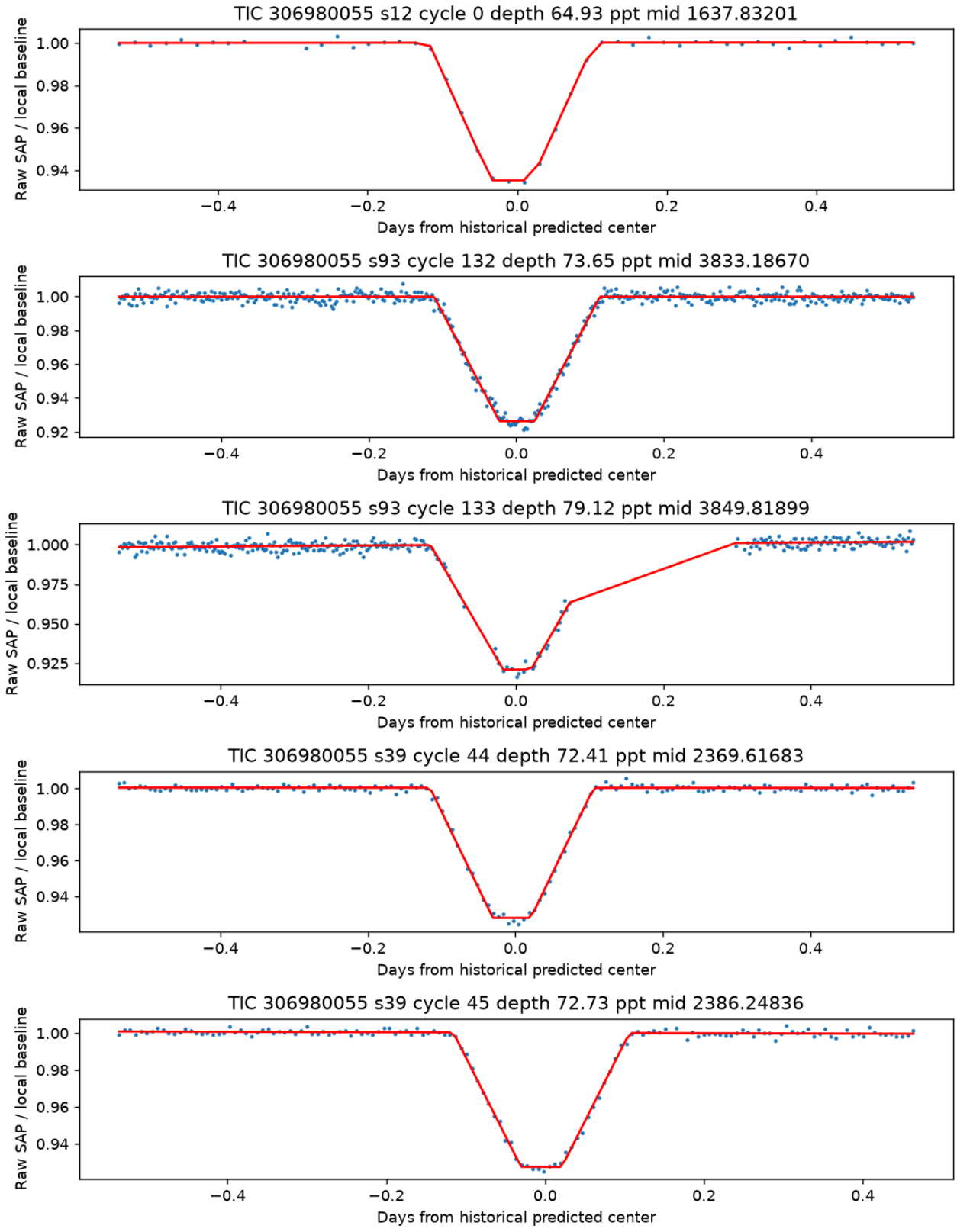
R484. TIC 306980055

physical_verdict: CONFIRMED

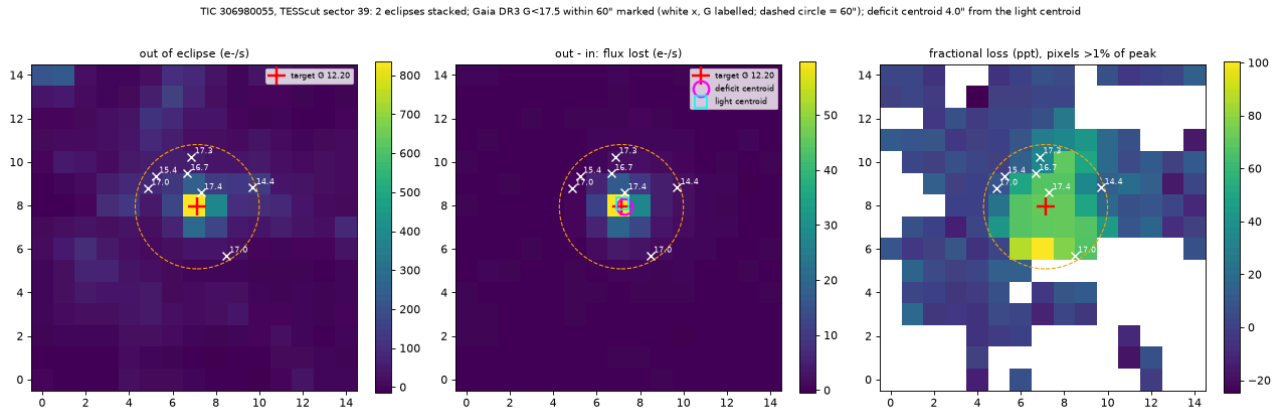
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R484; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R484; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 306980055: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 5 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 12, 39, 93. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 16.6314800623 ± 0.0000277694 d from actual midpoints, epoch BTJD 1637.8317347, cycle 0, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-39 TESScut difference centroid is 3.01 arcsec from the target coordinate; peak pixel [7, 8], target pixel [7, 8]. Fractional image-model residual 0.328 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 71.49/69.24/62.01 ppt across 2 stacked events, versus median raw QLP SAP fit depth 72.73 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 5911278044857572480 has orbital P = 16.6305032216 d, e = 0.36931. Closest edge-on conjunction is offset by -0.13935 d (-0.00838 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.29122 d. The rough central eclipse duration is 7.65 h versus observed median 5.32 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Measured sector-39 and sector-93 adjacent pairs span the long baseline. The capable 55.8-arcsec Gaia neighbor is distinct from the target-centered difference core.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
12	0	1637.8320133	0.010417	64.93	used
39	44	2369.6168339	0.003472	72.41	used
39	45	2386.2483563	0.003472	72.73	used
93	132	3833.1866950	0.001157	73.65	used
93	133	3849.8189880	0.001157	79.12	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5909777043986816768	55.81	14.437	0.578	54.0

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s39.json`, `historical/diff_s39.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json)`. `Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R484. Public evidence files: diff_s39.json, diff_s39.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

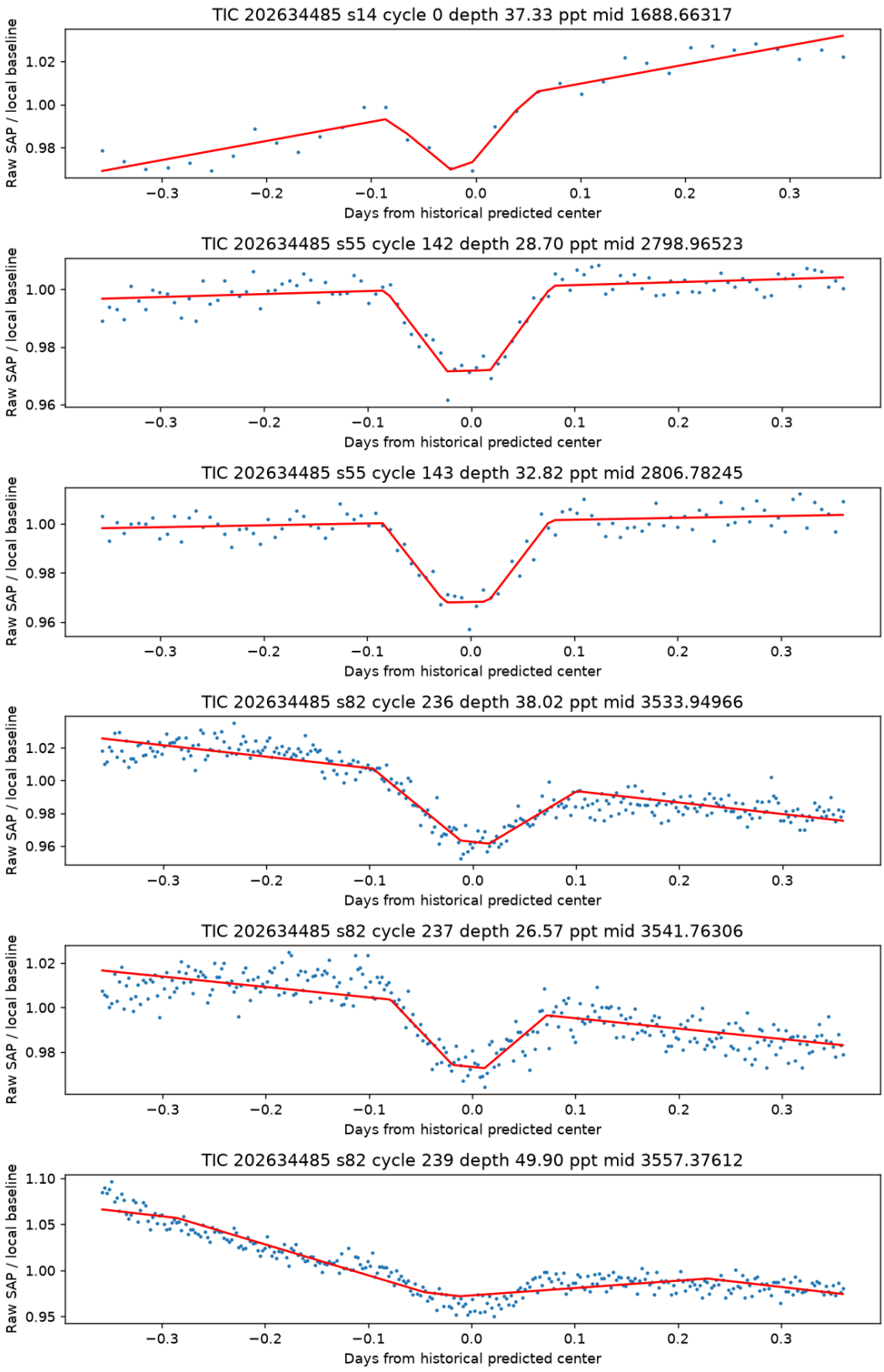
R485. TIC 202634485

physical_verdict: PROBABLE

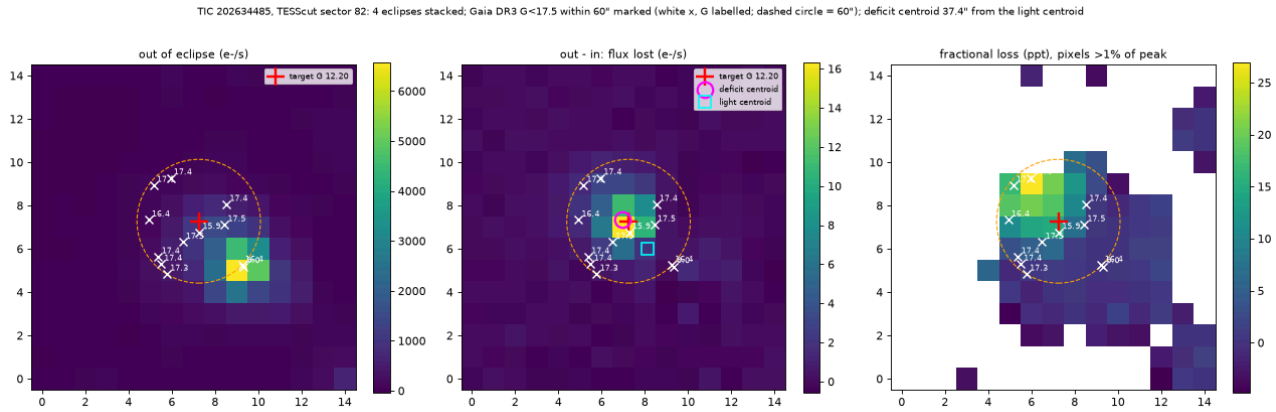
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source pixel-response fitting of the 10.1-arcsec pair.



R485; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R485; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 202634485: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 8 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 14, 55, 82. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 7.8189267254 ± 0.0000356216 d from actual midpoints, epoch BTJD 1688.6785071, cycle 0, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** unresolved. The historical sector-82 TESScut difference centroid is 5.99 arcsec from the target coordinate; peak pixel [7, 7], target pixel [7, 7]. Fractional image-model residual 0.968 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 14.95/8.35/3.26 ppt across 4 stacked events, versus median raw QLP SAP fit depth 33.47 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 2055792762025890176 has orbital P = 7.8183725267 d, e = 0.24161. Closest edge-on conjunction is offset by 0.06916 d (0.00885 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.27464 d. The rough central eclipse duration is 6.77 h versus observed median 3.87 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Sector-55 has three consecutive clean raw eclipses. The sector-82 raw baseline is structured; its same-photon custom-aperture signal is much shallower than QLP SAP, with crowding, aperture and baseline sensitivity likely contributing. A Gaia neighbor 10.1 arcsec away could in principle supply the event if nearly fully eclipsed, and the image-model residual is 0.968. Target attribution remains unresolved.

Next check: Resolved eclipse photometry or multi-source pixel-response fitting of the 10.1-arcsec pair.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
14	0	1688.6631740	0.033133	37.33	used
55	142	2798.9652268	0.003472	28.70	used
55	143	2806.7824513	0.003472	32.82	used
55	144	2814.6078397	0.003472	28.32	used
82	236	3533.9496625	0.006967	38.02	used
82	237	3541.7630557	0.005726	26.57	used
82	238	3549.5839019	0.003034	34.12	used
82	239	3557.3761215	0.012302	49.90	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
2055792766332050688	10.09	15.945	0.938	13.9
2055792727666151552	58.25	9.031	0.030	67.2

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s82.json`, `historical/diff_s82.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R485. Public evidence files: diff_s82.json, diff_s82.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

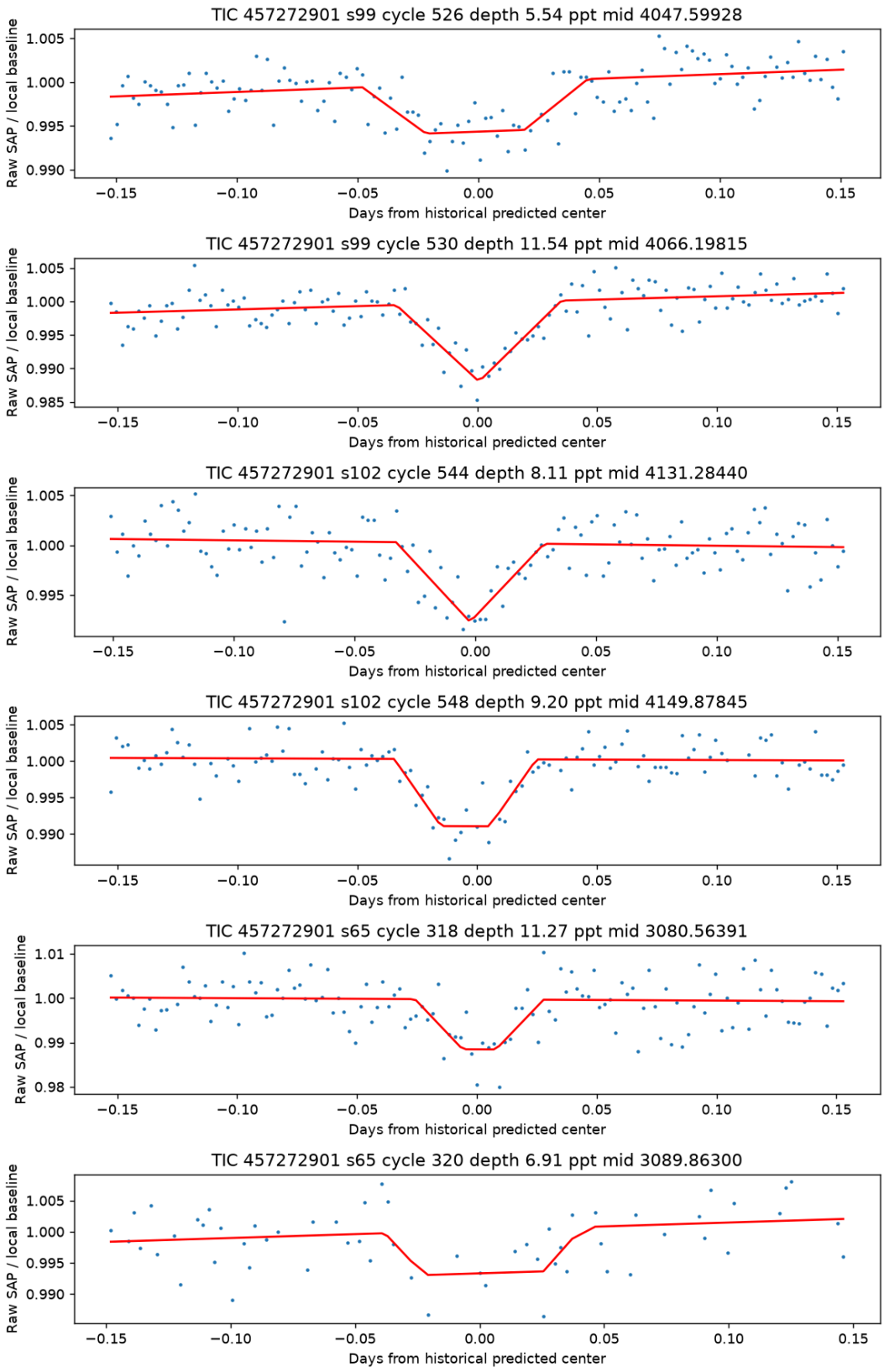
R486. TIC 457272901

physical_verdict: CONFIRMED

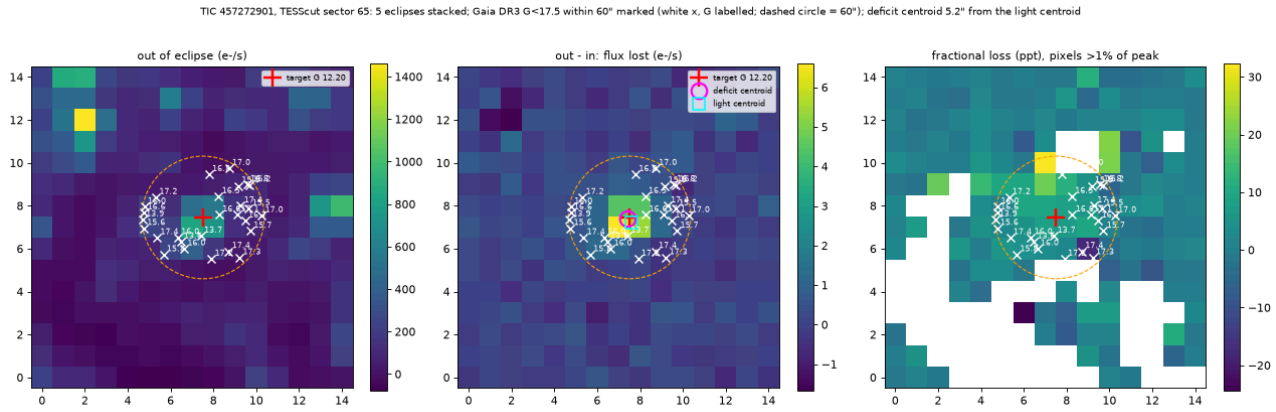
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R486; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R486; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 457272901: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 11 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 65, 99, 102. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 4.6492170041 ± 0.0000077651 d from actual midpoints, epoch BTJD 3071.2640710, cycle 316, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-65 TESScut difference centroid is 2.05 arcsec from the target coordinate; peak pixel [7, 7], target pixel [8, 7]. Fractional image-model residual 0.474 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 7.12/6.92/6.01 ppt across 5 stacked events, versus median raw QLP SAP fit depth 8.09 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 5869945139392446336 has orbital P = 4.6494664348 d, e = 0.06671. Closest edge-on conjunction is offset by -0.13084 d (-0.02814 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.36286 d. The rough central eclipse duration is 3.53 h versus observed median 1.47 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: The added sector-65 raw events anchor the 4.649217-day timing baseline; later sector-99/102 troughs repeat. The difference centroid is 2.1 arcsec from the target and the closest capable neighbor is about 17 arcsec from that centroid. Twenty-seven capable Gaia sources lie within 60 arcsec; the crowded image and uncalibrated residual limit precision, but the spatial deficit favors the target.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
65	316	3071.2627207	0.001744	9.13	used
65	318	3080.5639080	0.001896	11.27	used
65	319	3085.2108687	0.001497	7.92	used
65	320	3089.8630025	0.003589	6.91	used
99	526	4047.5992838	0.002607	5.54	used
99	527	4052.2519805	0.001235	7.51	used
99	530	4066.1981510	0.001357	11.54	used
99	531	4070.8470307	0.001829	8.09	used
102	544	4131.2844012	0.001576	8.11	used
102	545	4135.9353243	0.001157	7.21	used
102	548	4149.8784550	0.001157	9.20	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5869942184454493952	15.81	16.636	0.461	18.0
5869945139392446080	18.08	13.674	0.037	17.1
5869945139392446592	25.28	16.453	0.390	27.5
5869939263875939200	30.92	15.973	0.254	29.4
5869939263875937536	31.84	13.624	0.036	30.3
5869942184454489344	33.36	17.052	0.672	35.8
5869939263875934336	35.40	16.042	0.270	34.0
5869942184454491520	38.07	17.429	0.947	40.8
5869942184454483712	40.77	16.711	0.493	43.2
5869936308938425216	41.37	17.197	0.767	40.9
5869942184389479424	41.48	17.384	0.910	42.0
5869945139392447744	42.24	16.755	0.513	43.7
5869942184454487552	44.96	15.517	0.169	47.7

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5869942218814236544	45.73	15.875	0.232	48.4
5869942184454478976	47.67	15.702	0.199	49.8
5869939332595420544	48.37	17.389	0.914	47.2
5869945203761021568	49.50	17.171	0.749	48.9
5869939332595414400	52.30	15.832	0.224	51.1
5869942184468745088	52.87	17.290	0.834	53.7
5869942218814234240	53.06	16.785	0.527	56.0
5869942218814243584	53.79	16.964	0.620	56.1
5869942218814234112	54.64	15.222	0.131	57.6
5869942115735005312	56.67	17.032	0.660	59.5
586993932822889 8560	56.94	16.642	0.463	56.2
5869939332595433344	57.44	16.012	0.262	56.7
5869939332595426432	57.93	15.601	0.182	57.0
5869939332595430144	58.16	13.894	0.044	57.3

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s65.json`, `historical/diff_s65.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R486. Public evidence files: diff_s65.json, diff_s65.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

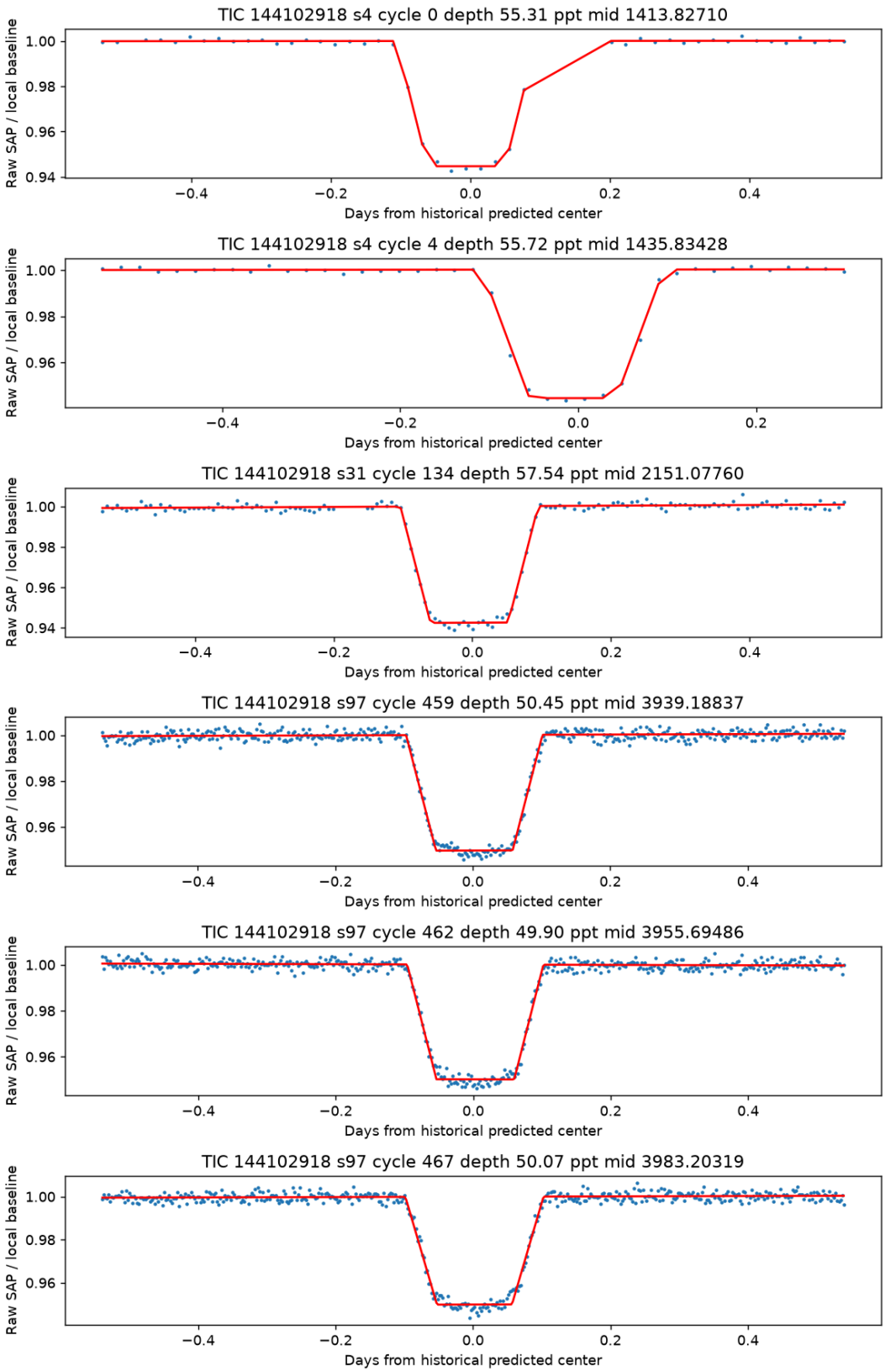
R487. TIC 144102918

physical_verdict: CONFIRMED

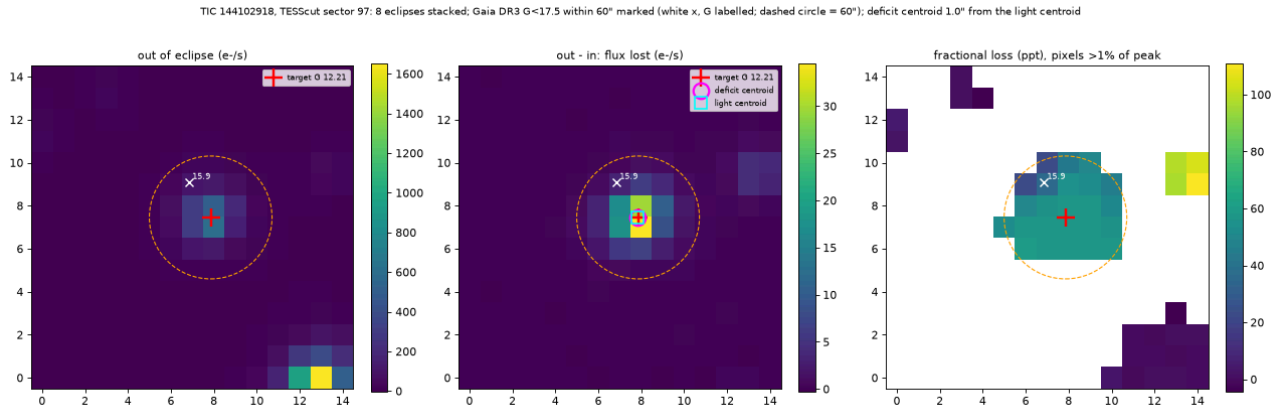
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R487; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R487; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 144102918: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 13 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 4, 31, 97. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 5.5018791815 ± 0.0000050961 d from actual midpoints, epoch BTJD 1413.8257574, cycle 0, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-97 TESScut difference centroid is 0.50 arcsec from the target coordinate; peak pixel [8, 7], target pixel [8, 7]. Fractional image-model residual 0.058 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 57.89/57.27/55.49 ppt across 8 stacked events, versus median raw QLP SAP fit depth 54.66 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 4841921897857941248 has orbital P = 5.5016102193 d, e = 0.07812. Closest edge-on conjunction is offset by 0.05387 d (0.00979 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.10114 d. The rough central eclipse duration is 4.99 h versus observed median 4.77 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Thirteen measured eclipses span sectors 4, 31 and 97, including adjacent cycles. Difference imaging is sharply target-centered with no capable Gaia neighbor in the 60-arcsec screen.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
4	0	1413.8270960	0.010417	55.31	used
4	3	1430.3331928	0.010417	54.66	used
4	4	1435.8342838	0.010417	55.72	used
31	133	2145.5757425	0.003472	57.84	used
31	134	2151.0776028	0.003472	57.54	used
31	136	2162.0809820	0.003472	58.09	used
31	137	2167.5827793	0.003472	58.95	used
97	459	3939.1883705	0.001157	50.45	used
97	460	3944.6897340	0.001157	50.13	used
97	462	3955.6948555	0.001157	49.90	used
97	464	3966.6976092	0.001157	49.73	used
97	465	3972.1992965	0.001157	50.07	used
97	467	3983.2031899	0.001157	50.07	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s97.json`, `historical/diff_s97.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R487. Public evidence files: diff_s97.json, diff_s97.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

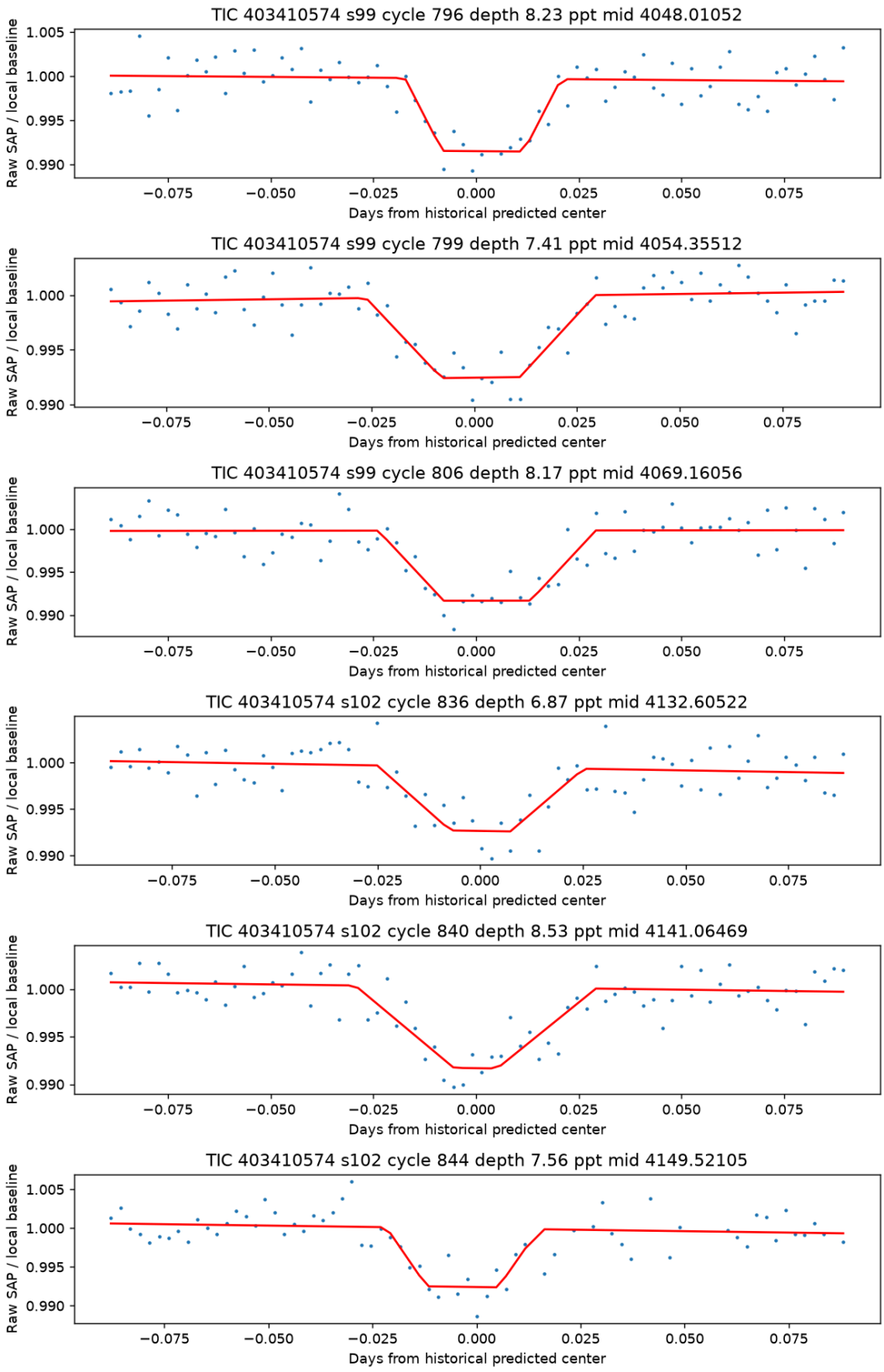
R488. TIC 403410574

physical_verdict: PROBABLE

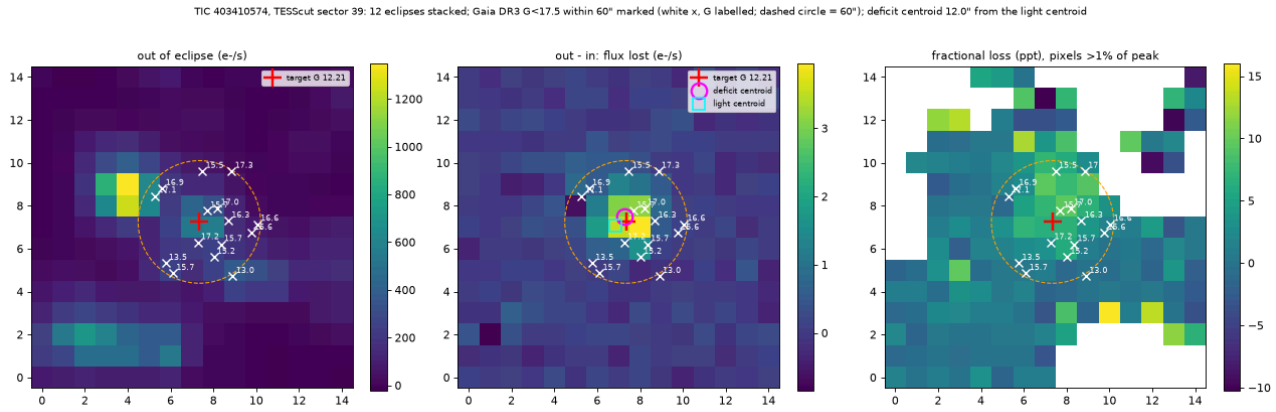
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source pixel-response fitting of the 12.2-arcsec pair.



R488; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R488; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 403410574: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 16 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 99, 102. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 2.1148563262 ± 0.0000186810 d from actual midpoints, epoch BTJD 4048.0101157, cycle 796, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** unresolved. The historical sector-39 TESScut difference centroid is 5.38 arcsec from the target coordinate; peak pixel [8, 7], target pixel [7, 7]. Fractional image-model residual 0.896 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 6.46/6.39/3.75 ppt across 12 stacked events, versus median raw QLP SAP fit depth 7.55 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 5792684548884156800 has orbital P = 2.1148150688 d, e = 0.11270. Closest edge-on conjunction is offset by 0.12513 d (0.05917 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.15197 d. The rough central eclipse duration is 3.34 h versus observed median 1.20 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Sixteen measured shallow but discrete raw eclipses in sectors 99 and 102 establish a repeated 2.11486-day signal. The early sector-39 cadence is too sparse for the local trapezoid timing criterion, although its stacked custom aperture shows the event. A capable Gaia neighbor 12.2 arcsec from the target lies only 10.9 arcsec from the deficit centroid; the residual is 0.896. TESS pixels do not settle attribution.

Next check: Resolved eclipse photometry or multi-source pixel-response fitting of the 12.2-arcsec pair.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
99	796	4048.0105152	0.001157	8.23	used
99	797	4050.1226481	0.001157	9.29	used
99	798	4052.2403352	0.001157	9.02	used
99	799	4054.3551161	0.001157	7.41	used
99	804	4064.9275859	0.001224	7.78	used
99	805	4067.0446358	0.001614	7.45	used
99	806	4069.1605614	0.001162	8.17	used
102	834	4128.3760132	0.001412	6.84	used
102	835	4130.4900083	0.001157	7.18	used
102	836	4132.6052166	0.001538	6.87	used
102	837	4134.7170281	0.001157	7.54	used
102	838	4136.8335795	0.002748	6.88	used
102	840	4141.0646915	0.001157	8.53	used
102	841	4143.1801215	0.001203	7.13	used
102	843	4147.4086149	0.001170	7.82	used
102	844	4149.5210460	0.001157	7.56	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5792684347027309312	12.23	15.744	0.202	10.9
5792684518826000768	18.96	17.228	0.771	25.2
5792684347027309184	21.08	16.978	0.614	21.3
5792684347027308416	27.46	16.298	0.332	30.4
5792684347027308032	29.35	15.674	0.190	35.4
5792684312667569408	35.36	15.218	0.128	42.6
5794185764514224512	46.47	15.531	0.168	44.8
5792684548884947072	47.39	16.886	0.565	44.2
5792684518826001408	49.08	13.476	0.032	55.2
5792684553185740928	49.17	17.116	0.697	46.4
5792684347027307008	50.48	15.561	0.172	54.8

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
5792684518826000896	52.86	15.681	0.191	60.4
5794185554054569856	55.03	17.267	0.800	55.8
5792684347027445120	55.22	16.618	0.443	59.0
5792684617604424064	55.68	17.504	0.992	53.2
5792684312667567744	59.84	13.035	0.024	67.8

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s39.json`, `historical/diff_s39.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R488. Public evidence files: diff_s39.json, diff_s39.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

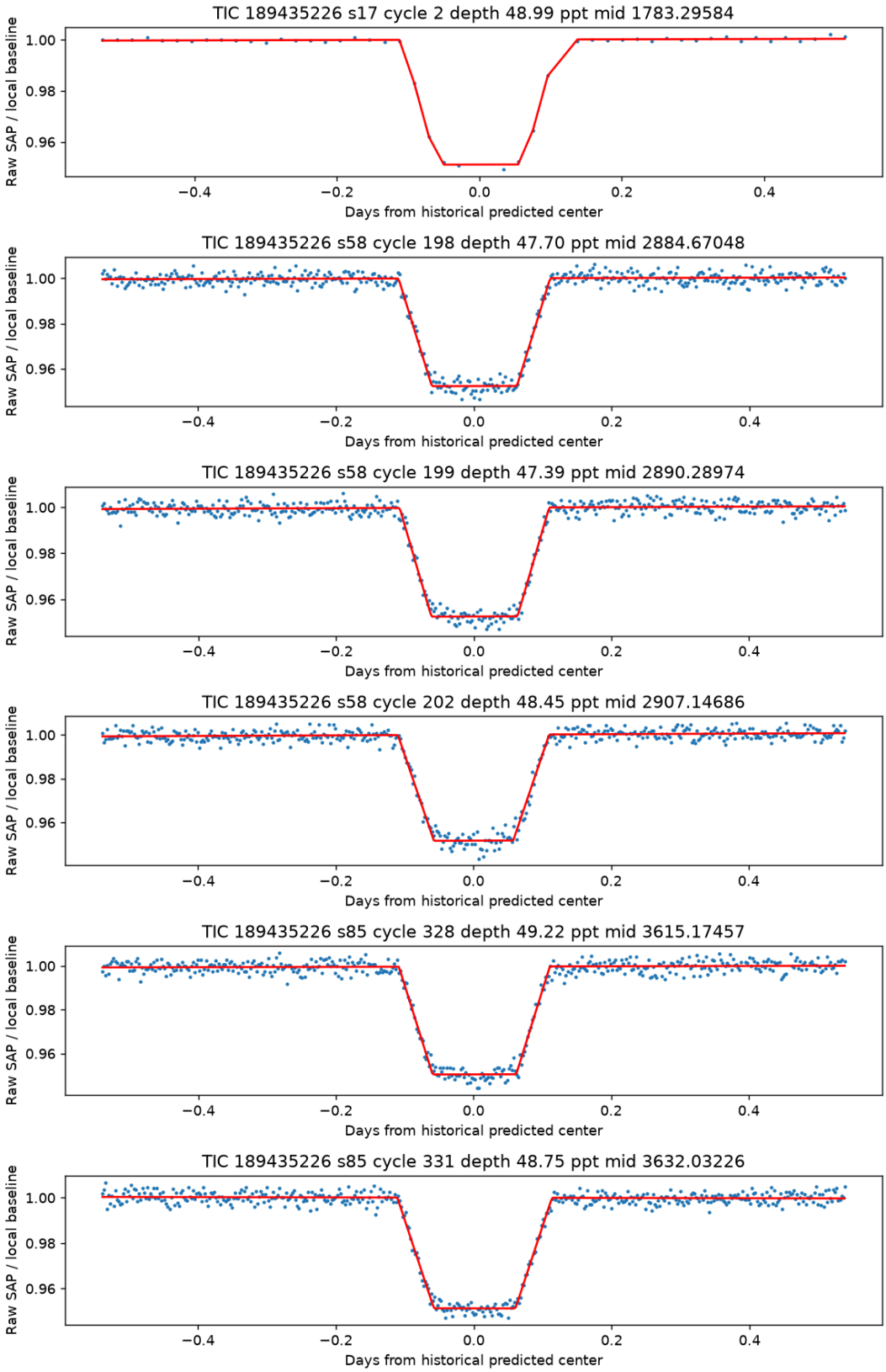
R489. TIC 189435226

physical_verdict: CONFIRMED

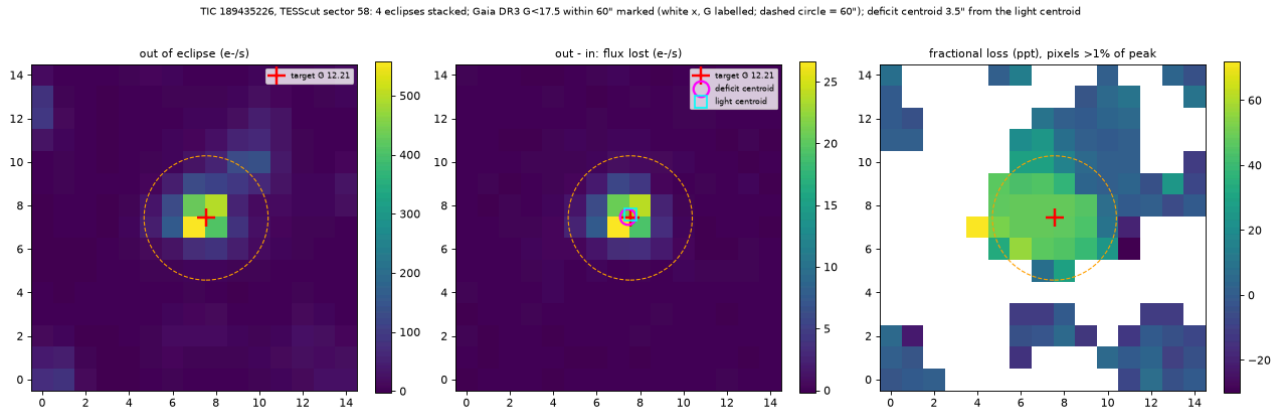
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R489; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R489; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 189435226: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate from the historical screening label.

1. **Raw morphology:** 8 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across sectors 17, 58, 85. Local trapezoids fit free centers, depths, widths and baselines. Predicted centers only selected windows; they were not timing points. The raw window CSV and plot preserve the observations. Flagged or uncovered windows are not clean nulls.
2. **Timing and alias:** P = 5.6192618708 ± 0.0000067407 d from actual midpoints, epoch BTJD 1783.2951463, cycle 2, kind: Selected repeating eclipse; deeper/secondary identity not independently established. Within-sector recurrence and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check rather than an exhaustive all-period search.
3. **Source attribution:** target favored at TESS resolution. The historical sector-58 TESScut difference centroid is 3.40 arcsec from the target coordinate; peak pixel [7, 7], target pixel [8, 7]. Fractional image-model residual 0.231 is not a calibrated localization error. Every capable Gaia neighbor in the 60-arcsec screen is listed below; its required intrinsic eclipse fraction is an approximate flux-budget bound, not a blended pixel fractional loss.
4. **Alternative reduction:** A separately extracted, target-centered, background-subtracted TESScut 1x1/3x3/5x5-pixel aperture gives 47.17/47.24/43.82 ppt across 4 stacked events, versus median raw QLP SAP fit depth 48.53 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Crowding, aperture dilution and different event selections limit depth equality.
5. **Gaia geometry:** DR3 source 323765286292704768 has orbital P = 5.6182753682 d, e = 0.05704. Closest edge-on conjunction is offset by 0.10708 d (0.01906 phase) at the measured epoch; full SB1 covariance gives descriptive linear timing error 0.13963 d. The rough central eclipse duration is 6.01 h versus observed median 5.27 h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility check, not a fitted binary model or a sigma veto. Agreement cannot identify an unresolved eclipsing neighbor.

Deciding interpretation and limitation: Eight raw eclipses span sectors 17, 58 and 85, with adjacent cycles in both later sectors. The difference image favors the target and no capable Gaia neighbor is listed within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: The central seven-group result is in `../../shared-catalogues/coverage-matrix.csv`: complete scoped checks and no identified prior eclipsing-binary match for this TIC. This does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
17	2	1783.2958411	0.010417	48.99	used
58	198	2884.6704822	0.001157	47.70	used
58	199	2890.2897400	0.001157	47.39	used
58	201	2901.5288785	0.001157	47.77	used
58	202	2907.1468576	0.001157	48.45	used
85	328	3615.1745664	0.001157	49.22	used
85	329	3620.7937892	0.001157	48.62	used
85	331	3632.0322602	0.001157	48.75	used

Capable Gaia neighbors in the historical difference-image 60-arcsec screen:

Gaia DR3 source	Target separation arcsec	G	Approximate required intrinsic eclipse fraction	Deficit-centroid separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../targeted-download-receipts.json`, `../gaia-geometry.json`, `../alias-audit.json`, `historical/diff_s58.json`, `historical/diff_s58.png`, `historical/gaia.json`, and `../helper-audit.md`. Original TESScut: `[local path omitted] (SHA256 in../source-hashes.json). Source code and retained evidence are listed in../SHA256SUMS.json`.

Stable review key: R489. Public evidence files: diff_s58.json, diff_s58.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

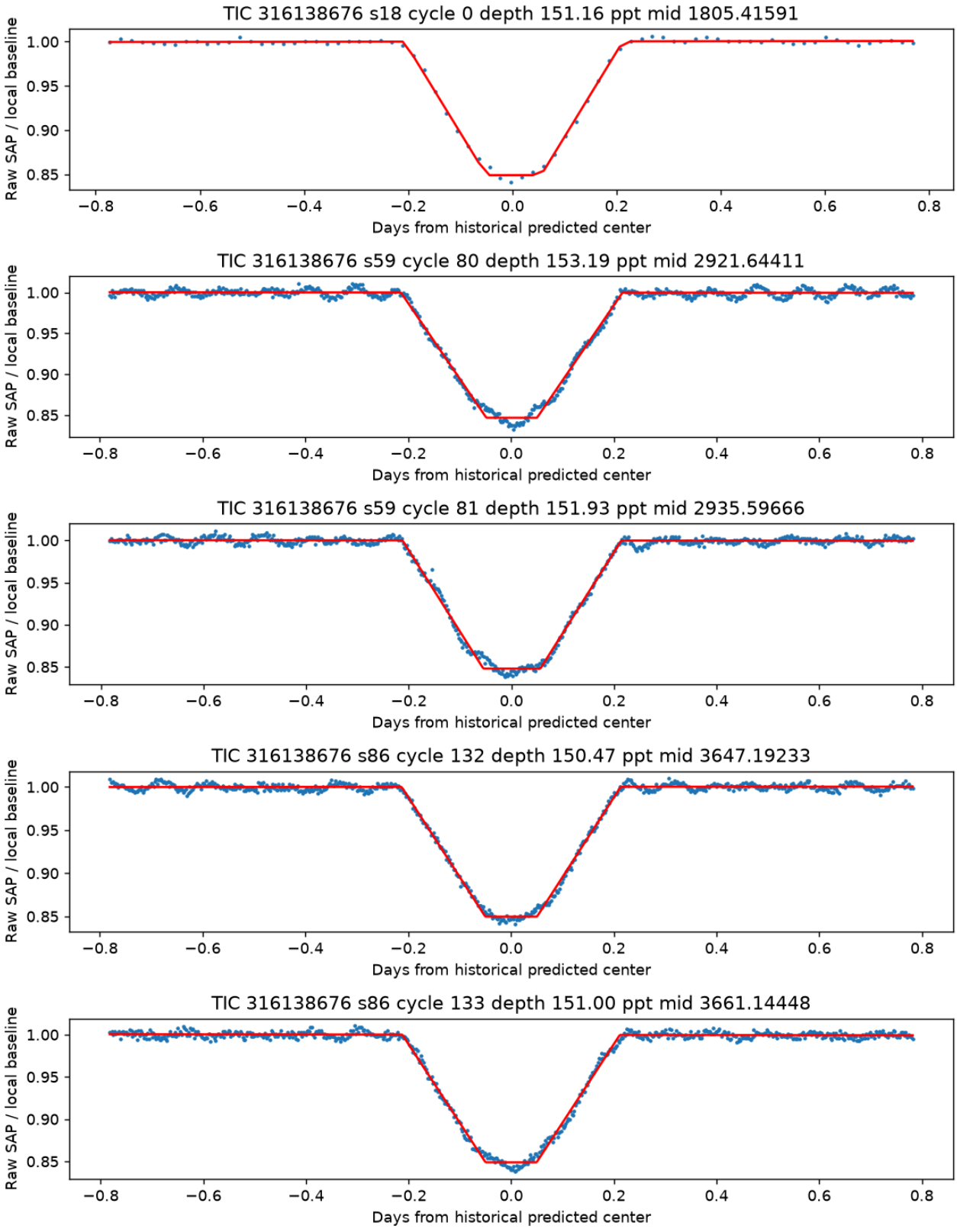
R490. TIC 316138676

physical_verdict: CONFIRMED

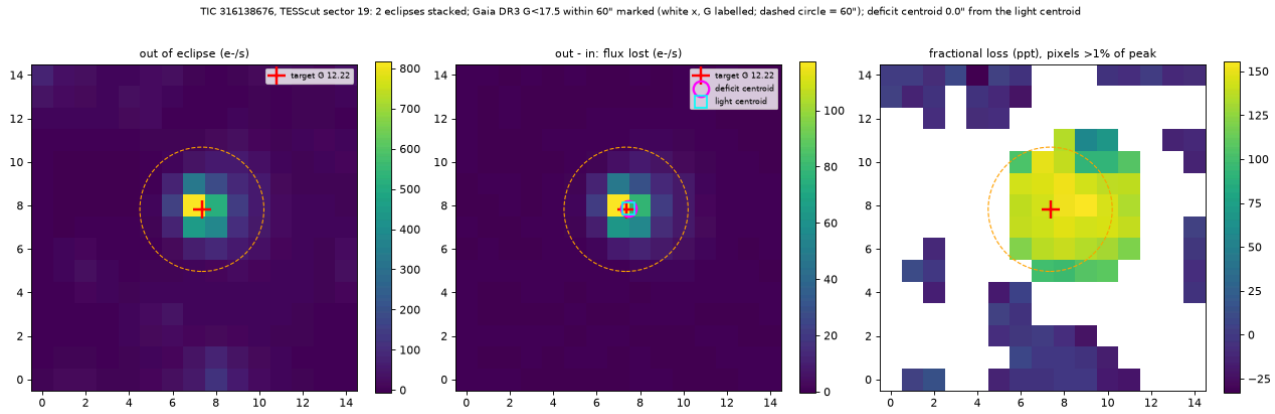
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R490; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R490; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 316138676: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 timing-usable measured eclipses in representative sectors [18, 59, 86]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 5 actual midpoints yield P = 13.9528465175 ± 0.0000217006 d and reference epoch BTJD 1805.4162403 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S19 cutout puts the deficit 2.82 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.050; this is not a calibrated localization error. Capable Gaia neighbors: none within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 143.67/145.57/144.91 ppt in 2 stacked events; median selected QLP raw fit depth 151.16 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 450156999764024576 has orbital P = 13.9484101055 d, e = 0.12110; closest edge-on conjunction offset at the measured epoch is 0.36270 d (0.02600 phase), with 0.36683 d linearly propagated full-SB1-covariance timing error. The rough central duration is 27.72 h versus observed median 10.23 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Five deep raw eclipses across S18/59/86, including consecutive S59 and S86 cycles. The deficit core is centered on the target; no capable Gaia neighbor is present under the approximate brightness screen. The long fitted duration is geometry dependent.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
18	0	1805.4159110	0.010417	151.16	used
59	80	2921.6441111	0.001157	153.19	used
59	81	2935.5966595	0.001157	151.93	used
86	132	3647.1923341	0.001157	150.47	used
86	133	3661.1444770	0.001157	151.00	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s19.json, historical/diff_s19.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R490. Public evidence files: diff_s19.json, diff_s19.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

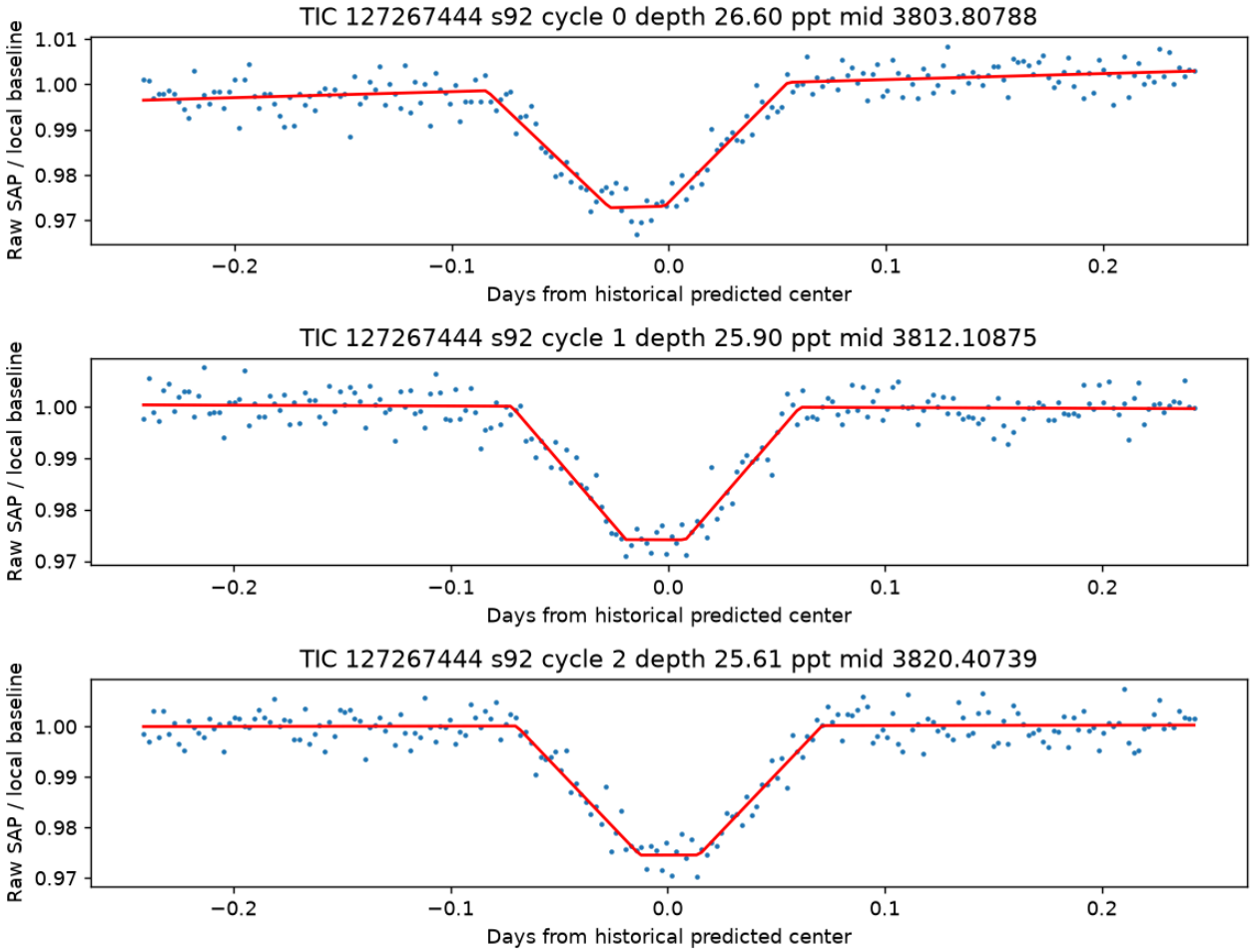
R491. TIC 127267444

physical_verdict: CONFIRMED

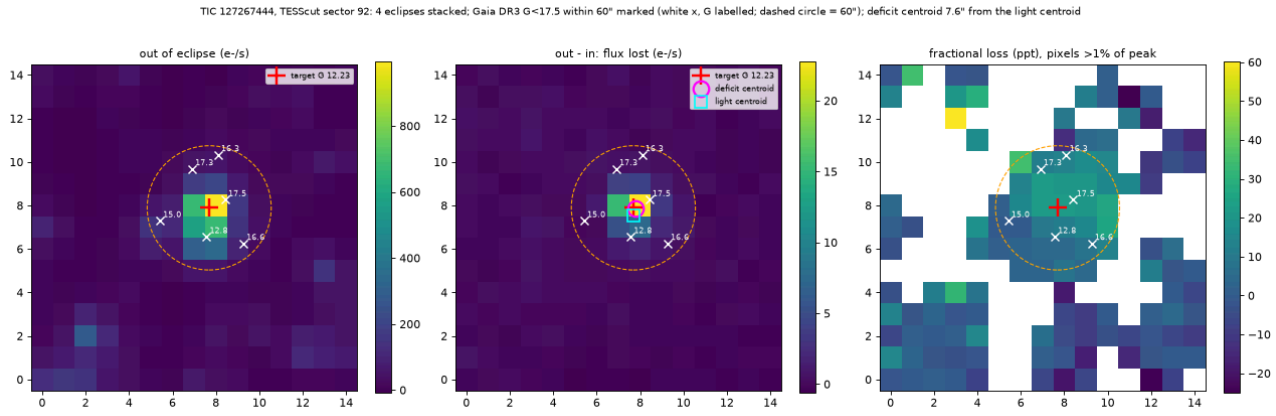
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R491; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R491; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 127267444: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 3 timing-usable measured eclipses in representative sectors [92]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 3 actual midpoints yield P = 8.2997438847 ± 0.0008290274 d and reference epoch BTJD 3803.8082712 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S92 cutout puts the deficit 1.73 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.388; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 6771875970694979712: 27.8 arcsec, G 12.830, approximate intrinsic eclipse requirement 0.066; Gaia 6771878959992220544: 49.5 arcsec, G 14.954, approximate intrinsic eclipse requirement 0.322. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 22.88/17.92/15.43 ppt in 4 stacked events; median selected QLP raw fit depth 25.90 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 6771876000757078144 has orbital P = 8.2991638272 d, e = 0.08323; closest edge-on conjunction offset at the measured epoch is 0.15077 d (0.01817 phase), with 0.61378 d linearly propagated full-SB1-covariance timing error. The rough central duration is 7.80 h versus observed median 3.33 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Three consecutive raw eclipses in S92 give a measured 8.299744-day period, close to the Gaia 8.299164-day orbit; the historical 8.29212-day screening estimate was inaccurate. Two capable neighbors at 27.8 and 49.5 arcsec are spatially distinct from the target-centered deficit. The pixel-model residual is substantial.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
92	0	3803.8078822	0.001187	26.60	used
92	1	3812.1087545	0.001157	25.90	used
92	2	3820.4073893	0.001157	25.61	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s92.json, historical/diff_s92.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R491. Public evidence files: diff_s92.json, diff_s92.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

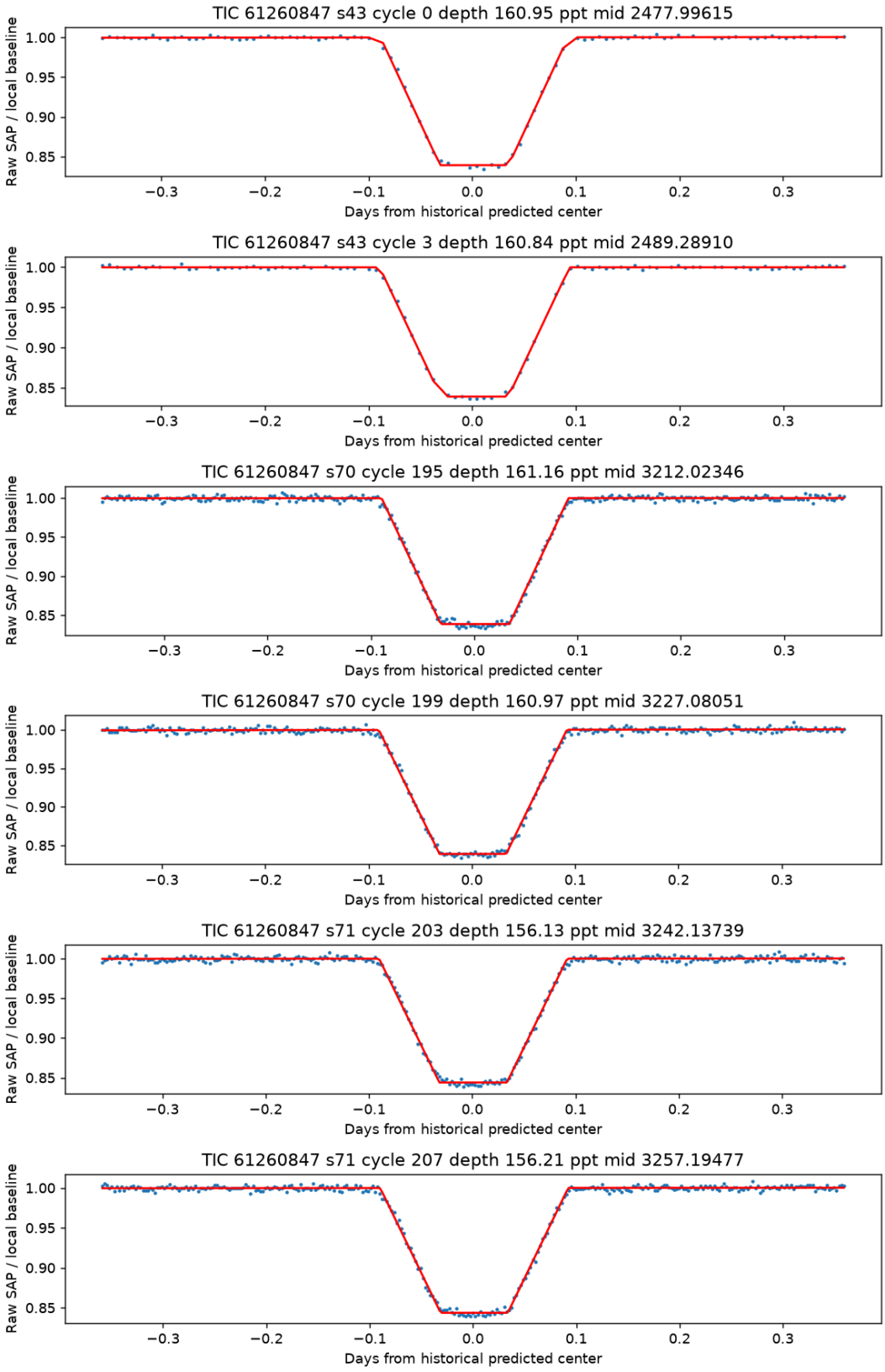
R492. TIC 61260847

physical_verdict: CONFIRMED

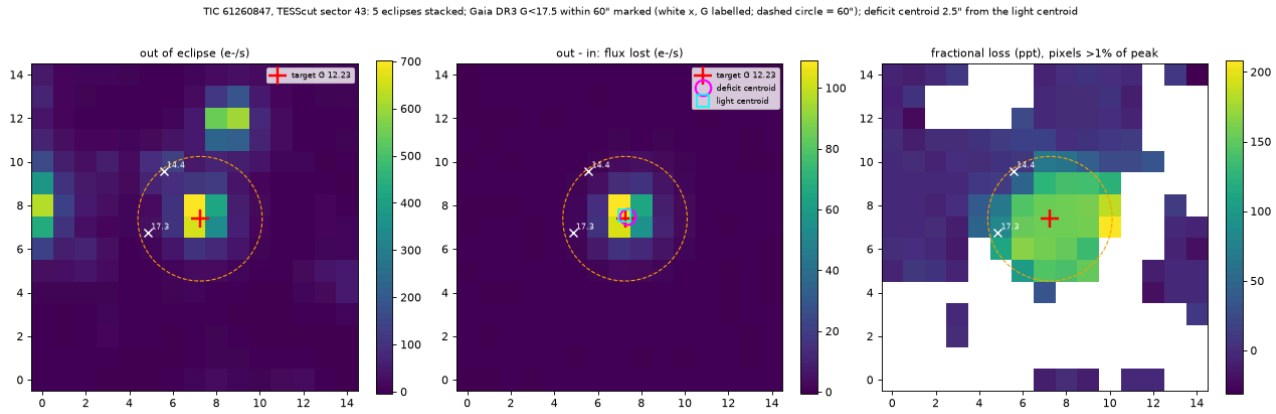
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R492; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R492; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 61260847: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 15 timing-usable measured eclipses in representative sectors [43, 70, 71]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 15 actual midpoints yield P = 3.7642439801 ± 0.0000080184 d and reference epoch BTJD 2477.9961067 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S43 cutout puts the deficit 2.54 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 7]. Image-model fractional residual 0.322; this is not a calibrated localization error. Capable Gaia neighbors: none within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 155.96/155.42/145.63 ppt in 5 stacked events; median selected QLP raw fit depth 160.34 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 3410845980312983680 has orbital P = 3.7643570975 d, e = 0.01600; closest edge-on conjunction offset at the measured epoch is -0.07963 d (-0.02115 phase), with 0.21627 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.75 h versus observed median 4.35 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Fifteen deep raw eclipses in S43/70/71 establish recurrence and a measured period. The cutout core favors the target with no capable Gaia neighbor; local image residuals limit precise centroid claims.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
43	0	2477.9961490	0.003473	160.95	used
43	1	2481.7601478	0.003473	159.16	used
43	3	2489.2890959	0.003473	160.84	used
43	4	2493.0530729	0.003473	159.18	used
43	5	2496.8173050	0.003473	160.37	used
70	195	3212.0234621	0.001158	161.16	used
70	196	3215.7881033	0.001158	160.34	used
70	198	3223.3164704	0.001158	161.07	used
70	199	3227.0805055	0.001158	160.97	used
70	200	3230.8449305	0.001158	161.27	used
71	202	3238.3734637	0.001157	156.10	used
71	203	3242.1373914	0.001157	156.13	used
71	204	3245.9020319	0.001157	155.95	used
71	205	3249.6660910	0.001157	156.03	used
71	207	3257.1947663	0.001157	156.21	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s43.json, historical/diff_s43.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R492. Public evidence files: diff_s43.json, diff_s43.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

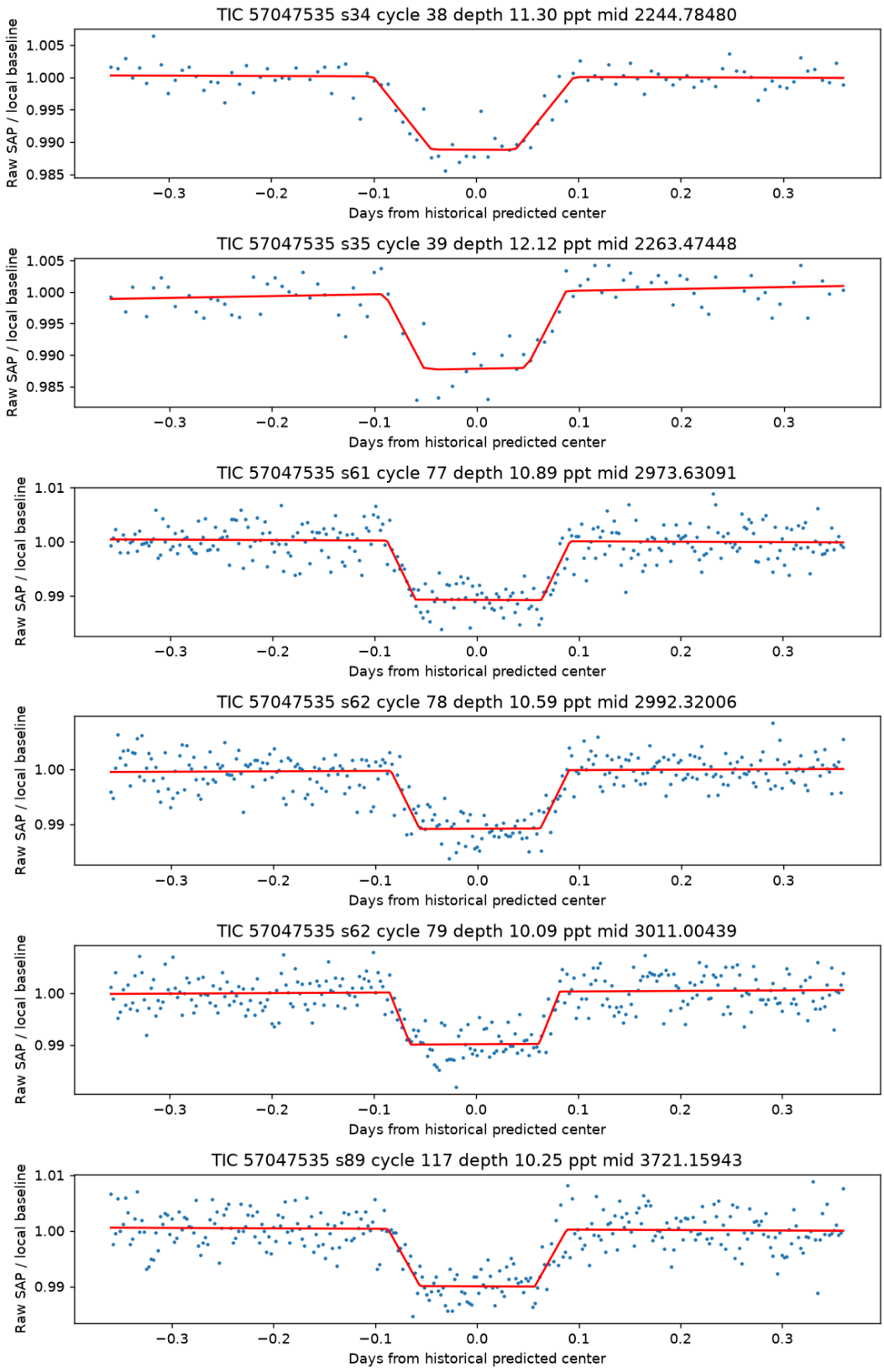
R493. TIC 57047535

physical_verdict: PROBABLE

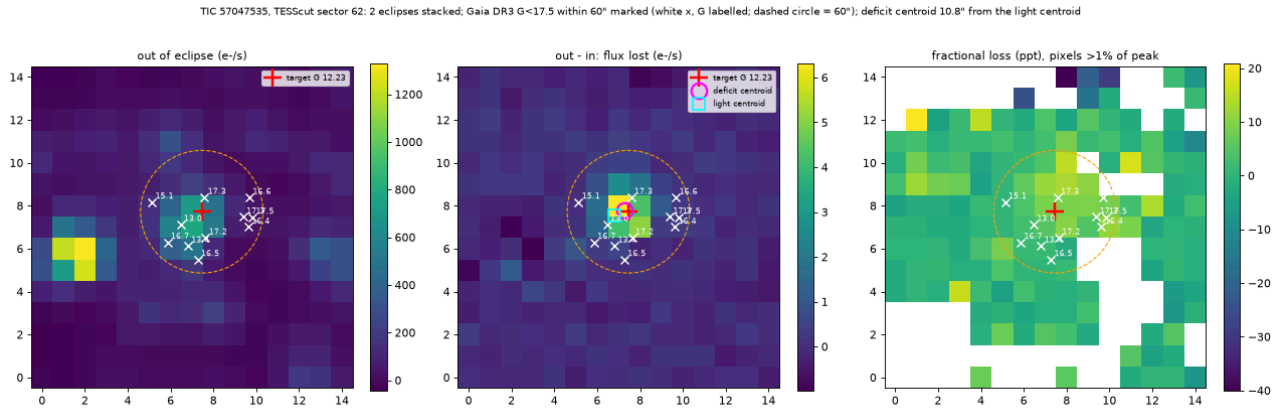
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization of the crowded field.



R493; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R493; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 57047535: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 7 timing-usable measured eclipses in representative sectors [34, 35, 61, 62, 88, 89]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 7 actual midpoints yield P = 18.6882586301 ± 0.0000306738 d and reference epoch BTJD 2244.7878116 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: unresolved after completed check.** Historical S62 cutout puts the deficit 4.30 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.674; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 5542738205536780288: 22.6 arcsec, G 12.965, approximate intrinsic eclipse requirement 0.031; Gaia 5542738205536774656: 33.5 arcsec, G 13.358, approximate intrinsic eclipse requirement 0.040; Gaia 5542738278564942848: 41.5 arcsec, G 16.683, approximate intrinsic eclipse requirement 0.647; Gaia 5542738209845466752: 44.2 arcsec, G 16.489, approximate intrinsic eclipse requirement 0.543; Gaia 5542737557010422784: 45.6 arcsec, G 16.553, approximate intrinsic eclipse requirement 0.576; Gaia 5542737453931213312: 46.2 arcsec, G 16.445, approximate intrinsic eclipse requirement 0.522; Gaia 5542738278566462336: 47.7 arcsec, G 15.131, approximate intrinsic eclipse requirement 0.163. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 8.93/7.09/5.36 ppt in 2 stacked events; median selected QLP raw fit depth 10.59 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate

observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.

5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5542738205536698240 has orbital P = 18.6663627993 d, e = 0.42196; closest edge-on conjunction offset at the measured epoch is 0.38371 d (0.02056 phase), with 1.66506 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.38 h versus observed median 4.22 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Seven raw eclipses across S34/35/61/62/88/89, including consecutive cycles 38/39, 77/78/79 and 116/117, resolve the initially sparse timing. Several capable neighbors make this a crowded field. The deficit core favors the target, but the closest capable Gaia source is only about 1.1 TESS pixels away and the image-model residual is large; attribution remains unresolved.

Next check: Resolved eclipse photometry or multi-source PRF localization of the crowded field.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
34	38	2244.7848021	0.003472	11.30	used
35	39	2263.4744847	0.003472	12.12	used
61	77	2973.6309138	0.001361	10.89	used
62	78	2992.3200617	0.001447	10.59	used
62	79	3011.0043901	0.001679	10.09	used
88	116	3702.4716187	0.001641	10.01	used
89	117	3721.1594292	0.001913	10.25	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s62.json, historical/diff_s62.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R493. Public evidence files: diff_s62.json, diff_s62.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 57047535-availability.json

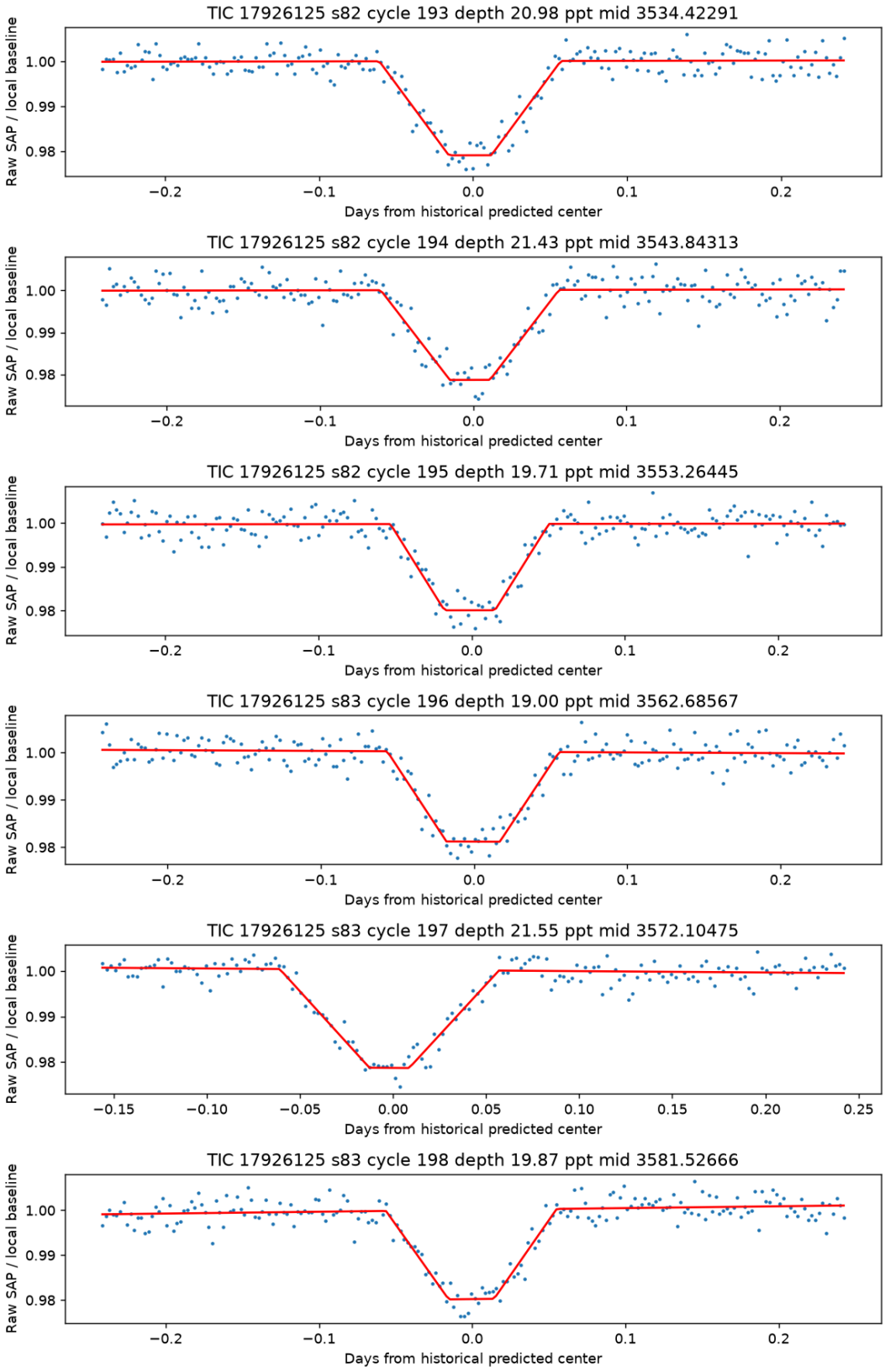
R494. TIC 17926125

physical_verdict: PROBABLE

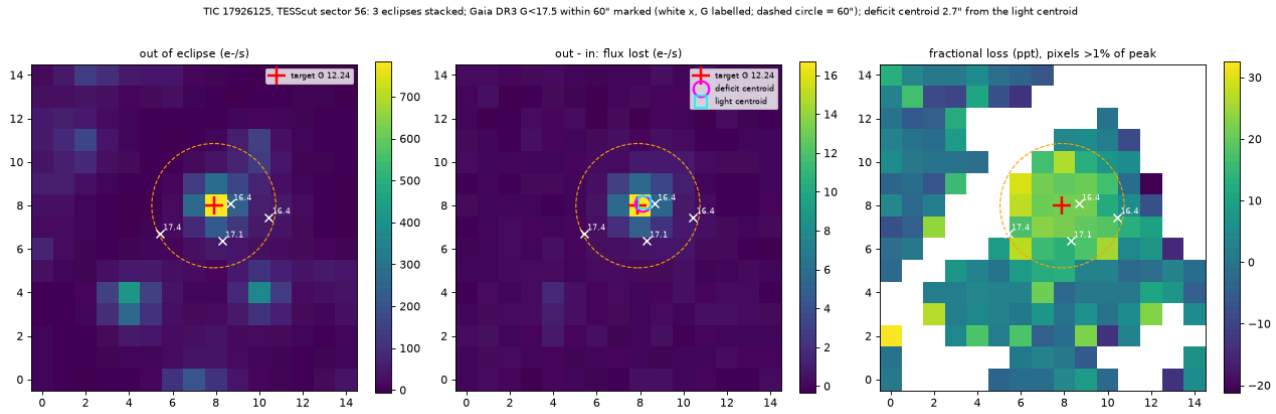
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization of the 15.7-arcsec pair.



R494; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R494; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 17926125: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured eclipses in representative sectors [82, 83]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 6 actual midpoints yield P = 9.4207091616 ± 0.0002766705 d and reference epoch BTJD 3534.4228208 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: unresolved after completed check.** Historical S56 cutout puts the deficit 4.79 arcsec from the target coordinate; peak [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.489; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 1896543625318721280: 15.7 arcsec, G 16.363, approximate intrinsic eclipse requirement 0.866; Gaia 1896543659678461312: 53.7 arcsec, G 16.358, approximate intrinsic eclipse requirement 0.862. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 21.32/20.96/19.82 ppt in 3 stacked events; median selected QLP raw fit depth 20.43 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1896543621022178048 has orbital P = 9.4199419993 d, e = 0.02747; closest edge-on conjunction offset at the measured epoch is 0.06074 d (0.00645 phase), with 0.52723 d linearly propagated full-SB1-covariance timing error. The rough central duration is 6.60 h versus observed median 2.74 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Six consecutive raw eclipses in S82/83 support the measured recurrence. The deficit core favors the target, but a capable G16.36 neighbor 15.7 arcsec away shares the central TESS pixel and could in principle produce the loss through an extremely deep intrinsic eclipse. Attribution remains unresolved at this resolution.

Next check: Resolved eclipse photometry or multi-source PRF localization of the 15.7-arcsec pair.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
82	193	3534.4229057	0.001157	20.98	used
82	194	3543.8431342	0.001157	21.43	used
82	195	3553.2644488	0.001157	19.71	used
83	196	3562.6856663	0.001157	19.00	used
83	197	3572.1047515	0.001157	21.55	used
83	198	3581.5266560	0.001157	19.87	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R494. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

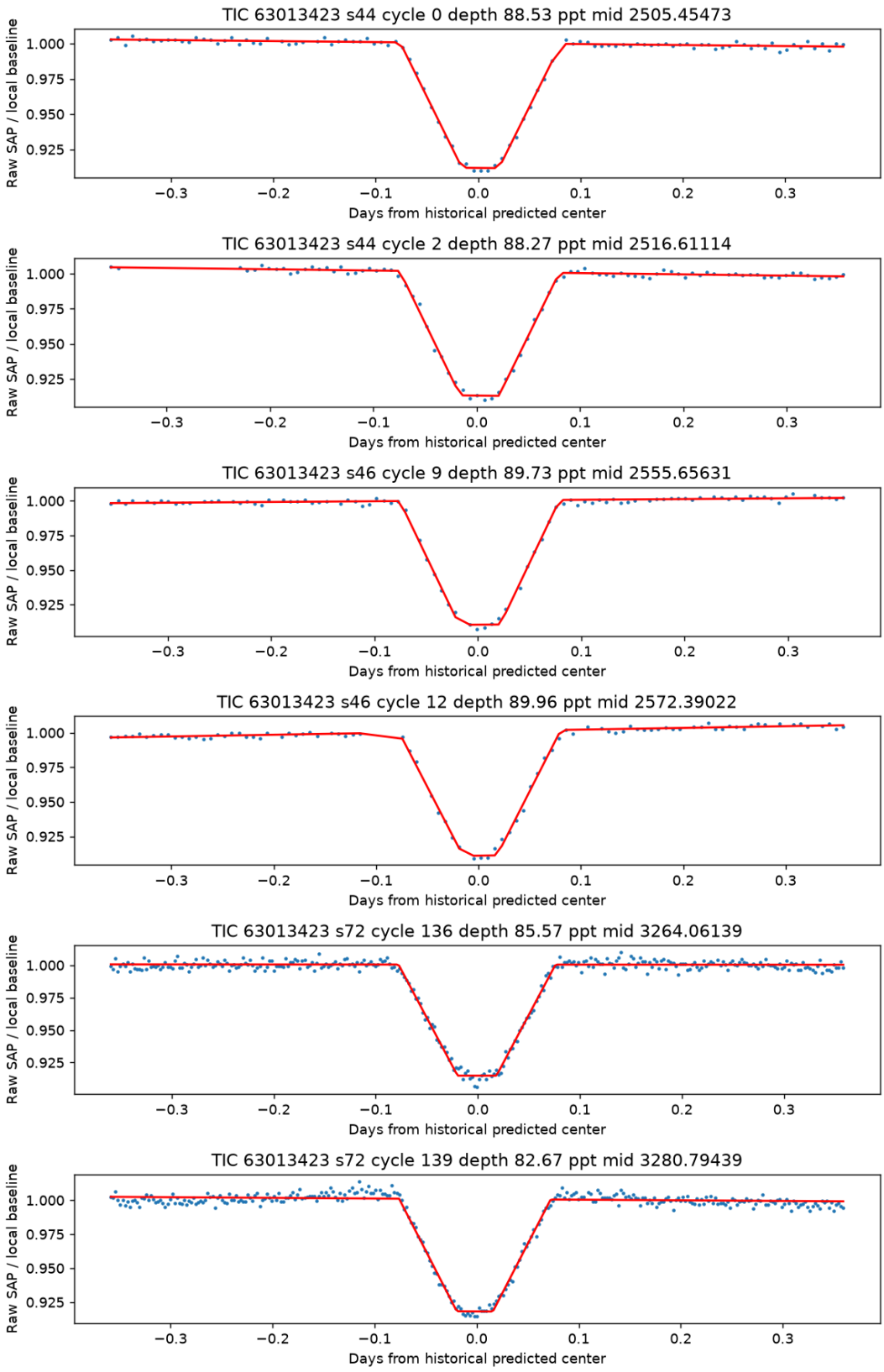
R495. TIC 63013423

physical_verdict: CONFIRMED

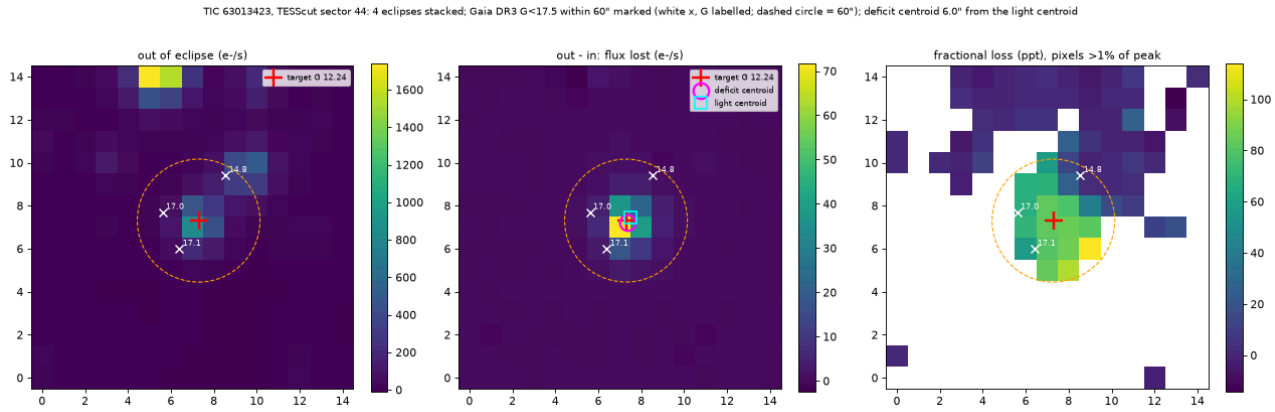
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R495; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R495; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 63013423: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 12 timing-usable measured eclipses in representative sectors [44, 46, 72]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 12 actual midpoints yield P = 5.5779862426 ± 0.0000103302 d and reference epoch BTJD 2505.4546770 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S44 cutout puts the deficit 2.57 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.569; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 675068326633876992: 49.1 arcsec, G 14.755, approximate intrinsic eclipse requirement 0.904. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 82.60/81.69/70.51 ppt in 4 stacked events; median selected QLP raw fit depth 87.58 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 675068532792307456 has orbital P = 5.5777973688 d, e = 0.04208; closest edge-on conjunction offset at the measured epoch is 0.13397 d (0.02402 phase), with 0.20174 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.22 h versus observed median 3.72 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Twelve raw eclipses in S44/46/72 show stable recurrence. The capable neighbor 49.1 arcsec away is distinct from the target-centered deficit; aperture and model residuals limit exact intrinsic depth.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
44	0	2505.4547268	0.003473	88.53	used
44	1	2511.0325313	0.003473	89.00	used
44	2	2516.6111421	0.003473	88.27	used
44	3	2522.1890296	0.003473	86.32	used
46	9	2555.6563082	0.003472	89.73	used
46	10	2561.2344079	0.003472	88.32	used
46	12	2572.3902221	0.003472	89.96	used
46	13	2577.9681930	0.003472	86.86	used
72	136	3264.0613877	0.001157	85.57	used
72	137	3269.6388279	0.001157	85.71	used
72	138	3275.2165537	0.001157	86.88	used
72	139	3280.7943908	0.001157	82.67	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s44.json, historical/diff_s44.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R495. Public evidence files: diff_s44.json, diff_s44.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

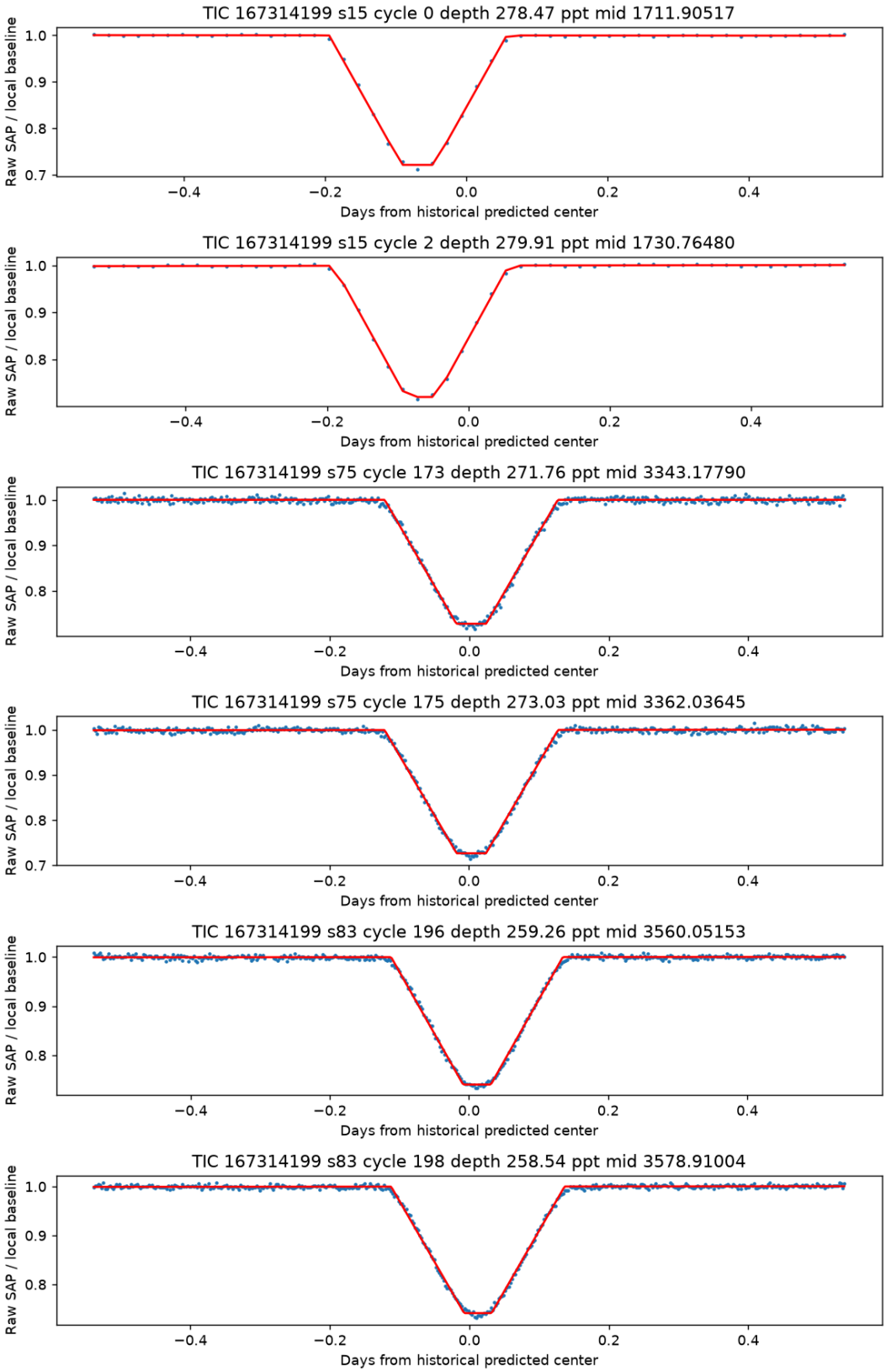
R496. TIC 167314199

physical_verdict: CONFIRMED

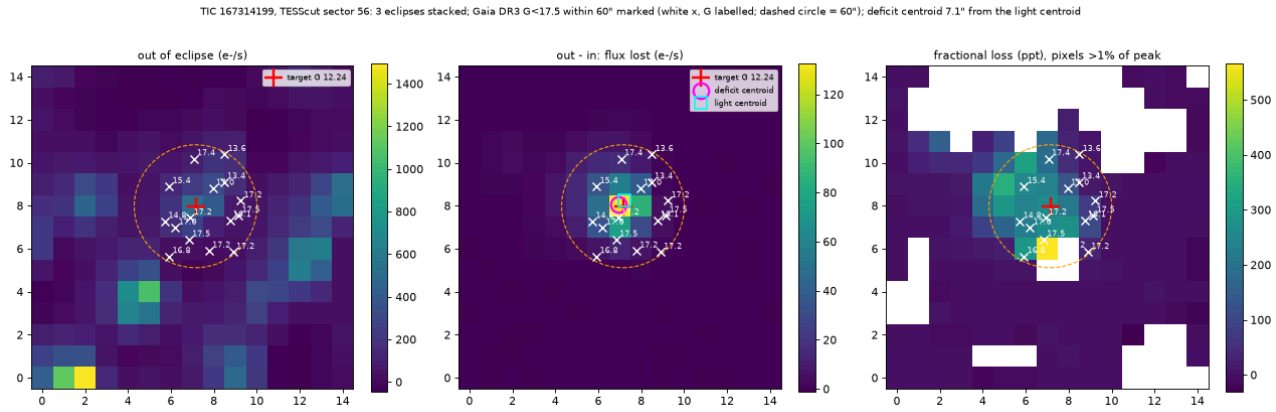
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R496; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R496; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 167314199: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 timing-usable measured eclipses in representative sectors [15, 75, 83]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 8 actual midpoints yield P = 9.4293015589 ± 0.0000286278 d and reference epoch BTJD 1711.9085480 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S56 cutout puts the deficit 5.06 arcsec from the target coordinate; peak [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.880; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 1964832780702837248: 35.7 arcsec, G 13.368, approximate intrinsic eclipse requirement 0.909. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 259.06/230.88/191.58 ppt in 3 stacked events; median selected QLP raw fit depth 272.15 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1964832776395084288 has orbital P = 9.4286060079 d, e = 0.02722; closest edge-on conjunction offset at the measured epoch is 0.14149 d (0.01501 phase), with 0.14582 d linearly propagated full-SB1-covariance timing error. The rough central duration is 7.89 h versus observed median 5.97 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Eight deep raw eclipses in S15/75/83 recur on the Gaia timescale. A capable 35.7-arcsec neighbor requires an extreme intrinsic eclipse and is distinct from the target-favored core; the image-model residual is high.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
15	0	1711.9051662	0.010417	278.47	used
15	2	1730.7647966	0.010417	279.91	used
75	173	3343.1779004	0.001157	271.76	used
75	174	3352.6073018	0.001157	272.54	used
75	175	3362.0364531	0.001157	273.03	used
83	196	3560.0515342	0.001157	259.26	used
83	197	3569.4807627	0.001157	256.44	used
83	198	3578.9100415	0.001157	258.54	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R496. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

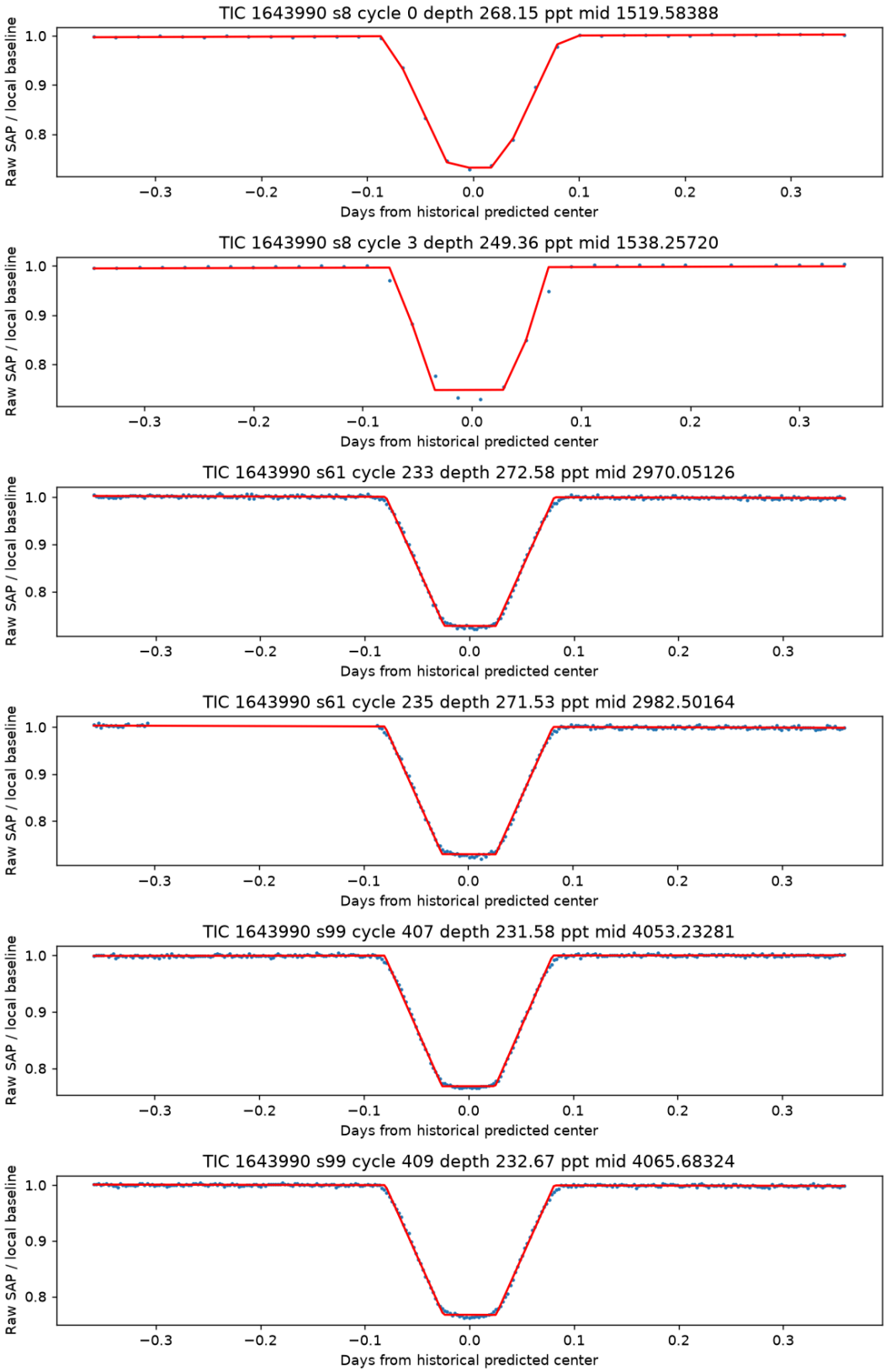
R497. TIC 1643990

physical_verdict: CONFIRMED

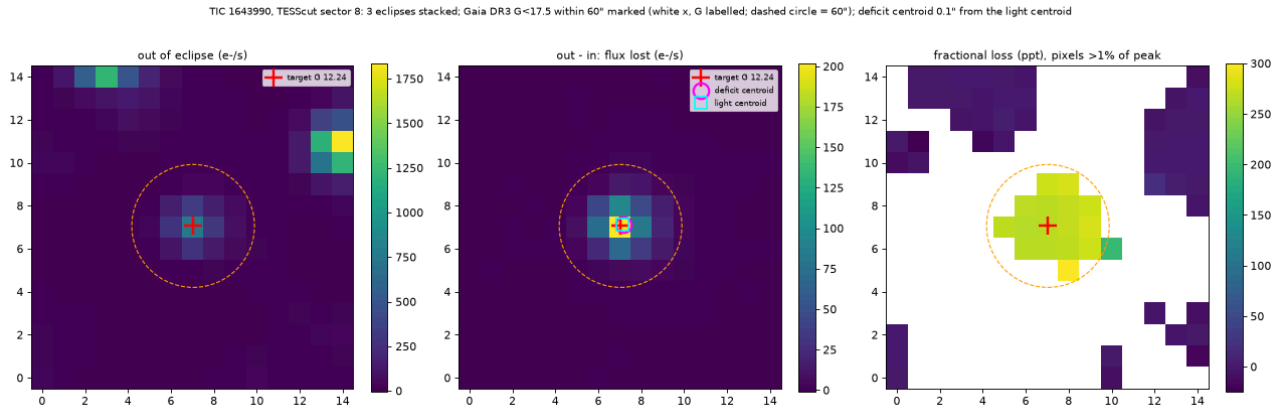
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R497; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R497; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 1643990: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 6 timing-usable measured eclipses in representative sectors [8, 61, 99]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 6 actual midpoints yield P = 6.2251817590 ± 0.0000063922 d and reference epoch BTJD 1519.5838843 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S8 cutout puts the deficit 2.55 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.046; this is not a calibrated localization error. Capable Gaia neighbors: none within 60 arcsec under the approximate G-band flux budget. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 267.11/268.43/270.89 ppt in 3 stacked events; median selected QLP raw fit depth 258.75 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5762102560669048704 has orbital P = 6.2252118769 d, e = 0.11362; closest edge-on conjunction offset at the measured epoch is -0.00339 d (-0.00055 phase), with 0.07702 d linearly propagated full-SB1-covariance timing error. The rough central duration is 4.43 h versus observed median 3.86 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Six deep raw eclipses in S8/61/99 give a long timing baseline. The cutout deficit lies on the target with low model residual and no capable Gaia neighbor under the brightness screen.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
8	0	1519.5838814	0.010417	268.15	used
8	3	1538.2572047	0.010417	249.36	used
61	233	2970.0512579	0.001157	272.58	used
61	235	2982.5016384	0.001157	271.53	used
99	407	4053.2328119	0.001157	231.58	used
99	409	4065.6832351	0.001157	232.67	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s08.json, historical/diff_s08.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R497. Public evidence files: diff_s08.json, diff_s08.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

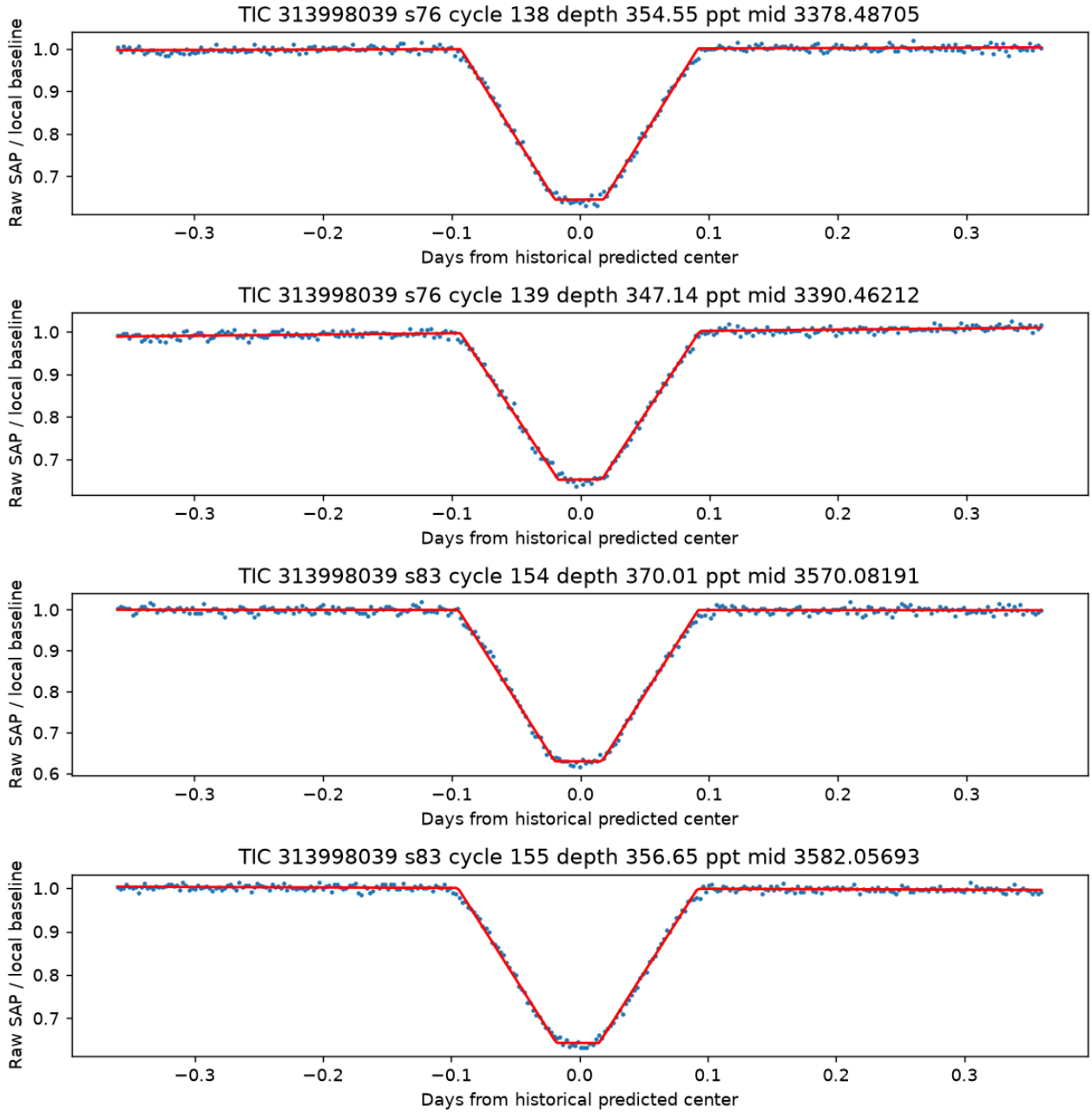
R498. TIC 313998039

physical_verdict: CONFIRMED

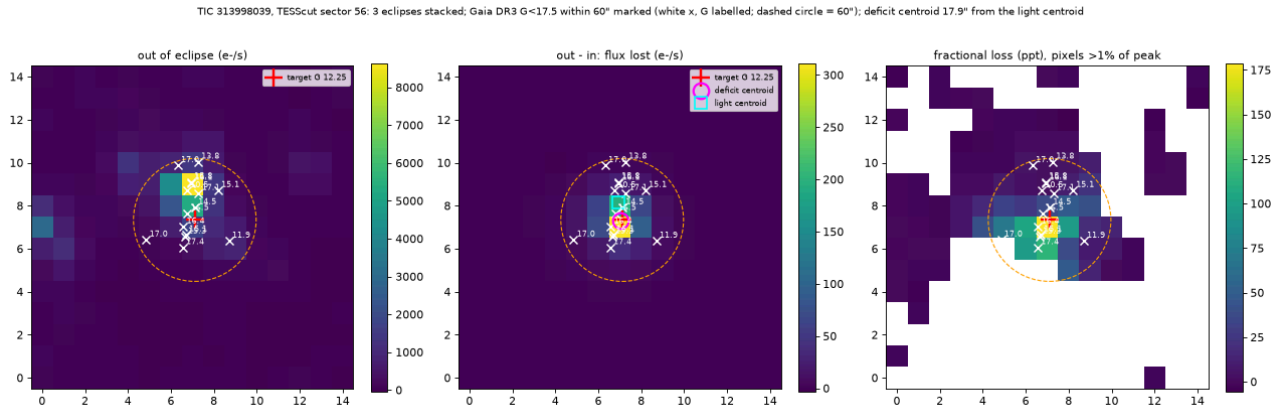
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.



R498; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R498; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 313998039: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 4 timing-usable measured eclipses in representative sectors [76, 83]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 4 actual midpoints yield P = 11.9746788325 ± 0.0000721956 d and reference epoch BTJD 3378.4872332 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: favors target at TESS image resolution.** Historical S56 cutout puts the deficit 2.09 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.874; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 2189517466440845568: 29.6 arcsec, G 10.549, approximate intrinsic eclipse requirement 0.405; Gaia 2189505711103804288: 39.1 arcsec, G 11.912, approximate intrinsic eclipse requirement 0.581. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 178.81/65.00/32.94 ppt in 3 stacked events; median selected QLP raw fit depth 355.60 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 2189517466440848256 has orbital P = 11.9737068648 d, e = 0.06505; closest edge-on conjunction offset at the measured epoch is 0.48689 d (0.04066 phase), with 0.44849 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.44 h versus observed median 4.47 h;

assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Four deep raw eclipses in S76/83 repeat consistently. Bright capable neighbors at 29.6 and 39.1 arcsec cause severe aperture dilution, but the deficit core favors the target location; high image residual prevents precise localization.

Next check: Optional resolved photometry and spectroscopy for precision binary parameters; no missing bounded admission check.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
76	138	3378.4870523	0.001157	354.55	used
76	139	3390.4621157	0.001157	347.14	used
83	154	3570.0819122	0.001157	370.01	used
83	155	3582.0569331	0.001157	356.65	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s56.json, historical/diff_s56.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R498. Public evidence files: diff_s56.json, diff_s56.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

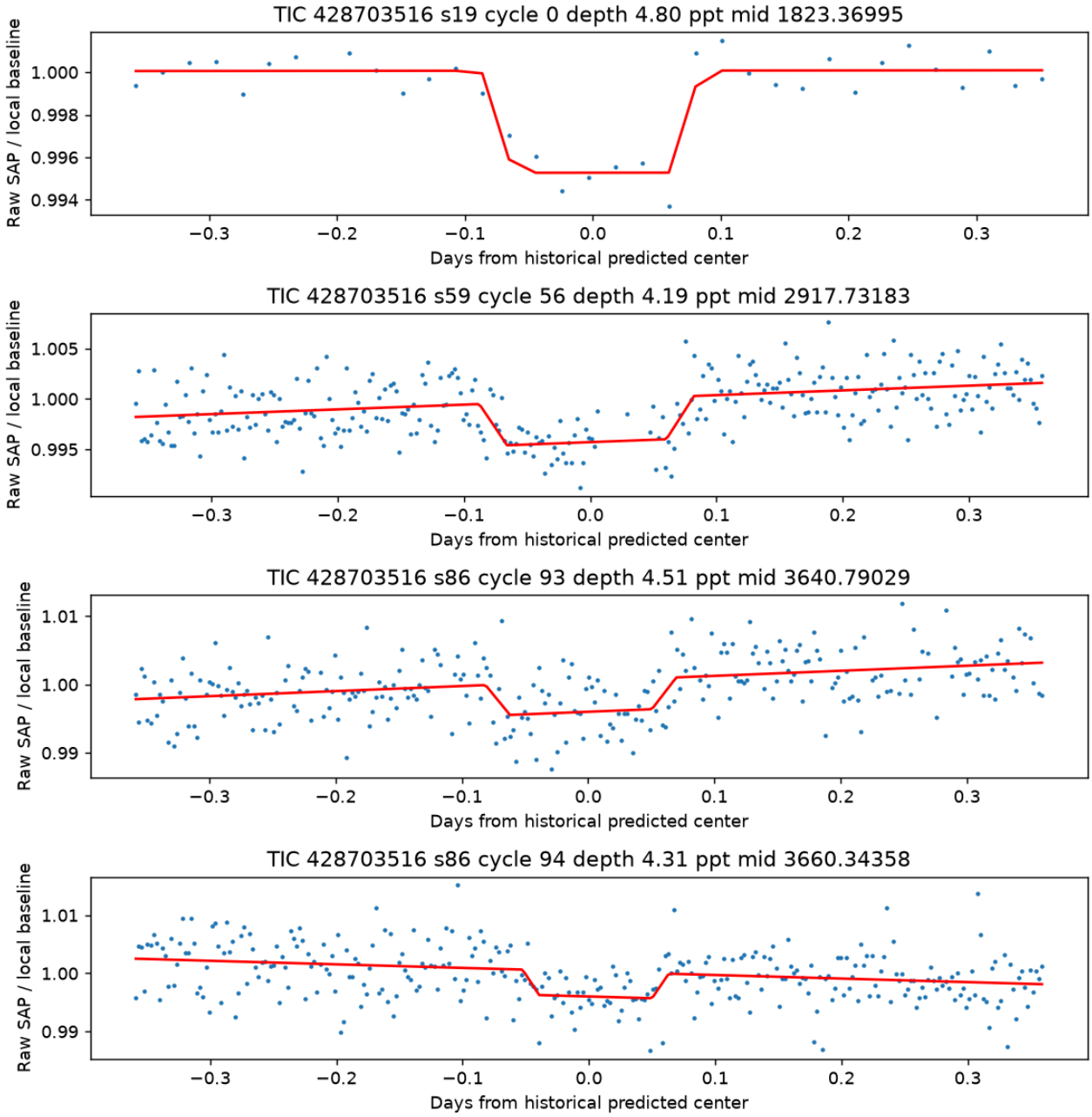
R499. TIC 428703516

physical_verdict: PROBABLE

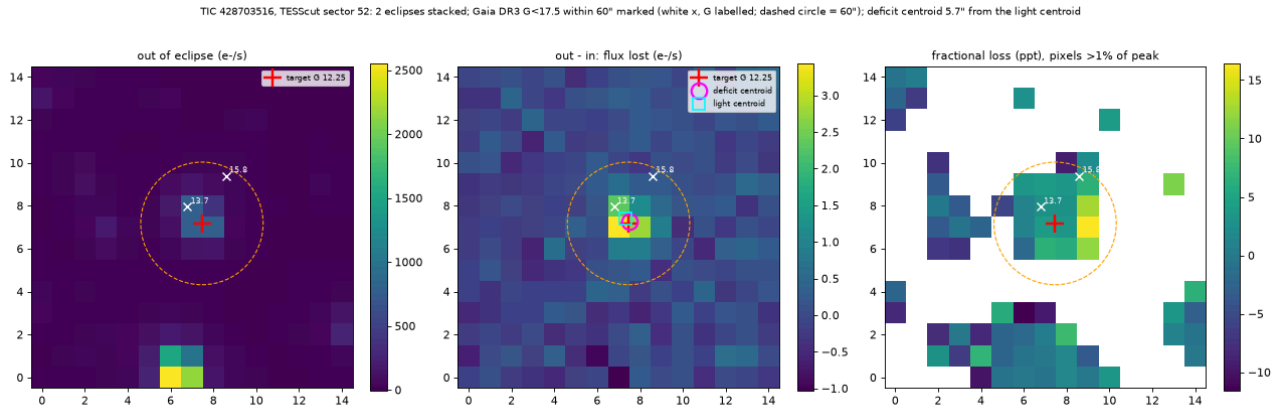
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization of the 11.2-arcsec pair.



R499; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R499; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 428703516: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are recorded separately. The historical screening label is not an admission verdict.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 4 timing-usable measured eclipses in representative sectors [19, 59, 86]. The raw plot and numeric windows retain the light-curve evidence. Local trapezoids have free centers, depths, durations and baselines. Historical predicted times guided windows only. Flagged or uncovered intervals are not null detections.
2. **Measured timing and alias: pass.** 4 actual midpoints yield P = 19.5422216389 ± 0.0001299315 d and reference epoch BTJD 1823.3679179 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). Same-kind cycle counts, within-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive blind all-period search.
3. **Source attribution: unresolved after completed check.** Historical S52 cutout puts the deficit 2.07 arcsec from the target coordinate; peak [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.488; this is not a calibrated localization error. Capable Gaia neighbors: Gaia 548058855130833792: 11.2 arcsec, G 18.035, approximate intrinsic eclipse requirement 0.888; Gaia 548058859427261184: 21.5 arcsec, G 13.709, approximate intrinsic eclipse requirement 0.021; Gaia 548047104100306688: 51.4 arcsec, G 17.903, approximate intrinsic eclipse requirement 0.787; Gaia 548058859427256832: 52.1 arcsec, G 15.785, approximate intrinsic eclipse requirement 0.116. The intrinsic requirement uses an approximate flux budget and is not compared directly with a blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 4.24/3.46/3.72 ppt in 2 stacked events; median selected QLP raw fit depth 4.41 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution and any mixed event kinds limit depth comparisons.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 548058859427458688 has orbital P = 19.5459437733 d, e = 0.02433; closest edge-on conjunction

offset at the measured epoch is -0.18196 d (-0.00931 phase), with 0.29026 d linearly propagated full-SB1-covariance timing error. The rough central duration is 5.97 h versus observed median 3.86 h; assumed TIC radius and minimum-mass companion are not a full fit. Geometry supports physical plausibility without an arbitrary sigma veto.

Deciding interpretation and limitation: Four shallow raw eclipses in S19/59/86, including consecutive S86 cycles, support recurrence. The deficit favors the target over the capable 21.5-arcsec neighbor, but a capable Gaia source 11.2 arcsec away is inside one TESS pixel and could in principle produce the 4-ppt loss with a deep intrinsic eclipse. Attribution remains unresolved at this resolution.

Next check: Resolved eclipse photometry or multi-source PRF localization of the 11.2-arcsec pair.

Catalogue novelty: central seven-group result is in ../../shared-catalogues/coverage-matrix.csv. A cone-source entry alone does not establish an exact target match or exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Timing status
19	0	1823.3699499	0.010417	4.80	used
59	56	2917.7318282	0.003696	4.19	used
86	93	3640.7902858	0.003409	4.51	used
86	94	3660.3435838	0.004222	4.31	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../gaia-geometry.json, ../alias-audit.json, historical/diff_s52.json, historical/diff_s52.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). Source code and all retained evidence are covered by ../SHA256SUMS.json.

Stable review key: R499. Public evidence files: diff_s52.json, diff_s52.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

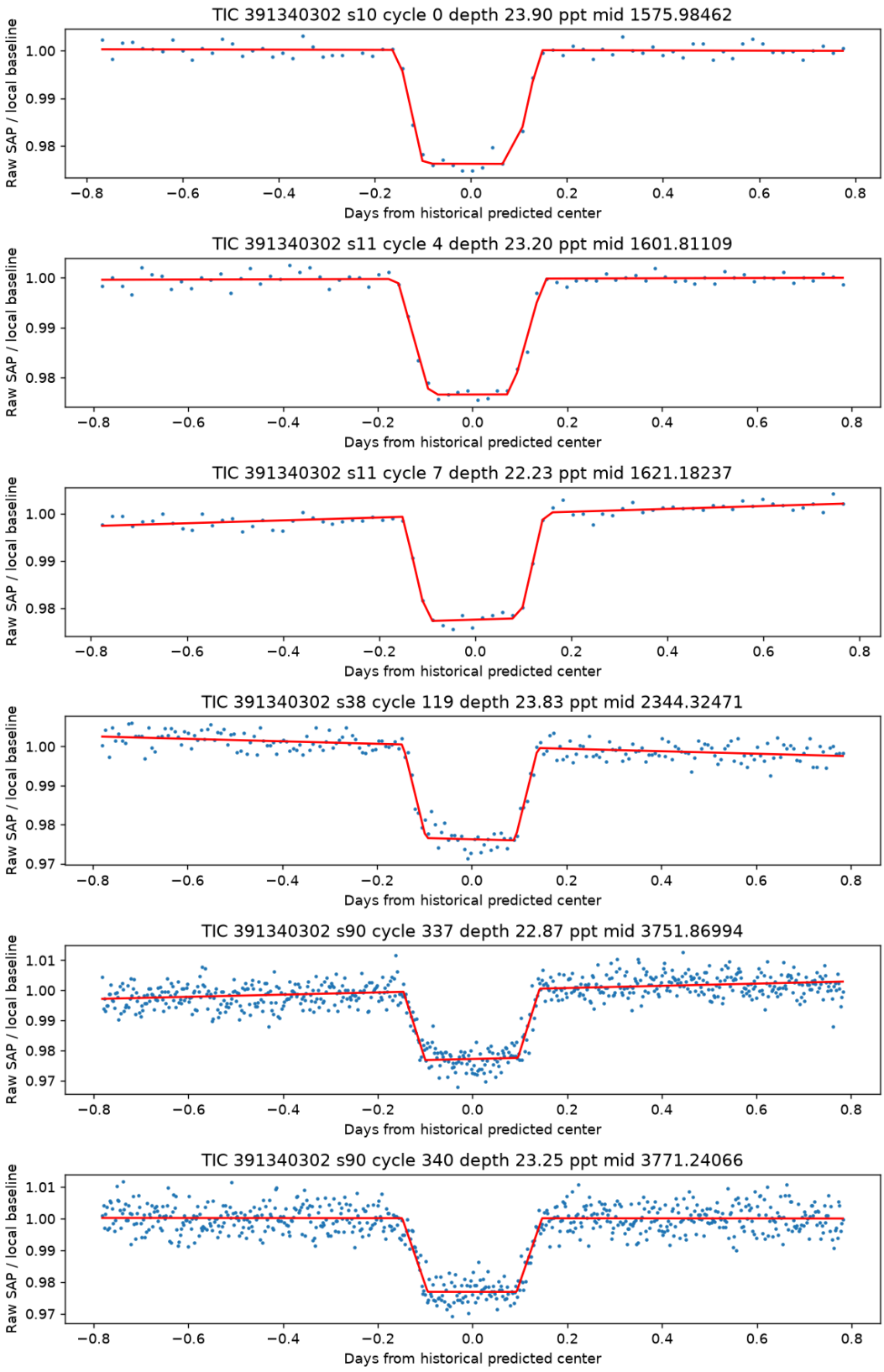
R500. TIC 391340302

physical_verdict: PROBABLE

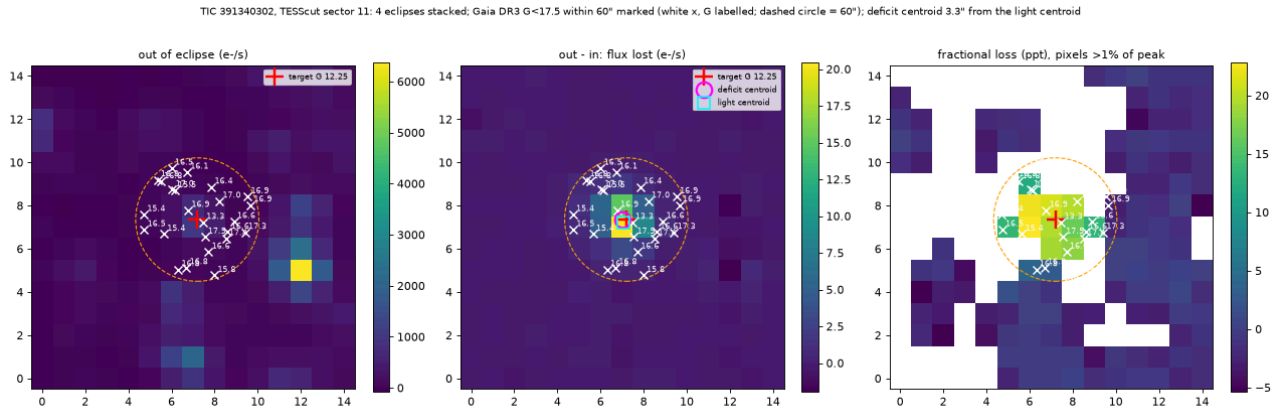
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry of the 6.1-arcsec pair.



R500; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R500; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 391340302: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 14 fitted representative eclipses in sectors [10, 11, 38, 90]; 14 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 14 same-kind actual midpoints yield P = 6.4566398472 ± 0.0000072175 d and reference epoch BTJD 1575.9846204 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: unresolved.** Historical sector 11 cutout places the deficit centroid 3.98 arcsec from the target; peak pixel [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.890 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 5239465552568592768: 6.1 arcsec, G 13.331, approximate intrinsic eclipse requirement 0.087; Gaia 5239465518208831104: 35.1 arcsec, G 15.493, approximate intrinsic eclipse requirement 0.487; Gaia 5239453801538064640: 35.2 arcsec, G 15.421, approximate intrinsic eclipse requirement 0.457; Gaia 5239465518208825984: 46.9 arcsec, G 16.057, approximate intrinsic eclipse requirement 0.802; Gaia 5239453767178311808: 51.5 arcsec, G 15.422, approximate intrinsic eclipse requirement 0.457; Gaia 5239454175171527424: 56.3 arcsec, G 15.756, approximate intrinsic eclipse requirement 0.614. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: supports recurring loss with unresolved host.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 19.49/19.83/18.92 ppt over 4 stacked events, compared with median selected QLP raw fit depth 23.22 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with

the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.

5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5239465548244669952 has P = 6.4572634522 d and e = 0.10891. The closest edge-on conjunction offset at the measured epoch is -0.08325 d (-0.01289 phase), with 0.16381 d linearly propagated full-SB1-covariance timing error. A rough central duration is 7.18 h versus observed median 7.04 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: The S11 difference image centers on the target pixel, but a Gaia neighbor 6.1 arcsec away lies in that same TESS core and can produce the observed loss. TESS pixels cannot assign the eclipses uniquely.

Next check: Resolved eclipse photometry of the 6.1-arcsec pair.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
10	0	1575.9846232	0.010417	23.90	used
10	2	1588.8979173	0.010417	23.10	used
11	4	1601.8110937	0.010417	23.20	used
11	5	1608.2682354	0.010417	22.15	used
11	6	1614.7223332	0.010417	23.13	used
11	7	1621.1823670	0.010417	22.23	used
38	118	2337.8680384	0.003472	24.38	used
38	119	2344.3247065	0.003472	23.83	used
38	120	2350.7815131	0.003472	23.58	used
38	121	2357.2381694	0.003472	24.48	used
90	337	3751.8699430	0.001301	22.87	used
90	338	3758.3315202	0.001186	23.05	used
90	339	3764.7857257	0.001157	24.18	used
90	340	3771.2406614	0.001513	23.25	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s11.json, historical/diff_s11.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R500. Public evidence files: diff_s11.json, diff_s11.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

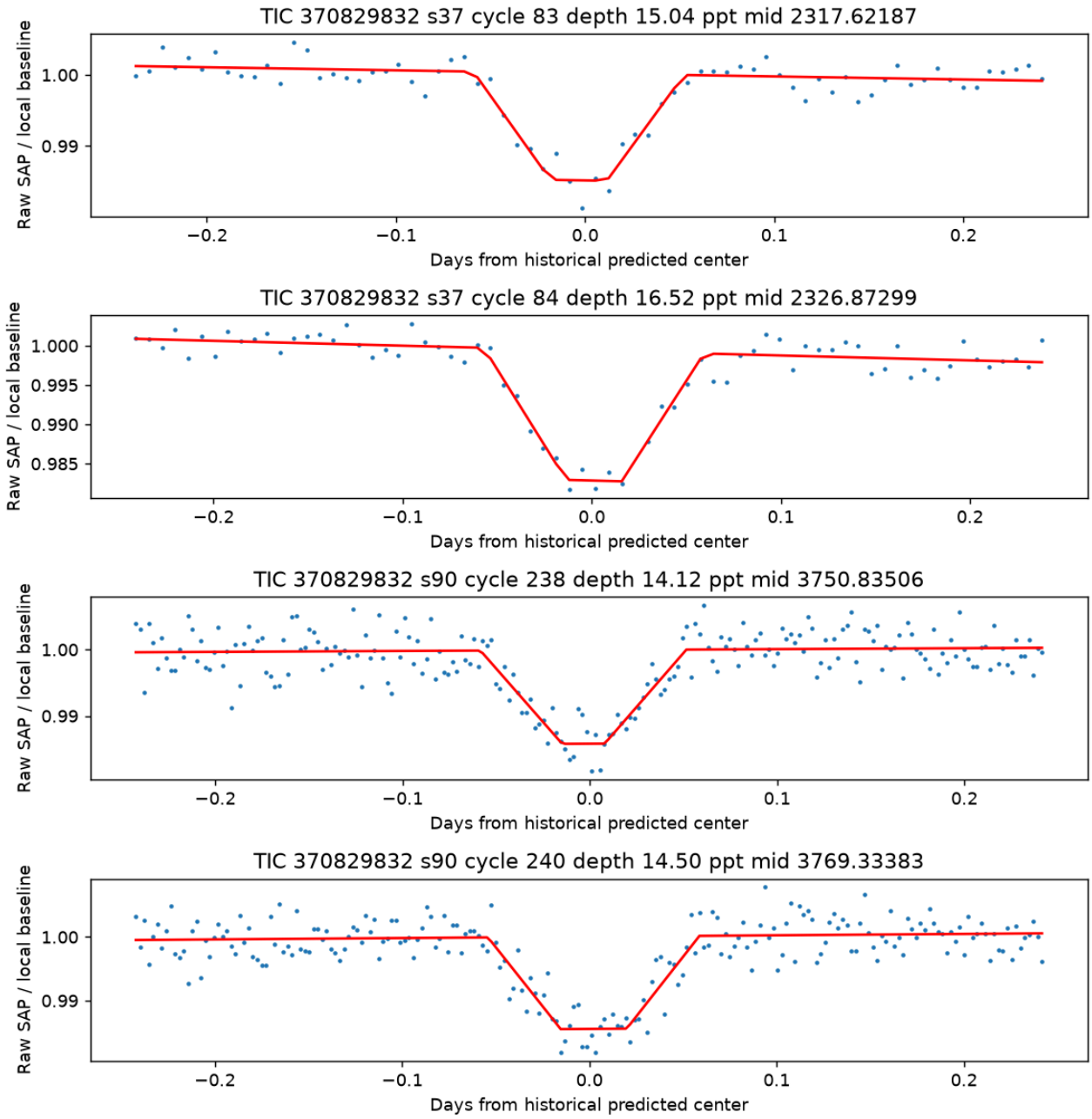
R501. TIC 370829832

physical_verdict: CONFIRMED

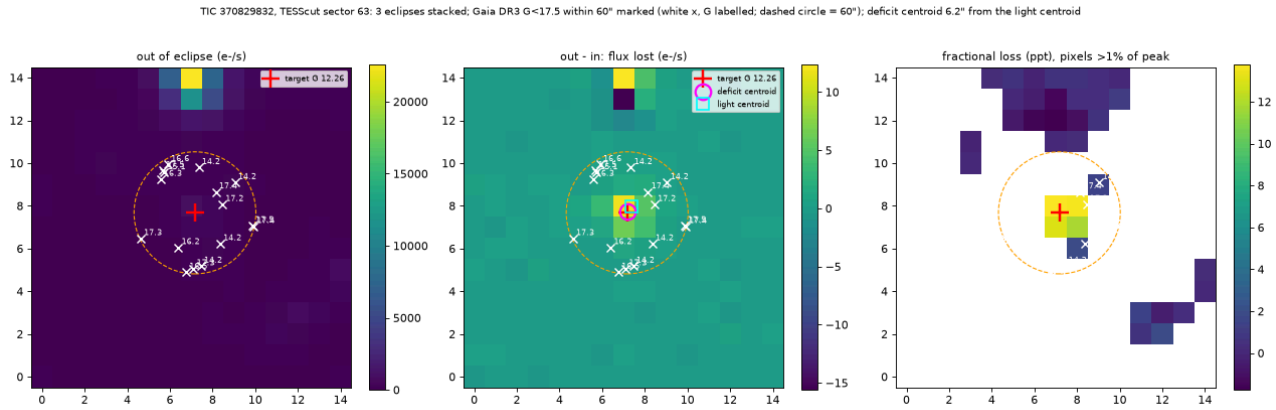
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional higher-resolution eclipse imaging and binary characterization.



R501; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R501; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 370829832: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 4 fitted representative eclipses in sectors [37, 90]; 4 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 4 same-kind actual midpoints yield P = 9.2465432321 ± 0.0000330778 d and reference epoch BTJD 2317.6240607 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 63 cutout places the deficit centroid 1.79 arcsec from the target; peak pixel [7, 14] versus target pixel [7, 8]. Image-model fractional residual 0.949 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 5249020789631628928: 37.8 arcsec, G 16.212, approximate intrinsic eclipse requirement 0.537; Gaia 5249020720912155520: 38.7 arcsec, G 14.177, approximate intrinsic eclipse requirement 0.094; Gaia 5249020858351113472: 44.5 arcsec, G 14.172, approximate intrinsic eclipse requirement 0.094; Gaia 5249020927070584832: 46.6 arcsec, G 16.304, approximate intrinsic eclipse requirement 0.583; Gaia 5249020858351116032: 49.0 arcsec, G 14.187, approximate intrinsic eclipse requirement 0.095; Gaia 5249020927070586112: 49.2 arcsec, G 15.096, approximate intrinsic eclipse requirement 0.201; Gaia 5249020789631629440: 52.4 arcsec, G 14.216, approximate intrinsic eclipse requirement 0.097; Gaia 5249020927070587392: 53.9 arcsec, G 16.631, approximate intrinsic eclipse requirement 0.783; Gaia 5249020789631627776: 58.8 arcsec, G 16.523, approximate intrinsic eclipse requirement 0.710. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.

4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 13.79/12.50/9.08 ppt over 3 stacked events, compared with median selected QLP raw fit depth 14.77 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5249020892710847488 has P = 9.2458347321 d and e = 0.18138. The closest edge-on conjunction offset at the measured epoch is -0.03335 d (-0.00361 phase), with 0.38441 d linearly propagated full-SB1-covariance timing error. A rough central duration is 6.19 h versus observed median 2.69 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: The S63 difference image has a remote single-pixel maximum at [7,14] coincident with a bright field source; its origin is uncertain. The multi-pixel eclipse deficit and centroid lie near the target [7,8]. The remote maximum and large image-model residual limit precision localization.

Next check: Optional higher-resolution eclipse imaging and binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
37	83	2317.6218651	0.003472	15.04	used
37	84	2326.8729892	0.003472	16.52	used
90	238	3750.8350612	0.001678	14.12	used
90	240	3769.3338324	0.001489	14.50	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s63.json, historical/diff_s63.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R501. Public evidence files: diff_s63.json, diff_s63.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

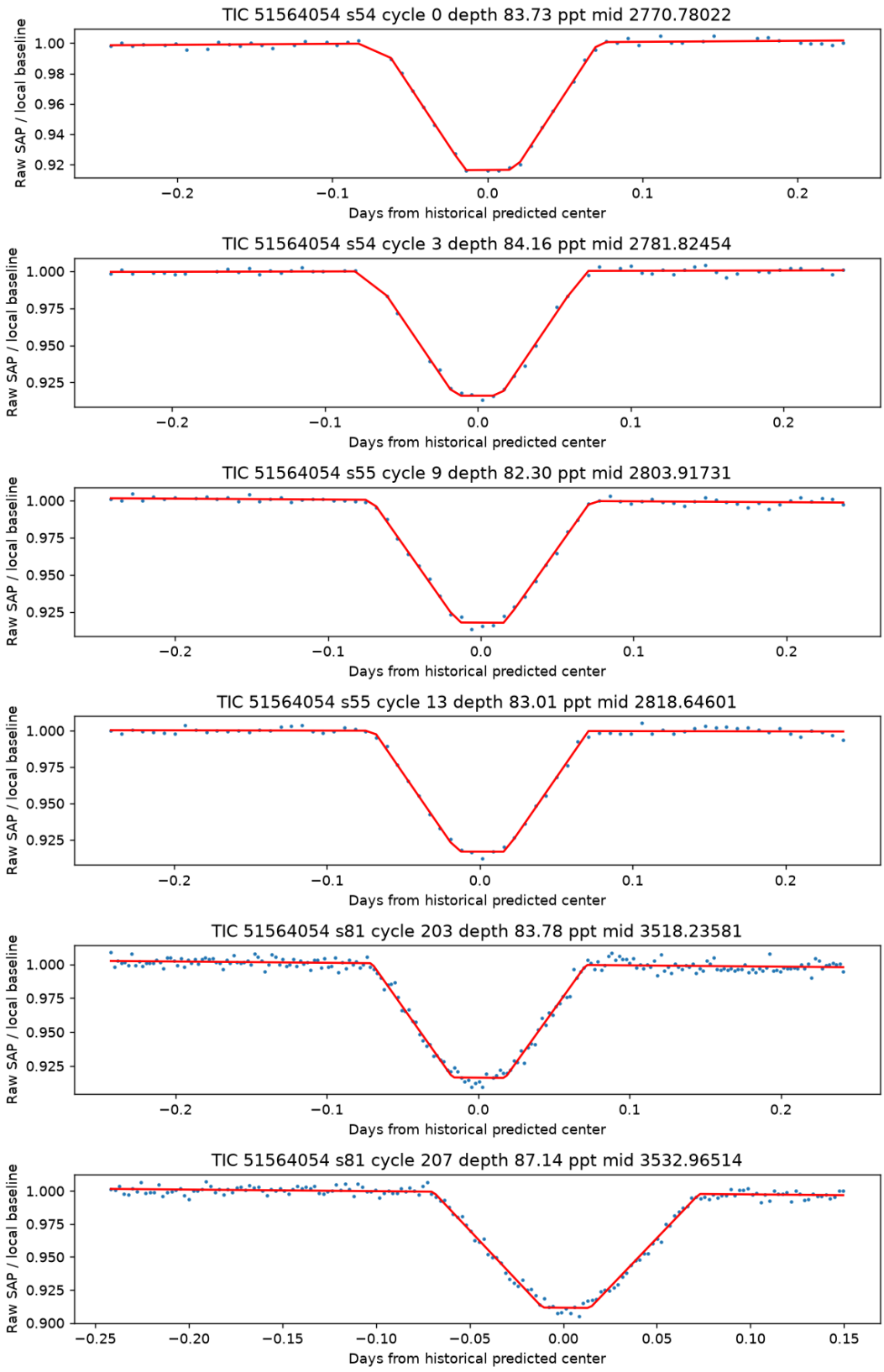
R502. TIC 51564054

physical_verdict: CONFIRMED

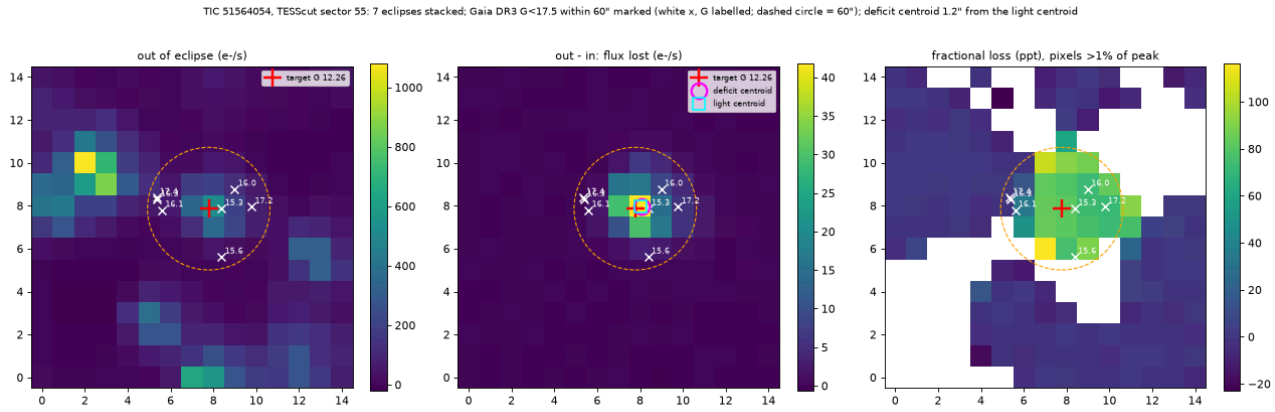
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R502; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R502; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 51564054: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 20 fitted representative eclipses in sectors [54, 55, 81]; 20 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 20 same-kind actual midpoints yield P = 3.6820558498 ± 0.0000053792 d and reference epoch BTJD 2770.7791920 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 55 cutout places the deficit centroid 6.58 arcsec from the target; peak pixel [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.567 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: None within 60 arcsec under the approximate G-band flux budget. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 81.07/80.04/77.97 ppt over 7 stacked events, compared with median selected QLP raw fit depth 83.28 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1805030134371636352 has P = 3.6821627017 d and e = 0.03959. The closest edge-on conjunction offset at the measured epoch is -0.09878 d (-0.02683 phase), with 0.16553 d linearly propagated full-SB1-covariance timing error. A rough central duration is 5.75 h versus observed median 3.39 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Twenty raw events across S54/55/81 recur at the measured period; the difference core lies on the target and no Gaia neighbor passes the approximate brightness screen.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
54	0	2770.7802194	0.003472	83.73	used
54	1	2774.4612189	0.003472	85.99	used
54	2	2778.1427547	0.003472	81.18	used
54	3	2781.8245378	0.003472	84.16	used
54	5	2789.1901975	0.003472	86.15	used
54	6	2792.8708408	0.003472	84.37	used
55	8	2800.2352509	0.003472	85.90	used
55	9	2803.9173093	0.003472	82.30	used
55	10	2807.5993533	0.003472	77.57	used
55	11	2811.2822047	0.003472	79.44	used
55	12	2814.9647274	0.003472	80.90	used
55	13	2818.6460093	0.003472	83.01	used
55	14	2822.3280330	0.003472	82.75	used
81	200	3507.1912335	0.001157	81.01	used
81	201	3510.8728408	0.001157	83.73	used
81	203	3518.2358094	0.001157	83.78	used
81	204	3521.9174461	0.001157	83.15	used
81	205	3525.6004124	0.001157	81.48	used
81	206	3529.2831125	0.001157	83.40	used
81	207	3532.9651413	0.001157	87.14	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s55.json, historical/diff_s55.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R502. Public evidence files: diff_s55.json, diff_s55.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

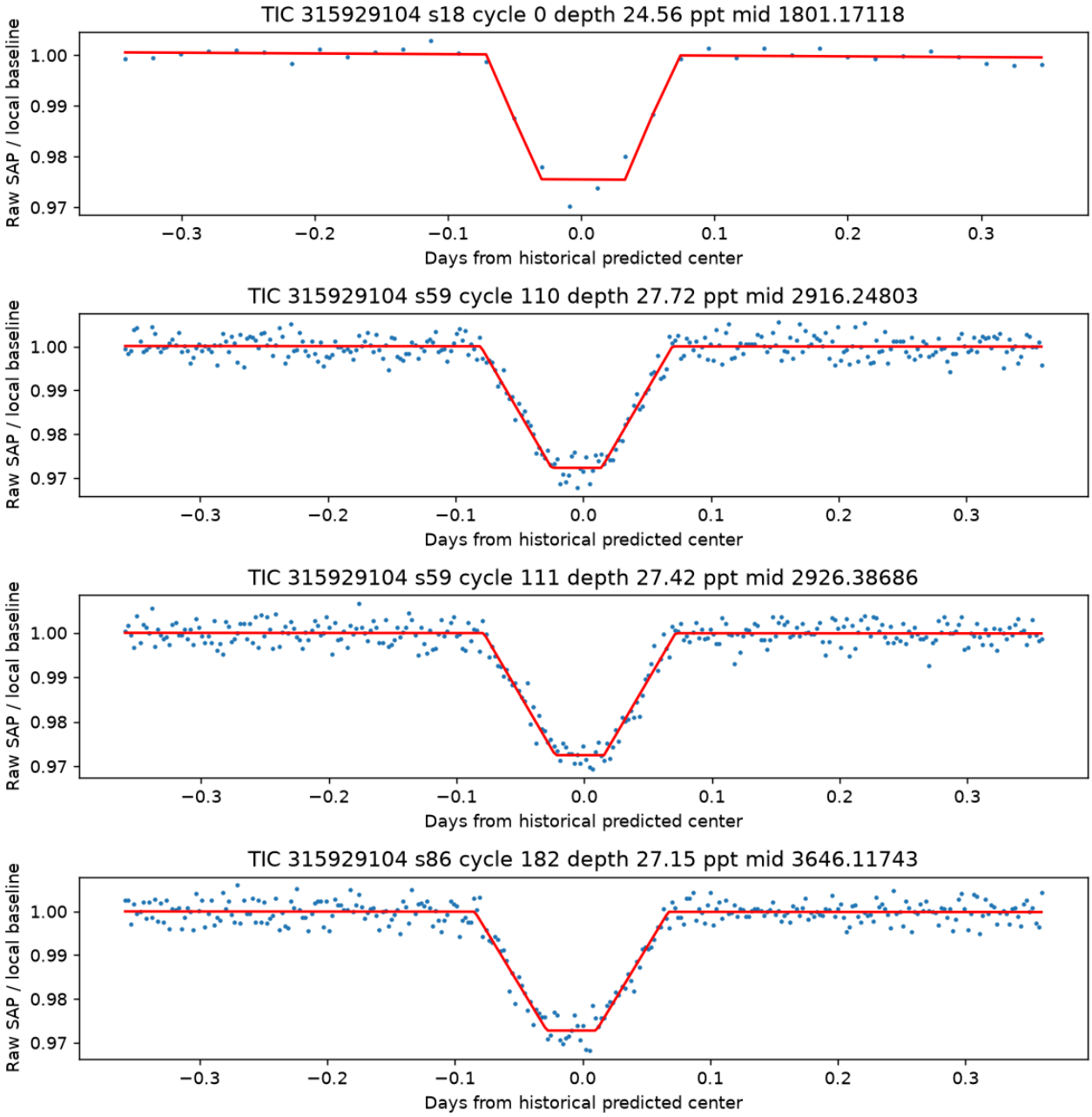
R503. TIC 315929104

physical_verdict: CONFIRMED

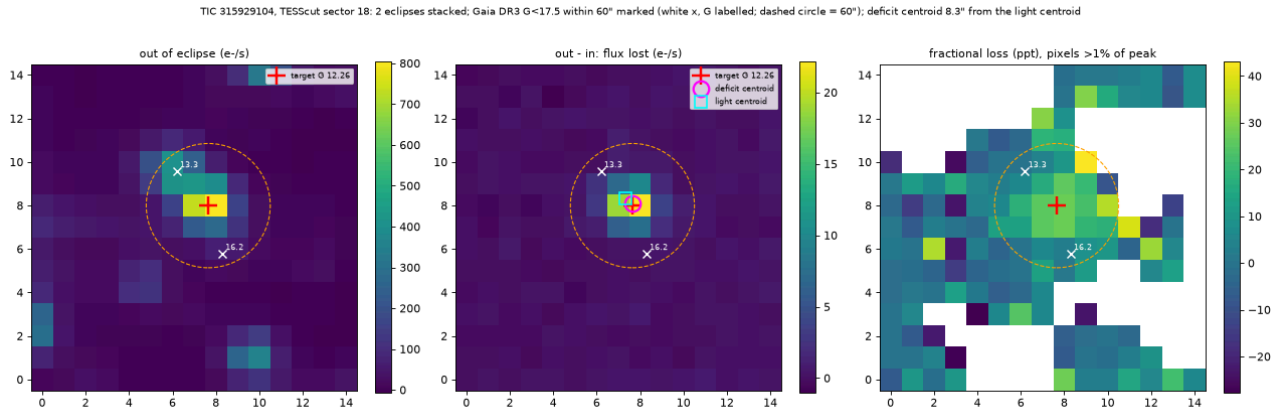
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R503; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R503; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 315929104: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 4 fitted representative eclipses in sectors [18, 59, 86]; 4 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 4 same-kind actual midpoints yield P = 10.1370634374 ± 0.0000192086 d and reference epoch BTJD 1801.1719141 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 18 cutout places the deficit centroid 1.70 arcsec from the target; peak pixel [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.460 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 449247978525955200: 43.0 arcsec, G 13.261, approximate intrinsic eclipse requirement 0.088; Gaia 449244959165948032: 46.1 arcsec, G 16.177, approximate intrinsic eclipse requirement 0.951. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 27.55/24.62/18.58 ppt over 2 stacked events, compared with median selected QLP raw fit depth 27.28 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 449245027885422592 has P = 10.1358765398 d and e = 0.08420. The closest edge-on conjunction offset at the measured epoch is 0.13186 d (0.01301 phase), with 0.16823 d linearly propagated full-

SB1-covariance timing error. A rough central duration is 8.05 h versus observed median 3.60 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Representative raw eclipses in S18/59/86 and a target-centered S18 difference core support the target; capable neighbors are tens of arcsec away.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
18	0	1801.1711761	0.010417	24.56	used
59	110	2916.2480272	0.001157	27.72	used
59	111	2926.3868563	0.001157	27.42	used
86	182	3646.1174333	0.001157	27.15	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s18.json, historical/diff_s18.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R503. Public evidence files: diff_s18.json, diff_s18.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

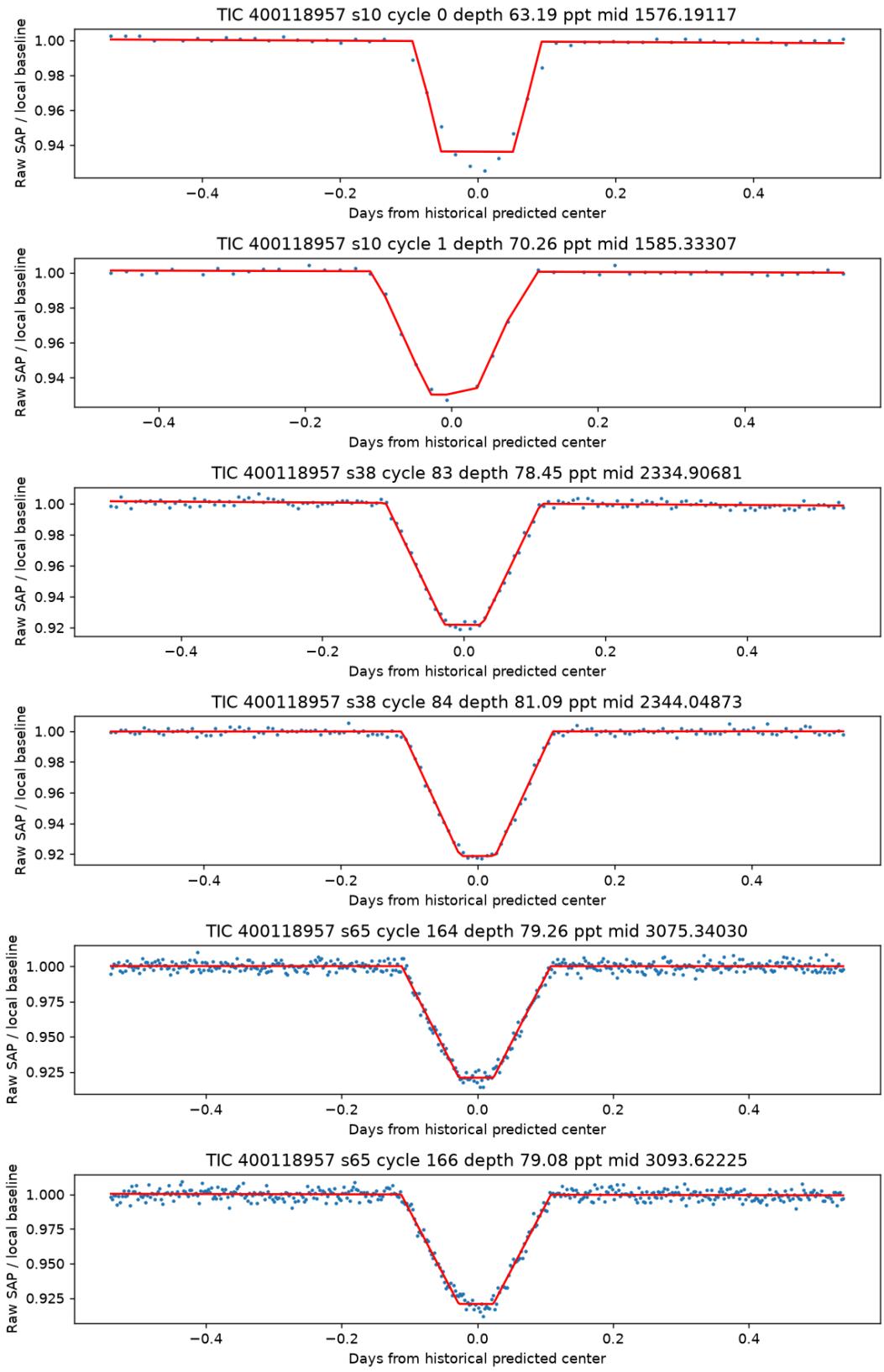
R504. TIC 400118957

physical_verdict: CONFIRMED

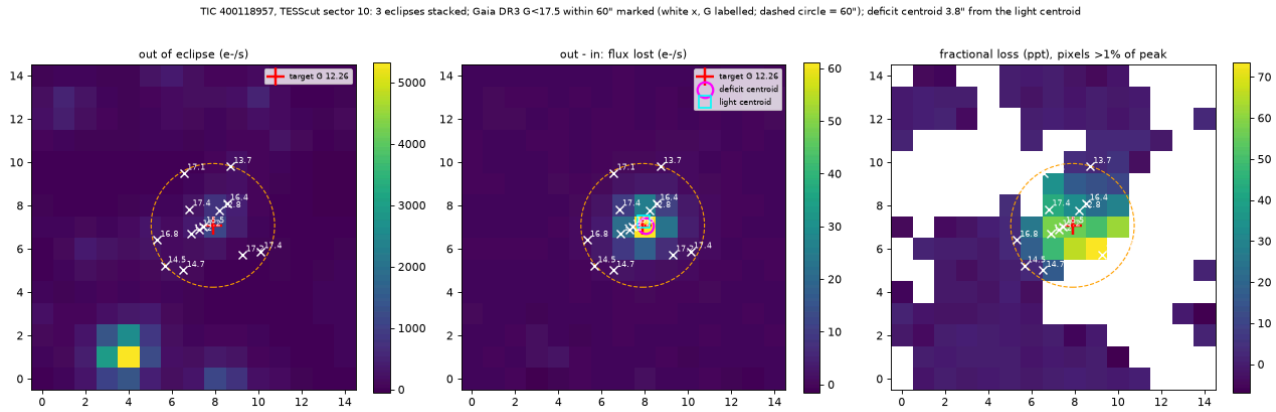
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R504; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R504; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 400118957: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 9 fitted representative eclipses in sectors [10, 38, 65]; 9 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 9 same-kind actual midpoints yield P = 9.1411474439 ± 0.0000216391 d and reference epoch BTJD 1576.1920454 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 10 cutout places the deficit centroid 2.32 arcsec from the target; peak pixel [8, 7] versus target pixel [8, 7]. Image-model fractional residual 0.514 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 5231653797315230336: 15.3 arcsec, G 12.789, approximate intrinsic eclipse requirement 0.188; Gaia 5231653969118047488: 50.8 arcsec, G 14.684, approximate intrinsic eclipse requirement 0.740; Gaia 5231653758651818240: 58.6 arcsec, G 13.675, approximate intrinsic eclipse requirement 0.336; Gaia 5231653969113936384: 59.0 arcsec, G 14.499, approximate intrinsic eclipse requirement 0.635. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 55.89/47.10/41.16 ppt over 3 stacked events, compared with median selected QLP raw fit depth 79.08 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5231653793011555840 has P = 9.1418612161 d and e = 0.18822. The closest edge-on conjunction

offset at the measured epoch is -0.06312 d (-0.00690 phase), with 0.11125 d linearly propagated full-SB1-covariance timing error. A rough central duration is 7.25 h versus observed median 5.27 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Raw depths change with sector aperture, while nine measured midpoints retain one recurrence. A capable neighbor 15.3 arcsec away is distinct from the target-centered difference core; crowding limits intrinsic depth claims.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
10	0	1576.1911659	0.010417	63.19	used
10	1	1585.3330745	0.010417	70.26	used
10	2	1594.4736317	0.010417	71.75	used
38	83	2334.9068107	0.003472	78.45	used
38	84	2344.0487326	0.003472	81.09	used
38	85	2353.1901072	0.003472	80.81	used
65	164	3075.3402986	0.001157	79.26	used
65	165	3084.4815508	0.001157	80.15	used
65	166	3093.6222526	0.001157	79.08	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s10.json, historical/diff_s10.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R504. Public evidence files: diff_s10.json, diff_s10.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

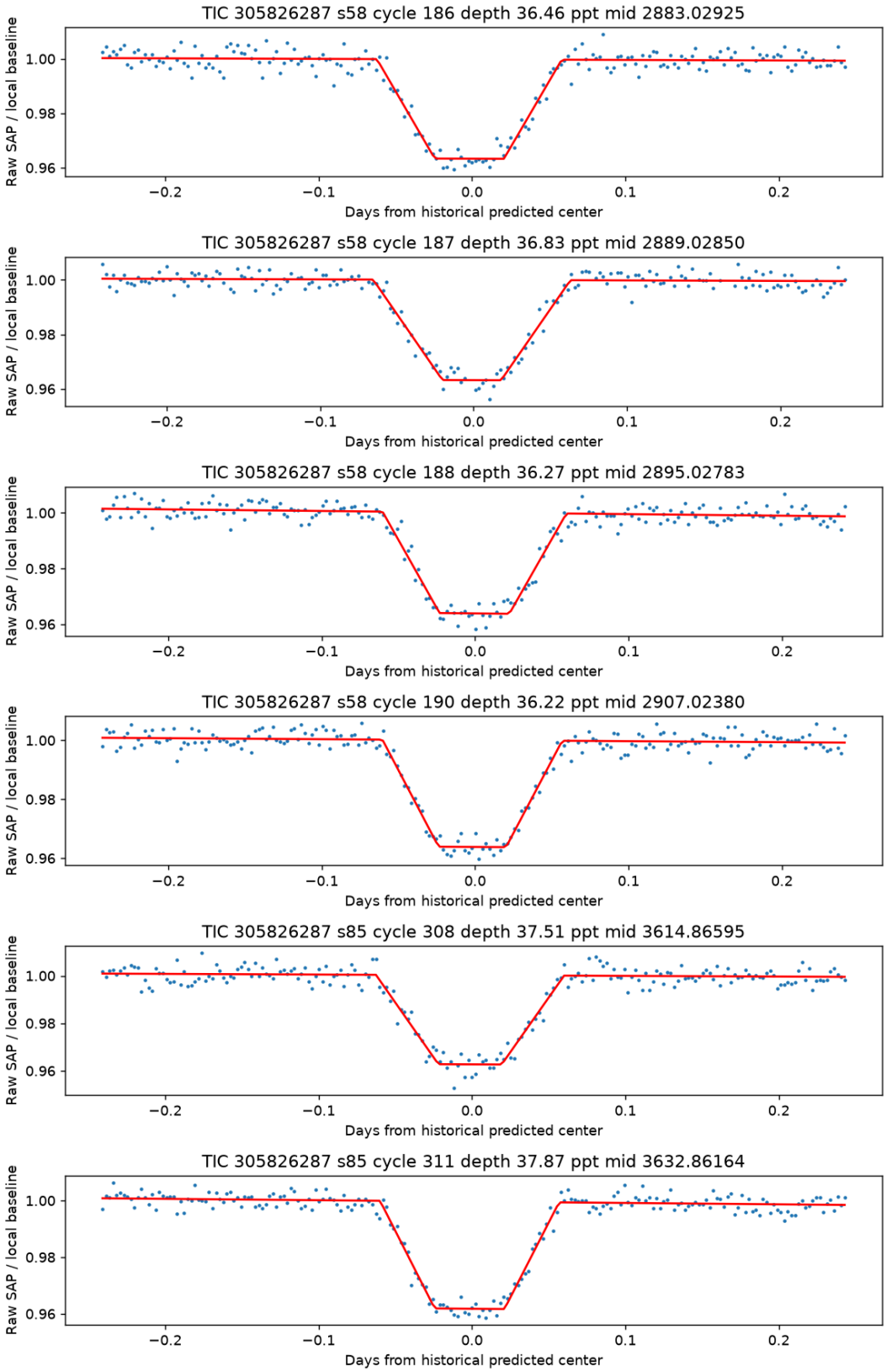
R505. TIC 305826287

physical_verdict: CONFIRMED

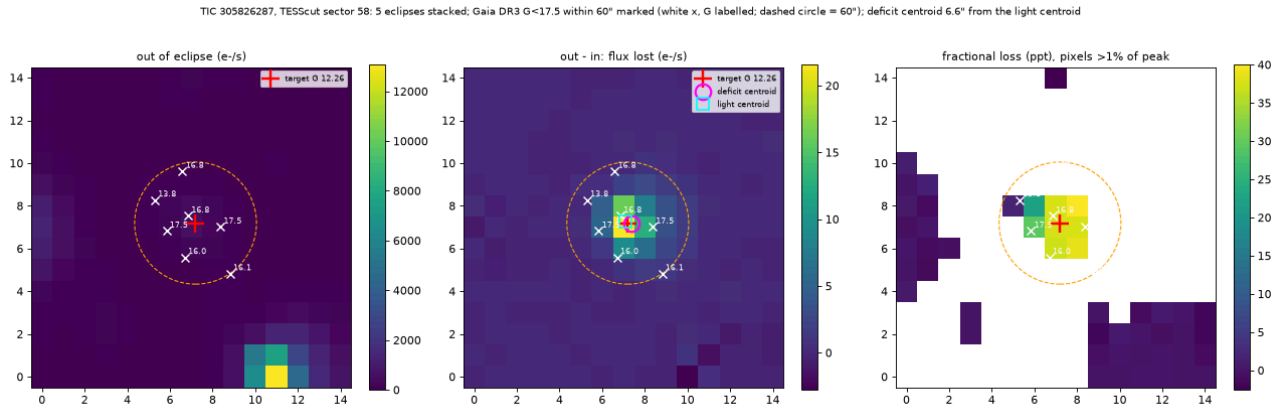
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R505; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R505; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 305826287: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 8 fitted representative eclipses in sectors [58, 85]; 8 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 8 same-kind actual midpoints yield P = 5.9986588262 ± 0.0000069644 d and reference epoch BTJD 2883.0297295 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 58 cutout places the deficit centroid 3.93 arcsec from the target; peak pixel [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.454 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 414950435582368640: 44.5 arcsec, G 13.760, approximate intrinsic eclipse requirement 0.186. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 37.50/35.53/32.80 ppt over 5 stacked events, compared with median selected QLP raw fit depth 36.65 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 414950435582368256 has P = 5.9989472733 d and e = 0.39238. The closest edge-on conjunction offset at the measured epoch is -0.14458 d (-0.02410 phase), with 0.64607 d linearly propagated full-SB1-covariance timing error. A rough central duration is 3.64 h versus observed median 2.88 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Eight raw events across S58/85 have stable depth and a target-centered difference core; the one capable listed neighbor is 44.5 arcsec away.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
58	186	2883.0292527	0.001157	36.46	used
58	187	2889.0284961	0.001157	36.83	used
58	188	2895.0278311	0.001157	36.27	used
58	189	2901.0258443	0.001157	36.28	used
58	190	2907.0238045	0.001157	36.22	used
85	308	3614.8659518	0.001157	37.51	used
85	309	3620.8653653	0.001157	37.55	used
85	311	3632.8616442	0.001157	37.87	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s58.json, historical/diff_s58.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R505. Public evidence files: diff_s58.json, diff_s58.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

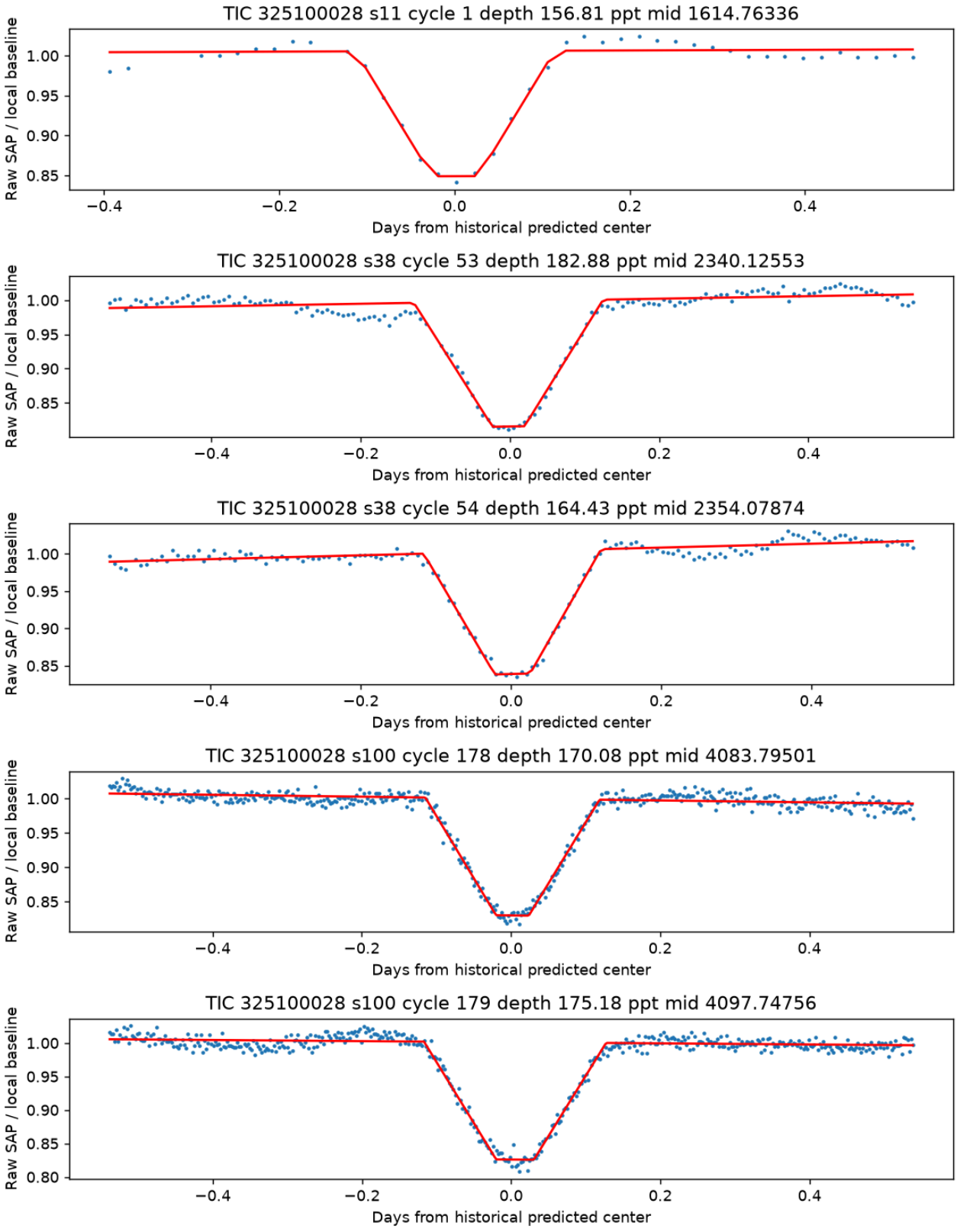
R506. TIC 325100028

physical_verdict: PROBABLE

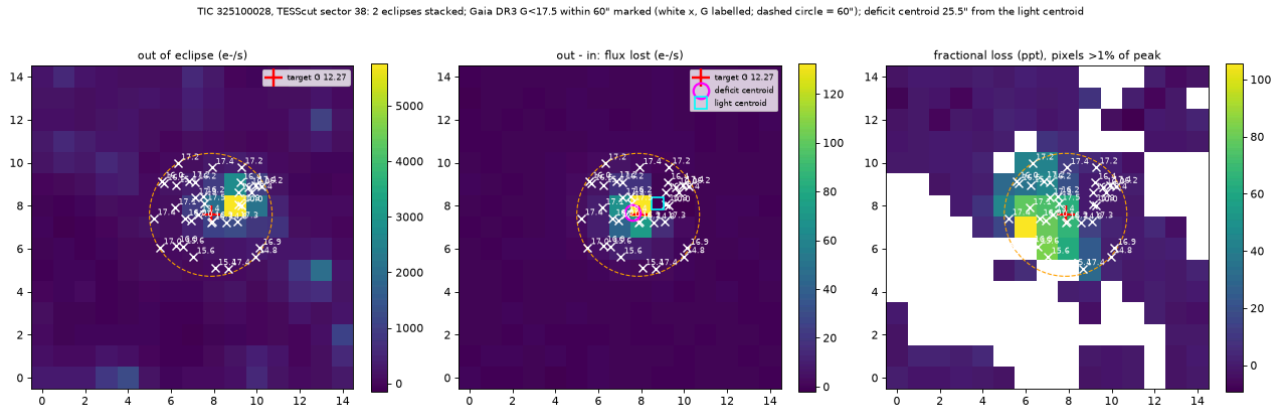
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse imaging and independent aperture/PRF characterization of the crowded field.



R506; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R506; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 325100028: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 5 fitted representative eclipses in sectors [11, 38, 100]; 5 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 5 same-kind actual midpoints yield P = 13.9493523851 ± 0.0000240409 d and reference epoch BTJD 1614.7612349 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: unresolved.** Historical sector 38 cutout places the deficit centroid 6.02 arcsec from the target; peak pixel [8, 8] versus target pixel [8, 8]. Image-model fractional residual 0.932 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 5864512143196081408: 12.3 arcsec, G 13.418, approximate intrinsic eclipse requirement 0.602; Gaia 5866013285774822528: 29.1 arcsec, G 11.757, approximate intrinsic eclipse requirement 0.252; Gaia 5866013285781269376: 31.6 arcsec, G 10.036, approximate intrinsic eclipse requirement 0.175. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: supports recurring loss with unresolved host.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 63.30/23.38/19.85 ppt over 2 stacked events, compared with median selected QLP raw fit depth 170.08 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 5866013354494300928 has P = 13.9475152836 d and e = 0.22338. The closest edge-on conjunction

offset at the measured epoch is 0.13013 d (0.00933 phase), with 0.22414 d linearly propagated full-SB1-covariance timing error. A rough central duration is 7.05 h versus observed median 5.67 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Five raw eclipses, including S38 events coincident with the historical TESScut stack, recur at one period. The S38 cutout shows a central deficit but its 3x3 depths of about 23 ppt disagree substantially with QLP raw fits of 164-183 ppt; dilution is plausible but not quantitatively demonstrated. A capable 12.3-arcsec Gaia neighbor remains viable at TESS resolution, so the host is unresolved.

Next check: Resolved eclipse imaging and independent aperture/PRF characterization of the crowded field.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
11	1	1614.7633607	0.010417	156.81	used
38	53	2340.1255290	0.003472	182.88	used
38	54	2354.0787364	0.003472	164.43	used
100	178	4083.7950061	0.001157	170.08	used
100	179	4097.7475570	0.001157	175.18	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s38.json, historical/diff_s38.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R506. Public evidence files: diff_s38.json, diff_s38.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

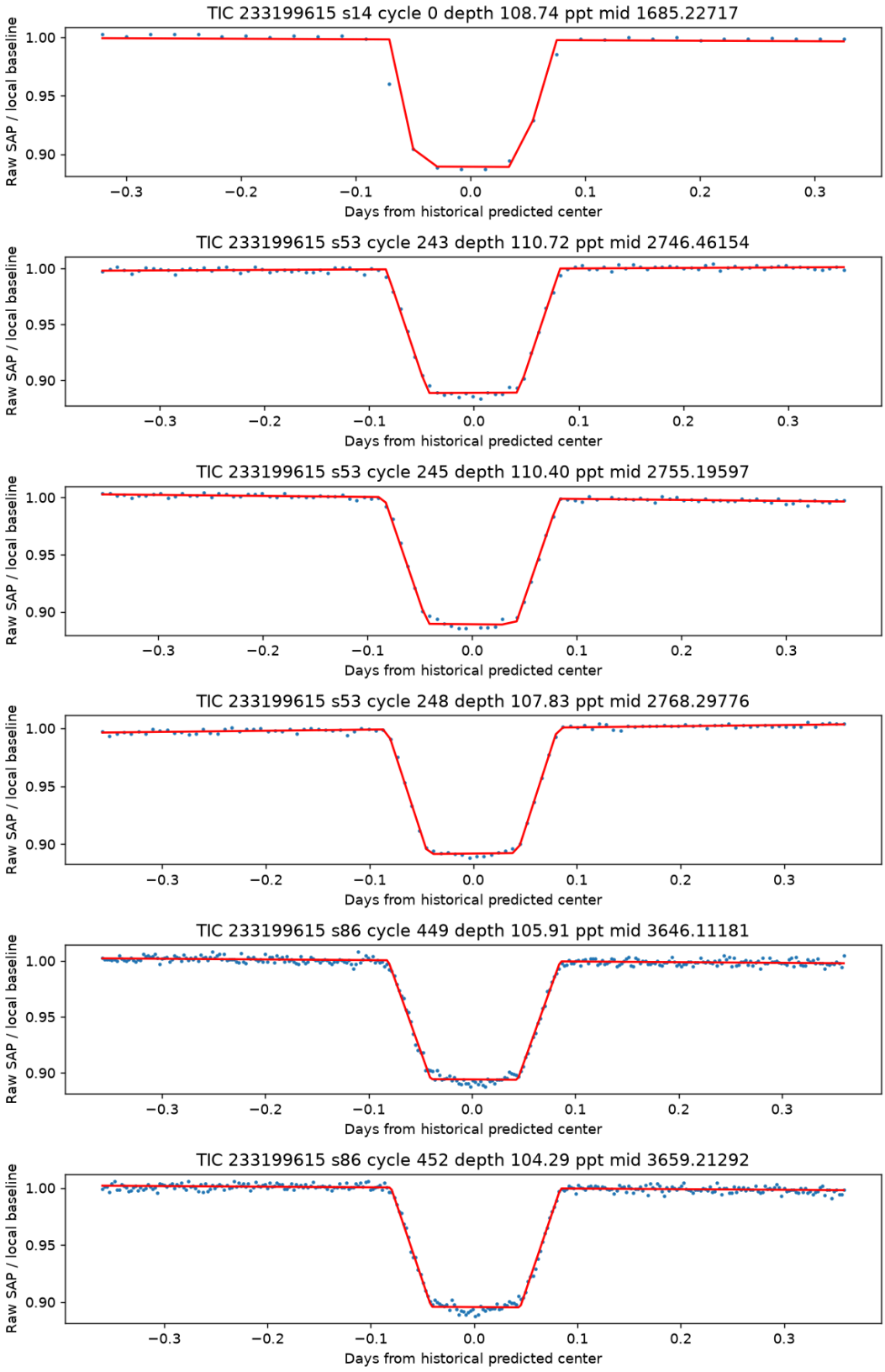
R507. TIC 233199615

physical_verdict: CONFIRMED

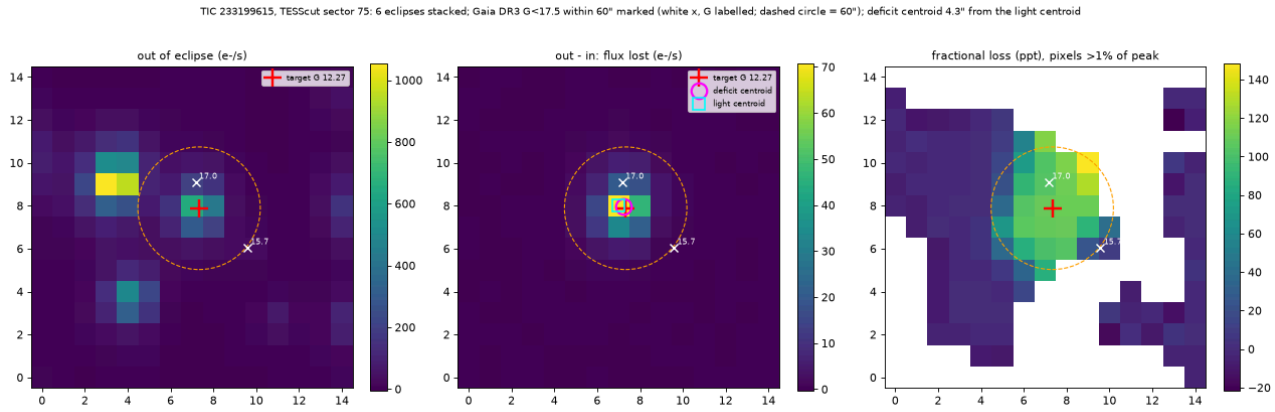
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision binary characterization.



R507; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R507; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 233199615: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 13 fitted representative eclipses in sectors [14, 53, 86]; 13 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 13 same-kind actual midpoints yield P = 4.3672327612 ± 0.0000067972 d and reference epoch BTJD 1685.2243747 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 75 cutout places the deficit centroid 3.05 arcsec from the target; peak pixel [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.884 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: None within 60 arcsec under the approximate G-band flux budget. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 109.47/108.17/97.70 ppt over 6 stacked events, compared with median selected QLP raw fit depth 107.83 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1638829129242739968 has P = 4.3672040925 d and e = 0.09279. The closest edge-on conjunction offset at the measured epoch is 0.05811 d (0.01331 phase), with 0.12907 d linearly propagated full-SB1-covariance timing error. A rough central duration is 4.84 h versus observed median 3.99 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Thirteen representative raw events across S14/53/86 repeat at the same period; the difference core is target-centered and no listed Gaia neighbor passes the approximate brightness screen.

Next check: Optional precision binary characterization.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
14	0	1685.2271693	0.010417	108.74	used
14	3	1698.3305035	0.010417	106.48	used
53	243	2746.4615423	0.003472	110.72	used
53	244	2750.8292655	0.003472	111.05	used
53	245	2755.1959690	0.003472	110.40	used
53	246	2759.5631421	0.003472	109.28	used
53	247	2763.9305927	0.003472	108.97	used
53	248	2768.2977583	0.003472	107.83	used
86	447	3637.3775447	0.001157	104.42	used
86	449	3646.1118101	0.001157	105.91	used
86	450	3650.4798914	0.001157	106.67	used
86	451	3654.8463029	0.001157	105.04	used
86	452	3659.2129190	0.001157	104.29	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s75.json, historical/diff_s75.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R507. Public evidence files: diff_s75.json, diff_s75.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

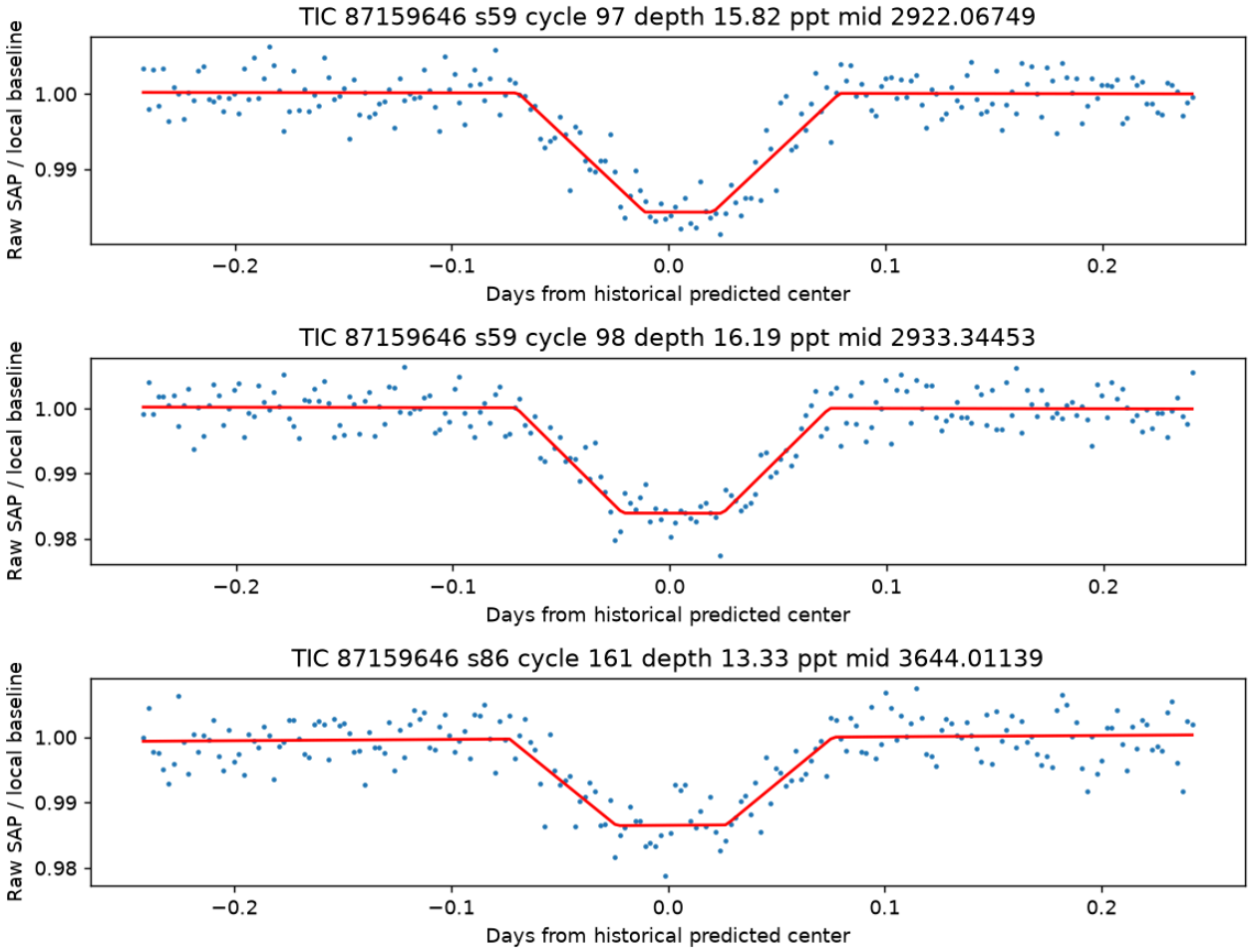
R508. TIC 87159646

physical_verdict: PROBABLE

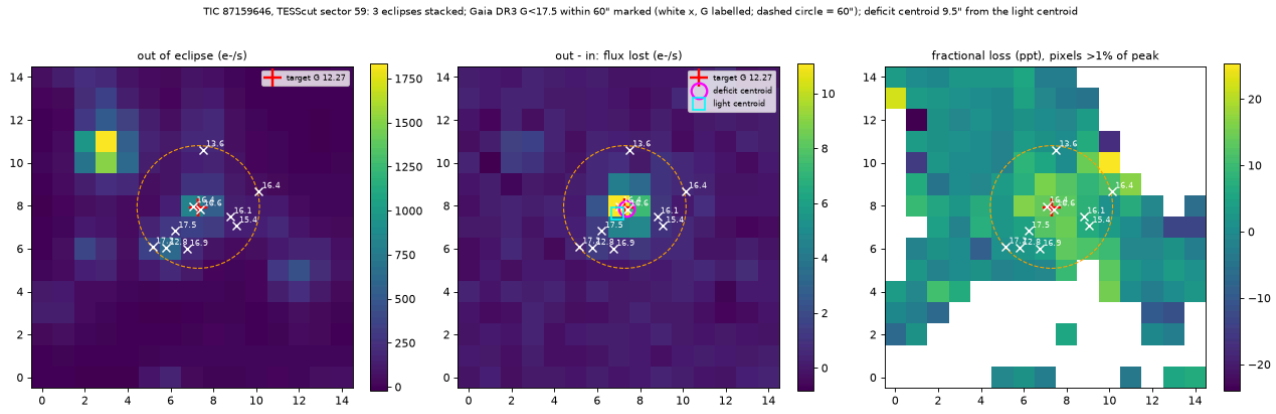
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry of the close Gaia sources.



R508; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R508; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 87159646: PROBABLE

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 3 fitted representative eclipses in sectors [59, 86]; 3 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 3 same-kind actual midpoints yield P = 11.2803975769 ± 0.0000584060 d and reference epoch BTJD 2922.0658972 (Selected repeating eclipse; primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; representative parity shows no persistent unequal alternation. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: unresolved.** Historical sector 59 cutout places the deficit centroid 1.83 arcsec from the target; peak pixel [7, 8] versus target pixel [7, 8]. Image-model fractional residual 0.861 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 469818814585955328: 3.0 arcsec, G 16.593, approximate intrinsic eclipse requirement 0.873; Gaia 469818810292668416: 4.9 arcsec, G 16.391, approximate intrinsic eclipse requirement 0.728; Gaia 469820223341588480: 32.6 arcsec, G 16.069, approximate intrinsic eclipse requirement 0.545; Gaia 469820223341588864: 40.8 arcsec, G 15.394, approximate intrinsic eclipse requirement 0.300; Gaia 469818745872854528: 49.1 arcsec, G 12.762, approximate intrinsic eclipse requirement 0.041; Gaia 469819020750741888: 54.5 arcsec, G 13.609, approximate intrinsic eclipse requirement 0.071; Gaia 469820223341582208: 59.6 arcsec, G 16.350, approximate intrinsic eclipse requirement 0.702. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.
4. **Alternative reduction: supports recurring loss with unresolved host.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 13.67/12.75/8.71 ppt over 3 stacked events, compared with median selected QLP raw fit depth 15.82 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 469818814592320896 has P = 11.2828845399 d and e = 0.14903. The closest edge-on conjunction offset at the measured epoch is -0.57231 d (-0.05072 phase), with 0.55955 d linearly propagated full-SB1-covariance timing error. A rough central duration is 8.87 h versus observed median 3.56 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Three raw eclipses repeat in S59/86, but two capable Gaia neighbors at 3.0 and 4.9 arcsec share the central TESS pixel with the target. The S59 deficit cannot resolve the host.

Next check: Resolved eclipse photometry of the close Gaia sources.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
59	97	2922.0674903	0.001435	15.82	used
59	98	2933.3445278	0.001500	16.19	used
86	161	3644.0113934	0.002044	13.33	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s59.json, historical/diff_s59.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R508. Public evidence files: diff_s59.json, diff_s59.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

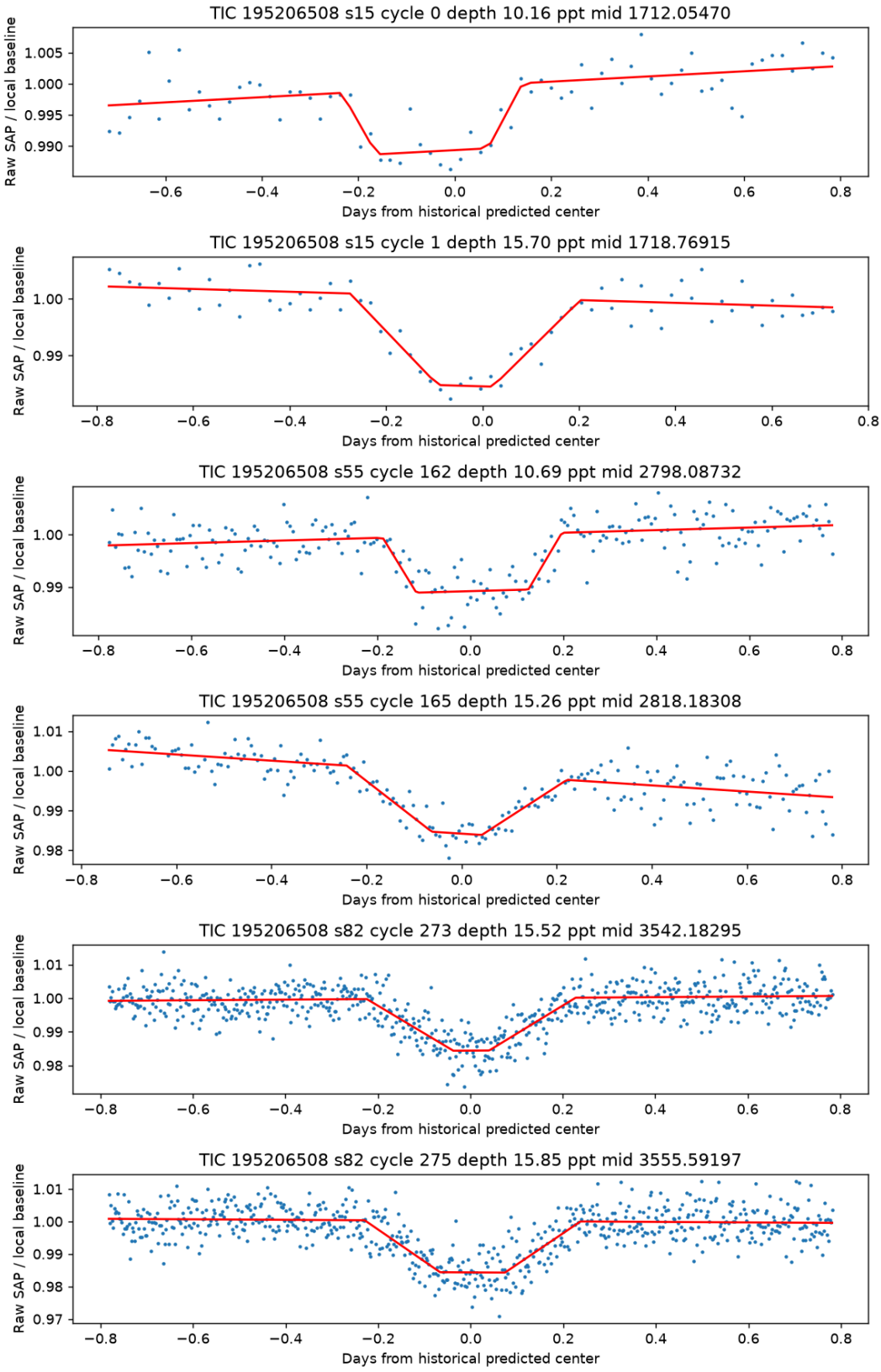
R509. TIC 195206508

physical_verdict: CONFIRMED

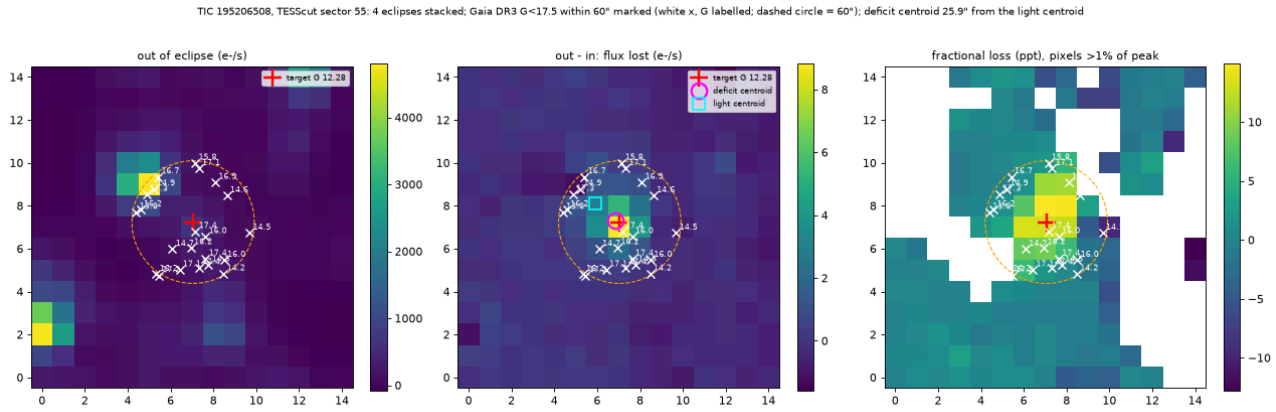
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved eclipse imaging and refined primary/secondary modeling.



R509; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R509; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 195206508: CONFIRMED

Full bounded review: **COMPLETE**. Physical outcome and catalogue novelty are separate.

1. **Raw repeated events: pass.** Quality-zero QLP SAP_FLUX contains 10 fitted representative eclipses in sectors [15, 55, 82]; 6 enter the reported timing fit. Local trapezoids have free centers, depths, durations and baselines. Predicted times guided windows only; flagged or uncovered intervals are not clean nulls. See raw-events.png, raw-windows.csv and measurements.json.
2. **Measured timing and aliases: pass.** 6 same-kind actual midpoints yield P = 13.4074536487 ± 0.0000554388 d and reference epoch BTJD 1718.7700294 (Deeper repeating eclipse, odd half-period cycles; primary by convention). Odd historical half-period cycles only; deeper repeating eclipse. Even cycles show a shallower secondary. Same-sector repeats and parity are in ../alias-audit.json. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Half/double alias assessment is bounded by representative raw morphology, not an exhaustive blind period search.
3. **Source attribution: favors target at TESS pixel resolution.** Historical sector 55 cutout places the deficit centroid 4.32 arcsec from the target; peak pixel [7, 7] versus target pixel [7, 7]. Image-model fractional residual 0.998 is not a calibrated localization error. Gaia neighbors capable under the approximate G-band flux budget: Gaia 1870085389842887808: 18.2 arcsec, G 16.013, approximate intrinsic eclipse requirement 0.405; Gaia 1870085389842888192: 24.0 arcsec, G 16.168, approximate intrinsic eclipse requirement 0.465; Gaia 1870085419893612288: 32.0 arcsec, G 14.661, approximate intrinsic eclipse requirement 0.125; Gaia 1870084801418161920: 40.4 arcsec, G 14.623, approximate intrinsic eclipse requirement 0.122; Gaia 1870085183684446336: 42.6 arcsec, G 16.907, approximate intrinsic eclipse requirement 0.906; Gaia 1870085389842895488: 43.0 arcsec, G 14.507, approximate intrinsic eclipse requirement 0.110; Gaia 1870085389842896384: 47.0 arcsec, G 16.029, approximate intrinsic eclipse requirement 0.411; Gaia 1870085149324698624: 49.8 arcsec, G 14.891, approximate intrinsic eclipse requirement 0.152; Gaia 1870085218044179712: 52.3 arcsec, G 16.164, approximate intrinsic eclipse requirement 0.463; Gaia 1870085149324696192: 55.0 arcsec, G 16.678, approximate intrinsic eclipse requirement 0.736; Gaia 1870085011885771648: 55.4 arcsec, G 14.453, approximate intrinsic eclipse requirement 0.106; Gaia 1870085114964960384: 55.6 arcsec, G 15.774, approximate intrinsic eclipse

requirement 0.327; Gaia 1870085218044179584: 55.8 arcsec, G 13.824, approximate intrinsic eclipse requirement 0.065; Gaia 1870085458562377600: 57.3 arcsec, G 14.222, approximate intrinsic eclipse requirement 0.088; Gaia 1870085424202627712: 59.9 arcsec, G 16.159, approximate intrinsic eclipse requirement 0.461. The neighbor intrinsic eclipse requirement is not directly comparable to blended pixel fractional loss.

4. **Alternative reduction: pass.** Target-centered background-subtracted TESScut 1x1/3x3/5x5 aperture depths are 13.13/11.62/3.32 ppt over 4 stacked events, compared with median selected QLP raw fit depth 15.78 ppt. QLP and TESScut products for the same sector share photons; distinct sectors are separate observing occasions with the same instrument. Aperture dilution or mixed eclipse kinds limit depth equality. and aperture dilution or mixed eclipse kinds limit depth equality.
5. **Gaia geometry: consistent within descriptive limits.** Matched DR3 source 1870085183684452864 has P = 13.4086220195 d and e = 0.03442. The closest edge-on conjunction offset at the measured epoch is -0.14065 d (-0.01049 phase), with 0.23479 d linearly propagated full-SB1-covariance timing error. A rough central duration is 27.40 h versus observed median 10.86 h; TIC radius and minimum-mass companion are approximations, and impact parameter is not fitted.

Deciding interpretation and limitation: Alternating shallow even and deeper odd eclipses across S15/55/82 set a 13.407-day orbital period. The deeper odd cycles alone define the reported epoch and timing fit. The S55 cutout has a central target deficit, but high image-model residual and neighboring field stars limit subpixel localization.

Next check: Optional resolved eclipse imaging and refined primary/secondary modeling.

Catalogue novelty is in the central seven-group ../../shared-catalogues/coverage-matrix.csv receipt; physical classification does not assert exhaustive priority.

Sector	Historical photometric cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt	Orbital timing status
15	0	1712.0546966	0.012570	10.16	excluded from orbital fit (secondary)
15	1	1718.7691468	0.010417	15.70	used
15	3	1732.1783285	0.010417	16.58	used
55	162	2798.0873194	0.004654	10.69	excluded from orbital fit (secondary)
55	163	2804.7689867	0.014417	20.51	used
55	165	2818.1830818	0.008534	15.26	used
82	272	3535.4816843	0.006814	12.26	excluded from orbital fit (secondary)
82	273	3542.1829509	0.003761	15.52	used
82	274	3548.8921402	0.004545	12.21	excluded from orbital fit (secondary)
82	275	3555.5919677	0.003868	15.85	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, ../download-receipts.json, ../alias-audit.json, ../gaia-geometry.json, historical/diff_s55.json, historical/diff_s55.png, historical/gaia.json and ../helper-audit.md. Original TESScut binary: [local path omitted] (hash in ../source-hashes.json). All evidence and source files are in ../SHA256SUMS.json.

Stable review key: R509. Public evidence files: diff_s55.json, diff_s55.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

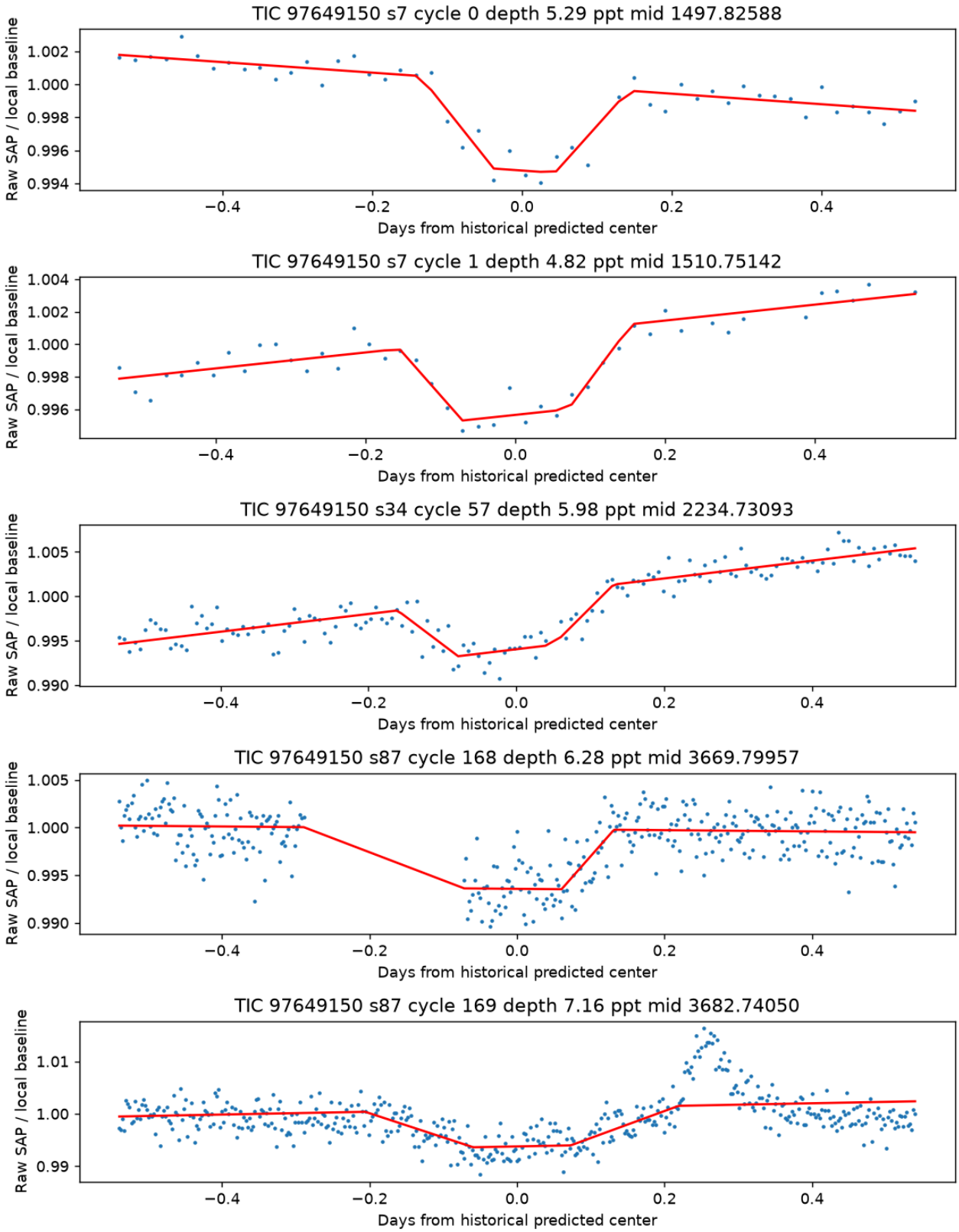
R510. TIC 97649150

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved time-series photometry.



R510; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 97649150: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 4 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [7, 34, 87]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 12.9285082280 ± 0.0000582502 d and epoch BTJD 1497.82197689 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 7 deficit centroid is about 2.44 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 6 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\97649150\diff_s07.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S7 aperture photometry recovers 2 stacked events at 3x3 depth 5.75 ppt, versus median selected raw QLP fit depth 5.63 ppt. TESScut S7 and representative QLP S7 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 5603413453346701184: orbital P = 12.927076847 d, e = 0.2541. Closest edge-on conjunction offset at the measured epoch is +0.3231 d (+0.0250 phase); linear full-NSS-covariance timing scale 0.3629 d. Approximate central duration 11.53 h versus measured median 7.02 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S7/34/87 raw eclipses agree. The closest capable neighbor is G17.24 at 13.7 arcsec and would need roughly 54% intrinsic loss; deficit is within 0.12 pixel of target. Faint close-source ambiguity remains below calibrated pixel precision.

Next check: Optional resolved time-series photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
7	0	1497.8258819	0.010417	5.29
7	1	1510.7514174	0.014168	4.82
34	57	2234.7309255	0.016170	5.98
87	169	3682.7404985	0.005530	7.16

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\97649150\diff_s07.json`, its PNG and Gaia-neighbor table; source and artifact hashes are

in `../SHA256SUMS.json`.

Stable review key: R510. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 97649150.png

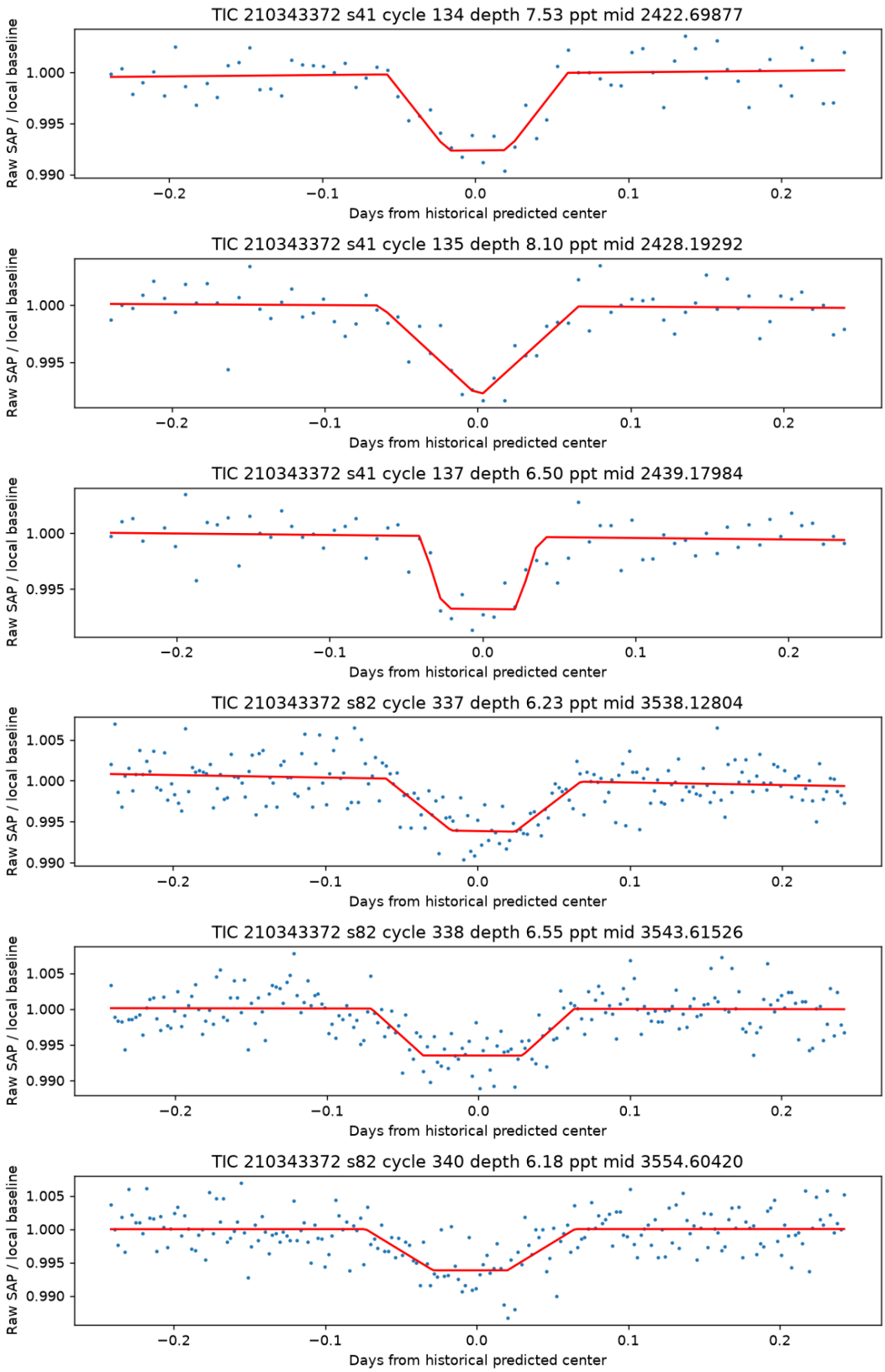
R511. TIC 210343372

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved time-series photometry.



R511; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 210343372: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 8 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [41, 82]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 5.4946996557 ± 0.0000125202 d and epoch BTJD 2422.69788125 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 55 deficit centroid is about 4.07 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 10 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\210343372\diff_s55.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S55 aperture photometry recovers 5 stacked events at 3x3 depth 5.84 ppt, versus median selected raw QLP fit depth 7.04 ppt. TESScut S55 and representative QLP sectors [41, 82] are separate TESS observing occasions on the same instrument; their photons are distinct. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 1842372749004992000: orbital P = 5.493566815 d, e = 0.0368. Closest edge-on conjunction offset at the measured epoch is +0.2292 d (+0.0417 phase); linear full-NSS-covariance timing scale 0.2725 d. Approximate central duration 8.20 h versus measured median 3.10 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S41/82 provide consecutive raw repeats. The closest capable G16.03 source lies 17.8 arcsec away and would require about 22% intrinsic loss; deficit is within 0.20 pixel of target. The Gaia phase offset is descriptive and within its large propagated orbital uncertainty.

Next check: Optional resolved time-series photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
41	134	2422.6987727	0.003472	7.53
41	135	2428.1929206	0.003472	8.10
41	137	2439.1798373	0.003472	6.50
41	138	2444.6774781	0.003472	8.89
82	337	3538.1280403	0.002663	6.23
82	338	3543.6152633	0.002550	6.55
82	339	3549.1081336	0.002577	9.30

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
82	340	3554.6042030	0.003268	6.18

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\210343372\diff_s55.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R511. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 210343372.png

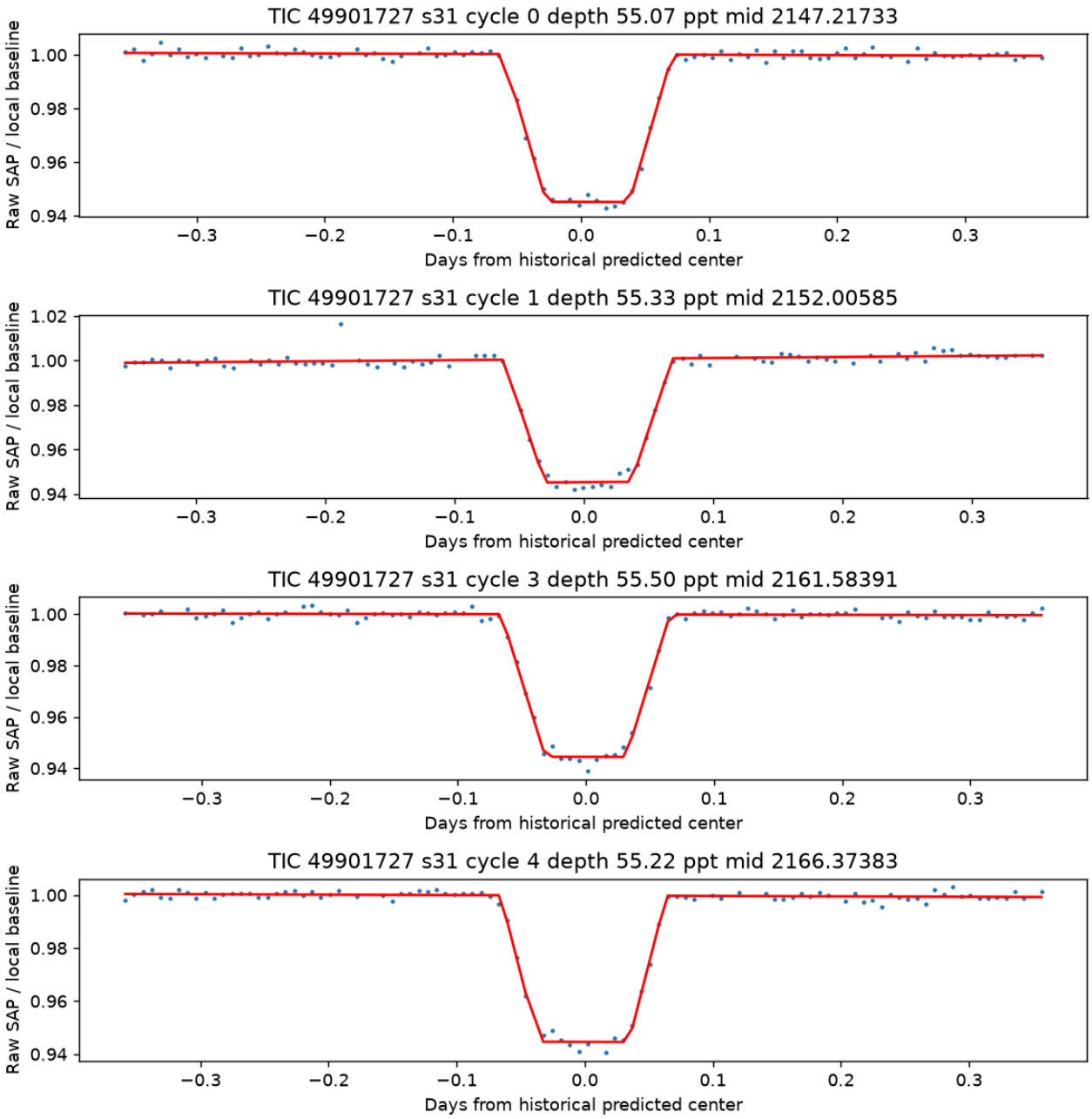
R512. TIC 49901727

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional-sector timing would refine the period.



R512; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 49901727: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 4 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [31]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 4.7891053447 ± 0.0010980248 d and epoch BTJD 2147.21701942 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 31 deficit centroid is about 2.74 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\49901727\diff_s31.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S31 aperture photometry recovers 4 stacked events at 3x3 depth 54.10 ppt, versus median selected raw QLP fit depth 55.27 ppt. TESScut S31 and representative QLP S31 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 3266000059384203136: orbital P = 4.788914677 d, e = 0.0281. Closest edge-on conjunction offset at the measured epoch is +0.1153 d (+0.0241 phase); linear full-NSS-covariance timing scale 0.3693 d. Approximate central duration 3.65 h versus measured median 3.15 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Four same-sector raw eclipses establish recurrence and a photometric period, but the short single-sector baseline leaves a comparatively broad period error. No capable Gaia neighbor within 60 arcsec; target-centered difference image.

Next check: Additional-sector timing would refine the period.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
31	0	2147.2173324	0.003472	55.07
31	1	2152.0058502	0.003472	55.33
31	3	2161.5839073	0.003472	55.50
31	4	2166.3738305	0.003472	55.22

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\49901727\diff_s31.json`, its PNG and Gaia-neighbor table; source and artifact hashes are

in `../SHA256SUMS.json`.

Stable review key: R512. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 49901727.png

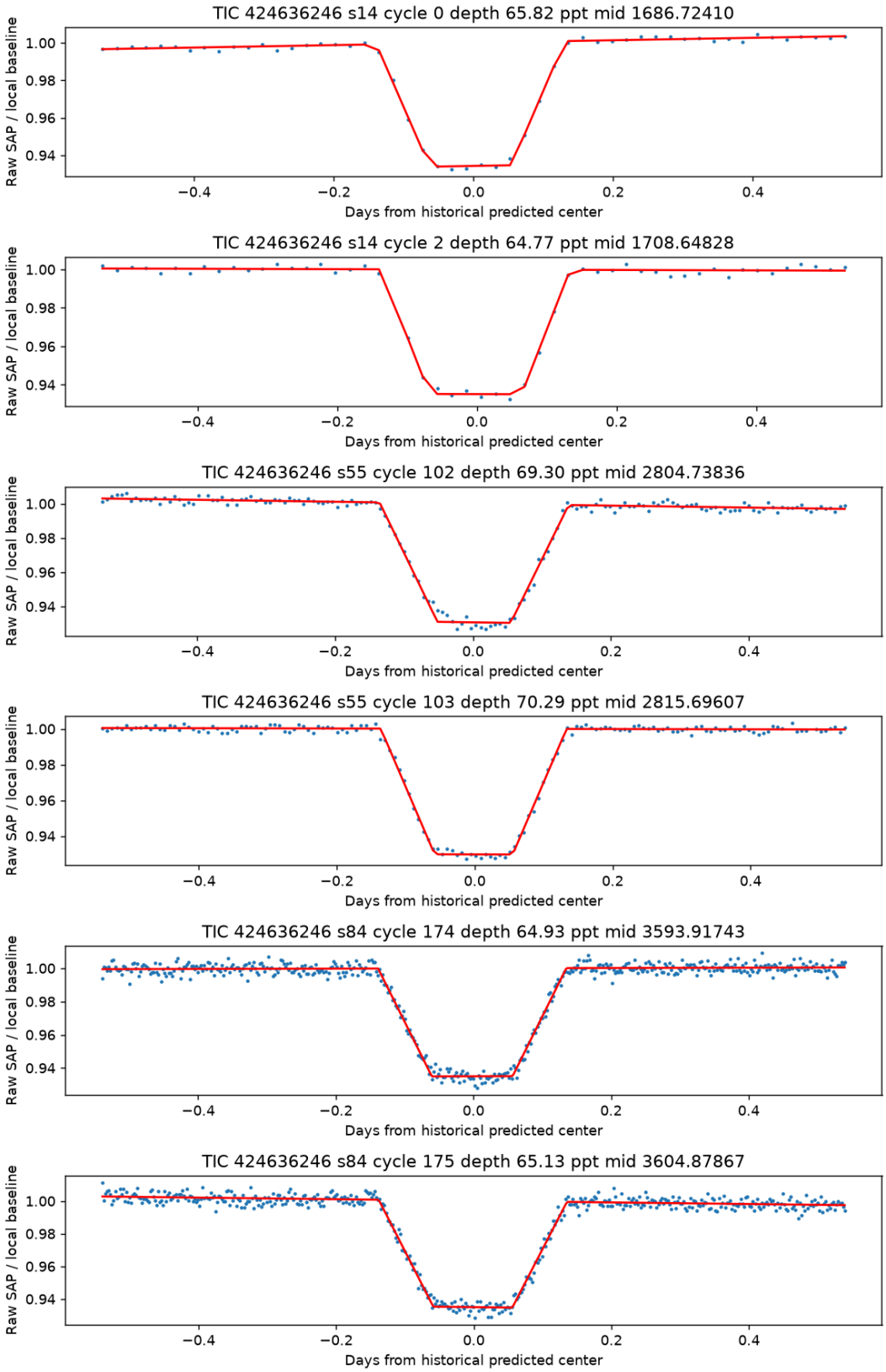
R513. TIC 424636246

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R513; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 424636246: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 6 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [14, 55, 84]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 10.9608574269 ± 0.0000279668 d and epoch BTJD 1686.72848166 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 14 deficit centroid is about 3.21 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\424636246\diff_s14.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S14 aperture photometry recovers 3 stacked events at 3x3 depth 64.40 ppt, versus median selected raw QLP fit depth 65.48 ppt. TESScut S14 and representative QLP S14 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 2139000984511908864: orbital P = 10.961251824 d, e = 0.0115. Closest edge-on conjunction offset at the measured epoch is -0.0947 d (-0.0086 phase); linear full-NSS-covariance timing scale 0.2147 d. Approximate central duration 8.37 h versus measured median 6.53 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S14/55/84 raw repeats and target-centered deficit. Capable catalogued neighbors are at least 50.6 arcsec away and need large intrinsic losses; no nearby source is favored by the deficit image.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
14	0	1686.7241028	0.010417	65.82
14	2	1708.6482791	0.010417	64.77
55	102	2804.7383553	0.003472	69.30
55	103	2815.6960722	0.003472	70.29
84	174	3593.9174280	0.001157	64.93
84	175	3604.8786671	0.001157	65.13

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\424636246\diff_s14.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R513. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 424636246.png

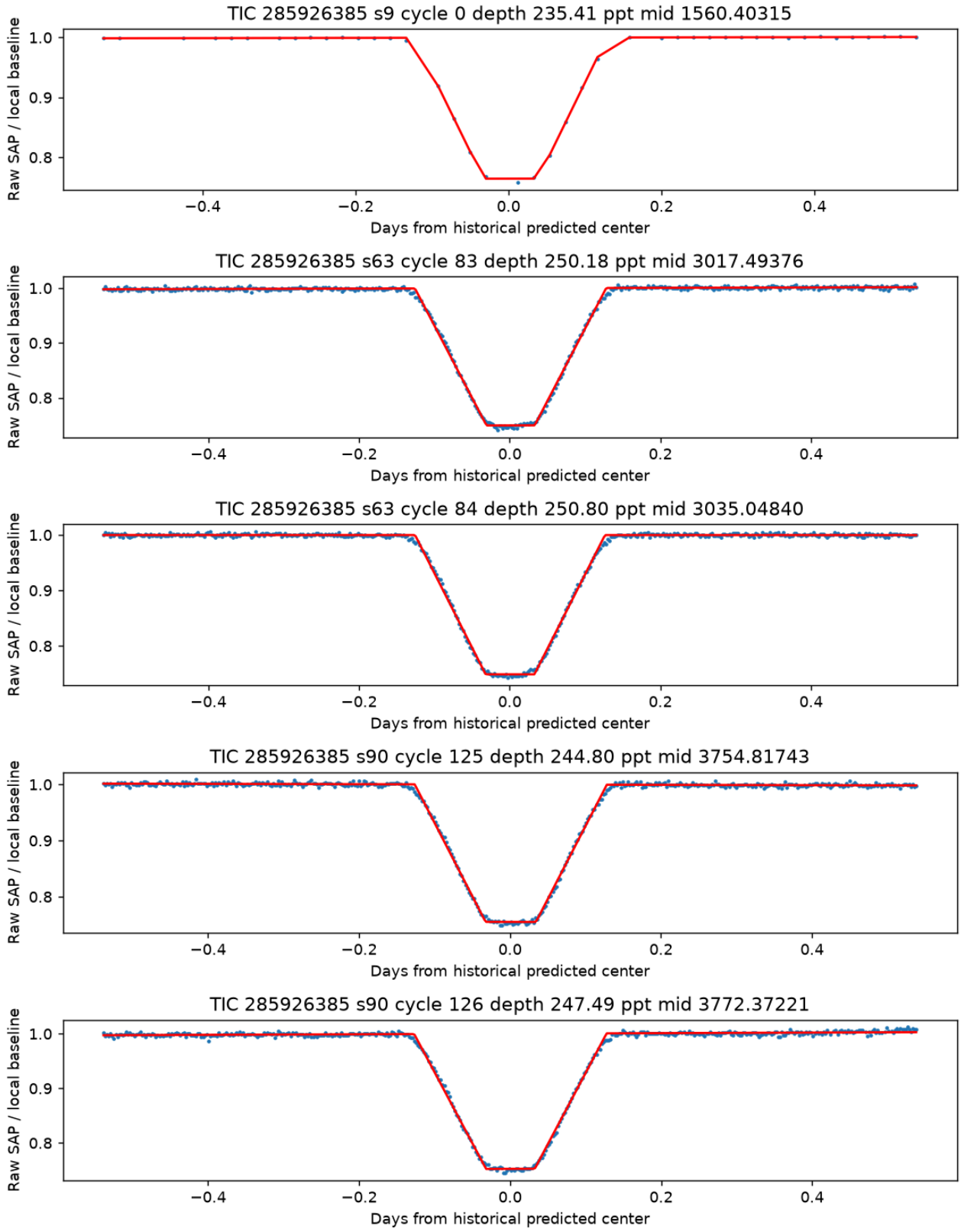
R514. TIC 285926385

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R514; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 285926385: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 5 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [9, 63, 90]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 17.5553255431 ± 0.0000265567 d and epoch BTJD 1560.40143870 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 63 deficit centroid is about 3.07 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\285926385\diff_s63.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S63 aperture photometry recovers 2 stacked events at 3x3 depth 246.47 ppt, versus median selected raw QLP fit depth 247.49 ppt. TESScut S63 and representative QLP S63 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 3589729714627304192: orbital P = 17.554087545 d, e = 0.4756. Closest edge-on conjunction offset at the measured epoch is +0.2965 d (+0.0169 phase); linear full-NSS-covariance timing scale 0.7498 d. Approximate central duration 8.01 h versus measured median 6.09 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Deep S9/63/90 raw eclipses recur at the measured period. Difference image favors the target and lists no capable Gaia neighbor within 60 arcsec.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
9	0	1560.4031478	0.010417	235.41
63	83	3017.4937647	0.001157	250.18
63	84	3035.0484007	0.001157	250.80
90	125	3754.8174316	0.001157	244.80
90	126	3772.3722138	0.001157	247.49

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\285926385\diff_s63.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R514. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 285926385.png

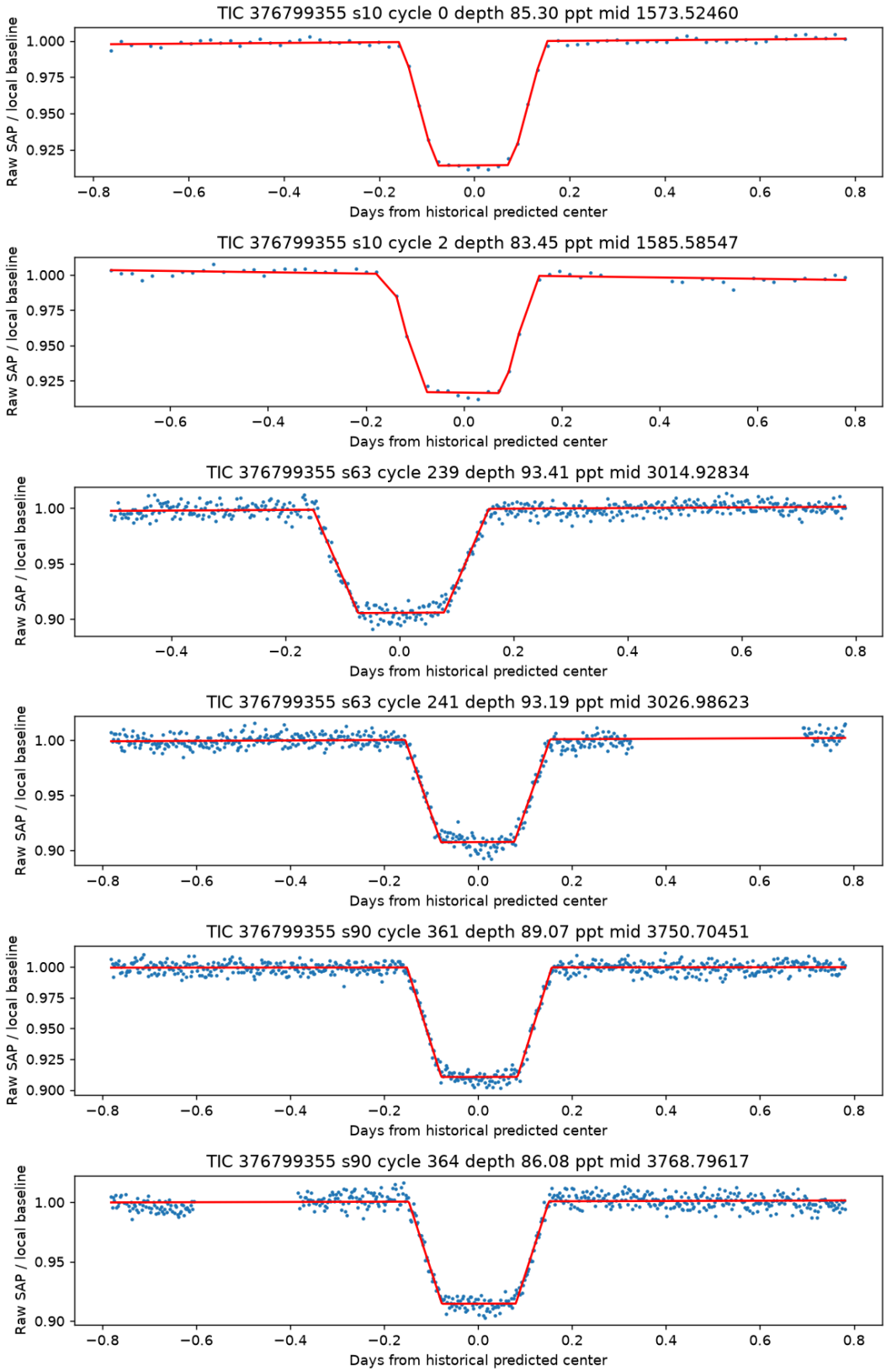
R515. TIC 376799355

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry across an eclipse or an independent high-resolution centroid.



R515; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 376799355: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 11 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [10, 63, 90]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 6.0309715740 ± 0.0000063815 d and epoch BTJD 1573.52342898 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 10 deficit centroid is about 3.49 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 6 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\376799355\diff_s10.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S10 aperture photometry recovers 4 stacked events at 3x3 depth 36.28 ppt, versus median selected raw QLP fit depth 87.69 ppt. TESScut S10 and representative QLP S10 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 5252574583035692416: orbital P = 6.030792509 d, e = 0.1048. Closest edge-on conjunction offset at the measured epoch is +0.0241 d (+0.0040 phase); linear full-NSS-covariance timing scale 0.3152 d. Approximate central duration 7.35 h versus measured median 7.30 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Raw eclipses repeat in S10/63/90 and the alternative 3x3 aperture also shows a deficit, diluted from roughly 88 ppt QLP to 36 ppt. The field is crowded: capable Gaia sources at 17.4 and 32.0 arcsec remain; the deficit centroid is only approximately target-centered and has no calibrated error. Attribution is unresolved.

Next check: Resolved photometry across an eclipse or an independent high-resolution centroid.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
10	0	1573.5245981	0.010417	85.30
10	2	1585.5854685	0.010417	83.45
10	3	1591.6168476	0.010417	86.74
63	239	3014.9283389	0.001157	93.41
63	240	3020.9555063	0.001157	95.32
63	241	3026.9862304	0.001157	93.19
63	242	3033.0181715	0.001157	93.14

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
90	361	3750.7045114	0.001157	89.07
90	362	3756.7359506	0.001157	86.28
90	363	3762.7659660	0.001157	87.69
90	364	3768.7961714	0.001157	86.08

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\376799355\diff_s10.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R515. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 376799355.png

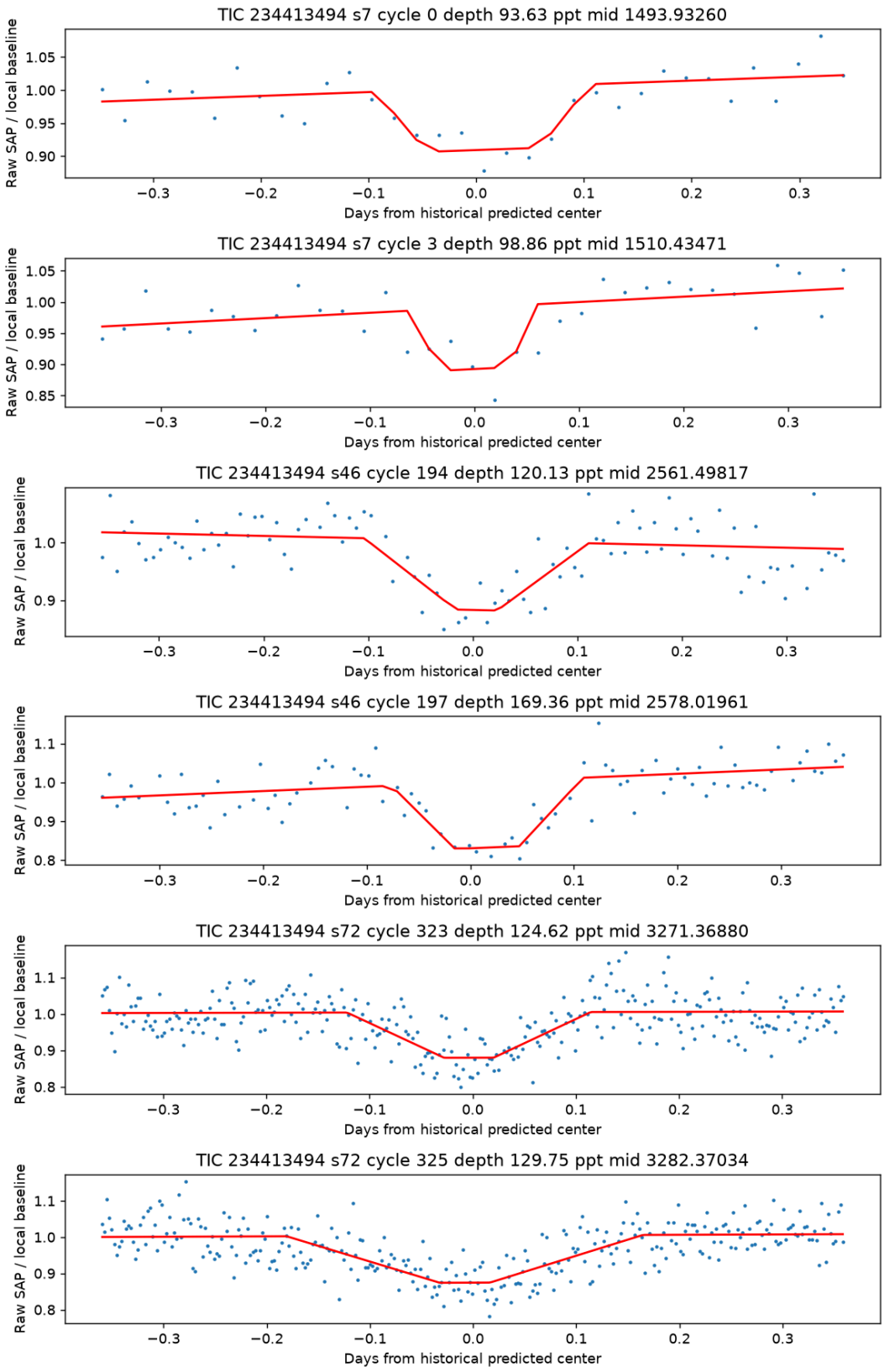
R516. TIC 234413494

physical_verdict: PROBABLE

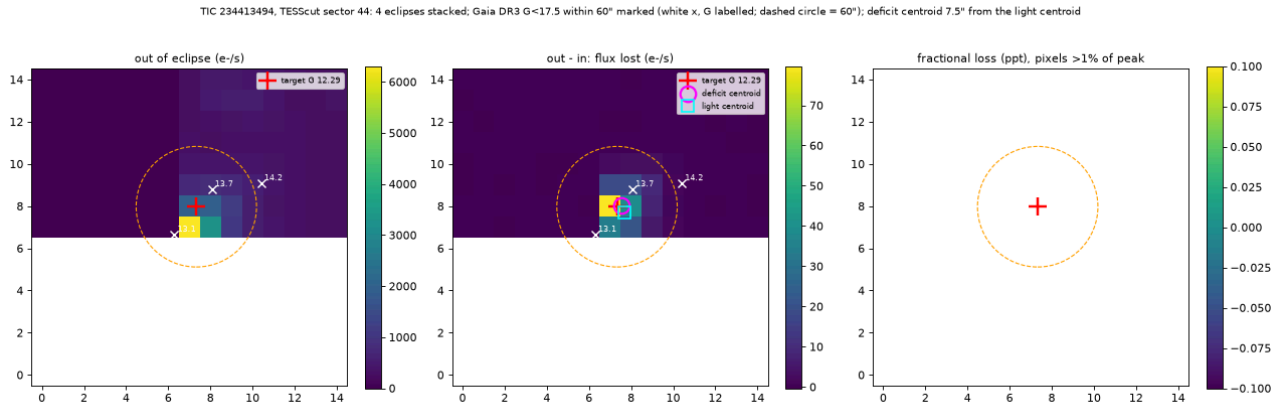
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Acquire a complete cutout or resolved eclipse photometry.



R516; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R516; Additional sector-44 eclipse-deficit stack, a separate TESS observing occasion from the historical sector.

Archived source: sanitized report text.

TIC 234413494: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 11 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [7, 46, 72]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 5.5028598185 ± 0.0000251813 d and epoch BTJD 1493.93985587 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 44 deficit centroid is about 4.75 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted]
4. **Alternative reduction.** Separately extracted S44 TESScut aperture photometry recovers 4 events at nominal 3x3 depth 14.03 ppt. The cutout is truncated, so this establishes a repeated signal in another extraction but cannot settle attribution or precise depth. TESScut S44 and representative QLP sectors [7, 46, 72] are separate TESS observing occasions on the same instrument; their photons are distinct.
5. **Gaia geometry.** DR3 source 3161704273321832064: orbital P = 5.503171903 d, e = 0.1043. Closest edge-on conjunction offset at the measured epoch is +0.0729 d (+0.0133 phase); linear full-NSS-covariance timing scale 0.1654 d. Approximate central duration 6.85 h versus measured median 4.76 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Raw eclipses repeat in S7/46/72. A separate S44 TESScut extraction recovers four events, but the available cutout is truncated and its peak/residual panel is corrupted. Capable neighbors at 19.3 and 30.4 arcsec remain, so attribution is unresolved.

Next check: Acquire a complete cutout or resolved eclipse photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
7	0	1493.9326038	0.010417	93.63
7	1	1499.4446516	0.010417	108.71
7	3	1510.4347053	0.010417	98.86
46	193	2555.9900823	0.004631	127.18
46	194	2561.4981707	0.006435	120.13
46	196	2572.4953342	0.007194	167.97
46	197	2578.0196107	0.005541	169.36
72	322	3265.8641418	0.003535	126.10
72	323	3271.3688033	0.004662	124.62
72	324	3276.8594444	0.002925	152.48
72	325	3282.3703365	0.005247	129.75

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted]`, its PNG and Gaia-neighbor table; `source and artifact hashes are in../SHA256SUMS.json`.

Stable review key: R516. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. diff_s44.json diff_s44.png gaia.json

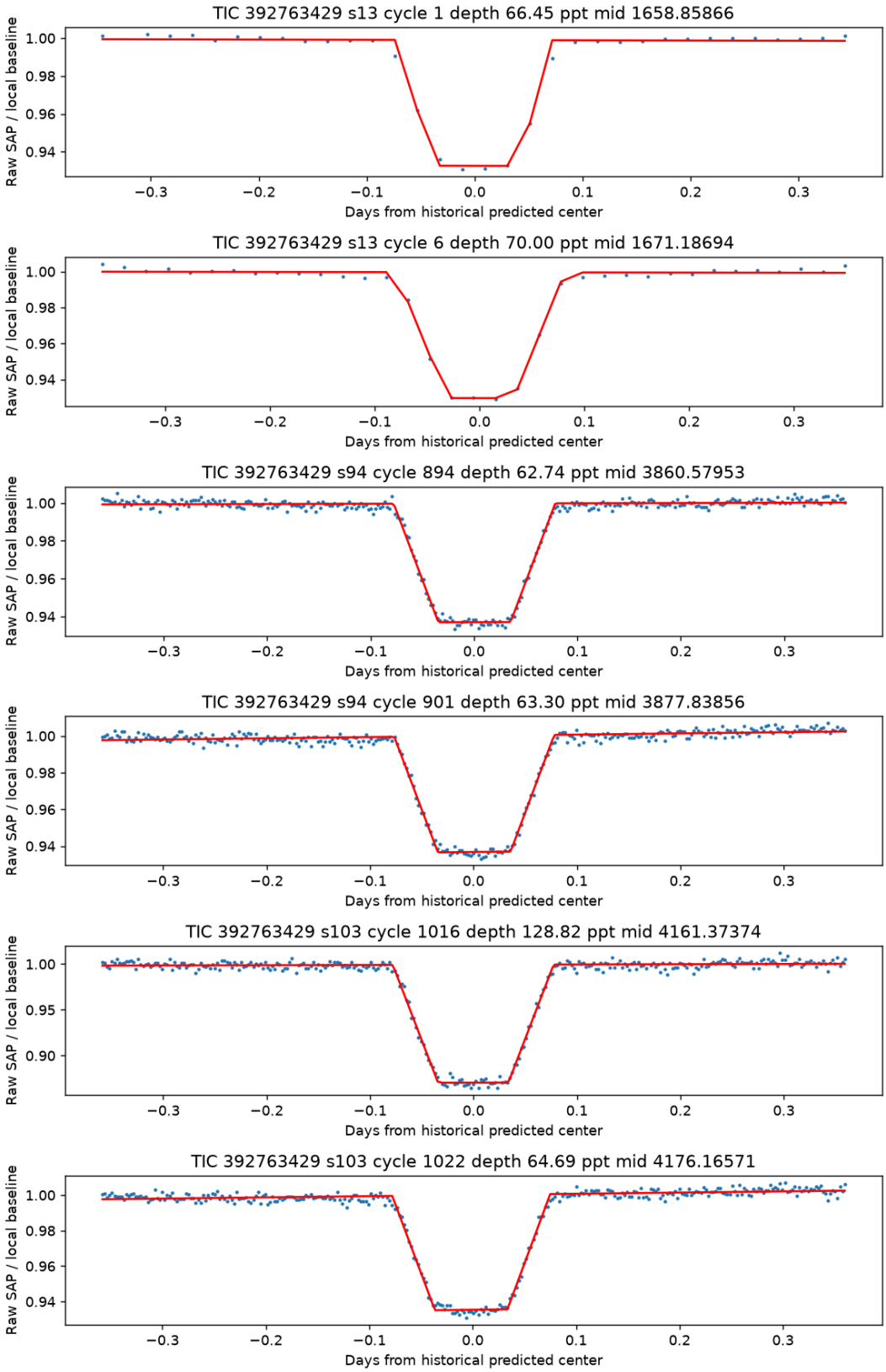
R517. TIC 392763429

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Investigate the S103 depth discontinuity with aperture diagnostics.



R517; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 392763429: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 24 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [13, 94, 103]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 2.4655285743 ± 0.0000030212 d and epoch BTJD 1658.86217686 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 13 deficit centroid is about 1.61 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\392763429\diff_s13.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S13 aperture photometry recovers 10 stacked events at 3x3 depth 73.91 ppt, versus median selected raw QLP fit depth 65.90 ppt. TESScut S13 and representative QLP S13 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 6355496414811552128: orbital P = 2.465425911 d, e = 0.0483. Closest edge-on conjunction offset at the measured epoch is +0.0286 d (+0.0116 phase); linear full-NSS-covariance timing scale 0.0496 d. Approximate central duration 4.20 h versus measured median 3.67 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S13/94/103 raw eclipses follow the same clock. A brief S103 depth doubling affects consecutive cycles, not stable odd/even alternation; its cause is unresolved. S13 difference image is target-centered with no capable Gaia neighbor within 60 arcsec.

Next check: Investigate the S103 depth discontinuity with aperture diagnostics.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
13	1	1658.8586557	0.010417	66.45
13	2	1661.3256533	0.010417	68.58
13	3	1663.7890780	0.010417	68.77
13	4	1666.2545928	0.010417	67.42
13	6	1671.1869404	0.010417	70.00
13	7	1673.6510554	0.010417	67.98
13	9	1678.5845727	0.010417	67.21
13	10	1681.0473863	0.010417	66.91

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
94	893	3858.1137926	0.001157	63.36
94	894	3860.5795276	0.001157	62.74
94	895	3863.0451407	0.001157	63.04
94	896	3865.5109645	0.001157	61.81
94	899	3872.9065859	0.001157	63.20
94	900	3875.3731737	0.001157	63.43
94	901	3877.8385592	0.001157	63.30
94	902	3880.3033415	0.001157	62.84
103	1013	4153.9778478	0.001157	125.72
103	1014	4156.4430929	0.001157	128.19
103	1016	4161.3737360	0.001157	128.82
103	1017	4163.8390246	0.001157	127.06
103	1018	4166.3038238	0.001157	62.23
103	1019	4168.7696539	0.001157	65.36
103	1021	4173.7004953	0.001157	65.11
103	1022	4176.1657093	0.001157	64.69

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\392763429\diff_s13.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R517. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 392763429.png

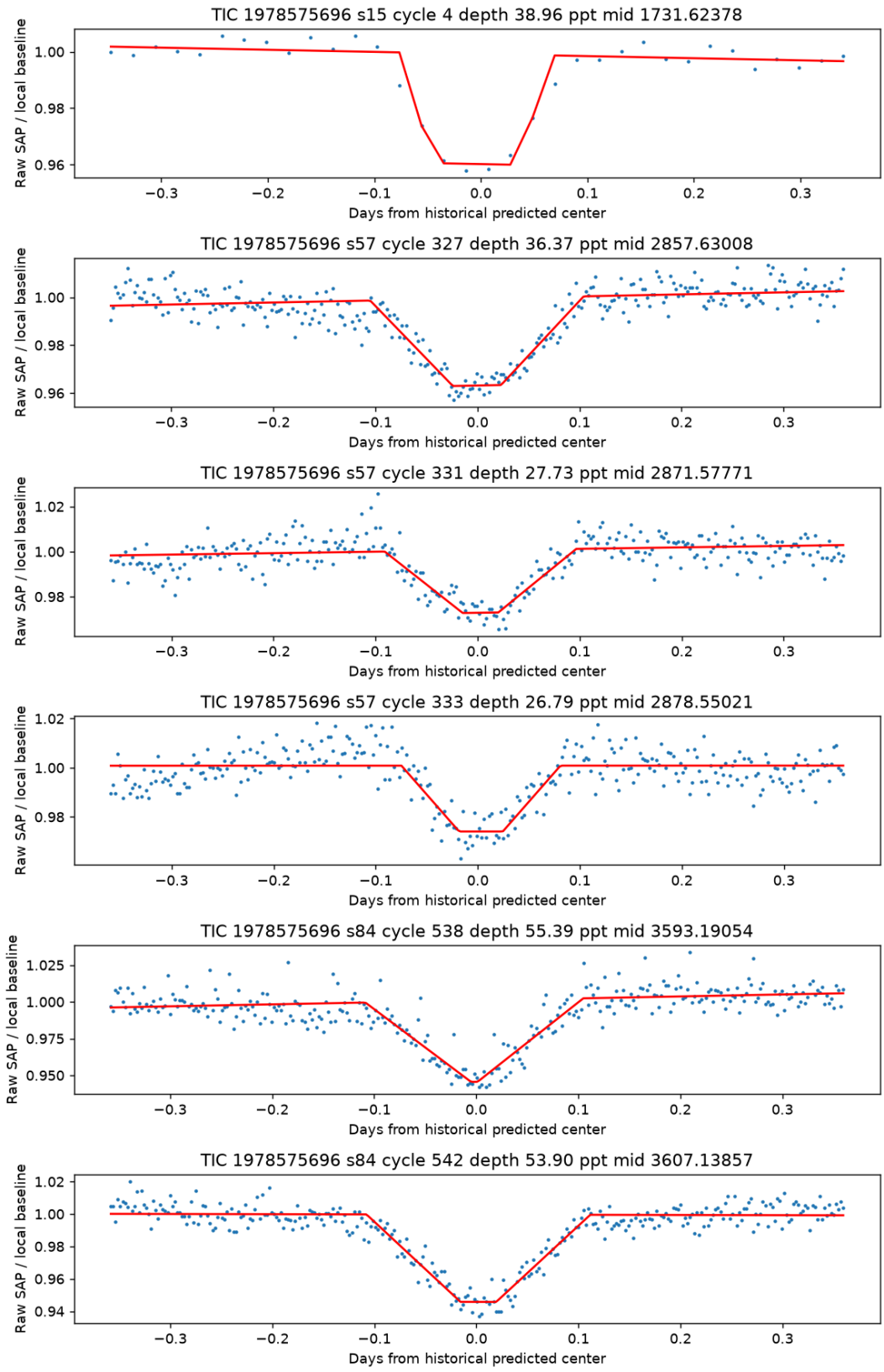
R518. TIC 1978575696

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and secondary-eclipse characterization.



R518; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 1978575696: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 9 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [15, 57, 84]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 6.9721607271 ± 0.0000141450 d and epoch BTJD 1731.62698790 (Deeper repeating eclipse; even historical half-period cycles). Even historical half-period cycles only: deeper same-kind eclipses; odd cycles show shallower secondary. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 41 deficit centroid is about 0.23 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 1 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\1978575696\diff_s41.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S41 aperture photometry recovers 8 stacked events at 3x3 depth 20.84 ppt, versus median selected raw QLP fit depth 51.39 ppt. TESScut S41 and representative QLP sectors [15, 57, 84] are separate TESS observing occasions on the same instrument; their photons are distinct. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 2185260393630495488: orbital P = 6.973173120 d, e = 0.0210. Closest edge-on conjunction offset at the measured epoch is -0.1824 d (-0.0262 phase); linear full-NSS-covariance timing scale 0.1588 d. Approximate central duration 10.30 h versus measured median 4.92 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Alternating raw depths identify the historical 3.486-day clock as half the orbital eclipse cycle. Deeper even cycles yield a measured 6.97216-day same-kind period, consistent with Gaia; S41 difference image favors the target. A capable G14.97 source at 23.9 arcsec is not favored by the target-centered deficit.

Next check: Optional resolved photometry and secondary-eclipse characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
15	4	1731.6237758	0.010417	38.96
57	326	2854.1448650	0.001798	50.18
57	328	2861.1157163	0.001157	51.39
57	330	2868.0906737	0.004100	52.16
57	332	2875.0627901	0.001210	56.85
84	536	3586.2278709	0.003264	43.05
84	538	3593.1905433	0.001658	55.39

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
84	540	3600.1703801	0.003626	41.44
84	542	3607.1385708	0.001179	53.90

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\1978575696\diff_s41.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R518. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 1978575696.png

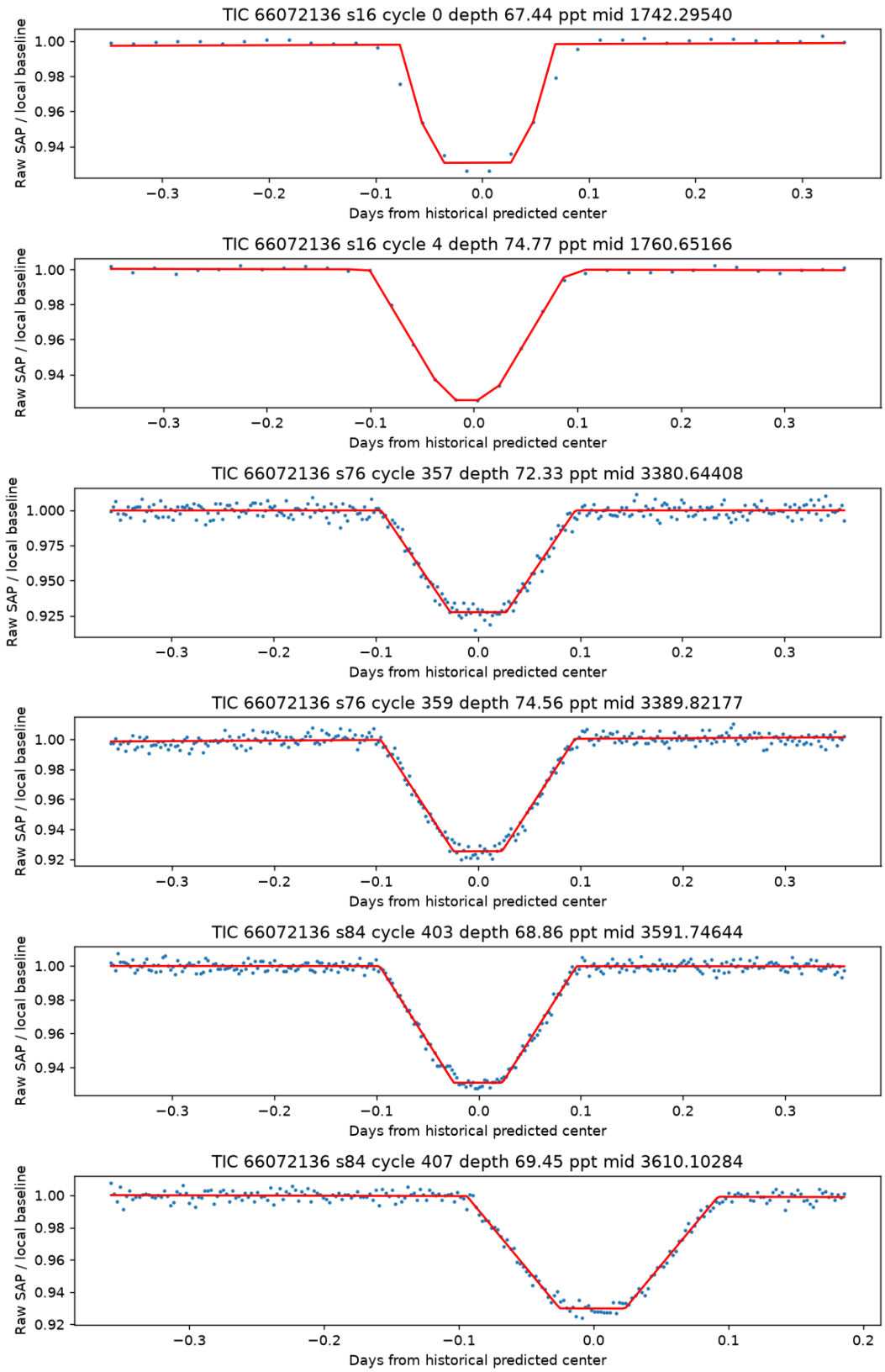
R519. TIC 66072136

physical_verdict: CONFIRMED

novelty_status: known prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolve the discordant VSX period for catalogue interpretation; no new-system count.



R519; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 66072136: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 14 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [16, 76, 84]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 4.5892005792 ± 0.0000108785 d and epoch BTJD 1742.29852360 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 76 deficit centroid is about 4.59 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\66072136\diff_s76.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut S76 aperture photometry recovers 6 stacked events at 3x3 depth 72.67 ppt, versus median selected raw QLP fit depth 71.44 ppt. TESScut S76 and representative QLP S76 are reductions of the same TESS observations and share photons. Aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 1989520523290696832: orbital P = 4.589208060 d, e = 0.0502. Closest edge-on conjunction offset at the measured epoch is +0.0446 d (+0.0097 phase); linear full-NSS-covariance timing scale 0.0990 d. Approximate central duration 6.32 h versus measured median 4.58 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S16/76/84 raw repeats and the alternative S76 aperture support a real target-centered eclipse. No capable Gaia neighbor within 60 arcsec. Central catalogue reconciliation identifies the close VSX EA V0693 Lac counterpart as the same object, so this is a known eclipsing-binary recovery, not a new-system addition. VSX lists 1.08916 d versus the measured same-kind 4.58920 d; the disagreement does not negate the measured raw eclipses.

Next check: Resolve the discordant VSX period for catalogue interpretation; no new-system count.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
16	0	1742.2954049	0.010417	67.44
16	3	1756.0651217	0.010417	66.89
16	4	1760.6516646	0.010417	74.77
76	355	3371.4642140	0.001157	78.54
76	356	3376.0541073	0.001157	73.67
76	357	3380.6440772	0.001157	72.33

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
76	358	3385.2334200	0.001157	73.42
76	359	3389.8217691	0.001157	74.56
76	360	3394.4095962	0.001157	74.35
84	402	3587.1568237	0.001157	70.55
84	403	3591.7464407	0.001157	68.86
84	405	3600.9241348	0.001157	70.36
84	406	3605.5144422	0.001157	69.97
84	407	3610.1028446	0.001157	69.45

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\66072136\diff_s76.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R519. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 66072136.png

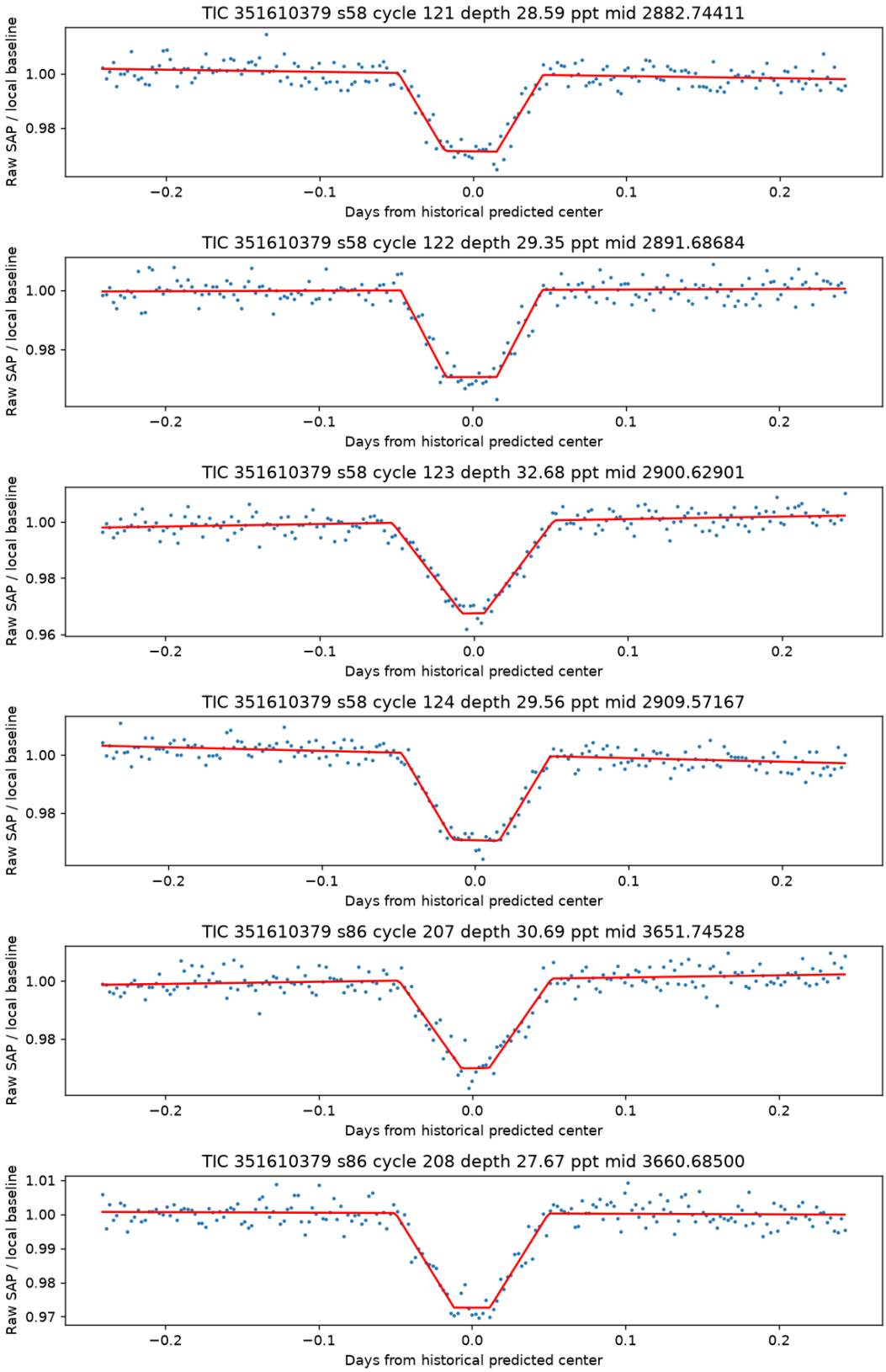
R520. TIC 351610379

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved eclipse photometry.



R520; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 351610379: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 6 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [58, 86]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 8.9418500033 ± 0.0000117888 d and epoch BTJD 2882.74512557 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 58 deficit centroid is about 4.80 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\351610379\diff_s58.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 4 stacked events at 3x3 depth 22.03 ppt, versus median selected raw QLP fit depth 29.46 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 536070020862283648: orbital P = 8.942678782 d, e = 0.0901. Closest edge-on conjunction offset at the measured epoch is -0.1852 d (-0.0207 phase); linear full-NSS-covariance timing scale 0.3265 d. Approximate central duration 4.19 h versus measured median 2.36 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S58/86 raw repeats and the S58 deficit favor the target. The nearest capable G15.21 neighbor is 20.3 arcsec away and would require about 42% intrinsic eclipse depth; a brighter capable source at 54.1 arcsec is spatially separate. Pixel localization has no calibrated error.

Next check: Optional resolved eclipse photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
58	121	2882.7441056	0.001157	28.59
58	122	2891.6868359	0.001157	29.35
58	123	2900.6290141	0.001157	32.68
58	124	2909.5716717	0.001157	29.56
86	207	3651.7452785	0.001157	30.69
86	208	3660.6849982	0.001157	27.67

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\351610379\diff_s58.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R520. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

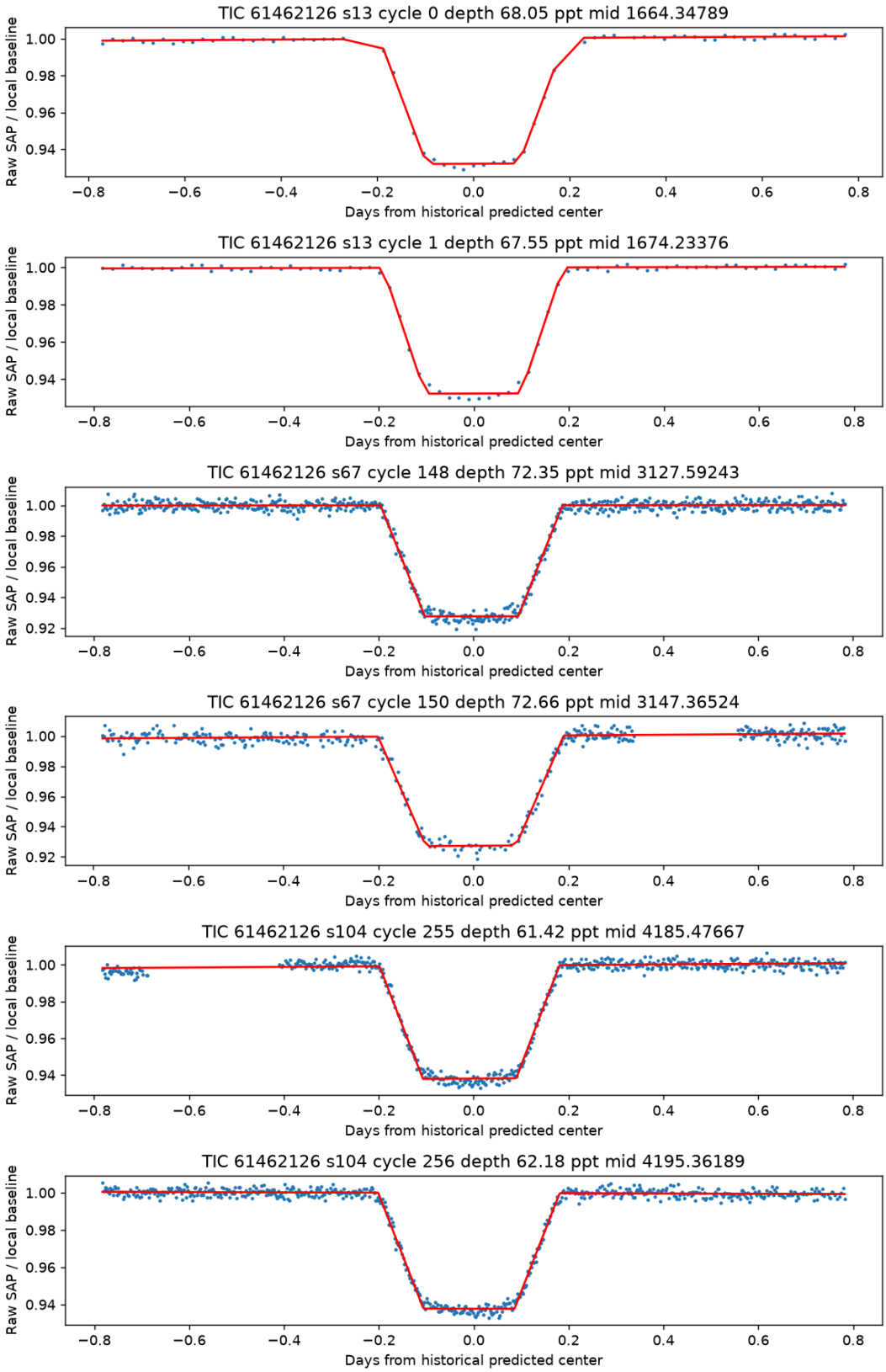
R521. TIC 61462126

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R521; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 61462126: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 6 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [13, 67, 104]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 9.8867660303 ± 0.0000104175 d and epoch BTJD 1664.35061007 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 27 deficit centroid is about 3.03 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\61462126\diff_s27.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 2 stacked events at 3x3 depth 70.42 ppt, versus median selected raw QLP fit depth 67.80 ppt. This historical TESScut sector is a separate observing occasion on the same instrument from the representative QLP sectors; extraction and aperture differences limit depth comparison.
5. **Gaia geometry.** DR3 source 6712693348579620608: orbital P = 9.884078396 d, e = 0.0419. Closest edge-on conjunction offset at the measured epoch is +0.2430 d (+0.0246 phase); linear full-NSS-covariance timing scale 0.2641 d. Approximate central duration 11.28 h versus measured median 9.20 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S13/67/104 raw eclipses repeat. The S27 alternative extraction recovers two events with a target-centered deficit; no Gaia neighbor within 60 arcsec passes the approximate flux-budget test.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
13	0	1664.3478919	0.010417	68.05
13	1	1674.2337609	0.010417	67.55
67	148	3127.5924308	0.001157	72.35
67	150	3147.3652395	0.001157	72.66
104	255	4185.4766718	0.001157	61.42
104	256	4195.3618949	0.001157	62.18

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted]` `5\made\sb1-scale\diffimg\61462126\diff_s27.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R521. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

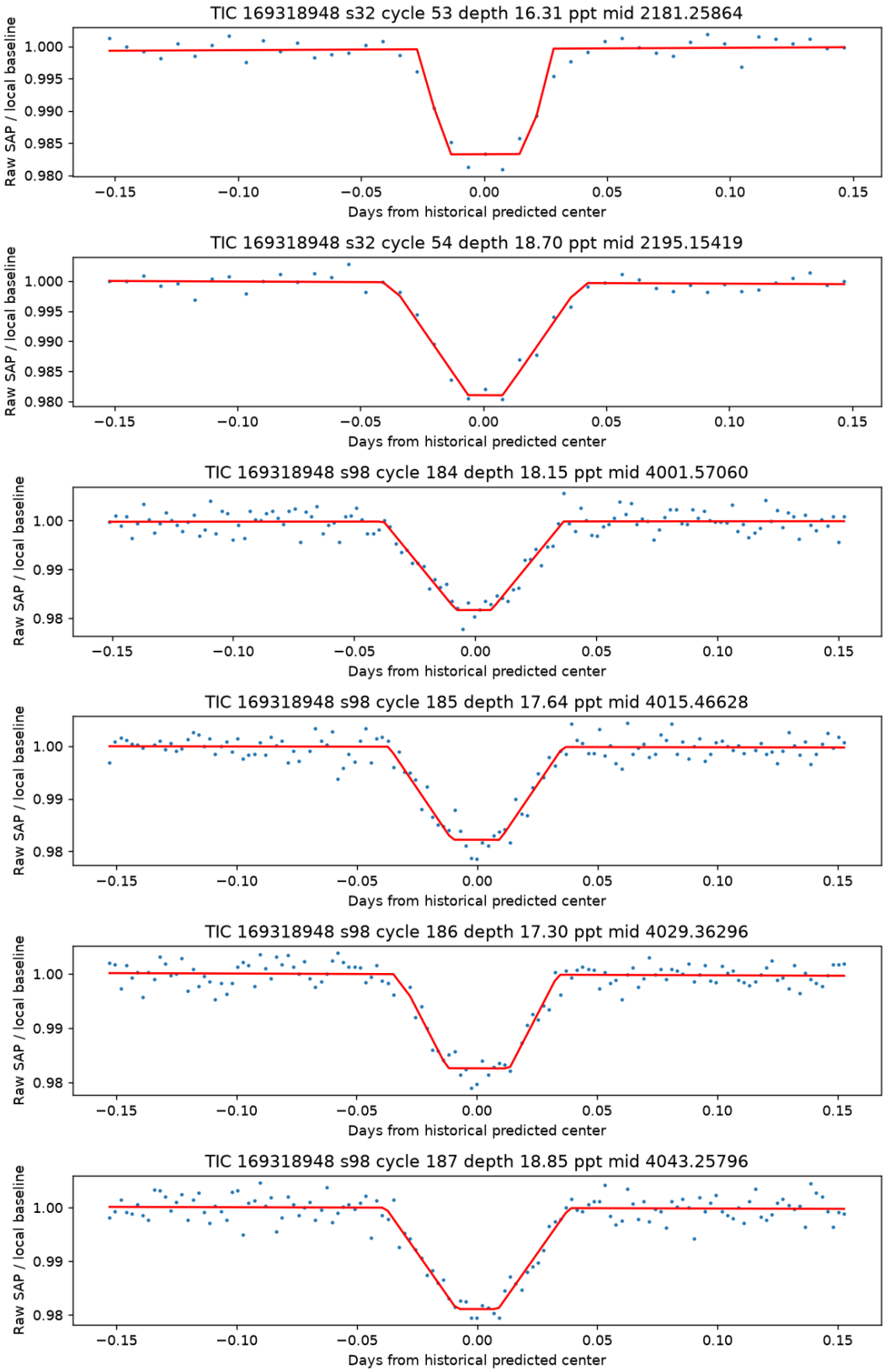
R522. TIC 169318948

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R522; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 169318948: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 6 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [32, 98]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 13.8955158737 ± 0.0000190968 d and epoch BTJD 2181.25859625 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 98 deficit centroid is about 1.05 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\169318948\diff_s98.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 4 stacked events at 3x3 depth 17.95 ppt, versus median selected raw QLP fit depth 17.89 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 2986427856265439488: orbital P = 13.896444944 d, e = 0.3989. Closest edge-on conjunction offset at the measured epoch is -0.2784 d (-0.0200 phase); linear full-NSS-covariance timing scale 0.4091 d. Approximate central duration 3.58 h versus measured median 1.76 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S32/98 raw repeated eclipses and the S98 alternative extraction agree. The difference image favors the target and lists no capable Gaia neighbor within 60 arcsec.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
32	53	2181.2586438	0.003472	16.31
32	54	2195.1541875	0.003472	18.70
98	184	4001.5705969	0.001157	18.15
98	185	4015.4662752	0.001157	17.64
98	186	4029.3629551	0.001157	17.30
98	187	4043.2579572	0.001157	18.85

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\169318948\diff_s98.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R522. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

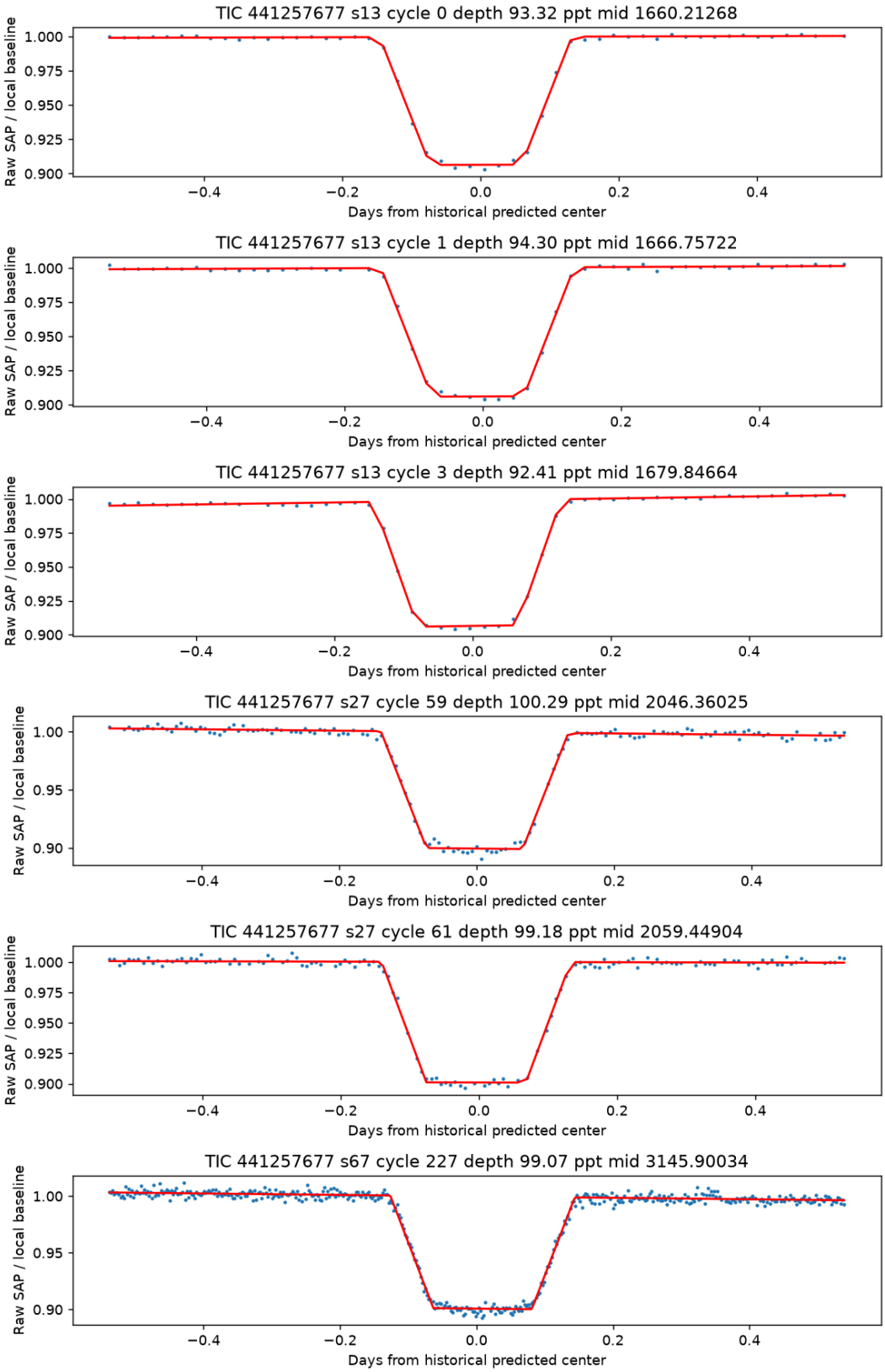
R523. TIC 441257677

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R523; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 441257677: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 10 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [13, 27, 67]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 6.5448839250 ± 0.0000106057 d and epoch BTJD 1660.21194066 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 13 deficit centroid is about 4.75 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\441257677\diff_s13.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 4 stacked events at 3x3 depth 100.87 ppt, versus median selected raw QLP fit depth 98.56 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 6739664574967468800: orbital P = 6.545173049 d, e = 0.1154. Closest edge-on conjunction offset at the measured epoch is +0.1321 d (+0.0202 phase); linear full-NSS-covariance timing scale 0.6016 d. Approximate central duration 8.15 h versus measured median 6.54 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S13/27/67 raw events recur with no persistent odd/even depth split. The S13 alternative extraction shows four events and a target-centered deficit with no capable Gaia neighbor within 60 arcsec.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
13	0	1660.2126763	0.010417	93.32
13	1	1666.7572200	0.010417	94.30
13	2	1673.3017353	0.010417	92.44
13	3	1679.8466357	0.010417	92.41
27	58	2039.8154346	0.003472	98.05
27	59	2046.3602493	0.003472	100.29
27	60	2052.9051908	0.003472	99.68
27	61	2059.4490443	0.003472	99.18

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
67	225	3132.8110855	0.001157	100.29
67	227	3145.9003392	0.001157	99.07

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\441257677\diff_s13.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R523. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

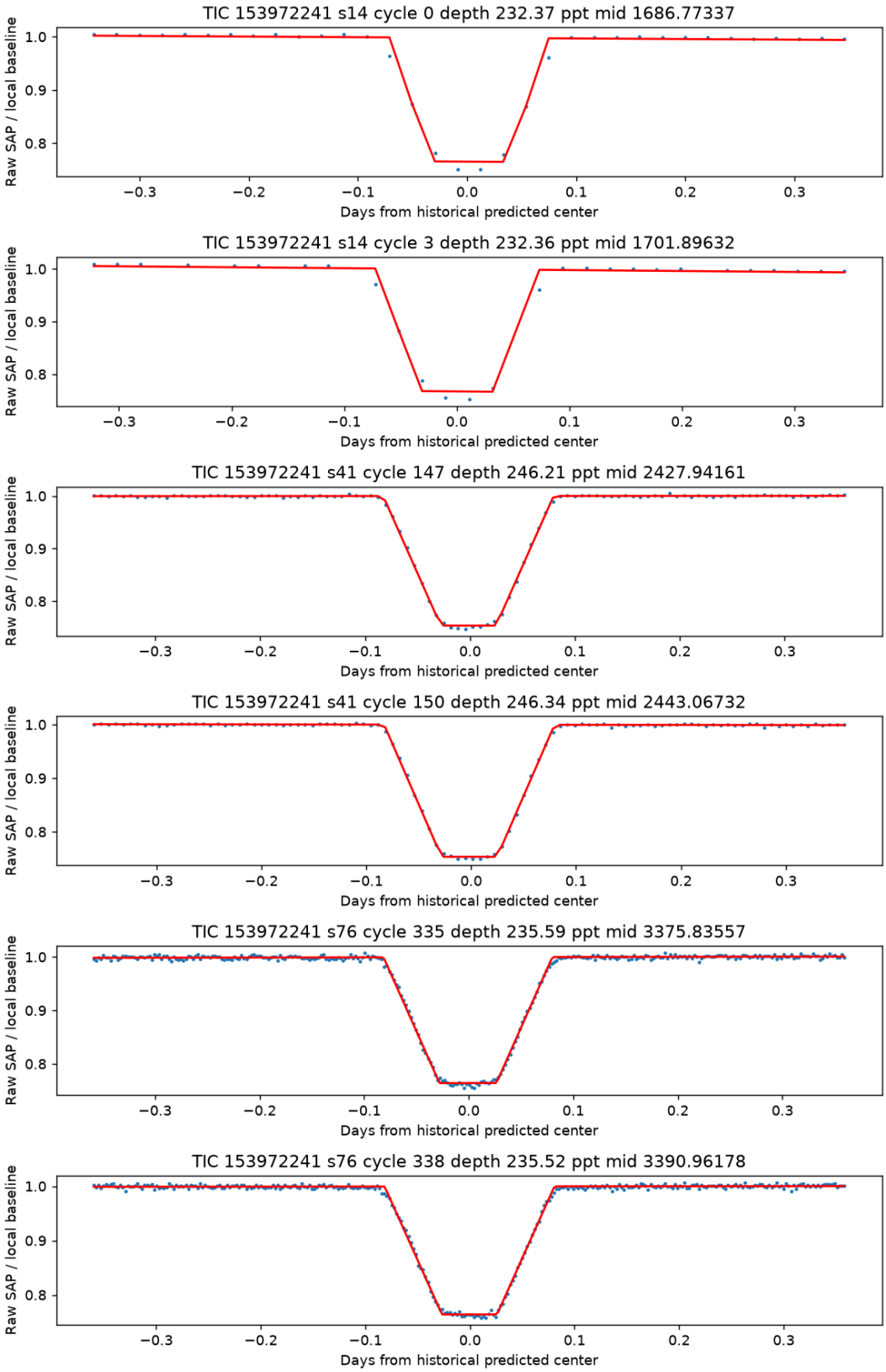
R524. TIC 153972241

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R524; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 153972241: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 11 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [14, 41, 76]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 5.0419880702 ± 0.0000086825 d and epoch BTJD 1686.76961209 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 75 deficit centroid is about 0.91 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\153972241\diff_s75.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 6 stacked events at 3x3 depth 243.75 ppt, versus median selected raw QLP fit depth 235.59 ppt. This historical TESScut sector is a separate observing occasion on the same instrument from the representative QLP sectors; extraction and aperture differences limit depth comparison.
5. **Gaia geometry.** DR3 source 1666322761092269568: orbital P = 5.041663243 d, e = 0.0256. Closest edge-on conjunction offset at the measured epoch is +0.0466 d (+0.0092 phase); linear full-NSS-covariance timing scale 0.0912 d. Approximate central duration 4.55 h versus measured median 3.87 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Deep S14/41/76 raw eclipses repeat; the S75 extraction recovers six events. Its target-centered difference image lists no capable Gaia neighbor within 60 arcsec.

Next check: Optional precision characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
14	0	1686.7733724	0.010416	232.37
14	1	1691.8152417	0.010416	232.32
14	3	1701.8963232	0.010416	232.36
41	146	2422.8992728	0.003472	246.28
41	147	2427.9416130	0.003472	246.21
41	149	2438.0255926	0.003472	248.01
41	150	2443.0673221	0.003472	246.34
76	334	3370.7935552	0.001157	236.10
76	335	3375.8355668	0.001157	235.59

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
76	336	3380.8776017	0.001157	234.76
76	338	3390.9617784	0.001157	235.52

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\153972241\diff_s75.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R524. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

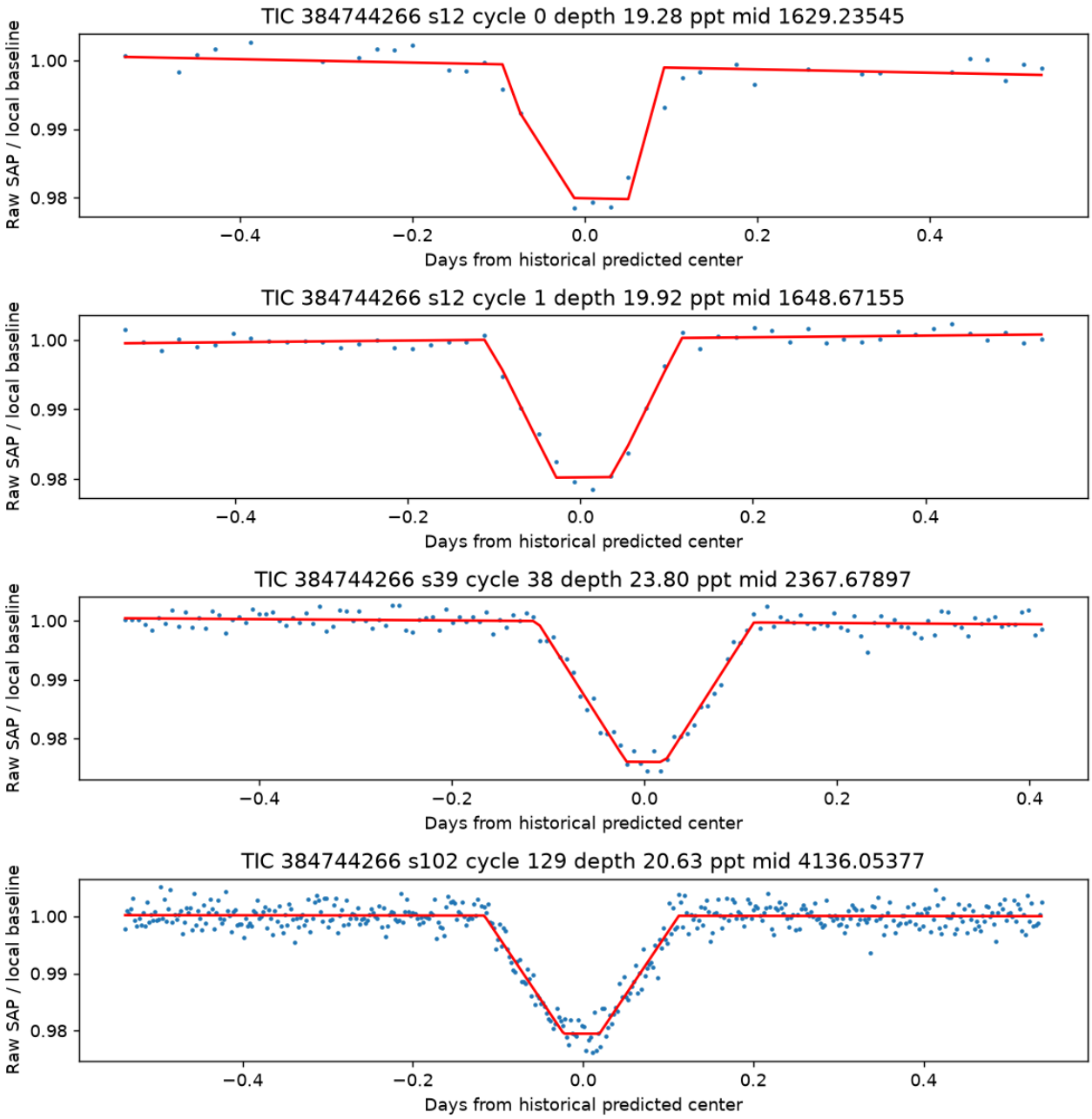
R525. TIC 384744266

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved eclipse photometry.



R525; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 384744266: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 4 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [12, 39, 102]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 19.4326891330 ± 0.0000338018 d and epoch BTJD 1629.23687168 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 12 deficit centroid is about 3.65 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\384744266\diff_s12.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 2 stacked events at 3x3 depth 20.23 ppt, versus median selected raw QLP fit depth 20.27 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 5776380960403754624: orbital P = 19.433184428 d, e = 0.2022. Closest edge-on conjunction offset at the measured epoch is -0.1733 d (-0.0089 phase); linear full-NSS-covariance timing scale 0.4624 d. Approximate central duration 10.72 h versus measured median 5.40 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S12/39/102 raw repeats agree. The S12 deficit is near the target. Three nominally capable neighbors lie 29–60 arcsec away, each requiring about 78–100% intrinsic loss, and none is favored by the observed deficit.

Next check: Optional resolved eclipse photometry.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
12	0	1629.2354465	0.010417	19.28
12	1	1648.6715529	0.010417	19.92
39	38	2367.6789719	0.003472	23.80
102	129	4136.0537725	0.001157	20.63

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\384744266\diff_s12.json`, its PNG and Gaia-neighbor table; source and artifact hashes

are in `../SHA256SUMS.json`.

Stable review key: R525. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

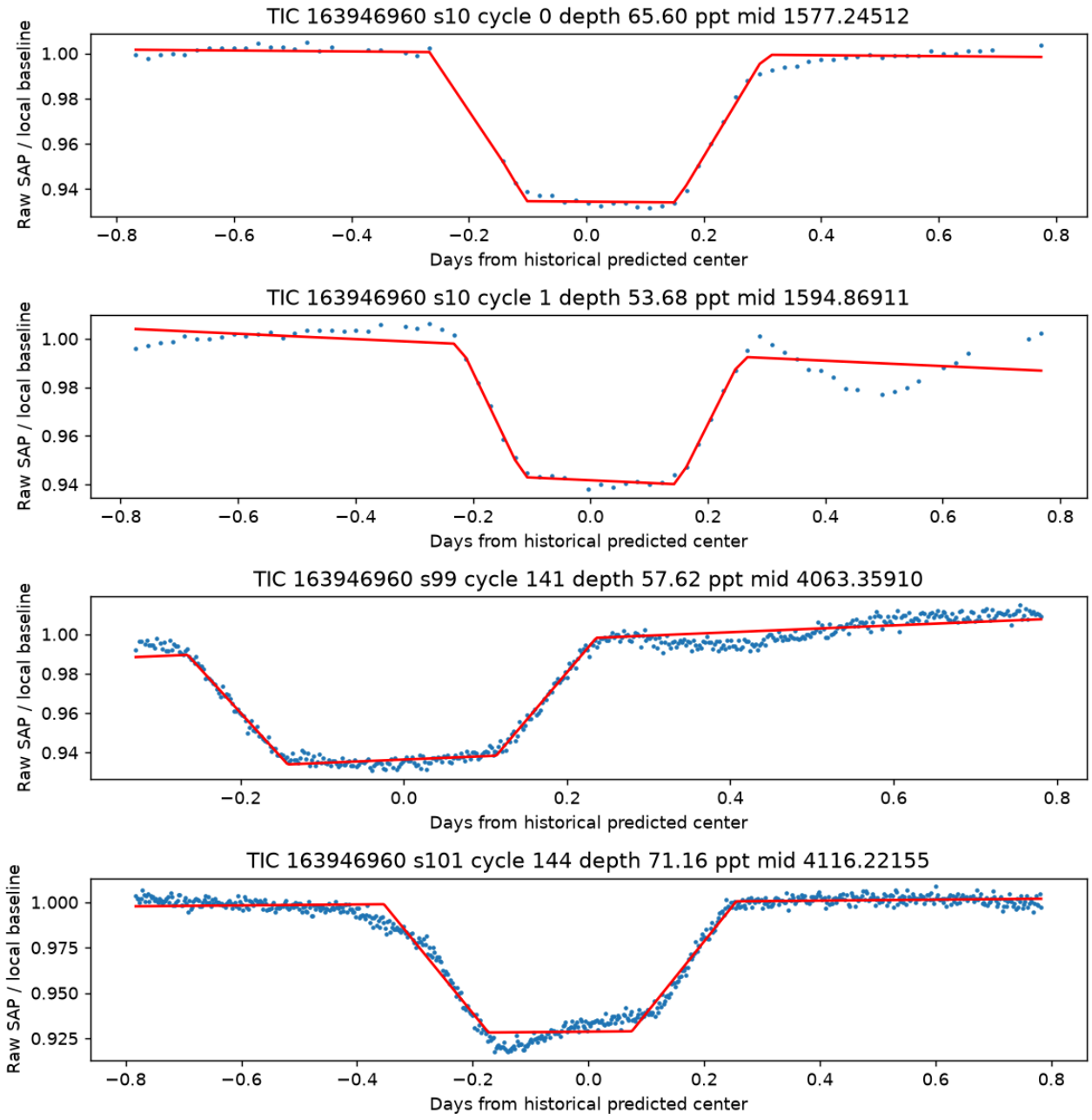
R526. TIC 163946960

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry would strengthen host attribution.



R526; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 163946960: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 4 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [10, 99, 101]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 17.6318158711 ± 0.0005396857 d and epoch BTJD 1577.24863847 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 10 deficit centroid is about 5.50 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 2 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\163946960\diff_s10.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 2 stacked events at 3x3 depth 51.83 ppt, versus median selected raw QLP fit depth 61.61 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 6078495037379889408: orbital P = 17.629414377 d, e = 0.2481. Closest edge-on conjunction offset at the measured epoch is +0.2046 d (+0.0116 phase); linear full-NSS-covariance timing scale 0.2688 d. Approximate central duration 15.88 h versus measured median 12.71 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S10/99/101 raw repeats and the S10 alternative extraction support the eclipse. The deficit lies about 0.26 pixel from the target, away from two capable Gaia neighbors at 15.8 and 29.7 arcsec; those require about 60% and 44% intrinsic losses. Localization is descriptive, without a calibrated uncertainty.

Next check: Resolved eclipse photometry would strengthen host attribution.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
10	0	1577.2451240	0.010417	65.60
10	1	1594.8691131	0.010417	53.68
99	141	4063.3590968	0.001933	57.62
101	144	4116.2215510	0.001157	71.16

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\163946960\diff_s10.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R526. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

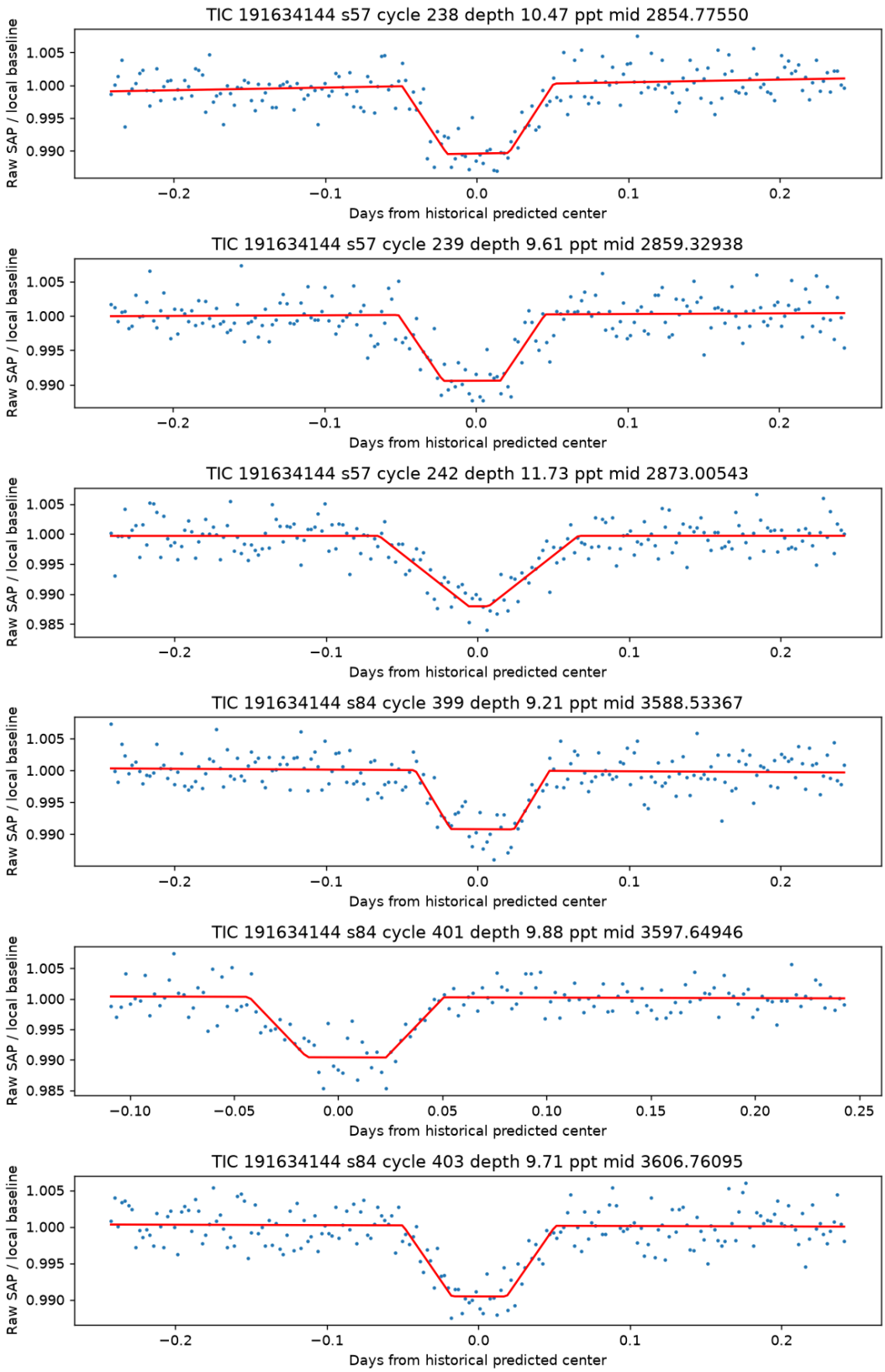
R527. TIC 191634144

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional-sector timing and resolved photometry could tighten this shallow case.



R527; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 191634144: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 10 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [57, 84]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 4.5575080623 ± 0.0000060351 d and epoch BTJD 2854.77452990 (Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 57 deficit centroid is about 3.11 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\191634144\diff_s57.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 5 stacked events at 3x3 depth 9.50 ppt, versus median selected raw QLP fit depth 10.47 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison.
5. **Gaia geometry.** DR3 source 375375060725718912: orbital P = 4.557098972 d, e = 0.0402. Closest edge-on conjunction offset at the measured epoch is +0.2181 d (+0.0479 phase); linear full-NSS-covariance timing scale 0.1720 d. Approximate central duration 6.73 h versus measured median 2.48 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S57/84 raw shallow eclipses recur. Odd/even fitted depths vary at roughly the local noise level without a stable half-period interpretation. The S57 difference image favors the target over capable neighbors at 23.9, 34.4 and 53.6 arcsec.

Next check: Additional-sector timing and resolved photometry could tighten this shallow case.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
57	238	2854.7755006	0.001157	10.47
57	239	2859.3293846	0.001316	9.61
57	241	2868.4473010	0.001556	10.92
57	242	2873.0054339	0.001722	11.73
57	243	2877.5638528	0.002082	10.48
84	399	3588.5336664	0.001462	9.21
84	400	3593.0908072	0.001500	12.34
84	401	3597.6494571	0.001157	9.88

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
84	402	3602.2053858	0.001671	11.23
84	403	3606.7609465	0.001728	9.71

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\191634144\diff_s57.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R527. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

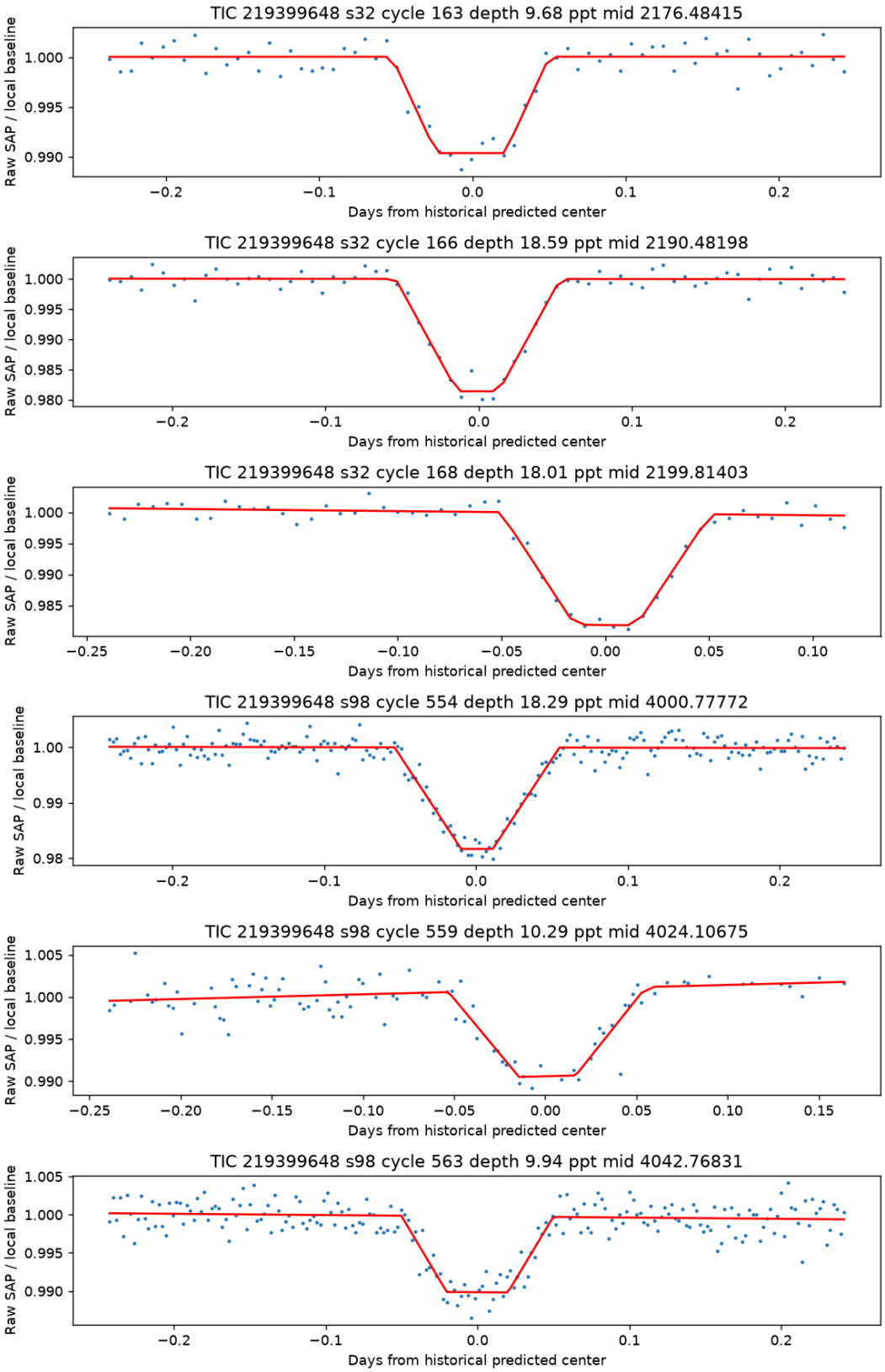
R528. TIC 219399648

physical_verdict: CONFIRMED

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional secondary-eclipse characterization.



R528; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 219399648: CONFIRMED

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 5 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [32, 98]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 9.3314205089 ± 0.0000110733 d and epoch BTJD 2181.15076257 (Deeper repeating eclipse; even historical half-period cycles). Even historical half-period cycles only: deeper same-kind eclipses; odd cycles show shallower secondary. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.
3. **Pixel attribution.** The alternative cutout sector 98 deficit centroid is about 3.27 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 0 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\219399648\diff_s98.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 11 stacked events at 3x3 depth 14.53 ppt, versus median selected raw QLP fit depth 18.29 ppt. The extraction and representative QLP raw measurement share TESS photons in this sector; aperture dilution limits depth comparison. The historical S98 stack mixes deeper and shallower half-period events, so its depth is not a same-kind depth comparison.
5. **Gaia geometry.** DR3 source 4797773555362406144: orbital P = 9.330097946 d, e = 0.0079. Closest edge-on conjunction offset at the measured epoch is +0.3396 d (+0.0364 phase); linear full-NSS-covariance timing scale 0.3330 d. Approximate central duration 6.49 h versus measured median 2.58 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: Alternating S32/98 raw depths show that the 4.66571-day SAP search branch is half the orbital eclipse cycle. Deeper even events give the 9.33142-day same-kind period consistent with Gaia. The S98 difference image favors the target with no capable Gaia neighbor within 60 arcsec.

Next check: Optional secondary-eclipse characterization.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
32	164	2181.1505425	0.003472	18.86
32	166	2190.4819811	0.003472	18.59
32	168	2199.8140342	0.003472	18.01
98	554	4000.7777232	0.001157	18.29
98	560	4028.7720610	0.001157	16.98

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\219399648\diff_s98.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R528. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

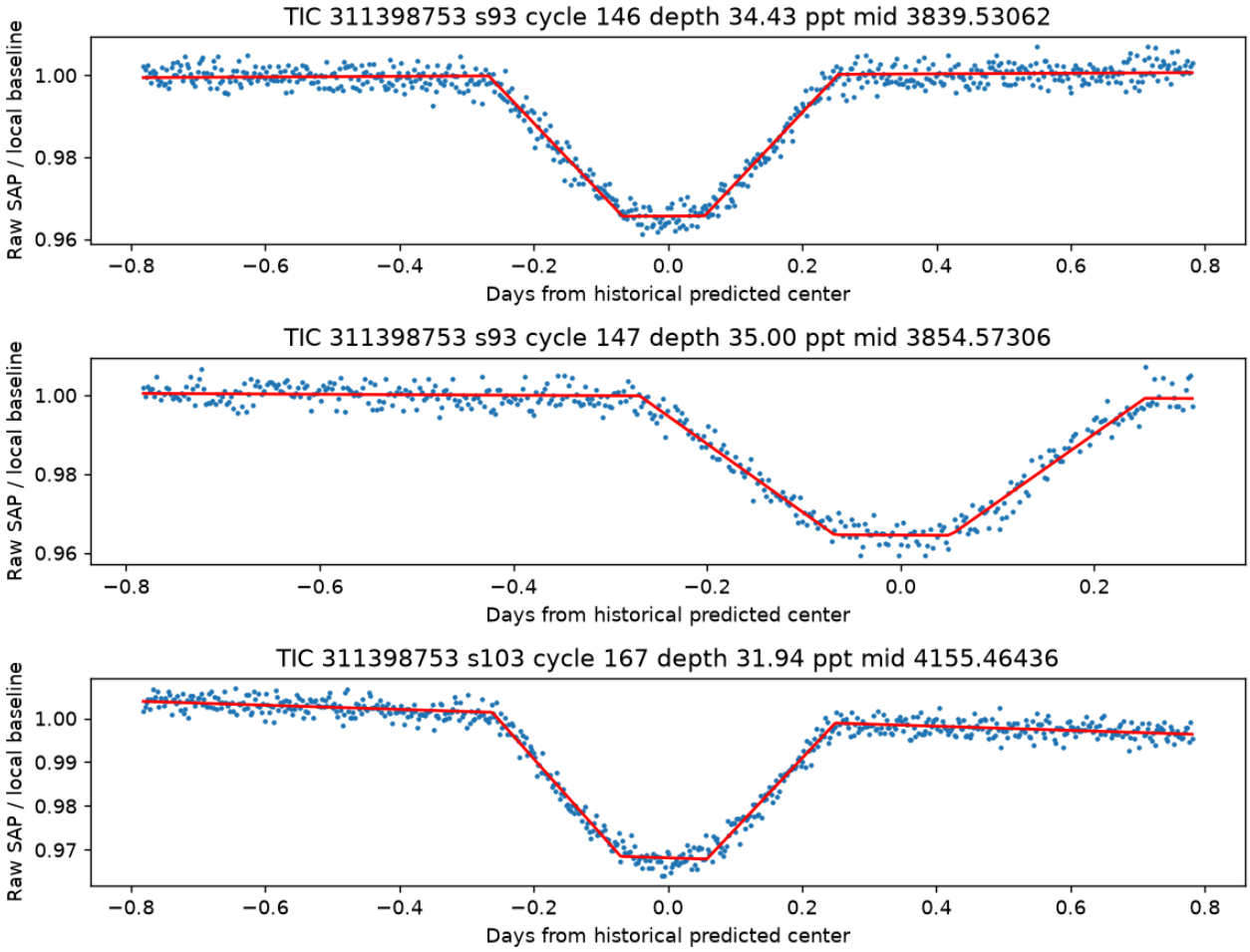
R529. TIC 311398753

physical_verdict: PROBABLE

novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved time-series photometry across an eclipse, plus refined Gaia-orbit comparison.



R529; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.

Archived source: sanitized report text.

TIC 311398753: PROBABLE

Full bounded review: COMPLETE. Physical outcome and catalogue novelty are separate; historical screening was a queue label.

1. **Raw repeated events.** 3 measured, timing-usable raw QLP SAP_FLUX events in representative sectors [93, 103]. Each local trapezoid has a free center, depth, duration and baseline. Predicted windows guided inspection; unobserved or flagged windows are not negative detections. See `measurements.json`, `raw-windows.csv` and `raw-events.png`.
2. **Timing and alias.** Same-kind midpoint fit gives P = 15.0445006706 ± 0.0001120670 d and epoch BTJD 3839.52974390 (Selected repeating eclipse, primary by convention; deeper/secondary identity

not independently established). All timing-usable historical photometric cycles; no persistent unequal alternating pattern seen in representative events. Within-sector recurrences and parity are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive period search.

3. **Pixel attribution.** The alternative cutout sector 13 deficit centroid is about 1.04 arcsec from the Gaia target coordinate; this is not a calibrated localization uncertainty. 3 Gaia neighbors within 60 arcsec are bright enough under the approximate flux-budget test to produce the selected event. Neighbor intrinsic eclipse requirements are not compared directly with blended pixel fractional losses. Source: `[local path omitted] 5\made\sb1-scale\diffimg\311398753\diff_s13.json`.
4. **Alternative reduction.** Separately extracted, target-centered TESScut aperture photometry recovers 2 stacked events at 3x3 depth 28.84 ppt, versus median selected raw QLP fit depth 34.43 ppt. This historical TESScut sector is a separate observing occasion on the same instrument from the representative QLP sectors; extraction and aperture differences limit depth comparison.
5. **Gaia geometry.** DR3 source 5804147301198863744: orbital P = 15.047545491 d, e = 0.0457. Closest edge-on conjunction offset at the measured epoch is -0.6848 d (-0.0455 phase); linear full-NSS-covariance timing scale 0.3346 d. Approximate central duration 21.27 h versus measured median 12.50 h. This is a plausibility check using assumed TIC radius/minimum-mass companion, not a fitted eclipse model or sigma veto.

Deciding interpretation and limitation: S93/103 raw eclipses repeat and the S13 alternative extraction recovers two. A capable G15.36 source lies only 9.65 arcsec from the target and shares the bright pixel; further capable sources lie at 20.8 and 38.2 arcsec. The target-near deficit cannot distinguish the closest host at uncalibrated pixel precision. Gaia conjunction offset is −0.685 d at the observed epoch versus a 0.335 d linear orbital timing scale; a projection near Gaia2016 gives +0.012 d with only a 0.026 d photometric period allowance, so phase agreement remains descriptive.

Next check: Resolved time-series photometry across an eclipse, plus refined Gaia-orbit comparison.

Catalogue novelty: central seven-group coverage is in `../../shared-catalogues/coverage-matrix.csv`. A cone-source entry alone does not settle exact counterpart identity or priority.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Depth ppt
93	146	3839.5306153	0.001157	34.43
93	147	3854.5730555	0.001319	35.00
103	167	4155.4643587	0.001760	31.94

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../alias-audit.json`, `../gaia-geometry.json`, `../download-receipts.json`, `[local path omitted] 5\made\sb1-scale\diffimg\311398753\diff_s13.json`, its PNG and Gaia-neighbor table; source and artifact hashes are in `../SHA256SUMS.json`.

Stable review key: R529. Public evidence files: measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

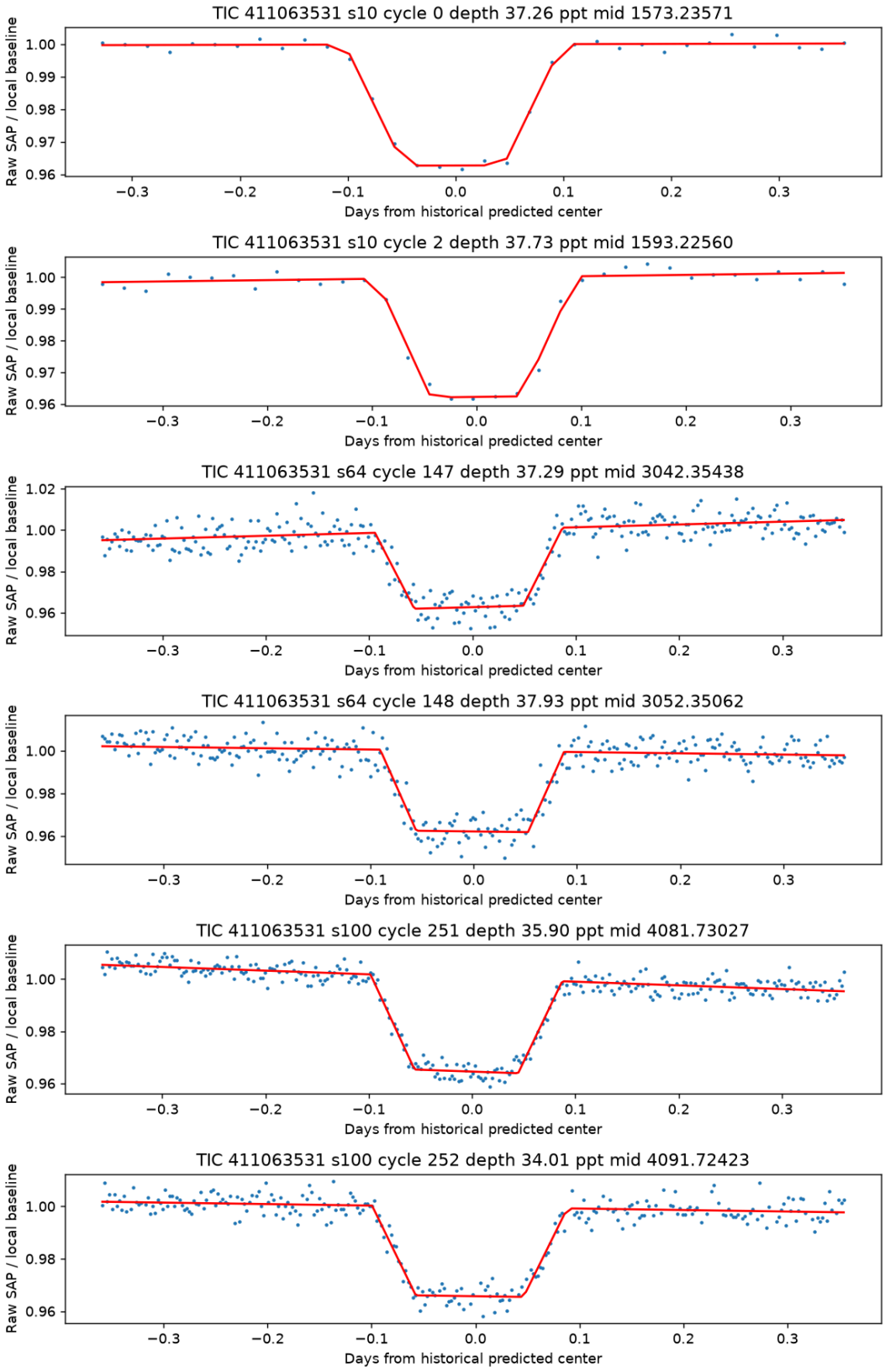
R530. TIC 411063531

physical_verdict: CONFIRMED

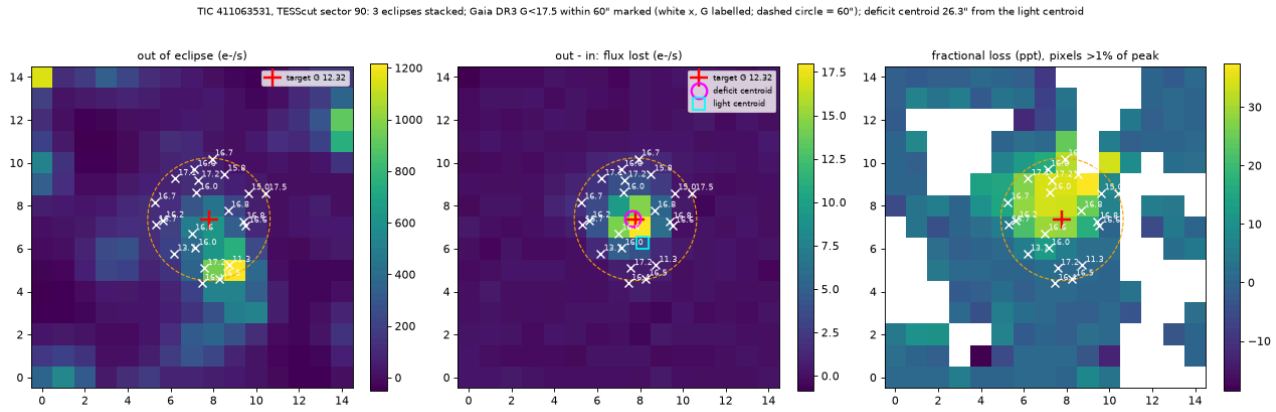
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse imaging would strengthen attribution in this crowded field.



R530; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R530; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 411063531: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 6 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [10, 64, 100]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 9.9939890066 ± 0.0000106933 d, reference BTJD 1573.2390596 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S90 TESScut deficit centroid is 3.14 arcsec from the target coordinate. Peak deficit pixel [8, 7]; target pixel [8, 7]. Image-model fractional residual 0.859 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 3 predicted event windows give 1x1/3x3/5x5 depths 28.30/18.60/12.01 ppt; median selected QLP raw fit depth is 37.27 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout sector is a separate TESS observing occasion from the selected QLP sectors; it is the same instrument with an alternative reduction, not independent-instrument confirmation. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 5354251091852534016 has orbital P = 9.994640884 d and e = 0.0745. The nearest edge-on conjunction is offset by +0.2248 d (+0.0225 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.4266 d. Approximate central duration 5.40 h versus observed median 4.48 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Six raw events span S10, S64 and S100, with adjacent pairs in both later sectors. The crowded S90 deficit centroid is 3.1 arcsec from the target while the light centroid is displaced 26.3 arcsec; four flux-capable catalogued neighbors are 44–47 arcsec from the deficit. The target is favored, but the image-model residual (0.86) limits localization precision.

Next check: Resolved eclipse imaging would strengthen attribution in this crowded field.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
10	0	1573.2357051	0.010417	37.26
10	2	1593.2255969	0.010417	37.73
64	147	3042.3543805	0.001173	37.29
64	148	3052.3506213	0.001157	37.93
100	251	4081.7302680	0.001157	35.90
100	252	4091.7242273	0.001157	34.01

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
5354250374612876928	44.15	15.755	0.934	47.0
5354250374612874496	44.88	15.020	0.493	48.3
5354249618698621696	45.66	11.275	0.052	51.3
5354249687418099072	46.48	13.728	0.176	46.7

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s90.json`, `historical/diff_s90.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R530. Public evidence files: diff_s90.json, diff_s90.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

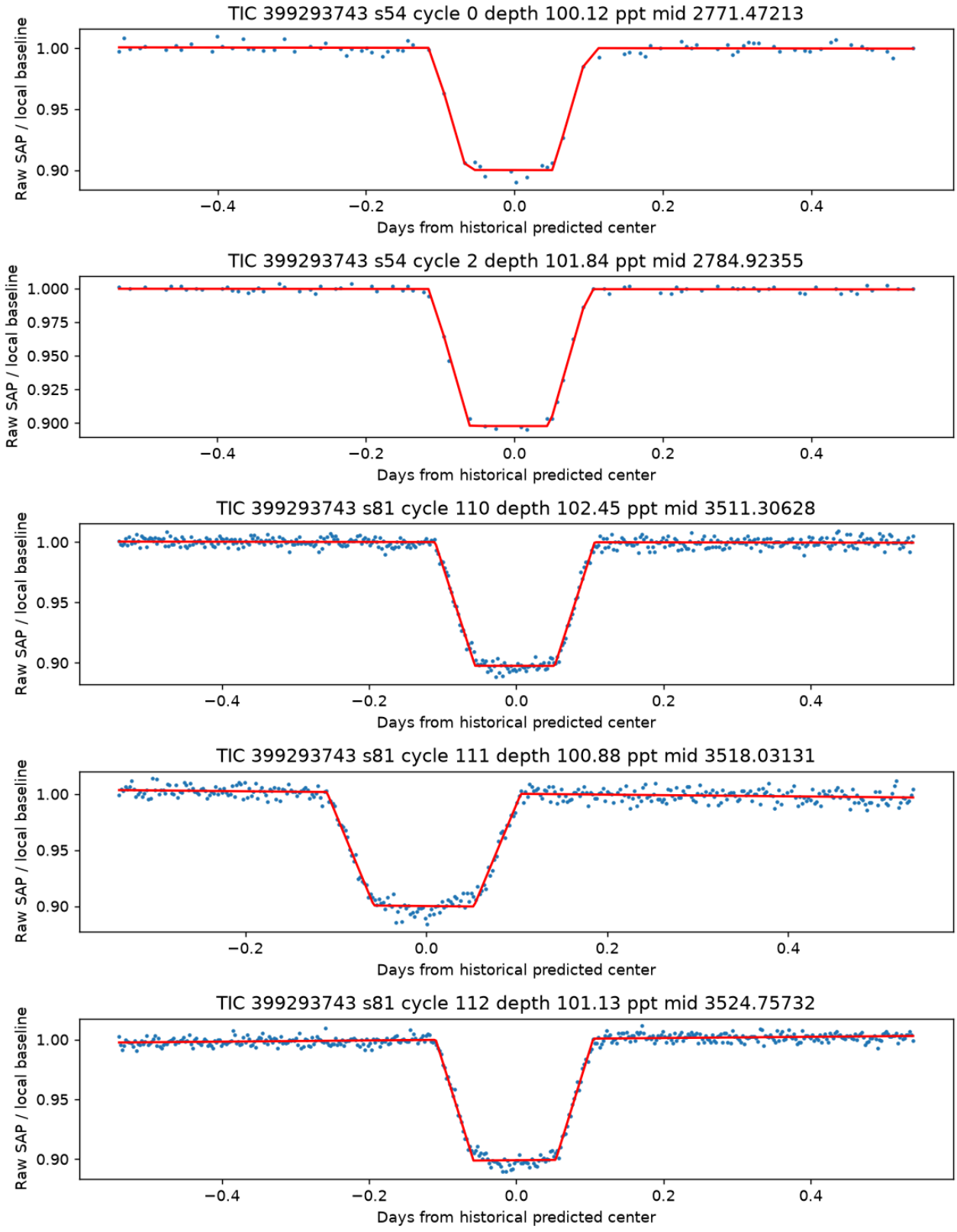
R531. TIC 399293743

physical_verdict: CONFIRMED

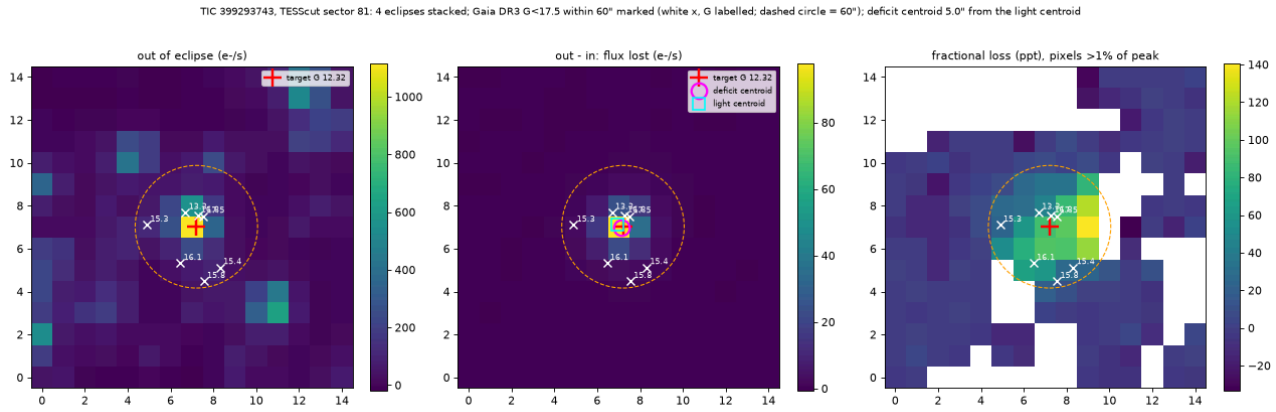
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy.



R531; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R531; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 399293743: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 5 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [54, 81]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 6.7257610294 ± 0.0000449821 d, reference BTJD 2771.4721584 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S81 TESScut deficit centroid is 1.35 arcsec from the target coordinate. Peak deficit pixel [7, 7]; target pixel [7, 7]. Image-model fractional residual 0.663 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 4 predicted event windows give 1x1/3x3/5x5 depths 87.65/74.64/68.29 ppt; median selected QLP raw fit depth is 101.13 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 4244542013422016256 has orbital P = 6.725623473 d and e = 0.0141. The nearest edge-on conjunction is offset by +0.1301 d (+0.0194 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.2912 d. Approximate central duration 5.87 h versus observed median 5.14 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Five raw events span S54 and S81, including three adjacent S81 cycles. The S81 deficit lies near the target, about 17.5 arcsec from the only flux-capable Gaia neighbor; the source is favored at TESS resolution.

Next check: Optional resolved photometry and spectroscopy.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
54	0	2771.4721285	0.006944	100.12
54	2	2784.9235520	0.006944	101.84
81	110	3511.3062759	0.001157	102.45
81	111	3518.0313096	0.001157	100.88
81	112	3524.7573169	0.001157	101.13

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
4244541940403576832	17.73	13.303	0.349	17.5

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s81.json`, `historical/diff_s81.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R531. Public evidence files: diff_s81.json, diff_s81.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

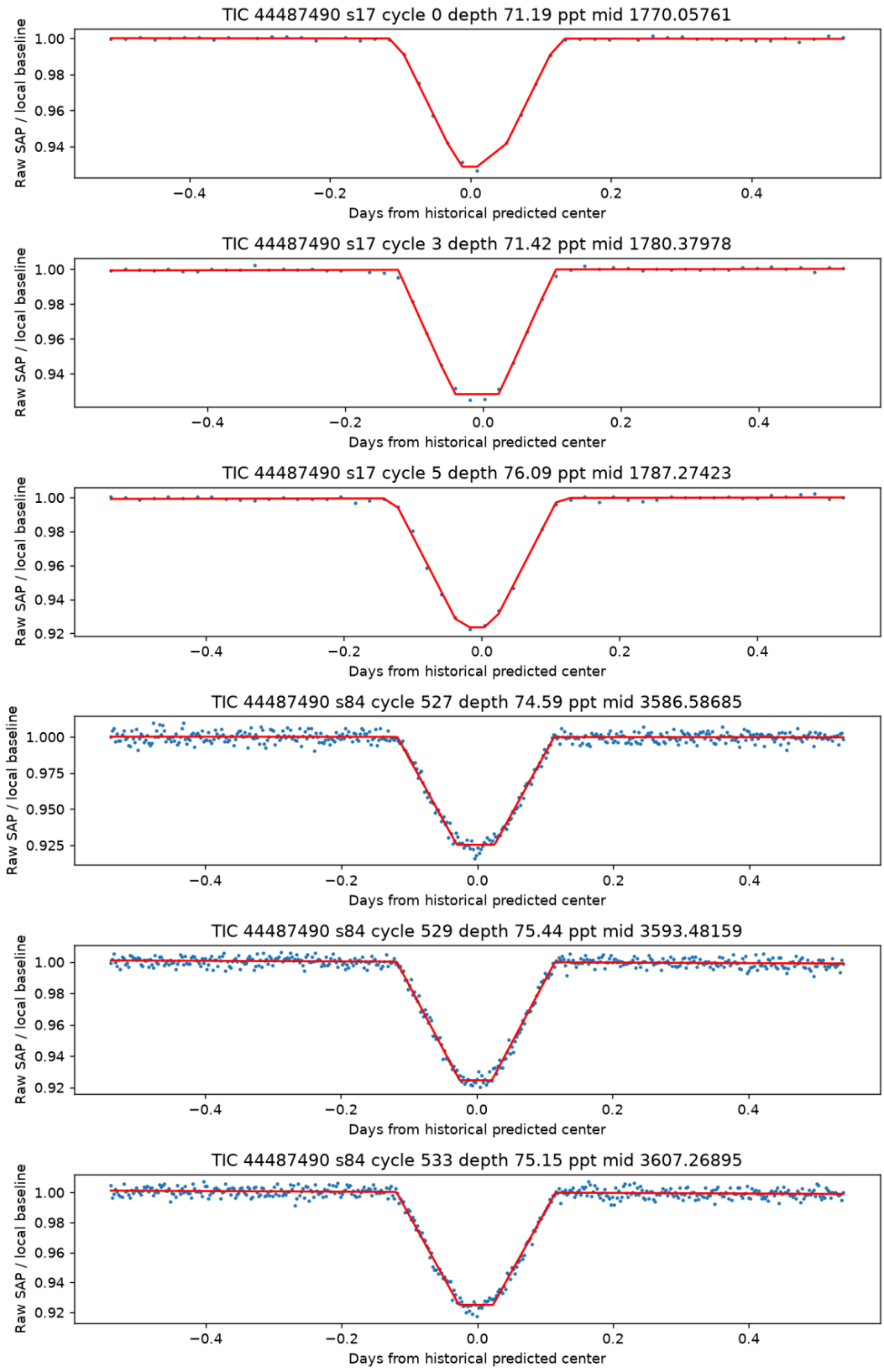
R532. TIC 44487490

physical_verdict: CONFIRMED

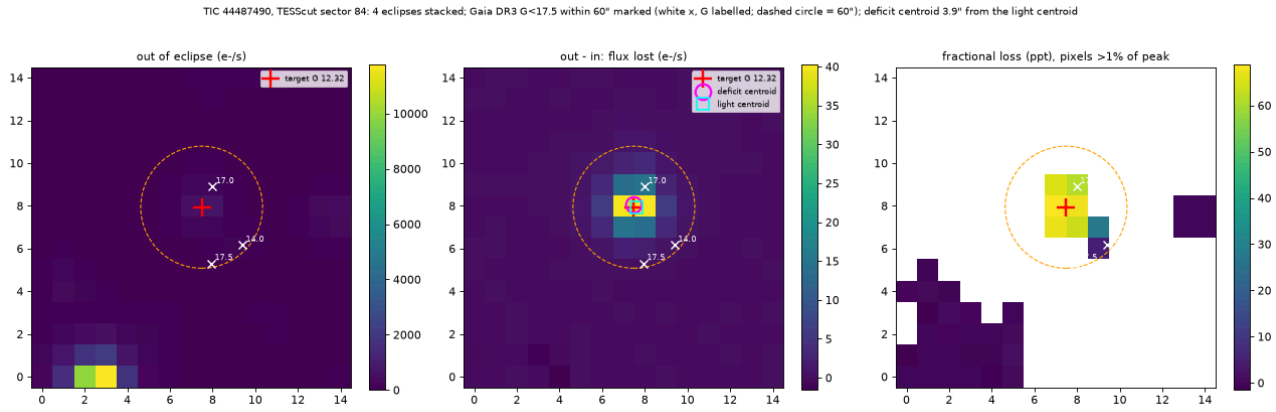
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional characterization of both eclipse kinds.



R532; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R532; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 44487490: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 5 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [17, 84]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 6.8939207479 ± 0.0000281001 d, reference BTJD 1780.3799981 (Odd-cycle, slightly deeper eclipse of the full orbit). Odd historical half-period cycles; the adjacent even events are a distinct eclipse kind with alternating timing offsets. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S84 TESScut deficit centroid is 2.39 arcsec from the target coordinate. Peak deficit pixel [8, 8]; target pixel [7, 8]. Image-model fractional residual 0.236 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 4 predicted event windows give 1x1/3x3/5x5 depths 68.92/66.68/60.44 ppt; median selected QLP raw fit depth is 75.15 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 2863142675459097216 has orbital P = 6.893921667 d and e = 0.0556. The nearest edge-on conjunction is offset by -0.0401 d (-0.0058 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.1483 d. Approximate central duration 8.28 h versus observed median 5.56 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Adjacent S17 and S84 dips alternate timing around the 3.44695-day photometric search clock. Odd cycles define the same-kind 6.89392-day orbit, matching Gaia; the even events are the other eclipse kind. The S84 image is target-centered, with the sole capable neighbor about 54 arcsec away.

Next check: Optional characterization of both eclipse kinds.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
17	3	1780.3797789	0.010416	71.42
17	5	1787.2742324	0.010416	76.09
84	527	3586.5868457	0.001157	74.59
84	529	3593.4815850	0.001157	75.44
84	533	3607.2689533	0.001157	75.15

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
2863142885913312128	53.96	13.988	0.382	57.0

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s84.json`, `historical/diff_s84.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R532. Public evidence files: diff_s84.json, diff_s84.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

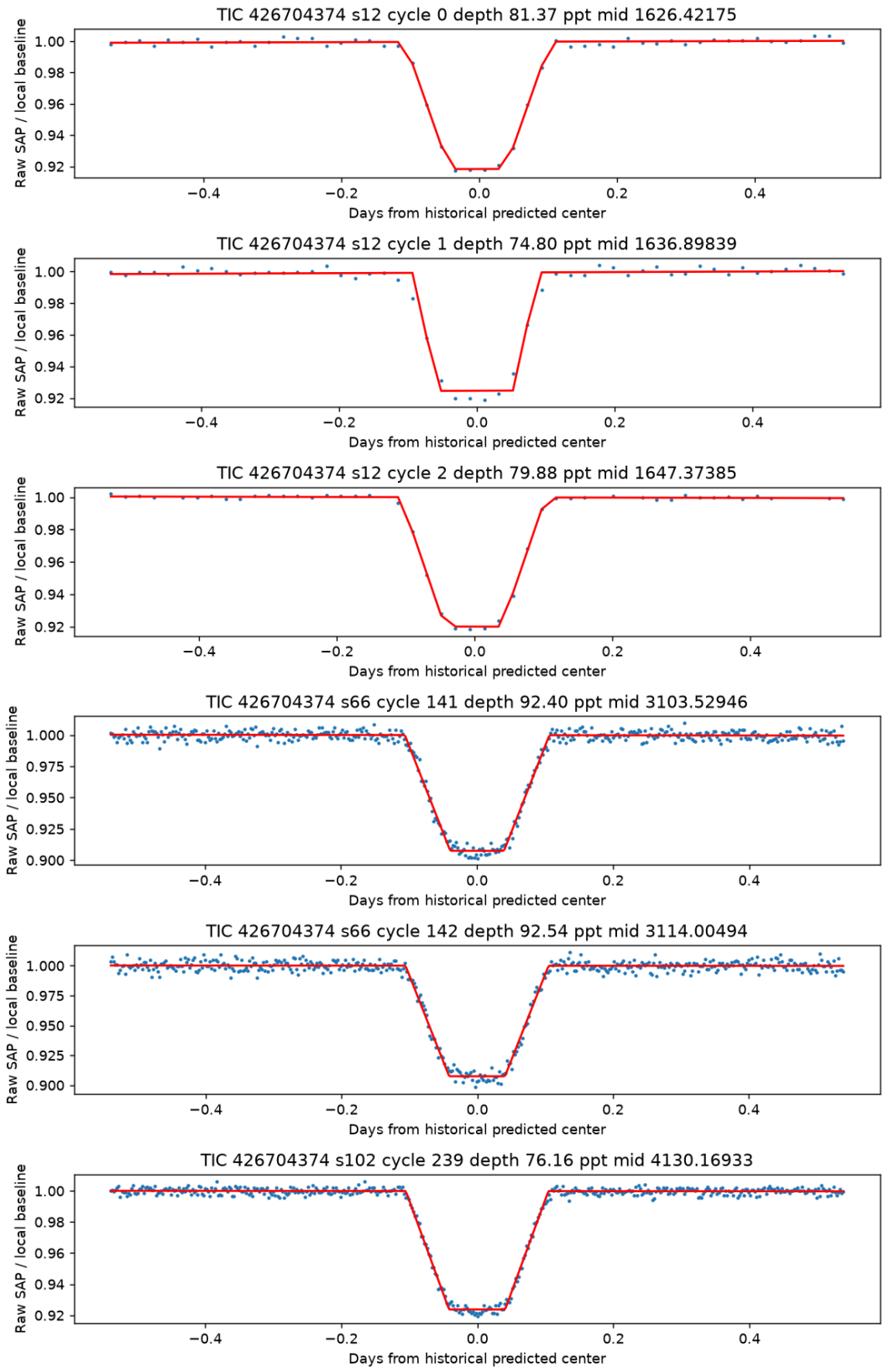
R533. TIC 426704374

physical_verdict: CONFIRMED

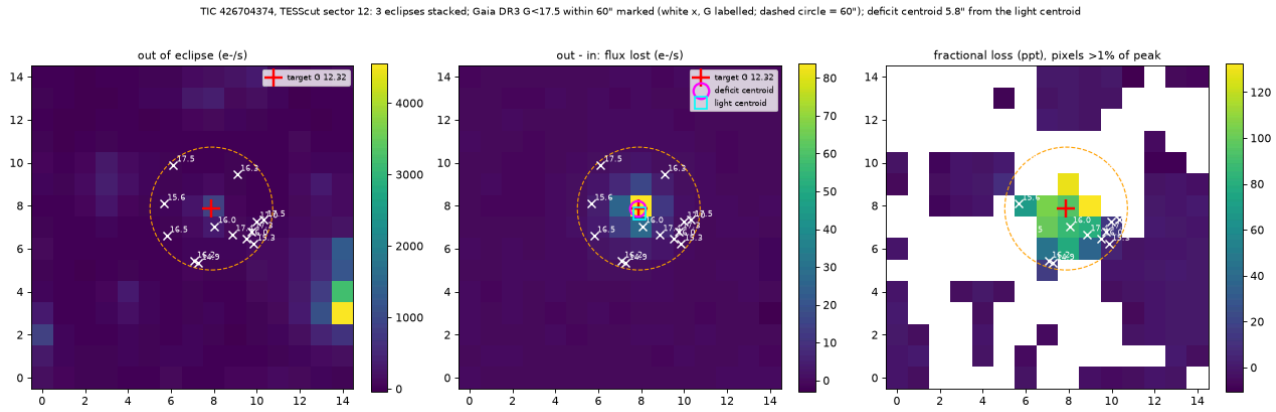
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R533; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R533; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 426704374: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 6 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [12, 66, 102]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 10.4759214103 ± 0.0000134247 d, reference BTJD 1626.4242223 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S12 TESScut deficit centroid is 1.59 arcsec from the target coordinate. Peak deficit pixel [8, 8]; target pixel [8, 8]. Image-model fractional residual 0.619 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 3 predicted event windows give 1x1/3x3/5x5 depths 96.32/102.01/94.06 ppt; median selected QLP raw fit depth is 80.63 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 5820688633321806080 has orbital P = 10.474613213 d and e = 0.0437. The nearest edge-on conjunction is offset by +0.0113 d (+0.0011 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.2625 d. Approximate central duration 6.59 h versus observed median 5.05 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Raw eclipses recur across S12, S66 and S102, including adjacent S12 and S66 cycles. The S12 deficit is target-centered and no Gaia neighbor in the 60-arcsec screen passes the flux-budget bound.

Next check: Optional precision characterization.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
12	0	1626.4217502	0.010417	81.37
12	1	1636.8983902	0.010417	74.80
12	2	1647.3738533	0.010417	79.88
66	141	3103.5294601	0.001157	92.40
66	142	3114.0049354	0.001157	92.54
102	239	4130.1693271	0.001157	76.16

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s12.json`, `historical/diff_s12.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R533. Public evidence files: diff_s12.json, diff_s12.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

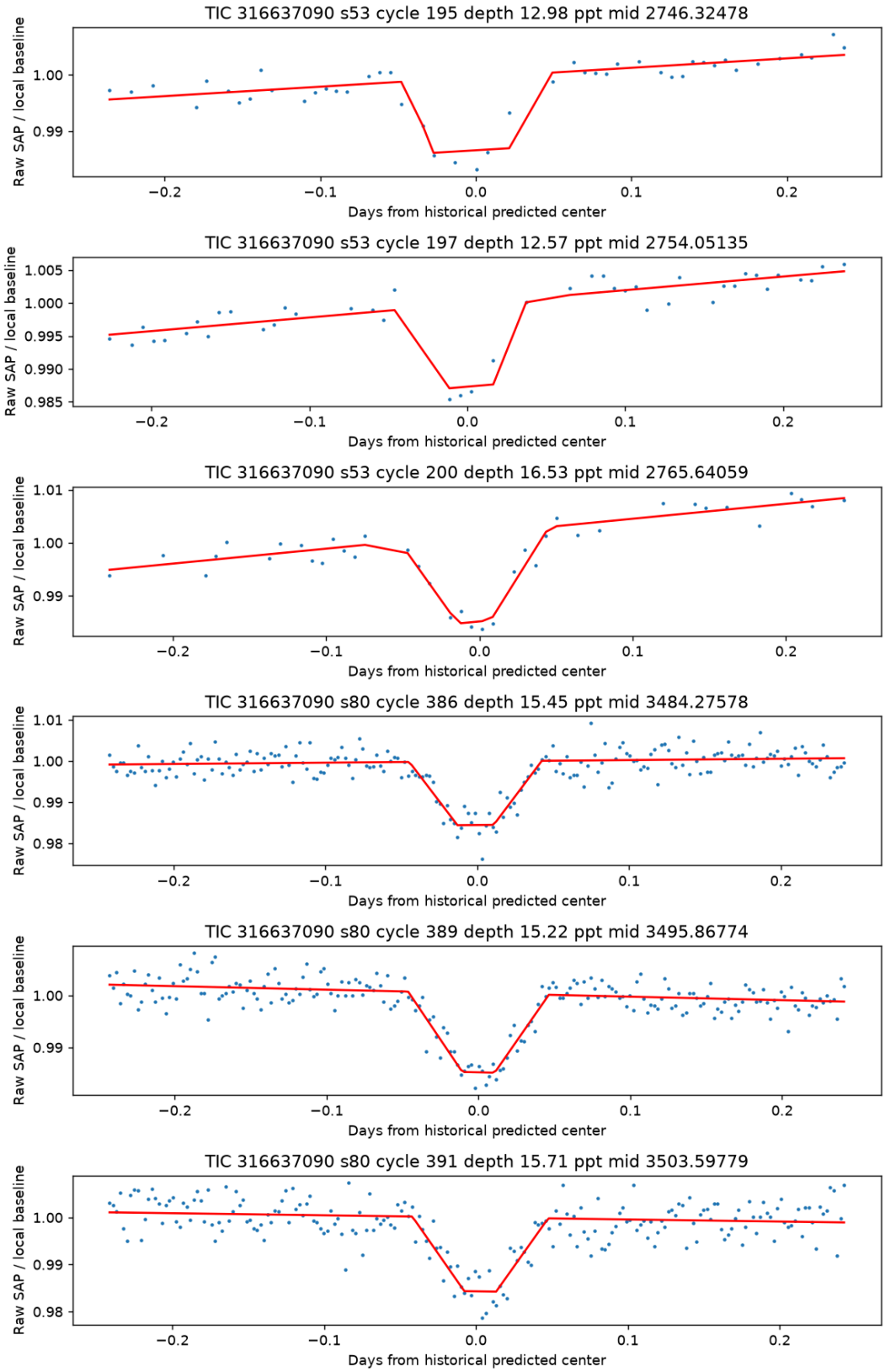
R534. TIC 316637090

physical_verdict: CONFIRMED

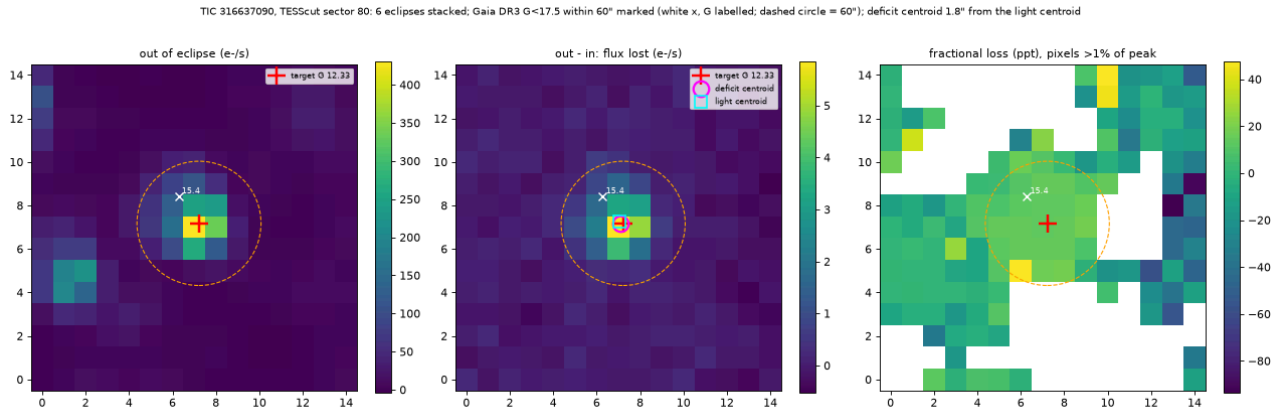
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional resolved photometry would strengthen this shallow attribution.



R534; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R534; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 316637090: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 10 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [53, 80]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 3.8636264942 ± 0.0000122316 d, reference BTJD 2746.3243407 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S80 TESScut deficit centroid is 3.12 arcsec from the target coordinate. Peak deficit pixel [7, 7]; target pixel [7, 7]. Image-model fractional residual 0.179 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 6 predicted event windows give 1x1/3x3/5x5 depths 13.58/13.92/13.64 ppt; median selected QLP raw fit depth is 15.34 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 4595458674208503424 has orbital P = 3.862983602 d and e = 0.1667. The nearest edge-on conjunction is offset by +0.2349 d (+0.0608 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.1301 d. Approximate central duration 4.61 h versus observed median 2.18 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Ten shallow raw events in S53 and S80 repeat; the S80 aperture extraction recovers six. The target-near deficit is roughly 31 arcsec from the only capable Gaia neighbor. The Gaia conjunction offset is descriptive and is not used as a significance veto.

Next check: Additional resolved photometry would strengthen this shallow attribution.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
53	195	2746.3247811	0.003472	12.98
53	197	2754.0513492	0.006473	12.57
53	198	2757.9189913	0.005698	14.81
53	200	2765.6405893	0.004227	16.53
80	385	3480.4125219	0.001253	16.41
80	386	3484.2757810	0.001157	15.45
80	388	3491.9986733	0.004023	10.24
80	389	3495.8677415	0.001157	15.22
80	390	3499.7325144	0.001157	16.71
80	391	3503.5977894	0.001547	15.71

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
4595458605489025280	29.26	15.390	0.246	30.8

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s80.json`, `historical/diff_s80.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R534. Public evidence files: diff_s80.json, diff_s80.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

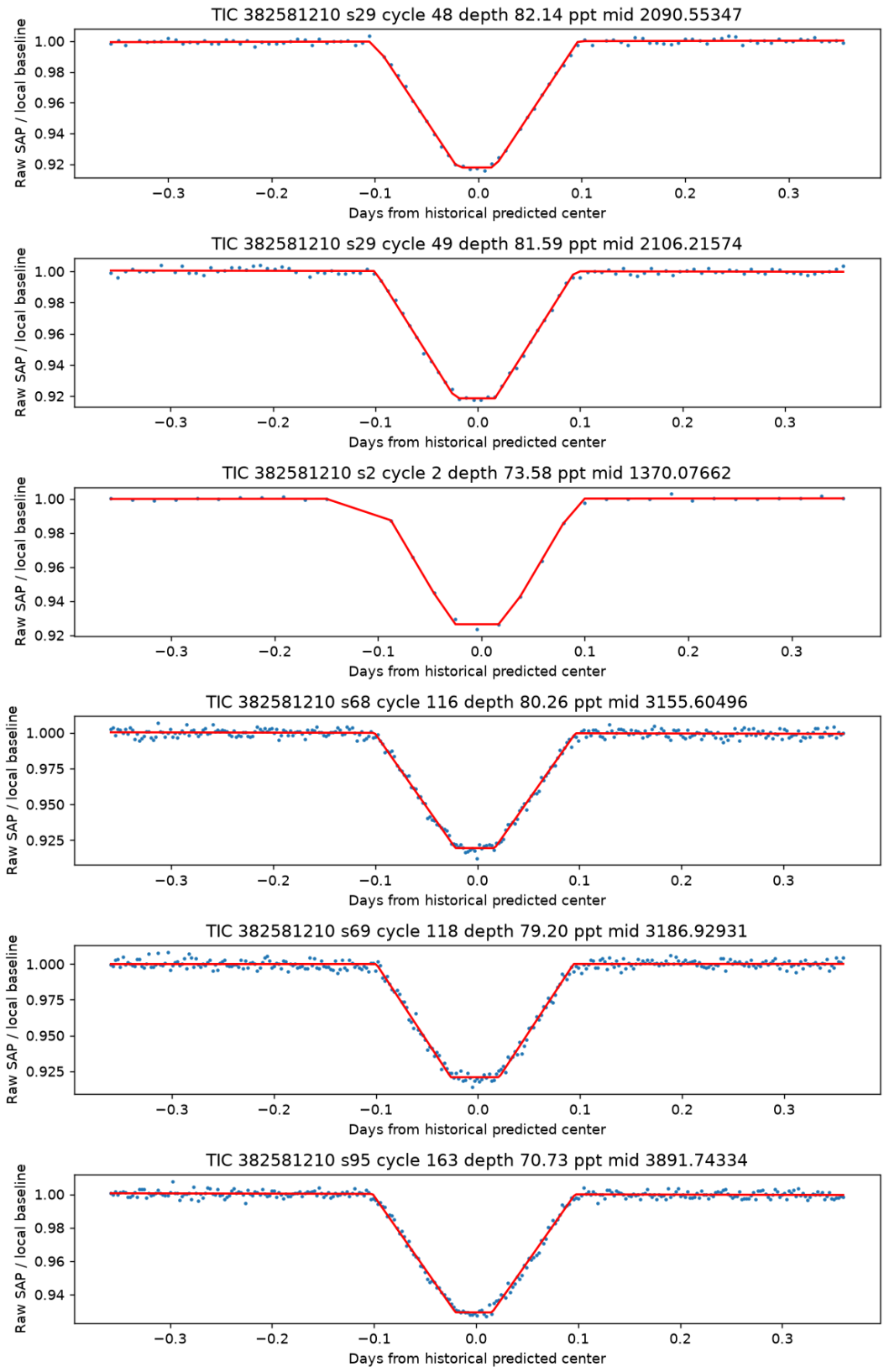
R535. TIC 382581210

physical_verdict: CONFIRMED

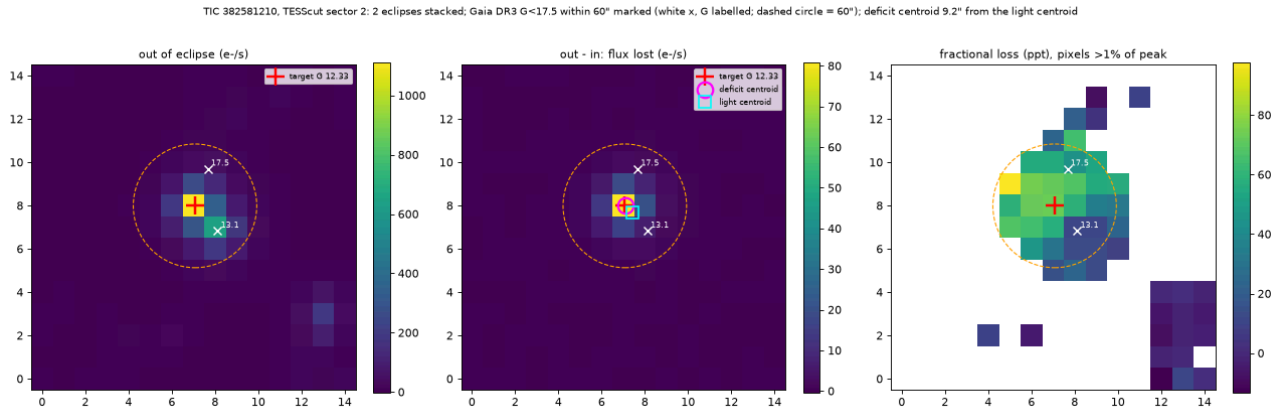
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional later-sector timing and resolved photometry.



R535; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R535; Representative eclipse-deficit image; source attribution retains the report limits.

Independent recurrence and coverage check. The S29 clean event pair and intermediate windows constrain equal-kind aliases; full numerical results and scripts are preserved in the public helper archive.

382581210-recurrence.csv (rounded display; full precision in linked CSV)

Sector	Cycle	Center BTJD	Good in window	Baseline	Deficit ppt
002	-47	1354.427	9	17	52.5
002	-46	1370.089	10	28	45.9
003	-45	1385.751	7	26	59.2
003	-44	1401.413	9	28	50.9
028	-1	2074.891	0	1	not covered
068	68	3155.588	87	259	48.7
068	69	3171.250	87	259	50.5
069	70	3186.913	87	259	52.9
069	71	3202.575	87	259	52.5
095	115	3891.715	86	259	43.5
095	116	3907.377	0	0	not covered

382581210-coverage.csv (rounded display; full precision in linked CSV)

Window	Phase	Center BTJD	Good	Baseline	Deficit ppt
event1	0.000	2090.553	27	83	49.8
quarter	0.250	2094.469	25	68	-0.3
third	0.333	2095.774	28	84	-0.1
half	0.500	2098.385	28	84	-1.1
two_thirds	0.667	2100.995	0	0	not covered
three_quarters	0.750	2102.300	0	0	not covered
event2	1.000	2106.216	29	84	52.7

Archived source: sanitized report text.

TIC 382581210: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 10 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [2, 3, 29, 68, 69, 95]. The local fit has a free center, depth, width and

baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.

2. **Measured timing and aliases.** P = 15.6625230978 ± 0.0000195721 d, reference BTJD 1354.4142823 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S2 TESScut deficit centroid is 0.97 arcsec from the target coordinate. Peak deficit pixel [7, 8]; target pixel [7, 8]. Image-model fractional residual 0.415 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 2 predicted event windows give 1x1/3x3/5x5 depths 72.76/56.61/50.49 ppt; median selected QLP raw fit depth is 78.95 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 4699704947887847296 has orbital P = 15.665541787 d and e = 0.2966. The nearest edge-on conjunction is offset by -0.2056 d (-0.0131 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.5483 d. Approximate central duration 7.01 h versus observed median 4.68 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: The current MAST QLP availability record lists nine sectors; all nine files were retrieved and checked. Ten measured quality-zero raw eclipses in S2, S3, S29, S68, S69 and S95 now span the long baseline, including adjacent S29 and S68 pairs. The S29 clean intermediate windows reject equal-kind half, third and quarter-period submultiples; flagged windows were not used as nulls. The historical S2 cutout separately stacks two sparse predicted windows into one target-centered deficit, which alone would not establish each window as an eclipse. Its capable Gaia neighbor is 33 arcsec from the target and 32.5 arcsec from the deficit.

Next check: Optional later-sector timing and resolved photometry.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
29	48	2090.5534699	0.003472	82.14
29	49	2106.2157431	0.003472	81.59
2	1	1354.4178976	0.010417	69.70
2	2	1370.0766181	0.010417	73.58
3	4	1401.4020768	0.010416	74.99
68	116	3155.6049624	0.001157	80.26
68	117	3171.2668610	0.001157	78.70
69	118	3186.9293125	0.001157	79.20
69	119	3202.5912876	0.001157	80.78

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
95	163	3891.7433424	0.001157	70.73

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
4699704986543565824	32.97	13.094	0.226	32.5

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s02.json`, `historical/diff_s02.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R535. Public evidence files: diff_s02.json, diff_s02.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index. 382581210-coverage.csv 382581210-coverage.py 382581210-downloads.json 382581210-recurrence.csv 382581210-recurrence.py

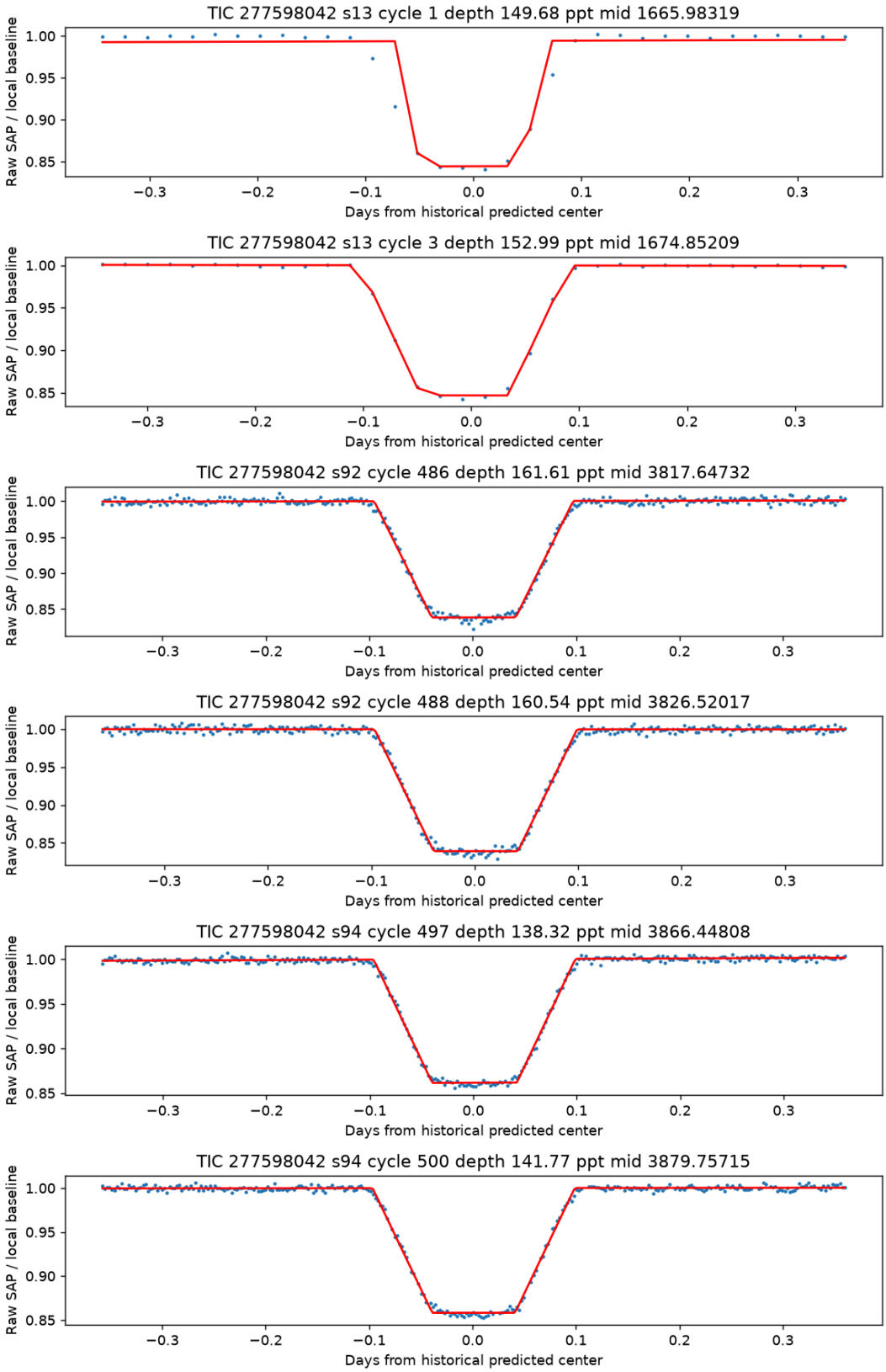
R536. TIC 277598042

physical_verdict: CONFIRMED

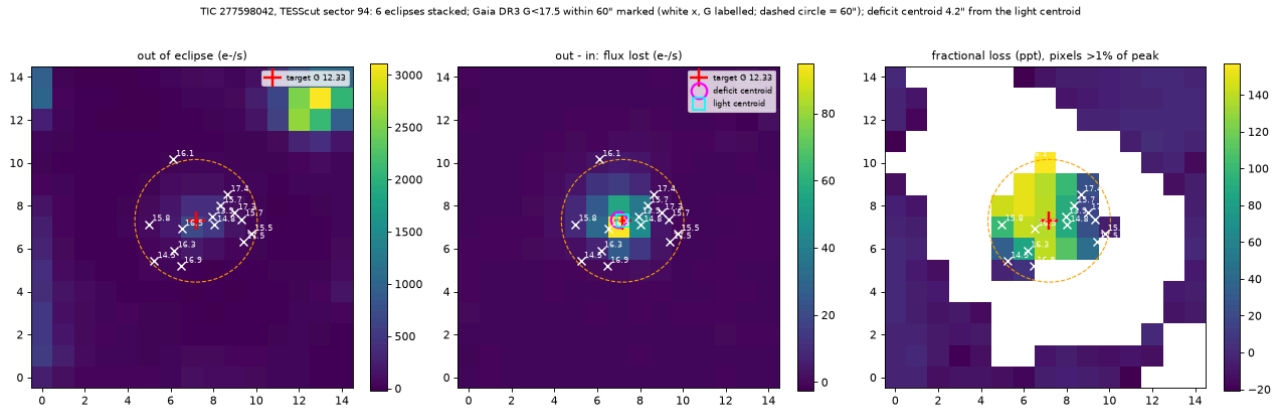
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry.



R536; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R536; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 277598042: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 14 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [13, 92, 94]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 4.4364264100 ± 0.0000120101 d, reference BTJD 1665.9805449 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S94 TESScut deficit centroid is 4.11 arcsec from the target coordinate. Peak deficit pixel [7, 7]; target pixel [7, 7]. Image-model fractional residual 0.279 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 6 predicted event windows give 1x1/3x3/5x5 depths 138.22/120.70/109.60 ppt; median selected QLP raw fit depth is 151.33 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 6732852653768547840 has orbital P = 4.435881632 d and e = 0.0840. The nearest edge-on conjunction is offset by +0.0447 d (+0.0101 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.1250 d. Approximate central duration 4.98 h versus observed median 4.68 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Fourteen raw events span S13, S92 and S94 with multiple adjacent cycles. The S94 difference core lies near the target and about 20 arcsec from the only capable Gaia neighbor; that neighbor would need an approximately 80% intrinsic eclipse.

Next check: Optional resolved photometry.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
13	1	1665.9831857	0.010417	149.68
13	2	1670.4154686	0.010417	155.48
13	3	1674.8520864	0.010417	152.99
92	483	3804.3384462	0.001157	162.28
92	484	3808.7745133	0.001157	162.52
92	486	3817.6473162	0.001157	161.61
92	487	3822.0834577	0.001157	161.63
92	488	3826.5201663	0.001157	160.54
94	495	3857.5750328	0.001157	142.98
94	496	3862.0116368	0.001157	139.06
94	497	3866.4480830	0.001157	138.32
94	498	3870.8844811	0.001157	141.72
94	499	3875.3211812	0.001157	140.58
94	500	3879.7571485	0.001157	141.77

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
6732864026841949952	15.82	13.858	0.797	19.9

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s94.json`, `historical/diff_s94.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R536. Public evidence files: diff_s94.json, diff_s94.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

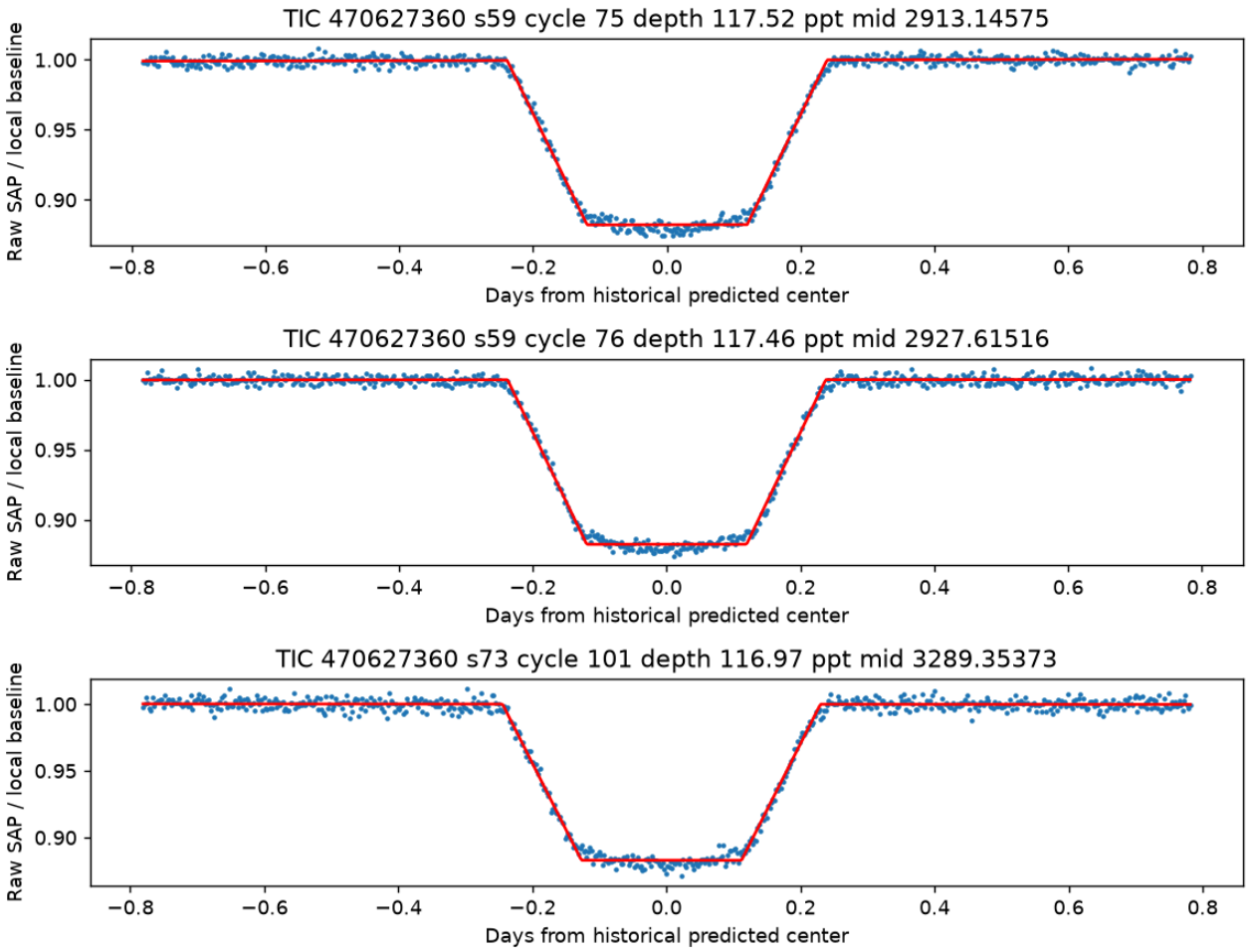
R537. TIC 470627360

physical_verdict: CONFIRMED

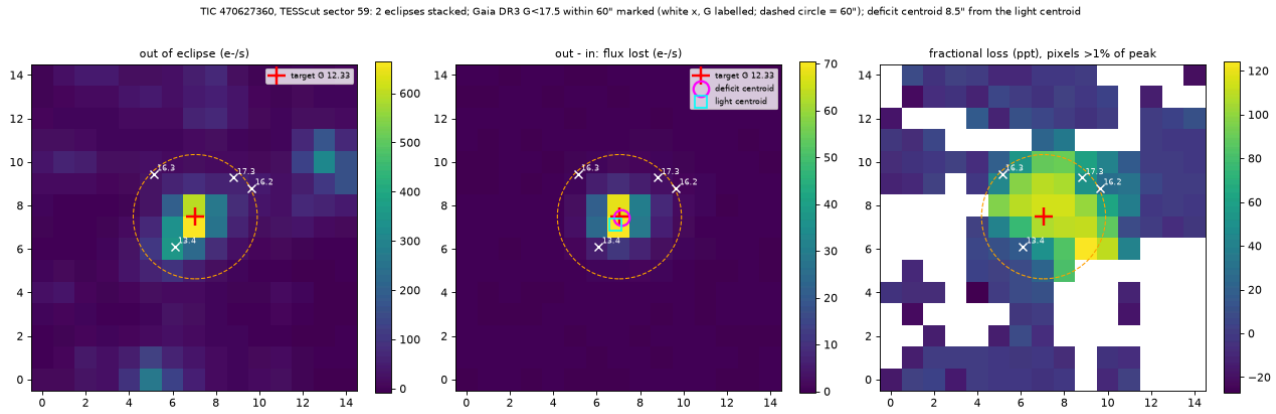
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Further-sector timing would refine the modest three-event ephemeris.



R537; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R537; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 470627360: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 3 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [59, 73]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 14.4695399219 ± 0.0000555576 d, reference BTJD 2913.1456895 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S59 TESScut deficit centroid is 1.66 arcsec from the target coordinate. Peak deficit pixel [7, 7]; target pixel [7, 7]. Image-model fractional residual 0.330 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 2 predicted event windows give 1x1/3x3/5x5 depths 106.04/90.28/83.80 ppt; median selected QLP raw fit depth is 117.46 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 197378967299172224 has orbital P = 14.476795633 d and e = 0.3774. The nearest edge-on conjunction is offset by -0.6255 d (-0.0432 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 1.0802 d. Approximate central duration 14.33 h versus observed median 11.41 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: S59 has two adjacent measured eclipses and S73 supplies a longer-baseline third. The S59 difference core is target-centered, about 35.5 arcsec from its only capable Gaia neighbor; the long event is also recovered by aperture extraction.

Next check: Further-sector timing would refine the modest three-event ephemeris.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
59	75	2913.1457540	0.001157	117.52
59	76	2927.6151624	0.001157	117.46
73	101	3289.3537301	0.001157	116.97

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
197378971597385472	34.28	13.428	0.445	35.5

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s59.json`, `historical/diff_s59.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R537. Public evidence files: diff_s59.json, diff_s59.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

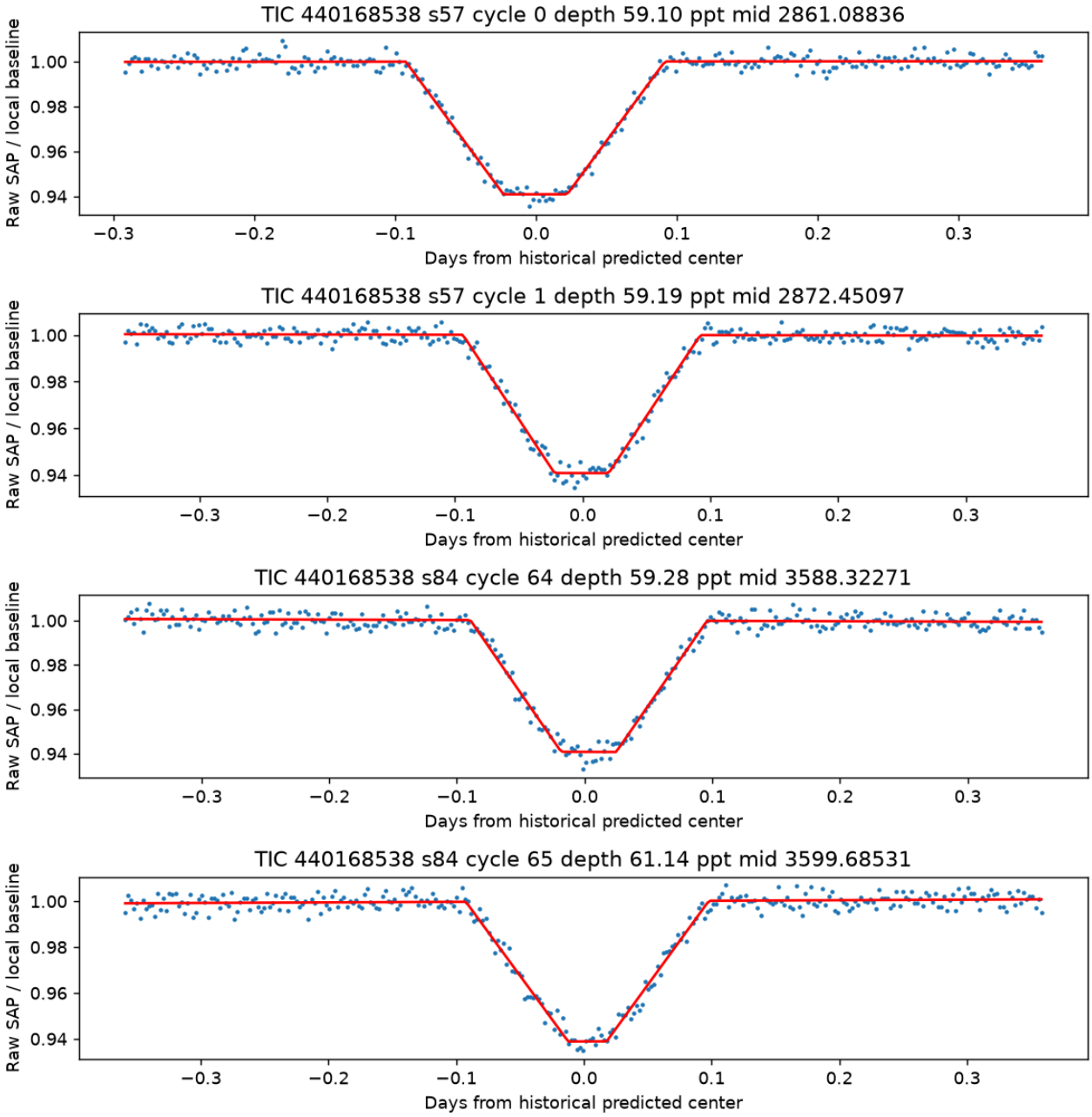
R538. TIC 440168538

physical_verdict: CONFIRMED

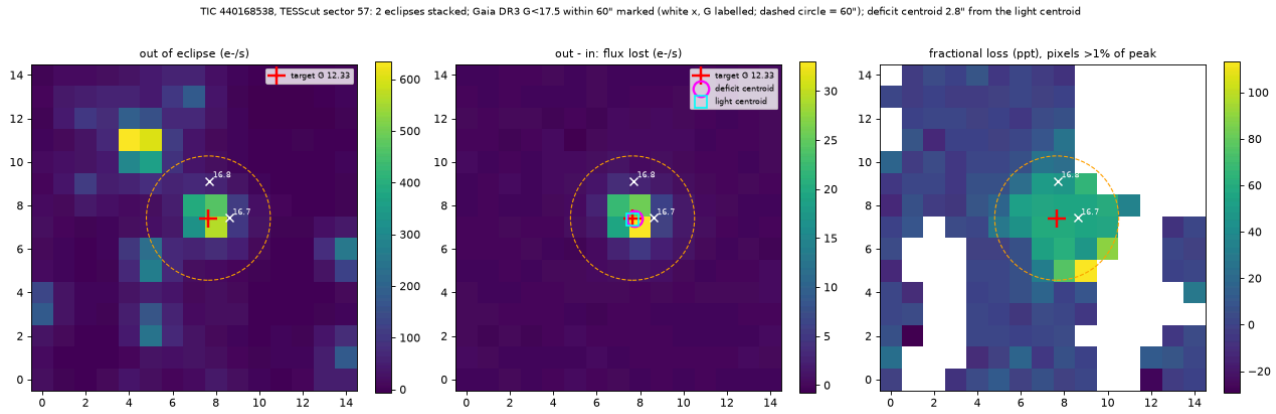
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry.



R538; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R538; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 440168538: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 4 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [57, 84]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 11.3630365737 ± 0.0000180823 d, reference BTJD 2861.0881487 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S57 TESScut deficit centroid is 2.24 arcsec from the target coordinate. Peak deficit pixel [8, 7]; target pixel [8, 7]. Image-model fractional residual 0.766 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 2 predicted event windows give 1x1/3x3/5x5 depths 58.62/56.89/53.38 ppt; median selected QLP raw fit depth is 59.24 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout and selected QLP observations use the same TESS sector and share photons; this is an alternative reduction, not an independent observing occasion. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 391817260687992960 has orbital P = 11.362526050 d and e = 0.0213. The nearest edge-on conjunction is offset by +0.2062 d (+0.0181 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.3661 d. Approximate central duration 7.55 h versus observed median 4.46 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Two adjacent raw eclipses in each of S57 and S84 anchor the recurrence. The S57 deficit favors the target; no Gaia neighbor within 60 arcsec passes the flux-budget screen. The image-model residual (0.77) limits localization precision.

Next check: Optional resolved photometry.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
57	0	2861.0883552	0.001157	59.10
57	1	2872.4509721	0.001157	59.19
84	64	3588.3227099	0.001157	59.28
84	65	3599.6853121	0.001157	61.14

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s57.json`, `historical/diff_s57.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R538. Public evidence files: diff_s57.json, diff_s57.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

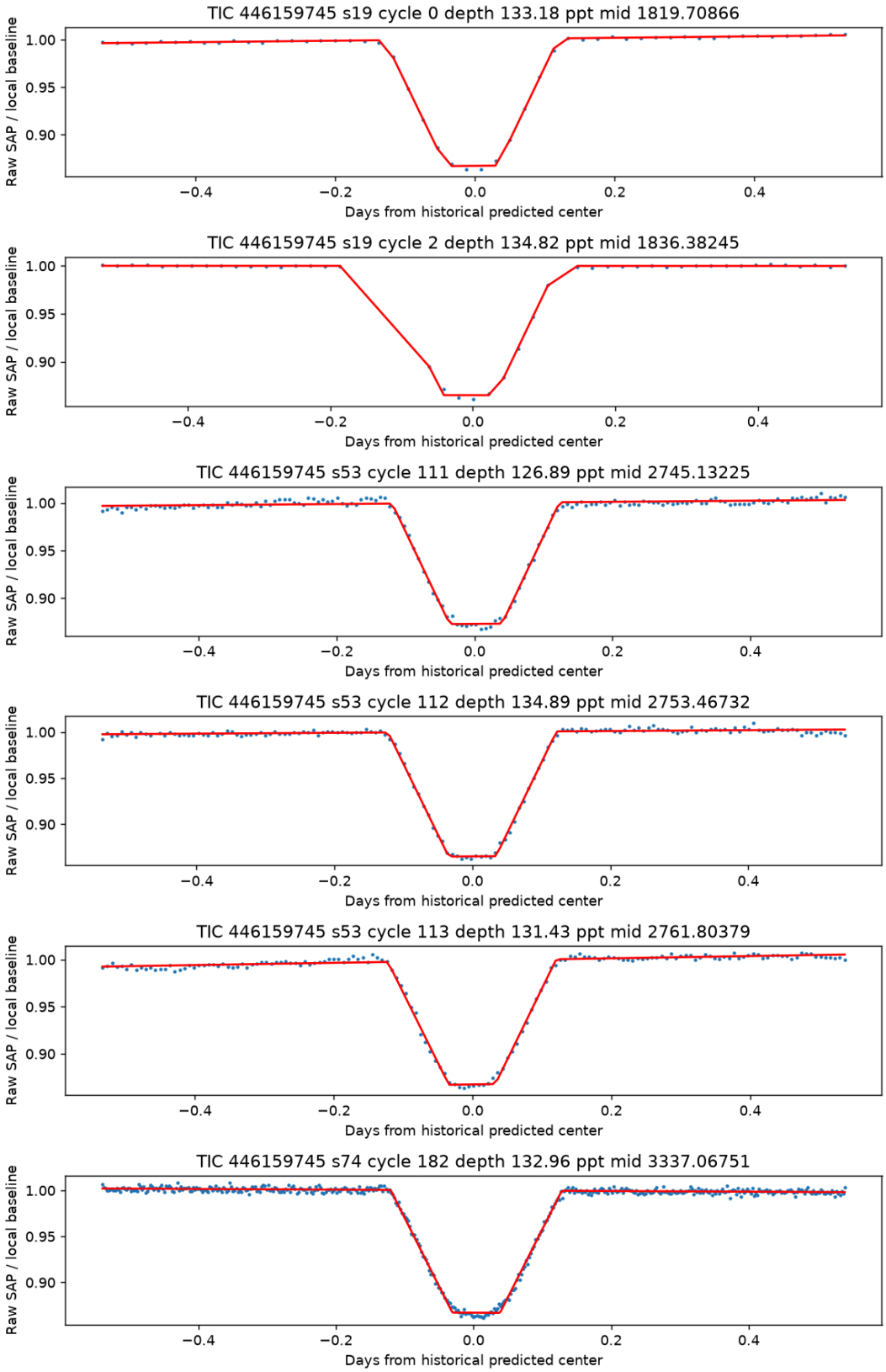
R539. TIC 446159745

physical_verdict: CONFIRMED

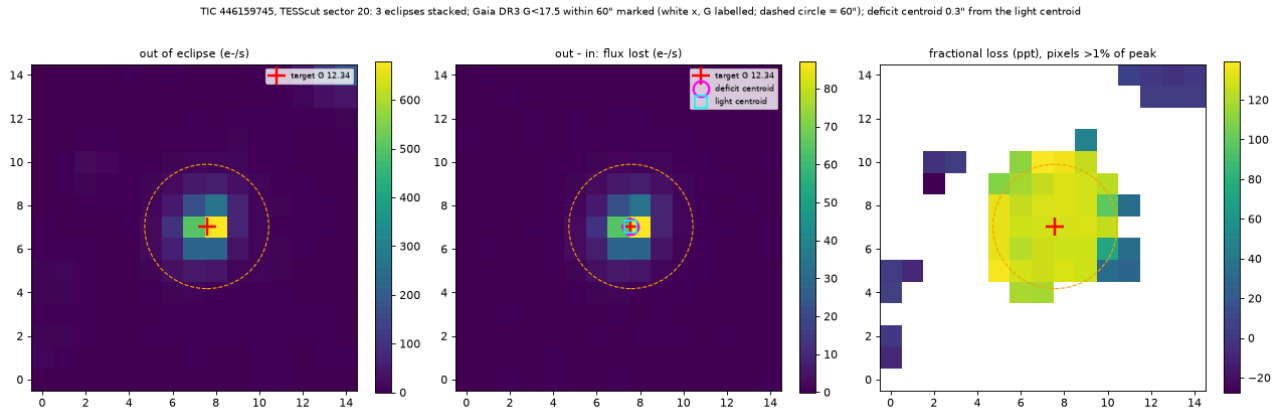
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional precision characterization.



R539; Measured raw-flux event windows; quality flags and timing exclusions are detailed in the report.



R539; Representative eclipse-deficit image; source attribution retains the report limits.

Archived source: sanitized report text.

TIC 446159745: CONFIRMED

Full bounded review: COMPLETE. Physical outcome, catalogue novelty and historical screening label are separate.

1. **Repeated raw morphology.** 7 timing-usable measured eclipses in quality-zero QLP SAP_FLUX across representative sectors [19, 53, 74]. The local fit has a free center, depth, width and baseline. Predicted centers select inspection windows; flagged and uncovered windows are not clean nulls. Raw points are preserved in `raw-windows.csv` and `raw-events.png`.
2. **Measured timing and aliases.** P = 8.3371405920 ± 0.0000254401 d, reference BTJD 1819.7081287 (Selected repeating eclipse; deeper/secondary identity not independently established). All timing-usable historical photometric cycles; deeper/secondary identity not independently established. Adjacent-cycle and parity summaries are in `../alias-audit.json`. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. This is a bounded alias check, not an exhaustive all-period search.
3. **Pixel attribution.** The S20 TESScut deficit centroid is 0.30 arcsec from the target coordinate. Peak deficit pixel [8, 7]; target pixel [8, 7]. Image-model fractional residual 0.023 is not a calibrated centroid error. The table below lists every Gaia neighbor in the 60-arcsec screen that can supply the selected depth under an approximate flux-budget calculation. Required intrinsic eclipse depth is not a blended pixel loss.
4. **Alternative reduction.** Target-centered, background-subtracted cutout apertures combining 3 predicted event windows give 1x1/3x3/5x5 depths 128.37/128.36/127.20 ppt; median selected QLP raw fit depth is 132.96 ppt. The stack does not by itself prove each cutout window contains an eclipse. The cutout sector is a separate TESS observing occasion from the selected QLP sectors; it is the same instrument with an alternative reduction, not independent-instrument confirmation. Aperture dilution and selected-event differences limit depth equality.
5. **Gaia geometry.** Source 1139354875446856704 has orbital P = 8.337129275 d and e = 0.0553. The nearest edge-on conjunction is offset by -0.0237 d (-0.0028 phase) at the measured reference epoch. Full source-specific NSS covariance gives a descriptive linear timing scale of 0.2423 d. Approximate central duration 7.25 h versus observed median 5.94 h. Assumed TIC radius and minimum-mass companion make this a plausibility check, not a fitted binary model or sigma veto.

Deciding interpretation and limitation: Seven raw eclipses span S19, S53 and S74, with an adjacent S53 triplet. The S20 alternative extraction is a separate observing occasion with three events; its deficit is sharply target-centered and no capable Gaia neighbor is listed.

Next check: Optional precision characterization.

Catalogue novelty: the central seven-group result in `../../shared-catalogues/coverage-matrix.csv` reports complete scoped checks with no identified prior EB match. It does not establish exhaustive literature priority or resolve embargoed identities.

Sector	Historical cycle	Measured midpoint BTJD	Input timing error d	Fitted depth ppt
19	0	1819.7086587	0.010417	133.18
19	2	1836.3824472	0.010417	134.82
53	111	2745.1322540	0.003472	126.89
53	112	2753.4673157	0.003472	134.89
53	113	2761.8037927	0.003472	131.43
74	180	3320.3936679	0.001157	132.55
74	182	3337.0675061	0.001157	132.96

Capable Gaia neighbors:

Gaia DR3 source	Target separation arcsec	G	Required intrinsic eclipse fraction	Deficit separation arcsec
None	—	—	—	—

Evidence: `measurements.json`, `raw-windows.csv`, `raw-events.png`, `../download-receipts.json`, `../alias-audit.json`, `../gaia-geometry.json`, `historical/diff_s20.json`, `historical/diff_s20.png`, `historical/gaia.json`, original `[local path omitted]`, and../helper-audit.md. `All file hashes are in`../SHA256SUMS.json`.

Stable review key: R539. Public evidence files: diff_s20.json, diff_s20.png, gaia.json, measurements.json, raw-events.png, raw-windows.csv. Batch-level audits and helper artifacts are linked in the batch index. Source and export hashes are in the manifest. FITS are not redistributed; source receipts retain hashes and acquisition records. Scripts are archival and may require original paths and external data.

Batch evidence and audit index.

Spectroscopic-orbit period folds: completion of the screened inventory

Completion of the screened period-fold inventory. The final 136 screened candidates, reviewed in increasing actual Gaia DR3 G (12.337704-12.742949), comprise 101 physically confirmed systems, 34 probable systems and one plausible signal. TIC 295771108 is a probable known recovery, excluded from additions. The 134 unique additions therefore comprise 101 confirmed and 33 probable systems. TIC 290749976 remains plausible because eclipse morphology and source attribution are insecure. This completes this screened inventory, while five weaker conjunction appointments, 58 lower-score survivors and 48 unexecuted exploratory pilot targets remain separate selections. The fixed balanced 120-target comparison is unchanged. These selected outcomes do not estimate blind discovery yield or completeness.

Measurement qualifications. Individual reports retain eclipse-kind, alias, timing-error and source-attribution limits. Reference epochs in BTJD are per-object measurements; formal local errors are not fully calibrated future-epoch uncertainties. Same-sector QLP and TESScut reductions share photons;

different sectors are separate TESS observing occasions. Current depth values are realized central depths of the fitted event models, rather than triangular amplitudes extrapolated beyond the event center. The correction leaves timings, physical verdicts, secondary-event values and capable-neighbour sets unchanged.